A Guided Course in Algebraic Topology

from surfaces to configuration spaces

Improved edition

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How to use this book

What this book is. This is a guided course in algebraic topology for a reader who has taken a first course and wants to reach the point where research papers — in particular papers on configuration spaces — can be read critically and extended. It is not a collection of problems with answers. Its exercises are part of the exposition: they sit next to the theory they use, each is followed at once by guidance (a hint, a sequence of hints, or a solution), and many later arguments refer back to them by number. Reading an exercise and its solution without attempting it first throws away most of its value; reading neither throws away part of the argument.

How the book is built. The book has six parts.

- *Parts I–II and Chapters 15–16* revise the foundations: fundamental groups and covers, homology and cohomology computed by hand, exact sequences, homological algebra, cofibrations and fibrations, Whitehead and Postnikov, the Serre spectral sequence and its multiplicative tools. These chapters assume that the reader has met most of the definitions before; they concentrate on what a first course tends to leave implicit — which isomorphisms are canonical, which hypotheses are load-bearing, where torsion comes from, and how a long exact sequence or a spectral sequence is actually operated.
- *From Chapter 17 (obstruction theory) onwards* the book becomes a self-contained textbook: every definition used is stated, foundational theorems are proved, and deep theorems are stated precisely with an honest indication of what is proved here, what is sketched, and where a complete proof can be found.
- *Part VI* is an introduction to configuration spaces, written for a reader who has finished the rest of the book. It covers the classical computations, integral homology and torsion, rational models, homotopy invariance, homological stability and the operadic structures that dominate current research, and it ends with a map of open problems whose status was checked against the literature in October 2026.

Exercises, hints, solutions. Exercises carry a difficulty marker (*easy*, *medium*, *hard*), which measures the thinking required rather than the length of the answer. Under each exercise a short line records its SOURCE: either a textbook or course whose problem it reproduces or adapts (with exact section and number), or "original". Hints are always printed. Solutions are printed in the full edition; setting `\showsolutionsfalse` in `preamble.tex` produces a student edition with hints only. In the first sixteen chapters every exercise has a complete solution. In later chapters a few optional, research-flavoured problems are followed by an *outline*, explicitly labelled as not a complete solution.

The six spines. Six chains of ideas run through the book, and most exercises belong to one of them:

S1. *Non-naturality and the limits of invariants.* Each invariant is pushed until it fails, and the failure names the next one: free versus based homotopy (Chapter 1), homology and the universal cover (Chapter 5), unnatural splittings (Chapters 6 and 8), cup products (Chapter 7), homotopy groups (Chapter 13), Steenrod operations (Chapter 18), Massey products (Chapter 31).

S2. *Torsion and Bocksteins*, from the Klein bottle to the integral homology of configuration spaces.

S3. *Cells, connectivity and obstruction*, from van Kampen to obstruction theory and horn filling.

S4. *Fibrations and spectral sequences*, from the Wang sequence to the Fadell–Neuwirth fibrations of configuration spaces.

S5. *Algebra of the abstract kind*: derived functors, Kan extensions, model categories, operads.

S6. *Rational homotopy*: Hopf algebras, Sullivan models, formality, and the models of configuration spaces.

Prerequisites. Point-set topology, a first course in algebraic topology (fundamental group, covering spaces, singular homology, CW complexes), linear algebra, and basic group theory and commutative algebra (finitely generated abelian groups, modules over a principal ideal domain, exact sequences). Some familiarity with smooth manifolds helps in Chapters 12 and 26 and Part VI.

Conventions. Spaces are compactly generated where it matters and usually of the homotopy type of CW complexes; maps are continuous. $H_*(X)$ means singular homology with coefficients in $\mathbb{Z}$ unless a coefficient ring is indicated; $\mathbb{F}_p$ is the field with p elements. "Simply connected" includes path-connected. Paths are concatenated left to right ($\gamma \cdot \delta$ traverses γ first). For configuration spaces we write $F_k(M)$ for ordered and $B_k(M) = F_k(M)/\mathfrak{S}_k$ for unordered configurations of k points in M; $\mathfrak{S}_k$ is the symmetric group.

Citations. References are given sparingly, where they help the reader find a theorem, an exercise or a fuller treatment; each part ends with suggestions for further reading, and the guide to the literature at the end says what each standard reference is good for.

Part I

Spaces, groups, and the first invariants

Chapter 1

Basepoints, conjugacy, and the change-of-basepoint action

Every first course proves that a path from x_0 to x_1 induces an isomorphism $\pi_1(X, x_1) \to \pi_1(X, x_0)$, and then quietly starts writing $\pi_1(X)$. This chapter is about the word *quietly*. The isomorphism is not canonical: it depends on the path, and two paths differ by conjugation by a loop. That ambiguity is not a blemish to be removed. It is the first appearance of a structure that organises the whole subject — in Chapter 3 it becomes the statement that conjugate subgroups classify isomorphic covering spaces, and in Chapter 15 it returns as the monodromy action attached to a fibration.

Throughout the book we use the standard results of a first course without comment: $\pi_1(S^1) \cong \mathbb{Z}$, the van Kampen theorem, the existence and uniqueness of lifts against a covering space. What such a course tends to leave as folklore — exactly which isomorphisms are canonical, and what happens to a basepoint during a homotopy — is what we set down first, because every later chapter leans on it.

Conventions. A path is a map $\gamma\colon I \to X$; we write $\gamma(0)$ for its initial and $\gamma(1)$ for its terminal point. The concatenation $\gamma \cdot \delta$, defined when $\gamma(1) = \delta(0)$, traverses γ first. The reverse path is $\bar{\gamma}(s) = \gamma(1 - s)$, and c_x is the constant path at x. Homotopies of paths are always rel $\{0, 1\}$ unless we say otherwise; homotopies of maps never are.

1.1 The change-of-basepoint isomorphism

Definition 1.1. Let $\gamma\colon I \to X$ satisfy $\gamma(0) = x_0$ and $\gamma(1) = x_1$. Define

$$\beta_\gamma\colon \pi_1(X, x_1) \longrightarrow \pi_1(X, x_0), \qquad \beta_\gamma[\alpha] = [\gamma \cdot \alpha \cdot \bar{\gamma}].$$

Note the direction: β_γ runs *backwards* along γ, from the group at the far end to the group at the near end. Getting this consistently right is worth a minute now and saves an hour later.

Why this is the natural definition. A loop at x_1 is something that happens "over there"; to compare it with loops at x_0 we must bring it home. The only tool available is a path, and the only way to turn a loop at x_1 into a loop at x_0 using a path γ is to walk out along γ, do the loop, and walk back along $\bar{\gamma}$. Every property in Proposition 1.4 below is a consequence of the fact that "walk out and walk back" cancels up to homotopy, $\gamma \cdot \bar{\gamma} \simeq c_{x_0}$.

Remark 1.2 (The fundamental groupoid). The cleanest way to organise this chapter is to stop privileging basepoints altogether. The *fundamental groupoid* $\Pi_1(X)$ is the category whose objects are the points of X and whose morphisms from x to y are the homotopy classes rel endpoints of paths from x to y, composed by concatenation. Every morphism is invertible, the inverse of $[\gamma]$ being $[\bar\gamma]$, and the automorphism group of the object x_0 is $\pi_1(X, x_0)$. In this language β_γ is simply *conjugation by the morphism* $[\gamma]$: it sends a loop at x_1 to "$[\gamma]$, then the loop, then $[\gamma]^{-1}$". The three statements of Proposition 1.4 become the facts that conjugation by a morphism is well defined on homotopy classes, compatible with composition, and invertible. In a groupoid the automorphism groups of objects in one connected component are all isomorphic, but the isomorphism depends on the morphism chosen — exactly the phenomenon this chapter studies. The groupoid point of view returns in Chapter 3 (for reasonable X, covering spaces are the same thing as functors $\Pi_1(X) \to \mathbf{Set}$) and is the natural home of the local coefficient systems of Chapter 15.

Warning 1.3 (Two classic slips). First, β_γ is defined for any path, and it is an isomorphism $\pi_1(X, x_1) \to \pi_1(X, x_0)$ between two *different* groups; only when γ is a loop does it become an automorphism of one group, and only then can one ask whether it is the identity. Second, the formula $\beta_{\gamma\cdot\delta} = \beta_\gamma\beta_\delta$ requires $\gamma(1) = \delta(0)$; it is a statement about composable paths, and with the opposite convention for concatenation the order of the factors reverses. Books differ on this convention, and so do the formulas copied from them.

Proposition 1.4. *β_γ is a well-defined group isomorphism, and:*

(a) β_γ depends only on the homotopy class of γ rel $\{0,1\}$;

(b) $\beta_{\gamma\cdot\delta} = \beta_\gamma \circ \beta_\delta$ whenever the concatenation is defined;

(c) $\beta_{c_{x_0}} = \mathrm{id}$, and $\beta_\gamma^{-1} = \beta_{\bar\gamma}$.

Proof. Well-definedness and the homomorphism property are the usual computations: a homotopy α_t rel endpoints yields the homotopy $\gamma\cdot\alpha_t\cdot\bar\gamma$, and $(\gamma\alpha\bar\gamma)(\gamma\beta\bar\gamma) \simeq \gamma(\alpha\beta)\bar\gamma$ because $\bar\gamma\cdot\gamma \simeq c_{x_1}$. For (b), $\beta_{\gamma\delta}[\alpha] = [(\gamma\delta)\alpha(\overline{\gamma\delta})] = [\gamma(\delta\alpha\bar\delta)\bar\gamma] = \beta_\gamma\beta_\delta[\alpha]$, using $\overline{\gamma\delta} = \bar\delta\bar\gamma$. Part (c) is immediate, and with (b) it shows β_γ is invertible. Part (a) is Exercise 1.5. □

Exercise 1.5 (easy)**.** Prove Proposition 1.4(a): if $\gamma \simeq \gamma'$ rel $\{0,1\}$ then $\beta_\gamma = \beta_{\gamma'}$.

SOURCE: standard; the same statement is [1, §1.1, Ex. 2]

Solution. Let γ_t be a homotopy rel endpoints from γ to γ'. For a loop α at x_1, the family $\gamma_t\cdot\alpha\cdot\bar\gamma_t$ is a homotopy of loops at x_0; it is rel $\{0,1\}$ because every γ_t starts at x_0. It is continuous in (s,t): on each third of the parameter interval it is a reparametrisation of γ_t, of α or of $\bar\gamma_t$, and the pieces agree at $s = \frac13, \frac23$ because $\gamma_t(1) = x_1 = \alpha(0)$ for every t — this is exactly where "rel endpoints" is used. Hence $[\gamma\alpha\bar\gamma] = [\gamma'\alpha\bar\gamma']$, that is, $\beta_\gamma[\alpha] = \beta_{\gamma'}[\alpha]$ for every $[\alpha]$.

Exercise 1.6 (easy)**.** Show that concatenation of paths has the cancellation property: if $f_0\cdot g_0 \simeq f_1\cdot g_1$ rel endpoints and $g_0 \simeq g_1$ rel endpoints, then $f_0 \simeq f_1$ rel endpoints.

SOURCE: [1, §1.1, Ex. 1], reproduced in substance

Hint. *Concatenate both sides on the right with $\bar g_1$.*

Solution. Homotopy rel endpoints is compatible with concatenation (concatenate the homotopies). From $g_0 \simeq g_1$ we get $g_0\cdot\bar g_1 \simeq g_1\cdot\bar g_1 \simeq c$, the constant path at $g_1(0) = f_1(1)$. Hence

$$f_0 \simeq f_0\cdot c \simeq f_0\cdot(g_0\cdot\bar g_1) \simeq (f_0\cdot g_0)\cdot\bar g_1 \simeq (f_1\cdot g_1)\cdot\bar g_1 \simeq f_1\cdot(g_1\cdot\bar g_1) \simeq f_1,$$

each step being one of: the reparametrisation homotopies expressing the unit and associativity laws, the hypothesis $f_0 g_0 \simeq f_1 g_1$ concatenated with $\bar{g}_1$, or $g_1 \bar{g}_1 \simeq c$. All of these are rel endpoints. In the language of Remark 1.2: $[f_0][g_0] = [f_1][g_1]$ and $[g_0] = [g_1]$ in the groupoid, and composing on the right with $[g_1]^{-1}$ gives $[f_0] = [f_1]$.

The first thing to extract is that a *loop* changes nothing except by conjugation.

Lemma 1.7. *If γ is a loop at x_0, then β_γ is conjugation by $[\gamma]$:*

$$\beta_\gamma[\alpha] = [\gamma][\alpha][\gamma]^{-1} \qquad \text{for all } [\alpha] \in \pi_1(X, x_0).$$

Consequently $\beta_\gamma = \mathrm{id}$ if and only if $[\gamma]$ lies in the centre of $\pi_1(X, x_0)$.

Proof. The displayed formula is Definition 1.1 read in the group $\pi_1(X, x_0)$, since $[\bar{\gamma}] = [\gamma]^{-1}$. Conjugation by g is the identity map precisely when g commutes with every element. □

Example 1.8. Let $X = S^1 \vee S^1$ with $\pi_1(X, x_0) = F(a, b)$ free on the two circles. Taking $\gamma = a$, a loop at x_0, Lemma 1.7 gives $\beta_a[b] = [aba^{-1}] \neq [b]$: the group is free, so aba^{-1} and b are distinct words. The change-of-basepoint map is genuinely not the identity even when it is an automorphism of one and the same group.

Corollary 1.9. *Let X be path-connected. The isomorphism $\pi_1(X, x_1) \cong \pi_1(X, x_0)$ is independent of the choice of path for every pair x_0, x_1 if and only if $\pi_1(X, x_0)$ is abelian.*

Proof. Two paths γ, δ from x_0 to x_1 give $\beta_\gamma \circ \beta_\delta^{-1} = \beta_\gamma \circ \beta_{\bar{\delta}} = \beta_{\gamma \cdot \bar{\delta}}$ by Proposition 1.4, and $\gamma \cdot \bar{\delta}$ is a loop at x_0. By Lemma 1.7 this is conjugation by $[\gamma\bar{\delta}]$. So all choices agree exactly when every loop at x_0 acts trivially by conjugation, i.e. when the group is abelian; and conversely, every class arises as some $[\gamma\bar{\delta}]$ (take δ fixed and $\gamma = \lambda \cdot \delta$ for a loop λ). □

Example 1.10 (Where the ambiguity is invisible, and where it is not)**.** For the torus T^2, $\pi_1 = \mathbb{Z}^2$ is abelian, so by Corollary 1.9 the fundamental groups at any two points are identified *canonically*; one may safely speak of "the" class of a loop in $\pi_1(T^2)$. For the figure eight of Example 1.8 the identification $\pi_1(X, x_1) \cong \pi_1(X, x_0)$ is defined only up to inner automorphisms of $F(a, b)$, and the subgroup $\langle b \rangle$ at one basepoint corresponds, along different paths, to the different subgroups $\langle b \rangle$ and $\langle aba^{-1} \rangle$ at the other. Configuration spaces of points in the plane (Part VI) have fundamental groups — braid groups — with trivial centre modulo a single infinite cyclic subgroup, so the basepoint configuration must always be named there.

Remark 1.11. This is the precise content of the abuse of notation $\pi_1(X)$: for path-connected X the fundamental group is well defined up to isomorphism, and canonically well defined only up to *inner* automorphism. Any statement one makes about $\pi_1(X)$ without naming a basepoint must therefore be invariant under inner automorphisms — being abelian, being finite, having a given conjugacy class of subgroups. Being *equal* to a particular subgroup is not such a statement, which is exactly why Chapter 3 classifies covering spaces by conjugacy classes of subgroups and not by subgroups.

Exercise 1.12 (medium)**.** Let $A \subseteq X$ be a path-connected subspace containing x_0. Show that $\pi_1(A, x_0) \to \pi_1(X, x_0)$ is surjective if and only if every path in X with both endpoints in A is homotopic rel endpoints to a path in A.

Source: [1, §1.1, Ex. 13], reproduced in substance

Hint. *For "⇐" a loop at x_0 is a path with endpoints in A. For "⇒", given a path f from a_0 to a_1 in A, choose paths u, v in A from x_0 to a_0 and from x_0 to a_1, and look at the loop $u \cdot f \cdot \bar{v}$.*

Solution. ($\Leftarrow$) A loop f at x_0 is a path with endpoints in A, so it is homotopic rel endpoints to a path in A, i.e. to a loop in A at x_0. Thus $[f]$ is in the image.

($\Rightarrow$) Take a path f in X joining $a_0 \in A$ to $a_1 \in A$. Since A is path-connected, choose paths u from x_0 to a_0 and v from x_0 to a_1, both inside A. Then $u \cdot f \cdot \bar{v}$ is a loop at x_0 in X; by surjectivity it is homotopic rel endpoints to a loop g in A. Then, all homotopies being rel endpoints,

$$f \simeq \bar{u} \cdot (u \cdot f \cdot \bar{v}) \cdot v \simeq \bar{u} \cdot g \cdot v,$$

and $\bar{u} \cdot g \cdot v$ is a path in A from a_0 to a_1. (The first homotopy cancels $\bar{u}u$ and $\bar{v}v$ by the groupoid laws.)

Exercise 1.13 (medium)**.** Let $f\colon X \to Y$ be a map, and let γ join x_0 to x_1 in X. Show that the square

$$\begin{array}{ccc} \pi_1(X, x_1) & \xrightarrow{f_*} & \pi_1(Y, f(x_1)) \\ {\scriptstyle \beta_\gamma}\downarrow & & \downarrow{\scriptstyle \beta_{f\circ\gamma}} \\ \pi_1(X, x_0) & \xrightarrow[f_*]{} & \pi_1(Y, f(x_0)) \end{array}$$

commutes.

Source: [1, §1.1, Ex. 15]

Hint. *There is nothing to prove beyond applying f to a concatenation.*

Solution. $f_*\beta_\gamma[\alpha] = [f \circ (\gamma\alpha\bar{\gamma})] = [(f\gamma)(f\alpha)\overline{(f\gamma)}] = \beta_{f\gamma} f_*[\alpha]$, since f commutes with concatenation and reversal of paths.

1.2 Free versus based homotopy

A loop at x_0 is a map $(S^1, s_0) \to (X, x_0)$; two such may be compared with or without keeping the basepoint fixed. The two notions differ, and the difference is measured by exactly the maps of the previous section.

Definition 1.14. Loops f, g in X are *freely homotopic* if they are homotopic as maps $S^1 \to X$, with no condition on basepoints. They are *based homotopic* (at x_0) if $f(s_0) = g(s_0) = x_0$ and the homotopy may be taken to fix s_0 throughout, i.e. if $[f] = [g]$ in $\pi_1(X, x_0)$.

One direction is a construction, and it is worth doing explicitly because the same picture — shrinking a path away — reappears whenever a basepoint has to be moved.

Lemma 1.15. *Let h be a path from x_0 to x_1 and let α be a loop at x_1. Then the loop $h \cdot \alpha \cdot \bar{h}$ at x_0 is freely homotopic to α.*

Proof. For $t \in I$ let $h_t(s) = h\big(t + (1-t)s\big)$, the tail of h from $h(t)$ to x_1, and set $\varphi_t = h_t \cdot \alpha \cdot \bar{h}_t$, a loop based at $h(t)$. The map $(s,t) \mapsto \varphi_t(s)$ is continuous, $\varphi_0 = h \cdot \alpha \cdot \bar{h}$, and $\varphi_1 = c_{x_1} \cdot \alpha \cdot c_{x_1}$, which is α up to reparametrisation. This is a homotopy of maps $S^1 \to X$; its basepoint travels along h, which is precisely what a free homotopy is allowed to do. □

The other direction — what a free homotopy does to a based class — follows from a lemma about homotopies of arbitrary maps, which we will use far more often than the loop case.

Lemma 1.16 (Tracking the basepoint)**.** *Let $\varphi_t\colon X \to Y$, $t \in I$, be a homotopy, and let $h(t) = \varphi_t(x_0)$ be the path traced by the image of the basepoint. Then*

$$\varphi_{0*} = \beta_h \circ \varphi_{1*}\colon \pi_1(X, x_0) \longrightarrow \pi_1\big(Y, \varphi_0(x_0)\big).$$

Proof. Let α be a loop at x_0 and define $H\colon I \times I \to Y$ by $H(s,t) = \varphi_t(\alpha(s))$. Reading the four edges of the square: the bottom is $\varphi_0 \circ \alpha$, the top is $\varphi_1 \circ \alpha$, and both vertical edges are h, because $\alpha(0) = \alpha(1) = x_0$.

Now work in the square before mapping it. The bottom edge of $I \times I$ and the concatenation (left edge)·(top edge)·(reversed right edge) are two paths in $I \times I$ with the same endpoints, hence homotopic rel endpoints, $I \times I$ being convex. Composing such a homotopy with H gives a homotopy rel endpoints in Y from $\varphi_0 \circ \alpha$ to $h \cdot (\varphi_1 \circ \alpha) \cdot \bar{h}$. Therefore $\varphi_{0*}[\alpha] = \beta_h \varphi_{1*}[\alpha]$. □

The picture behind the proof. Draw the square $I \times I$ with s horizontal and t vertical. The homotopy H paints the bottom edge with the loop $\varphi_0\alpha$, the top edge with $\varphi_1\alpha$, and *both* vertical edges with the same path h, because the corners $(0,t)$ and $(1,t)$ both go to $\varphi_t(x_0)$. A square is contractible, so "bottom" is homotopic rel corners to "left, top, right backwards". That sentence is the lemma; everything else is reading off which path is which. The same square will reappear in Chapter 2 (where it produces the relation $tat^{-1} = f_*(a)$ in a mapping torus) and in Chapter 15 (where it produces the monodromy action of a fibration).

Remark 1.17. Two features of Lemma 1.16 are worth naming, because both get used. First, it is a statement about *any* homotopy, not only one between maps that agree somewhere. Second, the correction term β_h is completely explicit: if one can show the track h is nullhomotopic, or that the target group is abelian, the correction disappears.

Example 1.18. Let $X = Y = S^1$ and let φ_t be rotation by $2\pi t$, so $\varphi_0 = \varphi_1 = \mathrm{id}$. The track h is a loop generating $\pi_1(S^1) \cong \mathbb{Z}$, and Lemma 1.16 asserts $\mathrm{id} = \beta_h$. That is no contradiction: by Lemma 1.7, β_h is conjugation by $[h]$, and $\mathbb{Z}$ is abelian. The lemma has extracted nothing here — as it must not, since the conclusion is about a group in which conjugation is invisible. To see it do work one needs a space with a nonabelian fundamental group, which is the subject of Chapter 2.

The first serious payoff of Lemma 1.16 is that homotopy equivalences behave as one wants them to, even though a homotopy inverse cannot usually be chosen to respect basepoints.

Proposition 1.19. *If $f\colon X \to Y$ is a homotopy equivalence, then $f_*\colon \pi_1(X, x_0) \to \pi_1(Y, f(x_0))$ is an isomorphism for every x_0.*

Proof. Let g be a homotopy inverse. Apply Lemma 1.16 to a homotopy from $g \circ f$ to id_X: writing h for the track of x_0, we get $(gf)_* = \beta_h \circ \mathrm{id}_* = \beta_h$, an isomorphism $\pi_1(X, x_0) \to \pi_1(X, gf(x_0))$. Since $(gf)_* = g_* \circ f_*$ is injective, f_* is injective at x_0; since it is surjective, g_* is surjective at $f(x_0)$.

Both conclusions hold at every basepoint. Apply the same argument to a homotopy from $f \circ g$ to id_Y at the basepoint $f(x_0)$: it shows g_* is injective at $f(x_0)$. Hence

$g_*\colon \pi_1(Y, f(x_0)) \to \pi_1(X, gf(x_0))$ is an isomorphism, and from $(gf)_* = g_* f_*$ being an isomorphism we conclude that f_* is one too. □

Exercise 1.20 (medium)**.** Show that a loop α at x_0 is freely homotopic to a constant map if and only if $[\alpha] = 1$ in $\pi_1(X, x_0)$.

Hint. *One direction is trivial. For the other, a free homotopy from α to a constant map is a homotopy $\varphi_t\colon S^1 \to X$; feed it to Lemma 1.16, taking X there to be S^1.*

Solution. If $[\alpha] = 1$ the based nullhomotopy is in particular a free one.

Conversely, let $\varphi_t\colon S^1 \to X$ be a homotopy with $\varphi_0 = \alpha$ and φ_1 constant, and let h be the track of s_0. Let $\iota \in \pi_1(S^1, s_0)$ be the class of the identity loop. By Lemma 1.16,

$$[\alpha] = \varphi_{0*}(\iota) = \beta_h\big(\varphi_{1*}(\iota)\big).$$

But φ_1 is constant, so φ_{1*} is the trivial homomorphism and $\varphi_{1*}(\iota) = 1$. As β_h is a homomorphism, $[\alpha] = 1$.

Exercise 1.21 (easy)**.** Show that for a space X the following are equivalent:

(a) every map $S^1 \to X$ is (freely) homotopic to a constant map;

(b) every map $S^1 \to X$ extends to a map $D^2 \to X$;

(c) $\pi_1(X, x_0) = 0$ for every $x_0 \in X$.

Deduce that a space is simply connected if and only if any two maps $S^1 \to X$ are freely homotopic. (Here "simply connected" means path-connected with trivial π_1.)

Source: [1, §1.1, Ex. 5], reproduced in substance

Hint. *For (a)⇔(b) use that D^2 is the cone on S^1: a map out of $D^2 = S^1 \times I / S^1 \times \{1\}$ is a homotopy that ends at a constant. For (a)⇔(c) use Exercise 1.20.*

Solution. (a)⇔(b). The quotient map $q\colon S^1 \times I \to D^2$, $q(z,t) = (1-t)z$, collapses $S^1 \times \{1\}$ to the centre and is a homeomorphism elsewhere; it is a closed map from a compact space onto a Hausdorff space, hence a quotient map. A map $F\colon D^2 \to X$ therefore corresponds exactly to a map $H = F \circ q\colon S^1 \times I \to X$ that is constant on $S^1 \times \{1\}$, i.e. a homotopy from $H_0 = F|_{S^1}$ to a constant map. So $f\colon S^1 \to X$ extends over D^2 iff f is homotopic to a constant map.

(a)⇒(c). Let α be a loop at x_0, viewed as a map $S^1 \to X$. By (a) it is freely nullhomotopic, so by Exercise 1.20 $[\alpha] = 1$.

(c)⇒(a). Let $f\colon S^1 \to X$ and put $x_0 = f(s_0)$. Then f is a loop at x_0, and $[f] = 1 \in \pi_1(X, x_0)$ by (c); a based nullhomotopy is in particular a free one.

For the last sentence: if X is simply connected, any two maps $S^1 \to X$ are both homotopic to constant maps, and any two constant maps are homotopic because X is path-connected (move the constant along a path). Conversely, if all maps $S^1 \to X$ are homotopic then in particular all constant maps are homotopic to each other, so X is path-connected (the track of a homotopy between the constants at x and y is a path from x to y), and every map is homotopic to a constant, so (c) holds by the equivalence just proved.

Exercise 1.22 (medium)**.** Suppose $\pi_1(X, x_0)$ is abelian. Show that two loops at x_0 which are freely homotopic are homotopic rel x_0.

Hint. *Run the argument of Exercise 1.20 without assuming the second loop is constant, and then use Lemma 1.7.*

Solution. Let $\varphi_t\colon S^1 \to X$ be a free homotopy from f to g, both loops at x_0, and let h be the track of s_0. Then h is a *loop* at x_0, since φ_0 and φ_1 both send s_0 to x_0. By Lemma 1.16 applied to the generator ι,

$$[f] = \beta_h[g] = [h][g][h]^{-1},$$

the last equality by Lemma 1.7. If $\pi_1(X, x_0)$ is abelian the right side is $[g]$.

Exercise 1.23 (medium)**.** In $X = S^1 \vee S^1$ exhibit two loops at x_0 that are freely homotopic but not homotopic rel x_0. Deduce that the hypothesis in Exercise 1.22 cannot be dropped.

Hint. *Lemma 1.15 manufactures free homotopies out of nothing. Apply it with h a loop rather than a path, and use that $F(a,b)$ is free to see the two classes are different.*

Solution. Take $f = b$ and $g = a\cdot b\cdot \bar{a}$, both loops at x_0. They are freely homotopic by Lemma 1.15 with $h = a$, $\alpha = b$. They are not based homotopic: in the free group $F(a,b)$ the words b and aba^{-1} are distinct, because $F(a,b)$ has trivial centre — or simply because reduced words in a free group are unique. Since $\pi_1(S^1 \vee S^1)$ is nonabelian, this is consistent with Exercise 1.22, and shows the hypothesis there is doing work.

Exercise 1.24 (medium)**.** Let X be path-connected and let $f, g\colon X \to Y$ be homotopic maps. Fix $x_0 \in X$. Show that the subgroups $f_*\pi_1(X, x_0) \le \pi_1(Y, f(x_0))$ and $g_*\pi_1(X, x_0) \le \pi_1(Y, g(x_0))$ correspond to each other under β_h for a suitable path h. Deduce that when $f(x_0) = g(x_0) = y_0$ the two images are *conjugate* subgroups of $\pi_1(Y, y_0)$, and give an example in which they are conjugate but not equal.

SOURCE: original

Hint. *Apply Lemma 1.16. For the example, take $X = S^1$, $Y = S^1 \vee S^1$, and conjugate a loop around the wedge point.*

Solution. Let φ_t be a homotopy from f to g and let $h(t) = \varphi_t(x_0)$, a path from $f(x_0)$ to $g(x_0)$. By Lemma 1.16, $f_* = \beta_h \circ g_*$ on $\pi_1(X, x_0)$, hence $f_*\pi_1(X, x_0) = \beta_h\big(g_*\pi_1(X, x_0)\big)$. If $f(x_0) = g(x_0) = y_0$ then h is a loop at y_0 and β_h is conjugation by $[h]$ (Lemma 1.7), so the two image subgroups are conjugate.

Example: in $Y = S^1 \vee S^1$ with $\pi_1 = F(a,b)$, let $g\colon S^1 \to Y$ be the loop b and f the loop $a \cdot b \cdot \bar{a}$ (both based at the wedge point y_0). They are freely homotopic by Lemma 1.15. Their images are $\langle b\rangle$ and $\langle aba^{-1}\rangle$. These are conjugate, and they are distinct: an element of $\langle b\rangle$ is a reduced word b^n, while aba^{-1} is a different reduced word, so $aba^{-1} \notin \langle b\rangle$.

Exercise 1.25 (medium)**.** Define $f\colon S^1 \times I \to S^1 \times I$ by $f(\theta, s) = (\theta + 2\pi s, s)$, writing points of S^1 as angles. So f is the identity on both boundary circles.

(a) Show that f is homotopic to the identity through maps that are the identity on the boundary circle $S^1 \times \{0\}$.

(b) Show that f is *not* homotopic to the identity through maps that are the identity on both boundary circles.

SOURCE: [1, §1.1, Ex. 7], reproduced in substance

Hint. *For (a), twist by a smaller amount. For (b), look at the path $p(s) = (\theta_0, s)$ crossing the annulus, and at the loop $(f \circ p)\cdot \bar{p}$.*

Solution. (a) Put $f_t(\theta, s) = (\theta + 2\pi t s, s)$ for $t \in I$. This is continuous, $f_0 = \mathrm{id}$, $f_1 = f$, and $f_t(\theta, 0) = (\theta, 0)$ for all t.

(b) Suppose f_t were a homotopy from id to f with every f_t the identity on both boundary circles. Fix θ_0 and let $p(s) = (\theta_0, s)$, a path from $(\theta_0, 0)$ to $(\theta_0, 1)$. Then $t \mapsto f_t \circ p$ is a homotopy of paths from p to $f \circ p$, and it is rel endpoints because the endpoints lie on the boundary circles, which every f_t fixes. Hence $(f \circ p) \cdot \bar{p} \simeq p \cdot \bar{p} \simeq c$ rel endpoints, so the loop $\lambda = (f \circ p) \cdot \bar{p}$ at $(\theta_0, 0)$ would be nullhomotopic.

But the projection $S^1 \times I \to S^1$ is a homotopy equivalence, inducing an isomorphism $\pi_1(S^1 \times I) \cong \pi_1(S^1) = \mathbb{Z}$, and it carries λ to the loop that first runs $s \mapsto \theta_0 + 2\pi s$ (once around the circle) and then stays at θ_0 (the image of $\bar{p}$ is constant). That loop has degree $1 \neq 0$. Contradiction.

1.3 Homotopies of maps, and the centre

Lemma 1.7 says that β_γ is the identity exactly when $[\gamma]$ is central. So any construction that forces β_γ to be the identity produces central elements. That is a small observation with a long reach: it is the mechanism behind the fact that H-spaces have abelian fundamental group, and — one step further, which we leave for the reader to take when they meet it — behind the constraint on loops of self-maps.

Definition 1.26. An *H-space* is a space X with a basepoint e and a map $\mu\colon X \times X \to X$ satisfying $\mu(e, e) = e$, such that $\mu(e, -)$ and $\mu(-, e)$ are homotopic to id_X. It has a *strict unit* if $\mu(e, x) = \mu(x, e) = x$ for all x; topological groups are the standard example.

Proposition 1.27 (Eckmann–Hilton)**.** *Let X be an H-space with a strict unit e. Then $\pi_1(X, e)$ is abelian.*

Proof. Besides concatenation $*$, the set $\pi_1(X, e)$ carries a second operation: $[f] \odot [g] = [\mu \circ (f, g)]$, the pointwise product, which is well defined because μ takes homotopies to homotopies. The constant loop c_e is a two-sided unit for $\odot$, strictly: $\mu(c_e, g) = g$.

These two operations satisfy the interchange law

$$([f] * [g]) \odot ([p] * [q]) = ([f] \odot [p]) * ([g] \odot [q]),$$

in fact on the nose: both sides are the loop that runs $\mu(f, p)$ on $[0, \frac{1}{2}]$ and $\mu(g, q)$ on $[\frac{1}{2}, 1]$.

Now the Eckmann–Hilton argument. Writing 1 for the class of c_e, which is a unit for both operations,

$$[f] * [g] = ([f] \odot 1) * (1 \odot [g]) = ([f] * 1) \odot (1 * [g]) = [f] \odot [g],$$

so the operations coincide; and

$$[f] * [g] = (1 \odot [f]) * ([g] \odot 1) = (1 * [g]) \odot ([f] * 1) = [g] \odot [f] = [g] * [f].$$

□

Example 1.28. S^1, S^3 and every topological group are H-spaces with strict units, so all have abelian fundamental group. S^3 is simply connected, so the statement is empty there; the content will come in Chapter 10, where the same multiplication is used on cohomology rather than on π_1 and computes the degree of a power map.

Exercise 1.29 (easy)**.** For pointed spaces (X, x_0), (Y, y_0) and loops f at x_0, g at y_0, the loops $\tilde{f} = (f, c_{y_0})$ and $\tilde{g} = (c_{x_0}, g)$ in $X \times Y$ represent commuting elements of $\pi_1(X \times Y, (x_0, y_0))$, because that group is $\pi_1(X) \times \pi_1(Y)$. Construct an *explicit* homotopy rel endpoints from $\tilde{f} \cdot \tilde{g}$ to $\tilde{g} \cdot \tilde{f}$.

Source: [1, §1.1, Ex. 10], reproduced in substance

Hint. *The map $F\colon I \times I \to X \times Y$, $F(u, v) = (f(u), g(v))$, is defined on a square. Both $\tilde{f} \cdot \tilde{g}$ and $\tilde{g} \cdot \tilde{f}$ are images of edge paths around that square from $(0, 0)$ to $(1, 1)$.*

Solution. Let $F(u, v) = (f(u), g(v))$. The edge path "right along the bottom, then up the right side" of $I \times I$, from $(0, 0)$ via $(1, 0)$ to $(1, 1)$, is mapped by F to $(f, y_0) \cdot (x_0, g) = \tilde{f} \cdot \tilde{g}$, using $g(0) = y_0$ and $f(1) = x_0$. The edge path "up the left side, then right along the top" is mapped to $\tilde{g} \cdot \tilde{f}$. Explicitly, for $\lambda \in I$ let σ_λ be the path in $I \times I$ from $(0, 0)$ to $(1, 1)$ that is the straight segment from $(0, 0)$ to $(1 - \lambda, \lambda)$ followed by the straight segment to $(1, 1)$; then σ_0 and σ_1 are the two edge paths, and $(\lambda, s) \mapsto F(\sigma_\lambda(s))$ is a homotopy rel endpoints from $\tilde{f} \cdot \tilde{g}$ to $\tilde{g} \cdot \tilde{f}$ (up to reparametrisation of each edge path, which is itself a homotopy rel endpoints). Its endpoints stay at $F(0, 0) = (x_0, y_0) = F(1, 1)$.

Exercise 1.30 (medium)**.** Prove Proposition 1.27 without assuming the unit is strict, using Lemma 1.16 instead.

Hint. *Do not repair the Eckmann–Hilton argument; replace it. The multiplication gives a homomorphism $\mu_*\colon \pi_1(X \times X, (e, e)) \to \pi_1(X, e)$, and $\pi_1(X \times X, (e, e)) \cong \pi_1(X, e) \times \pi_1(X, e)$.*

Hint. *In a product group, $(u, 1)$ and $(1, v)$ commute. Evaluate μ_* on each of them separately — each evaluation is the induced map of a map homotopic to id_X, so Lemma 1.16 applies.*

Solution. Since $\mu(e, e) = e$ we have a homomorphism $\mu_*\colon \pi_1(X \times X, (e, e)) \to \pi_1(X, e)$, and the projections identify the source with $\pi_1(X, e) \times \pi_1(X, e)$.

Evaluate on the two axes. The class $([f], 1)$ is represented by the loop $s \mapsto (f(s), e)$, so

$$\mu_*([f], 1) = [\, s \mapsto \mu(f(s), e)\,] = \big(\mu(-, e)\big)_*[f].$$

Now $\mu(-, e) \simeq \mathrm{id}_X$, and the track of e under such a homotopy is a *loop* h' at e, because $\mu(e, e) = e$. By Lemma 1.16, $\big(\mu(-, e)\big)_* = \beta_{h'}$, an automorphism of $\pi_1(X, e)$. Likewise $\mu_*(1, [g]) = \beta_h[g]$ for a loop h.

In the group $\pi_1(X, e) \times \pi_1(X, e)$ the elements $([f], 1)$ and $(1, [g])$ commute, and μ_* is a homomorphism, so

$$\beta_{h'}[f] \cdot \beta_h[g] = \mu_*\big([f], [g]\big) = \beta_h[g] \cdot \beta_{h'}[f].$$

Given arbitrary $u, v \in \pi_1(X, e)$, apply this with $[f] = \beta_{h'}^{-1}(u)$ and $[g] = \beta_h^{-1}(v)$, which is legitimate because $\beta_{h'}$ and β_h are bijections. It gives $uv = vu$.

Remark 1.31. The second proof is shorter, needs no interchange law, and generalises: the only thing it used about μ is that it is a map $X \times X \to X$ restricting to something invertible on π_1 along each axis. Proposition 1.27 is kept because the interchange argument itself reappears — in Chapter 10, where the same two products act on cohomology instead.

Exercise 1.32 (medium)**.** Construct infinitely many retractions $S^1 \vee S^1 \to S^1$ onto the first circle that are pairwise not homotopic, even without fixing basepoints.

Source: [1, §1.1, Ex. 17], reproduced in substance

Hint. *Send the second circle around the first n times. To distinguish the maps, use Lemma 1.16: freely homotopic maps into a space with abelian π_1 induce the same homomorphism.*

Solution. Write $X = A \vee B$ with A the first circle, $\pi_1(X, x_0) = F(a, b)$, and $\pi_1(A, x_0) = \mathbb{Z}$. For $n \in \mathbb{Z}$ define $r_n \colon X \to A$ to be the identity on A and the degree-n map $z \mapsto z^n$ on B, identifying both circles with the unit circle with basepoint 1. Since both pieces send the wedge point to x_0, r_n is continuous, and it is a retraction. On π_1 it sends $a \mapsto 1$ and $b \mapsto n$.

Suppose $r_m \simeq r_n$ by a free homotopy, with track h of the wedge point, a loop in A (both maps send x_0 to x_0). By Lemma 1.16, $r_{m*} = \beta_h \circ r_{n*}$, and β_h is conjugation in the abelian group $\pi_1(A) = \mathbb{Z}$, hence the identity (Lemma 1.7). So $m = r_{m*}(b) = r_{n*}(b) = n$. Thus the r_n are pairwise non-homotopic.

Remark 1.33. One step remains unclimbed, and it is the reader's. Suppose a loop of self-maps of X starts and ends at the identity. Lemma 1.16 computes the induced map on π_1 in terms of the track of the basepoint; Lemma 1.7 converts the answer into a statement about the centre. Putting the two together constrains which loops can arise this way. It is a good test of whether the chapter has been absorbed: the argument is two lines and uses nothing that is not on this page.

1.4 What to carry forward

- β_γ (Definition 1.1) and its four formal properties (Proposition 1.4); for a loop it is conjugation (Lemma 1.7).
- The tracking lemma (Lemma 1.16): a homotopy that moves the basepoint changes the induced map by β_h, and the correction is explicit. This is used again in Chapter 11 for homotopy extension arguments and in Chapter 13 for maps between towers.
- Homotopy equivalences induce isomorphisms on π_1 (Proposition 1.19) — the reason we may compute with any convenient model of a space.
- Free and based homotopy of loops differ exactly by conjugation, in the two directions supplied by Lemma 1.15 and Lemma 1.16.

The ambiguity isolated here is about to become an organising principle rather than a nuisance. Chapter 2 builds spaces whose fundamental groups are nonabelian enough for it to be visible, and Chapter 3 shows that conjugacy classes of subgroups — not subgroups — are what covering spaces actually see.

Chapter 2

Presentations from gluing data: van Kampen and CW structures

A space presented by gluing instructions — a polygon with edge identifications, a cube with face identifications, a cylinder with its ends matched by a map — has a fundamental group presented by generators and relations, and the dictionary between the two presentations is mechanical. This chapter sets up that dictionary and then does what a first course usually has no time for: uses it on spaces where the answer is not obvious, and uses it backwards, to prove that certain maps *cannot* exist.

The mechanical part is worth internalising because it is the only computational access to π_1 we have. The subtle part is that the dictionary is not injective: several gluings give the same group, and π_1 will turn out not to determine the space. The chapter ends there, and Chapters 4 and 7 supply the finer invariants.

2.1 From cells to generators and relations

We freely use the van Kampen theorem, but since every computation in this chapter rests on it, it is worth having the precise statement in front of us.

Theorem 2.1 (van Kampen; recalled)**.** *Let $X = \bigcup_\alpha A_\alpha$ be a union of open, path-connected subspaces, each containing the basepoint x_0.*

(a) If every intersection $A_\alpha \cap A_\beta$ is path-connected, the homomorphism $\Phi\colon *_\alpha \pi_1(A_\alpha, x_0) \to \pi_1(X, x_0)$ from the free product, induced by the inclusions, is surjective.

(b) If moreover every triple intersection $A_\alpha \cap A_\beta \cap A_\gamma$ is path-connected, the kernel of Φ is the normal subgroup generated by the elements $i_{\alpha\beta}(w)\, i_{\beta\alpha}(w)^{-1}$, where $w \in \pi_1(A_\alpha \cap A_\beta, x_0)$ and $i_{\alpha\beta}$ is induced by $A_\alpha \cap A_\beta \hookrightarrow A_\alpha$.

In the case of two sets $X = A \cup B$ with $A \cap B$ path-connected this says $\pi_1(X) \cong \pi_1(A) *_{\pi_1(A\cap B)} \pi_1(B)$, the amalgamated free product. Three things are easy to forget. The sets must be *open* (or deformation retracts of open sets), so in practice one thickens a decomposition before applying the theorem. The intersections must be path-connected; for the circle written as the union of two arcs the intersection has two components and the theorem, applied naively, would give the trivial group. And the basepoint must lie in every piece, which is why decompositions are usually chosen around a common point. When the intersection is not connected, the right tool is the fundamental groupoid of

Remark 1.2 with several basepoints, or the HNN-extension description that Proposition 2.13 below produces for a mapping torus.

What we need in practice is the specialisation of Theorem 2.1 to CW complexes, and one normalisation.

Proposition 2.2. *Every connected CW complex is homotopy equivalent to a CW complex with a single* 0*-cell.*

For a graph this is the familiar move of collapsing a spanning tree; in general one collapses a spanning tree of the 1-skeleton. The justification — that collapsing a contractible subcomplex does not change the homotopy type — is a consequence of the homotopy extension property, and we prove it in Chapter 11. We use it now and pay for it later.

Proposition 2.3. *Let X be a connected CW complex with a single* 0*-cell p, with* 1*-cells $(a_i)_{i\in I}$ and* 2*-cells attached along loops w_j, $j \in J$. Then*

$$\pi_1(X,p) \;\cong\; \big\langle\, a_i \ (i \in I) \;\big|\; w_j \ (j \in J) \,\big\rangle,$$

where w_j is read as a word in the $a_i^{\pm 1}$ by recording the 1*-cells the attaching map traverses.*

Proof sketch. The 1-skeleton is a wedge of circles, with free fundamental group on the a_i. Attaching a 2-cell along w kills exactly the normal closure of $[w]$, by van Kampen applied to the union of X^1 (thickened) with the open cell. Cells of dimension ≥ 3 do not change π_1, again by van Kampen, because the attaching spheres are simply connected.

In more detail, let Y be path-connected and $Z = Y \cup_\varphi e^n$ with $\varphi\colon S^{n-1} \to Y$. Take $A = Z \setminus \{c\}$, where c is the centre of the cell, and B the open cell. Radial projection deformation retracts A onto Y; B is contractible; $A \cap B$ is an open cell minus its centre, homotopy equivalent to S^{n-1}. To have a common basepoint, join a point of $A \cap B$ to the basepoint of Y by a path in A and enlarge B by a thin neighbourhood of that path (this changes no homotopy type). For $n = 2$, $\pi_1(A \cap B) = \mathbb{Z}$ is generated by a small circle around c, which pushed radially out to the boundary becomes the loop φ conjugated by the connecting path; van Kampen gives $\pi_1(Z) = \pi_1(Y)/\langle\!\langle [\varphi] \rangle\!\rangle$. For $n \geq 3$, $A \cap B$ is simply connected and $\pi_1(Z) \cong \pi_1(Y)$. For infinitely many cells the same argument works one cell at a time, together with the fact that a loop or a homotopy of loops, being compact, meets only finitely many cells. □

Example 2.4 (The sphere with two points identified)**.** Let X be S^2 with its north and south poles identified. A CW structure: two 0-cells would be wasteful, so take one 0-cell p (the identified poles), one 1-cell e (a meridian arc from pole to pole, which becomes a loop), and one 2-cell (the complement of the arc, an open disc). The boundary of the disc runs along the arc in one direction and back in the other, so its attaching word is $e\,e^{-1}$, which is trivial in the free group. By Proposition 2.3, $\pi_1(X) = \langle e \mid ee^{-1} \rangle \cong \mathbb{Z}$. The attaching map is nullhomotopic in $X^1 = S^1$, so in fact $X \simeq S^1 \vee S^2$: attaching a cell along a nullhomotopic map is, up to homotopy, wedging on a sphere (compare [1, §1.2, Ex. 7]; this is Exercise 11.10 in disguise; the homotopy invariance of attaching along homotopic maps follows from the homotopy extension property of Chapter 11). The relator is visible to π_1 only through its being trivial; the extra sphere is visible to H_2 and π_2, not to π_1.

Remark 2.5. There is a small subtlety hidden in the phrase "read as a word". An attaching map $S^1 \to X^1$ is a *free* loop: it comes with no basepoint, so its class in

$\pi_1(X^1, p)$ is only well defined up to conjugation, and up to inversion if one reverses the orientation. This does no harm, because the normal closure of $[w]$ is unchanged by conjugating w. It does mean that a relator is only ever well defined up to conjugacy — which is the reason presentations coming from geometry so often look different from the ones in a group theory book. Compare Exercise 1.20: whether a relator is trivial does not depend on the basepoint either.

Example 2.6 (Surfaces)**.** The closed orientable surface M_g of genus $g \geq 1$ is the quotient of a $4g$-gon by the edge word $a_1 b_1 a_1^{-1} b_1^{-1} \cdots a_g b_g a_g^{-1} b_g^{-1}$. All $4g$ vertices are identified to a single point, so this is a CW structure with one 0-cell, $2g$ 1-cells and one 2-cell, and Proposition 2.3 gives

$$\pi_1(M_g) = \Big\langle a_1, b_1, \ldots, a_g, b_g \;\Big|\; \prod_{i=1}^{g} [a_i, b_i] \Big\rangle.$$

Abelianising kills the single relator, since it is a product of commutators, so the abelianisation is free abelian of rank $2g$. That one observation — the relator dies on abelianisation — is the whole reason the genus is visible to homology, and it is what makes Exercise 2.22 below possible.

Exercise 2.7 (easy)**.** Show that $\pi_1(M_g)$ surjects onto the free group F_g, and deduce that it is not abelian for $g \geq 2$. Conclude that $M_g \not\simeq M_h$ for $g \neq h$, and that M_g is not homotopy equivalent to a product of two circles for $g \geq 2$.

Source: original

Hint. *Send each b_i to 1. To distinguish M_g from M_h use abelianisation.*

Solution. Define ψ on generators by $a_i \mapsto a_i$, $b_i \mapsto 1$, into $F(a_1, \ldots, a_g)$. It kills the relator: $\psi\big(\prod [a_i, b_i]\big) = \prod a_i 1 a_i^{-1} 1 = 1$. So ψ is a well-defined homomorphism, and it is surjective since the a_i generate F_g. A quotient of an abelian group is abelian, and F_g is not abelian for $g \geq 2$ ($a_1 a_2 \neq a_2 a_1$), so $\pi_1(M_g)$ is not abelian.

By Example 2.6 the abelianisation of $\pi_1(M_g)$ is $\mathbb{Z}^{2g}$, so $M_g \simeq M_h$ forces $2g = 2h$ (homotopy equivalences induce isomorphisms on π_1 by Proposition 1.19, hence on abelianisations). And $\pi_1(S^1 \times S^1) = \mathbb{Z}^2$ is abelian, so $M_g \not\simeq S^1 \times S^1$ for $g \geq 2$.

Exercise 2.8 (medium)**.** Compute $G = \pi_1(\mathbb{RP}^2 \vee \mathbb{RP}^2)$. Show that G is infinite, although it is generated by two elements of order two, and that it contains an infinite cyclic subgroup of index two.

Source: original

Hint. *Use Proposition 2.3 or van Kampen. Every element of a free product $A * B$ is uniquely a reduced alternating word in nontrivial elements of A and B.*

Solution. $\mathbb{RP}^2 \vee \mathbb{RP}^2$ has a CW structure with one 0-cell, two 1-cells a, b and two 2-cells attached along a^2 and b^2, so $G = \langle a, b \mid a^2, b^2 \rangle = \mathbb{Z}/2 * \mathbb{Z}/2$. In a free product of two copies of $\mathbb{Z}/2$ the reduced words are the alternating words in a and b, and distinct reduced words are distinct elements. Hence $(ab)^n = abab \cdots ab$ (length $2n$) is nontrivial for $n \neq 0$: ab has infinite order and G is infinite.

Let $\varepsilon \colon G \to \mathbb{Z}/2$ send $a, b \mapsto 1$; it kills a^2, b^2, so it is a homomorphism, and its kernel has index two. A reduced word lies in the kernel iff it has even length, and the alternating words of even length are exactly $(ab)^k$ and $(ba)^k = (ab)^{-k}$, $k \geq 0$. So $\ker \varepsilon = \langle ab \rangle \cong \mathbb{Z}$. Geometrically, the universal cover of $\mathbb{RP}^2 \vee \mathbb{RP}^2$ is an infinite chain of 2-spheres joined at points, on which G acts by reflections — a picture worth drawing.

2.2 Mapping tori

Definition 2.9. Let $f\colon X \to X$ be a map. The *mapping torus* is

$$T_f = \big(X \times [0,1]\big)\big/\big((x,0) \sim (f(x),1)\big).$$

We fix this convention once and never change it: the *bottom* of the cylinder is glued into the *top* by f. Every sign and every direction below depends on it, and the commonest error in this subject is to prove a correct theorem about the other convention. Exercise 2.19 asks the reader to redo the computation with the roles exchanged, precisely so that the dependence is seen rather than feared.

The seam — the image of $X \times \{0\} \cup X \times \{1\}$ — is a copy of X, namely the image of $X \times \{1\}$, since every bottom point has been identified with a top point. We regard $X \subseteq T_f$ this way, and take the basepoint $p \in X$ with $f(p) = p$.

Exercise 2.10 (easy)**.** Let X be obtained from two tori $S^1 \times S^1$ by identifying the circle $S^1 \times \{x_0\}$ in one with the corresponding circle in the other. Compute $\pi_1(X)$, and decide whether it is abelian.

Source: [1, §1.2, Ex. 8], reproduced in substance

Hint. *Give each torus its standard structure (one* 0*-cell,* 1*-cells* a, b_i*, one* 2*-cell) with the common circle as the* 1*-cell* a*.*

Solution. Let the common circle be the 1-cell a with 0-cell $p = (x_0, x_0)$. The two tori contribute 1-cells b_1, b_2 (the circles $\{x_0\} \times S^1$) and 2-cells attached along $ab_1a^{-1}b_1^{-1}$ and $ab_2a^{-1}b_2^{-1}$. So X has one 0-cell, three 1-cells and two 2-cells, and Proposition 2.3 gives

$$\pi_1(X) = \langle a, b_1, b_2 \mid [a,b_1],\, [a,b_2]\rangle \cong \mathbb{Z} \times F(b_1, b_2),$$

the isomorphism sending a to the generator of $\mathbb{Z}$: the relations say exactly that a commutes with both b_i, and nothing else. The group is not abelian, since $F(b_1, b_2)$ is not ($b_1b_2 \neq b_2b_1$ as reduced words). Equivalently, by van Kampen, $\pi_1(X) = \mathbb{Z}^2 *_{\mathbb{Z}} \mathbb{Z}^2$ amalgamated along the common $\mathbb{Z} = \langle a \rangle$.

Exercise 2.11 (medium)**.** Let $L \subseteq \mathbb{R}^3$ be the union of $n \geq 1$ distinct lines through the origin. Compute $\pi_1(\mathbb{R}^3 \setminus L)$.

Source: [1, §1.2, Ex. 4], reproduced in substance

Hint. *Radial projection deformation retracts* $\mathbb{R}^3 \setminus L$ *onto the unit sphere minus the points where it meets* L*.*

Solution. The origin lies on L, so $\mathbb{R}^3 \setminus L \subseteq \mathbb{R}^3 \setminus \{0\}$, and the straight-line homotopy $H_t(x) = (1-t)x + t\,x/|x|$ stays inside $\mathbb{R}^3 \setminus L$: it moves x along the ray through x, and a ray from the origin either lies in L (excluded) or meets L only at 0. So $\mathbb{R}^3 \setminus L$ deformation retracts onto $S^2 \setminus (S^2 \cap L)$. Each line meets S^2 in two antipodal points, and distinct lines give disjoint pairs, so $S^2 \cap L$ consists of $2n$ points. Removing one point from S^2 leaves a plane (stereographic projection), so S^2 minus $2n$ points is $\mathbb{R}^2$ minus $2n-1$ points, which deformation retracts onto a wedge of $2n-1$ circles. Hence $\pi_1(\mathbb{R}^3 \setminus L) \cong F_{2n-1}$, the free group of rank $2n-1$.

Exercise 2.12 (medium)**.** Show that the complement of a finite set of points in $\mathbb{R}^n$ is simply connected if $n \geq 3$. What goes wrong for $n = 2$?

Source: [1, §1.2, Ex. 3], reproduced in substance

Hint. *Induct on the number k of points. Separate one point from the rest by a hyperplane and cover the complement by two overlapping open half-spaces.*

Solution. Write $Q = \{q_1, \dots, q_k\}$. For $k = 0$ there is nothing to prove, and for $k = 1$, $\mathbb{R}^n \setminus \{q_1\} \simeq S^{n-1}$ is simply connected because $n - 1 \geq 2$.

Let $k \geq 2$. Choose a linear coordinate x on $\mathbb{R}^n$ taking distinct values on the q_i (all but finitely many directions work), and number the points so that $x(q_k)$ is the largest. Choose c with $x(q_{k-1}) < c < x(q_k)$ and $\delta > 0$ so small that the slab $\{|x - c| < \delta\}$ contains no point of Q. Put

$$U = \{x < c + \delta\} \setminus Q, \qquad V = \{x > c - \delta\} \setminus Q.$$

These are open, $U \cup V = \mathbb{R}^n \setminus Q$, and $U \cap V$ is the slab, which is convex, hence path-connected and simply connected. The half-space $\{x < c+\delta\}$ is homeomorphic to $\mathbb{R}^n$ by a homeomorphism respecting the points, so U is homeomorphic to $\mathbb{R}^n$ minus $k - 1$ points and is simply connected by induction; likewise V is homeomorphic to $\mathbb{R}^n$ minus one point. Both are path-connected (a half-space minus finitely many points, $n \geq 2$). By Theorem 2.1, $\pi_1(\mathbb{R}^n \setminus Q)$ is a quotient of $\pi_1(U) * \pi_1(V) = 1$, hence trivial.

For $n = 2$ the base case fails: $\mathbb{R}^2 \setminus \{q\} \simeq S^1$ has $\pi_1 = \mathbb{Z}$, and the same induction (now with nontrivial groups but the same simply connected intersection) gives $\pi_1(\mathbb{R}^2 \setminus Q) \cong F_k$. This difference between $n = 2$ and $n \geq 3$ is the reason braid groups appear for configurations in the plane and not in higher dimensions.

Proposition 2.13. *Let X be a connected CW complex with a single 0-cell p, and let f be cellular with $f(p) = p$. Write $\pi_1(X, p) = \langle a_i \mid r_j \rangle$ as in Proposition 2.3. Then*

$$\pi_1(T_f, p) \;\cong\; \big\langle\, a_i,\, t \;\big|\; r_j,\; t\, a_i\, t^{-1} = f_*(a_i) \,\big\rangle,$$

where t is the loop $s \mapsto (p, s)$. If f_* is an automorphism this is the semidirect product $\pi_1(X, p) \rtimes_{f_*} \mathbb{Z}$.

Proof. Give T_f the CW structure whose cells are the cells of the seam X together with one cell $e \times (0, 1)$ for each cell e of X. There is one 0-cell, p; the 1-cells are the a_i together with $t = p \times (0, 1)$; the 2-cells are those of X, contributing the relators r_j, together with one square $a_i \times (0, 1)$ for each i.

It remains to read the boundary word of that square, and for this it is quicker to use Lemma 1.16 than to stare at the picture. Let α be a loop at p in X and consider the homotopy of loops

$$\varphi_u \colon S^1 \to T_f, \qquad \varphi_u(s) = \big(\alpha(s), u\big), \qquad u \in [0, 1].$$

Its track is the path $u \mapsto (p, u)$, which is the loop t. At the top, $u = 1$, the loop φ_1 is α sitting in the seam. At the bottom, $u = 0$, the loop φ_0 is $s \mapsto (\alpha(s), 0) = (f(\alpha(s)), 1)$, which is the loop $f \circ \alpha$ in the seam. So Lemma 1.16 reads

$$f_*[\alpha] = [\varphi_0] = \beta_t[\varphi_1] = [t][\alpha][t]^{-1},$$

using Lemma 1.7 for the last equality. Thus the square $a_i \times (0, 1)$ contributes exactly the relation $t\, a_i\, t^{-1} = f_*(a_i)$. Cells of dimension ≥ 3 do not affect π_1. □

Remark 2.14. The proposition is the algebraic shadow of a geometric fact: T_f fibres over S^1 with fibre X, and t is the monodromy. Conjugation by the monodromy *is* the induced map f_*. In Chapter 5 the same fibration is run through homology instead and produces the Wang sequence, with $1 - f_*$ in place of conjugation by t.

Example 2.15 (The degenerate case)**.** If $f = \mathrm{id}$ then $T_f = X \times S^1$ and the relations say that t is central: $\pi_1 = \pi_1(X) \times \mathbb{Z}$, as it must be. Always check a formula against the case where nothing happens.

Exercise 2.16 (easy)**.** Let $f\colon S^1 \to S^1$ have degree d. Show $\pi_1(T_f) \cong \langle a, t \mid tat^{-1} = a^d \rangle$. Identify the space for $d = 1$ and for $d = -1$.

Solution. $\pi_1(S^1) = \langle a \mid \rangle$ and $f_*(a) = a^d$, so Proposition 2.13 gives the presentation at once. For $d = 1$ the group is $\mathbb{Z}^2$ and $T_f = S^1 \times S^1$ by Example 2.15. For $d = -1$ the relation is $tat^{-1} = a^{-1}$ and T_f is the Klein bottle.

Exercise 2.17 (medium)**.** Let $G = \langle a, t \mid tat^{-1} = a^{-1} \rangle$ be the fundamental group of the Klein bottle from Exercise 2.16.

(a) Show that every element of G can be written uniquely as $a^m t^n$ with $m, n \in \mathbb{Z}$, and that $G \cong \mathbb{Z} \rtimes \mathbb{Z}$ with t acting by -1.

(b) Show that G is not abelian, that t^2 is central, and that the subgroup $\langle a, t^2 \rangle$ has index two and is isomorphic to $\mathbb{Z}^2$.

(c) Compute the abelianisation of G.

SOURCE: original

Hint. *Use Proposition 2.13: f_* is the automorphism $a \mapsto a^{-1}$ of $\mathbb{Z}$, so G is a semidirect product. In $\mathbb{Z} \rtimes \mathbb{Z}$, compute $(m, n)(m', n')$.*

Solution. (a) By Proposition 2.13 with $f_* = -1$ invertible, $G \cong \mathbb{Z} \rtimes_\phi \mathbb{Z}$, where $\phi(n)$ is multiplication by $(-1)^n$; the element (m, n) corresponds to $a^m t^n$, and the multiplication is $(m, n)(m', n') = (m + (-1)^n m',\, n + n')$, which is the rule $t^n a^{m'} = a^{(-1)^n m'} t^n$ obtained by iterating $ta = a^{-1}t$. Every element of a semidirect product $N \rtimes Q$ is uniquely a product of an element of N and an element of Q, so the normal form $a^m t^n$ exists and is unique.

(b) $ta = a^{-1}t \neq at$, because by uniqueness of normal forms $a^{-1}t = at$ would force $-1 = 1$. The relation $t^2 a = a^{(-1)^2} t^2 = at^2$ shows t^2 commutes with a, and obviously with t, so it is central. The map $G \to \mathbb{Z}/2$, $a^m t^n \mapsto n \bmod 2$, is a homomorphism (the t-exponents add); its kernel is $\{a^m t^{2k}\} = \langle a, t^2 \rangle$, of index two. On this kernel the multiplication is $(m, 2k)(m', 2k') = (m + m', 2k + 2k')$, so $a^m t^{2k} \mapsto (m, k)$ is an isomorphism with $\mathbb{Z}^2$.

(c) Abelianising, the relation $tat^{-1} = a^{-1}$ becomes $a = -a$, i.e. $2a = 0$, so $G^{\mathrm{ab}} = \langle a, t \mid 2a \rangle \cong \mathbb{Z}/2 \oplus \mathbb{Z}$, with a generating the torsion. This is the group $H_1(K)$ computed three more times in this book (Examples 4.2, 5.16 and 15.12).

Exercise 2.18 (medium)**.** Let $f\colon \mathbb{RP}^2 \to \mathbb{RP}^2$ be any map with $f(p) = p$ that induces an isomorphism on π_1. Show that $\pi_1(T_f) \cong \mathbb{Z}/2 \times \mathbb{Z}$, whatever f is.

Hint. *What are the automorphisms of $\mathbb{Z}/2$?*

Solution. $\pi_1(\mathbb{RP}^2) = \mathbb{Z}/2 = \langle a \mid a^2 \rangle$, whose only automorphism is the identity, so $f_*(a) = a$. Proposition 2.13 gives $\langle a, t \mid a^2,\ tat^{-1} = a \rangle$, i.e. $\mathbb{Z}/2 \times \mathbb{Z}$.

Exercise 2.19 (medium)**.** Redo the proof of Proposition 2.13 for the opposite convention $(x, 1) \sim (f(x), 0)$ and show that the relation becomes $t^{-1} a_i t = f_*(a_i)$. Conclude that the two conventions give presentations differing by $t \leftrightarrow t^{-1}$, and that the resulting groups are isomorphic.

Solution. With the other convention the seam is the image of $X \times \{0\}$, and in the homotopy $\varphi_u(s) = (\alpha(s), u)$ it is now φ_0 that is α in the seam while φ_1 is $f \circ \alpha$. Lemma 1.16 gives $[\alpha] = \beta_t\big(f_*[\alpha]\big) = [t]f_*[\alpha][t]^{-1}$, that is $t^{-1}at = f_*(a)$. Substituting $t \mapsto t^{-1}$ carries one presentation to the other, so the groups are isomorphic; geometrically, the two mapping tori are homeomorphic by reversing the I coordinate.

Exercise 2.20 (hard)**.** Let $G = \langle a, t \mid tat^{-1} = a^2 \rangle$, the group of Exercise 2.16 with $d = 2$, and let $\varphi\colon G \to \mathbb{Z}$ be the homomorphism with $\varphi(t) = 1$, $\varphi(a) = 0$.

(a) Show that $\ker\varphi$ is the increasing union of the cyclic groups generated by $a_n := t^{-n}at^n$, and that $a_{n-1} = a_n^2$.

(b) Deduce $\ker\varphi \cong \mathbb{Z}[\frac{1}{2}]$, the additive group of dyadic rationals. In particular $\ker\varphi$ is not finitely generated.

(c) Conclude that $\pi_1(T_f)$ is *not* of the form $\pi_1(S^1) \rtimes \mathbb{Z}$, even though T_f fibres over S^1 with fibre S^1.

Hint. *For (a), conjugate the defining relation by powers of t. For (b), the tower $\mathbb{Z} \xrightarrow{2} \mathbb{Z} \xrightarrow{2} \cdots$ has a familiar colimit.*

Hint. *For (c), a degree-2 map $S^1 \to S^1$ is not a homotopy equivalence, so nothing forces the inclusion of the fibre to be injective with normal image; use (b) to see that it is not even surjective onto* $\ker\varphi$.

Solution. (a) Set $a_n = t^{-n}at^n$, so $a_0 = a$. Conjugating $tat^{-1} = a^2$ by t^{-n} gives $t^{-(n-1)}at^{n-1} = \big(t^{-n}at^n\big)^2$, that is $a_{n-1} = a_n^2$. Each a_n lies in $\ker\varphi$, and $\langle a_{n-1}\rangle \subseteq \langle a_n\rangle$. Every element of $\ker\varphi$ is a word in the $t^{-n}a^{\pm1}t^n$, hence lies in some $\langle a_n\rangle$; so $\ker\varphi = \bigcup_n \langle a_n\rangle$, and the union is increasing.

(b) The subgroup $\langle a_n\rangle$ is infinite cyclic and contains $\langle a_{n-1}\rangle$ with index 2. So $\ker\varphi$ is the colimit of $\mathbb{Z} \xrightarrow{2} \mathbb{Z} \xrightarrow{2} \cdots$, which is $\mathbb{Z}[\frac{1}{2}]$: send $a_n \mapsto 2^{-n}$. A finitely generated subgroup would lie in some $\langle a_n\rangle$ and be proper.

(c) In a semidirect product $N \rtimes \mathbb{Z}$ with quotient map to $\mathbb{Z}$, the kernel is N. Here the kernel is $\mathbb{Z}[\frac{1}{2}]$, not $\mathbb{Z} = \pi_1(S^1)$. The image of π_1(fibre) is the subgroup $\langle a\rangle = \langle a_0\rangle$, which is proper in the kernel and not normal in G — indeed $t^{-1}at = a_1 \notin \langle a_0\rangle$.

2.3 Retractions, and how to rule them out

The presentations above are useful in the negative direction too. The basic principle occupies one line.

Lemma 2.21. *If $A \subseteq X$ and $r\colon X \to A$ is a retraction, then $i_*\colon \pi_1(A,a) \to \pi_1(X,a)$ is injective and* split*: there is a homomorphism r_* with $r_*i_* = \mathrm{id}$. The same holds after abelianising.*

Proof. Apply π_1 to $r \circ i = \mathrm{id}_A$; functoriality gives $r_*i_* = \mathrm{id}$, and abelianisation is a functor. □

Injectivity alone is a weak conclusion, and the temptation is to look for its failure. Usually it does not fail; what fails is splitting.

Exercise 2.22 (medium)**.** Let M be the Möbius band and ∂M its boundary circle. Show that $\pi_1(\partial M) \to \pi_1(M)$ is injective, but that M does not retract onto ∂M.

Hint. *M deformation retracts onto its core circle c. Under that equivalence, what does the boundary circle become?*

Solution. The deformation retraction onto the core gives $\pi_1(M) = \mathbb{Z}\langle c\rangle$, and the boundary circle runs twice around the core, so $i_*\colon \mathbb{Z} \to \mathbb{Z}$ is multiplication by 2. That is injective, so no injectivity obstruction exists.

But a retraction would give $r_*\colon \mathbb{Z} \to \mathbb{Z}$ with $r_*(2x) = x$ for all x; taking $x = 1$ forces $2r_*(1) = 1$ in $\mathbb{Z}$, which is impossible. Hence there is no retraction.

The positive direction needs a way to turn a homomorphism into a map. For targets that are aspherical this is possible, and for S^1 it is elementary.

Proposition 2.23. *Let X be a connected CW complex with basepoint p. Every homomorphism $\varphi\colon \pi_1(X, p) \to \mathbb{Z}$ is induced by a map $g\colon X \to S^1$.*

Proof sketch. Normalise X to have a single 0-cell (Proposition 2.2) and send it to the basepoint of S^1. Send the 1-cell a_i to a loop of degree $\varphi(a_i)$. Over a 2-cell with attaching word w_j, the composite $S^1 \to X^1 \to S^1$ has degree $\varphi(w_j) = 0$ because φ is a homomorphism killing the relators; a degree-zero self-map of S^1 is nullhomotopic, so the map extends over the cell. For cells of dimension $n \geq 3$ the obstruction to extending lies in $\pi_{n-1}(S^1) = 0$, so the extension continues. □

Remark 2.24. The last two sentences of that proof are obstruction theory in embryo: extend skeleton by skeleton, and at each stage the obstruction is a homotopy class of the attaching map composed with what is already built. Chapter 11 makes the first half precise (the extension lemma) and Chapter 17 makes all of it precise, with the obstructions organised into cohomology classes.

Exercise 2.25 (medium)**.** Let K be the Klein bottle, with $\pi_1(K) = \langle a, b \mid abab^{-1}\rangle$, and let $A \subseteq K$, $B \subseteq K$ be the circles representing a and b.

(a) Show K does not retract onto A.

(b) Show that there is a map $K \to S^1$ restricting to a degree-one map on B.

Hint. *Abelianise the presentation. The relator $abab^{-1}$ becomes $2a$.*

Solution. Abelianising, $\pi_1(K)^{\mathrm{ab}} = \langle a, b \mid 2a\rangle \cong \mathbb{Z}/2 \oplus \mathbb{Z}$, with a generating the torsion summand.

(a) A retraction onto A would give, after abelianising (Lemma 2.21), a homomorphism $\mathbb{Z}/2 \oplus \mathbb{Z} \to \mathbb{Z}$ sending the class of a to a generator of $\mathbb{Z}$. But the class of a has order 2 and $\mathbb{Z}$ is torsion-free, so its image is 0. Contradiction. (Note that a itself has infinite order in $\pi_1(K)$, so once again injectivity is not the obstruction.)

(b) Define $\varphi\colon \pi_1(K) \to \mathbb{Z}$ by $\varphi(a) = 0$, $\varphi(b) = 1$. This respects the relator: $\varphi(abab^{-1}) = 0 + 1 + 0 - 1 = 0$. By Proposition 2.23 there is $g\colon K \to S^1$ inducing φ, and its restriction to B induces multiplication by 1 on $\pi_1(S^1)$, hence has degree one.

Remark 2.26. Part (b) produces a map that is a retraction *up to homotopy*: $g|_B$ is homotopic to a homeomorphism $B \to S^1$, not necessarily equal to one. Upgrading a homotopy to an equality is exactly what the homotopy extension property does, and Chapter 11 will let us convert such a g into an honest retraction. Until then we are careful to claim only what has been proved.

Exercise 2.27 (medium)**.** Let $(p, q) \in \mathbb{Z}^2$ be primitive, i.e. $\gcd(p, q) = 1$, and let $C \subseteq T^2$ be the image of the line through the origin of slope (p, q) in $T^2 = \mathbb{R}^2/\mathbb{Z}^2$, a circle. Show that there is a map $g\colon T^2 \to C$ whose restriction to C is homotopic to the identity of C. What goes wrong if (p, q) is not primitive?

SOURCE: original

Hint. *Find $\varphi\colon \mathbb{Z}^2 \to \mathbb{Z}$ with $\varphi(p,q) = 1$; this is where primitivity enters. Realise φ by Proposition 2.23 and compose with an identification $S^1 \cong C$.*

Solution. Since $\gcd(p,q) = 1$ there are integers r, s with $sp - rq = 1$. Let $\varphi(x,y) = sx - ry$, a homomorphism $\pi_1(T^2) = \mathbb{Z}^2 \to \mathbb{Z}$ with $\varphi(p,q) = 1$. The circle C is the image of $u \mapsto u\,(p,q)$, $u \in [0,1]$, which is a loop representing $(p,q) \in \pi_1(T^2)$; it is embedded because (p,q) is primitive (the points $u(p,q)$, $0 \le u < 1$, are distinct mod $\mathbb{Z}^2$). Let $h\colon S^1 \to C$ be this homeomorphism.

By Proposition 2.23 there is $G\colon T^2 \to S^1$ inducing φ. Then $G \circ h\colon S^1 \to S^1$ induces $1 \mapsto \varphi(p,q) = 1$ on π_1, so it has degree one and is homotopic to id_{S^1}. Put $g = h \circ G$. Then $g|_C = h \circ G \circ h \circ h^{-1} \simeq h \circ h^{-1} = \mathrm{id}_C$. (In fact g can be chosen explicitly: $G(x,y) = sx - ry \bmod 1$ is already a map $T^2 \to \mathbb{R}/\mathbb{Z}$ inducing φ.)

If $(p,q) = d(p',q')$ with $d \ge 2$, the loop $u \mapsto u(p,q)$ is not embedded (it runs d times around the circle C' of slope (p',q')), and the analogous question for its image C' asks for φ with $\varphi(p',q') = 1$ — the primitive case again. But no homomorphism $\mathbb{Z}^2 \to \mathbb{Z}$ sends (p,q) to 1, since $\varphi(p,q) = d\varphi(p',q')$ is divisible by d: a loop representing a non-primitive class can never be the core of a retraction up to homotopy.

2.4 Identification complexes in dimension three

The dictionary of Proposition 2.3 applies verbatim to solids with identified faces, provided one first does the bookkeeping: which vertices become the same point, which edges become the same edge. That step is where the errors happen, and it is worth doing slowly on an example where the answer is known.

Example 2.28 (The three-torus)**.** Let X be the cube I^3 with each face identified with the opposite face by the perpendicular translation.

Vertices. The translations identify all eight vertices to one point: the three translations generate a group acting transitively on the vertices of the cube modulo the identifications, and following any vertex under the three translations reaches every other. So there is one 0-cell.

Edges. The twelve edges fall into three orbits of four, one for each coordinate direction: the four edges parallel to the x-axis are identified by the y- and z-translations. Call the resulting 1-cells a, b, c.

Faces. The six faces are identified in opposite pairs, leaving three 2-cells. The face perpendicular to the z-axis is a square whose boundary reads $aba^{-1}b^{-1}$, and similarly for the others.

So X has one 0-cell, three 1-cells, three 2-cells and one 3-cell, and Proposition 2.3 gives $\pi_1(X) = \langle a, b, c \mid [a,b], [b,c], [a,c] \rangle \cong \mathbb{Z}^3$, as it must, since $X = S^1 \times S^1 \times S^1$.

Remark 2.29. Two features of Example 2.28 are accidents of the translations. First, all vertices became one point; second, each edge orbit had exactly four members. Compose the identifications with a rotation of the face and both counts can change: a rotation carries an edge of one face to a *different* edge of the opposite face, so the edge orbits are permuted in longer cycles and the vertex orbits can split. Any such computation must therefore begin by listing the vertex orbits and then the edge orbits, in that order, before a single relation is written down. Example 2.30 is a first case where the answer is not $\mathbb{Z}^n$.

Example 2.30 (Lens spaces)**.** Fix coprime integers $p > q \geq 1$. Realise S^3 as the unit sphere in $\mathbb{C}^2$ and let $\mathbb{Z}/p$ act by

$$\zeta \cdot (z_1, z_2) = (\zeta z_1,\ \zeta^q z_2), \qquad \zeta = e^{2\pi i/p}.$$

The action is free, so the quotient $L(p, q)$ is a closed 3-manifold with universal cover S^3, and since S^3 is simply connected, $\pi_1(L(p, q)) \cong \mathbb{Z}/p$.

Equivalently, and this is the picture to keep: $L(p, q)$ is the ball D^3 with its upper hemisphere identified with its lower hemisphere, by the rotation through $2\pi q/p$ about the vertical axis followed by reflection in the equatorial plane. That description gives a CW structure with a single cell in each of the four dimensions, and $L(2, 1) = \mathbb{RP}^3$.

Exercise 2.31 (medium)**.** Verify the cell counts in the ball model of $L(p, q)$: the boundary sphere becomes one 2-cell whose attaching map runs p times around a circle built from one 0-cell and one 1-cell. Deduce the presentation $\pi_1 = \langle a \mid a^p \rangle$ directly, without using the covering space.

Hint. *The rotation by $2\pi q/p$ has order p on the equator, so the equator's marked points fall into orbits of size p; a single 1-cell survives, and the boundary of the 2-cell winds around it once for each of the p sectors.*

Solution. Mark p equally spaced points on the equator of D^3, cutting it into p arcs. The gluing rotates by $2\pi q/p$, which permutes the marked points cyclically and permutes the arcs cyclically; so all p marked points become a single 0-cell and all p arcs a single 1-cell a. The upper hemisphere is glued to the lower, so the two hemispheres become a single 2-cell, and its attaching map runs once along each of the p arcs, i.e. traverses a exactly p times. The interior of the ball is the 3-cell. Proposition 2.3 now gives $\langle a \mid a^p \rangle = \mathbb{Z}/p$, agreeing with Example 2.30.

Remark 2.32 (Where this chapter stops)*.* The lens spaces expose the limit of the method. For fixed p, every $L(p, q)$ has fundamental group $\mathbb{Z}/p$, so π_1 cannot tell them apart. Nor can homology: we will compute in Chapter 4 that $H_*(L(p, q))$ is also independent of q. Yet they are genuinely different spaces — $L(5, 1)$ and $L(5, 2)$ are not even homotopy equivalent. Detecting that needs a product structure rather than a group structure, and we return to it in Chapter 7.

This is the first instance of the pattern that runs through the book: build an invariant, compute with it until it fails, and let the failure name the next invariant.

Remark 2.33 (Further reading)*.* Van Kampen and cell structures: [1, §1.2, §1.B]. Presentations and HNN extensions, which is what Proposition 2.13 produces when f_* is not invertible: Lyndon and Schupp [3]. The classification of lens spaces quoted in Remark 2.32: see Warning 7.19.

2.5 What to carry forward

- Cells to presentations (Proposition 2.3), and the licence to assume a single 0-cell (Proposition 2.2, proved in Chapter 11).
- Relators are well defined only up to conjugacy (Remark 2.5) — the chapter-one ambiguity, showing up as a computational convenience.
- The mapping torus presentation (Proposition 2.13), derived from the tracking lemma, and the warning that it is a semidirect product only when f_* is invertible (Exercise 2.20).

- Retractions split π_1 (Lemma 2.21); abelianisation turns that into an arithmetic obstruction, and Proposition 2.23 turns a homomorphism to $\mathbb{Z}$ back into a map.
- Orbit bookkeeping for face identifications: vertices first, then edges, then relations (Example 2.28 and Remark 2.29).

Chapter 3

Covering spaces as a Galois theory

The classification of covering spaces is the first place where the ambiguity of Chapter 1 becomes an organising principle: covers correspond to subgroups, and isomorphic covers correspond to *conjugate* subgroups, because a cover has no preferred basepoint upstairs. Everything in this chapter is an instance of the same dictionary, and the point of the exercises is to make the dictionary usable in both directions — reading a group off a space, and reading a space off a group.

We take the classification itself as known and concentrate on what a first course usually has no time for: computing with it.

3.1 The dictionary

Theorem 3.1 (Classification of covers; recalled)**.** *Let X be path-connected, locally path-connected and semilocally simply connected, with basepoint x_0.*

(a) Sending $(\widetilde{X}, \tilde{x}_0) \mapsto p_*\pi_1(\widetilde{X}, \tilde{x}_0)$ is a bijection between isomorphism classes of path-connected pointed covers and subgroups of $\pi_1(X, x_0)$.

(b) Forgetting the basepoint upstairs, isomorphism classes of path-connected covers correspond to conjugacy classes *of subgroups.*

(c) The cover corresponding to H has deck transformation group $N(H)/H$, where $N(H)$ is the normaliser. It is a normal cover exactly when $H \trianglelefteq \pi_1(X, x_0)$, and then the deck group is $\pi_1(X, x_0)/H$.

Part (b) is the reason Corollary 1.9 was worth proving. A different choice of basepoint upstairs changes H by conjugation and nothing else; a cover, being a space rather than a pointed space, therefore remembers only the conjugacy class.

Two results from a first course do all the work behind Theorem 3.1, and it is worth having them stated, because every computation below is an application of one of them.

Proposition 3.2 (Lifting criterion; recalled)**.** *Let $p\colon (\widetilde{X}, \tilde{x}_0) \to (X, x_0)$ be a covering and $f\colon (Y, y_0) \to (X, x_0)$ a map with Y path-connected and locally path-connected. Then f lifts to $\tilde{f}\colon (Y, y_0) \to (\widetilde{X}, \tilde{x}_0)$ if and only if $f_*\pi_1(Y, y_0) \subseteq p_*\pi_1(\widetilde{X}, \tilde{x}_0)$. A lift with prescribed value at one point is unique when Y is connected.*

Proposition 3.3 (Covers as π_1-sets; recalled)**.** *Let X be as in Theorem 3.1. For a covering $p\colon \widetilde{X} \to X$, the group $\pi_1(X, x_0)$ acts on the fibre $p^{-1}(x_0)$: a loop γ sends $\tilde{x}$ to the endpoint of the lift of γ starting at $\tilde{x}$ (a right action with our convention for concatenation). This gives an equivalence between coverings of X and sets with a right $\pi_1(X, x_0)$-action. Connected covers correspond to transitive actions; the stabiliser of $\tilde{x}$ is $p_*\pi_1(\widetilde{X}, \tilde{x})$; and moving $\tilde{x}$ within the fibre conjugates the stabiliser.*

Proposition 3.3 is the form of the theory that generalises. A covering is the simplest case of a fibration, and the way loops in the base permute the fibre is the simplest case of a *monodromy* action: in Chapter 15 the same loops act on the homology of the fibre of an arbitrary fibration, and the resulting *local coefficient system* is exactly what makes the Klein bottle's spectral sequence produce torsion. From the groupoid viewpoint of Remark 1.2, Proposition 3.3 says that a cover is a functor from $\Pi_1(X)$ to sets, and a local coefficient system will be a functor from $\Pi_1(X)$ to abelian groups.

Example 3.4 (The model case)**.** $\mathbb{Z}$ acts on $\mathbb{R}$ by translation and $\mathbb{R}/\mathbb{Z} = S^1$. The subgroups of $\pi_1(S^1) = \mathbb{Z}$ are the $n\mathbb{Z}$, and the corresponding covers are $\mathbb{R}/n\mathbb{Z} \to \mathbb{R}/\mathbb{Z}$, the n-fold covers of the circle, together with $\mathbb{R}$ itself for the trivial subgroup. Every subgroup is normal, every cover is normal, and the deck group of the n-fold cover is $\mathbb{Z}/n\mathbb{Z}$. Nothing in this example is ambiguous, because the group is abelian — which is exactly why it is a poor guide to the general case.

Exercise 3.5 (easy)**.** Classify the connected two-fold covers of $S^1 \vee S^1$ up to isomorphism, and draw them.

Hint. *An index-two subgroup of a group G is the kernel of a surjection $G \to \mathbb{Z}/2$, and index-two subgroups are automatically normal.*

Solution. Surjections $F(a, b) \to \mathbb{Z}/2$ are determined by the images of a and b and must be nonzero, so there are three: $(1, 0)$, $(0, 1)$, $(1, 1)$. Each kernel is normal of index two, and distinct kernels give non-isomorphic covers because the subgroups are distinct and normal, hence in distinct conjugacy classes.

Each cover has two vertices and four edges. For $(1, 0)$: the a-edges join the two vertices, and each vertex carries a b-loop. For $(0, 1)$ the roles are exchanged. For $(1, 1)$: both the a-edges and the b-edges join the two vertices, giving a circle of four edges (alternately a and b).

Exercise 3.6 (medium)**.** Let $H \leq F(a, b)$ be the subgroup corresponding to the three-fold cover with vertices v_1, v_2, v_3 in which a acts by the cycle $(v_1 v_2 v_3)$ and b fixes every vertex. Compute the deck group. Then modify the example so that the deck group is trivial.

Hint. *Deck transformations of a connected cover are determined by where they send one point, and they permute the fibre. A deck transformation exists carrying $\tilde{x}$ to $\tilde{x}'$ exactly when the two corresponding subgroups agree.*

Solution. The described cover is the one associated with the action of $F(a, b)$ on $\{v_1, v_2, v_3\}$ by $a \mapsto (v_1 v_2 v_3)$, $b \mapsto \mathrm{id}$; it is the cover corresponding to the stabiliser H of v_1. Since the image of $F(a, b)$ in S_3 is the cyclic group generated by a 3-cycle, which is transitive and abelian, all point stabilisers coincide, so H is normal and the deck group is $\mathbb{Z}/3$.

For a trivial deck group, take instead the action $a \mapsto (v_1 v_2)$, $b \mapsto (v_1 v_3)$. The image is all of S_3, and the stabiliser of v_1 is a non-normal subgroup of index three; its normaliser is itself, so $N(H)/H$ is trivial and the cover admits no nontrivial deck transformation.

Exercise 3.7 (medium)**.** Construct a nonnormal covering of the Klein bottle K by a torus, and a nonnormal covering of K by a Klein bottle.

SOURCE: [1, §1.3, Ex. 20], reproduced in substance

Hint. *Write $G = \pi_1(K) = \{a^m t^n\}$ as in Exercise 2.17 and let $N = \langle a, t^2\rangle \cong \mathbb{Z}^2$. Conjugation by t acts on N by $(m,k) \mapsto (-m,k)$ in the coordinates $a^m t^{2k} \leftrightarrow (m,k)$. A subgroup of N is normal in G iff it is invariant under this map. For the Klein bottle cover, try $\langle a^3, t\rangle$.*

Solution. *A torus.* Since N is abelian and $G = N \cup tN$, a subgroup $H \le N$ is normal in G exactly when $tHt^{-1} = H$, i.e. when H is invariant under $\sigma(m,k) = (-m,k)$ (indeed $ta^m t^{2k} t^{-1} = a^{-m}t^{2k}$). Let $H \le N$ be the lattice spanned by $(1,1)$ and $(0,3)$; it has index $\det\left(\begin{smallmatrix}1 & 1\\ 0 & 3\end{smallmatrix}\right) = 3$ in N, hence index 6 in G, and $H \cong \mathbb{Z}^2$. Now $\sigma(1,1) = (-1,1)$, and $(-1,1) = \lambda(1,1) + \mu(0,3)$ forces $\lambda = -1$, $3\mu = 2$, impossible in $\mathbb{Z}$. So H is not normal. The corresponding six-sheeted cover $\widetilde{K}$ is a closed surface (a finite cover of a closed surface is one) with $\pi_1 \cong \mathbb{Z}^2$, hence a torus.

A Klein bottle. Let $H = \langle a^3, t\rangle$. Since $ta^3t^{-1} = a^{-3}$, $H = \{a^{3j}t^n\}$, which has index 3 (the cosets are $a^r H$, $r = 0,1,2$). It is not normal: $ata^{-1} = a \cdot (ta^{-1}t^{-1}) \cdot t = a \cdot a \cdot t = a^2 t \notin H$. The corresponding three-sheeted cover is a closed surface with fundamental group $H \cong \langle a', t \mid ta't^{-1} = a'^{-1}\rangle$, $a' = a^3$, the Klein bottle group; since it is non-orientable (the loop t reverses orientation) and has Euler characteristic $3\chi(K) = 0$, it is a Klein bottle.

Exercise 3.8 (easy)**.** Given a group G and a normal subgroup N, show that there is a normal covering $\widetilde{X} \to X$ of connected CW complexes with $\pi_1(X) \cong G$, $\pi_1(\widetilde{X}) \cong N$, and deck group G/N.

SOURCE: [1, §1.3, Ex. 17], reproduced in substance

Hint. *Build a 2-complex from a presentation of G (Proposition 2.3) and apply Theorem 3.1.*

Solution. Choose a presentation $G = \langle S \mid R\rangle$ (e.g. $S = G$ and R the multiplication table). Let X be the CW complex with one 0-cell, one 1-cell for each $s \in S$ and one 2-cell attached along each word $r \in R$. By Proposition 2.3, $\pi_1(X) \cong G$. CW complexes are locally contractible, so the hypotheses of Theorem 3.1 hold, and there is a connected cover $\widetilde{X} \to X$ with $p_*\pi_1(\widetilde{X}) = N$; p_* is injective (covers induce injections on π_1), so $\pi_1(\widetilde{X}) \cong N$. Since N is normal the cover is normal with deck group G/N by Theorem 3.1(c). The cells of X lift to give $\widetilde{X}$ a CW structure.

3.2 Covers of graphs

Graphs make the dictionary completely computable, because for them π_1 is free and its rank is an Euler characteristic.

Proposition 3.9. *A connected graph Γ is homotopy equivalent to a wedge of circles, and if Γ is finite then $\pi_1(\Gamma)$ is free of rank $1 - \chi(\Gamma) = E - V + 1$. If $\Gamma' \to \Gamma$ is a k-sheeted cover then $\chi(\Gamma') = k\,\chi(\Gamma)$.*

Proof. Collapsing a spanning tree is a homotopy equivalence (Proposition 2.2) and leaves one vertex and $E - V + 1$ edges, so π_1 is free of that rank by Proposition 2.3. A k-sheeted cover has kV vertices and kE edges. □

Example 3.10 (An index-two subgroup, with a basis)**.** Let $\varphi\colon F(a,b) \to \mathbb{Z}/2$ send $a \mapsto 1$, $b \mapsto 0$, and let $H = \ker\varphi$. The corresponding cover is the first graph of Exercise 3.5: vertices v_0, v_1; two a-edges, from v_0 to v_1 and from v_1 to v_0; a b-loop at each vertex.

Here $V = 2$, $E = 4$, so H is free of rank 3. To name a basis, take the spanning tree consisting of the a-edge from v_0 to v_1. The three remaining edges give three loops at v_0, obtained by travelling through the tree, crossing the edge, and returning:

$$b, \qquad a\,b\,a^{-1}, \qquad a^2.$$

So $H = \langle b,\ aba^{-1},\ a^2 \rangle$, free on those three elements. Note that the rank went *up*: a proper subgroup of finite index in F_2 always has rank greater than 2, by Exercise 3.11. Subgroups of free groups are not smaller; they are freer.

Exercise 3.11 (medium)**.** Let $H \leq F_r$ have index $k < \infty$. Show H is free of rank $1 + k(r-1)$.

Hint. *Realise F_r as π_1 of a wedge of r circles and H as the fundamental group of a k-sheeted cover; then use Proposition 3.9.*

Solution. The wedge of r circles has $\chi = 1 - r$. The cover corresponding to H is a graph with $\chi = k(1-r)$, and H is its fundamental group, hence free of rank $1 - k(1-r) = 1 + k(r-1)$.

Exercise 3.12 (medium)**.** Deduce that F_2 contains subgroups isomorphic to F_n for every finite n.

Solution. Apply Exercise 3.11 with $r = 2$: a subgroup of index k is free of rank $k+1$. Index-k subgroups exist for every k — for instance the preimage of the trivial subgroup under $F_2 \to \mathbb{Z}/k$, $a \mapsto 1$, $b \mapsto 0$. Taking $k = n - 1$ gives F_n.

The finite-index formula is seductive, and the next exercise is the antidote.

Exercise 3.13 (hard)**.** Let $\varphi\colon F(a,b) \to \mathbb{Z}$ send $a \mapsto 1$ and $b \mapsto 0$, and let $K = \ker \varphi$.

(a) Describe the covering graph of $S^1 \vee S^1$ corresponding to K.

(b) Show that K is free of infinite rank, with basis $\{\, a^n b a^{-n} \ :\ n \in \mathbb{Z} \,\}$.

(c) Conclude that a finitely generated group can have a subgroup that is not finitely generated, and that the formula of Exercise 3.11 has no infinite-index analogue.

Hint. *The cover has one vertex for each coset of K, i.e. one for each integer. Where do the a-edges go, and where do the b-edges go?*

Hint. *Choose the spanning tree to be the whole line of a-edges. What is left?*

Solution. (a) The vertices are the cosets Ka^n, indexed by $n \in \mathbb{Z}$. The edge labelled a leaving the vertex n ends at $n+1$, since $\varphi(a) = 1$; the edge labelled b leaving n returns to n, since $\varphi(b) = 0$. So the cover is a line of a-edges with one b-loop attached at every integer.

(b) The line of a-edges is a spanning tree: it is connected, contains every vertex, and has no circuit. It is contractible, so collapsing it is a homotopy equivalence (Proposition 2.2), and what remains is a wedge of one circle for each leftover edge — that is, one for each b-loop, indexed by $n \in \mathbb{Z}$. Hence K is free on that set. Travelling out along the tree to the vertex n, round the loop, and back gives the element $a^n b a^{-n}$.

(c) F_2 is generated by two elements, while K has infinite rank, so K is not finitely generated. If a formula of the shape "rank $= 1 + k(r-1)$" held for infinite index it would have to give a finite answer for every subgroup of a finitely generated free group, which (b) contradicts.

Remark 3.14. Exercise 3.13 is the easy member of a family. The same method — take the cover determined by a surjection $F_2 \to A$ onto an abelian group, choose a spanning tree in the resulting graph, and read the leftover edges — computes a free basis for the kernel in every case. The kernel of $F_2 \to \mathbb{Z}^2$ is the commutator subgroup, and the picture there is the integer grid in the plane. That computation is a good test of the method once this chapter is finished; the only new difficulty is choosing a spanning tree of the grid.

3.3 Group actions and their quotients

A covering space is often presented not by a subgroup but by a group action, and the translation between the two is worth having explicitly.

Definition 3.15. An action of a group G on a space Y is a *covering space action* if every $y \in Y$ has a neighbourhood U such that the translates gU, $g \in G$, are pairwise disjoint.

Proposition 3.16. *Let Y be path-connected and locally path-connected, and let G act on it by a covering space action. Then $Y \to Y/G$ is a normal covering with deck group G, and there is a short exact sequence*

$$1 \longrightarrow \pi_1(Y, y_0) \longrightarrow \pi_1(Y/G, \bar{y}_0) \longrightarrow G \longrightarrow 1.$$

In particular, if Y is simply connected then $\pi_1(Y/G) \cong G$.

Proof sketch. Disjointness of the translates makes the quotient map a covering, and G acts by deck transformations transitively on each fibre, so the cover is normal with deck group G by Theorem 3.1(c). The exact sequence is the long exact sequence of the covering: the map to G sends a loop to the deck transformation carrying the start of its lift to the end. □

Example 3.17. The $\mathbb{Z}/p$ action on S^3 of Example 2.30 is free, and a free action of a finite group on a Hausdorff space is a covering space action. Since S^3 is simply connected, $\pi_1(L(p,q)) \cong \mathbb{Z}/p$ — the computation of Exercise 2.31, now obtained without a single cell.

Exercise 3.18 (medium)**.** Classify the connected covering spaces of the torus T^2.

Hint. *$\pi_1(T^2) = \mathbb{Z}^2$ is abelian, so conjugacy classes of subgroups are subgroups, and every cover is normal. Every subgroup of $\mathbb{Z}^2$ is free abelian of rank 0, 1 or 2.*

Solution. Write $T^2 = \mathbb{R}^2/\mathbb{Z}^2$. Subgroups $\Lambda \leq \mathbb{Z}^2$ come in three kinds.

- Rank 2: Λ has finite index k, and the cover is the torus $\mathbb{R}^2/\Lambda \to \mathbb{R}^2/\mathbb{Z}^2$, a k-sheeted cover of T^2 by T^2.
- Rank 1: $\Lambda = \mathbb{Z}v$ for a nonzero vector v (any multiple of a primitive vector); the cover is $\mathbb{R}^2/\Lambda \cong S^1 \times \mathbb{R}$, an infinite cylinder.
- Rank 0: the universal cover $\mathbb{R}^2$.

Since $\mathbb{Z}^2$ is abelian, all of these are normal covers, with deck group $\mathbb{Z}^2/\Lambda$. Every connected cover arises this way: the universal cover of T^2 is $\mathbb{R}^2$ with deck group $\mathbb{Z}^2$ acting by translations, and by Theorem 3.1 the cover corresponding to Λ is the intermediate quotient $\mathbb{R}^2/\Lambda$. Distinct subgroups give non-isomorphic covers because conjugacy classes in an abelian group are single subgroups.

Exercise 3.19 (easy)**.** Let a group G act on a Hausdorff space Y freely and properly discontinuously: every y has a neighbourhood U with $\{g : U \cap gU \neq \varnothing\}$ finite. Show that the action is a covering space action in the sense of Definition 3.15. In particular, a free action of a finite group on a Hausdorff space is a covering space action.

SOURCE: [1, §1.3, Ex. 23], reproduced in substance

Hint. *Let $g_1, \dots, g_r$ be the nonidentity elements with $U \cap g_i U \neq \varnothing$. Use the Hausdorff property to separate y from each $g_i y$ and shrink U.*

Solution. Fix y and U as in the hypothesis, and let $g_1, \dots, g_r$ be the elements $g \neq e$ with $U \cap gU \neq \varnothing$. Since the action is free, $g_i y \neq y$, and by the Hausdorff property there are disjoint open sets $V_i \ni y$, $W_i \ni g_i y$. Put

$$U' = U \cap \bigcap_{i=1}^{r} \left(V_i \cap g_i^{-1} W_i\right),$$

an open neighbourhood of y (note $g_i^{-1} W_i \ni y$). Then $g_i U' \subseteq W_i$ and $U' \subseteq V_i$, so $U' \cap g_i U' = \varnothing$ for each i; and for $g \neq e$ not among the g_i, $U' \cap gU' \subseteq U \cap gU = \varnothing$. Finally if $g_1 U' \cap g_2 U' \neq \varnothing$ with $g_1 \neq g_2$, then $U' \cap g_1^{-1} g_2 U' \neq \varnothing$, which we have just excluded. So the translates of U' are pairwise disjoint. For finite G proper discontinuity is automatic.

Exercise 3.20 (medium)**.** Let $n \geq 2$. The group $\mathbb{Z}/2$ acts on $F_2(\mathbb{R}^n) = \{(x, y) \in \mathbb{R}^n \times \mathbb{R}^n : x \neq y\}$ by swapping the coordinates; write $B_2(\mathbb{R}^n)$ for the quotient.

(a) Show that $(x, y) \mapsto \left(\frac{x+y}{2},\, y - x\right)$ is a $\mathbb{Z}/2$-equivariant homeomorphism $F_2(\mathbb{R}^n) \cong \mathbb{R}^n \times (\mathbb{R}^n \setminus \{0\})$, where $\mathbb{Z}/2$ acts on the right by $(c, v) \mapsto (c, -v)$.

(b) Deduce $F_2(\mathbb{R}^n) \simeq S^{n-1}$ and $B_2(\mathbb{R}^n) \simeq \mathbb{RP}^{n-1}$, and compute $\pi_1(B_2(\mathbb{R}^n))$ for all $n \geq 2$.

SOURCE: original

Hint. *For (b), the homotopy $(c, v) \mapsto \big((1-t)c,\, v/((1-t) + t|v|)\big)$ is equivariant. Then use Proposition 3.16 and Exercise 3.19.*

Solution. (a) The map $\Phi(x, y) = ((x + y)/2,\, y - x)$ is linear and invertible on $\mathbb{R}^n \times \mathbb{R}^n$, with inverse $(c, v) \mapsto (c - v/2,\, c + v/2)$; it carries the diagonal $\{x = y\}$ onto $\{v = 0\}$, so it restricts to a homeomorphism $F_2(\mathbb{R}^n) \cong \mathbb{R}^n \times (\mathbb{R}^n \setminus \{0\})$. Swapping x, y fixes $(x + y)/2$ and negates $y - x$, so Φ is equivariant for the stated action.

(b) The homotopy $H_t(c, v) = \big((1-t)c,\, v/((1-t) + t|v|)\big)$ is well defined (the denominator is positive), commutes with $v \mapsto -v$, starts at the identity and ends at the retraction onto $\{0\} \times S^{n-1}$, which it fixes throughout. So $F_2(\mathbb{R}^n)$ equivariantly deformation retracts onto S^{n-1} with the antipodal action, and passing to quotients (an equivariant deformation retraction descends to the orbit spaces) $B_2(\mathbb{R}^n) \simeq S^{n-1}/\pm = \mathbb{RP}^{n-1}$.

The swap is free on $F_2(\mathbb{R}^n)$ and $\mathbb{Z}/2$ is finite, so by Exercise 3.19 and Proposition 3.16 there is an exact sequence $1 \to \pi_1(F_2(\mathbb{R}^n)) \to \pi_1(B_2(\mathbb{R}^n)) \to \mathbb{Z}/2 \to 1$. For $n \geq 3$, $F_2(\mathbb{R}^n) \simeq S^{n-1}$ is simply connected and $\pi_1(B_2(\mathbb{R}^n)) \cong \pi_1(\mathbb{RP}^{n-1}) \cong \mathbb{Z}/2$. For $n = 2$, $B_2(\mathbb{R}^2) \simeq \mathbb{RP}^1 \cong S^1$, so $\pi_1(B_2(\mathbb{R}^2)) \cong \mathbb{Z}$, and the exact sequence reads $0 \to \mathbb{Z} \xrightarrow{2} \mathbb{Z} \to \mathbb{Z}/2 \to 0$: the generator of $\pi_1(B_2(\mathbb{R}^2))$ is the loop in which the two points exchange places by a half-turn, and its square is the full twist, which generates $\pi_1(F_2(\mathbb{R}^2))$.

Exercise 3.21 (medium)**.** Let $\varphi\colon \mathbb{R}^2 \to \mathbb{R}^2$, $\varphi(x, y) = (2x, y/2)$, and let $\mathbb{Z}$ act on $X = \mathbb{R}^2 \setminus \{0\}$ by powers of φ. Show that this is a covering space action, compute $\pi_1(X/\mathbb{Z})$, and show that $X/\mathbb{Z}$ is not Hausdorff.

SOURCE: [1, §1.3, Ex. 25], first three parts, reproduced in substance

Hint. *For the covering condition, use $|x|$ near points off the y-axis and $|y|$ near points off the x-axis. For π_1, use Proposition 3.16 and the fact that φ is isotopic to the identity through the maps $(x,y)\mapsto(2^s x, 2^{-s}y)$. For non-Hausdorffness, compare the images of $(1,0)$ and $(0,1)$.*

Solution. *Covering space action.* Let $(x_0,y_0)\neq 0$, say $x_0\neq 0$. Let U be the set of points of X with $|x|\in(|x_0|/\sqrt{2},\sqrt{2}|x_0|)$. For $n\neq 0$, $\varphi^n(U)$ consists of points with $|x|\in(2^n|x_0|/\sqrt{2},2^n\sqrt{2}|x_0|)$, an interval disjoint from the previous one because $2^{|n|}\geq 2$. So the translates of U are pairwise disjoint. If $x_0=0$ then $y_0\neq 0$ and the same argument with $|y|$ works.

Fundamental group. X is path-connected and locally path-connected, so Proposition 3.16 gives $1\to\pi_1(X,x_0)\to\pi_1(X/\mathbb{Z},\bar{x}_0)\to\mathbb{Z}\to 1$. Take $x_0=(1,1)$, let c be the circle of radius $\sqrt{2}$ through x_0, a loop generating $\pi_1(X,x_0)\cong\mathbb{Z}$, and let $q\colon X\to X/\mathbb{Z}$ be the quotient map. Put $\varphi_s(x,y)=(2^s x,2^{-s}y)$ for $s\in[0,1]$ and let $\tau(s)=\varphi_s(x_0)$, a path from x_0 to $\varphi(x_0)$; then $t=q\circ\tau$ is a loop at $\bar{x}_0$ which maps to the generator of $\mathbb{Z}$ (it lifts to a path from x_0 to $\varphi(x_0)$). Now $s\mapsto q\circ\varphi_s\circ c$ is a free homotopy of loops in $X/\mathbb{Z}$ from $q\circ c$ to $q\circ\varphi\circ c=q\circ c$, whose track is t. By Lemma 1.16 and Lemma 1.7, $[qc]=t\,[qc]\,t^{-1}$: the elements $[qc]$ and t commute. The image of $\pi_1(X,x_0)$ is generated by $[qc]$, and together with t it generates $\pi_1(X/\mathbb{Z})$ (the extension is generated by a generator of the kernel and a lift of a generator of the quotient). Hence $\pi_1(X/\mathbb{Z})$ is abelian, an extension of $\mathbb{Z}$ by $\mathbb{Z}$, so $\pi_1(X/\mathbb{Z})\cong\mathbb{Z}^2$.

Not Hausdorff. Let $p=(1,0)$ and $q=(0,1)$; they lie in different orbits (orbits of points on an axis stay on that axis). Let $U\ni p$ and $V\ni q$ be open in X. For large n, U contains $(1,2^{-n})$ and V contains $(2^{-n},1)$, and $\varphi^{-n}(1,2^{-n})=(2^{-n},1)$. So the saturations $\bigcup_n\varphi^n U$ and $\bigcup_n\varphi^n V$ meet, and the images of p and q have no disjoint neighbourhoods in $X/\mathbb{Z}$.

Exercise 3.22 (medium)**.** Show that the closed orientable surface M_g has a connected normal covering space with deck group $\mathbb{Z}^n$ if and only if $n\leq 2g$.

SOURCE: [1, §1.3, Ex. 19], first sentence, reproduced in substance

Hint. *A normal cover with deck group D corresponds to a surjection $\pi_1(M_g)\to D$. If D is abelian, such a surjection factors through the abelianisation.*

Solution. By Theorem 3.1(c), connected normal covers with deck group D correspond to normal subgroups N with $\pi_1(M_g)/N\cong D$, i.e. to surjections $\pi_1(M_g)\twoheadrightarrow D$. For $D=\mathbb{Z}^n$ abelian, every homomorphism $\pi_1(M_g)\to\mathbb{Z}^n$ factors through $\pi_1(M_g)^{\mathrm{ab}}=\mathbb{Z}^{2g}$ (Example 2.6), and is surjective iff the induced map $\mathbb{Z}^{2g}\to\mathbb{Z}^n$ is. A surjection $\mathbb{Z}^{2g}\to\mathbb{Z}^n$ exists iff $n\leq 2g$: if $n\leq 2g$ project onto the first n coordinates; conversely a surjection becomes, after $\otimes\mathbb{Q}$, a surjective linear map $\mathbb{Q}^{2g}\to\mathbb{Q}^n$, forcing $n\leq 2g$.

Remark 3.23 (Where this chapter stops). Covering space theory is a complete theory of π_1 and nothing else. It cannot distinguish two simply connected spaces, and it says nothing whatever about π_n for $n\geq 2$ — indeed a covering induces isomorphisms on all higher homotopy groups, which is what makes the universal cover such a useful place to work later (Chapter 13).

The next two chapters therefore change invariant. Chapter 4 builds homology, which sees simply connected spaces; and Chapter 5 produces two spaces with the same homology whose universal covers differ, so that the two invariants of Part I are seen to be genuinely independent.

Remark 3.24 (Further reading). The classification and lifting theorems with complete proofs: [1, §1.3]; graphs and subgroups of free groups: [1, §1.A] and Serre's *Trees* [5]. The fundamental groupoid viewpoint of Remark 1.2, in which Proposition 3.3 becomes an equivalence of categories: May [4, Ch. 2–3].

3.4 What to carry forward

- Covers correspond to subgroups, and without basepoints only to their conjugacy classes (Theorem 3.1); the deck group is $N(H)/H$.
- For graphs everything is computable: the rank is $1 - \chi$, the Euler characteristic multiplies (Proposition 3.9), and a basis is read off a spanning tree (Example 3.10).
- Finite index behaves arithmetically (Exercise 3.11); infinite index does not (Exercise 3.13).
- A covering space action gives $1 \to \pi_1(Y) \to \pi_1(Y/G) \to G \to 1$ (Proposition 3.16), which is how a group is realised as a fundamental group.

Chapter 4

Homology you can compute: Δ-complexes, cellular chains, degree

Homology is computable in a way that π_1 is not: for a space presented by cells, the chain complex is finite in each degree and the boundary maps are integer matrices. The price is bookkeeping — orientations, orderings, signs — and the purpose of this chapter is to fix conventions and then run them on enough examples that the reader trusts their own arithmetic.

Two threads start here and run through the book. The first is *torsion*: it appears as a cokernel, produced by a gluing that wraps a cell around a cycle more than once, and it is the reason homology is not merely a count of holes. The second is *degree*, which converts a geometric symmetry into an integer, and which will reappear as the cellular boundary, as the Hopf invariant (Chapter 16) and as the transgression (Chapter 19).

4.1 Δ-complexes, and the conventions that make them work

Definition 4.1. A Δ-*complex* structure on X is a collection of maps $\sigma_\alpha\colon \Delta^n \to X$, each restricting to a homeomorphism from the interior of Δ^n onto its image, such that every point of X is in exactly one such image, the restriction of each σ_α to a face of Δ^n is one of the maps of the collection, and a set is open iff its preimage under every σ_α is open.

The simplicial chain complex has $\Delta_n(X)$ free abelian on the n-simplices and

$$\partial\sigma = \sum_{i=0}^{n} (-1)^i \, \sigma|_{[v_0,\dots,\hat{v}_i,\dots,v_n]}.$$

Everything below depends on the ordering of the vertices of each simplex, so:

Convention. Before computing, write down the vertex ordering of every simplex, and write down which face of which simplex each lower simplex is. The overwhelming majority of errors in this subject are made before any homology is computed, by counting simplices whose identifications have not been recorded.

Example 4.2 (Torus and Klein bottle)**.** Give the square the usual Δ-structure with two triangles U (upper) and L (lower) meeting along a diagonal c, and identify the sides in pairs.

For the torus, all four vertices become one vertex v; the sides give two edges a, b; with the diagonal there are three edges and two triangles. With vertices ordered so that $\partial U = b - c + a$ and $\partial L = a - c + b$ we get

$$\Delta_2 = \mathbb{Z}^2 \xrightarrow{\partial_2} \Delta_1 = \mathbb{Z}^3 \xrightarrow{\partial_1 = 0} \Delta_0 = \mathbb{Z},$$

with $\partial_2(U) = \partial_2(L) = a + b - c$ and $\partial_1 = 0$ because every edge is a loop. Hence $H_0 = \mathbb{Z}$, $H_1 = \mathbb{Z}^3/\langle a + b - c\rangle \cong \mathbb{Z}^2$ and $H_2 = \ker \partial_2 \cong \mathbb{Z}$, generated by $U - L$.

For the Klein bottle the same picture with one side reversed gives $\partial_2(U) = a + b - c$ and $\partial_2(L) = a - b + c$, whence ∂_2 has matrix $\left(\begin{smallmatrix} 1 & 1 \\ 1 & -1 \\ -1 & 1 \end{smallmatrix}\right)$ in the basis a, b, c. Now $\ker \partial_2 = 0$, so $H_2 = 0$; and $H_1 = \mathbb{Z}^3/\langle a + b - c,\ a - b + c\rangle \cong \mathbb{Z} \oplus \mathbb{Z}/2$, the torsion coming from the sum of the two relators, $2a$.

The last computation is worth pausing over. The two surfaces have identical cell counts and differ only in the sign of one identification, and that sign is the entire difference between $\mathbb{Z}^2$ and $\mathbb{Z} \oplus \mathbb{Z}/2$. Torsion is what a gluing produces when it wraps around with the wrong orientation.

To see the $\mathbb{Z}/2$ emerge mechanically rather than by inspection, change basis in Δ_1 to a, b, $c' = a + b - c$ (an invertible change, since $c = a + b - c'$). In the new basis $\partial_2(U) = c'$ and $\partial_2(L) = a - b + c = 2a - c'$, so the image of ∂_2 is spanned by c' and $2a$, and $\Delta_1/\operatorname{im}\partial_2 = \langle a, b, c'\rangle/\langle c', 2a\rangle \cong \mathbb{Z}/2 \oplus \mathbb{Z}$. That manoeuvre — change bases until the matrix of the boundary map is diagonal — works for every finite chain complex of free abelian groups, and Section 4.3 below turns it into an algorithm.

Exercise 4.3 (easy)**.** Let X be the Δ-complex obtained from Δ^n by identifying all faces of the same dimension, so that X has exactly one k-simplex σ_k for each $0 \le k \le n$. Compute $H_*(X)$.

Source: [1, §2.1, Ex. 9], reproduced in substance

Hint. *Every face of σ_k is σ_{k-1}, so $\partial\sigma_k = \sum_{i=0}^{k}(-1)^i\sigma_{k-1}$.*

Solution. The chain complex has $\Delta_k(X) = \mathbb{Z}\sigma_k$ for $0 \le k \le n$, and $\partial\sigma_k = \big(\sum_{i=0}^{k}(-1)^i\big)\sigma_{k-1}$, which is σ_{k-1} for k even and 0 for k odd ($k \ge 1$). So the complex is

$$\cdots \xrightarrow{1} \mathbb{Z} \xrightarrow{0} \mathbb{Z} \xrightarrow{1} \mathbb{Z} \xrightarrow{0} \mathbb{Z},$$

in degrees $\ldots, 3, 2, 1, 0$, with $\partial_k = 1$ for even $k \ge 2$ and $\partial_k = 0$ for odd k. For $0 < k < n$: if k is odd, $\ker\partial_k = \mathbb{Z}$ and $\operatorname{im}\partial_{k+1} = \mathbb{Z}$ ($k+1$ even), so $H_k = 0$; if k is even, ∂_k is injective and $H_k = 0$. $H_0 = \mathbb{Z}/\operatorname{im}\partial_1 = \mathbb{Z}$. In the top degree, $H_n = \ker\partial_n$ is $\mathbb{Z}$ for n odd and 0 for n even. Hence $H_*(X) \cong H_*(S^n)$ for n odd and $H_*(X) \cong H_*(\mathrm{pt})$ for n even. (Indeed X is contractible for n even and has the homology of S^n for n odd; the case $n = 1$ is a circle.)

Exercise 4.4 (easy)**.** Compute $H_*(\mathbb{RP}^2; \mathbb{Z})$ from the CW structure that has a single cell in dimensions 0, 1 and 2, with the 2-cell attached along a^2. Then describe a Δ-complex structure on $\mathbb{RP}^2$ with two triangles, and say — without computing — how many vertex classes it has.

Hint. *For the second half:* $\mathbb{RP}^2$ *is the square with edge word abab. Follow the four corners around and see which of them are identified.*

Solution. The cellular complex is $\mathbb{Z} \xrightarrow{2} \mathbb{Z} \xrightarrow{0} \mathbb{Z}$, by Example 4.9 with $n = 2$. So $H_0 = \mathbb{Z}$, $H_1 = \mathbb{Z}/2$ generated by the 1-cell, and $H_2 = \ker(\times 2) = 0$: the surface is closed but carries no fundamental class over $\mathbb{Z}$, which is orientability making its first appearance.

For the Δ-structure, cut the square along a diagonal. Going round the boundary, the corners alternate between two classes, so *two* vertices survive, not one — unlike the torus and the Klein bottle of Example 4.2, where all four corners were identified. That is exactly the kind of fact that must be recorded before any boundary map is written, and missing it is the standard way to get a wrong answer with a plausible shape.

Remark 4.5. The parenthetical in that solution is not an evasion, it is the moral of the section. Δ-complexes are the right setting for *defining* simplicial homology and for proving it agrees with singular homology, and they are a poor setting for computing, because the number of simplices grows quickly and each one carries an ordering. The next section replaces them.

4.2 Cellular chains

Theorem 4.6 (Cellular homology; recalled)**.** *Let X be a CW complex. The groups $C_n^{\mathrm{CW}}(X) = H_n(X^n, X^{n-1})$ are free abelian on the n-cells, and with the boundary maps below their homology is $H_*(X)$.*

Proposition 4.7 (Cellular boundary formula)**.** $d_n(e_\alpha^n) = \sum_\beta d_{\alpha\beta}\, e_\beta^{n-1}$*, where $d_{\alpha\beta}$ is the degree of the composite*

$$S^{n-1} \xrightarrow{\varphi_\alpha} X^{n-1} \longrightarrow X^{n-1}/X^{n-2} = \bigvee_\beta S^{n-1} \longrightarrow S_\beta^{n-1},$$

in which φ_α is the attaching map of e_α^n and the last map collapses all wedge summands but the β-th.

So the boundary map is a matrix of degrees, and computing homology of a CW complex is computing degrees of maps between spheres. Two model cases carry most of the weight.

How the degrees are found in practice. In dimension one, $d_1(e^1) = v_{\mathrm{end}} - v_{\mathrm{start}}$. In dimension two, the attaching map of a 2-cell is a word in the 1-cells, and the degree $d_{\alpha\beta}$ is the *exponent sum* of the letter β in the word α — each traversal of e_β^1 in the positive direction contributes $+1$ and each traversal against it -1. This is why a 2-cell attached along a commutator contributes zero, and along a^n contributes n. In higher dimensions one uses the local degree formula: if the composite in Proposition 4.7 is, near finitely many preimages of a regular value, a homeomorphism onto a small disc, its degree is the sum of ± 1 according to whether each local homeomorphism preserves or reverses orientation. The projective-space computation below is exactly a sum of two local degrees.

Warning 4.8 (What the formula needs). Proposition 4.7 requires the cells to be *oriented* — a generator of $H_n(D^n, \partial D^n)$ chosen for each cell — and the answer depends on those choices only up to the sign of rows and columns, which does not change the homology.

What it does not tolerate is identifications that have not been recorded: if two 1-cells become one in a quotient, their columns must be added before computing, and forgetting this is the most common source of wrong answers with the right shape.

Example 4.9 (Projective spaces)**.** $\mathbb{RP}^n$ has one cell in each dimension $0, \dots, n$, the k-cell attached by the double cover $S^{k-1} \to \mathbb{RP}^{k-1}$. The composite in Proposition 4.7 is

$$S^{k-1} \longrightarrow \mathbb{RP}^{k-1} \longrightarrow \mathbb{RP}^{k-1}/\mathbb{RP}^{k-2} \cong S^{k-1},$$

which collapses the equator and folds the two hemispheres onto S^{k-1}, the second one composed with the antipodal map. Since the antipodal map of S^{k-1} has degree $(-1)^k$, the degree of the composite is $1 + (-1)^k$. Hence

$$d_k = \begin{cases} 2, & k \text{ even}, \\ 0, & k \text{ odd}, \end{cases}$$

and therefore

$$H_k(\mathbb{RP}^n; \mathbb{Z}) = \begin{cases} \mathbb{Z}, & k = 0, \\ \mathbb{Z}/2, & k \text{ odd},\ 0 < k < n, \\ 0, & k \text{ even},\ 0 < k < n, \\ \mathbb{Z}, & k = n \text{ odd}, \\ 0, & k = n \text{ even}. \end{cases}$$

The alternation of 0s and 2s is not a curiosity: it is the parity of the antipodal map, and it will return in Exercise 4.26.

Example 4.10 (Lens spaces)**.** Take the CW structure on $L(p, q)$ from Exercise 2.31, with a single cell in each of the dimensions 0 through 3. The 2-cell is attached along a^p, so $d_2 = p$. The 3-cell is attached by a map that carries the upper and lower hemispheres of ∂D^3 each homeomorphically onto the 2-cell, and the identification reverses orientation, so the two contributions have opposite signs and $d_3 = 0$. Finally $d_1 = 0$, there being one 0-cell. The complex

$$\mathbb{Z} \xrightarrow{0} \mathbb{Z} \xrightarrow{p} \mathbb{Z} \xrightarrow{0} \mathbb{Z}$$

gives

$$H_*(L(p, q); \mathbb{Z}) = \big(\mathbb{Z},\ \mathbb{Z}/p,\ 0,\ \mathbb{Z}\big).$$

Remark 4.11 (The promise of Remark 2.32, kept). The answer does not involve q. So homology joins π_1 in failing to distinguish $L(5, 1)$ from $L(5, 2)$, even though those spaces are not homotopy equivalent. Two invariants have now failed on the same pair of spaces, and both failed for the same reason: they are built from the cells one at a time, and the spaces differ in how the cells are *linked*. Chapter 7 introduces the product that sees linking.

Example 4.12 (Surfaces)**.** The CW structure of Example 2.6 on M_g has one 0-cell, $2g$ 1-cells and one 2-cell attached along $\prod[a_i, b_i]$. Each generator occurs once with each sign in that word, so every degree $d_{\alpha\beta}$ is $1 - 1 = 0$ and $d_2 = 0$. Hence $H_*(M_g) = (\mathbb{Z},\ \mathbb{Z}^{2g},\ \mathbb{Z})$. Compare Example 2.6: the relator dies on abelianisation, and $d_2 = 0$ is the same statement one dimension up.

Exercise 4.13 (medium)**.** Let $X_n = S^1 \cup_{a^n} e^2$, the complex with one 0-cell, one 1-cell and a 2-cell attached by a degree-n map. Compute $H_*(X_n)$, and compute $\pi_1(X_n)$. Which of the two invariants is stronger here?

Solution. The cellular complex is $\mathbb{Z} \xrightarrow{n} \mathbb{Z} \xrightarrow{0} \mathbb{Z}$, so $H_0 = \mathbb{Z}$, $H_1 = \mathbb{Z}/n$, $H_2 = \ker(n) = 0$. By Proposition 2.3, $\pi_1(X_n) = \langle a \mid a^n \rangle = \mathbb{Z}/n$. Here the two agree, because the group is abelian; in general H_1 is the abelianisation of π_1, so π_1 is at least as strong, and Example 2.6 is a case where it is strictly stronger.

Exercise 4.14 (easy)**.** Let X be a finite CW complex and $p\colon \widetilde{X} \to X$ an n-sheeted covering. Show $\chi(\widetilde{X}) = n\,\chi(X)$. Deduce that if the closed orientable surface M_g covers M_h with n sheets, then $g = n(h-1)+1$.

SOURCE: [1, §2.2, Ex. 22 and 23], reproduced in substance

Hint. *The cells of X lift: each open k-cell is simply connected, so its preimage is n disjoint open k-cells. Then use Exercise 4.15 for both spaces.*

Solution. Each open cell e of X is contractible, hence its preimage under the covering is a disjoint union of n open sets each mapped homeomorphically onto e (a cover over a simply connected, locally path-connected space is trivial). The characteristic maps lift as well (lift $D^k \to X$, D^k being simply connected), and these lifts give $\widetilde{X}$ a CW structure with exactly n cells over each cell of X. So $\widetilde{X}$ has nc_k cells of dimension k and $\chi(\widetilde{X}) = \sum(-1)^k n c_k = n\chi(X)$, the Euler characteristic being computable from cell counts by Exercise 4.15 (stated just below).

For surfaces, $\chi(M_g) = 2 - 2g$ (one 0-cell, $2g$ 1-cells, one 2-cell), so $2 - 2g = n(2-2h)$, i.e. $g = n(h-1)+1$.

Exercise 4.15 (medium)**.** Let X be a finite CW complex with c_n cells of dimension n. Show that $\sum_n (-1)^n c_n = \sum_n (-1)^n \operatorname{rank} H_n(X)$.

Hint. *For a chain complex of finitely generated free abelian groups, $\operatorname{rank} C_n = \operatorname{rank} Z_n + \operatorname{rank} B_{n-1}$ where Z and B are cycles and boundaries.*

Solution. Write $Z_n = \ker d_n$, $B_n = \operatorname{im} d_{n+1}$. From $0 \to Z_n \to C_n \to B_{n-1} \to 0$, which splits as B_{n-1} is free, $\operatorname{rank} C_n = \operatorname{rank} Z_n + \operatorname{rank} B_{n-1}$. From $0 \to B_n \to Z_n \to H_n \to 0$, $\operatorname{rank} Z_n = \operatorname{rank} B_n + \operatorname{rank} H_n$. Hence $\sum(-1)^n \operatorname{rank} C_n = \sum(-1)^n(\operatorname{rank} H_n + \operatorname{rank} B_n + \operatorname{rank} B_{n-1})$, and the B terms cancel in pairs.

Exercise 4.16 (medium)**.** Let X be the quotient of S^2 obtained by identifying $x \sim -x$ for x on the equator. Compute $H_*(X)$. Do the same for the quotient of S^3 in which antipodal points of the equatorial S^2 are identified.

SOURCE: [1, §2.2, Ex. 10], reproduced in substance

Hint. *The equator becomes $\mathbb{RP}^1$, resp. $\mathbb{RP}^2$, and each hemisphere becomes a cell attached by the double cover $S^{k-1} \to \mathbb{RP}^{k-1}$.*

Solution. In the first case the equator becomes $\mathbb{RP}^1$, a circle with one 0-cell v and one 1-cell a; the two hemispheres become 2-cells e_+, e_-, each attached by the double cover $S^1 \to \mathbb{RP}^1$, of degree 2. So the cellular complex is $\mathbb{Z}^2 \xrightarrow{(2\ 2)} \mathbb{Z} \xrightarrow{0} \mathbb{Z}$. Hence $H_0 = \mathbb{Z}$, $H_1 = \mathbb{Z}/2$, $H_2 = \ker(2\ 2) = \mathbb{Z}(e_+ - e_-) \cong \mathbb{Z}$. (With suitable orientations; reorienting e_- changes the matrix to $(2\ {-2})$ and the generator to $e_+ + e_-$, with the same groups.)

In the second case the equator becomes $\mathbb{RP}^2$ with cells v, a, b and the two hemispheres become 3-cells $e_\pm$ attached by $S^2 \to \mathbb{RP}^2$. By Example 4.9 the cellular boundary of such a cell onto the 2-cell of $\mathbb{RP}^2$ has degree $1 + (-1)^3 = 0$. So the complex is

$$\mathbb{Z}^2 \xrightarrow{0} \mathbb{Z} \xrightarrow{2} \mathbb{Z} \xrightarrow{0} \mathbb{Z},$$

and $H_* = (\mathbb{Z},\ \mathbb{Z}/2,\ 0,\ \mathbb{Z}^2)$. The parity flip between the two cases is the parity of the antipodal map again.

Exercise 4.17 (medium)**.** For $0 \le m < n$ compute $H_*(\mathbb{RP}^n/\mathbb{RP}^m)$ using the CW structure of Example 4.9.

SOURCE: [1, §2.2, Ex. 19], reproduced in substance

Hint. *Collapsing $\mathbb{RP}^m$ leaves one 0-cell and one k-cell for each $m < k \le n$, with the boundary maps of Example 4.9 except at the bottom.*

Solution. $\mathbb{RP}^n/\mathbb{RP}^m$ has cells e^0 and e^k for $m+1 \le k \le n$, and the cellular boundary $d_k\colon C_k \to C_{k-1}$ is $1+(-1)^k$ for $m+2 \le k \le n$ (the same attaching maps, composed with a collapse that does not touch the cells involved), while $d_{m+1} = 0$ because $C_m = 0$ (for $m \ge 1$; for $m = 0$ the target is the 0-cell and $d_1 = 0$ as well). Therefore:

- $H_{m+1} = \mathbb{Z}/\operatorname{im} d_{m+2}$, which is $\mathbb{Z}/2$ if m is even (then $m+2$ is even and $d_{m+2} = 2$) and $\mathbb{Z}$ if m is odd, provided $m+2 \le n$; if $n = m+1$ then $H_{m+1} = \mathbb{Z}$;
- for $m+1 < k < n$: $H_k = \mathbb{Z}/2$ for k odd and 0 for k even;
- $H_n = \mathbb{Z}$ for n odd and 0 for n even (when $n > m+1$);
- $H_0 = \mathbb{Z}$ and all other groups vanish.

For example $\mathbb{RP}^4/\mathbb{RP}^1$ has cells in dimensions $0, 2, 3, 4$ with $d_3 = 0$, $d_4 = 2$, so $H_* = (\mathbb{Z}, 0, \mathbb{Z}, \mathbb{Z}/2, 0)$ in degrees 0 to 4, while $\mathbb{RP}^4/\mathbb{RP}^2$ has cells in dimensions $0, 3, 4$ and $H_* = (\mathbb{Z}, 0, 0, \mathbb{Z}/2, 0)$: the bottom cell carries $\mathbb{Z}$ or $\mathbb{Z}/2$ according to the parity of m.

4.3 Smith normal form: homology as integer linear algebra

Every homology computation in this chapter ended with a cokernel of an integer matrix. There is one algorithm that computes all of them, and it is worth knowing explicitly: it is how computer programs compute integral homology, it explains *where* torsion comes from, and in Part VI it is the tool for testing conjectures about torsion in configuration spaces on small cases.

Theorem 4.18 (Smith normal form)**.** *Let A be an $m \times n$ integer matrix. There are invertible integer matrices $P \in \mathrm{GL}_m(\mathbb{Z})$, $Q \in \mathrm{GL}_n(\mathbb{Z})$ such that*

$$PAQ = \begin{pmatrix} D & 0 \\ 0 & 0 \end{pmatrix}, \qquad D = \operatorname{diag}(d_1, \dots, d_r), \quad d_i \ge 1, \quad d_1 \mid d_2 \mid \cdots \mid d_r.$$

The invariant factors *d_i are unique: $d_1 d_2 \cdots d_k$ is the greatest common divisor of the $k \times k$ minors of A, and r is the rank of A.*

Proof sketch. Row and column operations over $\mathbb{Z}$ — swapping two rows, adding an integer multiple of one row to another, multiplying a row by -1, and the same for columns — are multiplication by invertible integer matrices. Move an entry of smallest nonzero absolute value to position $(1,1)$; divide every other entry of the first row and column by it with remainder and subtract; if a nonzero remainder appears it is smaller than the pivot, and one repeats. This terminates because the pivot decreases. When the first row and column are clear, if some entry of the remaining block is not divisible by the pivot, add its row to the first row and repeat; this again decreases the pivot. Finally induct on the size. Uniqueness: the gcd of the $k \times k$ minors is unchanged by invertible row and column operations (the minors of PA are integer combinations of those of A, and conversely), and for a diagonal matrix it equals $d_1 \cdots d_k$. □

Corollary 4.19 (Homology from Smith normal form)**.** *Let C_* be a chain complex of finitely generated free abelian groups, and let $d_{k+1}\colon C_{k+1} \to C_k$ have rank r_{k+1} and nontrivial invariant factors $d_1, \dots, d_s$ (those > 1). Then*

$$H_k(C) \cong \mathbb{Z}^{\operatorname{rank} C_k - r_k - r_{k+1}} \oplus \bigoplus_{i=1}^{s} \mathbb{Z}/d_i.$$

For a prime p, $\dim_{\mathbb{F}_p} H_k(C \otimes \mathbb{F}_p) = \operatorname{rank} C_k - \operatorname{rank}_{\mathbb{F}_p} \bar{d}_k - \operatorname{rank}_{\mathbb{F}_p} \bar{d}_{k+1}$, where $\bar{d}$ is the reduction mod p, and $\operatorname{rank}_{\mathbb{F}_p} \bar{d}$ equals the number of invariant factors of d not divisible by p.

Proof. Choose bases making d_{k+1} diagonal as in Theorem 4.18. Then $\operatorname{im} d_{k+1}$ is spanned by $d_1 e_1, \dots, d_r e_r$ for part of a basis $e_1, \dots, e_{\operatorname{rank} C_k}$ of C_k. The cycles $Z_k = \ker d_k$ form a direct summand of C_k of rank $\operatorname{rank} C_k - r_k$ (a kernel of a map to a free group is a direct summand, the image being free) containing $\operatorname{im} d_{k+1}$, and the elements $e_1, \dots, e_r$ lie in Z_k: if $d_i e_i \in Z_k$ then $d_i\, d_k(e_i) = 0$, so $d_k(e_i) = 0$ since C_{k-1} is torsion-free. Choosing the basis of C_k adapted to both (possible because $Z_k/\langle e_1, \dots, e_r\rangle$ is free, being a subgroup of the free group $C_k/\langle e_1, \dots, e_r\rangle$), $H_k = Z_k/\operatorname{im} d_{k+1} \cong \bigoplus \mathbb{Z}/d_i \oplus \mathbb{Z}^{\operatorname{rank} Z_k - r}$. The mod p statement follows because P, Q remain invertible mod p, and a diagonal entry d_i becomes zero mod p exactly when $p \mid d_i$. □

Remark 4.20 (Reading torsion off a matrix). Corollary 4.19 makes two facts used throughout the book visible. First, the torsion of H_k depends only on d_{k+1}: torsion in degree k is created by $(k+1)$-cells whose boundaries wrap around k-cycles a multiple number of times — the gcd of Remark 5.17 in Chapter 5 is the first invariant factor of a 1-column matrix. Second, comparing the integral and mod-p ranks detects p-torsion: $\dim H_k(C \otimes \mathbb{F}_p)$ exceeds the Betti number $\operatorname{rank} H_k(C)$ by the number of invariant factors of d_k and of d_{k+1} divisible by p. That count is the universal coefficient theorem of Chapter 6 seen at the level of matrices, and the distinction between $\mathbb{Z}/p$ and $\mathbb{Z}/p^2$ — invisible to that count — is what the Bockstein of Chapter 9 detects.

Example 4.21 (The Klein bottle by Smith normal form)**.** The matrix of ∂_2 in Example 4.2 is $A = \left(\begin{smallmatrix} 1 & 1 \\ 1 & -1 \\ -1 & 1 \end{smallmatrix}\right)$. The gcd of its entries is 1 and the gcd of its 2×2 minors $\{-2, 2, 0\}$ is 2, so the invariant factors are $d_1 = 1$, $d_2 = 2$. With $\partial_1 = 0$ and $\operatorname{rank} \Delta_1 = 3$, Corollary 4.19 gives $H_1 \cong \mathbb{Z}^{3-0-2} \oplus \mathbb{Z}/2 = \mathbb{Z} \oplus \mathbb{Z}/2$ and $H_2 = \ker \partial_2 = 0$. Mod 2 the matrix has rank 1 (the invariant factor 2 dies), so $H_1(K; \mathbb{F}_2) \cong \mathbb{F}_2^{3-1} = \mathbb{F}_2^2$ and $H_2(K; \mathbb{F}_2) \cong \mathbb{F}_2^{2-1} = \mathbb{F}_2$: mod 2 the Klein bottle has a fundamental class, as Theorem 7.11(b) in Chapter 7 requires of every closed surface.

Exercise 4.22 (medium)**.** Let X be the 2-complex with one 0-cell, three 1-cells a, b, c and two 2-cells attached along the words a^2b^4 and a^4b^2. Compute $H_*(X)$ and $H_*(X; \mathbb{F}_p)$ for every prime p. Is $H_1(X)$ cyclic modulo its free part?

SOURCE: original; answer checked with `code/snf.py`

Hint. *The cellular boundary d_2 has columns $(2, 4, 0)$ and $(4, 2, 0)$. Compute the gcd of the entries and of the 2×2 minors.*

Solution. By Proposition 4.7 each letter contributes its exponent sum to the column of its 2-cell (as in Exercise 4.13), so with respect to the bases a, b, c and the two 2-cells, $d_2 = \left(\begin{smallmatrix} 2 & 4 \\ 4 & 2 \\ 0 & 0 \end{smallmatrix}\right)$, and

$d_1 = 0$. The gcd of the entries is 2, and the 2×2 minors are $2 \cdot 2 - 4 \cdot 4 = -12$, 0, 0, with gcd 12. So $d_1 = 2$, $d_1 d_2 = 12$, $d_2 = 6$. By Corollary 4.19,

$$H_0 = \mathbb{Z}, \qquad H_1 \cong \mathbb{Z}^{3-0-2} \oplus \mathbb{Z}/2 \oplus \mathbb{Z}/6 = \mathbb{Z} \oplus \mathbb{Z}/2 \oplus \mathbb{Z}/6, \qquad H_2 = \ker d_2 = 0.$$

The torsion $\mathbb{Z}/2 \oplus \mathbb{Z}/6 \cong (\mathbb{Z}/2)^2 \oplus \mathbb{Z}/3$ is not cyclic (it has two independent elements of order 2), although its order is 12: one must not conclude "$\mathbb{Z}/12$" from the determinant alone.

Mod p: for $p = 2$ both invariant factors die, $\operatorname{rank}_{\mathbb{F}_2} \bar{d}_2 = 0$, so $H_1(X;\mathbb{F}_2) = \mathbb{F}_2^3$ and $H_2(X;\mathbb{F}_2) = \mathbb{F}_2^2$. For $p = 3$ only $d_2 = 6$ dies, the rank is 1, so $H_1(X;\mathbb{F}_3) = \mathbb{F}_3^2$, $H_2(X;\mathbb{F}_3) = \mathbb{F}_3$. For $p \geq 5$ the rank is 2, $H_1(X;\mathbb{F}_p) = \mathbb{F}_p$, $H_2(X;\mathbb{F}_p) = 0$. In each case $\dim H_1(X;\mathbb{F}_p) - 1$ equals the number of cyclic p-primary summands of the torsion, as Remark 4.20 predicts. (The script `code/snf.py` reproduces these invariant factors.)

Exercise 4.23 (medium)**.** Let X be the quotient of $T^2 = S^1 \times S^1$ obtained by identifying points of the circle $S^1 \times \{x_0\}$ that differ by rotation through $2\pi/m$, and points of $\{x_0\} \times S^1$ that differ by rotation through $2\pi/n$ $(m, n \geq 1)$. Compute $H_*(X)$ and a presentation of $\pi_1(X)$. Are X and T^2 homotopy equivalent when $m, n \geq 2$?

SOURCE: [1, §2.2, Ex. 9(d)] (homology part), extended

Hint. *The quotient of a circle by rotation through $2\pi/m$ is a circle, and the quotient map has degree m. Find the attaching word of the single 2-cell.*

Solution. Give T^2 the standard structure: 0-cell (x_0, x_0), 1-cells a, b, and a 2-cell attached along $aba^{-1}b^{-1}$. The identification on a is the quotient of a circle by a free rotation of order m; the quotient is a circle a' and a maps onto it with degree m, i.e. a becomes the loop a'^m. Likewise b becomes b'^n. (The basepoint (x_0, x_0) is identified with other points of the two circles; choosing the CW structure on a' and b' with a single 0-cell at the image of (x_0, x_0), the wedge point, keeps one 0-cell.) So X is $a' \vee b'$ with one 2-cell attached along $a'^m b'^n a'^{-m} b'^{-n}$.

By Proposition 2.3, $\pi_1(X) = \langle a', b' \mid a'^m b'^n = b'^n a'^m \rangle$. The exponent sums of a' and b' in the relator are $m - m = 0$ and $n - n = 0$, so $d_2 = 0$ and $H_*(X) = (\mathbb{Z}, \mathbb{Z}^2, \mathbb{Z})$, the homology of the torus.

For $m, n \geq 2$ the group $\pi_1(X)$ is not abelian. To see it, map it onto a group known to be nonabelian: the assignment $a' \mapsto \bar{a}$, $b' \mapsto \bar{b}$ into the free product $\mathbb{Z}/m * \mathbb{Z}/n = \langle \bar{a}, \bar{b} \mid \bar{a}^m, \bar{b}^n \rangle$ kills the relator (each of a'^m and b'^n goes to 1), so it defines a surjective homomorphism; and $\mathbb{Z}/m * \mathbb{Z}/n$ is nonabelian, since $\bar{a}\bar{b}$ and $\bar{b}\bar{a}$ are distinct reduced words. Hence $\pi_1(X) \ncong \mathbb{Z}^2$ and $X \not\simeq T^2$, although $H_*(X) \cong H_*(T^2)$.

4.4 Degree

Definition 4.24. For $f\colon S^n \to S^n$ with $n \geq 1$, the *degree* $\deg f$ is the integer with $f_*(z) = (\deg f)\, z$ for z a generator of $H_n(S^n) \cong \mathbb{Z}$; the definition is independent of the generator.

We use without proof the standard properties: degree is a homotopy invariant and detects homotopy classes ($f \simeq g \iff \deg f = \deg g$); $\deg(fg) = \deg f \cdot \deg g$; the antipodal map of S^n has degree $(-1)^{n+1}$; and a map with a regular value having finitely many preimages has degree the sum of the local degrees.

Exercise 4.25 (easy)**.** Construct, for every $n \geq 1$ and every $k \in \mathbb{Z}$, a map $S^n \to S^n$ of degree k.

Hint. *For $n = 1$ use $z \mapsto z^k$. For larger n, suspend.*

Solution. On S^1, $z \mapsto z^k$ has degree k. Suspension multiplies nothing: the suspension Σf of a degree-k map has degree k, because $H_{n+1}(\Sigma S^n) \cong H_n(S^n)$ naturally. Iterating gives a degree-k map of S^n for every n. Degree 0 is realised by a constant map and degree -1 by a reflection.

The next computation is the one that makes symmetry arguments possible: it says what the double cover does to the top homology class.

Exercise 4.26 (medium)**.** Let $q\colon S^n \to \mathbb{RP}^n$ be the quotient map, $n \geq 1$. Show that

$$q_*\colon H_n(S^n) \longrightarrow H_n(\mathbb{RP}^n)$$

is multiplication by 2 when n is odd, and is zero when n is even.

Hint. *Give S^n the CW structure with two cells in each dimension permuted by the antipodal map τ, so that q identifies the two cells of each dimension.*

Hint. *Write the generator of $H_n(S^n)$ as a cellular cycle $z = e_+ + \varepsilon e_-$ and determine ε using $\tau_* z = (\deg \tau) z$.*

Solution. In the antipodal CW structure, $C_n^{\mathrm{CW}}(S^n) = \mathbb{Z}e_+ \oplus \mathbb{Z}e_-$ with τ exchanging e_+ and e_-. A generator of $H_n(S^n) \cong \mathbb{Z}$ is a cycle $z = e_+ + \varepsilon e_-$ for some $\varepsilon = \pm 1$. Applying τ, which has degree $(-1)^{n+1}$,

$$\varepsilon e_+ + e_- = \tau_* z = (-1)^{n+1}\big(e_+ + \varepsilon e_-\big),$$

so $\varepsilon = (-1)^{n+1}$.

The map q sends both n-cells of S^n to the single n-cell e of $\mathbb{RP}^n$, so on cellular chains $q_\#(z) = \big(1 + (-1)^{n+1}\big)e$. For n odd this is $2e$, and $H_n(\mathbb{RP}^n) = \mathbb{Z}\langle e\rangle$ by Example 4.9, so q_* is multiplication by 2. When n is even we have $H_n(\mathbb{RP}^n) = 0$ by Example 4.9, and there is nothing to prove; consistently, $q_\#(z) = 0$.

Remark 4.27. The use of Exercise 4.26 is always the same. Suppose a self-map f of S^n takes the same value at antipodal points. Then f is constant on the fibres of q, so it factors as $f = g \circ q$ for some $g\colon \mathbb{RP}^n \to S^n$ — the quotient by a free action being a covering, the induced g is continuous. Applying H_n and Exercise 4.26 turns that factorisation into a divisibility statement about $\deg f$, with the parity of n deciding which statement. Carrying this out, in both parities, and then asking which degrees actually occur, is a good measure of whether this chapter has been absorbed; Exercise 4.25 supplies the construction needed for the second half.

Remark 4.28 (Further reading). Simplicial, singular and cellular homology with complete proofs: [1, §§2.1–2.2]. The Smith normal form over a principal ideal domain and its uniqueness: any graduate algebra text, e.g. [2, Ch. 12]. Computer algebra systems compute integral homology this way; the script `code/snf.py` accompanying this book is a minimal implementation, used again in Part VI.

4.5 What to carry forward

- Fix orderings and identifications *before* computing (Definition 4.1 and the convention after it); the standard error is made at that stage.
- Torsion is a cokernel produced by an orientation-reversing gluing (Example 4.2).
- The cellular boundary is a matrix of degrees (Proposition 4.7); $\mathbb{RP}^n$ (Example 4.9) and the lens spaces (Example 4.10) are the two model computations.

- Homology fails on lens spaces exactly where π_1 failed (Remark 4.11) — the second invariant of the book to run out, and the reason for Chapter 7.
- Euler characteristic as a free consistency check (Exercise 4.15), multiplicative in finite covers (Exercise 4.14).
- Smith normal form computes integral and mod-p homology of any finite complex (Theorem 4.18 and Corollary 4.19); torsion in H_k is read off the invariant factors of d_{k+1} (Remark 4.20).
- The double cover multiplies the top class by 2 in odd dimensions and kills it in even ones (Exercise 4.26).

Chapter 5

Exact sequences as machines

An exact sequence is not a theorem to be admired but a device to be operated. In each of the three sections below the pattern is the same: a geometric decomposition produces a long exact sequence, and the sequence produces an answer by an entirely mechanical procedure — a cokernel at one end, a kernel at the other, and an extension problem in between. The skill being trained is running the machine confidently, and knowing which of its hypotheses is load-bearing.

The chapter closes by looking at what all this computation cannot see. Homology is about to be the third invariant of Part I to run out of power, and the way it fails — it cannot see the universal cover — points straight at Chapter 13.

5.1 Where long exact sequences come from

Every long exact sequence in this chapter — and the Wang, Gysin and Serre sequences of later chapters — is produced by one algebraic lemma, and knowing how its connecting map is built is what makes the sequences computable rather than merely exact.

Lemma 5.1 (The zig-zag lemma)**.** *A short exact sequence of chain complexes* $0 \to A_* \xrightarrow{i} B_* \xrightarrow{j} C_* \to 0$ *induces a long exact sequence*

$$\cdots \to H_n(A) \xrightarrow{i_*} H_n(B) \xrightarrow{j_*} H_n(C) \xrightarrow{\partial} H_{n-1}(A) \xrightarrow{i_*} H_{n-1}(B) \to \cdots ,$$

natural with respect to maps of short exact sequences. The connecting map is computed as follows: represent a class of $H_n(C)$ *by a cycle* c*; lift it to* $b \in B_n$ *with* $j(b) = c$*; then* $j(db) = d(jb) = 0$*, so* $db = i(a)$ *for a unique* $a \in A_{n-1}$*, which is a cycle; put* $\partial[c] = [a]$*.*

The recipe "lift, differentiate, pull back" is the only thing to remember. For the pair (X, A) it reads: a relative cycle is a chain on X whose boundary lies in A, and ∂ takes that boundary. For Mayer–Vietoris, applied to $0 \to C_*(A \cap B) \to C_*(A) \oplus C_*(B) \to C_*^{\{A,B\}}(X) \to 0$ (chains made of small simplices, which compute $H_*(X)$ by excision), it reads: write a cycle of X as a sum $z = a + b$ of chains in A and in B; then $\partial[z] = [da]$, a cycle in $A \cap B$ because $da = -db$. In the Heegaard splittings of Exercises 5.19 and 5.20 below, ∂ carries the fundamental class of the closed 3-manifold to the fundamental class of the splitting torus: cut the 3-cycle along the torus and take the boundary of one half.

Warning 5.2 (Exactness is not splitting). A long exact sequence determines each group only up to an *extension problem*: from $0 \to \operatorname{coker} \alpha \to H_n \to \ker \beta \to 0$ one learns H_n only up to extension unless the right-hand group is free (then the sequence splits) or some

further structure decides. The standard example: $0 \to \mathbb{Z}/2 \to E \to \mathbb{Z}/2 \to 0$ is satisfied by both $E = \mathbb{Z}/4$ and $E = \mathbb{Z}/2 \oplus \mathbb{Z}/2$. Exercise 5.3 below classifies the possibilities in the cases one meets most often; the same problem returns in every spectral sequence of Chapter 15, where it is one of the three things the reader must always check.

Exercise 5.3 (medium)**.** (a) Decide whether there is a short exact sequence $0 \to \mathbb{Z}/4 \to \mathbb{Z}/8 \oplus \mathbb{Z}/2 \to \mathbb{Z}/4 \to 0$.

(b) Let p be prime and $m, n \geq 1$. Show that an abelian group A fits into a short exact sequence $0 \to \mathbb{Z}/p^m \to A \to \mathbb{Z}/p^n \to 0$ if and only if $A \cong \mathbb{Z}/p^a \oplus \mathbb{Z}/p^{m+n-a}$ for some a with $\max(m,n) \leq a \leq m+n$.

(c) Show that A fits into $0 \to \mathbb{Z} \to A \to \mathbb{Z}/n \to 0$ if and only if $A \cong \mathbb{Z} \oplus \mathbb{Z}/d$ for some divisor d of n.

Source: [1, §2.1, Ex. 14], reproduced in substance

Hint. *For (b), A has order p^{m+n} and is generated by two elements, so it is $\mathbb{Z}/p^a \oplus \mathbb{Z}/p^b$ with $a \geq b$; compare exponents. For the converse try the subgroup generated by $(p^{a-m}, 1)$. For (c), the torsion subgroup of A meets the image of $\mathbb{Z}$ trivially.*

Solution. (a) Yes. Define $\mathbb{Z}/4 \to \mathbb{Z}/8 \oplus \mathbb{Z}/2$ by $1 \mapsto (2,1)$; it is injective, since $x(2,1) = 0$ forces $2x \equiv 0 \pmod 8$, i.e. $x \equiv 0 \pmod 4$. The quotient has order $16/4 = 4$, and the class of $(1,0)$ has order 4: if $k(1,0) = x(2,1)$ then x is even and $k \equiv 2x \equiv 0 \pmod 4$. A group of order 4 with an element of order 4 is $\mathbb{Z}/4$.

(b) Suppose A fits. Then $|A| = p^{m+n}$, and A is generated by a generator of the subgroup and a lift of a generator of the quotient, so $A \cong \mathbb{Z}/p^a \oplus \mathbb{Z}/p^b$ with $a \geq b$, $a + b = m + n$. The exponent p^a of A is at least p^m (it contains an element of order p^m) and at least p^n (the quotient has an element of order p^n, and orders can only drop in quotients). So $a \geq \max(m,n)$, and $a \leq a + b = m + n$.

Conversely let $A = \mathbb{Z}/p^a \oplus \mathbb{Z}/p^b$ with $b = m + n - a$, $\max(m,n) \leq a \leq m+n$, so $0 \leq b \leq \min(m,n)$. Let C be generated by $c = (p^{a-m}, 1)$. Its order is $\max(p^m, p^b) = p^m$, so $C \cong \mathbb{Z}/p^m$. The quotient A/C has order $p^{a+b-m} = p^n$, and the class of $(1,0)$ has order p^n: $k(1,0) = jc$ forces $p^b \mid j$ and $k \equiv jp^{a-m} \pmod{p^a}$, so the least positive k is $p^{b+a-m} = p^n$. Hence $A/C \cong \mathbb{Z}/p^n$.

(c) If $A = \mathbb{Z} \oplus \mathbb{Z}/d$ with $d \mid n$, let C be generated by $(n/d, 1)$; it is infinite cyclic, and in A/C the class of $(0,1)$ equals that of $(-n/d, 0)$, so A/C is generated by the class of $(1,0)$, whose order is the least $k > 0$ with $(k,0) = j(n/d, 1)$, i.e. $d \mid j$ and $k = jn/d$: $k = n$. So $A/C \cong \mathbb{Z}/n$. Conversely, if $0 \to \mathbb{Z} \to A \to \mathbb{Z}/n \to 0$ is exact, A is finitely generated of rank one, so $A \cong \mathbb{Z} \oplus T$ with T its torsion subgroup. Since $T \cap \mathrm{im}(\mathbb{Z}) = 0$, T maps injectively into $\mathbb{Z}/n$, so T is cyclic of order $d \mid n$.

Proposition 5.4 (Five lemma; recalled)**.** *In a commutative diagram of abelian groups with exact rows*

$$\begin{array}{ccccccccc} A & \longrightarrow & B & \longrightarrow & C & \longrightarrow & D & \longrightarrow & E \\ \downarrow{\scriptstyle\alpha} & & \downarrow{\scriptstyle\beta} & & \downarrow{\scriptstyle\gamma} & & \downarrow{\scriptstyle\delta} & & \downarrow{\scriptstyle\varepsilon} \\ A' & \longrightarrow & B' & \longrightarrow & C' & \longrightarrow & D' & \longrightarrow & E' \end{array}$$

if β, δ are isomorphisms, α is surjective and ε is injective, then γ is an isomorphism.

This is how a map of long exact sequences — for instance the map of Mayer–Vietoris sequences induced by a map of spaces respecting decompositions — is shown to be an isomorphism in the middle. The hypothesis is that $\alpha, \beta, \delta, \varepsilon$ be given *maps* making the diagram commute; abstract isomorphisms of the outer groups are useless.

Exercise 5.5 (easy)**.** Give a commutative diagram with exact rows, as in Proposition 5.4, all of whose groups are 0 or $\mathbb{Z}$, in which $\alpha, \beta, \delta, \varepsilon$ are all *zero* but γ is an isomorphism. Then give one in which the four outer maps are zero and γ is zero although $C \cong C'$.

Source: [1, §2.1, Ex. 31], reproduced in substance

Hint. *Use the rows* $0 \to 0 \to \mathbb{Z} \xrightarrow{1} \mathbb{Z} \to 0$ *and* $0 \to \mathbb{Z} \xrightarrow{1} \mathbb{Z} \to 0 \to 0$*, suitably aligned.*

Solution. Take the top row $0 \to 0 \to \mathbb{Z} \xrightarrow{\mathrm{id}} \mathbb{Z} \to 0$ (in positions A, B, C, D, E) and the bottom row $0 \to \mathbb{Z} \xrightarrow{\mathrm{id}} \mathbb{Z} \to 0 \to 0$. Both are exact. Let $\gamma = \mathrm{id}\colon \mathbb{Z} \to \mathbb{Z}$ and all other vertical maps zero (they have source or target 0, or are $\delta\colon D = \mathbb{Z} \to D' = 0$). The squares commute: the B–C square has $B = 0$; the C–D square sends $c \mapsto (\text{top } c) \mapsto 0$ and $c \mapsto c \mapsto 0$ in $D' = 0$. So γ is an isomorphism while all four outer maps vanish.

For the second diagram take the same two rows and $\gamma = 0$: every square still commutes, since each composite through C' or out of C that could be nonzero now passes through $\gamma = 0$ or through a zero group. So knowing the outer maps (here all zero) and the abstract groups ($C \cong C' \cong \mathbb{Z}$) does not determine γ; the five lemma needs the outer maps to be isomorphisms (or, at the ends, surjective/injective), and then it is γ itself that it controls.

5.2 Pairs, quotients, and a hypothesis that matters

5.3 Pairs, quotients, and a hypothesis that matters

The long exact sequence of a pair (X, A),

$$\cdots \to H_n(A) \to H_n(X) \to H_n(X, A) \xrightarrow{\partial} H_{n-1}(A) \to \cdots,$$

is useless without a way of identifying $H_n(X, A)$, and the identification one reaches for is $H_n(X, A) \cong \widetilde{H}_n(X/A)$. That isomorphism is a theorem with a hypothesis.

Theorem 5.6 (Good pairs; recalled)**.** *If A is a nonempty closed subspace that is a deformation retract of some neighbourhood in X — for instance if (X, A) is a CW pair — then the quotient map induces an isomorphism $H_n(X, A) \cong \widetilde{H}_n(X/A)$ for all n.*

Exercise 5.7 (easy)**.** For an exact sequence $A \xrightarrow{f} B \xrightarrow{g} C \xrightarrow{h} D \xrightarrow{k} E$, show that $C = 0$ if and only if f is surjective and k is injective. Deduce that for a pair (X, A) the inclusion induces isomorphisms $H_n(A) \cong H_n(X)$ for all n if and only if $H_n(X, A) = 0$ for all n.

Source: [1, §2.1, Ex. 15], reproduced in substance

Solution. If $C = 0$ then $\operatorname{im} f = \ker g = B$ (since g maps into $C = 0$), so f is onto; and $\ker k = \operatorname{im} h = 0$, so k is injective. Conversely, if f is onto then $\ker g = \operatorname{im} f = B$, so $g = 0$ and $\ker h = \operatorname{im} g = 0$; if k is injective then $\operatorname{im} h = \ker k = 0$, so $h = 0$ and $\ker h = C$. Together $C = \ker h = 0$.

For the pair, apply this to the segments $H_n(A) \to H_n(X) \to H_n(X, A) \to H_{n-1}(A) \to H_{n-1}(X)$ of the long exact sequence: $H_n(X, A) = 0$ for all n iff every $i_*\colon H_n(A) \to H_n(X)$ is surjective and every $i_*\colon H_{n-1}(A) \to H_{n-1}(X)$ injective, i.e. iff all i_* are isomorphisms.

Exercise 5.8 (medium)**.** Show that $H_0(X, A) = 0$ if and only if A meets every path component of X, and that $H_1(X, A) = 0$ if and only if $H_1(A) \to H_1(X)$ is surjective and each path component of X contains at most one path component of A.

SOURCE: [1, §2.1, Ex. 16], reproduced in substance

Hint. *Use the end* $H_1(A) \to H_1(X) \to H_1(X,A) \to H_0(A) \to H_0(X) \to H_0(X,A) \to 0$ *and the description of* H_0 *as the free abelian group on path components.*

Solution. $H_0(X) = \mathbb{Z}[\pi_0 X]$ and $H_0(A) = \mathbb{Z}[\pi_0 A]$, and i_* sends a component of A to the component of X containing it. Exactness of $H_0(A) \to H_0(X) \to H_0(X,A) \to 0$ gives $H_0(X,A) = \operatorname{coker} i_*$, which is free on the components of X not meeting A; so $H_0(X,A) = 0$ iff every component of X meets A.

For H_1, apply Exercise 5.7-style reasoning to $H_1(A) \to H_1(X) \to H_1(X,A) \to H_0(A) \xrightarrow{i_*} H_0(X)$: $H_1(X,A) = 0$ iff $H_1(A) \to H_1(X)$ is onto and i_* on H_0 is injective. A homomorphism $\mathbb{Z}[\pi_0 A] \to \mathbb{Z}[\pi_0 X]$ sending basis elements to basis elements is injective iff it is injective on the bases, i.e. iff no two components of A lie in the same component of X.

Exercise 5.9 (easy)**.** Compute $H_n(X,A)$ when X is S^2 or T^2 and $A \subseteq X$ is a set of $k \geq 1$ points.

SOURCE: [1, §2.1, Ex. 17(a)], reproduced in substance

Solution. $H_n(A) = \mathbb{Z}^k$ for $n = 0$ and 0 otherwise, and $i_*\colon H_0(A) \to H_0(X) = \mathbb{Z}$ sends each point to the generator, so it is onto with kernel $\mathbb{Z}^{k-1}$. The long exact sequence gives $H_n(X,A) \cong H_n(X)$ for $n \geq 2$, $H_0(X,A) = \operatorname{coker}(i_*) = 0$, and an exact sequence $0 \to H_1(X) \to H_1(X,A) \to \ker(i_*|_{H_0}) = \mathbb{Z}^{k-1} \to 0$, which splits since $\mathbb{Z}^{k-1}$ is free. So:

$$X = S^2 :\ H_*(X,A) = (0,\ \mathbb{Z}^{k-1},\ \mathbb{Z}); \qquad X = T^2 :\ H_*(X,A) = (0,\ \mathbb{Z}^{k+1},\ \mathbb{Z}).$$

Consistently, $X/A \simeq X \vee \bigvee_{k-1} S^1$ (identify the points one at a time, each identification of two points in a connected CW complex wedging on a circle up to homotopy), and $\widetilde{H}_*(X/A)$ agrees with the above by Theorem 5.6.

Exercise 5.10 (easy)**.** Verify the hypothesis for a CW pair: exhibit a neighbourhood of A in X that deformation retracts to A.

Hint. *Work cell by cell: in a cell* e^n *attached to* A*, push away from the centre.*

Solution. Build N skeleton by skeleton. Having a neighbourhood N^{n-1} of $A \cap X^{n-1}$ in X^{n-1} with a deformation retraction to it, extend over each n-cell: in the characteristic map $D^n \to X$, take the preimage of N^{n-1} in ∂D^n, thicken it radially inwards to an open collar, and deformation retract that collar radially. The union over all cells is the required neighbourhood, and the deformation retractions assemble because they agree on the boundaries and CW topology is determined cell by cell.

Two facts about the failure of Theorem 5.6 are worth having, because both are counterintuitive and both are used to build counterexamples.

Example 5.11 (Zero-dimensional homology counts components, one at a time)**.** If A is a space with infinitely many path components, then $H_0(A) \cong \bigoplus_{\pi_0(A)} \mathbb{Z}$, a direct *sum*, never a product. The reason is that a singular chain is a *finite* linear combination of simplices. So for the compact space $A = \{0\} \cup \{1/n : n \geq 1\} \subseteq \mathbb{R}$, which has countably many components, $H_0(A)$ is countable — even though A is compact and looks like a limit of finite sets.

Example 5.12 (The shrinking wedge)**.** Let $\mathbb{H} \subseteq \mathbb{R}^2$ be the union of the circles C_n of radius $1/n$ centred at $(1/n, 0)$, all passing through the origin, with the subspace topology. For each n there is a retraction $r_n\colon \mathbb{H} \to C_n$ collapsing every other circle to the origin, and it is continuous precisely because the circles shrink.

Together the r_n give a homomorphism

$$\Phi\colon H_1(\mathbb{H}) \longrightarrow \prod_{n\geq 1} H_1(C_n) = \prod_{n\geq 1} \mathbb{Z},$$

and Φ is *surjective*. To see it, prescribe $(k_n) \in \prod \mathbb{Z}$ and build a single loop which winds k_1 times around C_1 during $[0, \frac{1}{2}]$, then k_2 times around C_2 during $[\frac{1}{2}, \frac{3}{4}]$, and so on. Continuity at the endpoint holds because the circles shrink to the origin, and applying r_{n*} to its class returns k_n.

Exercise 5.13 (hard)**.** Deduce that $\mathbb{H}$ is not homotopy equivalent to a countable wedge of circles, even though it is a countable union of circles glued at a point.

Hint. *Compare the cardinality of the two homology groups. What is H_1 of a countable wedge of circles?*

Solution. A countable wedge $\bigvee_n S^1$ is a CW complex whose cellular chain complex has $C_1 = \bigoplus_n \mathbb{Z}$ and $C_0 = \mathbb{Z}$ with zero boundary, so $H_1\big(\bigvee_n S^1\big) = \bigoplus_n \mathbb{Z}$, which is countable.

By Example 5.12, $H_1(\mathbb{H})$ surjects onto $\prod_n \mathbb{Z}$, which is uncountable. A surjection from a countable group onto an uncountable one is impossible, so $H_1(\mathbb{H})$ is uncountable and the two groups are not isomorphic. Homology being a homotopy invariant, the spaces are not homotopy equivalent.

Remark 5.14. Examples 5.11 and 5.12 are the two halves of a machine for breaking Theorem 5.6. Take a pair (X, A) with X contractible and A compact, totally disconnected and infinite. The long exact sequence computes $H_1(X, A) \cong \widetilde{H}_0(A)$, a countable direct sum by Example 5.11; while X/A can easily be a space of shrinking-wedge type, with uncountable H_1 by Example 5.12. The hypothesis of Theorem 5.6 is therefore not removable, and the reader now has both computations needed to say so with a specific pair in hand.

5.4 Mayer–Vietoris

Theorem 5.15 (Mayer–Vietoris; recalled)**.** *If $X = \mathring{A} \cup \mathring{B}$ then there is a long exact sequence*

$$\cdots \to H_n(A\cap B) \xrightarrow{(i_*,-j_*)} H_n(A)\oplus H_n(B) \xrightarrow{k_*+l_*} H_n(X) \xrightarrow{\partial} H_{n-1}(A\cap B) \to \cdots$$

Operating it means, in each degree, computing one map of finitely generated abelian groups and taking a cokernel and a kernel. The example to keep in mind is one where the interesting behaviour comes from a gluing of degree two.

Example 5.16 (The Klein bottle from two Möbius bands)**.** The Klein bottle K is the union of two Möbius bands M_1, M_2 glued along their boundary circles; thickening slightly, $A\cap B \simeq S^1$ and $A \simeq B \simeq S^1$, the equivalences being the deformation retractions onto the core circles c_1, c_2.

The boundary circle of a Möbius band runs twice around the core, so with z a generator of $H_1(A\cap B)$,

$$\varphi\colon H_1(A\cap B) = \mathbb{Z} \longrightarrow \mathbb{Z}c_1 \oplus \mathbb{Z}c_2, \qquad \varphi(z) = (2c_1, -2c_2).$$

This is injective, so $H_2(K) = \ker \varphi = 0$; and

$$H_1(K) = \operatorname{coker} \varphi = \mathbb{Z}^2/\langle (2,-2) \rangle \cong \mathbb{Z} \oplus \mathbb{Z}/2,$$

since $(2,-2) = 2 \cdot (1,-1)$ with $(1,-1)$ primitive. So $H_*(K) = (\mathbb{Z},\ \mathbb{Z} \oplus \mathbb{Z}/2,\ 0)$, agreeing with the Δ-complex computation of Example 4.2. Two independent routes to the same answer is the cheapest insurance available.

Remark 5.17. The arithmetic in Example 5.16 deserves naming, because it is what decides whether torsion appears. A cokernel $\mathbb{Z}^k/\langle v \rangle$ is free exactly when v is primitive, i.e. when the gcd of its entries is 1; if $v = d \cdot v'$ with v' primitive and $d > 1$, the cokernel has a $\mathbb{Z}/d$ summand. So when gluing along a circle, the question is never "is there a Möbius band?" but "what is the gcd of the entries of the gluing vector?".

Exercise 5.18 (medium)**.** Let N_h be the connected sum of $h \geq 1$ copies of $\mathbb{RP}^2$, so $N_1 = \mathbb{RP}^2$ and $N_2 = K$. Using Mayer–Vietoris for the decomposition of N_h into N_{h-1} minus an open disc and a Möbius band, show by induction that

$$H_*(N_h) = \big(\mathbb{Z},\ \mathbb{Z}^{h-1} \oplus \mathbb{Z}/2,\ 0\big).$$

Check the answer against the cellular chain complex of the CW structure with one 0-cell, h 1-cells and one 2-cell attached along $a_1^2 a_2^2 \cdots a_h^2$.

SOURCE: original

Hint. *N_{h-1} minus a disc deformation retracts onto a wedge of $h-1$ circles, whose H_1 is free on $a_1, \dots, a_{h-1}$, and the boundary circle represents $2(a_1 + \cdots + a_{h-1})$ there. The Möbius band contributes a core circle c with the boundary going to $2c$.*

Solution. *Cellular check first.* The 2-cell is attached along $a_1^2 \cdots a_h^2$, so $d_2 = (2, 2, \dots, 2)^T$ and $d_1 = 0$. The invariant factor of this column is $\gcd = 2$, so by Corollary 4.19 $H_1 \cong \mathbb{Z}^{h-1} \oplus \mathbb{Z}/2$ and $H_2 = \ker d_2 = 0$.

Mayer–Vietoris. Write $N_h = U \cup V$, with $U \simeq N_{h-1} \setminus \mathring{D}$ and V a Möbius band, $U \cap V \simeq S^1$ the common boundary circle. Removing an open disc from the 2-cell of N_{h-1} leaves a space that deformation retracts onto the 1-skeleton $\bigvee_{h-1} S^1$, and the boundary circle of the hole is homotopic to the attaching map $a_1^2 \cdots a_{h-1}^2$, so in $H_1(U) = \mathbb{Z}^{h-1}$ it represents $2(a_1 + \cdots + a_{h-1})$. In $H_1(V) = \mathbb{Z}c$ the boundary represents $2c$. The relevant part of the sequence is

$$0 \to H_2(N_h) \to H_1(U \cap V) = \mathbb{Z} \xrightarrow{\varphi} \mathbb{Z}^{h-1} \oplus \mathbb{Z}c \to H_1(N_h) \to \widetilde{H}_0(U \cap V) = 0,$$

with $H_2(U) = H_2(V) = 0$ and $\varphi(z) = \big(2(a_1 + \cdots + a_{h-1}), -2c\big)$. Since φ is injective, $H_2(N_h) = 0$. The vector $\varphi(z) = 2 \cdot (1, \dots, 1, -1)$ is twice a primitive vector, so by Remark 5.17 its cokernel is $\mathbb{Z}^{h-1} \oplus \mathbb{Z}/2$. Hence $H_1(N_h) \cong \mathbb{Z}^{h-1} \oplus \mathbb{Z}/2$, agreeing with the cellular computation. (For $h = 1$, U is a disc and the formula reads $H_1 = \mathbb{Z}/2$: $\mathbb{RP}^2$ is a Möbius band with a disc glued on.)

Exercise 5.19 (medium)**.** Let X be the union of two solid tori V_1, V_2 glued along their boundary tori by a homeomorphism carrying the meridian of V_1 to the meridian of V_2 and the longitude to the longitude. Compute $H_*(X)$ and identify X.

Hint. *A solid torus deformation retracts onto its core circle. In H_1 of the boundary torus, the meridian bounds a disc in the solid torus and the longitude maps to the core.*

Solution. Write $H_1(T^2) = \mathbb{Z}m \oplus \mathbb{Z}l$ and $H_1(V_i) = \mathbb{Z}c_i$. Then $m \mapsto 0$ and $l \mapsto c_i$ in each solid torus, so

$$\varphi(m) = (0, 0), \qquad \varphi(l) = (c_1, -c_2).$$

Also $H_2(V_1) \oplus H_2(V_2) = 0$. Mayer–Vietoris gives

$$0 \to H_3(X) \to H_2(T^2) = \mathbb{Z} \to 0, \qquad 0 \to H_2(X) \to H_1(T^2) \xrightarrow{\varphi} \mathbb{Z}^2 \to H_1(X) \to 0.$$

Hence $H_3(X) \cong \mathbb{Z}$; $H_2(X) = \ker\varphi = \mathbb{Z}m \cong \mathbb{Z}$; and $H_1(X) = \operatorname{coker}\varphi = \mathbb{Z}^2/\langle(1,-1)\rangle \cong \mathbb{Z}$. So $H_*(X) = (\mathbb{Z},\mathbb{Z},\mathbb{Z},\mathbb{Z})$, which is the homology of $S^1 \times S^2$ — and indeed $X \cong S^1 \times S^2$, the two solid tori being the two halves of $S^1 \times S^2$ cut along $S^1 \times S^1$.

Exercise 5.20 (hard)**.** Now glue the two solid tori by a homeomorphism whose effect on H_1 of the boundary carries the meridian m of V_1 to $p\,l_2 + q\,m_2$, with $p \geq 1$ and $\gcd(p,q) = 1$. Compute $H_*(X)$.

Hint. *Only the images in H_1 matter. Write the gluing on $H_1(T^2)$ as a matrix in $GL_2(\mathbb{Z})$, with $m \mapsto p\,l_2 + q\,m_2$ and $l \mapsto r\,l_2 + s\,m_2$, where $ps - qr = \pm 1$.*

Hint. *In V_2, the meridian m_2 dies and l_2 becomes the core c_2. So the second component of φ only sees the l_2-coefficients, namely p and r.*

Solution. With the matrix as in the hint, $\varphi(m) = (0, -p\,c_2)$ and $\varphi(l) = (c_1, -r\,c_2)$. Then

$$H_1(X) = \operatorname{coker}\varphi = \mathbb{Z}^2/\big\langle(0,-p),\,(1,-r)\big\rangle \cong \mathbb{Z}/p,$$

because the second generator lets one eliminate the first coordinate, leaving $\mathbb{Z}/\langle p\rangle$. And φ is injective: $a(0,-p) + b(1,-r) = 0$ forces $b = 0$ and then $ap = 0$, so $a = 0$. Hence $H_2(X) = \ker\varphi = 0$, while $H_3(X) \cong H_2(T^2) = \mathbb{Z}$ as before. So

$$H_*(X) = \big(\mathbb{Z},\ \mathbb{Z}/p,\ 0,\ \mathbb{Z}\big),$$

which is the homology of the lens space $L(p,q)$ computed in Example 4.10. Indeed X *is* $L(p,q)$: the genus-one Heegaard splitting is the standard second description of a lens space.

5.5 The Wang sequence

A mapping torus is glued from one piece along two copies of the fibre, and Mayer–Vietoris turns that into a single long exact sequence in which the monodromy appears as $1 - f_*$.

Proposition 5.21 (Wang sequence)**.** *For the mapping torus T_f of $f\colon X \to X$ there is a long exact sequence*

$$\cdots \to H_n(X) \xrightarrow{1-f_*} H_n(X) \to H_n(T_f) \to H_{n-1}(X) \xrightarrow{1-f_*} H_{n-1}(X) \to \cdots$$

Equivalently, for each n there is a short exact sequence

$$0 \to \operatorname{coker}\big(1 - f_*\ \text{on}\ H_n(X)\big) \to H_n(T_f) \to \ker\big(1 - f_*\ \text{on}\ H_{n-1}(X)\big) \to 0, \tag{5.1}$$

which splits whenever the kernel is free abelian.

Proof sketch. Cut T_f along a fibre. Both pieces — a neighbourhood of the seam, and the complement of the seam — deformation retract onto copies of X, and their intersection is two copies of X. The Mayer–Vietoris map $H_n(X) \oplus H_n(X) \to H_n(X) \oplus H_n(X)$ is $(u,v) \mapsto (u - v,\ u - f_*v)$ up to sign; changing basis by $(u,v) \mapsto (u - v, v)$ turns it into a map whose cokernel and kernel are those of $1 - f_*$, and the long exact sequence collapses to the one displayed. Exactness of (5.1) is a restatement, and a short exact sequence with free abelian right-hand term splits. □

Exercise 5.22 (medium)**.** Check Proposition 5.21 against $f = \mathrm{id}$, where the answer is known independently.

Solution. For $f = \mathrm{id}$ we have $T_f = X \times S^1$ and $1 - f_* = 0$, so (5.1) reads $0 \to H_n(X) \to H_n(X \times S^1) \to H_{n-1}(X) \to 0$. That is exactly the Künneth splitting for $X \times S^1$, since $H_*(S^1) = (\mathbb{Z}, \mathbb{Z})$ and there are no Tor terms.

Example 5.23 (Mapping tori of self-maps of a sphere)**.** Let $f\colon S^k \to S^k$ have degree d, with $k \geq 2$. Then f_* is the identity on H_0 and multiplication by d on H_k, so $1 - f_*$ is 0 on H_0 and $1 - d$ on H_k. Feeding this into (5.1):

$$H_0(T_f) = \mathbb{Z}, \qquad H_1(T_f) = \mathbb{Z}, \qquad H_k(T_f) = \mathbb{Z}/(d-1), \qquad H_{k+1}(T_f) = \ker(1-d),$$

and all other groups vanish. For $d \neq 1$ the last group is 0; for $d = 1$ it is $\mathbb{Z}$, recovering $H_*(S^k \times S^1)$.

The torsion is the point. The fibre S^k has no torsion, the base S^1 has no torsion, and the total space has $\mathbb{Z}/(d-1)$: it was manufactured by the monodromy, exactly as in Remark 5.17 the torsion was manufactured by the gluing vector.

Exercise 5.24 (medium)**.** Let $X = S^1 \vee S^1$ and let f exchange the two circles. Compute $H_*(T_f)$.

Hint. *On $H_1(X) = \mathbb{Z}^2$ the map f_* is the exchange matrix, so $1 - f_*$ has matrix $\left(\begin{smallmatrix} 1 & -1 \\ -1 & 1 \end{smallmatrix}\right)$.*

Solution. On H_0, $1 - f_* = 0$; on $H_1 = \mathbb{Z}^2$, $1 - f_*$ has the matrix above, with $\ker \cong \mathbb{Z}$ spanned by $(1,1)$ and $\mathrm{coker} = \mathbb{Z}^2/\langle(1,-1)\rangle \cong \mathbb{Z}$.

Degree 0: $H_0(T_f) = \mathrm{coker}(0 \text{ on } H_0) = \mathbb{Z}$. Degree 1: $0 \to \mathrm{coker}(1 - f_*|H_1) \to H_1(T_f) \to \ker(1 - f_*|H_0) \to 0$ is $0 \to \mathbb{Z} \to H_1(T_f) \to \mathbb{Z} \to 0$, which splits, so $H_1(T_f) \cong \mathbb{Z}^2$. Degree 2: $0 \to 0 \to H_2(T_f) \to \ker(1 - f_*|H_1) = \mathbb{Z} \to 0$, so $H_2(T_f) \cong \mathbb{Z}$. All higher groups vanish. Thus $H_*(T_f) = (\mathbb{Z}, \mathbb{Z}^2, \mathbb{Z})$ — the homology of a torus. But T_f is *not* a torus, and not even homotopy equivalent to one: by Proposition 2.13, $\pi_1(T_f) = \langle a, b, t \mid tat^{-1} = b,\ tbt^{-1} = a\rangle = \langle a, t \mid t^2 a t^{-2} = a\rangle$, which is not abelian (send $t \mapsto (1\,2)$ and $a \mapsto (1\,2\,3)$ in $\mathfrak{S}_3$: the relation holds because $t^2 \mapsto 1$, while the images of a and t do not commute). Nor is T_f a surface: along the circle traced by the wedge point four sheets meet. The Wang sequence computed homology correctly; it simply cannot see the difference. (The original edition of this book asserted $T_f \cong T^2$; that was wrong.)

5.6 What homology cannot see

Every computation above was insensitive to one thing: the way the pieces are attached in a way that is invisible on the level of chains. The sharpest form of this is that homology does not see the universal cover.

Example 5.25 (The universal cover of a wedge)**.** Let $X = S^1 \vee S^2$. Its universal cover $\widetilde{X}$ is the real line with a copy of S^2 attached at each integer: the line covers the circle, and each sheet of the covering carries its own copy of the sphere. Hence

$$H_2(\widetilde{X}) \cong \bigoplus_{n \in \mathbb{Z}} \mathbb{Z},$$

free abelian of countably infinite rank, and $H_1(\widetilde{X}) = 0$.

Note the extra structure: the deck group $\mathbb{Z}$ permutes the spheres, so $H_2(\widetilde{X})$ is a free module of rank one over the group ring $\mathbb{Z}[t, t^{-1}] = \mathbb{Z}[\pi_1 X]$. That module structure is invisible downstairs and will be the subject of Chapter 13.

Exercise 5.26 (medium)**.** Using the Hurewicz theorem in the form $\pi_2(Y) \cong H_2(Y)$ for simply connected Y, together with $\pi_n(\widetilde{X}) \cong \pi_n(X)$ for $n \geq 2$, show that $\pi_2(S^1 \vee S^2)$ is not finitely generated as an abelian group.

Solution. $\widetilde{X}$ is simply connected — it is a universal cover — so Hurewicz gives $\pi_2(\widetilde{X}) \cong H_2(\widetilde{X}) \cong \bigoplus_{\mathbb{Z}} \mathbb{Z}$ by Example 5.25. A covering induces isomorphisms on π_n for $n \geq 2$, so $\pi_2(X) \cong \bigoplus_{\mathbb{Z}} \mathbb{Z}$, which is not finitely generated.

Exercise 5.27 (medium)**.** For a pair (X, A) of CW complexes let $X \cup CA$ be X with the cone on A attached.

(a) Show that X is a retract of $X \cup CA$ if and only if the inclusion $A \hookrightarrow X$ is nullhomotopic.

(b) Show that in that case $H_n(X, A) \cong \widetilde{H}_n(X) \oplus \widetilde{H}_{n-1}(A)$ for all n.

Source: [1, §2.2, Ex. 26], reproduced in substance

Hint. *A map $CA \to X$ extending the inclusion of A is the same as a nullhomotopy of that inclusion. For (b), use the long exact sequence of $(X \cup CA, X)$ and $(X \cup CA)/X = \Sigma A$, the (unreduced) suspension.*

Solution. (a) Write points of CA as $[a, t]$ with $t \in I$, the base $t = 0$ glued to A and the apex at $t = 1$. A retraction $r\colon X \cup CA \to X$ restricts on CA to a map $f(a, t) = r[a, t]$ with $f(a, 0) = a$ and $f(-, 1)$ constant: a nullhomotopy of the inclusion. Conversely a nullhomotopy f_t defines r by $r = \mathrm{id}$ on X and $r[a, t] = f_t(a)$, continuous because it agrees on A and the cone is a quotient of $A \times I$.

(b) First, $H_n(X, A) \cong H_n(X \cup CA, CA)$ by excision (deformation retract the lower half of the cone onto A), and $H_n(X \cup CA, CA) \cong \widetilde{H}_n(X \cup CA)$ by the long exact sequence of the pair, CA being contractible. Next, in the long exact sequence of the good pair $(X \cup CA, X)$,

$$\cdots \to \widetilde{H}_n(X) \xrightarrow{i_*} \widetilde{H}_n(X \cup CA) \to H_n(X \cup CA, X) \to \widetilde{H}_{n-1}(X) \xrightarrow{i_*} \cdots,$$

i_* has the left inverse r_* by (a), so it is injective, the connecting maps vanish, and the sequence breaks into split short exact sequences $0 \to \widetilde{H}_n(X) \to \widetilde{H}_n(X \cup CA) \to H_n(X \cup CA, X) \to 0$. Finally $H_n(X \cup CA, X) \cong \widetilde{H}_n\big((X \cup CA)/X\big) = \widetilde{H}_n(SA) \cong \widetilde{H}_{n-1}(A)$ (suspension shifts reduced homology by Mayer–Vietoris on the two cones). Combining, $H_n(X, A) \cong \widetilde{H}_n(X) \oplus \widetilde{H}_{n-1}(A)$.

Remark 5.28 (Where this chapter stops)*.* $H_*(X)$ is a quotient of information about X that has already forgotten the covering. Two spaces can therefore agree in every homology group and disagree about their universal covers, hence about π_2; and the two ingredients for producing such a pair are now both available — the universal cover of a wedge (Example 5.25) and the fact that a torus is aspherical. Constructing the pair, and proving the two spaces are not homotopy equivalent, is the natural exercise to attempt at this point.

The systematic repair is Chapter 13: replace $H_*(X)$ by $H_*(\widetilde{X})$ as a module over $\mathbb{Z}[\pi_1 X]$, and the resulting invariant *does* detect homotopy equivalence for CW complexes.

5.7 What to carry forward

- $H_n(X, A) \cong \widetilde{H}_n(X/A)$ needs a neighbourhood deformation retract (Theorem 5.6); Examples 5.11 and 5.12 are the standard tools for showing it fails without one.

- Mayer–Vietoris is arithmetic: torsion appears exactly when the gluing vector is imprimitive (Remark 5.17).
- Heegaard gluings of two solid tori produce $S^1 \times S^2$ and the lens spaces (Exercises 5.19 and 5.20) — the same spaces as Chapter 2, by a different route.
- The Wang sequence (Proposition 5.21): $\operatorname{coker}(1 - f_*)$ below, $\ker(1 - f_*)$ above, and check the splitting before using it.
- Homology does not see the universal cover (Remark 5.28); the module structure over $\mathbb{Z}[\pi_1]$ is what is missing.

Chapter 6

Cohomology, coefficients, and the failure of naturality

Cohomology is not merely homology with the arrows reversed. The universal coefficient theorem says how much of H^* is determined by H_* — almost all of it — and the residue, an Ext term sitting one degree up, is exactly what makes cohomology a strictly finer invariant of *maps* than homology.

The chapter is built around one distinction, and it is worth stating before any mathematics: a short exact sequence can be natural, and can split, without the splitting being natural. Readers meet this first as a piece of pedantry and later as the reason a whole subject exists. It recurs in Chapter 8 for the Künneth sequence, and the invariant that non-naturality has been concealing is finally constructed in Chapter 18.

6.1 The universal coefficient theorem

Definitions, and the point of dualising. For a space X and an abelian group G, the singular cochains are $C^n(X;G) = \mathrm{Hom}(C_n(X), G)$, functions on singular n-simplices with values in G, and $\delta\colon C^n \to C^{n+1}$ is the transpose of ∂: $(\delta\varphi)(\sigma) = \varphi(\partial\sigma)$. Then $H^n(X;G) = \ker\delta/\operatorname{im}\delta$. For a CW complex one may use cellular cochains, the dual of the cellular chain complex, exactly as for homology. The reasons to dualise are structural: H^* is contravariant, so a map $f\colon X \to Y$ pulls classes back and the cup product of Chapter 7 makes f^* a ring map; and a cochain is a *function* on chains, so a cocycle can be evaluated on a cycle. The second point is what the universal coefficient theorem measures: how far a cohomology class is from being a function on homology classes.

Where the Ext **term comes from.** Let C_* be a chain complex of free abelian groups, with cycles Z_n and boundaries B_n. The sequence $0 \to Z_n \to C_n \xrightarrow{\partial} B_{n-1} \to 0$ splits (B_{n-1} is free, as a subgroup of a free abelian group), so it stays exact after applying $\mathrm{Hom}(-,G)$, and the resulting short exact sequence of cochain complexes gives, via Lemma 5.1 in its cohomological form, a long exact sequence

$$\begin{aligned}\cdots \to \mathrm{Hom}(B_n,G) \xrightarrow{\iota^*} \mathrm{Hom}(Z_n,G) \to H^n(C;G)\\ \to \mathrm{Hom}(B_{n-1},G) \xrightarrow{\iota^*} \mathrm{Hom}(Z_{n-1},G) \to \cdots\end{aligned}$$

in which the connecting maps are restrictions along $\iota\colon B \hookrightarrow Z$. Now $0 \to B_n \to Z_n \to H_n \to 0$ is a free resolution of H_n, so the cokernel of $\iota^*\colon \operatorname{Hom}(Z_n, G) \to \operatorname{Hom}(B_n, G)$ is by definition $\operatorname{Ext}(H_n, G)$ and its kernel is $\operatorname{Hom}(H_n, G)$. Breaking the long exact sequence into short ones gives Theorem 6.1. The proof shows where every term lives: the Hom term consists of cocycles seen as functions on cycles, and the Ext term consists of cocycles that vanish on all cycles but are not coboundaries — they detect a cycle z with $mz = \partial c$ by taking a nonzero value on c.

Theorem 6.1 (Universal coefficients; recalled)**.** *For a chain complex of free abelian groups and any abelian group G there is a short exact sequence, natural in the complex,*

$$0 \longrightarrow \operatorname{Ext}\big(H_{n-1}(X), G\big) \longrightarrow H^n(X; G) \xrightarrow{\;h\;} \operatorname{Hom}\big(H_n(X), G\big) \longrightarrow 0,$$

where h evaluates a cocycle on a cycle. The sequence splits, but — see Section 6.3 — not naturally.

The practical content is that H^n is Hom of H_n plus Ext of H_{n-1}: free parts stay where they are, torsion moves up one degree.

Example 6.2 (Projective spaces)**.** Dualise the cellular chain complex of Example 4.9. Since $d_k = 1 + (-1)^k$, the cochain differential d^k is the transpose of d_{k+1}, so $d^k = 2$ for k odd and $d^k = 0$ for k even. Hence

$$H^k(\mathbb{RP}^n; \mathbb{Z}) = \begin{cases} \mathbb{Z}, & k = 0, \\ \mathbb{Z}/2, & k \text{ even},\ 0 < k \le n, \\ 0, & k \text{ odd},\ k < n, \\ \mathbb{Z}, & k = n \text{ odd}. \end{cases}$$

Compare with Example 4.9: the $\mathbb{Z}/2$'s have moved up one degree. That is $\operatorname{Ext}(\mathbb{Z}/2, \mathbb{Z}) = \mathbb{Z}/2$ appearing in degree k because H_{k-1} was $\mathbb{Z}/2$.

Example 6.3 (The Ext term, seen on a cochain)**.** For $\mathbb{RP}^2$ with one cell in each dimension, $d_2 = 2$, so the 1-cell a is a cycle with $2a = \partial e^2$. The cellular cochain φ with $\varphi(e^2) = 1$ is a 2-cocycle (there are no 3-cells). It vanishes on every 2-cycle (there are none), so its image in $\operatorname{Hom}(H_2, \mathbb{Z}) = 0$ is zero; but it is not a coboundary, since $\delta\psi(e^2) = \psi(\partial e^2) = 2\psi(a)$ is even. Its class is the generator of $H^2(\mathbb{RP}^2; \mathbb{Z}) = \operatorname{Ext}(\mathbb{Z}/2, \mathbb{Z}) = \mathbb{Z}/2$: it "remembers" that $2a$ bounds, by taking the value 1 on the chain that $2a$ bounds.

Exercise 6.4 (easy)**.** Compute $H^*(K; \mathbb{Z})$ for the Klein bottle from Example 5.16, and note where each summand comes from.

Solution. $H_*(K) = (\mathbb{Z},\ \mathbb{Z} \oplus \mathbb{Z}/2,\ 0)$. So

$$H^0 = \operatorname{Hom}(\mathbb{Z}, \mathbb{Z}) = \mathbb{Z}, \qquad H^1 = \operatorname{Hom}(\mathbb{Z} \oplus \mathbb{Z}/2, \mathbb{Z}) \oplus \operatorname{Ext}(\mathbb{Z}, \mathbb{Z}) = \mathbb{Z},$$

$$H^2 = \operatorname{Hom}(0, \mathbb{Z}) \oplus \operatorname{Ext}(\mathbb{Z} \oplus \mathbb{Z}/2, \mathbb{Z}) = \mathbb{Z}/2.$$

So $H^*(K; \mathbb{Z}) = (\mathbb{Z},\ \mathbb{Z},\ \mathbb{Z}/2)$. Note $H^2 \neq 0$ while $H_2 = 0$: the $\mathbb{Z}/2$ is the Ext of the torsion in H_1, and it lives one degree higher than the torsion that produced it.

Exercise 6.5 (easy)**.** Compute $H^*(K; \mathbb{Z}/2)$ and $H^*(\mathbb{RP}^2; \mathbb{Z}/2)$ for the Klein bottle and the projective plane in two ways: from the cellular cochain complex with $\mathbb{Z}/2$ coefficients, and from Theorem 6.1 applied to the integral homology.

SOURCE: [1, §3.1, Ex. 6(b)], $\mathbb{Z}/2$ part, using cellular instead of simplicial cochains

Hint. *Over $\mathbb{Z}/2$ every cellular boundary of these surfaces is 0 mod 2.*

Solution. *Cochains.* The Klein bottle has a CW structure with one 0-cell, 1-cells a, b and one 2-cell attached along $abab^{-1}$; the integral boundaries are $d_1 = 0$ and $d_2 = (2, 0)^T$. Mod 2 all coboundaries vanish, so $H^*(K; \mathbb{Z}/2) = (\mathbb{Z}/2, (\mathbb{Z}/2)^2, \mathbb{Z}/2)$. For $\mathbb{RP}^2$, $d_2 = 2 \equiv 0$ and $H^*(\mathbb{RP}^2; \mathbb{Z}/2) = (\mathbb{Z}/2, \mathbb{Z}/2, \mathbb{Z}/2)$.

Universal coefficients. $H_*(K) = (\mathbb{Z}, \mathbb{Z} \oplus \mathbb{Z}/2, 0)$. Using $\mathrm{Hom}(\mathbb{Z}, \mathbb{Z}/2) = \mathbb{Z}/2$, $\mathrm{Hom}(\mathbb{Z}/2, \mathbb{Z}/2) = \mathbb{Z}/2$, $\mathrm{Ext}(\mathbb{Z}, \mathbb{Z}/2) = 0$ and $\mathrm{Ext}(\mathbb{Z}/2, \mathbb{Z}/2) = \mathbb{Z}/2$ (Example 8.5): $H^0 = \mathbb{Z}/2$; $H^1 = \mathrm{Hom}(\mathbb{Z} \oplus \mathbb{Z}/2, \mathbb{Z}/2) \oplus \mathrm{Ext}(\mathbb{Z}, \mathbb{Z}/2) = (\mathbb{Z}/2)^2$; $H^2 = \mathrm{Hom}(0, \mathbb{Z}/2) \oplus \mathrm{Ext}(\mathbb{Z} \oplus \mathbb{Z}/2, \mathbb{Z}/2) = \mathbb{Z}/2$. For $\mathbb{RP}^2$, $H_* = (\mathbb{Z}, \mathbb{Z}/2, 0)$ gives $H^1 = \mathrm{Hom}(\mathbb{Z}/2, \mathbb{Z}/2) = \mathbb{Z}/2$ and $H^2 = \mathrm{Ext}(\mathbb{Z}/2, \mathbb{Z}/2) = \mathbb{Z}/2$. Both methods agree. Note that the top class $H^2(-; \mathbb{Z}/2) \cong \mathbb{Z}/2$ of each non-orientable surface comes entirely from an Ext term: it is the mod-2 fundamental class whose existence Theorem 7.11(b) asserts.

Exercise 6.6 (easy)**.** Show that if $f\colon S^n \to S^n$ has degree d, then $f^*\colon H^n(S^n; G) \to H^n(S^n; G)$ is multiplication by d, for every abelian group G.

SOURCE: [1, §3.1, Ex. 9], reproduced in substance

Solution. $H_{n-1}(S^n)$ is free (0 or, for $n = 1$, $H_0 = \mathbb{Z}$), so the Ext term vanishes and the evaluation map $h\colon H^n(S^n; G) \to \mathrm{Hom}(H_n(S^n), G)$ is an isomorphism; it is natural (Exercise 6.16), so $h \circ f^* = \mathrm{Hom}(f_*, G) \circ h$. Since f_* is multiplication by d on $H_n(S^n) = \mathbb{Z}$, $\mathrm{Hom}(f_*, G)$ sends φ to $\varphi \circ (d\cdot) = d\varphi$. Hence f^* is multiplication by d.

Exercise 6.7 (easy)**.** Show that if A is a retract of X then $H^n(X; G) \cong H^n(A; G) \oplus H^n(X, A; G)$ for all n.

SOURCE: [1, §3.1, Ex. 8(c)], reproduced in substance

Solution. Let $r\colon X \to A$ be a retraction and $i\colon A \to X$ the inclusion. Then $i^* r^* = (ri)^* = \mathrm{id}$, so $i^*\colon H^n(X; G) \to H^n(A; G)$ is surjective for every n. In the long exact sequence $\cdots \to H^{n-1}(X) \xrightarrow{i^*} H^{n-1}(A) \xrightarrow{\delta} H^n(X, A) \to H^n(X) \xrightarrow{i^*} H^n(A) \to \cdots$ surjectivity of i^* in degree $n-1$ gives $\delta = 0$, so $0 \to H^n(X, A) \to H^n(X) \xrightarrow{i^*} H^n(A) \to 0$ is exact, and it splits via r^*.

Exercise 6.8 (medium)**.** Regard a cochain $\varphi \in C^1(X; G)$ as a function on paths (a path is a singular 1-simplex). Show that if φ is a cocycle then (a) $\varphi(f \cdot g) = \varphi(f) + \varphi(g)$; (b) φ vanishes on constant paths; (c) $\varphi(f) = \varphi(g)$ if $f \simeq g$ rel endpoints; (d) φ is a coboundary iff $\varphi(f)$ depends only on the endpoints of f. Deduce a homomorphism $H^1(X; G) \to \mathrm{Hom}(\pi_1(X, x_0), G)$.

SOURCE: [1, §3.1, Ex. 5], reproduced in substance

Hint. *Apply the cocycle condition $\varphi(\partial\sigma) = 0$ to well-chosen singular 2-simplices: one whose edges are f, g and $f \cdot g$; a constant one; and the two triangles into which a homotopy square can be cut.*

Solution. For $\sigma\colon \Delta^2 \to X$ with edges $\sigma|_{[v_0,v_1]} = f$, $\sigma|_{[v_1,v_2]} = g$, $\sigma|_{[v_0,v_2]} = h$, the cocycle condition reads $0 = \delta\varphi(\sigma) = \varphi(g) - \varphi(h) + \varphi(f)$.

(a) Let $\lambda\colon \Delta^2 \to I$ be the affine map with $\lambda(v_0) = 0$, $\lambda(v_1) = \frac{1}{2}$, $\lambda(v_2) = 1$, and $\sigma = (f \cdot g) \circ \lambda$. Its edges are exactly f (since $(f \cdot g)(s/2) = f(s)$), g, and $f \cdot g$. Hence $\varphi(f \cdot g) = \varphi(f) + \varphi(g)$.

(b) The constant 2-simplex at x has all three edges equal to the constant path c_x, so $\varphi(c_x) - \varphi(c_x) + \varphi(c_x) = 0$, i.e. $\varphi(c_x) = 0$.

(c) Let $H\colon I \times I \to X$ be a homotopy rel endpoints from f to g. Divide the square by a diagonal into two triangles and regard H on each as a singular 2-simplex. Their edges are: f,

the constant path at the end, the diagonal Δ; and the constant path at the start, g, Δ. The two cocycle relations give $\varphi(\Delta) = \varphi(f) + \varphi(c) = \varphi(f)$ and $\varphi(\Delta) = \varphi(c) + \varphi(g) = \varphi(g)$, using (b). So $\varphi(f) = \varphi(g)$.

(d) If $\varphi = \delta\psi$ then $\varphi(f) = \psi(f(1)) - \psi(f(0))$ depends only on endpoints. Conversely, if it does, fix a basepoint in each path component, set $\psi(x) = \varphi(\gamma_x)$ for any path γ_x from the base point to x; this is well defined by hypothesis, and $\varphi(f) = \psi(f(1)) - \psi(f(0))$ because $\varphi(\gamma_{f(0)} \cdot f) = \varphi(\gamma_{f(0)}) + \varphi(f)$ by (a) and the left side equals $\psi(f(1))$. So $\varphi = \delta\psi$.

By (a) and (c), a cocycle defines a homomorphism $\pi_1(X, x_0) \to G$, $[\gamma] \mapsto \varphi(\gamma)$; by (d) coboundaries give the zero homomorphism on loops (a loop has equal endpoints). Hence a well-defined homomorphism $H^1(X; G) \to \mathrm{Hom}(\pi_1(X, x_0), G)$.

6.2 Moore spaces

To probe a theorem about H_* one wants spaces with prescribed homology, and they exist.

Proposition 6.9. *For every abelian group A and every $n \geq 1$ there is a CW complex $M(A, n)$, with cells only in dimensions n and $n+1$ besides a single 0-cell, such that*

$$\widetilde{H}_i\big(M(A,n)\big) = \begin{cases} A, & i = n, \\ 0, & i \neq n. \end{cases}$$

Proof. Choose a free resolution $0 \to F_1 \xrightarrow{\alpha} F_0 \to A \to 0$; over $\mathbb{Z}$ every subgroup of a free abelian group is free, so a resolution of length one exists for *every* A, finitely generated or not. Let $X_0 = \bigvee_{F_0} S^n$, a wedge with one sphere for each basis element of F_0, so that $H_n(X_0) = F_0$. For each basis element of F_1 attach an $(n+1)$-cell along a map $S^n \to X_0$ representing the corresponding element of $\pi_n(X_0) \to H_n(X_0) = F_0$, namely its image α(basis element); such a map exists because for $n \geq 2$ the Hurewicz map $\pi_n(X_0) \to H_n(X_0)$ is onto, and for $n = 1$ one uses that H_1 is the abelianisation of a free group.

The cellular chain complex of the result is $F_1 \xrightarrow{\alpha} F_0$ concentrated in degrees $n+1$ and n, whose homology is A in degree n and $\ker\alpha = 0$ in degree $n+1$. □

Remark 6.10. Two features will be used repeatedly. First, $M(A, n)$ exists for *arbitrary* A — there is no finite generation hypothesis, because $\mathbb{Z}$ is a principal ideal domain and resolutions have length one. Second, $H_{n+1}(M(A, n)) = 0$ by construction, which is what makes these spaces useful as test objects: they have homology in exactly one degree, so any invariant that is nonzero elsewhere is detecting something other than homology. Chapter 8 uses both facts to prove a theorem of pure algebra by topological means.

Exercise 6.11 (easy)**.** Compute $H^*\big(M(\mathbb{Z}/m, n); \mathbb{Z}\big)$ for $n \geq 1$, $m \geq 2$.

Hint. *The cellular cochain complex is $\mathbb{Z} \xrightarrow{m} \mathbb{Z}$ in degrees n and $n+1$.*

Solution. By Proposition 6.9 the chain complex is $\mathbb{Z} \xrightarrow{m} \mathbb{Z}$ in degrees $n+1, n$, so dualising gives $\mathbb{Z} \xrightarrow{m} \mathbb{Z}$ in degrees $n, n+1$. Hence $H^n = \ker(m) = 0$ and $H^{n+1} = \mathbb{Z}/m$, all other reduced groups vanishing. Via Theorem 6.1: H^n has $\mathrm{Hom}(\mathbb{Z}/m, \mathbb{Z}) = 0$ and no Ext contribution, while H^{n+1} is $\mathrm{Ext}(\mathbb{Z}/m, \mathbb{Z}) = \mathbb{Z}/m$.

6.3 A natural sequence with an unnatural splitting

What exactly is being claimed when one says the splitting is not natural? The following turns the claim into something testable.

Proposition 6.12. *Suppose the splitting of Theorem 6.1 could be chosen naturally, i.e. so that for every map $f\colon X \to Y$ the induced maps commute with the chosen isomorphisms $H^n(-;\mathbb{Z}) \cong \mathrm{Hom}(H_n,\mathbb{Z}) \oplus \mathrm{Ext}(H_{n-1},\mathbb{Z})$. Then any map f with $\widetilde{H}_*(f) = 0$ would satisfy $\widetilde{H}^*(f) = 0$.*

Proof. A natural splitting would identify f^* with $\mathrm{Hom}(f_*,\mathbb{Z}) \oplus \mathrm{Ext}(f_*,\mathbb{Z})$. Each summand is a functor evaluated at f_*; so if f_* vanishes in all positive degrees, both summands vanish, and therefore so does f^*. □

So it suffices to exhibit one map that kills reduced homology and does not kill reduced cohomology. The smallest such map is already available.

Example 6.13 (A map that homology cannot see)**.** Let $q\colon \mathbb{RP}^2 \to \mathbb{RP}^2/\mathbb{RP}^1 \cong S^2$ collapse the 1-skeleton.

On homology q is zero. By Example 4.9, $\widetilde{H}_*(\mathbb{RP}^2)$ is $\mathbb{Z}/2$ in degree 1 and 0 elsewhere, while $\widetilde{H}_*(S^2)$ is $\mathbb{Z}$ in degree 2 and 0 elsewhere. In degree 1 the target is 0; in degree 2 the source is 0. So $q_* = 0$ in every positive degree, for want of anywhere to go.

On cohomology q is not zero. On cellular chains, q carries the 2-cell of $\mathbb{RP}^2$ homeomorphically to the 2-cell of S^2, so $q_\#\colon C_2(\mathbb{RP}^2) \to C_2(S^2)$ is an isomorphism, and therefore so is $q^\#\colon C^2(S^2) \to C^2(\mathbb{RP}^2)$. Passing to cohomology, and using Example 6.2, the map

$$q^*\colon H^2(S^2;\mathbb{Z}) = \mathbb{Z} \longrightarrow H^2(\mathbb{RP}^2;\mathbb{Z}) = \mathbb{Z}/2$$

is the quotient map $\mathbb{Z} \to \mathbb{Z}/2$, since a generating cocycle of $C^2(S^2)$ pulls back to a generating cocycle of $C^2(\mathbb{RP}^2)$. In particular $q^* \neq 0$.

Corollary 6.14. *The splitting in Theorem 6.1 cannot be chosen naturally.*

Proof. Example 6.13 contradicts Proposition 6.12. □

Warning 6.15. None of this contradicts Theorem 6.1. What is natural is the *sequence*: for every f, the two squares

$$\begin{array}{ccccccccc} 0 & \longrightarrow & \mathrm{Ext}(H_{n-1}Y,G) & \longrightarrow & H^n(Y;G) & \longrightarrow & \mathrm{Hom}(H_nY,G) & \longrightarrow & 0 \\ & & \downarrow & & \downarrow{\scriptstyle f^*} & & \downarrow & & \\ 0 & \longrightarrow & \mathrm{Ext}(H_{n-1}X,G) & \longrightarrow & H^n(X;G) & \longrightarrow & \mathrm{Hom}(H_nX,G) & \longrightarrow & 0 \end{array}$$

commute, the outer vertical maps being $\mathrm{Ext}(f_*,G)$ and $\mathrm{Hom}(f_*,G)$. And for each fixed X the sequence splits. What fails is the possibility of choosing the splittings compatibly for all X at once. Example 6.13 shows exactly how: there f^* is nonzero although both outer maps are zero, which a splitting compatible with f would forbid.

Exercise 6.16 (medium)**.** Verify directly that the evaluation map $h\colon H^n(X;G) \to \mathrm{Hom}(H_n(X),G)$ of Theorem 6.1 is natural, and explain why this does not conflict with Corollary 6.14.

Hint. *Write h on cocycles and cycles and chase $f^{\#}$ and $f_{\#}$.*

Solution. For a cocycle $\varphi \in C^n(Y;G)$ and a cycle $z \in C_n(X)$,

$$h\big(f^*[\varphi]\big)([z]) = \big(f^{\#}\varphi\big)(z) = \varphi\big(f_{\#}z\big) = h([\varphi])\big(f_*[z]\big) = \mathrm{Hom}(f_*,G)\big(h[\varphi]\big)([z]),$$

so $h \circ f^* = \mathrm{Hom}(f_*,G) \circ h$. This is a statement about the map $H^n \to \mathrm{Hom}(H_n,G)$, which exists canonically. A splitting is a map $\mathrm{Hom}(H_n,G) \to H^n$ going the other way; nothing in the computation above produces one, and Corollary 6.14 says no natural choice exists. In Example 6.13 the map h is zero on both spaces — both Hom terms vanish — and all the content of q^* lives in the Ext subgroup, which is precisely where a splitting would have had to place it compatibly.

Remark 6.17 (What to do with the mechanism). Example 6.13 is the smallest instance of a general construction: take a space whose reduced homology is a single torsion group, collapse the part of it carrying the torsion, and compare. Exercise 6.11 has already computed the cohomology of $M(\mathbb{Z}/m,n)$ for every m and n, and Proposition 6.9 supplies the collapse map $M(\mathbb{Z}/m,n) \to S^{n+1}$; running the argument of Example 6.13 in those dimensions, and then running it backwards along the inclusion of the bottom sphere, is the natural way to consolidate this section.

Exercise 6.18 (medium)**.** Show that the functors $h^n(X) = \mathrm{Hom}(H_n(X),\mathbb{Z})$ cannot be extended to a cohomology theory on CW pairs: there are no relative groups $h^n(X,A)$ for which the long exact sequences of pairs, with the maps $h^n(X) \to h^n(A)$ induced by inclusion, are exact and $h^n(X,A) \cong \widetilde{h}^n(X/A)$ for CW pairs.

Source: [1, §3.1, Ex. 7], reproduced in substance

Hint. *Use the mapping cylinder X of a degree-two map $S^1 \to S^1$, with A the source circle. Then $X \simeq S^1$ and $X/A \cong \mathbb{RP}^2$.*

Solution. Let $f\colon S^1 \to S^1$ have degree 2, $X = M_f$ its mapping cylinder and $A = S^1 \times \{0\} \subseteq X$ the source end. Then (X,A) is a CW pair, X deformation retracts onto the target circle, and X/A is the mapping cone of f, which is $\mathbb{RP}^2$ (Example 4.9: the 2-cell of $\mathbb{RP}^2$ is attached by a degree-two map). The inclusion $A \hookrightarrow X$ composed with the retraction is f, so it induces multiplication by 2 on $H_1 \cong \mathbb{Z}$, hence multiplication by 2 on $h^1 = \mathrm{Hom}(H_1,\mathbb{Z}) \cong \mathbb{Z}$.

If a theory as in the statement existed, the segment $h^1(X) \xrightarrow{\times 2} h^1(A) \to h^2(X,A) \to h^2(X)$ would be exact, with $h^2(X) = \mathrm{Hom}(H_2(S^1),\mathbb{Z}) = 0$. Hence $h^2(X,A) \cong \mathrm{coker}(\times 2) = \mathbb{Z}/2$. But $h^2(X,A) \cong \widetilde{h}^2(\mathbb{RP}^2) = \mathrm{Hom}(H_2(\mathbb{RP}^2),\mathbb{Z}) = 0$. Contradiction. Genuine cohomology passes this test because $H^2(\mathbb{RP}^2;\mathbb{Z}) = \mathrm{Ext}(\mathbb{Z}/2,\mathbb{Z}) = \mathbb{Z}/2$.

Exercise 6.19 (medium)**.** Compute $H^*(L(p,q);G)$ for $G = \mathbb{Z}$, $\mathbb{Q}$, $\mathbb{Z}/m$ (any $m \geq 2$), using the cellular cochain complex, and verify the answers with Theorem 6.1.

Source: [1, §3.1, Ex. 10], specialised to the 3-dimensional lens spaces $L(p,q)$ of Example 2.30

Hint. *The cellular chain complex is $\mathbb{Z} \xrightarrow{0} \mathbb{Z} \xrightarrow{p} \mathbb{Z} \xrightarrow{0} \mathbb{Z}$ (Example 4.10). Dualise and tensor with G.*

Solution. The cellular cochain complex with coefficients G is $G \xrightarrow{0} G \xrightarrow{p} G \xrightarrow{0} G$ in degrees $0,1,2,3$ (the transpose of $\mathbb{Z} \xrightarrow{0} \mathbb{Z} \xrightarrow{p} \mathbb{Z} \xrightarrow{0} \mathbb{Z}$, tensored with G). So

$$H^0 = G, \quad H^1 = \ker(p\colon G \to G) = {}_pG, \quad H^2 = G/pG, \quad H^3 = G,$$

where ${}_pG$ is the p-torsion of G. For $G = \mathbb{Z}$: $(\mathbb{Z},0,\mathbb{Z}/p,\mathbb{Z})$. For $G = \mathbb{Q}$: $(\mathbb{Q},0,0,\mathbb{Q})$, the rational cohomology of S^3. For $G = \mathbb{Z}/m$, with $d = \gcd(p,m)$: ${}_p(\mathbb{Z}/m) \cong \mathbb{Z}/d$ and $(\mathbb{Z}/m)/p(\mathbb{Z}/m) \cong \mathbb{Z}/d$, so $(\mathbb{Z}/m,\ \mathbb{Z}/d,\ \mathbb{Z}/d,\ \mathbb{Z}/m)$.

By Theorem 6.1 with $H_* = (\mathbb{Z}, \mathbb{Z}/p, 0, \mathbb{Z})$: $H^1 = \mathrm{Hom}(\mathbb{Z}/p, G) \oplus \mathrm{Ext}(\mathbb{Z}, G) = {}_pG$; $H^2 = \mathrm{Hom}(0, G) \oplus \mathrm{Ext}(\mathbb{Z}/p, G) = G/pG$ (from the resolution $0 \to \mathbb{Z} \xrightarrow{p} \mathbb{Z} \to \mathbb{Z}/p \to 0$); $H^3 = \mathrm{Hom}(\mathbb{Z}, G) \oplus \mathrm{Ext}(0, G) = G$. These agree.

Remark 6.20 (Forward). The same phenomenon occurs one theorem later. The Künneth sequence for $H_*(X \times Y)$ is natural and splits, and its splitting is not natural either; Chapter 8 treats it, and by then the reader should be able to predict both the statement and the shape of the counterexample before reading them.

The deeper point is that the Ext summand carries information about X that $H_*(X)$ alone does not determine as a functor. Recovering it systematically — finding invariants of a space finer than its cohomology groups — is what cohomology operations do, and Chapter 18 constructs them.

6.4 What to carry forward

- H^n is $\mathrm{Hom}(H_n, G)$ plus $\mathrm{Ext}(H_{n-1}, G)$: torsion moves up one degree (Theorem 6.1 and Example 6.2).
- Moore spaces exist for every abelian group, with cells in two dimensions only (Proposition 6.9 and Remark 6.10); they are the standard probes.
- Natural sequence, natural evaluation map, unnatural splitting (Warning 6.15); the test for non-naturality is Proposition 6.12, and the smallest witness is Example 6.13.
- Cohomology distinguishes maps that homology cannot — the first evidence that the extra structure on H^* is worth having, which Chapter 7 turns into a product.

Chapter 7

Products and duality

Three invariants have now failed on the same pair of spaces. In Chapter 2 the fundamental group did not distinguish $L(5,1)$ from $L(5,2)$; in Chapter 4 their homology turned out not to involve q at all. The reason, in both cases, was structural: those invariants are assembled from the cells one at a time, and lens spaces differ in how their cells are *linked*.

This chapter introduces the first invariant with a multiplication. It succeeds where the groups fail — distinguishing spaces with identical cohomology groups — and it also fails, on the same lens spaces, in a way that will take until Chapter 18 to repair. Both halves are the point.

7.1 The cup product

We take the construction as known: $H^*(X;R) = \bigoplus_n H^n(X;R)$ carries an associative product, natural in X, with unit $1 \in H^0$, satisfying

$$\alpha \smile \beta = (-1)^{|\alpha||\beta|}\, \beta \smile \alpha.$$

Naturality means $f^*(\alpha \smile \beta) = f^*\alpha \smile f^*\beta$: a map of spaces induces a map of rings, and that is where all the power comes from.

The formula, and the cross product. On singular cochains with coefficients in a commutative ring R, the cup product of $\varphi \in C^k$ and $\psi \in C^\ell$ is the cochain $(\varphi \smile \psi)(\sigma) = \varphi\big(\sigma|_{[v_0,\dots,v_k]}\big)\,\psi\big(\sigma|_{[v_k,\dots,v_{k+\ell}]}\big)$: evaluate φ on the front k-face and ψ on the back ℓ-face. It satisfies $\delta(\varphi \smile \psi) = \delta\varphi \smile \psi + (-1)^k \varphi \smile \delta\psi$, so it descends to cohomology. It is associative on the nose, but commutative only up to homotopy: the front-face/back-face recipe treats the two factors asymmetrically, and the homotopy correcting this asymmetry is the germ of the Steenrod operations of Chapter 18. The *cross product* $a \times b = p_1^* a \smile p_2^* b \in H^{k+\ell}(X \times Y; R)$ of $a \in H^k(X;R)$ and $b \in H^\ell(Y;R)$ is the form in which products are usually computed.

Theorem 7.1 (Künneth for cohomology rings; recalled)**.** *Let X, Y be CW complexes and suppose $H^k(Y;R)$ is a finitely generated free R-module for every k (for instance, R a field and Y with finitely many cells in each dimension). Then*

$$\times \colon H^*(X;R) \otimes_R H^*(Y;R) \xrightarrow{\;\cong\;} H^*(X \times Y; R)$$

is an isomorphism of graded rings, where the tensor product of graded-commutative rings has $(a \otimes b)(a' \otimes b') = (-1)^{|b||a'|} aa' \otimes bb'$ [1, Thm. 3.15].

The sign in the multiplication rule is the first appearance of the Koszul sign rule: moving an element of degree $|b|$ past one of degree $|a'|$ costs $(-1)^{|b||a'|}$. Chapter 28 explains where it comes from.

Example 7.2 (The torus)**.** $H^*(T^2;\mathbb{Z})$ is the exterior algebra $\Lambda(a,b)$ on two generators of degree one: $a^2 = b^2 = 0$ by graded commutativity, $ab = -ba$, and ab generates $H^2(T^2) \cong \mathbb{Z}$. The last assertion — that the product of the two one-dimensional generators is a generator, not merely some element — is the content, and it is what Theorem 7.11 will explain.

Example 7.3 (Complex projective plane against a wedge)**.** $H^*(\mathbb{CP}^2;\mathbb{Z}) = \mathbb{Z}[x]/(x^3)$ with $|x| = 2$, so the groups are $\mathbb{Z}, 0, \mathbb{Z}, 0, \mathbb{Z}$ in degrees $0, \dots, 4$ and x^2 generates H^4.

The wedge $S^2 \vee S^4$ has exactly the same cohomology *groups*. But its ring is different: writing y for the generator of $H^2(S^2 \vee S^4)$, we have $y^2 = 0$. Indeed y restricts to a generator on S^2 and to 0 on S^4, so y^2 restricts to 0 on both — on S^2 because $H^4(S^2) = 0$ — and restriction to the two summands is injective on $\widetilde{H}^*(S^2 \vee S^4)$.

So $\mathbb{CP}^2 \not\simeq S^2 \vee S^4$: the groups cannot see the difference and the ring can.

The mechanism in that example is worth isolating, because it is the only thing one ever uses about wedges.

Exercise 7.4 (easy)**.** Show that $\widetilde{H}^*(X \vee Y; R) \cong \widetilde{H}^*(X;R) \oplus \widetilde{H}^*(Y;R)$ as graded groups, and that under this isomorphism the product of a class coming from X with a class coming from Y is zero.

Hint. *Use the retractions $X \vee Y \to X$ and $X \vee Y \to Y$, together with the inclusions, and naturality of the product.*

Solution. The inclusions $i\colon X \to X \vee Y$, $j\colon Y \to X \vee Y$ and the retractions $r\colon X \vee Y \to X$, $s\colon X \vee Y \to Y$ collapsing the other summand satisfy $ri = \mathrm{id}$, $sj = \mathrm{id}$, $rj = \mathrm{const}$, $si = \mathrm{const}$. Hence (i^*, j^*) is a split surjection with splitting $r^* + s^*$, giving the direct sum decomposition, with the summand "coming from X" being $\operatorname{im} r^*$.

Now let $\alpha = r^*\alpha'$ and $\beta = s^*\beta'$ with α', β' of positive degree. Then $i^*(\alpha \smile \beta) = i^*\alpha \smile i^*\beta$, and $i^*\beta = i^*s^*\beta' = (si)^*\beta' = 0$ since si is constant and β' has positive degree. So $i^*(\alpha \smile \beta) = 0$, and likewise $j^*(\alpha \smile \beta) = 0$. As (i^*, j^*) is injective on $\widetilde{H}^*$, the product vanishes.

Exercise 7.5 (medium)**.** Show that $\mathbb{RP}^3$ is not homotopy equivalent to $\mathbb{RP}^2 \vee S^3$, although the two spaces have isomorphic homology and cohomology groups with every coefficient group.

Source: [1, §3.2, Ex. 7], reproduced in substance

Hint. *$H^*(\mathbb{RP}^3;\mathbb{Z}/2) = \mathbb{Z}/2[a]/(a^4)$ (quoted from Chapter 12, or computed by the method of Example 6.2 together with Poincaré duality). What is a^3 in each space?*

Solution. Both spaces have one cell in each dimension $0, 1, 2, 3$, and the cellular chain complexes are $\mathbb{Z} \xrightarrow{0} \mathbb{Z} \xrightarrow{2} \mathbb{Z} \xrightarrow{0} \mathbb{Z}$ for both (for $\mathbb{RP}^3$ by Example 4.9; for $\mathbb{RP}^2 \vee S^3$ because the 3-cell is attached to the basepoint). So all homology and cohomology groups agree, with any coefficients.

In $H^*(\mathbb{RP}^3;\mathbb{Z}/2) = \mathbb{Z}/2[a]/(a^4)$ the generator $a \in H^1$ has $a^3 \neq 0$. In $H^*(\mathbb{RP}^2 \vee S^3;\mathbb{Z}/2)$ the generator of H^1 is a class b pulled back from $\mathbb{RP}^2$ along the retraction, and b^3 is pulled back from $H^3(\mathbb{RP}^2;\mathbb{Z}/2) = 0$, so $b^3 = 0$. A homotopy equivalence would induce a ring isomorphism carrying the unique nonzero class of H^1 to the unique nonzero class of H^1, hence a^3 to b^3 — impossible. (The ring $H^*(\mathbb{RP}^n;\mathbb{Z}/2) = \mathbb{Z}/2[a]/(a^{n+1})$ is proved in [1, Thm. 3.19]; for $n = 3$ one can also argue: Poincaré duality mod 2 forces $a \cdot a^2 \neq 0$, since a^2 generates H^2 by the case $n = 2$ of the same theorem.)

Exercise 7.6 (medium)**.** Let p be a prime, $X = \mathbb{CP}^2 \cup_\phi e^3$ with $\phi\colon S^2 \to \mathbb{CP}^1 \subseteq \mathbb{CP}^2$ of degree p, and $Y = M(\mathbb{Z}/p, 2) \vee S^4$. Show that $H^*(X;\mathbb{Z}) \cong H^*(Y;\mathbb{Z})$ as rings, but $H^*(X;\mathbb{Z}/p) \ncong H^*(Y;\mathbb{Z}/p)$ as rings.

SOURCE: [1, §3.2, Ex. 8], reproduced in substance

Hint. *Compute both integral rings: there is no room for nonzero products. Mod p, use the inclusion $\mathbb{CP}^2 \subseteq X$.*

Solution. *Integral rings.* X has cells e^0, e^2, e^3, e^4 with $d_3(e^3) = pe^2$ and $d_4 = 0$ (the 4-cell is attached inside $\mathbb{CP}^1$, missing e^3). So $H_*(X) = (\mathbb{Z}, 0, \mathbb{Z}/p, 0, \mathbb{Z})$ and by Theorem 6.1 $H^*(X;\mathbb{Z}) = (\mathbb{Z}, 0, 0, \mathbb{Z}/p, \mathbb{Z})$ in degrees 0–4. Y has the same cells and boundaries, hence the same groups. In both, a product of two positive-degree classes would have degree $\geq 3 + 3$ (the only nonzero positive degrees are 3 and 4), so all such products vanish. Hence the rings are isomorphic.

Mod p rings. With $\mathbb{Z}/p$ coefficients all cellular coboundaries vanish, so both spaces have $H^*(-;\mathbb{Z}/p)$ equal to $\mathbb{Z}/p$ in degrees $0, 2, 3, 4$. Let x generate H^2. The inclusion $j\colon \mathbb{CP}^2 \to X$ is an isomorphism on cellular cochains in degrees 2 and 4 (the same cells), hence on H^2 and H^4 mod p. Since $H^*(\mathbb{CP}^2;\mathbb{Z}/p) = \mathbb{Z}/p[y]/(y^3)$, $j^*(x^2) = (j^*x)^2 = (\pm y)^2 \neq 0$ up to a unit, so $x^2 \neq 0$ in X. In Y, x comes from $M(\mathbb{Z}/p, 2)$ and x^2 restricts to zero on both wedge summands (degree reasons on $M(\mathbb{Z}/p, 2)$, and $x|_{S^4} = 0$), so $x^2 = 0$ by Exercise 7.4. A ring isomorphism would carry x to a unit multiple of x and preserve squares, so the mod-p rings differ. Hence $X \not\simeq Y$, although no integral cup product distinguishes them.

Remark 7.7 (When the integral ring does decide)*.* If $H_n(X;\mathbb{Z})$ is free for every n then $H^*(X;\mathbb{Z}/p) \cong H^*(X;\mathbb{Z}) \otimes \mathbb{Z}/p$ as rings [1, §3.2, Ex. 9]: reduction mod p is a ring map, and it is surjective with kernel pH^* exactly when there are no Ext terms to contribute classes that are not reductions. Exercise 7.6 is the counterexample when torsion is present; the mod-p class x there is the image of no integral class.

Exercise 7.8 (medium)**.** Show that $S^2 \times S^2$ and $S^2 \vee S^2 \vee S^4$ have isomorphic cohomology groups in every degree but non-isomorphic cohomology rings, and deduce that they are not homotopy equivalent.

Hint. *By Künneth, $H^*(S^2 \times S^2)$ has generators a, b in degree 2 with $a^2 = b^2 = 0$. What is ab?*

Solution. For the product, Künneth gives $H^*(S^2 \times S^2;\mathbb{Z}) = \mathbb{Z}[a, b]/(a^2, b^2)$ with $|a| = |b| = 2$, so the groups are $\mathbb{Z}, 0, \mathbb{Z}^2, 0, \mathbb{Z}$ and ab generates H^4. For the wedge, the groups are the same, but by Exercise 7.4 every product of positive-degree classes vanishes: the classes from the two S^2's multiply to zero, and each squares to zero because $H^4(S^2) = 0$.

A ring isomorphism would have to carry a pair of degree-2 generators to a pair of degree-2 generators and preserve their product, which is a generator of H^4 on one side and 0 on the other. So the rings differ, and since $H^*(-;\mathbb{Z})$ is a homotopy-invariant functor to rings, the spaces are not homotopy equivalent.

Exercise 7.9 (medium)**.** (a) Show that for $n > m \geq 1$ there is no map $\mathbb{RP}^n \to \mathbb{RP}^m$ inducing a nonzero map $H^1(\mathbb{RP}^m;\mathbb{Z}/2) \to H^1(\mathbb{RP}^n;\mathbb{Z}/2)$.

(b) Deduce the Borsuk–Ulam theorem: for every map $f\colon S^n \to \mathbb{R}^n$ there is x with $f(x) = f(-x)$.

SOURCE: [1, §3.2, Ex. 3], reproduced in substance

Hint. *For (a), $a_n^{m+1} \neq 0$ in $\mathbb{Z}/2[a_n]/(a_n^{n+1})$ but $a_m^{m+1} = 0$. For (b), if $f(x) \neq f(-x)$ everywhere, $g(x) = (f(x) - f(-x))/|f(x) - f(-x)|$ is an odd map $S^n \to S^{n-1}$; show the induced map*

$\mathbb{RP}^n \to \mathbb{RP}^{n-1}$ is nonzero on $H^1(-;\mathbb{Z}/2)$ by looking at what it does to a path lifting to an arc from x to $-x$.

Solution. (a) Let $h\colon \mathbb{RP}^n \to \mathbb{RP}^m$ with $h^*(a_m) = a_n$ (the only nonzero value). Then $h^*(a_m^{m+1}) = a_n^{m+1}$, which is nonzero because $m+1 \le n$; but $a_m^{m+1} = 0$ in $H^{m+1}(\mathbb{RP}^m;\mathbb{Z}/2) = 0$. Contradiction.

(b) Suppose $f(x) \neq f(-x)$ for all x, and define g as in the hint. Then $g(-x) = -g(x)$, so g induces $\bar g\colon \mathbb{RP}^n \to \mathbb{RP}^{n-1}$. By (a), $\bar g^* = 0$ on $H^1(-;\mathbb{Z}/2)$. Now $H^1(\mathbb{RP}^k;\mathbb{Z}/2) = \operatorname{Hom}(\pi_1(\mathbb{RP}^k),\mathbb{Z}/2)$ for $k \ge 2$ (and for $k = 1$ similarly with $\pi_1 = \mathbb{Z}$), and the nonzero class takes the value 1 on a loop exactly when the loop lifts to a path in the sphere joining a point to its antipode. Choose a path γ in S^n joining x to $-x$; its image is a loop $\bar\gamma$ in $\mathbb{RP}^n$ on which a_n is nonzero. Then $g\circ\gamma$ is a path from $g(x)$ to $g(-x) = -g(x)$, so $\bar g\circ\bar\gamma$ is a loop on which a_{n-1} is nonzero, i.e. $(\bar g^* a_{n-1})(\bar\gamma) \neq 0$. So $\bar g^* \neq 0$, contradicting (a). (For $n = 1$ the statement is the intermediate value theorem applied to $f(x) - f(-x)$, which changes sign.)

Exercise 7.10 (medium)**.** Let $k, \ell > 0$. Show that every map $f\colon S^{k+\ell} \to S^k \times S^\ell$ induces the zero map on $H_{k+\ell}$.

SOURCE: [1, §3.2, Ex. 11], reproduced in substance

Solution. By Theorem 7.1, $H^{k+\ell}(S^k \times S^\ell;\mathbb{Z})$ is generated by $a \times b = p_1^* a \smile p_2^* b$ with a, b generators. Then $f^*(a\times b) = f^*p_1^* a \smile f^* p_2^* b = 0$, since $f^*p_1^* a \in H^k(S^{k+\ell}) = 0$. Since $H_{k+\ell-1}(S^k \times S^\ell)$ is free, the evaluation map identifies $H^{k+\ell}$ with the dual of $H_{k+\ell}$ naturally (Exercise 6.6), so $f^* = 0$ on $H^{k+\ell}$ forces $\operatorname{Hom}(f_*,\mathbb{Z}) = 0$, i.e. $f_* = 0$ on $H_{k+\ell} \cong \mathbb{Z}$.

7.2 Poincaré duality

Theorem 7.11 (Poincaré duality; recalled)**.** *Let M be a closed connected n-manifold.*

(a) If M is orientable, capping with the fundamental class gives an isomorphism

$$H^k(M;\mathbb{Z}) \;\cong\; H_{n-k}(M;\mathbb{Z}) \qquad \text{for all } k.$$

(b) With no orientability hypothesis at all, the same holds over $\mathbb{F}_2$:

$$H^k(M;\mathbb{F}_2) \;\cong\; H_{n-k}(M;\mathbb{F}_2).$$

In the orientable case the pairing $H^k(M;\mathbb{Z}) \otimes H^{n-k}(M;\mathbb{Z}) \to H^n(M;\mathbb{Z}) \cong \mathbb{Z}$ given by the cup product is nondegenerate modulo torsion; with $\mathbb{F}_2$ coefficients it is nondegenerate.

The nondegeneracy statement is what makes the cup product computable on manifolds: it says a nonzero class in H^k *must* have a partner it pairs nontrivially with. Example 7.2 is the smallest instance.

The ingredients, briefly. An n-manifold M has local homology $H_n(M, M\setminus x;\mathbb{Z}) \cong \mathbb{Z}$ at each point; a *local orientation* is a choice of generator, and an orientation is a choice varying continuously with x. For closed connected M, an orientation exists iff $H_n(M;\mathbb{Z}) \cong \mathbb{Z}$, and then a *fundamental class* $[M] \in H_n(M;\mathbb{Z})$ restricts to the chosen generator at every point. Over $\mathbb{F}_2$ every manifold is orientable, and $[M] \in H_n(M;\mathbb{F}_2)$ always exists. The duality isomorphism of Theorem 7.11 is the *cap product* $\alpha \mapsto [M] \frown \alpha$, where for a singular simplex σ and a k-cochain α one puts $\sigma \frown \alpha =$

$\alpha(\sigma|_{[v_0,\dots,v_k]})\,\sigma|_{[v_k,\dots,v_n]}$; the cup and cap products are related by $\beta(\sigma \frown \alpha) = (\alpha \smile \beta)(\sigma)$, which is what turns duality into the nondegeneracy of the cup pairing. In practice one uses the following form: for a closed connected R-orientable n-manifold and R a field, the pairing $H^k(M;R) \times H^{n-k}(M;R) \to R$, $(\alpha,\beta) \mapsto \langle \alpha \smile \beta, [M]\rangle$, is perfect.

Exercise 7.12 (easy)**.** Show that $H^1(X;\mathbb{Z})$ is torsion-free for every space X, and free abelian if $H_1(X)$ is finitely generated.

Hint. *Apply Theorem 6.1 in degree one.*

Solution. By Theorem 6.1, $H^1(X;\mathbb{Z})$ is an extension of $\mathrm{Hom}(H_1X,\mathbb{Z})$ by $\mathrm{Ext}(H_0X,\mathbb{Z})$, and H_0X is free abelian, so $\mathrm{Ext}(H_0X,\mathbb{Z}) = 0$ and $H^1(X;\mathbb{Z}) \cong \mathrm{Hom}(H_1X,\mathbb{Z})$. A group of homomorphisms into $\mathbb{Z}$ is torsion-free, since $n\varphi = 0$ with $n \neq 0$ forces $\varphi = 0$; and Hom of a finitely generated group into $\mathbb{Z}$ is finitely generated and torsion-free, hence free.

Corollary 7.13. *For a closed orientable n-manifold M, the group $H_{n-1}(M;\mathbb{Z})$ is free abelian.*

Proof. $H_{n-1}(M;\mathbb{Z}) \cong H^1(M;\mathbb{Z})$ by Theorem 7.11(a), which is free by Exercise 7.12, the homology of a closed manifold being finitely generated. □

Exercise 7.14 (medium)**.** Show that $\chi(M) = 0$ for every closed manifold of odd dimension, orientable or not.

Hint. *The Euler characteristic can be computed with coefficients in any field (Exercise 4.15 applies verbatim to the $\mathbb{F}_2$ chain complex). Then use duality in the form that needs no orientation.*

Solution. Compute $\chi(M) = \sum_k (-1)^k \dim_{\mathbb{F}_2} H_k(M;\mathbb{F}_2)$, which agrees with the integral Euler characteristic because both equal the alternating sum of the numbers of cells. By Theorem 7.11(b), $\dim_{\mathbb{F}_2} H_k(M;\mathbb{F}_2) = \dim_{\mathbb{F}_2} H^{n-k}(M;\mathbb{F}_2) = \dim_{\mathbb{F}_2} H_{n-k}(M;\mathbb{F}_2)$, the last equality because over a field cohomology is the dual of homology. So the terms k and $n-k$ have equal dimensions, and for n odd they have opposite signs, so they cancel in pairs.

Exercise 7.15 (medium)**.** For closed orientable surfaces show that a map $f\colon M_g \to M_h$ of degree one (i.e. $f_*[M_g] = [M_h]$) exists if and only if $g \geq h$.

Source: [1, §3.3, Ex. 11]; the solution given here uses cup products instead of the covering-space argument suggested there

Hint. *For "if", collapse the extra handles. For "only if", show f^* is injective on H^1 using the cup pairing and the fact that f^* is an isomorphism on H^2.*

Solution. *If.* Write $M_g = M_h \# M_{g-h}$ (with $M_0 = S^2$ and the convention $M\#S^2 = M$). Collapsing the complement of an open disc in the M_{g-h} summand — a subcomplex in a suitable CW structure, namely M_{g-h} minus a disc, which contracts onto a wedge of circles, collapsing which leaves M_h — gives a map $c\colon M_g \to M_h$ that is a homeomorphism over the M_h part away from a small disc. By the local degree formula (Chapter 4) its degree is ± 1; composing with an orientation-reversing homeomorphism if needed gives degree 1.

Only if. Let f have degree 1. By the naturality of evaluation, $\langle f^*\gamma, [M_g]\rangle = \langle \gamma, f_*[M_g]\rangle = \langle \gamma, [M_h]\rangle$ for $\gamma \in H^2(M_h;\mathbb{Z})$, so f^* is an isomorphism on $H^2 \cong \mathbb{Z}$. Let $\alpha \in H^1(M_h;\mathbb{Q})$ be nonzero. By nondegeneracy over $\mathbb{Q}$ there is β with $\langle \alpha \smile \beta, [M_h]\rangle \neq 0$. Then $\langle f^*\alpha \smile f^*\beta, [M_g]\rangle = \langle \alpha \smile \beta, [M_h]\rangle \neq 0$, so $f^*\alpha \neq 0$. Thus $f^*\colon H^1(M_h;\mathbb{Q}) \to H^1(M_g;\mathbb{Q})$ is injective and $2h \leq 2g$.

Exercise 7.16 (easy)**.** For closed n-manifolds M_1, M_2 show that $\chi(M_1\#M_2) = \chi(M_1)+\chi(M_2)-\chi(S^n)$. Deduce $\chi(M_g) = 2-2g$ and $\chi(N_h) = 2-h$ for the surfaces of Exercise 5.18.

SOURCE: [1, §3.3, Ex. 6(b)], reproduced in substance

Hint. *For finite CW complexes, $\chi(A\cup B) = \chi(A)+\chi(B)-\chi(A\cap B)$ (count cells, or use Mayer–Vietoris with field coefficients). Apply it twice.*

Solution. For $X = A\cup B$ a union of subcomplexes, every cell of X is counted in $\chi(A)+\chi(B)$ once or, if it lies in $A\cap B$, twice, so $\chi(X) = \chi(A)+\chi(B)-\chi(A\cap B)$ by Exercise 4.15. Let M_i° be M_i minus an open ball; then $M_i = M_i^\circ \cup D^n$ with intersection S^{n-1}, so $\chi(M_i^\circ) = \chi(M_i)-1+\chi(S^{n-1})$. And $M_1\#M_2 = M_1^\circ \cup M_2^\circ$ with intersection S^{n-1}, so

$$\chi(M_1\#M_2) = \chi(M_1)+\chi(M_2)-2+2\chi(S^{n-1})-\chi(S^{n-1}) = \chi(M_1)+\chi(M_2)-\big(2-\chi(S^{n-1})\big),$$

and $2-\chi(S^{n-1}) = 2-(1+(-1)^{n-1}) = 1+(-1)^n = \chi(S^n)$. (The manifolds admit CW structures compatible with these decompositions; alternatively use Mayer–Vietoris over $\mathbb{Q}$, which gives the same formula without cells.) For surfaces, $\chi(T^2) = 0$ and $\chi(\mathbb{RP}^2) = 1$ give inductively $\chi(M_g) = \chi(M_{g-1})+0-2$ with $\chi(M_1) = 0$, so $\chi(M_g) = 2-2g$; and $\chi(N_h) = \chi(N_{h-1})+1-2$ with $\chi(N_1) = 1$, so $\chi(N_h) = 2-h$.

Remark 7.17. Put the last three results together for a closed connected 3-manifold M and almost nothing is left undetermined. Write $H_1(M;\mathbb{Z}) = \mathbb{Z}^r \oplus F$ with F finite. Then $H_0 = \mathbb{Z}$; H_3 is $\mathbb{Z}$ or 0 according to orientability; Exercise 7.14 fixes the rank of H_2 once the others are known; and Corollary 7.13 settles the torsion of H_2 in the orientable case. Only the nonorientable torsion needs a further idea, namely the Bockstein of the coefficient sequence $0\to\mathbb{Z}\xrightarrow{2}\mathbb{Z}\to\mathbb{F}_2\to 0$ applied to the $\mathbb{F}_2$-fundamental class. Assembling this is the natural way to finish the chapter's duality section, and every ingredient is now on the page.

7.3 Where the ring fails

Example 7.18 (Lens spaces, again)**.** By Example 4.10 and Theorem 6.1,

$$H^*(L(p,q);\mathbb{Z}) = \big(\mathbb{Z},\ 0,\ \mathbb{Z}/p,\ \mathbb{Z}\big)$$

in degrees $0,1,2,3$. The multiplication is forced and carries no information: the only products of positive-degree classes to compute land in $H^4 = 0$ or pair $H^2\otimes H^2\to H^4 = 0$. So the integral cohomology ring of $L(p,q)$, like its homology and its fundamental group, is independent of q.

Yet $L(5,1)$ and $L(5,2)$ are not homotopy equivalent. This is a classical theorem: $L(p,q)$ and $L(p,q')$ are homotopy equivalent if and only if $qq' \equiv \pm n^2 \pmod p$ for some integer n, and for $p=5$ the squares mod 5 are $\{0,1,4\}$, which is closed under negation, while $qq' = 2$.

Warning 7.19. The book does not prove the classification just quoted, and the reader should not pretend otherwise. What separates these spaces is a refinement of the cup product called the *linking form*, a $\mathbb{Q}/\mathbb{Z}$-valued pairing on the torsion of H_1, which is invisible to the integral ring because that ring is trivial in the relevant degrees. What is worth extracting here is only the shape of the failure: a product can be forced to vanish for degree reasons and therefore carry no information, while a finer operation on the same cohomology does.

Remark 7.20 (The linking form, defined). For a closed oriented 3-manifold M and torsion classes $x, y \in TH_1(M;\mathbb{Z})$, choose $m \neq 0$ with $my = 0$, a 2-chain c with $\partial c = m \cdot$ (a cycle representing y), and set $\mathrm{lk}(x,y) = \frac{1}{m}(x \cdot c) \in \mathbb{Q}/\mathbb{Z}$, where $x \cdot c$ is the intersection number of a cycle representing x with c. It is a well-defined, symmetric, nonsingular bilinear form $TH_1 \times TH_1 \to \mathbb{Q}/\mathbb{Z}$. Cohomologically: $TH_1 \cong TH^2$ by duality, and lk can be computed from the Bockstein of $0 \to \mathbb{Z} \to \mathbb{Q} \to \mathbb{Q}/\mathbb{Z} \to 0$ followed by the cup product; so it is a "product" that uses $\mathbb{Q}/\mathbb{Z}$ coefficients and a connecting map, which is why the integral ring does not see it. For $L(p,q)$ and a suitable generator g of $H_1 \cong \mathbb{Z}/p$, one finds $\mathrm{lk}(g,g) = \pm\bar{q}/p$, where $\bar{q}$ is q or its inverse modulo p depending on conventions; either way, replacing g by ng multiplies the value by n^2, and the form up to isomorphism and sign records exactly the class of q modulo $\pm$ squares — which is the criterion of Example 7.18. That the linking form is a homotopy invariant, and that it is the *only* obstruction for lens spaces, are the parts left unproved here. Hatcher outlines a proof of the homotopy classification in [1, §3.E, Ex. 2 and §4.2, Ex. 29] using exactly this "cup product with a Bockstein": for x generating $H^1(L;\mathbb{Z}/p)$, the product $x \smile \beta(x)$ is a multiple t of the top class, and t modulo $\pm$ squares is a homotopy invariant; see also [7] for the simple-homotopy classification.

Remark 7.21 (The question Part I ends on). Example 7.3 and Exercise 7.8 showed that the cohomology ring is strictly stronger than the cohomology groups. Example 7.18 shows it is not complete. There are even pairs of spaces with *isomorphic cohomology rings*, over every coefficient ring at once, that are not homotopy equivalent; producing one is not hard once one knows where to look, and recognising that the ring has run out is the first step.

What replaces it is an operation that is natural but not induced by any multiplication: a family of maps $H^n(X;\mathbb{F}_2) \to H^{n+i}(X;\mathbb{F}_2)$ commuting with f^* for every f. These are the Steenrod squares, and Chapter 18 constructs them and uses them to settle a case that the ring provably cannot. Part II builds the machinery they need.

Remark 7.22 (Further reading). Cup and cap products, the Künneth formula and Poincaré duality with complete proofs: [1, §§3.2–3.3, 3.B]; a geometric treatment of duality via differential forms, with intersection and linking numbers: Bott–Tu [6]; the homotopy classification of lens spaces: [7].

7.4 What to carry forward

- The cup product makes f^* a ring map; that naturality is the whole source of its strength.
- On a wedge, classes from different summands multiply to zero (Exercise 7.4) — the standard obstruction.
- $\mathbb{CP}^2$ against $S^2 \vee S^4$, and $S^2 \times S^2$ against $S^2 \vee S^2 \vee S^4$: same groups, different rings (Example 7.3 and Exercise 7.8).
- Duality (Theorem 7.11): orientable over $\mathbb{Z}$, always over $\mathbb{F}_2$; the cup pairing is nondegenerate, so nonzero classes have partners.
- H^1 is torsion-free (Exercise 7.12), H_{n-1} of a closed orientable manifold is free (Corollary 7.13), and $\chi = 0$ in odd dimensions (Exercise 7.14).

- The ring is not a complete invariant either (Example 7.18 and Remark 7.21); the repair is Chapter 18.

Part II

Homological algebra and the homotopy toolkit

Chapter 8

Derived functors in practice: Ext and Tor

Part I used Ext and Tor as bookkeeping: a term in the universal coefficient sequence, a term in Künneth. This chapter treats them as what they are — derived functors — and the change of viewpoint pays off twice. It explains which of their properties are theorems and which are accidents of the ring $\mathbb{Z}$; and it makes them computable in situations where no exact sequence is handed to us.

The second half of the chapter runs the traffic in the unusual direction. A space with prescribed homology is an algebraic device: build one, compute its homology in two ways, and out comes a theorem about abelian groups. That manoeuvre is the point of Section 8.3, and it is worth more than either of the particular results proved with it.

8.1 Resolutions, and what is special about $\mathbb{Z}$

Definition 8.1. A *free resolution* of an R-module A is an exact sequence

$$\cdots \to F_2 \to F_1 \to F_0 \to A \to 0$$

with each F_i free. Applying $\mathrm{Hom}_R(-, B)$ to the deleted resolution and taking cohomology gives $\mathrm{Ext}^n_R(A, B)$; applying $- \otimes_R B$ and taking homology gives $\mathrm{Tor}^R_n(A, B)$.

Proposition 8.2 (Comparison; recalled)**.** *Any two free resolutions of A are chain homotopy equivalent, by an equivalence lifting the identity of A and unique up to chain homotopy. Consequently $\mathrm{Ext}^n_R(A, B)$ and $\mathrm{Tor}^R_n(A, B)$ do not depend on the resolution chosen.*

Over $\mathbb{Z}$ something special happens, and it is worth naming so that it can later be taken away.

Definition 8.3. A ring is *hereditary* if every submodule of a free module is projective.

Example 8.4 ($\mathbb{Z}$ is hereditary)**.** Every subgroup of a free abelian group is free abelian. Hence for every abelian group A the kernel F_1 of a surjection $F_0 \twoheadrightarrow A$ from a free group is itself free, and

$$0 \longrightarrow F_1 \longrightarrow F_0 \longrightarrow A \longrightarrow 0$$

is a free resolution of length one. Therefore

$$\mathrm{Ext}^n_{\mathbb{Z}}(A, B) = 0 \quad \text{and} \quad \mathrm{Tor}^{\mathbb{Z}}_n(A, B) = 0 \qquad \text{for all } n \geq 2,$$

which is why Part I could write Ext and Tor with no index at all.

Example 8.5 (The standard computations)**.** From $0 \to \mathbb{Z} \xrightarrow{m} \mathbb{Z} \to \mathbb{Z}/m \to 0$ one reads off

$$\operatorname{Ext}(\mathbb{Z}/m, \mathbb{Z}) \cong \mathbb{Z}/m, \qquad \operatorname{Ext}(\mathbb{Z}/m, \mathbb{Z}/n) \cong \mathbb{Z}/d, \qquad \operatorname{Tor}(\mathbb{Z}/m, \mathbb{Z}/n) \cong \mathbb{Z}/d,$$

where $d = \gcd(m, n)$; and both vanish as soon as the first variable is free.

The rules one actually computes with. For abelian groups the following suffice for every finitely generated case, and each is a one-line consequence of Definition 8.1 and Proposition 8.2:

- additivity: $\operatorname{Ext}(A \oplus A', B) \cong \operatorname{Ext}(A, B) \oplus \operatorname{Ext}(A', B)$, $\operatorname{Ext}(A, B \oplus B') \cong \operatorname{Ext}(A, B) \oplus \operatorname{Ext}(A, B')$, and the same for Tor (finite sums; Tor also commutes with arbitrary direct sums and with filtered colimits, Ext turns sums in the first variable into products);
- cyclic groups: $\operatorname{Ext}(\mathbb{Z}/m, G) \cong G/mG$ and $\operatorname{Tor}(\mathbb{Z}/m, G) \cong {}_mG = \{g : mg = 0\}$, from the resolution $0 \to \mathbb{Z} \xrightarrow{m} \mathbb{Z} \to \mathbb{Z}/m \to 0$;
- long exact sequences: a short exact sequence $0 \to A' \to A \to A'' \to 0$ gives $0 \to \operatorname{Hom}(A'', B) \to \operatorname{Hom}(A, B) \to \operatorname{Hom}(A', B) \to \operatorname{Ext}(A'', B) \to \operatorname{Ext}(A, B) \to \operatorname{Ext}(A', B) \to 0$, and $0 \to \operatorname{Tor}(A', B) \to \operatorname{Tor}(A, B) \to \operatorname{Tor}(A'', B) \to A' \otimes B \to A \otimes B \to A'' \otimes B \to 0$ (the sequences stop because $\operatorname{Ext}^2 = \operatorname{Tor}_2 = 0$ over $\mathbb{Z}$).

In particular $\operatorname{Tor}(-, B)$ is left exact over $\mathbb{Z}$: a subgroup $A' \subseteq A$ gives $\operatorname{Tor}(A', B) \subseteq \operatorname{Tor}(A, B)$. That one fact is used repeatedly below.

Theorem 8.6 (Universal coefficients for homology; recalled)**.** *For a chain complex C_* of free abelian groups and an abelian group G there is a natural short exact sequence*

$$0 \to H_n(C) \otimes G \to H_n(C \otimes G) \to \operatorname{Tor}\big(H_{n-1}(C), G\big) \to 0,$$

which splits, though not naturally. For a space, $H_n(X; G) \cong H_n(X) \otimes G \oplus \operatorname{Tor}(H_{n-1}(X), G)$.

The proof is the homology twin of the argument given for Theorem 6.1 in Chapter 6: tensor the split sequences $0 \to Z_n \to C_n \to B_{n-1} \to 0$ with G and run the zig-zag lemma. For $G = \mathbb{F}_p$ the theorem turns into a counting rule which is the main tool for detecting torsion from field coefficients (Exercise 8.14).

Now remove the hypothesis.

Exercise 8.7 (medium)**.** Let k be a field and $R = k[x]/(x^2)$, the ring of dual numbers, and regard $k = R/(x)$ as an R-module. Show that

$$\operatorname{Ext}^n_R(k, k) \cong k \qquad \text{for every } n \geq 0.$$

Hint. *Multiplication by x has kernel exactly (x), which is also its image. What does that say about the sequence $\cdots \to R \xrightarrow{x} R \xrightarrow{x} R$?*

Solution. In R we have $\ker(\cdot x) = (x) = \operatorname{im}(\cdot x)$, so

$$\cdots \longrightarrow R \xrightarrow{\cdot x} R \xrightarrow{\cdot x} R \longrightarrow k \longrightarrow 0$$

is exact: the augmentation $R \to k = R/(x)$ has kernel (x), which is the image of the first $\cdot x$, and exactness at every later spot is the displayed equality. So k admits a free resolution which repeats itself with period one.

Apply $\mathrm{Hom}_R(-,k)$. Each $\mathrm{Hom}_R(R,k) = k$, and the induced map is multiplication by x on k, which is zero because x acts as 0 on $R/(x)$. So the complex computing Ext is $k \xrightarrow{0} k \xrightarrow{0} k \to \cdots$, and $\mathrm{Ext}^n_R(k,k) = k$ for every $n \geq 0$.

Remark 8.8. A periodic resolution produces periodic Ext, and periodicity is common: it happens for $k[x]/(x^2)$ above, for cyclic groups over $\mathbb{Z}$ in group cohomology (Chapter 13 uses $H^*(\mathbb{Z}/p)$ being nonzero in every degree), and for any ring where some element has annihilator equal to its own ideal. The general lesson is that "how long is a resolution" is a question about the ring, and Example 8.4 is the accident that lets one forget it over $\mathbb{Z}$.

8.2 Tor, flatness, and balance

Tor may be computed by resolving either variable, and the resolution need not be free.

Theorem 8.9 (Balance and flat resolutions; recalled)**.** $\mathrm{Tor}^R_n(A,B)$ *may be computed from a projective resolution of* A *or of* B*, with canonically isomorphic results. More generally it may be computed from a resolution of either variable by* flat *modules.*

Example 8.10. Compute $\mathrm{Tor}(\mathbb{Z}/m, \mathbb{Z}/n)$ twice. Resolving the first variable by $0 \to \mathbb{Z} \xrightarrow{m} \mathbb{Z} \to \mathbb{Z}/m \to 0$ and tensoring with $\mathbb{Z}/n$ gives $\mathbb{Z}/n \xrightarrow{m} \mathbb{Z}/n$, whose kernel is $\mathbb{Z}/\gcd(m,n)$. Resolving the second variable gives $\mathbb{Z}/m \xrightarrow{n} \mathbb{Z}/m$, whose kernel is again $\mathbb{Z}/\gcd(m,n)$. The two computations look different and agree, which is what Theorem 8.9 promises.

The flat-resolution clause is not a technicality; it is what makes the non-finitely-generated cases computable at all, because the useful resolutions there are rarely free.

Exercise 8.11 (medium)**.** Show that a torsion-free abelian group A is flat, and deduce $\mathrm{Tor}(A,B) = 0$ for all B. In particular $\mathbb{Q}$ is flat, so

$$0 \longrightarrow \mathbb{Z} \longrightarrow \mathbb{Q} \longrightarrow \mathbb{Q}/\mathbb{Z} \longrightarrow 0$$

is a flat resolution of $\mathbb{Q}/\mathbb{Z}$ of length one.

Hint. *A finitely generated torsion-free abelian group is free. And* A *is the filtered union of its finitely generated subgroups.*

Solution. Write $A = \varinjlim A_i$ over its finitely generated subgroups, a filtered colimit. Each A_i is finitely generated and torsion-free, hence free, hence flat. Tensor products commute with filtered colimits and filtered colimits are exact, so $A \otimes -$ is exact: A is flat. Equivalently $\mathrm{Tor}_1(A,B) = \varinjlim \mathrm{Tor}_1(A_i,B) = 0$.

$\mathbb{Q}$ is torsion-free, so it is flat, and the displayed sequence is then a resolution of $\mathbb{Q}/\mathbb{Z}$ by flat modules, of length one. By Theorem 8.9 it computes $\mathrm{Tor}(-,\mathbb{Q}/\mathbb{Z})$.

Exercise 8.12 (easy)**.** Show that the kernel of $A \to A \otimes \mathbb{Q}$, $a \mapsto a \otimes 1$, is exactly the torsion subgroup of A.

Solution. If $na = 0$ with $n \neq 0$ then $a \otimes 1 = a \otimes \frac{n}{n} = na \otimes \frac{1}{n} = 0$, so torsion elements die. Conversely $A \otimes \mathbb{Q}$ is the localisation of A at $\mathbb{Z} \setminus \{0\}$, in which $a \otimes 1 = 0$ precisely when $na = 0$ for some $n \neq 0$; so the kernel consists of torsion elements only.

Remark 8.13. Exercises 8.11 and 8.12 fit together. Applying $A\otimes -$ to the flat resolution $0\to\mathbb{Z}\to\mathbb{Q}\to\mathbb{Q}/\mathbb{Z}\to 0$ makes $\operatorname{Tor}(A,\mathbb{Q}/\mathbb{Z})$ the kernel of $A\otimes\mathbb{Z}\to A\otimes\mathbb{Q}$, and Exercise 8.12 identifies that kernel. Carrying this out, and then deducing a characterisation of torsion-freeness in terms of the vanishing of Tor, is the natural exercise to set oneself here; both ingredients are on this page.

Exercise 8.14 (medium)**.** Let X be a space with $H_n(X;\mathbb{Z})$ finitely generated for every n. For a prime p write $b_n=\operatorname{rank}H_n(X)$ and let t_n be the number of cyclic summands of order a power of p in a decomposition of $H_n(X)$ into cyclic groups. Show that

$$\dim_{\mathbb{F}_p}H_n(X;\mathbb{F}_p)=b_n+t_n+t_{n-1}.$$

Deduce that $H_*(X;\mathbb{Z})$ has p-torsion if and only if $\dim_{\mathbb{F}_p}H_n(X;\mathbb{F}_p)>\dim_{\mathbb{Q}}H_n(X;\mathbb{Q})$ for some n, and that the $\mathbb{F}_p$-Betti numbers do *not* distinguish $\mathbb{Z}/p$ from $\mathbb{Z}/p^2$.

Source: original

Hint. *Use Theorem 8.6 with $G=\mathbb{F}_p$ and compute $\mathbb{Z}/p^a\otimes\mathbb{F}_p$ and $\operatorname{Tor}(\mathbb{Z}/p^a,\mathbb{F}_p)$, and the same for $\mathbb{Z}$ and for $\mathbb{Z}/q^a$ with $q\neq p$.*

Solution. By Theorem 8.6, $H_n(X;\mathbb{F}_p)\cong H_n(X)\otimes\mathbb{F}_p\oplus\operatorname{Tor}(H_{n-1}(X),\mathbb{F}_p)$. Both functors are additive, so it suffices to evaluate them on cyclic groups: $\mathbb{Z}\otimes\mathbb{F}_p=\mathbb{F}_p$, $\operatorname{Tor}(\mathbb{Z},\mathbb{F}_p)=0$; $\mathbb{Z}/p^a\otimes\mathbb{F}_p=\mathbb{F}_p$ and $\operatorname{Tor}(\mathbb{Z}/p^a,\mathbb{F}_p)={}_{p^a}\mathbb{F}_p=\mathbb{F}_p$ for $a\geq 1$; and for $q\neq p$, $\mathbb{Z}/q^a\otimes\mathbb{F}_p=0=\operatorname{Tor}(\mathbb{Z}/q^a,\mathbb{F}_p)$ since q^a is invertible mod p. Hence $\dim H_n(X)\otimes\mathbb{F}_p=b_n+t_n$ and $\dim\operatorname{Tor}(H_{n-1}(X),\mathbb{F}_p)=t_{n-1}$, giving the formula. Since $\dim_{\mathbb{Q}}H_n(X;\mathbb{Q})=b_n$, the difference is t_n+t_{n-1}, which is positive for some n iff some $t_n>0$. Finally a summand $\mathbb{Z}/p^a$ contributes the same amount (1 to degree n and 1 to degree $n+1$) whatever $a\geq 1$ is, so the $\mathbb{F}_p$-Betti numbers cannot see the exponent.

Exercise 8.15 (easy)**.** Show that if $H_*(X;\mathbb{Z})$ is finitely generated (in total), then $\chi(X)=\sum_n(-1)^n\operatorname{rank}H_n(X;\mathbb{Z})$ equals $\sum_n(-1)^n\dim_F H_n(X;F)$ for every field F.

Source: [1, §3.A, Ex. 1], reproduced in substance

Solution. For F of characteristic 0, $H_n(X;F)\cong H_n(X)\otimes F$ has dimension b_n (torsion dies, $\operatorname{Tor}(-,F)=0$ since F is torsion-free, hence flat by Exercise 8.11). For F of characteristic p, F is a vector space over $\mathbb{F}_p$ and $H_n(X;F)\cong H_n(X;\mathbb{F}_p)\otimes_{\mathbb{F}_p}F$ (the chain complex $C\otimes F=(C\otimes\mathbb{F}_p)\otimes_{\mathbb{F}_p}F$ and F is flat over $\mathbb{F}_p$), so $\dim_F H_n(X;F)=b_n+t_n+t_{n-1}$ by Exercise 8.14. In the alternating sum the terms t_n and t_{n-1} appear in adjacent degrees with opposite signs and cancel (all but finitely many are zero), leaving $\sum(-1)^n b_n$.

Exercise 8.16 (medium)**.** Show that if $\widetilde{H}_n(X;\mathbb{Q})=0$ and $\widetilde{H}_n(X;\mathbb{F}_p)=0$ for all n and all primes p, then $\widetilde{H}_n(X;\mathbb{Z})=0$ for all n, and hence $\widetilde{H}_n(X;G)=0$ for every G. No finiteness assumption is needed.

Source: [1, §3.A, Ex. 3], reproduced in substance

Hint. *Use Theorem 8.6 for the reduced complex, with $G=\mathbb{Q}$ and $G=\mathbb{F}_p$. A torsion group with no element of prime order is zero.*

Solution. Write $A_n=\widetilde{H}_n(X;\mathbb{Z})$. From $G=\mathbb{Q}$: $A_n\otimes\mathbb{Q}=0$, so every element of A_n is torsion (Exercise 8.12). From $G=\mathbb{F}_p$ in degree $n+1$: $\operatorname{Tor}(A_n,\mathbb{F}_p)={}_pA_n=0$, so A_n has no element of order p. This holds for every prime p, and a nonzero torsion element a of order m has a multiple $(m/p)a$ of prime order $p\mid m$. Hence $A_n=0$. Then for any G, Theorem 8.6 gives $\widetilde{H}_n(X;G)=0$.

Exercise 8.17 (medium)**.** Show that $\operatorname{Tor}(A, B)$ is always a torsion group, and that it contains an element of order n if and only if both A and B do.

SOURCE: [1, §3.A, Ex. 6], reproduced in substance

Hint. *Reduce to finitely generated subgroups using* $\operatorname{Tor}(\varinjlim A_\alpha, B) = \varinjlim \operatorname{Tor}(A_\alpha, B)$*, then use* $\operatorname{Tor}(\mathbb{Z}/a, \mathbb{Z}/b) = \mathbb{Z}/\gcd(a, b)$*. For "if", use left exactness.*

Solution. *If.* If A and B contain elements of order n, they contain copies of $\mathbb{Z}/n$, and by left exactness (twice) $\mathbb{Z}/n = \operatorname{Tor}(\mathbb{Z}/n, \mathbb{Z}/n) \subseteq \operatorname{Tor}(\mathbb{Z}/n, B) \subseteq \operatorname{Tor}(A, B)$.

Torsion, and only if. Every element of $\operatorname{Tor}(A, B)$ comes from $\operatorname{Tor}(A', B')$ for finitely generated subgroups $A' \subseteq A$, $B' \subseteq B$ (filtered colimits, applied in each variable). Write $A' = \mathbb{Z}^r \oplus \bigoplus \mathbb{Z}/a_i$ and $B' = \mathbb{Z}^s \oplus \bigoplus \mathbb{Z}/b_j$ with prime-power orders. Then $\operatorname{Tor}(A', B') = \bigoplus_{i,j} \mathbb{Z}/\gcd(a_i, b_j)$, a finite group, so $\operatorname{Tor}(A, B)$ is torsion. If $x \in \operatorname{Tor}(A, B)$ has order $n = \prod p^{e_p}$, choose A', B' as above containing a preimage x'; the order of x' is a multiple of n (mapping can only lower orders). For each $p \mid n$ the p-primary part of x' has order at least p^{e_p}, so some summand $\mathbb{Z}/\gcd(a_i, b_j)$ with a_i, b_j powers of p has gcd $\geq p^{e_p}$, hence $a_i, b_j \geq p^{e_p}$. So A and B contain elements of order p^{e_p} for each $p \mid n$, and the sum of such elements for the different primes has order exactly n.

8.3 Spaces as algebraic devices

Moore spaces (Proposition 6.9) have homology concentrated in one degree, for an arbitrary prescribed group. That makes them instruments: compute an invariant of a Moore space in two different ways, and the comparison is a theorem of algebra.

Example 8.18 (Tor realised as a homology group)**.** Let $n, m \geq 1$ and put $X = M(A, n)$, $Y = M(B, m)$. Künneth gives

$$H_k(X \times Y) \cong \bigoplus_{i+j=k} H_i(X) \otimes H_j(Y) \;\oplus\; \bigoplus_{i+j=k-1} \operatorname{Tor}\big(H_i(X), H_j(Y)\big).$$

The only nonzero reduced homology of X is A in degree n, and of Y is B in degree m. In degree $k = n + m + 1$ the tensor summands all vanish — the available degrees are 0, n, m and $n + m$, and $n + m + 1$ is none of them — while the Tor summands require $i + j = n + m$, leaving exactly $\operatorname{Tor}(A, B)$. Hence

$$H_{n+m+1}\big(M(A, n) \times M(B, m)\big) \;\cong\; \operatorname{Tor}(A, B).$$

Exercise 8.19 (medium)**.** Deduce that $\operatorname{Tor}(A, B) \cong \operatorname{Tor}(B, A)$ for all abelian groups A and B, without using a resolution.

Hint. *$X \times Y$ and $Y \times X$ are homeomorphic.*

Solution. Take $n = m$ in Example 8.18, so that $\operatorname{Tor}(A, B) \cong H_{2n+1}(M(A, n) \times M(B, n))$ and $\operatorname{Tor}(B, A) \cong H_{2n+1}(M(B, n) \times M(A, n))$. The swap map $(u, v) \mapsto (v, u)$ is a homeomorphism between the two spaces, so it induces an isomorphism on homology in every degree, and in particular in degree $2n + 1$.

Remark 8.20 (The method, stated plainly)*.* The manoeuvre has three steps: realise an algebraic functor as a homology group of a space built to order; find a second computation of that homology group; and read the comparison as an algebraic identity.

Example 8.18 realises Tor; the second computation in Exercise 8.19 was a homeomorphism. Other second computations are available — coefficients in a field, where the universal coefficient theorem loses its Ext term, is the most useful one, and it converts a vanishing statement about the cohomology of a Moore space directly into the vanishing of an Ext group. Proposition 6.9 builds the space for any group at all, so nothing here requires finite generation.

8.4 Künneth, and non-naturality again

Theorem 8.21 (Künneth; recalled)**.** *For CW complexes X, Y there is a short exact sequence, natural in both variables,*

$$0 \to \bigoplus_{i+j=k} H_i(X) \otimes H_j(Y) \to H_k(X \times Y) \to \bigoplus_{i+j=k-1} \operatorname{Tor}\big(H_i(X), H_j(Y)\big) \to 0,$$

and it splits — but not naturally.

Example 8.22 (A Künneth drill)**.** $H_*(\mathbb{RP}^2) = (\mathbb{Z},\ \mathbb{Z}/2,\ 0)$, so for $X = Y = \mathbb{RP}^2$:

$$H_*(\mathbb{RP}^2 \times \mathbb{RP}^2) = \big(\mathbb{Z},\ (\mathbb{Z}/2)^2,\ \mathbb{Z}/2,\ \mathbb{Z}/2,\ 0\big).$$

Degrees 1 and 2 come from tensor summands, and the $\mathbb{Z}/2$ in degree 3 comes from $\operatorname{Tor}(\mathbb{Z}/2, \mathbb{Z}/2)$ alone: there is no tensor contribution in that degree at all. A class living in a degree where the tensor part vanishes is exactly what a natural splitting would have to keep track of.

Exercise 8.23 (medium)**.** Compute $H_*(\mathbb{RP}^2 \times \mathbb{RP}^3; \mathbb{Z})$ and $H^*(\mathbb{RP}^2 \times \mathbb{RP}^3; \mathbb{Z}/2)$, and check the answers against Exercise 8.15.

Source: [1, §3.B, Ex. 1], specialised to $m = 2$, $n = 3$

Hint. *$H_*(\mathbb{RP}^2) = (\mathbb{Z}, \mathbb{Z}/2, 0)$ and $H_*(\mathbb{RP}^3) = (\mathbb{Z}, \mathbb{Z}/2, 0, \mathbb{Z})$. Over $\mathbb{Z}/2$ use Theorem 7.1.*

Solution. Write $P = H_*(\mathbb{RP}^2)$, $Q = H_*(\mathbb{RP}^3)$. Tensor terms $\bigoplus_{i+j=k} P_i \otimes Q_j$ and Tor terms $\bigoplus_{i+j=k-1} \operatorname{Tor}(P_i, Q_j)$:

- $k = 0$: $\mathbb{Z}$.
- $k = 1$: $P_1 \otimes Q_0 \oplus P_0 \otimes Q_1 = (\mathbb{Z}/2)^2$; no Tor.
- $k = 2$: $P_1 \otimes Q_1 = \mathbb{Z}/2$; Tor for $i + j = 1$: $\operatorname{Tor}(P_0, Q_1) = \operatorname{Tor}(P_1, Q_0) = 0$. Total $\mathbb{Z}/2$.
- $k = 3$: $P_0 \otimes Q_3 = \mathbb{Z}$, $P_2 \otimes Q_1 = 0$; $\operatorname{Tor}(P_1, Q_1) = \mathbb{Z}/2$. Total $\mathbb{Z} \oplus \mathbb{Z}/2$.
- $k = 4$: $P_1 \otimes Q_3 = \mathbb{Z}/2$; Tor for $i + j = 3$: $\operatorname{Tor}(P_0, Q_3) = 0$, $\operatorname{Tor}(P_2, Q_1) = 0$. Total $\mathbb{Z}/2$.
- $k = 5$: $P_2 \otimes Q_3 = 0$; $\operatorname{Tor}(P_1, Q_3) = 0$. Total 0.

So $H_*(\mathbb{RP}^2 \times \mathbb{RP}^3) = (\mathbb{Z}, (\mathbb{Z}/2)^2, \mathbb{Z}/2, \mathbb{Z} \oplus \mathbb{Z}/2, \mathbb{Z}/2, 0)$. Mod 2, $H^* = \mathbb{F}_2[a]/(a^3) \otimes \mathbb{F}_2[b]/(b^4)$ with $|a| = |b| = 1$, of dimensions $1, 2, 3, 3, 2, 1$ in degrees 0–5. Check: the integral Euler characteristic is $1 - 0 + 0 - 1 + 0 - 0 = 0$, and $1 - 2 + 3 - 3 + 2 - 1 = 0 \bmod 2$; also $\chi(\mathbb{RP}^2)\chi(\mathbb{RP}^3) = 1 \cdot 0$.

Proposition 8.24. *Suppose the splitting of Theorem 8.21 could be chosen naturally in both variables. Then for maps $f\colon X \to X'$ and $g\colon Y \to Y'$ the induced map $(f \times g)_*$ would decompose as $\big(\bigoplus f_* \otimes g_*\big) \oplus \big(\bigoplus \operatorname{Tor}(f_*, g_*)\big)$. In particular, were f_* to vanish on $\widetilde{H}_*$, every summand built from positive-degree homology of X would be killed by $(f \times g)_*$.*

Proof. A natural splitting identifies the middle term with the sum of the two outer ones, compatibly with induced maps. On each outer term the induced map is the functor $\otimes$ or Tor evaluated at the pair (f_*, g_*), and both vanish as soon as f_* does. □

Remark 8.25 (What to do with it). Proposition 8.24 is the exact analogue of Proposition 6.12, and it is refuted in exactly the same way: exhibit one map that is zero on reduced homology and one product in which it is not zero. Chapter 6 supplied both the collapse map $M(\mathbb{Z}/m, n) \to S^{n+1}$ and the computation showing what it does to cohomology; Example 8.22 shows where in a product a Tor class can sit with no tensor class beside it. Crossing such a collapse with the identity and asking what happens to the Tor summand is the natural experiment, and the reader who ran Remark 6.17 has already done its harder half.

The pattern is now established twice, and it is worth stating as a slogan: an Ext or Tor summand records information about a space that its homology groups do not determine *as a functor*. Everything in Chapter 18 is an attempt to name that information.

8.5 What to carry forward

- Resolutions compute derived functors and the answer is resolution-independent (Proposition 8.2); over $\mathbb{Z}$ they have length one, and that is a property of $\mathbb{Z}$ (Example 8.4 and Exercise 8.7).
- Tor is balanced and may be computed by *flat* resolutions (Theorem 8.9); torsion-free abelian groups are flat (Exercise 8.11).
- A Moore space realises Tor as a homology group (Example 8.18); comparing two computations of that group proves algebra (Remark 8.20).
- Künneth is natural and splits, unnaturally (Theorem 8.21 and Proposition 8.24) — the second instance of the pattern opened in Chapter 6.

Chapter 9

Towers, inverse limits, and Bocksteins

Two constructions in this chapter look unrelated and are not. Both concern a sequence of approximations to an object, and both measure the failure of an obvious guess.

The inverse limit of a tower of groups is not right exact, and its first derived functor $\varprojlim^1$ is the error term. It appears whenever the cohomology of an infinite complex is assembled from its skeleta, and it is usually zero — but only for a reason, and the reason is worth knowing.

The Bockstein is the error term of a different reduction: $H^*(X;\mathbb{F}_p)$ does not determine $H^*(X;\mathbb{Z})$, and β is the first approximation to the difference. It is a differential on mod-p cohomology whose homology detects which torsion has order exactly p. Chapter 18 will identify β with the first Steenrod square, and Chapter 26 will use that identification to write an integral characteristic class as a Bockstein.

9.1 The derived functor of the inverse limit, as a cokernel

Let (G_i, α_i) be a tower of abelian groups,

$$\cdots \longrightarrow G_2 \xrightarrow{\alpha_2} G_1 \xrightarrow{\alpha_1} G_0.$$

Definition 9.1. Let $\Delta\colon \prod_i G_i \to \prod_i G_i$ be given by $\Delta(x)_i = x_i - \alpha_{i+1}(x_{i+1})$. Then $\varprojlim G_i = \ker \Delta$ and $\varprojlim^1 G_i = \operatorname{coker} \Delta$.

So $\varprojlim^1$ measures the failure of the system of equations $x_i - \alpha_{i+1}(x_{i+1}) = y_i$ to be solvable. That is the only description we need: every computation below either solves those equations or shows they cannot be solved.

Exercise 9.2 (easy)**.** Show that $\varprojlim^1 G_i = 0$ if every α_i is surjective.

Hint. *Solve the equations recursively, choosing a preimage at each stage.*

Solution. Given $y \in \prod G_i$, set $x_0 = 0$ and choose inductively x_{i+1} with $\alpha_{i+1}(x_{i+1}) = x_i - y_i$, which is possible because α_{i+1} is onto. Then $\Delta(x) = y$, so Δ is surjective and its cokernel vanishes.

Exercise 9.3 (medium)**.** Call a tower *pro-zero* if for every i some composite $G_{i+n} \to G_i$ is zero. Show that a pro-zero tower has $\varprojlim G_i = 0$ and $\varprojlim^1 G_i = 0$.

Hint. *Write $\alpha^{(n)}$ for the n-fold composite $G_{i+n} \to G_i$ and try $x_i = \sum_{n\geq 0} \alpha^{(n)}(y_{i+n})$.*

Solution. The sum is finite, since $\alpha^{(n)}$ into G_i vanishes for large n. With that definition,

$$\alpha_{i+1}(x_{i+1}) = \sum_{n\geq 0} \alpha^{(n+1)}(y_{i+1+n}) = \sum_{m\geq 1} \alpha^{(m)}(y_{i+m}),$$

so $x_i - \alpha_{i+1}(x_{i+1}) = \alpha^{(0)}(y_i) = y_i$, and Δ is onto. For the limit, a compatible family satisfies $x_i = \alpha^{(n)}(x_{i+n}) = 0$ once n is large.

Theorem 9.4 (Six-term sequence; recalled)**.** *A short exact sequence of towers $0 \to A \to B \to C \to 0$ induces an exact sequence*

$$0 \to \varprojlim A \to \varprojlim B \to \varprojlim C \to \varprojlim{}^1 A \to \varprojlim{}^1 B \to \varprojlim{}^1 C \to 0.$$

Definition 9.5. Write $G_i^{(n)} = \operatorname{im}\big(G_{i+n} \to G_i\big)$, a descending chain of subgroups of G_i. The tower satisfies the *Mittag-Leffler condition* if for each i that chain is eventually constant, with stable value G_i^{∞}.

Warning 9.6. The condition does not say that the maps are eventually surjective. It says that inside each *fixed* G_i the images of the groups far above stop shrinking. A tower whose maps are eventually zero satisfies it, with $G_i^{\infty} = 0$, and no map in it need be surjective.

Remark 9.7 (The step that remains). The subgroups G_i^{∞} form a subtower whose transition maps are surjective, and under Mittag-Leffler the quotient tower G_i/G_i^{∞} is pro-zero: for large n the image of G_{i+n} in G_i is already G_i^{∞}, so the composite into G_i/G_i^{∞} vanishes. Feeding that short exact sequence of towers into Theorem 9.4, and applying Exercises 9.2 and 9.3 to the two outer terms, yields the standard vanishing criterion in a line. Assembling it is the natural exercise to set oneself here; the only thing to watch is Warning 9.6.

For later reference we record the outcome, since later chapters use it.

Theorem 9.8 (Mittag-Leffler)**.** *If a tower of abelian groups satisfies the Mittag-Leffler condition, then $\varprojlim{}^1 G_i = 0$. In particular this holds if every G_i is finite, or if every G_i is a finite-dimensional vector space over a field and the maps are linear.*

Proof. The argument is the one outlined in Remark 9.7: the stable images form a subtower with surjective maps, the quotient tower is pro-zero, and Theorem 9.4 with Exercises 9.2 and 9.3 gives the vanishing. (For the surjectivity of the maps $G_{i+1}^{\infty} \to G_i^{\infty}$ one uses that the images stabilise: an element of G_i^{∞} is the image of an element of G_{i+n} for large n, whose image in G_{i+1} lies in G_{i+1}^{∞}.) For finite groups or finite-dimensional vector spaces a descending chain of subgroups or subspaces stabilises, so Mittag-Leffler holds automatically. □

Exercise 9.2 might suggest that $\varprojlim{}^1$ is a formality.

Example 9.9 (A $\varprojlim{}^1$ that is uncountable)**.** Fix a prime p, let $G_i = \mathbb{Z}$ for all i, and let every transition map be multiplication by p. Then $\varprojlim G_i = 0$, since a compatible family has x_0 divisible by p^n for every n.

For the derived functor, consider $\Sigma\colon \prod_i \mathbb{Z} \to \mathbb{Z}_p$ into the p-adic integers, $\Sigma(y) = \sum_i p^i y_i$. The series converges p-adically, and Σ is surjective because every p-adic integer has such an expansion. Composing with Δ telescopes:

$$\Sigma\big(\Delta(x)\big) = \sum_i p^i\big(x_i - p x_{i+1}\big) = \lim_{N\to\infty}\big(x_0 - p^N x_N\big) = x_0,$$

because $p^N x_N \to 0$ in $\mathbb{Z}_p$. So Σ maps $\operatorname{im}\Delta$ onto $\mathbb{Z} \subseteq \mathbb{Z}_p$, and it descends to an isomorphism

$$\varprojlim{}^1\,\mathbb{Z} \;\cong\; \mathbb{Z}_p/\mathbb{Z},$$

an uncountable group. A tower of copies of $\mathbb{Z}$ with the simplest possible maps already has an enormous $\varprojlim^1$.

Theorem 9.10 (Milnor sequence; recalled)**.** *Let the CW complex X be the union of an increasing sequence of subcomplexes $X_0 \subseteq X_1 \subseteq \cdots$. For every n there is a short exact sequence*

$$0 \to \varprojlim{}^1 H^{n-1}(X_i; G) \to H^n(X; G) \to \varprojlim H^n(X_i; G) \to 0.$$

Exercise 9.11 (easy)**.** Use Theorem 9.10 to compute $H^*(\mathbb{CP}^\infty;\mathbb{Z})$ from the groups $H^*(\mathbb{CP}^n;\mathbb{Z})$, and say which hypothesis makes the $\varprojlim^1$ term disappear.

Solution. Each restriction $H^k(\mathbb{CP}^{n+1};\mathbb{Z}) \to H^k(\mathbb{CP}^n;\mathbb{Z})$ is surjective: for k even and $k \le 2n$ both groups are $\mathbb{Z}$ and the generator restricts to the generator, and otherwise the target is 0. So the tower of cohomology groups has surjective transition maps and $\varprojlim^1 = 0$ by Exercise 9.2. Hence $H^k(\mathbb{CP}^\infty;\mathbb{Z}) \cong \varprojlim_n H^k(\mathbb{CP}^n;\mathbb{Z})$, which is $\mathbb{Z}$ for k even and 0 for k odd.

Exercise 9.12 (medium)**.** Let X be the mapping telescope of $S^1 \xrightarrow{p} S^1 \xrightarrow{p} S^1 \to \cdots$, the union of the mapping cylinders of degree-p maps, filtered by the finite telescopes $X_i \simeq S^1$. Compute $H_*(X;\mathbb{Z})$ and $H^*(X;\mathbb{Z})$, using Theorem 9.10 for the latter, and check the answer against the universal coefficient theorem.

Source: original (compare the discussion of telescopes in [1, §3.F])

Hint. *Homology commutes with the increasing union: $H_1(X) = \varinjlim(\mathbb{Z} \xrightarrow{p} \mathbb{Z} \to \cdots) = \mathbb{Z}[\frac{1}{p}]$. For cohomology, the tower $H^1(X_i)$ is $\cdots \xrightarrow{p} \mathbb{Z} \xrightarrow{p} \mathbb{Z}$.*

Solution. Each finite telescope X_i deformation retracts onto its last circle, so $H_*(X_i) = (\mathbb{Z},\mathbb{Z})$, and the inclusion $X_i \subseteq X_{i+1}$ induces multiplication by p on H_1. Since every compact subset of X (in particular the image of a singular chain) lies in some X_i, $H_n(X) = \varinjlim H_n(X_i)$; so $H_0(X) = \mathbb{Z}$, $H_1(X) = \varinjlim(\mathbb{Z} \xrightarrow{p} \mathbb{Z} \xrightarrow{p} \cdots) \cong \mathbb{Z}[\frac{1}{p}]$ (send the generator of the i-th copy to p^{-i}), and $H_n(X) = 0$ for $n \ge 2$. So X is a Moore space $M(\mathbb{Z}[\frac{1}{p}], 1)$.

For cohomology, Theorem 9.10 gives $0 \to \varprojlim^1 H^{n-1}(X_i) \to H^n(X) \to \varprojlim H^n(X_i) \to 0$. The tower $H^1(X_i)$ is $\mathbb{Z} \leftarrow \mathbb{Z} \leftarrow \cdots$ with all maps multiplication by p (restriction is the dual of the inclusion), and $H^0(X_i) = \mathbb{Z}$ with identity maps. So $H^0(X) = \mathbb{Z}$; $H^1(X) = \varprojlim(\cdots \xrightarrow{p} \mathbb{Z} \xrightarrow{p} \mathbb{Z}) = 0$ (Example 9.9) plus $\varprojlim^1 H^0 = 0$ (surjective tower); and $H^2(X) = \varprojlim^1 H^1(X_i) \cong \mathbb{Z}_p/\mathbb{Z}$ by Example 9.9, the $\varprojlim$ term vanishing as $H^2(X_i) = 0$.

Check: by Theorem 6.1, $H^1(X) = \operatorname{Hom}(\mathbb{Z}[\frac{1}{p}],\mathbb{Z}) = 0$ (an element of $\mathbb{Z}[\frac{1}{p}]$ divisible by all powers of p must map to an integer divisible by all powers of p, i.e. to 0), and $H^2(X) = \operatorname{Ext}(\mathbb{Z}[\frac{1}{p}],\mathbb{Z})$. So the computation proves $\operatorname{Ext}(\mathbb{Z}[\frac{1}{p}],\mathbb{Z}) \cong \mathbb{Z}_p/\mathbb{Z}$, an uncountable group, although $\mathbb{Z}[\frac{1}{p}]$ is countable and torsion-free — a second appearance of the "spaces as devices" method of Section 8.3.

Exercise 9.13 (medium)**.** Granting that $\mathbb{Q}$ and $\mathbb{Q}/\mathbb{Z}$ are injective abelian groups (so that $\operatorname{Ext}(A,\mathbb{Q}) = \operatorname{Ext}(A,\mathbb{Q}/\mathbb{Z}) = 0$ for every A; divisible groups are injective), show that $\operatorname{Ext}(A,\mathbb{Z}) \cong \operatorname{coker}\big(\operatorname{Hom}(A,\mathbb{Q}) \to \operatorname{Hom}(A,\mathbb{Q}/\mathbb{Z})\big)$, and deduce that $\operatorname{Ext}(\mathbb{Z}_{p^\infty},\mathbb{Z}) \cong \mathbb{Z}_p$, where $\mathbb{Z}_{p^\infty} = \mathbb{Z}[\frac{1}{p}]/\mathbb{Z}$ is the Prüfer group and $\mathbb{Z}_p$ the p-adic integers.

Source: [1, §3.F, Ex. 5], reproduced in substance

Hint. *Apply* $\operatorname{Hom}(A,-)$ *to* $0 \to \mathbb{Z} \to \mathbb{Q} \to \mathbb{Q}/\mathbb{Z} \to 0$. *For* $A = \mathbb{Z}_{p^\infty} = \varinjlim \mathbb{Z}/p^n$, *compute* $\operatorname{Hom}(A,\mathbb{Q}/\mathbb{Z}) = \varprojlim \operatorname{Hom}(\mathbb{Z}/p^n,\mathbb{Q}/\mathbb{Z})$.

Solution. The long exact sequence of $\operatorname{Hom}(A,-)$ applied to $0 \to \mathbb{Z} \to \mathbb{Q} \to \mathbb{Q}/\mathbb{Z} \to 0$ reads $0 \to \operatorname{Hom}(A,\mathbb{Z}) \to \operatorname{Hom}(A,\mathbb{Q}) \to \operatorname{Hom}(A,\mathbb{Q}/\mathbb{Z}) \to \operatorname{Ext}(A,\mathbb{Z}) \to \operatorname{Ext}(A,\mathbb{Q}) = 0$, which gives the formula.

For $A = \mathbb{Z}_{p^\infty}$: $\operatorname{Hom}(A,\mathbb{Q}) = 0$ because A is torsion and $\mathbb{Q}$ is torsion-free. And $\operatorname{Hom}(\varinjlim \mathbb{Z}/p^n, \mathbb{Q}/\mathbb{Z}) = \varprojlim \operatorname{Hom}(\mathbb{Z}/p^n,\mathbb{Q}/\mathbb{Z}) = \varprojlim \mathbb{Z}/p^n$, with the transition maps the reductions (a homomorphism $\mathbb{Z}/p^{n+1} \to \mathbb{Q}/\mathbb{Z}$ is determined by the image k/p^{n+1} of 1, and restricting to $\mathbb{Z}/p^n = p\mathbb{Z}/p^{n+1}$ gives k/p^n, i.e. reduction of k mod p^n). Hence $\operatorname{Ext}(\mathbb{Z}_{p^\infty},\mathbb{Z}) \cong \varprojlim \mathbb{Z}/p^n = \mathbb{Z}_p$.

9.2 The Bockstein as a differential

Definition 9.14. The coefficient sequence $0 \to \mathbb{Z}/p \to \mathbb{Z}/p^2 \to \mathbb{Z}/p \to 0$ induces a natural connecting homomorphism

$$\beta\colon H^n(X;\mathbb{F}_p) \longrightarrow H^{n+1}(X;\mathbb{F}_p),$$

the *Bockstein*. The sequence $0 \to \mathbb{Z} \xrightarrow{p} \mathbb{Z} \to \mathbb{F}_p \to 0$ induces the *integral Bockstein* $\tilde\beta\colon H^n(X;\mathbb{F}_p) \to H^{n+1}(X;\mathbb{Z})$, of which β is the mod-p reduction.

Exercise 9.15 (medium)**.** Show that $\beta^2 = 0$ and that β is a derivation: $\beta(xy) = \beta(x)\,y + (-1)^{|x|}x\,\beta(y)$.

Hint. *For the first, the composite* $\mathbb{Z}/p \hookrightarrow \mathbb{Z}/p^2 \twoheadrightarrow \mathbb{Z}/p$ *occurring in two consecutive connecting maps is zero. For the second, lift cocycles to* $\mathbb{Z}/p^2$*-cochains and differentiate the product of the lifts.*

Solution. Applying the connecting map twice factors through $\mathbb{Z}/p \hookrightarrow \mathbb{Z}/p^2 \twoheadrightarrow \mathbb{Z}/p$, which is zero; hence $\beta^2 = 0$.

For the derivation rule, represent x, y by $\mathbb{F}_p$-cocycles a, b and choose $\mathbb{Z}/p^2$-cochain lifts $\tilde a, \tilde b$, so that $\delta\tilde a = p\,a'$ and $\delta\tilde b = p\,b'$ with a', b' representing $\beta x, \beta y$. The product $\tilde a\tilde b$ lifts ab, and

$$\delta(\tilde a\tilde b) = (\delta\tilde a)\,\tilde b + (-1)^{|a|}\,\tilde a\,(\delta\tilde b) = p\big(a'\tilde b + (-1)^{|a|}\tilde a\,b'\big).$$

Reducing mod p gives the stated formula.

Since $\beta^2 = 0$, one may take homology with respect to it.

Definition 9.16. The *Bockstein homology* of X is $H\big(H^*(X;\mathbb{F}_p),\beta\big)$.

Proposition 9.17 (What β detects)**.** *Suppose* $H^n(X;\mathbb{Z})$ *is finitely generated for every* n*, and decompose each* $H^n(X;\mathbb{Z})$ *as* $\mathbb{Z}^{b_n} \oplus \bigoplus_{r\ge1}(\mathbb{Z}/p^r)^{t_{n,r}} \oplus (\text{torsion prime to } p)$*. Then the Bockstein homology of* X *in degree* n *has dimension*

$$b_n + \sum_{r\ge2}\big(t_{n,r} + t_{n+1,r}\big).$$

In particular it equals $\big(H^n(X;\mathbb{Z})/\text{torsion}\big)\otimes\mathbb{F}_p$ *if and only if no* p*-primary torsion of order* $\geq p^2$ *occurs in degrees* n *and* $n+1$*; and the Bockstein homology vanishes in all positive degrees exactly when the free part of* $H^*(X;\mathbb{Z})$ *sits in degree* 0 *and every* p*-primary torsion class has order exactly* p*.*

Warning 9.18 (A correction). The original edition of this book stated that the Bockstein homology is always $(H^*/\text{torsion})\otimes\mathbb{F}_p$. That is false, and Example 9.19 below is a counterexample: for $M(\mathbb{Z}/p^2,n)$ the Bockstein is zero, so its Bockstein homology is $\mathbb{F}_p$ in degrees n and $n+1$, while the free part of the cohomology is zero. The correct general statement is Theorem 9.22: $(H^*/\text{torsion})\otimes\mathbb{F}_p$ is the *last* page of the Bockstein spectral sequence, and the Bockstein homology is only its second page.

Proof of Proposition 9.17. Work with a cochain complex of free abelian groups computing $H^*(X;\mathbb{Z})$. By the universal coefficient theorem for the reduction mod p (the cochain version of Theorem 8.6), $H^n(X;\mathbb{F}_p)\cong H^n(X;\mathbb{Z})\otimes\mathbb{F}_p\oplus\operatorname{Tor}(H^{n+1}(X;\mathbb{Z}),\mathbb{F}_p)$, and the decomposition can be taken compatible with a splitting of the cochain complex into elementary pieces: up to chain homotopy equivalence, a finite-type cochain complex of free abelian groups is a direct sum of complexes of the forms $\mathbb{Z}$ (in degree n) and $\mathbb{Z}\xrightarrow{m}\mathbb{Z}$ (in degrees $n,n+1$), by Theorem 4.18 applied to each coboundary. The Bockstein is computed piece by piece. A piece $\mathbb{Z}$ in degree n contributes $\mathbb{F}_p$ in degree n with $\beta=0$. A piece $\mathbb{Z}\xrightarrow{m}\mathbb{Z}$ with $p\nmid m$ contributes nothing mod p. A piece $\mathbb{Z}\xrightarrow{p^r u}\mathbb{Z}$ $(p\nmid u)$, whose integral cohomology is $\mathbb{Z}/p^r$ in degree $n+1$, contributes $\mathbb{F}_p$ in degrees n and $n+1$; its Bockstein $\mathbb{F}_p\to\mathbb{F}_p$ is computed exactly as in Example 9.19 and is an isomorphism for $r=1$ and zero for $r\geq 2$. Adding up gives the dimension formula; the remaining statements are read off from it. □

Example 9.19 (β on a Moore space)**.** For $X=M(\mathbb{Z}/p,n)$ the reduced mod-p cohomology is $\mathbb{F}_p$ in degrees n and $n+1$, and $\beta\colon H^n\to H^{n+1}$ is an isomorphism: by Exercise 6.11 the integral cohomology is $\mathbb{Z}/p$ in degree $n+1$, on which multiplication by p is zero, so the connecting map of $0\to\mathbb{Z}\xrightarrow{p}\mathbb{Z}\to\mathbb{F}_p\to 0$ is nonzero.

For $X=M(\mathbb{Z}/p^2,n)$ the mod-p cohomology consists of exactly the same two groups, but now $\beta=0$: the integral torsion has order p^2, and multiplication by p on $\mathbb{Z}/p^2$ is not injective, which kills the connecting map.

Two spaces with identical mod-p cohomology groups are thus separated by β. It sees the *order* of the torsion, not merely its presence.

Example 9.20 (Infinite projective space)**.** $H^*(\mathbb{RP}^\infty;\mathbb{F}_2)=\mathbb{F}_2[x]$ with $|x|=1$, and $\beta x=x^2$ — Example 9.19 in degree one, $\mathbb{RP}^2=M(\mathbb{Z}/2,1)$ being the 2-skeleton. By Exercise 9.15,

$$\beta(x^k)=k\,x^{k+1},$$

which is x^{k+1} for k odd and 0 for k even. So $\ker\beta$ has basis $1,x^2,x^4,\dots$ and $\operatorname{im}\beta$ has basis $x^2,x^4,\dots$, and the Bockstein homology is $\mathbb{F}_2$ in degree 0 and zero in every positive degree.

Read through Proposition 9.17, that says the free part of $H^*(\mathbb{RP}^\infty;\mathbb{Z})$ lives in degree 0 and every torsion class has order exactly 2 — both true by Example 6.2.

Exercise 9.21 (medium)**.** For p odd, $H^*(L(p,q);\mathbb{F}_p)$ has basis $1,x,y,xy$ in degrees $0,1,2,3$, with $\beta x=y$. Compute the Bockstein homology, and check it against Proposition 9.17 and the integral computation of Example 4.10.

Hint. *Get βy from $\beta^2 = 0$, and $\beta(xy)$ from the derivation rule. Note which degree y^2 would live in.*

Solution. From $\beta x = y$ and $\beta^2 = 0$ we get $\beta y = 0$. By Exercise 9.15, $\beta(xy) = \beta(x)\,y - x\,\beta(y) = y^2$, which vanishes because it lies in $H^4(L(p,q);\mathbb{F}_p) = 0$.

So $\ker\beta$ has basis $1, y, xy$ while $\operatorname{im}\beta$ has basis y. The Bockstein homology is $\mathbb{F}_p$ in degree 0, zero in degrees 1 and 2, and $\mathbb{F}_p$ in degree 3.

This agrees with Proposition 9.17: by Example 4.10 the integral cohomology is $\mathbb{Z}, 0, \mathbb{Z}/p, \mathbb{Z}$, whose free part sits in degrees 0 and 3, and whose only torsion is one copy of $\mathbb{Z}/p$ — of order exactly p, which is why degree 2 contributes nothing.

9.3 The Bockstein spectral sequence

The Bockstein separates $\mathbb{Z}/p$ from $\mathbb{Z}/p^r$, $r \geq 2$, but does not tell $\mathbb{Z}/p^2$ from $\mathbb{Z}/p^3$. The repair is to iterate, and the iteration is the first spectral sequence of the book. We state it in homology, the form used in Part VI; the cohomological version is identical with the arrows reversed.

Let C_* be a chain complex of free abelian groups of finite type (finitely generated in each degree). The short exact sequence $0 \to C \xrightarrow{p} C \to C \otimes \mathbb{F}_p \to 0$ gives a long exact sequence

$$\cdots \to H_n(C) \xrightarrow{p} H_n(C) \xrightarrow{\rho} H_n(C;\mathbb{F}_p) \xrightarrow{\partial} H_{n-1}(C) \xrightarrow{p} \cdots$$

and the homology Bockstein is $\beta_1 = \rho \circ \partial \colon H_n(C;\mathbb{F}_p) \to H_{n-1}(C;\mathbb{F}_p)$.

Theorem 9.22 (The Bockstein spectral sequence)**.** *There are $\mathbb{F}_p$-vector spaces E^r_n, $r \geq 1$, and differentials $\beta_r \colon E^r_n \to E^r_{n-1}$ with $\beta_r^2 = 0$ and $E^{r+1} = H(E^r, \beta_r)$, such that:*

(a) $E^1_n = H_n(C;\mathbb{F}_p)$ and β_1 is the Bockstein;

(b) $E^r_n \cong \big(p^{r-1}H_n(C)\big) \otimes \mathbb{F}_p \;\oplus\; \operatorname{Tor}\big(p^{r-1}H_{n-1}(C), \mathbb{F}_p\big)$, where $p^{r-1}H$ denotes the image of multiplication by p^{r-1} on $H_*(C)$ modulo torsion prime to p; concretely, E^r "remembers" a summand $\mathbb{Z}$ in degree n as an $\mathbb{F}_p$ in degree n, and a summand $\mathbb{Z}/p^s$ of $H_{n-1}(C)$ as an $\mathbb{F}_p$ in degrees n and $n-1$ for as long as $r \leq s$;

(c) a summand $\mathbb{Z}/p^s$ of $H_{n-1}(C)$ contributes a pair of classes in E^r_n and E^r_{n-1} connected by β_r exactly when $r = s$, after which both die;

(d) $E^\infty_n = E^r_n$ for r larger than the exponent of the p-torsion, and $E^\infty_n \cong \big(H_n(C)/\mathit{torsion}\big) \otimes \mathbb{F}_p$.

In words: β_r is nonzero precisely on the torsion of order exactly p^r, and the number of summands $\mathbb{Z}/p^r$ in $H_{n-1}(C)$ equals the rank of $\beta_r \colon E^r_n \to E^r_{n-1}$.

Proof sketch. As in the proof of Proposition 9.17, C is chain homotopy equivalent to a direct sum of elementary complexes $\mathbb{Z}$ (in one degree) and $\mathbb{Z} \xrightarrow{m} \mathbb{Z}$; everything is additive, so it suffices to treat these. For $\mathbb{Z}$ in degree n all β_r vanish and $\mathbb{F}_p$ survives in degree n. For $\mathbb{Z} \xrightarrow{p^s u} \mathbb{Z}$ in degrees $n, n-1$ ($p \nmid u$), $H_{n-1} = \mathbb{Z}/p^s u$, mod p we get $\mathbb{F}_p$ in degrees n and $n-1$, and one defines β_r on a class x by lifting to an integral chain c, writing $\partial c = p^r c'$, and setting $\beta_r(x) = [c' \bmod p]$ — defined when ∂c is divisible by p^r, which for the generator happens for $r \leq s$, giving a nonzero answer exactly when $r = s$. The general construction uses the exact couple formed by the long exact sequence

above, with $D = H_*(C)$, $E = H_*(C;\mathbb{F}_p)$ and the map "multiplication by p"; E^r is the r-th derived couple. See [8, Ch. 2] for the exact-couple formalism, [8, Ch. 10] for the Bockstein spectral sequence itself, and [1, §3.E] for the cohomological version with full proofs. □

Remark 9.23 (What the spectral sequence proves, and what it does not). To prove that all p-torsion of $H_*(C)$ has exponent exactly p, it suffices to show $E^2 = E^\infty$, i.e. that the Bockstein homology already equals the free part mod p; by Proposition 9.17 this can be checked degree by degree. To *count* the summands one needs the ranks of β_1, i.e. the mod-p homology and the Bockstein. To show that a particular summand is $\mathbb{Z}/p^2$ one must exhibit a nonzero β_2. None of this requires knowing the integral homology in advance — which is why the Bockstein spectral sequence is the main tool for torsion questions about configuration spaces in Part VI, where the mod-p homology is often computable while the integral homology is not.

Exercise 9.24 (medium)**.** Let p be a prime and $X = M(\mathbb{Z}/p,1) \vee M(\mathbb{Z}/p^2,2) \vee M(\mathbb{Z}/p^3,2) \vee S^3$. Write down the homology Bockstein spectral sequence of X: the groups E^r_n and the nonzero differentials β_r for $r = 1,2,3$, and E^∞. Then explain how one would recover $H_*(X;\mathbb{Z})$ (the p-primary part) from the spectral sequence alone.

Source: original

Hint. *$H_*(X) = (\mathbb{Z},\ \mathbb{Z}/p,\ \mathbb{Z}/p^2 \oplus \mathbb{Z}/p^3,\ \mathbb{Z})$. Each $\mathbb{Z}/p^s$ in degree $n-1$ gives $\mathbb{F}_p$'s in degrees $n-1$ and n joined by β_s.*

Solution. By Theorem 8.6 and Exercise 8.14, $E^1 = H_*(X;\mathbb{F}_p)$ has dimensions: degree 0: 1; degree 1: 1 (from $\mathbb{Z}/p$); degree 2: 1 (Tor of the $\mathbb{Z}/p$ in H_1) $+2$ (from $\mathbb{Z}/p^2, \mathbb{Z}/p^3$) $= 3$; degree 3: 2 (Tor of the two summands of H_2) $+1$ (from S^3) $= 3$; total $1,1,3,3$.

β_1: nonzero only on the pair coming from $\mathbb{Z}/p$, mapping the class in degree 2 to the class in degree 1. So E^2 has dimensions $1,0,2,3$.

β_2: nonzero only on the pair from $\mathbb{Z}/p^2$, from degree 3 to degree 2. So E^3 has dimensions $1,0,1,2$.

β_3: nonzero only on the pair from $\mathbb{Z}/p^3$, from degree 3 to degree 2. So $E^4 = E^\infty$ has dimensions $1,0,0,1$, which is $(H_*/\text{torsion})\otimes\mathbb{F}_p$ (the classes of H_0 and of S^3), as Theorem 9.22(d) requires.

Conversely, given only the pages: the free ranks are the dimensions of E^∞; the number of summands $\mathbb{Z}/p^r$ in H_{n-1} is the rank of $\beta_r\colon E^r_n \to E^r_{n-1}$. Here: rank of β_1 from degree 2 is 1 (one $\mathbb{Z}/p$ in H_1), of β_2 from degree 3 is 1 (one $\mathbb{Z}/p^2$ in H_2), of β_3 from degree 3 is 1 (one $\mathbb{Z}/p^3$ in H_2). This recovers the p-primary part of $H_*(X)$ completely.

Exercise 9.25 (medium)**.** Using Example 9.20 and the derivation property, compute the Bockstein homology of $H^*(\mathbb{RP}^\infty \times \mathbb{RP}^\infty;\mathbb{F}_2) = \mathbb{F}_2[x,y]$. Deduce that every torsion class in $H^*(\mathbb{RP}^\infty \times \mathbb{RP}^\infty;\mathbb{Z})$ has order exactly 2 and that there is no free part in positive degrees.

Source: original

Hint. *β is a derivation and acts on $x\otimes 1$ and $1\otimes y$ as on the factors, so $(\mathbb{F}_2[x,y],\beta) \cong (\mathbb{F}_2[x],\beta)\otimes(\mathbb{F}_2[y],\beta)$ as differential graded algebras. Use the Künneth theorem for complexes over a field.*

Solution. The cross product identifies $H^*(\mathbb{RP}^\infty \times \mathbb{RP}^\infty;\mathbb{F}_2)$ with $\mathbb{F}_2[x]\otimes\mathbb{F}_2[y]$ (Theorem 7.1), and β is natural and a derivation (Exercise 9.15); since $\beta(x\otimes 1) = x^2\otimes 1$ and $\beta(1\otimes y) = 1\otimes y^2$ (naturality along the projections), the derivation rule gives $\beta = \beta\otimes 1 + 1\otimes\beta$ on the tensor product (with signs irrelevant mod 2). Over a field, the homology of a tensor product of complexes is the

tensor product of the homologies (algebraic Künneth, no Tor term over a field). By Example 9.20 each factor has Bockstein homology $\mathbb{F}_2$ in degree 0, so the Bockstein homology of the product is $\mathbb{F}_2$ in degree 0.

The integral cohomology is of finite type (finitely many cells in each degree in the product CW structure), and it is all 2-primary in positive degrees (by the integral Künneth theorem, all positive-degree groups of the factors are 2-torsion or zero). By Proposition 9.17, vanishing of the Bockstein homology in positive degrees means: no free part in positive degrees and every 2-primary torsion class of order exactly 2.

Exercise 9.26 (easy)**.** Show that the mod-p Bockstein is $\beta = \rho \circ \tilde{\beta}$, where $\rho\colon H^{n+1}(X;\mathbb{Z}) \to H^{n+1}(X;\mathbb{F}_p)$ is reduction, and that $\tilde{\beta}(x) = 0$ if and only if x is the reduction of an integral class. Deduce that the generator of $H^1(\mathbb{RP}^2;\mathbb{F}_2)$ is not the reduction of an integral class, while the generator of $H^2(\mathbb{RP}^2;\mathbb{F}_2)$ is.

SOURCE: original

Solution. The map of coefficient sequences

$$\begin{array}{ccccccccc} 0 & \longrightarrow & \mathbb{Z} & \xrightarrow{p} & \mathbb{Z} & \longrightarrow & \mathbb{F}_p & \longrightarrow & 0 \\ & & \downarrow{\scriptstyle \rho} & & \downarrow & & \| & & \\ 0 & \longrightarrow & \mathbb{Z}/p & \longrightarrow & \mathbb{Z}/p^2 & \longrightarrow & \mathbb{F}_p & \longrightarrow & 0 \end{array}$$

(the middle map reduction mod p^2) induces a map of long exact sequences, and naturality of connecting homomorphisms gives $\beta = \rho \circ \tilde{\beta}$. In the long exact sequence $\cdots \to H^n(X;\mathbb{Z}) \xrightarrow{\rho} H^n(X;\mathbb{F}_p) \xrightarrow{\tilde{\beta}} H^{n+1}(X;\mathbb{Z}) \to \cdots$, exactness at $H^n(X;\mathbb{F}_p)$ says $\ker\tilde{\beta} = \operatorname{im}\rho$. For $\mathbb{RP}^2$: $H^1(\mathbb{RP}^2;\mathbb{Z}) = 0$, so the reduction map to $H^1(\mathbb{RP}^2;\mathbb{F}_2) = \mathbb{F}_2$ is zero and the generator does not lift; consistently $\tilde{\beta}$ maps it isomorphically onto $H^2(\mathbb{RP}^2;\mathbb{Z}) = \mathbb{Z}/2$. In degree 2, $\rho\colon H^2(\mathbb{RP}^2;\mathbb{Z}) = \mathbb{Z}/2 \to H^2(\mathbb{RP}^2;\mathbb{F}_2) = \mathbb{F}_2$ is an isomorphism (the cellular cochain complex mod 2 has zero differentials, and the cellular cocycle e^{2*} reduces to itself), so the generator lifts.

Remark 9.27 (Three continuations). First, β is only the first approximation. Torsion of order p^r is detected by an r-th Bockstein defined on the previous Bockstein homology, and the family assembles into the Bockstein spectral sequence of Theorem 9.22, whose E_∞-page is the free part mod p and whose r-th differential counts torsion of order p^r. Example 9.19 is the case $r = 1$ against $r = 2$.

Second, because β is a derivation squaring to zero, the mod-p cohomology of a product is a tensor product of complexes and its Bockstein homology is the tensor product of the Bockstein homologies. Applied to a smash power of a Moore space, and read back through Proposition 9.17, that determines the integral cohomology of the smash power completely. It is the natural consolidating computation for this chapter, and everything it needs is now available.

Third, β is not merely a connecting homomorphism: it is a stable cohomology operation, natural in X and commuting with suspension. Chapter 18 identifies it with Sq^1, and that identification is what lets Chapter 26 express an integral Euler class as a Bockstein.

9.4 What to carry forward

- $\varprojlim^1$ is the cokernel of $1 - \text{shift}$ (Definition 9.1). It vanishes for surjective towers (Exercise 9.2) and for pro-zero ones (Exercise 9.3); those two facts and Theorem 9.4 give the general criterion (Remark 9.7).

- Mittag-Leffler is about images stabilising, not about surjectivity (Warning 9.6).
- $\varprojlim^1$ can be uncountable (Example 9.9), and it is the error term in the Milnor sequence (Theorem 9.10).
- β is a derivation with $\beta^2 = 0$ (Exercise 9.15), so mod-p cohomology is a complex, and its homology detects the free part together with the order of the torsion (Proposition 9.17, Examples 9.19 and 9.20, and Exercise 9.21).

Chapter 10

H-spaces and Hopf algebras

A multiplication on a space induces a comultiplication on its cohomology, and the resulting structure — a Hopf algebra — is rigid enough to answer questions that look purely geometric. This chapter is the first place in the book where an algebraic structure *forbids* something: Hopf's theorem will show that most spheres carry no multiplication at all.

The other half of the chapter concerns maps rather than spaces. The multiplication on Y makes $[X, Y]$ into an algebraic object, and the k-th power map is then k times the identity in a precise sense. Where exactly that "k times" survives into cohomology is the delicate point, and Example 10.19 shows that the naive guess is false.

10.1 The cohomology of an H-space

Recall Definition 1.26: an H-space is a space X with basepoint e and a map $\mu\colon X \times X \to X$, with $\mu(e, e) = e$, for which $\mu(e, -)$ and $\mu(-, e)$ are homotopic to the identity.

Let k be a field and suppose $H^*(X; k)$ is finite-dimensional in each degree, so that the Künneth theorem gives $H^*(X \times X; k) \cong H^*(X; k) \otimes H^*(X; k)$. Then μ induces

$$\mu^*\colon H^*(X; k) \longrightarrow H^*(X; k) \otimes H^*(X; k),$$

a *coproduct*. The unit conditions say that μ^* is counital, and $H^*(X; k)$ becomes a graded Hopf algebra: a graded-commutative algebra with a coassociative coproduct that is a map of algebras.

Definition 10.1. A *connected graded Hopf algebra* over a field k is a graded k-algebra $A = \bigoplus_{n \geq 0} A^n$ with $A^0 = k$, together with an algebra homomorphism $\Delta\colon A \to A \otimes A$ (the tensor product algebra with the Koszul sign, as in Theorem 7.1) such that $\Delta(a) = a \otimes 1 + 1 \otimes a + \sum a' \otimes a''$ with $|a'|, |a''| > 0$ for every a of positive degree. (Coassociativity, which holds when μ is homotopy-associative, is sometimes included in the definition; Hopf's theorem below does not need it.)

The condition on $\Delta(a)$ is the counit axiom: restricting μ to $X \times \{e\}$ and $\{e\} \times X$ gives maps homotopic to the identity, and on cohomology these restrictions are the maps $A \otimes A \to A$ killing the positive-degree part of one factor. The fact that Δ is a ring map is just naturality of the cup product applied to μ; it is what makes the definition restrictive, because Δ of a product is forced once Δ is known on generators.

The dual picture. Homology of an H-space carries a product, the *Pontryagin product* $H_*(X;k)\otimes H_*(X;k)\xrightarrow{\times} H_*(X\times X;k)\xrightarrow{\mu_*} H_*(X;k)$, and a coproduct induced by the diagonal. When $H^*(X;k)$ is of finite type the two Hopf algebras H_* and H^* are dual to each other: the Pontryagin product is the transpose of μ^*, and the cup product is the transpose of the diagonal coproduct. For loop spaces the Pontryagin ring is often the more natural object — by the Milnor–Moore theorem of Chapter 25, $H_*(\Omega X;\mathbb{Q})$ is the universal enveloping algebra of the rational homotopy Lie algebra of X.

Example 10.2 (Odd spheres)**.** $H^*(S^n;k)=\Lambda(x)$ with $|x|=n$. There is nothing in degrees strictly between 0 and n, so the only possible terms in μ^*x are $x\otimes 1$ and $1\otimes x$, and the counit conditions force both coefficients to be 1:

$$\mu^*x = x\otimes 1 + 1\otimes x.$$

For n odd this is consistent: $x^2=0$ automatically by graded commutativity. For n even it is not, as Exercise 10.5 shows.

Example 10.3 (The torus and infinite projective space)**.** T^2 is a topological group, and $H^*(T^2;k)=\Lambda(a,b)$ with $|a|=|b|=1$ and $\mu^*a = a\otimes 1+1\otimes a$, similarly for b. Since μ^* is a ring map,

$$\mu^*(ab) = (a\otimes 1+1\otimes a)(b\otimes 1+1\otimes b) = ab\otimes 1 + a\otimes b - b\otimes a + 1\otimes ab,$$

the sign coming from moving b of degree one past a of degree one. So ab has extra terms: not every class behaves like a and b do.

$\mathbb{CP}^\infty$ is also an H-space — it is $K(\mathbb{Z},2)$, and the multiplication classifies the addition of degree-two cohomology classes — with $H^*(\mathbb{CP}^\infty;k)=k[c]$, $|c|=2$, and $\mu^*c = c\otimes 1+1\otimes c$. Again $\mu^*(c^2)=c^2\otimes 1+2\,c\otimes c+1\otimes c^2$ has an extra term.

Theorem 10.4 (Hopf; recalled)**.** *Let A be a connected graded Hopf algebra over a field of characteristic zero which is finite-dimensional as a vector space. Then A is an exterior algebra $\Lambda(x_1,\dots,x_r)$ on generators of odd degree.*

Exercise 10.5 (medium)**.** Deduce that S^n is not an H-space for any even $n\geq 2$.

Hint. *What would $H^*(S^n;\mathbb{Q})$ have to be, as an algebra, if Theorem 10.4 applied?*

Solution. Suppose S^n were an H-space with n even and $n\geq 2$. Then $A=H^*(S^n;\mathbb{Q})=\mathbb{Q}\{1,x\}$ with $|x|=n$ even and $x^2=0$ would be a connected finite-dimensional graded Hopf algebra over $\mathbb{Q}$, so by Theorem 10.4 it would be an exterior algebra on odd generators. But A has a nonzero element in the even degree n and none in odd degrees, so its generators would have to include one of even degree n, which Theorem 10.4 forbids.

Equivalently and more concretely: for a free graded-commutative algebra a generator of even degree generates a polynomial algebra, which is infinite-dimensional, contradicting $x^2=0$.

Exercise 10.6 (medium)**.** Let X be a connected H-space which is a finite CW complex, and suppose $\widetilde{H}^*(X;\mathbb{Q})\neq 0$. Show that $\chi(X)=0$.

Source: [1, §3.C, Ex. 9], adapted: Hatcher assumes only $\widetilde{H}^*(X;\mathbb{Z})\neq 0$ and uses Borel's theorem over $\mathbb{F}_p$ as well; with rational hypotheses Hopf's theorem suffices

Hint. *Compute $\chi(X)$ with $\mathbb{Q}$ coefficients (Exercise 8.15) and evaluate the Poincaré polynomial of $\Lambda(x_1,\dots,x_r)$ at $t=-1$.*

Solution. $H^*(X;\mathbb{Q})$ is finite-dimensional (X is a finite complex) and, by the Künneth theorem over $\mathbb{Q}$, a connected graded Hopf algebra. By Theorem 10.4 it is an exterior algebra $\Lambda(x_1,\dots,x_r)$ with all $|x_i|$ odd, and $r \geq 1$ because $\widetilde{H}^*(X;\mathbb{Q}) \neq 0$. Its Poincaré polynomial is $\prod_{i=1}^{r}(1+t^{|x_i|})$, so $\chi(X) = \sum_n (-1)^n \dim H^n(X;\mathbb{Q}) = \prod_i (1+(-1)^{|x_i|}) = 0$, each factor being $1-1=0$. (Here χ computed over $\mathbb{Q}$ equals the Euler characteristic by Exercise 8.15.) For example $\chi(M_g) = 2-2g \neq 0$ for $g \neq 1$, so those surfaces admit no H-space structure, while $T^2 = M_1$ is a group.

Exercise 10.7 (easy)**.** Show that a retract A of an H-space (X, e, μ) with $e \in A$ is an H-space.

SOURCE: [1, §3.C, Ex. 2], reproduced in substance

Solution. Let $r\colon X \to A$ be a retraction, $i\colon A \to X$ the inclusion, and define $\mu_A = r \circ \mu \circ (i \times i)\colon A \times A \to A$. Then $\mu_A(e,e) = r(e) = e$. If h_t is a homotopy from $\mu(-,e)$ to id_X, then $r \circ h_t \circ i$ is a homotopy from $\mu_A(-,e) = r\mu(i(-),e)$ to $ri = \mathrm{id}_A$; similarly on the other side. (If the homotopies in the definition are required to fix e, so are these, since $r(e) = e$.)

Remark 10.8. S^1, S^3 and S^7 *are* H-spaces, being the unit spheres of the complex numbers, the quaternions and the octonions. Exercise 10.5 excludes the even spheres, and a much harder theorem — proved with the machinery of Chapter 16 and completed by Adams — excludes every odd sphere except those three.

10.2 Primitives

Definition 10.9. An element $\alpha \in H^*(X;k)$ of positive degree is *primitive* if

$$\mu^*\alpha = \alpha \otimes 1 + 1 \otimes \alpha.$$

The primitives form a graded subspace $P\,H^*(X;k)$.

Lemma 10.10 (Primitivity for degree reasons)**.** *If $H^i(X;k) = 0$ for all $0 < i < n$, then every element of $H^n(X;k)$ is primitive.*

Proof. Write $\mu^*\alpha = \sum \alpha' \otimes \alpha''$ with the terms graded by $(|\alpha'|, |\alpha''|)$ summing to n. The counit conditions supply the terms $\alpha \otimes 1$ and $1 \otimes \alpha$; every other term has both factors of positive degree less than n, hence lies in a group that vanishes. □

Example 10.11. By Lemma 10.10, the generator x of $H^n(S^n;k)$ is primitive. By the computations of Example 10.3, the primitives of $\Lambda(a,b) = H^*(T^2;k)$ are exactly the span of a and b: the class ab is not primitive, since $\mu^*(ab)$ contains $a \otimes b$. Likewise the primitives of $k[c] = H^*(\mathbb{CP}^\infty;k)$ in the case $\operatorname{char} k = 0$ are exactly the span of c: for $r \geq 2$,

$$\mu^*(c^r) = \sum_{i=0}^{r} \binom{r}{i} c^i \otimes c^{r-i},$$

which has nonzero middle terms.

Exercise 10.12 (medium)**.** Let p be a prime and let $A = \mathbb{F}_p[x]$, with x of even degree (any degree if $p = 2$), be the Hopf algebra with x primitive. Determine the primitive elements of A. Compare with Example 10.11.

SOURCE: [1, §3.C, Ex. 7], reproduced in substance

Hint. *Δ preserves degree and each $A^{n|x|}$ is spanned by x^n; compute $\Delta(x^n)$ and use that $\binom{n}{i} \equiv 0 \pmod p$ for all $0 < i < n$ iff n is a power of p.*

Solution. Since Δ is an algebra map and x is primitive, $\Delta(x^n) = (x\otimes 1 + 1\otimes x)^n = \sum_i \binom{n}{i} x^i \otimes x^{n-i}$ (no signs: x has even degree, or $p = 2$). So x^n is primitive iff $\binom{n}{i} \equiv 0 \bmod p$ for $0 < i < n$. This holds iff $n = p^k$: if $n = p^k$, Frobenius gives $(1+t)^{p^k} = 1 + t^{p^k}$ in $\mathbb{F}_p[t]$; if $n = p^k m$ with $p \nmid m$, $m > 1$, then $(1+t)^n = (1+t^{p^k})^m$ has coefficient $m \not\equiv 0$ at t^{p^k}, with $0 < p^k < n$. Since Δ preserves degree and each homogeneous component is one-dimensional, a homogeneous element cx^n ($c \neq 0$) is primitive iff x^n is, and a sum of primitives is primitive; since primitivity of a sum of homogeneous components forces each component to be primitive (Δ is graded), the primitives are exactly the span of $x, x^p, x^{p^2}, \dots$. Over $\mathbb{Q}$ the same computation leaves only x, as in Example 10.11.

Exercise 10.13 (medium)**.** Show that the Pontryagin ring $H_*(T^n;\mathbb{Z})$ of the torus, with the group structure of $T^n = (S^1)^n$, is the exterior algebra $\Lambda_{\mathbb{Z}}[x_1,\dots,x_n]$ with $|x_i| = 1$.

SOURCE: [1, §3.C, Ex. 11], reproduced in substance

Hint. *Let x_i be the image of the fundamental class of the i-th circle factor. Show $x_{i_1}\cdots x_{i_k}$ ($i_1 < \cdots < i_k$) is the image of the fundamental class of the corresponding subtorus, using that the group multiplication restricted to complementary coordinate subtori is a homeomorphism onto their product.*

Solution. Write $T^n = S^1_1 \times \cdots \times S^1_n$ and let $x_i \in H_1(T^n)$ be the image of the fundamental class of the coordinate circle S^1_i (all other coordinates 1). For a subset $I = \{i_1 < \cdots < i_k\}$ let T_I be the coordinate subtorus and $[T_I] = [S^1_{i_1}] \times \cdots \times [S^1_{i_k}] \in H_k(T^n)$ the image of its fundamental class. By the Künneth theorem (no Tor terms, all groups free) the $[T_I]$ form a basis of $H_*(T^n)$.

The Pontryagin product of $[T_I]$ and $[T_J]$ is $\mu_*([T_I]\times[T_J])$, where μ restricted to $T_I \times T_J$ is coordinatewise multiplication. If $I \cap J \neq \varnothing$, $\mu|_{T_I\times T_J}$ factors through a torus of dimension $< |I| + |J|$ (namely $T_{I\cup J}$), so the image of a class of degree $|I|+|J|$ is zero. If $I \cap J = \varnothing$, μ maps $T_I \times T_J$ homeomorphically onto $T_{I\cup J}$ by a map that is a permutation of coordinates, so $[T_I]\cdot[T_J] = \pm[T_{I\cup J}]$, the sign being that of the shuffle putting $I \cup J$ in increasing order (Lemma 28.5 in Chapter 28, or directly: a transposition of two circle factors reverses orientation of their product). Hence $x_i x_j = -x_j x_i$, $x_i^2 = 0$, and $x_{i_1}\cdots x_{i_k} = [T_I]$: the map $\Lambda_{\mathbb{Z}}[x_1,\dots,x_n] \to H_*(T^n)$ is a ring homomorphism carrying the monomial basis to the basis $\{[T_I]\}$, hence an isomorphism.

Exercise 10.14 (easy)**.** Let $f\colon X \to Y$ be a map of H-spaces which commutes with the multiplications up to homotopy, i.e. $f\circ\mu_X \simeq \mu_Y \circ (f\times f)$. Show that f^* carries primitives to primitives.

Solution. Let $\alpha \in P\,H^*(Y;k)$. Then

$$\mu_X^*\big(f^*\alpha\big) = \big(f\circ\mu_X\big)^*\alpha = \big(\mu_Y\circ(f\times f)\big)^*\alpha = (f^*\otimes f^*)\mu_Y^*\alpha,$$

using homotopy invariance for the middle equality and the Künneth identification for the last. Since $\mu_Y^*\alpha = \alpha\otimes 1 + 1\otimes\alpha$, the right-hand side is $f^*\alpha\otimes 1 + 1\otimes f^*\alpha$.

10.3 Maps into an H-space, and power maps

Definition 10.15. For maps $f, g\colon X \to Y$ into an H-space Y, their *pointwise product* is

$$fg = \mu\circ(f,g)\colon X \longrightarrow Y, \qquad (fg)(x) = \mu\big(f(x), g(x)\big).$$

The *k-th power map* of Y is $p_k\colon Y \to Y$, the k-fold pointwise product of the identity with itself.

The pointwise product descends to homotopy classes, so $[X, Y]$ carries a binary operation with a two-sided unit. If the multiplication is homotopy-associative and admits a homotopy inverse — as it does for a topological group — then $[X, Y]$ is a group, naturally in X.

Remark 10.16. The case to keep in mind is $Y = K(G, n)$. There $[X, K(G, n)] \cong H^n(X; G)$, a fact proved in Chapter 17 (Theorem 17.19), and under that identification the pointwise product is the addition of cohomology classes. The group structure on cohomology and the H-space structure on an Eilenberg–MacLane space are the same thing seen from two sides.

Example 10.17 (The baby case)**.** For $Y = S^1$ with its group structure, p_k is $z \mapsto z^k$. On $\pi_1(S^1) = \mathbb{Z}$ it is multiplication by k: the loop $t \mapsto e^{2\pi i t}$ is sent to $t \mapsto e^{2\pi i k t}$, which winds k times. Hence $\deg p_k = k$.

That computation generalises, and the general statement is about homotopy groups.

Exercise 10.18 (medium)**.** Let G be a topological group (or any homotopy-associative H-space with a homotopy inverse). Show that for every $n \geq 1$ the k-th power map induces multiplication by k on $\pi_n(G)$.

Hint. *$\pi_n(G)$ carries two products, concatenation and the pointwise one, and by the Eckmann–Hilton argument of Chapter 1 they coincide. Which one is $p_k \circ f$ obviously built from?*

Solution. Let $\alpha \in \pi_n(G)$ be represented by $f\colon S^n \to G$. By definition of the power map, $p_k \circ f$ is the k-fold pointwise product $f \cdot f \cdots f$, so $[p_k \circ f]$ is the k-fold product of $[f]$ under the pointwise operation on $\pi_n(G)$. By Proposition 1.27 — whose proof applies verbatim to π_n — the pointwise operation agrees with concatenation, which is the group law of $\pi_n(G)$. Hence $p_{k*}(\alpha) = k\alpha$.

Now the point of the chapter. The corresponding statement in cohomology is *false* in general, and it fails in the smallest example.

Example 10.19 (The power map on a torus)**.** Take $G = T^2$ and $H^*(T^2; \mathbb{Z}) = \Lambda(a, b)$ as in Example 10.3. The power map p_k is $(z, w) \mapsto (z^k, w^k)$, so by Example 10.17 applied in each factor,

$$p_k^* a = k a, \qquad p_k^* b = k b, \qquad p_k^*(ab) = (ka)(kb) = k^2\, ab.$$

So p_k acts by k in degree one and by k^2 in degree two; its degree as a self-map of the surface T^2 is k^2, not k.

Compare Example 10.11: the classes on which p_k^* is multiplication by k are exactly a and b, the primitives, and ab — the one class that failed to be primitive — is exactly the one where the factor is k^2 instead.

Remark 10.20 (The pattern, and what to do with it). Example 10.19 is not an accident of the torus. The pointwise product of two maps is $\mu \circ (f, g)$, so computing $(fg)^*$ means evaluating μ^* and then $f^* \otimes g^*$; and $\mu^*\alpha$ has exactly two terms precisely when α is primitive. On a primitive class the two contributions add; on a non-primitive class the cross terms survive and produce something else — k^2 in the example above.

Turning that observation into a proposition, and then reading off what the k-th power map does to the primitives of a specific H-space, is the natural continuation of this chapter. Lemma 10.10 identifies the primitives of a sphere in one line, and Example 10.17 shows what the answer must specialise to. Note that Exercise 10.18 gives no information here beyond the top degree, since the higher homotopy of an H-space and its cohomology are quite different objects; the cohomological statement has to be proved cohomologically.

Remark 10.21 (Forward). The Hopf algebra structure of this chapter is used three more times. In Chapter 26 the H-space structure of $\mathbb{CP}^\infty = K(\mathbb{Z}, 2)$ is what makes the Euler class of a tensor product of line bundles additive. In Chapter 16 the failure of most spheres to be H-spaces is refined into the Hopf invariant one problem. And in Chapter 31 the same structure reappears rationally, where Hopf's theorem becomes the statement that the rational homotopy of an H-space is its space of primitives.

10.4 What to carry forward

- A multiplication on X makes $H^*(X; k)$ a Hopf algebra (Examples 10.2 and 10.3); Hopf's theorem then constrains which spaces admit one (Theorem 10.4 and Exercise 10.5).
- Primitives (Definition 10.9); low degrees are primitive for free (Lemma 10.10); H-maps preserve them (Exercise 10.14).
- The k-th power map is multiplication by k on homotopy groups (Exercise 10.18) but not on cohomology (Example 10.19); primitivity is exactly the condition that repairs the discrepancy (Remark 10.20).

Chapter 11

Cofibrations, approximation, and connectivity

This chapter pays a debt. Twice in Part I we collapsed a contractible subcomplex and asserted that nothing was lost — once to normalise a CW complex to a single 0-cell (Proposition 2.2), and once to read a free basis off a spanning tree (Example 3.10). The justification is the homotopy extension property, and it is proved in Section 11.1.

The rest of the chapter is about replacing a space by a better model of itself. Two theorems do all the work — cellular approximation and the extension lemma — and between them they turn statements about maps into statements about cell dimensions. That translation is the technique of Chapter 17, met here in its simplest form.

11.1 The homotopy extension property

Definition 11.1. A pair (X, A) has the *homotopy extension property* (HEP) if every map $X \times \{0\} \cup A \times I \to Y$ extends to $X \times I \to Y$. An inclusion $A \hookrightarrow X$ with the HEP is a *cofibration*.

Concretely: a homotopy defined on A and a map defined on X that agree at time zero can be combined into a homotopy on all of X.

Why this is the right notion. Homotopy theory wants to treat a subspace $A \subseteq X$ "up to homotopy": deform a map on A and drag the rest of the map along. The HEP is precisely the condition that this is always possible, and it is a property of how A sits in X, not of A or X separately: $\{0\} \cup \{1/n\} \subseteq [0,1]$ fails it, while every subcomplex of a CW complex has it (Exercise 11.3). There is a useful reformulation: if A is closed in X, then (X, A) has the HEP if and only if $X \times \{0\} \cup A \times I$ is a retract of $X \times I$ (Exercise 11.2). Every map $f\colon A \to Y$ can be replaced by a cofibration up to homotopy equivalence, namely the inclusion of A into the *mapping cylinder* $M_f = (A \times I) \sqcup Y / (a, 1) \sim f(a)$, which deformation retracts onto Y; this is the mechanism behind the homotopy pushouts of Chapter 21.

Exercise 11.2 (medium)**.** Let $A \subseteq X$ be closed. Show that (X, A) has the homotopy extension property if and only if $X \times \{0\} \cup A \times I$ is a retract of $X \times I$.

Source: [1, §0, Ex. 26] (first part) and the discussion preceding statement [1, Prop. 0.16], reproduced in substance

Hint. *For "⇒" apply the HEP with $Y = X \times \{0\} \cup A \times I$ and the identity map. For "⇐" compose with the retraction. Where is closedness of A used?*

Solution. ($\Rightarrow$) Let $Y = X \times \{0\} \cup A \times I$ and take the identity $Y \to Y$, which is a map $X \times \{0\} \cup A \times I \to Y$. The HEP extends it to $r\colon X \times I \to Y$, a retraction.

($\Leftarrow$) Let r be a retraction and $F\colon X \times \{0\} \cup A \times I \to Z$ a map. Then $F \circ r$ extends F over $X \times I$. The only subtlety is that F must be continuous on $X \times \{0\} \cup A \times I$ when it is continuous on the two pieces; this holds because both pieces are closed in $X \times I$ when A is closed (pasting lemma). That is where closedness enters.

Exercise 11.3 (easy)**.** Show that every CW pair has the homotopy extension property.

Hint. *It suffices to construct a retraction $D^n \times I \to D^n \times \{0\} \cup \partial D^n \times I$, and then to induct over cells.*

Solution. For a single cell, project radially from the point $(0, 2) \in D^n \times \mathbb{R}$: the ray from that point through any $(x, t) \in D^n \times I$ meets $D^n \times \{0\} \cup \partial D^n \times I$ in exactly one point, and this defines a retraction r.

Now let (X, A) be a CW pair and suppose given F on $X \times \{0\} \cup A \times I$. Extend over the cells of $X \setminus A$ by induction on dimension: for an n-cell with characteristic map $\Phi\colon D^n \to X$, the map F is already defined on $D^n \times \{0\} \cup \partial D^n \times I$ after composing with $\Phi \times \mathrm{id}$, and precomposing with the retraction r extends it over $D^n \times I$. The extensions agree on overlaps because they are defined cell by cell, and the CW topology makes the resulting function continuous.

Proposition 11.4. *Let (X, A) be a CW pair with A contractible. Then the quotient map $q\colon X \to X/A$ is a homotopy equivalence.*

Proof. Let $h_t\colon A \to A$ be a contraction, $h_0 = \mathrm{id}_A$ and h_1 constant at a_0. Regard it as a homotopy $A \times I \to X$ and combine it with id_X at time zero; by Exercise 11.3 there is a homotopy $H_t\colon X \to X$ with $H_0 = \mathrm{id}_X$ and $H_t|_A = h_t$.

At time one, H_1 maps A to the single point a_0, so it factors through the quotient: $H_1 = g \circ q$ for a map $g\colon X/A \to X$. We claim g is a homotopy inverse for q.

In one direction, $g \circ q = H_1 \simeq H_0 = \mathrm{id}_X$.

In the other, consider $q \circ g\colon X/A \to X/A$. For each t the map H_t sends A into A, so $q \circ H_t$ factors through q, giving a homotopy $\overline{H}_t\colon X/A \to X/A$ with $q \circ H_t = \overline{H}_t \circ q$. Then $\overline{H}_0 = \mathrm{id}$ and $\overline{H}_1 \circ q = q \circ H_1 = q \circ g \circ q$; since q is surjective this gives $\overline{H}_1 = q \circ g$, so $q \circ g \simeq \mathrm{id}_{X/A}$. □

Example 11.5 (The debt, paid)**.** A spanning tree T of a connected graph Γ is a contractible subcomplex, so $\Gamma \to \Gamma/T$ is a homotopy equivalence and Γ/T is a wedge of circles — one for each edge outside T. That is exactly the move made in Example 3.10 and Exercise 3.13, and in the proof of Proposition 3.9.

More generally, a spanning tree of the 1-skeleton of a connected CW complex is a contractible subcomplex, and collapsing it produces a homotopy equivalent complex with a single 0-cell. This is Proposition 2.2.

Warning 11.6. The proof used the HEP exactly once, to promote the contraction of A to a homotopy of X. Nothing else in the argument would survive its absence: without some hypothesis relating A to X, the contraction cannot be made to move the rest of the space, and H_1 never gets defined. The natural generalisation — replacing "CW pair" by "cofibration" throughout — costs nothing, since Exercise 11.3 was the only place the CW structure was used.

Proposition 11.7 (Attaching along homotopic maps)**.** *Let (X_1, A) be a CW pair with the HEP and let $f \simeq g\colon A \to X_0$. Then $X_0 \cup_f X_1 \simeq X_0 \cup_g X_1$ rel X_0. In particular, attaching a cell along a nullhomotopic map gives, up to homotopy equivalence rel X_0, the wedge of X_0 with a sphere.*

Proof sketch. Let $F\colon A\times I \to X_0$ be a homotopy from f to g, and form $W = X_0 \cup_F (X_1 \times I)$, gluing $A \times I$ to X_0 by F. It contains $X_0 \cup_f X_1$ and $X_0 \cup_g X_1$ as the two ends. Since $(X_1 \times I, X_1 \times \{0\} \cup A \times I)$ is a deformation retraction pair (the retraction of Exercise 11.2 can be chosen to be a deformation retraction for CW pairs), W deformation retracts onto $X_0 \cup_f X_1$, and likewise onto $X_0 \cup_g X_1$. Hence both ends are homotopy equivalent to W rel X_0. For the last assertion, attaching D^n along a constant map gives $X_0 \vee S^n$. Full details: [1, Prop. 0.18]. □

Exercise 11.8 (easy)**.** Show that the space obtained from S^2 by attaching n 2-cells along any n circles in S^2 (images of maps $S^1 \to S^2$) is homotopy equivalent to the wedge of $n + 1$ 2-spheres.

Source: [1, §0, Ex. 19], reproduced in substance

Solution. Every map $S^1 \to S^2$ is nullhomotopic, since $\pi_1(S^2) = 0$ and a based nullhomotopy is a free one (Exercise 1.20). Attach the cells one at a time. After attaching k cells the space is, by Proposition 11.7, homotopy equivalent rel S^2 to $S^2 \vee \bigvee_k S^2$, and the next attaching map still lands in the original S^2, where it is nullhomotopic; by the same proposition the $(k + 1)$-st attachment yields $S^2 \vee \bigvee_{k+1} S^2$ up to homotopy. (The equivalences are rel S^2, so the image of the next attaching circle is unaffected.) After n steps we have $\bigvee_{n+1} S^2$.

Exercise 11.9 (easy)**.** Show that a CW complex $X = A \cup B$, where A, B and $A \cap B$ are contractible subcomplexes, is contractible.

Source: [1, §0, Ex. 23], reproduced in substance

Solution. By Proposition 11.4, $X \to X/A$ is a homotopy equivalence. Now $X/A = B/(A \cap B)$ as spaces (every point of X outside A lies in $B \setminus (A \cap B)$, and the quotient topologies agree for CW complexes), and $B \to B/(A \cap B)$ is a homotopy equivalence by the same proposition since $A \cap B$ is a contractible subcomplex of B. So $X \simeq X/A \cong B/(A \cap B) \simeq B \simeq *$.

Exercise 11.10 (medium)**.** Let (X, A) be a CW pair. Show that the mapping cone $X \cup_A CA$ is homotopy equivalent to X/A.

Hint. *CA is contractible. Which pair is Proposition 11.4 being applied to?*

Solution. Inside $X \cup_A CA$ the cone CA is a contractible subcomplex, so by Proposition 11.4 the quotient map $X \cup_A CA \to (X \cup_A CA)/CA$ is a homotopy equivalence. Collapsing the cone identifies all of A to a point, so the target is X/A.

11.2 Better models

Theorem 11.11 (Cellular approximation; recalled)**.** *Every map between CW complexes is homotopic to a cellular one, and a homotopy between cellular maps may be taken cellular.*

Exercise 11.12 (easy)**.** Deduce that the inclusion $X^n \hookrightarrow X$ of the n-skeleton induces an isomorphism on π_k for $k < n$ and a surjection for $k = n$.

Solution. Let $k < n$. Any map $S^k \to X$ is homotopic to a cellular one by Theorem 11.11, hence lands in $X^k \subseteq X^n$: this gives surjectivity. If two maps $S^k \to X^n$ become homotopic in X, the homotopy $S^k \times I \to X$ may be taken cellular, so it lands in $X^{k+1} \subseteq X^n$ when $k+1 \leq n$: this gives injectivity for $k < n$. For $k = n$ only the surjectivity argument applies.

Theorem 11.13 (Connectivity-adapted models; recalled)**.** *Every m-connected CW complex is homotopy equivalent to one whose cells occupy only dimension 0 — a single cell there — and dimensions strictly greater than m.*

Proof sketch. Build a CW approximation from the bottom. Start with a point mapping to the basepoint; having built a complex Z_k with a map to X inducing isomorphisms on π_i for $i < k$ and a surjection on π_k, attach cells of dimension $k+1$ to kill the kernel on π_k and cells of dimension $k+1$ to hit new generators of π_{k+1}. When X is m-connected the first m stages require no cells at all, because there is nothing to hit and nothing to kill; the construction therefore produces cells only in dimension 0 and in dimensions $> m$. □

Example 11.14. S^n is $(n-1)$-connected, and the standard CW structure — one 0-cell and one n-cell — is already of the shape promised by Theorem 11.13. A simply connected complex may always be modelled with no 1-cells, which is why arguments about simply connected spaces so often begin "we may assume there are no cells in dimension one".

11.3 The extension lemma

Everything above is preparation for a single statement, which converts a question about extending maps into a question about the dimensions of cells.

Theorem 11.15 (Extension lemma)**.** *Let (X, A) be a CW pair and Y a space with $\pi_{k-1}(Y) = 0$ for every k such that $X \setminus A$ contains a cell of dimension k. Then every map $A \to Y$ extends to X.*

Proof sketch. Extend over cells by induction on dimension. To extend over a k-cell with attaching map $\varphi\colon S^{k-1} \to X^{k-1}$, one must show the composite $S^{k-1} \to X^{k-1} \to Y$ is nullhomotopic; its class lies in $\pi_{k-1}(Y)$, which vanishes by hypothesis, and a nullhomotopy is precisely an extension over the cell. □

Exercise 11.16 (easy)**.** Use Theorem 11.15 to show that a CW complex retracts onto any contractible subcomplex.

Source: [1, §4.1, Ex. 13], reproduced in substance

Solution. Let $A \subseteq X$ be a contractible subcomplex. All homotopy groups of A vanish, so the hypothesis of Theorem 11.15 holds for the pair (X, A) and the target $Y = A$, whatever cells $X \setminus A$ has. Hence $\mathrm{id}_A\colon A \to A$ extends to $r\colon X \to A$, which is a retraction.

Exercise 11.17 (medium)**.** Let X and Y be homotopy equivalent CW complexes, neither of which has cells of dimension $n+1$. Show that their n-skeleta are homotopy equivalent.

Source: [1, §4.1, Ex. 14], reproduced in substance

Hint. *Make the equivalences and the homotopies cellular (Theorem 11.11).*

Solution. Let $f\colon X \to Y$ and $g\colon Y \to X$ be inverse homotopy equivalences; by Theorem 11.11 we may take them cellular, so they restrict to $f_n\colon X^n \to Y^n$, $g_n\colon Y^n \to X^n$. A homotopy $H\colon X \times I \to X$ from gf to id is a homotopy between cellular maps, so it may be taken cellular (relative to $X \times \partial I$, where it is already cellular). Then H maps $X^n \times I$, which has dimension $n+1$, into $X^{n+1} = X^n$ (no $(n+1)$-cells). So $g_n f_n \simeq \mathrm{id}_{X^n}$. Symmetrically $f_n g_n \simeq \mathrm{id}_{Y^n}$.

Remark 11.18. That proof is obstruction theory in embryo, and it is worth naming the shape now: an extension is built skeleton by skeleton; at each stage the obstruction to continuing is the homotopy class of a composite; and if the relevant homotopy group vanishes, the obstruction vanishes with it. Chapter 17 keeps track of the obstructions when they do *not* vanish, and organises them into cohomology classes with coefficients in $\pi_{k-1}(Y)$. The proof of Proposition 2.23 in Chapter 2 was the same argument, run against $Y = S^1$ with $\pi_k(S^1) = 0$ for $k \geq 2$.

Exercise 11.19 (medium)**.** Let Y be n-connected and let X be a CW complex of dimension at most n. Show that every map $X \to Y$ is nullhomotopic.

Hint. *A nullhomotopy of $f\colon X \to Y$ is an extension over $X \times I$ of the map defined on $X \times \{0,1\}$ by f and by a constant. What are the cells of $X \times I$ relative to $X \times \{0,1\}$, and what dimensions do they have?*

Solution. Put $W = X \times I$ and $A = X \times \{0,1\}$, a CW pair. Define g on A by f on $X \times \{0\}$ and by the constant map on $X \times \{1\}$. A cell of $W \setminus A$ is $e \times (0,1)$ for a cell e of X, so its dimension is $\dim e + 1 \leq n+1$. The extension lemma therefore requires $\pi_{k-1}(Y) = 0$ for $k \leq n+1$, i.e. $\pi_j(Y) = 0$ for $j \leq n$, which is exactly n-connectedness. So g extends over $X \times I$, and the extension is a homotopy from f to a constant map.

Remark 11.20 (Where connectivity arguments go next)*.* Two directions open here, and both are used repeatedly later.

The first is to feed a space into Exercise 11.19 as both source and target, or as source and target after replacing one of them by a connectivity-adapted model from Theorem 11.13. Such arguments turn a numerical coincidence between connectivity and dimension into a statement about the space itself, and they need nothing beyond this page.

The second is to estimate the connectivity of a pair by counting the dimensions of its relative cells. Theorem 11.13 is what makes this work: after replacing X and Y by good models, the relative cells of a product are products of cells, and their dimensions add. That is the standard route to connectivity estimates for products and smash products, and the elementary shadow of the Blakers–Massey theorem.

11.4 What to carry forward

- CW pairs have the HEP (Exercise 11.3); collapsing a contractible subcomplex is a homotopy equivalence (Proposition 11.4), which is the promise made in Chapters 2 and 3.
- The HEP is used exactly once in that proof, to move a homotopy of A out into X (Warning 11.6).

- Mapping cone $\simeq$ quotient for a CW pair (Exercise 11.10).
- Cellular approximation makes homotopy groups skeletal (Exercise 11.12); connectivity can be built into the model (Theorem 11.13).
- The extension lemma (Theorem 11.15) trades homotopy groups against cell dimensions; it is obstruction theory with all obstructions assumed to vanish (Remark 11.18).

Chapter 12

Vector bundles and Stiefel–Whitney classes

Characteristic classes attach cohomology to a bundle, turning geometric questions — is this bundle trivial? does this manifold admit q independent vector fields? — into computations in a ring one already knows. This chapter sets up the mod-2 theory, which is the easiest and the sharpest of its kind, and shows it deciding a question about projective spaces.

The pattern to watch is that a single structural identity about the tangent bundle, fed through a multiplicative axiom, converts a topological question into an arithmetic one about binomial coefficients. That conversion is the whole technique, and Chapter 26 runs it again over $\mathbb{Z}$ where the arithmetic is harder and the conclusions are stronger.

12.1 Bundles

Definition 12.1. A *real vector bundle* of rank n over X is a map $p\colon E \to X$ with a real vector space structure on each fibre $p^{-1}(x)$, locally trivial: every point of X has a neighbourhood U with a homeomorphism $p^{-1}(U) \cong U \times \mathbb{R}^n$ that is linear on fibres. We write ε^n for the trivial bundle $X \times \mathbb{R}^n$.

Bundles pull back along maps, add under Whitney sum $E \oplus F$, and are classified: over a paracompact base, isomorphism classes of rank-n real bundles correspond to homotopy classes of maps $X \to BO(n)$, with the universal bundle pulling back to the given one.

Example 12.2 (The tautological line bundle)**.** Over $\mathbb{RP}^n$ let γ be the bundle whose fibre over a line $\ell \subseteq \mathbb{R}^{n+1}$ is ℓ itself. It is the basic example, and it is not trivial for $n \geq 1$: a nowhere-zero section would be a continuous choice of nonzero vector $s(\ell) \in \ell$ in each line. Composing with the quotient $S^n \to \mathbb{RP}^n$, write $s([x]) = \lambda(x)\, x$ for $x \in S^n$; then $\lambda\colon S^n \to \mathbb{R} \setminus \{0\}$ is continuous ($\lambda(x) = \langle s([x]), x\rangle$) and, since $[x] = [-x]$, $\lambda(-x)(-x) = \lambda(x)x$, i.e. $\lambda(-x) = -\lambda(x)$. A continuous nowhere-zero function on the connected space S^n has constant sign, which contradicts $\lambda(-x) = -\lambda(x)$. So γ has no nowhere-zero section, and a trivial line bundle does.

Proposition 12.3. *$T\mathbb{RP}^q \oplus \varepsilon^1 \cong (q+1)\gamma$, the Whitney sum of $q+1$ copies of the tautological line bundle.*

Proof. Write $\gamma^\perp$ for the orthogonal complement bundle, so $\gamma \oplus \gamma^\perp = \varepsilon^{q+1}$. At a line ℓ, a tangent vector to $\mathbb{RP}^q$ is a linear map $\ell \to \ell^\perp$: moving the line means tilting it, and the tilt is recorded by such a map. Hence $T\mathbb{RP}^q \cong \operatorname{Hom}(\gamma, \gamma^\perp)$. Adding $\operatorname{Hom}(\gamma,\gamma) \cong \varepsilon^1$ gives

$$T\mathbb{RP}^q \oplus \varepsilon^1 \cong \operatorname{Hom}\left(\gamma, \gamma^\perp \oplus \gamma\right) = \operatorname{Hom}\left(\gamma, \varepsilon^{q+1}\right) \cong (q+1)\operatorname{Hom}(\gamma, \varepsilon^1),$$

and $\operatorname{Hom}(\gamma,\varepsilon^1) \cong \gamma$ because a real line bundle is isomorphic to its dual, via any fibrewise inner product. □

Classifying spaces, concretely. The space $BO(n)$ can be taken to be the infinite Grassmannian $\operatorname{Gr}_n(\mathbb{R}^\infty)$ of n-planes in $\mathbb{R}^\infty = \bigcup_N \mathbb{R}^N$, with the universal bundle γ_n whose fibre over a plane is the plane itself. For $n = 1$ this is $\mathbb{RP}^\infty$ with the tautological line bundle. A bundle $E \to X$ over a paracompact base is classified by a map $X \to \operatorname{Gr}_n(\mathbb{R}^N)$ as soon as E embeds in a trivial bundle $X \times \mathbb{R}^N$, which is always possible for compact X; the classifying map sends x to the fibre $E_x \subseteq \mathbb{R}^N$. All characteristic classes are pulled back from the cohomology of $BO(n)$, which is why its cohomology ring (Theorem 12.4) controls everything below.

12.2 Stiefel–Whitney classes

Theorem 12.4 (Existence and axioms; recalled)**.** *There are classes $w_i(E) \in H^i(X;\mathbb{F}_2)$, for a real bundle E over X, with $w_0(E) = 1$ and $w_i(E) = 0$ for $i > \operatorname{rank} E$, such that:*

(a) Naturality: $w_i(f^*E) = f^* w_i(E)$.

(b) Whitney formula: $w(E \oplus F) = w(E)\,w(F)$, *where $w = 1 + w_1 + w_2 + \cdots$ is the total class.*

(c) Normalisation: $w(\gamma) = 1 + a$ *over $\mathbb{RP}^\infty$, where a generates $H^1(\mathbb{RP}^\infty;\mathbb{F}_2)$.*

These properties determine the classes, and $H^*(BO(n);\mathbb{F}_2) = \mathbb{F}_2[w_1,\dots,w_n]$.

The last statement is what makes the axioms legitimate rather than optimistic: the classes exist because the cohomology of the classifying space is a polynomial ring on them, and every characteristic class is a polynomial in them.

Example 12.5 (Trivial bundles, and orientability)**.** A trivial bundle is pulled back from a point, so by naturality $w(\varepsilon^n) = 1$; and by the Whitney formula, adding a trivial summand does not change the total class. Consequently w is an invariant of the *stable* isomorphism class of a bundle, which is exactly the right level of generality for Proposition 12.3 to be useful.

In rank one, w_1 detects orientability: $w_1(E) = 0$ if and only if E is orientable. Over $\mathbb{RP}^1 = S^1$ the bundle γ is the Möbius line bundle and $w(\gamma) = 1 + a$ with $a \neq 0$, so it is non-orientable — the Möbius band having one side is a statement about w_1.

Example 12.6 (Projective spaces)**.** $H^*(\mathbb{RP}^q;\mathbb{F}_2) = \mathbb{F}_2[a]/(a^{q+1})$. Restricting the normalisation of Theorem 12.4 along $\mathbb{RP}^q \subseteq \mathbb{RP}^\infty$ gives $w(\gamma) = 1 + a$ there, and Proposition 12.3 with the Whitney formula gives the total class of the tangent bundle:

$$w(\mathbb{RP}^q) := w\big(T\mathbb{RP}^q\big) = w\big(T\mathbb{RP}^q \oplus \varepsilon^1\big) = (1+a)^{q+1} \in \mathbb{F}_2[a]/(a^{q+1}).$$

Everything about the tangent bundle that w can see is now a question about expanding a power of $1 + a$ in a truncated polynomial ring.

Warning 12.7. The truncation is not decoration. In $\mathbb{F}_2[a]$ the element $(1+a)^{q+1}$ is never 1 for $q \geq 1$; it is only after killing a^{q+1} that the question becomes interesting. Every computation below must be carried out in $\mathbb{F}_2[a]/(a^{q+1})$, and the commonest error is to expand in the polynomial ring and forget to truncate.

Example 12.8 (The first three cases)**.** In $\mathbb{F}_2[a]/(a^{q+1})$:

$$q=1:\ (1+a)^2 = 1+a^2 = 1; \qquad q=2:\ (1+a)^3 = 1+a+a^2; \qquad q=3:\ (1+a)^4 = 1.$$

For $q=1$ we used $a^2=0$ and $2a=0$; for $q=2$, that $3\equiv 1 \bmod 2$ and $a^3=0$; for $q=3$, that all of $\binom{4}{1}, \binom{4}{2}, \binom{4}{3}$ are even.

So $w(\mathbb{RP}^2) = 1+a+a^2 \neq 1$, while $w(\mathbb{RP}^1) = w(\mathbb{RP}^3) = 1$.

Exercise 12.9 (easy)**.** Deduce from Example 12.8 that $\mathbb{RP}^2$ is not parallelisable, and indeed that $T\mathbb{RP}^2$ does not become trivial after adding any trivial bundle.

Solution. If $T\mathbb{RP}^2 \oplus \varepsilon^k$ were trivial then its total class would be 1 by Example 12.5; but adding a trivial summand does not change w, so $w(\mathbb{RP}^2)$ would be 1, contradicting $w(\mathbb{RP}^2) = 1+a+a^2$. In particular $T\mathbb{RP}^2$ itself is not trivial, so $\mathbb{RP}^2$ is not parallelisable.

Note that $\mathbb{RP}^1 = S^1$ and $\mathbb{RP}^3$ pass this test, which does not prove they are parallelisable — only that w raises no objection. Both in fact are, being Lie groups.

Exercise 12.10 (easy)**.** Show that $(1+x)^{2^k} = 1 + x^{2^k}$ in $\mathbb{F}_2[x]$, and deduce that $\binom{2^k}{i}$ is even for $0 < i < 2^k$.

Hint. *Squaring is a ring homomorphism in characteristic* 2*.*

Solution. In characteristic 2 the Frobenius $y \mapsto y^2$ is a ring homomorphism, so $(1+x)^2 = 1+x^2$, and iterating k times gives $(1+x)^{2^k} = 1 + x^{2^k}$. Comparing coefficients with the binomial expansion, $\binom{2^k}{i} \equiv 0 \bmod 2$ for $0 < i < 2^k$.

Whitney duality and immersions. If a closed n-manifold M immerses in $\mathbb{R}^{n+k}$, the normal bundle ν has rank k and $TM \oplus \nu$ is the trivial bundle ε^{n+k} (the restriction of $T\mathbb{R}^{n+k}$). By the Whitney formula $w(TM)\,w(\nu) = 1$, so $w(\nu) = w(TM)^{-1}$ — the inverse exists because $w(TM) = 1 + (\text{nilpotent})$ — and $w_i(\nu) = 0$ for $i > k$. So the dual class $\bar{w} = w(TM)^{-1}$ gives a lower bound for the codimension of any immersion: $k \geq \max\{i : \bar{w}_i \neq 0\}$. This is the first of many places where characteristic classes give a necessary condition for a geometric construction; Smale–Hirsch theory shows that for immersions the bundle-theoretic condition is also sufficient.

Characteristic numbers. For a closed n-manifold M with mod-2 fundamental class $[M]$ and a monomial $w_{i_1}\cdots w_{i_r}$ of total degree n, the *Stiefel–Whitney number* $\langle w_{i_1}(TM)\cdots w_{i_r}(TM), [M]\rangle \in \mathbb{F}_2$ is an invariant of M. Pontryagin and Thom showed that these numbers vanish when M is a boundary, and Thom that their vanishing characterises boundaries (unoriented cobordism); see [10, §4]. The top class $w_n(TM)$ evaluates to $\chi(M) \bmod 2$.

Exercise 12.11 (medium)**.** Show that $\mathbb{RP}^{2^k}$ does not immerse in $\mathbb{R}^{2^{k+1}-2}$.

SOURCE: [4, Ch. 23, Problem 2], reproduced in substance

Hint. *With $n = 2^k$, $w(\mathbb{RP}^n) = (1+a)^{n+1} = (1+a)(1+a^n)$ in $\mathbb{F}_2[a]/(a^{n+1})$. Invert it.*

Solution. Let $n = 2^k$. By Example 12.6 and Exercise 12.10, $w(\mathbb{RP}^n) = (1+a)^{n+1} = (1+a)(1+a)^{2^k} = (1+a)(1+a^n)$ in $\mathbb{F}_2[a]/(a^{n+1})$. In this ring $(1+a^n)^{-1} = 1+a^n$ (as $a^{2n}=0$) and $(1+a)^{-1} = \sum_{i=0}^{n} a^i$. Hence

$$\bar{w} = (1+a^n)\sum_{i=0}^{n} a^i = \sum_{i=0}^{n} a^i + a^n = \sum_{i=0}^{n-1} a^i,$$

all products $a^n \cdot a^i$ with $i \geq 1$ vanishing. So $\bar{w}_{n-1} = a^{n-1} \neq 0$. If $\mathbb{RP}^n$ immersed in $\mathbb{R}^{n+k'}$, the normal bundle would have rank k' and $\bar{w}_i = 0$ for $i > k'$, forcing $k' \geq n-1$. So $\mathbb{RP}^n$ does not immerse in $\mathbb{R}^{n+(n-2)} = \mathbb{R}^{2^{k+1}-2}$.

Exercise 12.12 (medium)**.** Show that all Stiefel–Whitney numbers of $\mathbb{RP}^q$ vanish if and only if q is odd.

SOURCE: [4, Ch. 23, Problem 3], reproduced in substance

Hint. *For q odd write $(1+a)^{q+1} = (1+a^2)^{(q+1)/2}$. For q even, look at w_q.*

Solution. If q is odd, $q+1 = 2m$ and $w(\mathbb{RP}^q) = (1+a)^{2m} = (1+a^2)^m$, so every w_i with i odd vanishes. A monomial $w_{i_1}\cdots w_{i_r}$ of odd total degree q must contain a factor of odd degree, hence vanishes; all Stiefel–Whitney numbers are zero. If q is even, $w_q = \binom{q+1}{q}a^q = (q+1)a^q = a^q$, which is the nonzero element of $H^q(\mathbb{RP}^q;\mathbb{F}_2)$, so $\langle w_q, [\mathbb{RP}^q]\rangle = 1$. (Consistently, $w_q[\mathbb{RP}^q] = \chi(\mathbb{RP}^q) \bmod 2 = 1$.)

Exercise 12.13 (medium)**.** Show that $w(TS^n) = 1$ for every n, but that TS^n is not trivial for n even. Conclude that Stiefel–Whitney classes cannot detect every nontrivial bundle, even stably-nontrivial ones aside.

SOURCE: original

Hint. *$TS^n \oplus \nu = \varepsilon^{n+1}$ with ν the normal line bundle, which is trivial. For nontriviality, a nowhere-zero section v of TS^n gives a homotopy from the identity to the antipodal map; use Definition 4.24.*

Solution. The normal bundle of $S^n \subseteq \mathbb{R}^{n+1}$ has the nowhere-zero section $x \mapsto x$, so it is trivial, and $TS^n \oplus \varepsilon^1 \cong \varepsilon^{n+1}$. By Example 12.5, $w(TS^n) = w(\varepsilon^{n+1}) = 1$.

If TS^n were trivial it would have a nowhere-zero section v, i.e. $v(x) \perp x$, $v(x) \neq 0$. Then $f_t(x) = \cos(\pi t)\, x + \sin(\pi t)\, v(x)/|v(x)|$ is a homotopy of maps $S^n \to S^n$ (the vectors x and $v(x)/|v(x)|$ are orthonormal) from the identity to the antipodal map. So the antipodal map would have degree 1; but it has degree $(-1)^{n+1} = -1$ for n even. Hence TS^n is nontrivial for n even, although $w(TS^n) = 1$ and indeed TS^n is stably trivial.

Exercise 12.14 (easy)**.** Over $\mathbb{RP}^1 = S^1$, show that $\gamma \oplus \gamma$ is trivial although γ is not, and verify that $w(\gamma\oplus\gamma) = 1$ there. Over $\mathbb{RP}^2$, show that $\gamma\oplus\gamma$ is nontrivial.

SOURCE: original

Solution. By the Whitney formula $w(\gamma\oplus\gamma) = (1+a)^2 = 1+a^2$. Over $\mathbb{RP}^1$, $a^2 = 0$, so $w(2\gamma) = 1$. To see triviality directly: a rank-2 bundle over S^1 is determined by the clutching of its restriction to an interval, i.e. by an element of $\pi_0(GL_2(\mathbb{R})) = \mathbb{Z}/2$, the sign of the determinant; for 2γ the clutching is $(-1)\oplus(-1) = -I$, which has positive determinant, so 2γ is trivial (it is orientable). Over $\mathbb{RP}^2$, $w_2(2\gamma) = a^2 \neq 0$, so 2γ is not trivial (trivial bundles have $w = 1$). And γ itself is nontrivial over every $\mathbb{RP}^n$, $n \geq 1$, by Example 12.2, consistently with $w_1(\gamma) = a \neq 0$.

Remark 12.15 (The question this sets up). Example 12.6 says that $w(\mathbb{RP}^q) = 1$ exactly when $(1+a)^{q+1} = 1$ in $\mathbb{F}_2[a]/(a^{q+1})$, i.e. exactly when $\binom{q+1}{i}$ is even for all $0 < i \leq q$. Exercise 12.10 settles one direction of the resulting arithmetic; the converse needs the general statement about binomial coefficients modulo 2 — Lucas' theorem, which reads off the parity of $\binom{m}{i}$ from a digitwise comparison of the binary expansions of i and m, the coefficient being odd exactly when no digit of i exceeds the corresponding digit of m. Assembling the two halves determines exactly which projective spaces pass the test of Exercise 12.9, and Example 12.8 already shows the pattern in the first three cases.

Two cautions for whoever carries that out. The conclusion concerns *stable* triviality, since w cannot distinguish $T\mathbb{RP}^q$ from $T\mathbb{RP}^q \oplus \varepsilon$; promoting it to parallelisability is a genuinely harder problem, solved by Adams. And the computation must respect Warning 12.7.

12.3 What the mod-2 theory cannot do

Remark 12.16. Stiefel–Whitney classes live in $\mathbb{F}_2$-cohomology, so they are blind to everything of odd order and to all sign information. Three examples of what they miss, each repaired later:

- An oriented bundle has an *Euler class* $e(E) \in H^n(X;\mathbb{Z})$ whose mod-2 reduction is $w_n(E)$ but which carries strictly more information; Chapter 26 computes with it.
- Complex bundles have Chern classes in integral cohomology, and the relation between c_i and w_{2i} is again reduction mod 2.
- The relation between the two theories is itself a cohomology operation: $\mathrm{Sq}^1 w_i$ is expressible in terms of the w_j by the Wu formula, and Sq^1 is a stable operation of degree one, so it converts mod-2 data into information about integral torsion — compare Chapter 9. That triangle — Stiefel–Whitney, Bockstein, Euler — is what Chapter 26 closes.

Remark 12.17 (Further reading). Milnor–Stasheff [10] is the classical reference for everything in this chapter (axioms, Grassmannians, Whitney duality, characteristic numbers); Hatcher's *Vector Bundles and K-Theory* [9] is closer in style to this book.

12.4 What to carry forward

- $T\mathbb{RP}^q \oplus \varepsilon \cong (q+1)\gamma$ (Proposition 12.3) — one structural identity, from which everything about $w(\mathbb{RP}^q)$ follows.
- The axioms (Theorem 12.4), and the fact that w is a stable invariant (Example 12.5).
- Compute in the truncated ring (Warning 12.7); the arithmetic is Frobenius and Lucas (Exercise 12.10).
- w_1 detects orientability; the mod-2 theory is blind to signs and to odd torsion (Remark 12.16), which is why Chapter 26 exists.

Chapter 13

Whitehead theorems, Postnikov towers, and what the invariants miss

Chapter 5 ended by observing that homology cannot see the universal cover. This chapter says what does. Whitehead's theorem is the statement that homotopy groups, *together with a map realising them*, form a complete invariant of CW homotopy type; the rest of the chapter is about how much of that completeness survives when one of those two ingredients is weakened.

Every result here has a hypothesis that looks removable and is not. The reader should finish the chapter able to name, for each theorem, the space or map that goes wrong when the hypothesis is dropped.

13.1 Whitehead's theorem, and the map it needs

Theorem 13.1 (Whitehead; recalled)**.** *A map $f\colon X \to Y$ between connected CW complexes that induces isomorphisms $\pi_n(X, x_0) \to \pi_n(Y, f(x_0))$ for all $n \geq 1$ is a homotopy equivalence.*

Warning 13.2 (The map is not optional). The theorem is about a map. Two spaces can have isomorphic homotopy groups in every degree and fail to be homotopy equivalent, and the book has already met such a pair.

Take the lens spaces $L(5,1)$ and $L(5,2)$ of Example 2.30. Both have fundamental group $\mathbb{Z}/5$ and universal cover S^3, so for $n \geq 2$

$$\pi_n\big(L(5,q)\big) \cong \pi_n(S^3)$$

by the covering-space isomorphism, independently of q; and π_1 agrees too. So all their homotopy groups are isomorphic. Their homology agrees (Example 4.10) and so does their cohomology ring (Example 7.18). They are nevertheless not homotopy equivalent.

What is missing is not an invariant but a map: no map between them induces those isomorphisms.

Exercise 13.3 (easy)**.** Let X be a connected CW complex with $\pi_n(X) = 0$ for all $n \geq 1$. Show X is contractible.

Solution. Apply Theorem 13.1 to the inclusion of a point $\{x_0\} \hookrightarrow X$, which induces isomorphisms $0 \to \pi_n(X)$ in every degree by hypothesis. Both spaces are CW complexes, so the inclusion is a homotopy equivalence.

Exercise 13.4 (medium)**.** Let X be a CW complex that is the union of an increasing sequence of subcomplexes $X_1 \subseteq X_2 \subseteq \cdots$ such that each inclusion $X_i \hookrightarrow X_{i+1}$ is nullhomotopic. Show that X is contractible. Deduce that $S^\infty = \bigcup_n S^n$ is contractible.

SOURCE: [1, §4.1, Ex. 11], reproduced in substance

Hint. *A map $S^n \to X$ has compact image, which lies in some X_i.*

Solution. A compact subset of a CW complex meets only finitely many cells, so lies in some finite subcomplex, hence in some X_i (each cell lies in some X_j, and the sequence increases). So every map $f\colon S^n \to X$ factors through some X_i, and composing with the nullhomotopic inclusion $X_i \hookrightarrow X_{i+1}$ shows that f is freely nullhomotopic in X. In particular X is path-connected (take $n = 0$). A free nullhomotopy of f is a map from the cone on S^n, i.e. an extension $F\colon D^{n+1} \to X$ of f. Such an extension gives a *based* nullhomotopy: the straight-line contraction $H_t(x) = (1-t)x + ts_0$ of D^{n+1} onto the basepoint $s_0 \in S^n$ fixes s_0, so $F \circ H_t|_{S^n}$ is a homotopy rel s_0 from f to a constant. Hence $\pi_n(X) = 0$ for all n, and X is contractible by Exercise 13.3. For S^∞, each inclusion $S^n \hookrightarrow S^{n+1}$ is nullhomotopic (it misses a point of S^{n+1}, and S^{n+1} minus a point is contractible).

13.2 Homology, and the universal cover

Homotopy groups are hard to compute and homology is easy, so one wants a homological criterion. There is one, with a hypothesis. The bridge is the Hurewicz theorem, which we state in the form used throughout the rest of the book.

Theorem 13.5 (Hurewicz; recalled)**.** *Let $n \geq 2$.*

(a) If X is $(n-1)$-connected, then $\widetilde{H}_i(X) = 0$ for $i < n$ and the Hurewicz map $h\colon \pi_n(X) \to H_n(X)$, $[f] \mapsto f_*[S^n]$, is an isomorphism.

(b) If (X, A) is an $(n-1)$-connected pair of path-connected spaces and A is simply connected, then $H_i(X, A) = 0$ for $i < n$ and $\pi_n(X, A) \cong H_n(X, A)$.

(c) For $n = 1$: $\pi_1(X)^{\mathrm{ab}} \cong H_1(X)$ for path-connected X.

This is [1, Thms. 4.32, 4.37 and 2A.1]. Without simple connectivity of A in (b) one must replace $\pi_n(X, A)$ by its quotient by the action of $\pi_1(A)$; that version is what makes the warning below possible.

Theorem 13.6 (Homological Whitehead; recalled)**.** *Let $f\colon X \to Y$ be a map of* simply connected *CW complexes inducing an isomorphism on $H_n(-;\mathbb{Z})$ for every n. Then f is a homotopy equivalence.*

Proof sketch. Replace f by an inclusion using the mapping cylinder. The homology of the pair vanishes, and the pair is simply connected, so the relative Hurewicz theorem gives $\pi_n(Y, X) = 0$ for all n; the long exact sequence of the pair then makes f_* an isomorphism on homotopy groups, and Theorem 13.1 applies. □

Warning 13.7 (Simple connectivity is essential). Let P be the Poincaré homology sphere: a closed orientable 3-manifold with $\pi_1(P)$ the binary icosahedral group of order 120, and with $H_*(P;\mathbb{Z}) \cong H_*(S^3;\mathbb{Z})$. Collapsing the complement of an open ball gives a degree-one map

$$c\colon P \longrightarrow P/(P \setminus B^3) \cong S^3,$$

which is an isomorphism on H_3 because it has degree one, and trivially so in the other degrees, both spaces having the homology of a sphere. So c_* is an isomorphism on all integral homology.

But c is not a homotopy equivalence: $\pi_1(P)$ has order 120 and $\pi_1(S^3)$ is trivial. The single hypothesis that fails is simple connectivity, and it fails on the source.

The Poincaré sphere fails on π_1. A subtler failure keeps π_1 correct and still defeats homology.

Exercise 13.8 (hard)**.** Let $\omega \in \pi_1(S^1 \vee S^2)$ and $\alpha \in \pi_2(S^1 \vee S^2)$ be represented by the inclusions of the two wedge summands, and let X be obtained by attaching a 3-cell along a map representing $2\alpha - \omega \cdot \alpha \in \pi_2(S^1 \vee S^2)$, where $\omega \cdot \alpha$ denotes the action of π_1 on π_2. Show that the inclusion $S^1 \hookrightarrow X$ induces isomorphisms on π_1 and on all homology groups, but that X is not weakly equivalent to S^1.

Source: MIT OCW 18.906 (Spring 2020, H. Miller), Problem Set III, Problem 12, reproduced in substance

Hint. *Use the universal cover of $S^1 \vee S^2$ from Example 5.25: $\pi_2(S^1 \vee S^2) \cong H_2(\widetilde{X}) \cong \mathbb{Z}[t,t^{-1}]$, with ω acting by t. Compute the cellular chain complex of X and of its universal cover.*

Solution. *π_1 and homology.* Attaching a 3-cell does not change π_1 (Proposition 2.3), so $\pi_1(X) = \pi_1(S^1 \vee S^2) = \mathbb{Z}$ and the inclusion of S^1 induces an isomorphism. Cellularly, X has cells e^0, e^1, e^2, e^3. The boundary $d_3(e^3)$ is the image of the attaching class under the Hurewicz map $\pi_2(S^1 \vee S^2) \to H_2(S^1 \vee S^2) = \mathbb{Z}e^2$, which ignores the action of π_1 (a based map and its translate by a loop are freely homotopic, so have the same image in homology): $h(\omega \cdot \alpha) = h(\alpha) = e^2$. So $d_3(e^3) = 2e^2 - e^2 = e^2$, and $d_2 = 0$, $d_1 = 0$. Hence $H_*(X) = (\mathbb{Z}, \mathbb{Z}, 0, 0) = H_*(S^1)$, and $S^1 \to X$ induces an isomorphism on homology.

Not a weak equivalence. The universal cover $\widetilde{X}$ is the line of spheres of Example 5.25 with one 3-cell attached for each $n \in \mathbb{Z}$, the n-th attached along the lift of $t^n(2\alpha - t\alpha)$. Its cellular chain complex is a complex of free $\mathbb{Z}[t,t^{-1}]$-modules of rank one,

$$\mathbb{Z}[t^{\pm 1}] \xrightarrow{2-t} \mathbb{Z}[t^{\pm 1}] \xrightarrow{0} \mathbb{Z}[t^{\pm 1}] \xrightarrow{t-1} \mathbb{Z}[t^{\pm 1}]$$

in degrees $3, 2, 1, 0$. So $H_2(\widetilde{X}) = \mathbb{Z}[t^{\pm 1}]/(2-t)$, which is isomorphic to $\mathbb{Z}[\frac{1}{2}]$ via $t \mapsto 2$ (a Laurent polynomial in t modulo $t = 2$ evaluates to a dyadic rational, and every dyadic rational $m/2^k$ is the image of mt^{-k}). In particular $H_2(\widetilde{X}) \neq 0$. Since $\widetilde{X}$ is simply connected, Theorem 13.5 gives $\pi_2(X) \cong \pi_2(\widetilde{X}) \cong H_2(\widetilde{X}) \cong \mathbb{Z}[\frac{1}{2}] \neq 0 = \pi_2(S^1)$. So the inclusion is not an isomorphism on π_2, and no weak equivalence $X \to S^1$ exists at all (π_2 differs).

The repair promised in Remark 5.28 is to work upstairs.

Proposition 13.9. *Let $f\colon X \to Y$ be a map of connected CW complexes inducing an isomorphism on π_1, and let $\tilde{f}\colon \widetilde{X} \to \widetilde{Y}$ be a lift to the universal covers. If $\tilde{f}_*$ is an isomorphism on homotopy groups, then f is a homotopy equivalence.*

Proof. A covering induces isomorphisms on π_n for $n \geq 2$, and the square formed by f, $\tilde{f}$ and the two covering projections commutes. So f_* is an isomorphism on π_n for $n \geq 2$, and by hypothesis also on π_1. Apply Theorem 13.1. □

Exercise 13.10 (easy)**.** Prove the fact used twice above: a covering $p\colon \widetilde{X} \to X$ induces isomorphisms $\pi_n(\widetilde{X}) \to \pi_n(X)$ for all $n \geq 2$.

Hint. *Lift maps $S^n \to X$ and homotopies $S^n \times I \to X$, using that S^n is simply connected for $n \geq 2$.*

Solution. For surjectivity, a map $S^n \to X$ with $n \geq 2$ lifts to $\widetilde{X}$ because S^n is simply connected, so the lifting criterion applies. For injectivity, if two lifts become homotopic downstairs, the homotopy $S^n \times I \to X$ lifts (again by simple connectivity of $S^n \times I$) to a homotopy upstairs starting at the given lift, and uniqueness of lifts identifies its other end with the second lift.

Exercise 13.11 (medium)**.** Show that a closed simply connected 3-manifold M is homotopy equivalent to S^3. (You may use that closed manifolds have the homotopy type of CW complexes.)

SOURCE: [1, §4.2, Ex. 15], reproduced in substance

Hint. *A simply connected manifold is orientable. Compute $H_*(M)$ by Theorem 7.11, then find a map $S^3 \to M$ with Theorem 13.5.*

Solution. M is orientable: its orientation double cover is a 2-sheeted cover of a simply connected space, hence disconnected, which means an orientation exists. By Theorem 7.11(a), $H_3(M) \cong H^0(M) = \mathbb{Z}$, $H_2(M) \cong H^1(M) = \mathrm{Hom}(H_1(M), \mathbb{Z}) = 0$ (as $H_1 = \pi_1^{\mathrm{ab}} = 0$), and $H_1 = 0$. So $\widetilde{H}_*(M) \cong \widetilde{H}_*(S^3)$. By Theorem 13.5(a) with $n = 2$, $\pi_2(M) \cong H_2(M) = 0$; so M is 2-connected and, with $n = 3$, $\pi_3(M) \cong H_3(M) \cong \mathbb{Z}$. Let $f\colon S^3 \to M$ represent a generator. By naturality of the Hurewicz map, $f_*[S^3] = h([f])$ generates $H_3(M)$, so f_* is an isomorphism on H_3, and trivially on all other $\widetilde{H}_i$ (both zero). Both spaces are simply connected and of CW type, so f is a homotopy equivalence by Theorem 13.6.

Exercise 13.12 (medium)**.** Show that every closed surface with infinite fundamental group is a $K(\pi, 1)$, i.e. has contractible universal cover. (You may use that a connected noncompact n-manifold has $H_n(-; \mathbb{Z}) = 0$.)

SOURCE: [1, §4.2, Ex. 16], reproduced in substance; the fact quoted is [1, Prop. 3.29]

Solution. Let S be a closed surface with $\pi_1(S)$ infinite and $\widetilde{S}$ its universal cover, a simply connected surface. The covering has infinitely many sheets, so $\widetilde{S}$ is not compact (the fibre over a point is an infinite discrete closed subset). Hence $H_2(\widetilde{S}) = 0$ by the quoted fact, $H_1(\widetilde{S}) = 0$ since $\widetilde{S}$ is simply connected, and $H_i(\widetilde{S}) = 0$ for $i \geq 3$ since $\widetilde{S}$ is a 2-dimensional CW complex (lift a CW structure of S). By Theorem 13.5, $\pi_2(\widetilde{S}) \cong H_2(\widetilde{S}) = 0$, then inductively $\pi_n(\widetilde{S}) \cong H_n(\widetilde{S}) = 0$ for all n. So $\widetilde{S}$ is contractible (Exercise 13.3), and $\pi_n(S) \cong \pi_n(\widetilde{S}) = 0$ for $n \geq 2$. The closed surfaces with finite π_1 are S^2 and $\mathbb{RP}^2$, which are not aspherical.

Exercise 13.13 (hard)**.** Let $f\colon X \to Y$ be a map between connected CW complexes of dimension at most n which induces isomorphisms on π_i for all $i \leq n$. Show that f is a homotopy equivalence.

SOURCE: [1, §4.2, Ex. 13], reproduced in substance

Hint. *Pass to universal covers and replace $\tilde{f}$ by an inclusion $\widetilde{X} \subseteq M$ into its mapping cylinder. Show the pair is n-connected, then use Theorem 13.5(b) in degree $n+1$ and the fact that $H_{n+1}(\widetilde{Y}) = 0$.*

Solution. By Proposition 13.9 it suffices to show that a lift $\tilde{f}\colon \widetilde{X} \to \widetilde{Y}$ is a homotopy equivalence; $\tilde{f}$ induces isomorphisms on π_i for $2 \leq i \leq n$ (covering isomorphisms) and $\widetilde{X}, \widetilde{Y}$ are simply

connected CW complexes of dimension $\leq n$. Let M be the mapping cylinder of $\tilde{f}$, which deformation retracts to $\widetilde{Y}$ and contains $\widetilde{X}$. In the long exact sequence of the pair $(M, \widetilde{X})$, the maps $\pi_i(\widetilde{X}) \to \pi_i(M)$ are isomorphisms for $i \leq n$, so $\pi_i(M, \widetilde{X}) = 0$ for $i \leq n$: the pair is n-connected, and $\pi_{n+1}(M) \to \pi_{n+1}(M, \widetilde{X})$ is surjective (its cokernel injects into $\ker(\pi_n \widetilde{X} \to \pi_n M) = 0$). By Theorem 13.5(b), $H_i(M, \widetilde{X}) = 0$ for $i \leq n$ and $\pi_{n+1}(M, \widetilde{X}) \cong H_{n+1}(M, \widetilde{X})$. The Hurewicz maps are natural, so the surjection $\pi_{n+1}(M) \to \pi_{n+1}(M, \widetilde{X}) \cong H_{n+1}(M, \widetilde{X})$ factors through $H_{n+1}(M) \cong H_{n+1}(\widetilde{Y}) = 0$ ($\widetilde{Y}$ has dimension $\leq n$). Hence $H_{n+1}(M, \widetilde{X}) = 0$. Together with $H_i(M, \widetilde{X}) = 0$ for $i \geq n+2$ (both spaces have dimension $\leq n+1$; M is the mapping cylinder, of dimension $\leq n+1$) all relative homology vanishes, so $\tilde{f}$ is a homology isomorphism, hence a homotopy equivalence by Theorem 13.6.

The action of π_1 on π_n. For $n \geq 1$ a path γ from x_0 to x_1 induces an isomorphism $\beta_\gamma \colon \pi_n(X, x_1) \to \pi_n(X, x_0)$, defined exactly as in Definition 1.1: shrink the domain of a map $(I^n, \partial I^n) \to (X, x_1)$ to a smaller concentric cube and fill the shell with the path γ running radially. For loops this gives an action of $\pi_1(X, x_0)$ on $\pi_n(X, x_0)$, by conjugation when $n = 1$. Under the isomorphism $\pi_n(X) \cong \pi_n(\widetilde{X})$ ($n \geq 2$) it corresponds to the action of $\pi_1(X)$ on $\widetilde{X}$ by deck transformations [1, §4.1, Ex. 4]. A space is *simple* when this action is trivial in every degree; Lie groups and H-spaces are simple, $S^1 \vee S^2$ is not (Example 5.25: π_2 is the free $\mathbb{Z}[t^{\pm 1}]$-module of rank one). The obstruction theory of Chapter 17 is stated for simple targets, and Part VI meets non-simple configuration spaces of surfaces.

Exercise 13.14 (medium)**.** Show that the action of $\pi_1(\mathbb{RP}^n) = \mathbb{Z}/2$ on $\pi_n(\mathbb{RP}^n) \cong \mathbb{Z}$ ($n \geq 2$) is trivial for n odd and is multiplication by -1 for n even.

Source: [1, §4.2, Ex. 2], reproduced in substance

Solution. The covering $S^n \to \mathbb{RP}^n$ identifies $\pi_n(\mathbb{RP}^n)$ with $\pi_n(S^n) = \mathbb{Z}$ (Exercise 13.10), and under this identification the nontrivial element of $\pi_1(\mathbb{RP}^n)$ acts through the nontrivial deck transformation, the antipodal map τ, as recalled above. The action of τ on $\pi_n(S^n) \cong H_n(S^n)$ (Hurewicz) is multiplication by $\deg \tau = (-1)^{n+1}$. (The identification uses basepoint changes along paths from the basepoint to its antipode; since S^n is simply connected for $n \geq 2$ these are canonical by Corollary 1.9 applied to π_n.) Hence the action is trivial for n odd and is -1 for n even.

Remark 13.15 (The invariant that works). Proposition 13.9 is stated in terms of homotopy groups upstairs, but the useful version replaces them by homology. The homology of $\widetilde{X}$ is not merely a graded abelian group: the deck group $\pi_1(X)$ acts on it, making it a module over the group ring $\mathbb{Z}[\pi_1 X]$ — this is what Example 5.25 exhibited for $S^1 \vee S^2$, where $H_2(\widetilde{X})$ was free of rank one over $\mathbb{Z}[t, t^{-1}]$. That module is homology with local coefficients, $H_*(X; \mathbb{Z}[\pi_1 X])$.

Combining Proposition 13.9 with Theorem 13.6 applied upstairs — the universal cover being simply connected — turns a homological hypothesis on $\widetilde{X}$ into a homotopy equivalence downstairs. Assembling that statement, and checking exactly which hypothesis on π_1 it needs, is the natural exercise at this point, and it is the invariant Chapter 5 promised.

13.3 Postnikov towers

Whitehead's theorem says the homotopy groups determine the homotopy type once a map is given. Postnikov towers organise the homotopy groups so that the *glue* between

them becomes visible as well.

Theorem 13.16 (Postnikov towers; recalled)**.** *For a connected CW complex X there are spaces X^n and maps forming a commutative tower under X,*

$$X \longrightarrow \cdots \longrightarrow X^n \longrightarrow X^{n-1} \longrightarrow \cdots \longrightarrow X^1,$$

such that $X \to X^n$ induces isomorphisms on π_i for $i \le n$, and $\pi_i(X^n) = 0$ for $i > n$. Each map $X^n \to X^{n-1}$ is a fibration with fibre $K(\pi_n X, n)$, and it is pulled back from the path fibration over $K(\pi_n X, n+1)$ along a map

$$k^n \colon X^{n-1} \longrightarrow K(\pi_n X,\, n+1),$$

*the k-*invariant*, a class in $H^{n+1}\big(X^{n-1}; \pi_n X\big)$. The stages and the tower are unique up to homotopy over X.*

Dually, the *n-connected cover* $X\langle n\rangle \to X$ kills the homotopy below degree $n+1$ and leaves the rest alone.

Exercise 13.17 (easy)**.** Let X be connected and let $X \to X^n$ be its n-th Postnikov stage (Theorem 13.16). Replace this map by a fibration (every map is homotopy equivalent to a fibration, [1, Prop. 4.64]) and let $X\langle n\rangle$ be its fibre. Show that the inclusion of the fibre induces isomorphisms on π_i for $i > n$, and that $\pi_i(X\langle n\rangle) = 0$ for $i \le n$. Then identify $X\langle 2\rangle$ when X is simply connected.

Solution. Write $F = X\langle n\rangle$ and $q \colon X \to X^n$. The long exact sequence of the fibration (Corollary 14.3) contains

$$\pi_{i+1}(X) \xrightarrow{q_*} \pi_{i+1}(X^n) \to \pi_i(F) \to \pi_i(X) \xrightarrow{q_*} \pi_i(X^n).$$

For $i > n$ both $\pi_i(X^n)$ and $\pi_{i+1}(X^n)$ vanish, so $\pi_i(F) \cong \pi_i(X)$, the isomorphism being induced by the inclusion of the fibre. For $i \le n$: q_* is an isomorphism on π_i, so $\pi_i(F) \to \pi_i(X)$ has image $\ker(q_*) = 0$; and $\pi_{i+1}(X^n) \to \pi_i(F)$ is onto with kernel the image of q_* on π_{i+1}, which is all of $\pi_{i+1}(X^n)$ (for $i+1 \le n$, since q_* is onto) or which is $\pi_{n+1}(X^n) = 0$ itself (for $i = n$). Either way $\pi_i(F) = 0$. (At $i = 0$ the same argument with pointed sets shows F is path-connected.)

For X simply connected and $n = 2$: $X^2 = K(\pi_2 X, 2)$, and the fibre sequence $X\langle 2\rangle \to X \to K(\pi_2 X, 2)$ exhibits $X\langle 2\rangle$ as the 2-connected cover, with π_i equal to $\pi_i(X)$ for $i \ge 3$ and zero below. For instance $S^2\langle 2\rangle \simeq S^3$: the Hopf fibration $S^1 \to S^3 \to S^2$ deloops to a fibration $S^3 \to S^2 \to \mathbb{CP}^\infty = K(\mathbb{Z}, 2)$ (Exercise 13.18).

Exercise 13.18 (medium)**.** Identify, up to homotopy equivalence, the 2-connected cover of $\mathbb{CP}^n$ and the second Postnikov stage of S^2.

Source: MIT OCW 18.906 (Spring 2020), Problem Set III, Problem 13, reproduced in substance

Hint. *For $\mathbb{CP}^n$ use the fibration $S^1 \to S^{2n+1} \to \mathbb{CP}^n$ and its classifying map $\mathbb{CP}^n \to \mathbb{CP}^\infty$. For S^2, use that $\mathbb{CP}^\infty$ is S^2 with cells of dimension ≥ 4 attached.*

Solution. *S^2.* $\mathbb{CP}^\infty$ is obtained from $\mathbb{CP}^1 = S^2$ by attaching cells of dimensions $4, 6, \ldots$, so the inclusion $S^2 \to \mathbb{CP}^\infty$ induces isomorphisms on π_i for $i \le 2$ (Exercise 11.12), and $\mathbb{CP}^\infty = K(\mathbb{Z}, 2)$ has no homotopy above degree 2 (Exercise 14.6). By the uniqueness in Theorem 13.16, $\mathbb{CP}^\infty$ is the second Postnikov stage of S^2.

$\mathbb{CP}^n$. The Hopf bundle $S^1 \to S^{2n+1} \xrightarrow{p} \mathbb{CP}^n$ is a fibration, and its long exact sequence, with $\pi_i(S^1) = 0$ for $i \ge 2$, shows that $p_* \colon \pi_i(S^{2n+1}) \to \pi_i(\mathbb{CP}^n)$ is an isomorphism for $i \ge 3$. Moreover S^{2n+1} is 2-connected ($2n+1 \ge 3$). So p is a map from a 2-connected space inducing

isomorphisms on π_i for $i \geq 3$: it has the defining property of the 2-connected cover, and by the uniqueness clause of Theorem 13.16 (in its dual form for connected covers, Remark 13.21) $\mathbb{CP}^n\langle 2\rangle \simeq S^{2n+1}$. (Equivalently, S^{2n+1} is the homotopy fibre of the second Postnikov stage $\mathbb{CP}^n \to \mathbb{CP}^\infty$, which is the inclusion, by the same argument as for S^2.)

Example 13.19 (A two-stage tower)**.** Let X be simply connected with $\pi_2 X = \pi_3 X = \mathbb{Z}$ and all other homotopy groups zero. Then $X^2 = K(\mathbb{Z}, 2) = \mathbb{CP}^\infty$, and $X = X^3$ sits in a fibration

$$K(\mathbb{Z}, 3) \longrightarrow X \longrightarrow \mathbb{CP}^\infty$$

classified by a single k-invariant

$$k \in H^4\big(\mathbb{CP}^\infty; \mathbb{Z}\big) = \mathbb{Z} \cdot \iota^2,$$

where $\iota \in H^2(\mathbb{CP}^\infty; \mathbb{Z})$ is the fundamental class and ι^2 its square. So such spaces are classified by an integer. The value $k = 0$ gives the product $\mathbb{CP}^\infty \times K(\mathbb{Z}, 3)$; the value $k = \iota^2$ gives the 3-type of S^2, and the appearance of a *cup square* as a k-invariant is the homotopy-theoretic shadow of the Hopf map.

Exercise 13.20 (medium)**.** Show that a two-stage Postnikov tower with vanishing k-invariant is a product of Eilenberg–MacLane spaces.

Hint. *The stage is the pullback of the path fibration along k. What is the pullback of a fibration along a constant map?*

Solution. By Theorem 13.16 the stage $X^n \to X^{n-1}$ is the pullback of the path fibration $PK(\pi, n+1) \to K(\pi, n+1)$ along k^n. If k^n is nullhomotopic the pullback is, up to homotopy, the pullback along a constant map, which is the product of the base with the fibre. With $X^{n-1} = K(\pi_{n-1}, n-1)$ that gives $X^n \simeq K(\pi_{n-1}, n-1) \times K(\pi_n, n)$.

Remark 13.21 (Manipulating the two truncations). Postnikov stages cut the homotopy groups off above a degree; connected covers cut them off below. Both are determined by universal properties — a map from X inducing isomorphisms in a range, respectively a map to X doing so — and both are unique up to homotopy under or over X by Theorem 13.16. Two such constructions applied in the two possible orders therefore have the same homotopy groups, namely those of X in a window and zero outside it; and by uniqueness each is characterised by a universal property that the other also satisfies.

Writing that argument out is the natural exercise here, and it is a good test of whether Warning 13.2 has been absorbed: isomorphic homotopy groups are not enough, so the argument must produce a comparison *map* and then invoke Theorem 13.1. The uniqueness clause of Theorem 13.16 is what supplies it.

13.4 What the invariants miss

Remark 13.22 (Spaces). Warning 13.2 is worth restating as a summary of the book so far. For $L(5, 1)$ and $L(5, 2)$:

- the fundamental groups agree (Example 2.30);
- the homology groups agree (Example 4.10);

- the cohomology rings agree (Example 7.18);
- all higher homotopy groups agree (Warning 13.2);

and the spaces are still different. Every invariant built so far is a functor applied to one space at a time; what distinguishes these two is how their cells are linked, which is visible only to an operation that mixes degrees. Chapter 18 constructs the first such operation.

Remark 13.23 (Maps). There is a second failure, one degree more subtle, and it concerns maps rather than spaces. A map $f\colon X \to Y$ induces f_* on homotopy groups and on homology. Both can vanish identically with f still essential.

The place to look for such an f is a two-stage Postnikov tower. If $\dim X \leq n$ then $[X, Y] \to [X, Y^n]$ is a bijection — by Exercise 11.12 and the extension lemma, nothing above stage n can be detected by a complex of dimension n — so classifying maps out of X reduces to classifying maps into a tower with two stages. Such a map is a cohomology class on X together with a lift, the obstruction to lifting is the k-invariant evaluated on that class, and the ambiguity in the lift is a quotient of a cohomology group of X. When the k-invariant is a cup square, as in Example 13.19, that obstruction is a quadratic condition, and quadratic conditions have solutions that linear invariants cannot see.

Carrying this out on a specific low-dimensional X is the natural place to test the chapter. Chapter 17 develops the machinery properly; what should be clear already is why π_* and H_* cannot be expected to suffice.

13.5 What to carry forward

- Whitehead's theorem needs a map (Theorem 13.1); Warning 13.2 is the standing counterexample to forgetting it.
- The homological form needs simple connectivity (Theorem 13.6), and the Poincaré homology sphere shows why (Warning 13.7).
- Work on the universal cover (Proposition 13.9); the right invariant is $H_*(\widetilde{X})$ as a $\mathbb{Z}[\pi_1 X]$-module (Remark 13.15) — the repair promised in Chapter 5.
- Postnikov towers and k-invariants (Theorem 13.16); the k-invariant measures non-splitting (Exercise 13.20), and a cup square as k-invariant is the Hopf map in disguise (Example 13.19).
- Neither π_* nor H_* classifies spaces (Remark 13.22) or maps (Remark 13.23).

Chapter 14

Fibrations and the first homotopy groups of spheres

A fibration converts a space into two simpler ones at the cost of a long exact sequence, and that trade is the only systematic access to higher homotopy groups we have. This chapter sets up the sequence, isolates the mechanism by which it splits, and runs it on the two bundles where the answer is a theorem rather than a tautology.

Everything here is preparation for Chapter 15, which replaces the long exact sequence by a spectral sequence and thereby handles fibrations whose fibre is not a sphere.

14.1 The long exact sequence

Definition 14.1. A map $p\colon E \to B$ is a *fibration* if it has the homotopy lifting property: every homotopy $X \times I \to B$ whose restriction to $X \times \{0\}$ lifts to E lifts entirely.

Fibre bundles over paracompact bases are fibrations, and that is how almost every example arises.

Serre versus Hurewicz. Definition 14.1 with X ranging over all spaces defines a *Hurewicz fibration*; with X ranging only over discs (equivalently CW complexes) it defines a *Serre fibration*. Every fibre bundle is a Serre fibration, with no hypothesis on the base, and everything in this chapter and the next needs only the Serre property. The point of the definition is that it makes fibres over different points of a path-connected base homotopy equivalent: lifting a path in the base transports one fibre to another. Over a loop this transport is a self-equivalence of the fibre, defined up to homotopy — the *monodromy*, which for a covering space is the action of Proposition 3.3 and for a mapping torus the map f itself (Exercise 15.15).

Every map is a fibration up to homotopy. For $f\colon A \to B$ let $E_f = \{(a,\gamma) \in A \times B^I : \gamma(0) = f(a)\}$ and $p(a,\gamma) = \gamma(1)$. Then $A \simeq E_f$ (shrink the paths), p is a fibration [1, Prop. 4.64], and p corresponds to f under this equivalence. Its fibre is the *homotopy fibre* of f, studied in Chapter 21. So the long exact sequence below applies to every map, with the homotopy fibre in place of the fibre.

Theorem 14.2 (Relative homotopy; recalled)**.** *Let $p\colon E \to B$ be a fibration with B path-connected, $b_0 \in B$, $F = p^{-1}(b_0)$ and $x_0 \in F$. Then*

$$p_*\colon \pi_n(E, F, x_0) \longrightarrow \pi_n(B, b_0)$$

is an isomorphism for all $n \geq 1$.

Corollary 14.3 (The long exact sequence)**.**

$$\cdots \to \pi_n(F) \xrightarrow{i_*} \pi_n(E) \xrightarrow{p_*} \pi_n(B) \xrightarrow{\partial} \pi_{n-1}(F) \to \cdots \to \pi_0(E).$$

Proof. Substitute Theorem 14.2 into the long exact sequence of the pair (E, F). □

Exercise 14.4 (easy)**.** Let $F \to E \to B$ be a fibration with E contractible. Show $\pi_n(B) \cong \pi_{n-1}(F)$ for all $n \geq 1$.

Solution. In Corollary 14.3 the terms $\pi_n(E)$ and $\pi_{n-1}(E)$ vanish, so exactness of $\pi_n(E) \to \pi_n(B) \xrightarrow{\partial} \pi_{n-1}(F) \to \pi_{n-1}(E)$ forces ∂ to be injective and surjective.

Example 14.5 (Path–loop)**.** For any based space B the path space PB of paths starting at b_0 is contractible, and evaluation at the endpoint is a fibration $PB \to B$ with fibre ΩB. By Exercise 14.4,

$$\pi_n(B) \cong \pi_{n-1}(\Omega B) \qquad (n \geq 1),$$

which is the statement that looping shifts homotopy groups down by one.

Exercise 14.6 (medium)**.** The unit sphere $S^\infty \subseteq \mathbb{C}^\infty$ is contractible, and the circle acts freely on it with quotient $\mathbb{CP}^\infty$. Deduce that $\mathbb{CP}^\infty$ is a $K(\mathbb{Z}, 2)$.

Hint. *A free action of a compact Lie group gives a fibre bundle, hence a fibration, with fibre the group.*

Solution. The action gives a fibre bundle $S^1 \to S^\infty \to \mathbb{CP}^\infty$, hence a fibration. Since S^∞ is contractible, Exercise 14.4 gives

$$\pi_n(\mathbb{CP}^\infty) \cong \pi_{n-1}(S^1).$$

Now $\pi_0(S^1) = 0$, $\pi_1(S^1) = \mathbb{Z}$ and $\pi_k(S^1) = 0$ for $k \geq 2$ — the last because the universal cover $\mathbb{R}$ is contractible and covers induce isomorphisms on π_k for $k \geq 2$ (Exercise 13.10). So $\pi_2(\mathbb{CP}^\infty) = \mathbb{Z}$ and all other homotopy groups vanish: $\mathbb{CP}^\infty$ is a $K(\mathbb{Z}, 2)$.

Exercise 14.7 (medium)**.** (a) Let $1 \to N \to G \to H \to 1$ be a short exact sequence of groups. Show that there is a map $K(G, 1) \to K(H, 1)$, inducing the given surjection on π_1, whose homotopy fibre is a $K(N, 1)$.

(b) Identify the homotopy fibre of a map $\mathbb{CP}^\infty \to \mathbb{CP}^\infty$ inducing multiplication by 2 on H^2.

Source: MIT OCW 18.906 (Spring 2020), Problem Set III, Problem 14 (part (a) and the example in (b)), reproduced in substance

Hint. *For (a), a homomorphism into π_1 of a $K(H, 1)$ is induced by a map from any CW complex with the given π_1 [1, Prop. 1B.9]. Then read off the homotopy groups of the fibre from the long exact sequence. For (b), the same.*

Solution. (a) Choose CW models of $K(G, 1)$ and $K(H, 1)$. Since $K(H, 1)$ has vanishing higher homotopy, the homomorphism $\pi_1(K(G, 1)) = G \to H$ is induced by a map $q \colon K(G, 1) \to K(H, 1)$ (extend over the cells of $K(G, 1)$ one at a time; the obstructions lie in $\pi_{k-1}(K(H, 1)) = 0$ for $k \geq 3$, and over 2-cells the relators are killed because the map is a homomorphism — the argument of Proposition 2.23). Replace q by a fibration with fibre F. The long exact sequence gives, for $i \geq 2$, $0 = \pi_{i+1}(K(H, 1)) \to \pi_i(F) \to \pi_i(K(G, 1)) = 0$, so $\pi_i(F) = 0$; and in low

degrees $0 = \pi_2(K(H,1)) \to \pi_1(F) \to G \to H \to \pi_0(F) \to 0$. Since $G \to H$ is onto, F is path-connected and $\pi_1(F) \cong \ker(G \to H) = N$. So F is a $K(N,1)$.

(b) Let $f\colon \mathbb{CP}^\infty \to \mathbb{CP}^\infty$ induce multiplication by 2 on H^2, hence on $\pi_2 \cong H_2$ (Hurewicz and the universal coefficient theorem). With F the homotopy fibre, the long exact sequence reads $0 \to \pi_2(F) \to \mathbb{Z} \xrightarrow{2} \mathbb{Z} \to \pi_1(F) \to 0$ and $\pi_i(F) = 0$ for $i \geq 3$ (both $\pi_i(\mathbb{CP}^\infty)$ and $\pi_{i+1}(\mathbb{CP}^\infty)$ vanish) and $\pi_0(F) = 0$. So $\pi_2(F) = \ker(2) = 0$ and $\pi_1(F) = \mathbb{Z}/2$: F is a $K(\mathbb{Z}/2,1)$, i.e. $F \simeq \mathbb{RP}^\infty$ (which is a $K(\mathbb{Z}/2,1)$ as the quotient of the contractible S^∞ by the free antipodal action). Geometrically: $S^1 \to S^1$, $z \mapsto z^2$, deloops to f, and its kernel $\mathbb{Z}/2$ deloops to the fibre.

Exercise 14.8 (easy)**.** Use homotopy groups to show that $\mathbb{RP}^n$ does not retract onto $\mathbb{RP}^k$ for $n > k > 0$.

SOURCE: [1, §4.2, Ex. 1], reproduced in substance

Solution. A retraction r would make $i_*\colon \pi_k(\mathbb{RP}^k) \to \pi_k(\mathbb{RP}^n)$ injective ($r_* i_* = \mathrm{id}$). For $k \geq 2$, $\pi_k(\mathbb{RP}^k) \cong \pi_k(S^k) \cong \mathbb{Z}$ while $\pi_k(\mathbb{RP}^n) \cong \pi_k(S^n) = 0$ since $k < n$ (Exercise 13.10 and cellular approximation). For $k = 1$, $\pi_1(\mathbb{RP}^1) = \mathbb{Z}$ while $\pi_1(\mathbb{RP}^n) = \mathbb{Z}/2$ for $n \geq 2$. In either case i_* is not injective.

14.2 When the sequence splits

A long exact sequence is most useful when it breaks into short pieces, and there are two standard hypotheses that make it do so. One is easy and is below; the other is harder and is the point of Remark 14.12.

Exercise 14.9 (medium)**.** Suppose the fibration $F \to E \xrightarrow{p} B$ admits a section, i.e. a map $s\colon B \to E$ with $p \circ s = \mathrm{id}_B$. Show that for $n \geq 2$

$$\pi_n(E) \cong \pi_n(F) \oplus \pi_n(B).$$

Hint. $p_* s_* = \mathrm{id}$ *already gives a splitting of one map in Corollary 14.3. What does that force* ∂ *to be?*

Solution. From $p \circ s = \mathrm{id}$ we get $p_* s_* = \mathrm{id}$ on $\pi_n(B)$, so p_* is surjective with a chosen splitting s_*. Exactness at $\pi_n(B)$ then makes $\partial = 0$, so the sequence breaks into

$$0 \to \pi_n(F) \xrightarrow{i_*} \pi_n(E) \xrightarrow{p_*} \pi_n(B) \to 0,$$

exact because ∂ vanishes on both sides. For $n \geq 2$ these are abelian groups and the sequence splits via s_*, giving $\pi_n(E) \cong \pi_n(F) \oplus \pi_n(B)$.

Warning 14.10 (Surjective does not mean split)*.* A short exact sequence of abelian groups splits when the right-hand term is free, and otherwise only when a splitting is produced. In Exercise 14.9 the splitting came from the section; with no section in sight, exhibiting one is work, and asserting it because a map happens to be surjective is an error.

Exercise 14.11 (medium)**.** The map $SO(n) \to S^{n-1}$ sending a matrix to its first column is a fibre bundle with fibre $SO(n-1)$. Show that it has a section if and only if S^{n-1} is parallelisable, i.e. TS^{n-1} is trivial. Decide the cases $n = 3$ and $n = 4$.

Source: MIT OCW 18.906 (Spring 2020), Problem Set I, Problem 4, reproduced in substance

Hint. *The columns $2, \dots, n$ of a matrix in $SO(n)$ with first column x are an orthonormal basis of $x^{\perp} = T_x S^{n-1}$.*

Solution. A section $s\colon S^{n-1} \to SO(n)$ assigns to x a matrix with first column x; its remaining columns $v_2(x), \dots, v_n(x)$ are orthonormal and orthogonal to x, i.e. a continuous orthonormal frame of $T_x S^{n-1}$, so TS^{n-1} is trivial. Conversely, if TS^{n-1} is trivial choose a frame and apply Gram–Schmidt (continuous in x) to get an orthonormal frame $v_2(x), \dots, v_n(x)$; the matrix $(x, v_2, \dots, v_n)$ is orthogonal with determinant ± 1, the sign being constant on the connected S^{n-1}; if it is -1 replace v_n by $-v_n$. This is a section.

$n = 3$: S^2 is not parallelisable (Exercise 12.13: it has no nowhere-zero vector field), so $SO(3) \to S^2$ has no section. $n = 4$: S^3 is the unit quaternions, and $x \mapsto (x, xi, xj, xk)$ is an orthonormal frame (left multiplication by a unit quaternion is an isometry of $\mathbb{H} = \mathbb{R}^4$), so $SO(4) \to S^3$ has a section, and by Exercise 14.9 $\pi_k(SO(4)) \cong \pi_k(SO(3)) \oplus \pi_k(S^3)$ for $k \geq 2$.

Remark 14.12 (The other splitting mechanism). The dual hypothesis to a section is that the *fibre inclusion* is nullhomotopic. It also collapses Corollary 14.3 into short exact sequences — $i_* = 0$ makes p_* injective and ∂ surjective — but now no splitting is handed over, and by Warning 14.10 one has to be built.

The construction is worth naming, because it is used again in Chapter 17. A nullhomotopy of a map $g\colon A \to E$ is exactly a map $CA \to E$ out of the cone extending g. So if $\alpha\colon S^{n-1} \to F$ is given and the inclusion $F \hookrightarrow E$ is nullhomotopic, composing the nullhomotopy with α produces a map of pairs

$$\left(D^n, S^{n-1}\right) \longrightarrow (E, F),$$

that is, a class in $\pi_n(E, F)$ — which Theorem 14.2 identifies with $\pi_n(B)$. Checking that this is a homomorphism, and that it is a section of ∂, is the natural exercise here, and it turns a structural hypothesis about the fibre into a direct sum decomposition of the homotopy groups of the base.

14.3 First computations

Example 14.13 (The Hopf bundle)**.** The Hopf map $p\colon S^3 \to S^2$, $(z_1, z_2) \mapsto [z_1 : z_2]$, is a fibre bundle with fibre S^1. Since $\pi_n(S^1) = 0$ for $n \geq 2$, the long exact sequence

$$\pi_n(S^1) \to \pi_n(S^3) \to \pi_n(S^2) \to \pi_{n-1}(S^1)$$

has vanishing outer terms for $n \geq 3$, giving

$$\pi_n(S^3) \cong \pi_n(S^2) \qquad (n \geq 3).$$

In particular $\pi_3(S^2) \cong \pi_3(S^3) = \mathbb{Z}$, generated by the Hopf map itself.

That is already a surprise worth registering: S^2 has nontrivial homotopy in a degree above its dimension, and it has infinitely many distinct homotopy classes of maps from S^3, while $H_3(S^2) = 0$ sees none of them.

Exercise 14.14 (easy)**.** Show that $\pi_n(T^k) = 0$ for all $n \geq 2$, and more generally that a space with contractible universal cover has vanishing higher homotopy groups.

Solution. The universal cover of T^k is $\mathbb{R}^k$, which is contractible. By Exercise 13.10, $\pi_n(T^k) \cong \pi_n(\mathbb{R}^k) = 0$ for $n \geq 2$. The same argument applies whenever the universal cover is contractible; such a space is a $K(\pi_1, 1)$.

Theorem 14.15 (Freudenthal; recalled)**.** *The suspension map $\pi_n(S^k) \to \pi_{n+1}(S^{k+1})$ is an isomorphism for $n < 2k-1$ and a surjection for $n = 2k-1$.*

Example 14.16 (The stable range)**.** Freudenthal makes $\pi_{n+k}(S^k)$ independent of k once $k > n+1$; the common value is the n-th *stable stem* π_n^s. The first few are

$$\pi_0^s = \mathbb{Z}, \qquad \pi_1^s = \mathbb{Z}/2, \qquad \pi_2^s = \mathbb{Z}/2, \qquad \pi_3^s = \mathbb{Z}/24.$$

Outside the stable range the groups are wilder and depend on k; the computation of Example 14.13 is unstable, $\pi_3(S^2)$ living well below the range where Freudenthal applies.

Exercise 14.17 (easy)**.** Granting $\pi_3(S^2) \cong \mathbb{Z}$ (Example 14.13), show that $\pi_{n+1}(S^n)$ is cyclic for every $n \geq 2$ and is independent of n for $n \geq 3$.

Source: original

Solution. Apply Theorem 14.15 to $\pi_{k+1}(S^k) \to \pi_{k+2}(S^{k+1})$: here the degree is $n = k+1$, and the map is an isomorphism when $k+1 < 2k-1$, i.e. $k \geq 3$, and a surjection when $k+1 = 2k-1$, i.e. $k = 2$. So $\pi_3(S^2) = \mathbb{Z} \to \pi_4(S^3)$ is surjective, making $\pi_4(S^3)$ cyclic, and $\pi_4(S^3) \cong \pi_5(S^4) \cong \pi_6(S^5) \cong \cdots$. All these groups are cyclic. (In fact they are $\mathbb{Z}/2$, generated by suspensions of the Hopf map, so the surjection $\mathbb{Z} \to \mathbb{Z}/2$ has a large kernel; proving this needs the tools of Chapter 18.)

Remark 14.18 (What the long exact sequence cannot do). Example 14.13 worked because the fibre was S^1, whose homotopy groups vanish above degree one, so two of every three terms in the sequence were zero. That is a special circumstance. For a fibration with fibre S^3, say, the outer terms do not vanish and the sequence determines its middle terms only up to extension — and in general not even that, since the maps are unknown.

There are two ways forward, and the book takes both. Chapter 17 keeps track of the ambiguity by organising it into cohomology classes. Chapter 15 replaces the long exact sequence by a spectral sequence, which computes the homology of the total space from that of the base and the fibre whatever they are, at the price of differentials that must themselves be determined.

Before leaving the long exact sequence it is worth noticing what it *can* do when a structural hypothesis is available: Exercise 14.9 and Remark 14.12 each convert a single geometric statement into a direct sum decomposition, and applied to the classical Hopf bundles $S^3 \to S^7 \to S^4$ and $S^7 \to S^{15} \to S^8$ they produce infinite cyclic summands in homotopy groups of spheres that no elementary argument reaches.

14.4 What to carry forward

- $\pi_n(E, F) \cong \pi_n(B)$ for a fibration (Theorem 14.2) — the source of the long exact sequence and of every splitting argument.
- Contractible total space turns the connecting map into an isomorphism (Exercise 14.4); this computes Ω and identifies $\mathbb{CP}^\infty = K(\mathbb{Z}, 2)$ (Example 14.5 and Exercise 14.6).
- A section splits the sequence (Exercise 14.9); a nullhomotopic fibre inclusion does too, but the splitting must be constructed from a cone (Warning 14.10 and Remark 14.12).

- $\pi_3(S^2) = \mathbb{Z}$ (Example 14.13); homotopy above the dimension is the normal state of affairs, and homology sees none of it.
- Freudenthal and the stable range (Theorem 14.15 and Example 14.16).

Part III

Spectral sequences, operations, and the rational beginning

Chapter 15

The Serre spectral sequence: setting it up and running it

Chapter 14 ended by admitting that the long exact sequence of a fibration computes almost nothing once the fibre is complicated. The replacement computes the homology of the total space from that of the base and the fibre, for any fibre at all — at the cost that the answer arrives in instalments, with differentials that have to be determined by other means.

Learning to use it is learning three habits. Draw the E^2-page and see which entries are forced to vanish. Decide what the answer *must* be, and read the differentials backwards from that. And keep the extension problem in view, since the spectral sequence computes an associated graded and not a group.

15.1 What a spectral sequence is

Before the Serre spectral sequence, the general notion — because every one of its features (pages, differentials, convergence, extension problems) comes from a single source: a filtered chain complex.

Definition 15.1. A *filtered chain complex* is a chain complex C_* with subcomplexes $\cdots \subseteq F_{p-1}C \subseteq F_pC \subseteq F_{p+1}C \subseteq \cdots$. It is *first quadrant* if $F_{-1}C = 0$, $F_pC_n = C_n$ for $p \geq n$, and $C_n = 0$ for $n < 0$. The filtration induces one on homology, $F_pH_n(C) = \operatorname{im}\big(H_n(F_pC) \to H_n(C)\big)$, and the *associated graded* groups are $F_pH_n/F_{p-1}H_n$.

The standard example is the skeletal filtration of a CW complex: $F_pC = C_*(X^p)$. Its homology is filtered, and $F_pH_n/F_{p-1}H_n$ measures the classes "born on the p-skeleton". For a fibration $p\colon E \to B$ over a CW complex the Serre spectral sequence comes from filtering E by the preimages $E_p = p^{-1}(B^p)$ of the skeleta of the base.

Theorem 15.2 (The spectral sequence of a filtered complex)**.** *A first-quadrant filtered chain complex determines groups $E^r_{p,q}$ for $r \geq 0$ and differentials $d^r\colon E^r_{p,q} \to E^r_{p-r,q+r-1}$ with $d^r \circ d^r = 0$ such that:*

(a) $E^0_{p,q} = F_pC_{p+q}/F_{p-1}C_{p+q}$ with d^0 induced by ∂, so $E^1_{p,q} = H_{p+q}(F_pC/F_{p-1}C)$, and d^1 is the connecting map of the triple (F_p, F_{p-1}, F_{p-2});

(b) $E^{r+1} = H(E^r, d^r)$: each page is the homology of the previous one;

(c) for fixed (p,q) the groups stabilise: $E^r_{p,q} = E^\infty_{p,q}$ for $r > \max(p, q+1)$;

(d) $E^{\infty}_{p,q} \cong F_pH_{p+q}(C)/F_{p-1}H_{p+q}(C)$.

A useful mental picture of $E^r_{p,q}$: chains c in filtration p whose boundary drops at least r steps in filtration, modulo boundaries coming from at most $r-1$ steps above; $d^r[c] = [\partial c]$, viewed in filtration $p-r$. A differential d^r records that a chain born in filtration p has boundary born in filtration $p-r$ — a "long" boundary that the cruder pages could not see.

Warning 15.3 (The three things to check). (1) *Pages*: what is E^2, exactly, including twisted coefficients? (2) *Differentials*: which can be nonzero for degree reasons, and what forces the others? (3) *Extensions*: E^{∞} gives the associated graded of H_n, so H_n is assembled from the $E^{\infty}_{p,n-p}$ by successive extensions $0 \to F_{p-1}H_n \to F_pH_n \to E^{\infty}_{p,n-p} \to 0$, each of which may or may not split. Skipping (3) is the commonest error; Warning 15.18 gives the standard example, and Example 15.13 below a fibration where the extension is genuinely non-split.

Exercise 15.4 (medium)**.** Let C be a first-quadrant filtered chain complex whose spectral sequence has $E^1 = 0$. Show directly that $H_*(C) = 0$: given a cycle $z \in F_sC_n$, construct a chain y with $\partial y = z$.

Source: MIT OCW 18.906 (Spring 2020), Problem Set V, Problem 23(a), reproduced in substance

Hint. *$E^1_{s,n-s} = H_n(F_sC/F_{s-1}C) = 0$, so z is a boundary modulo F_{s-1}: find $y_s \in F_sC_{n+1}$ with $z - \partial y_s \in F_{s-1}C_n$. Iterate.*

Solution. Let $z \in F_sC_n$ be a cycle. Its image in $F_sC/F_{s-1}C$ is a cycle in degree n, and $H_n(F_sC/F_{s-1}C) = E^1_{s,n-s} = 0$, so there is $y_s \in F_sC_{n+1}$ with $z - \partial y_s \in F_{s-1}C_n$. The element $z_1 = z - \partial y_s$ is again a cycle, now in $F_{s-1}C_n$. Repeating, we get $y_{s-1}, \dots, y_0$ with $z - \partial(y_s + \cdots + y_{s-k}) \in F_{s-k-1}C_n$; after $s+1$ steps it lies in $F_{-1}C_n = 0$. So $z = \partial(y_s + \cdots + y_0)$. Every cycle lies in some F_sC_n (first quadrant: $F_nC_n = C_n$), so $H_*(C) = 0$. The first-quadrant hypothesis is used twice: to start ($z \in F_s$ for some finite s) and to stop ($F_{-1} = 0$).

15.2 The machine

Theorem 15.5 (Serre; recalled)**.** *Let $F \to E \xrightarrow{p} B$ be a fibration with B path-connected. There is a first-quadrant spectral sequence with*

$$E^2_{p,q} = H_p\big(B;\, H_q(F)\big) \implies H_{p+q}(E),$$

with differentials $d^r \colon E^r_{p,q} \to E^r_{p-r,\,q+r-1}$, the coefficients being local if $\pi_1(B)$ acts non-trivially on $H_*(F)$. Convergence means: there is a filtration of $H_n(E)$ whose associated graded is $\bigoplus_{p+q=n} E^{\infty}_{p,q}$.

There is a cohomological version,

$$E_2^{p,q} = H^p\big(B;\, H^q(F)\big) \implies H^{p+q}(E),$$

with differentials $d_r \colon E_r^{p,q} \to E_r^{p+r,\,q-r+1}$.

Example 15.6 (The degenerate case)**.** For a trivial bundle $E = B \times F$ with B simply connected, the spectral sequence collapses at E^2: it must, since the answer is already known by Künneth to be $\bigoplus H_p(B) \otimes H_q(F)$ plus Tor terms, and that is the size of the E^2-page. Collapse is the normal state of affairs only for products; expecting it in general is the standard error, and Example 15.9 is the antidote.

The first real computation uses the shape of the page rather than any knowledge of the differentials: when only two columns are nonzero, everything is forced.

Example 15.7 (The homology of a loop space)**.** Apply Theorem 15.5 to the path–loop fibration $\Omega S^n \to PS^n \to S^n$ of Example 14.5, with $n \geq 2$. The base is simply connected, so the coefficients are untwisted, and $H_p(S^n)$ vanishes except for $p = 0, n$. So the E^2-page has only two nonzero columns, and the only possible differential is

$$d^n \colon E^n_{n,q} = A_q \longrightarrow E^n_{0,\,q+n-1} = A_{q+n-1}, \qquad A_q := H_q(\Omega S^n).$$

The total space is contractible, so E^∞ must vanish outside total degree 0, where it is $\mathbb{Z}$. In total degree $k > 0$ the only possible entries are $E_{0,k}$ and $E_{n,k-n}$; the first must be hit and the second must die, so every d^n is an isomorphism. Starting from $A_0 = \mathbb{Z}$ and iterating,

$$H_q(\Omega S^n) = \begin{cases} \mathbb{Z}, & q \equiv 0 \pmod{n-1}, \\ 0, & \text{otherwise.} \end{cases}$$

No differential was computed; the answer was forced by what had to vanish.

Warning 15.8. That is the characteristic move of the subject, and it is worth stating explicitly because it inverts the usual logic. One does not compute differentials and deduce the answer. One knows something about the answer — often only that certain groups vanish — and deduces the differentials. A reader waiting for a formula for d^r will wait indefinitely.

Example 15.9 (A spectral sequence that does not collapse)**.** Take the Hopf fibration $S^1 \to S^3 \to S^2$. The E^2-page is concentrated in columns $p = 0, 2$ and rows $q = 0, 1$, with $\mathbb{Z}$ in each of the four spots $(0,0), (2,0), (0,1), (2,1)$.

The total space is S^3, with homology $\mathbb{Z}, 0, 0, \mathbb{Z}$. In total degree 1 the sole entry is $E_{0,1} = \mathbb{Z}$, and it has to disappear. Just one differential is capable of removing it,

$$d^2 \colon E^2_{2,0} \longrightarrow E^2_{0,1},$$

which must therefore be surjective. Total degree 2 contains only $E_{2,0}$, which must also die, so d^2 must be injective. Hence d^2 is an isomorphism $\mathbb{Z} \to \mathbb{Z}$, and what survives is $E^\infty_{0,0} = \mathbb{Z}$ and $E^\infty_{2,1} = \mathbb{Z}$, giving $H_*(S^3) = (\mathbb{Z}, 0, 0, \mathbb{Z})$ as required.

Had one assumed collapse, the answer would have been $H_*(S^2 \times S^1)$, which is wrong. The Hopf fibration is not a product, and the spectral sequence says so by having a nonzero differential.

15.3 Local coefficients

When $\pi_1(B)$ acts nontrivially on $H_*(F)$ the E^2-page is homology with local coefficients, and ignoring this gives wrong answers. We first say what the coefficients are, then run the smallest example — one the book has already computed twice by other means — in full detail.

Definition 15.10. Let B be path-connected with basepoint b_0, $\pi = \pi_1(B, b_0)$. A *local coefficient system* on B is an abelian group M with an action of π, i.e. a $\mathbb{Z}[\pi]$-module. (Equivalently, a functor from the fundamental groupoid $\Pi_1(B)$ of Remark 1.2

to abelian groups.) If B has a universal cover $\widetilde{B}$, the *homology and cohomology with local coefficients* are

$$H_*(B;M) = H_*\big(C_*(\widetilde{B}) \otimes_{\mathbb{Z}[\pi]} M\big), \qquad H^*(B;M) = H^*\big(\operatorname{Hom}_{\mathbb{Z}[\pi]}(C_*(\widetilde{B}), M)\big),$$

where π acts on the chains of $\widetilde{B}$ by deck transformations (as a right action via $g \mapsto g^{-1}$ where needed). For a CW complex one may use the cellular chains of $\widetilde{B}$, a complex of free $\mathbb{Z}[\pi]$-modules with one generator for each cell of B.

If π acts trivially, $C_*(\widetilde{B}) \otimes_{\mathbb{Z}[\pi]} M = C_*(B) \otimes M$ and one recovers ordinary homology with coefficients in M. The cases needed below:

- $B = S^1$, $\pi = \mathbb{Z} = \langle t \rangle$: $\widetilde{B} = \mathbb{R}$ has one 0-cell and one 1-cell per integer, and $C_*(\mathbb{R})$ is $\mathbb{Z}[t^{\pm 1}] \xrightarrow{t-1} \mathbb{Z}[t^{\pm 1}]$. Hence, for a module M on which t acts by φ,

  $$H_0(S^1;M) = M/(1-\varphi)M = M_\pi \text{ (coinvariants)},$$
  $$H_1(S^1;M) = \ker(1-\varphi) = M^\pi \text{ (invariants)},$$

  and dually $H^0(S^1;M) = M^\pi$, $H^1(S^1;M) = M_\pi$.
- In general $H_0(B;M) = M_\pi$ and $H^0(B;M) = M^\pi$.

Theorem 15.11 (Serre spectral sequence with local coefficients; recalled)**.** *For a fibration $F \to E \to B$ with B a path-connected CW complex, $\pi_1(B)$ acts on $H_q(F)$ by transport of fibres along loops (the monodromy), making $H_q(F)$ a local coefficient system $\mathcal{H}_q(F)$, and there is a spectral sequence as in Theorem 15.5 with $E^2_{p,q} = H_p(B;\mathcal{H}_q(F)) \Rightarrow H_{p+q}(E)$. The filtration of $H_n(E)$ is $F_pH_n = \operatorname{im}\big(H_n(p^{-1}(B^p)) \to H_n(E)\big)$.*

Example 15.12 (The Klein bottle, a third time)**.** We compute $H_*(K;\mathbb{Z})$ from a fibration, making every step explicit.

Step 1: the fibration. The Klein bottle is the mapping torus of the reflection $r\colon S^1 \to S^1$, $r(z) = \bar{z}$ (Exercise 2.16): $K = S^1 \times [0,1]/(z,0) \sim (\bar{z},1)$. The projection $(z,s) \mapsto e^{2\pi i s}$ is a fibre bundle $S^1 \to K \to S^1$, hence a fibration, with fibre $F = S^1$ over the base point.

Step 2: the monodromy. By Exercise 15.15, the generator t of $\pi_1(S^1)$ acts on $H_*(F)$ through r_*. On $H_0(F) = \mathbb{Z}$ it acts trivially (as every map does on H_0 of a connected space); on $H_1(F) = \mathbb{Z}$ it acts by $\deg r = -1$. So the local systems are $\mathcal{H}_0 = \mathbb{Z}$ (trivial) and $\mathcal{H}_1 = \mathbb{Z}^-$, the group $\mathbb{Z}$ with t acting by -1.

Step 3: the E^2-page. Using the formulas for $B = S^1$:

$q=1$	$H_0(S^1;\mathbb{Z}^-) = \mathbb{Z}/(1-(-1))\mathbb{Z} = \mathbb{Z}/2$	$H_1(S^1;\mathbb{Z}^-) = \ker(\times 2) = 0$
$q=0$	$H_0(S^1;\mathbb{Z}) = \mathbb{Z}$	$H_1(S^1;\mathbb{Z}) = \mathbb{Z}$
	$p=0$	$p=1$

All other entries vanish: $p \geq 2$ because S^1 has no cells there, $q \geq 2$ because $H_q(S^1) = 0$.

Step 4: differentials. d^r has bidegree $(-r, r-1)$ with $r \geq 2$; it moves at least two columns to the left. With only the columns $p = 0, 1$ occupied, every d^r has zero source or zero target. So $E^2 = E^\infty$. (This is not luck: any fibration over S^1 has a two-column spectral sequence, Exercise 15.19.)

Step 5: filtration and extension. In total degree 1 the filtration of $H_1(K)$ has two steps:

$$0 \subseteq F_0H_1 = E^\infty_{0,1} = \mathbb{Z}/2 \subseteq F_1H_1 = H_1(K), \qquad H_1(K)/F_0H_1 \cong E^\infty_{1,0} = \mathbb{Z}.$$

So $0 \to \mathbb{Z}/2 \to H_1(K) \to \mathbb{Z} \to 0$ is exact. The extension problem is solved by algebra: $\mathbb{Z}$ is free, so the sequence splits and $H_1(K) \cong \mathbb{Z} \oplus \mathbb{Z}/2$. Here F_0H_1 is the image of H_1 of the fibre, and the quotient $\mathbb{Z}$ is detected by the projection to the base. In total degree 2 the only possible entries are $E^\infty_{1,1} = 0$ and $E^\infty_{2,0} = E^\infty_{0,2} = 0$, so $H_2(K) = 0$; and $H_0(K) = \mathbb{Z}$. Hence

$$H_0(K) = \mathbb{Z}, \qquad H_1(K) \cong \mathbb{Z} \oplus \mathbb{Z}/2, \qquad H_2(K) = 0,$$

agreeing with the Δ-complex computation of Example 4.2 and the Mayer–Vietoris computation of Example 5.16.

Where the torsion came from, and why it is easy to miss. Every input of the computation is torsion-free: the base is S^1, the fibre is S^1. The $\mathbb{Z}/2$ is a group of *coinvariants*: the fibre class $[F]$ and its image under the monodromy, $-[F]$, become equal in K (a loop carrying the fibre once around the base turns it over), so $2[F] = 0$. It lives in E^2 because of the local coefficients and nowhere else; no differential and no extension produces it. A reader who forgets that $\pi_1(B)$ acts and writes the untwisted $E^2_{p,q} = H_p(S^1; H_q(S^1))$ gets $\mathbb{Z}$ in all four spots and the answer $H_* = (\mathbb{Z}, \mathbb{Z}^2, \mathbb{Z})$ — the homology of the torus, which is the mapping torus of the identity. Conversely, the vanishing of $H_2(K)$ is also a local-coefficients phenomenon: $E^2_{1,1} = H_1(S^1; \mathbb{Z}^-)$ consists of the *invariant* fibre classes, and there are none, which is the homological shadow of non-orientability.

Division of labour. The local coefficients produced both the $\mathbb{Z}/2$ in $E^2_{0,1}$ and the vanishing of $E^2_{1,1}$. The spectral sequence had no differentials to contribute. The filtration and the extension problem contributed the identification of H_1 as an extension of $\mathbb{Z}$ by $\mathbb{Z}/2$, which splits for the purely algebraic reason that $\mathbb{Z}$ is free — Example 15.13 shows that in general it need not.

Example 15.13 (A non-split extension)**.** The central extension $0 \to \mathbb{Z}/2 \to \mathbb{Z}/4 \to \mathbb{Z}/2 \to 0$ gives a fibration $K(\mathbb{Z}/2,1) \to K(\mathbb{Z}/4,1) \to K(\mathbb{Z}/2,1)$ (Exercise 14.7), i.e. $\mathbb{RP}^\infty \to L \to \mathbb{RP}^\infty$ with $L = K(\mathbb{Z}/4,1)$ an infinite lens space. The monodromy is conjugation in $\mathbb{Z}/4$, which is trivial (the group is abelian), so the coefficients are untwisted. In total degree 1 the E^2-page has $E^2_{1,0} = H_1(\mathbb{RP}^\infty) = \mathbb{Z}/2$ and $E^2_{0,1} = H_1(\mathbb{RP}^\infty) = \mathbb{Z}/2$. No differential can enter or leave these spots except $d^2\colon E^2_{2,0} \to E^2_{0,1}$, and $E^2_{2,0} = H_2(\mathbb{RP}^\infty) = 0$. So E^∞ has two copies of $\mathbb{Z}/2$ in total degree 1. But $H_1(L) = \pi_1(L)^{\mathrm{ab}} = \mathbb{Z}/4$: the extension $0 \to \mathbb{Z}/2 \to H_1(L) \to \mathbb{Z}/2 \to 0$ is the non-split one. The spectral sequence by itself cannot decide this; here the answer is known from the group, and in general one needs extra input such as a Bockstein (Theorem 9.22: a nonzero β_2 detects $\mathbb{Z}/4$).

Exercise 15.14 (easy)**.** Compute $H_*(S^1; \mathbb{Z}^-)$ and $H^*(S^1; \mathbb{Z}^-)$, and $H_*(S^1; M)$ for $M = \mathbb{F}_2$ with the (necessarily trivial) twisted action. Use the results to compute $H^*(K; \mathbb{Z})$ and $H_*(K; \mathbb{F}_2)$ from the Serre spectral sequence of the Klein bottle, and compare with Exercises 6.4 and 6.5.

SOURCE: first part: [1, §3.H, Ex. 1], reproduced in substance; the Klein bottle part is original

Solution. With t acting by -1: $H_0(S^1; \mathbb{Z}^-) = \mathbb{Z}/(1-(-1)) = \mathbb{Z}/2$, $H_1(S^1; \mathbb{Z}^-) = \ker(\times 2) = 0$; dually $H^0(S^1; \mathbb{Z}^-) = (\mathbb{Z}^-)^\pi = \ker(1-(-1)) = 0$ and $H^1(S^1; \mathbb{Z}^-) = \mathbb{Z}/2$. Over $\mathbb{F}_2$, $-1 = 1$, so the action is trivial and $H_*(S^1; \mathbb{F}_2) = (\mathbb{F}_2, \mathbb{F}_2)$.

Cohomology Serre spectral sequence of K: $E_2^{p,q} = H^p(S^1; \mathcal{H}^q(F))$ with $\mathcal{H}^0 = \mathbb{Z}$, $\mathcal{H}^1 = \mathbb{Z}^-$ (the monodromy acts on $H^1(F) = \mathrm{Hom}(H_1F, \mathbb{Z})$ by the transpose, again -1). So $E_2^{0,0} = \mathbb{Z}$, $E_2^{1,0} = \mathbb{Z}$, $E_2^{0,1} = 0$, $E_2^{1,1} = \mathbb{Z}/2$; no differentials (two columns), no extension problems (each total degree has one nonzero entry). Hence $H^*(K; \mathbb{Z}) = (\mathbb{Z}, \mathbb{Z}, \mathbb{Z}/2)$, matching Exercise 6.4.

Homology with $\mathbb{F}_2$ coefficients: all four entries $E^2_{p,q}$, $p,q \in \{0,1\}$, are $\mathbb{F}_2$; no differentials; extensions of $\mathbb{F}_2$-vector spaces split. So $H_*(K;\mathbb{F}_2) = (\mathbb{F}_2, \mathbb{F}_2^2, \mathbb{F}_2)$, matching Exercise 6.5 (over a field homology and cohomology have the same dimensions).

Exercise 15.15 (medium)**.** Let $T_f \to S^1$ be the mapping torus fibration of Definition 2.9. Show that the action of $\pi_1(S^1)$ on $H_*(F)$ is generated by f_*.

Hint. *Transport the fibre once around the base circle. In the mapping torus, going once around t is exactly applying f.*

Solution. The action of a loop γ in the base on $H_*(F)$ is induced by the homotopy equivalence of F obtained by lifting γ to a homotopy of the fibre. For T_f with the convention of Definition 2.9, the loop t lifts to the homotopy $\varphi_u(x) = (x,u)$ carrying the fibre at the seam to itself, and at $u = 1$ the identification $(x,0) \sim (f(x),1)$ makes that self-map f. Hence t acts by f_*, and $\pi_1(S^1) = \langle t \rangle$ acts through the powers of f_*.

Remark 15.16 (When the action is trivial)*.* Three common situations give untwisted coefficients: B simply connected; F having homology on which every automorphism in the image of the monodromy acts trivially; and, for an aspherical fibration coming from a group extension, the case where the monodromy — which is conjugation — is inner-trivial. That last case is worth working out for oneself, since the monodromy of a fibration $K(A,1) \to K(B,1) \to K(C,1)$ can be identified with the conjugation action of B on A by exactly the transport argument of Exercise 15.15, and the condition on the extension that makes it trivial is then a purely group-theoretic one.

15.4 Forcing differentials, and the extension problem

When the groups are finite, counting orders replaces knowing maps.

Exercise 15.17 (easy)**.** Suppose all groups $E^\infty_{p,q}$ with $p+q = n$ are finite. Show $|H_n(E)| = \prod_{p+q=n} |E^\infty_{p,q}|$.

Solution. Convergence gives a finite filtration $0 = F_{-1} \subseteq F_0 \subseteq \cdots \subseteq F_n = H_n(E)$ with $F_p/F_{p-1} \cong E^\infty_{p,n-p}$. For finite groups the order is multiplicative in short exact sequences, so $|H_n(E)| = \prod_p |F_p/F_{p-1}|$.

Warning 15.18 (E^∞ is not the answer)*.* The spectral sequence computes the associated graded of a filtration, not the group. Two groups with the same associated graded need not agree: filter $\mathbb{Z}/4$ by $\{0\} \subseteq \mathbb{Z}/2 \subseteq \mathbb{Z}/4$ and the associated graded is $\mathbb{Z}/2 \oplus \mathbb{Z}/2$. So an E^∞-page consisting of two copies of $\mathbb{Z}/2$ in one total degree is consistent with $H_n(E)$ being either $\mathbb{Z}/2 \oplus \mathbb{Z}/2$ or $\mathbb{Z}/4$, and the spectral sequence alone cannot decide.

The ambiguity is real and it is often where the interesting content sits: a non-split extension in E^∞ is the spectral sequence's way of recording that the total space is glued together in a way the pieces do not see. When the answer is known independently — as for a fibration of Eilenberg–MacLane spaces coming from a group extension — the comparison runs the other way, and the extension problem is solved by the group theory rather than by the topology.

Exercise 15.19 (medium)**.** Let $F \to E \to S^1$ be a fibration over the circle. Show that the Serre spectral sequence collapses at E^2 and reduces to short exact sequences

$$0 \to H_0\big(S^1; H_n(F)\big) \to H_n(E) \to H_1\big(S^1; H_{n-1}(F)\big) \to 0,$$

and identify these with the Wang sequence of Proposition 5.21.

Hint. *How many nonzero columns does the E^2-page have? And for a local system M over S^1 with monodromy φ, what are $H_0(S^1;M)$ and $H_1(S^1;M)$?*

Solution. $H_p(S^1;-)$ vanishes for $p \geq 2$, so only the columns $p = 0, 1$ survive. Every differential d^r with $r \geq 2$ has source or target in a vanishing column, so $E^2 = E^\infty$, and the two-step filtration of $H_n(E)$ gives the displayed short exact sequence.

For a local system M over S^1 with monodromy φ, homology is computed by the complex $M \xrightarrow{1-\varphi} M$, so $H_0 = \operatorname{coker}(1-\varphi)$ and $H_1 = \ker(1-\varphi)$. By Exercise 15.15 the monodromy of a mapping torus is f_*, so the sequence reads

$$0 \to \operatorname{coker}\left(1 - f_* \text{ on } H_n(F)\right) \to H_n(E) \to \ker\left(1 - f_* \text{ on } H_{n-1}(F)\right) \to 0,$$

which is (5.1).

Remark 15.20 (How a computation of this kind is organised). For a fibration whose total space has known homology — typically because base, fibre and total space are all classifying spaces of groups in a short exact sequence — the procedure is fixed:

(i) decide whether the coefficients are twisted (Remark 15.16);

(ii) write out the E^2-page from the homology of the base and fibre;

(iii) compare $\prod |E^2_{p,q}|$ in each total degree with the known $|H_n(E)|$ using Exercise 15.17;

(iv) where the orders exceed the target, differentials must act, and their ranks are determined by how much has to be killed;

(v) where the orders match, no differential acts — but the extension problem remains, and Warning 15.18 says it must be settled separately.

Step (v) is the one readers skip. A total degree in which the orders already agree is precisely a total degree in which the spectral sequence has told you nothing about how the group is assembled.

Exercise 15.21 (medium)**.** Use the homology Serre spectral sequence of $S^1 \to S^{2n+1} \to \mathbb{CP}^n$ to compute $H_*(\mathbb{CP}^n;\mathbb{Z})$, given only that $\mathbb{CP}^n$ is simply connected and that $H_k(\mathbb{CP}^n) = 0$ for $k > 2n$ (it is a $2n$-dimensional CW complex).

SOURCE: original

Hint. *The page has two rows, $q = 0, 1$, both equal to $H_*(\mathbb{CP}^n)$, and only $d^2\colon E^2_{p,0} \to E^2_{p-2,1}$ can act. Show by induction on p that d^2 is an isomorphism for $2 \leq p \leq 2n$.*

Solution. $\mathbb{CP}^n$ is simply connected, so $E^2_{p,q} = H_p(\mathbb{CP}^n) \otimes H_q(S^1)$ with no twisting and no Tor term ($H_q(S^1)$ is free): two identical rows $q = 0, 1$ equal to $A_p := H_p(\mathbb{CP}^n)$. The only possible differential is $d^2\colon A_p = E^2_{p,0} \to E^2_{p-2,1} = A_{p-2}$, and $E^3 = E^\infty$. The total space S^{2n+1} has $H_k = 0$ for $0 < k < 2n+1$.

Total degree 1: $E^\infty_{1,0} = A_1$ and $E^\infty_{0,1} = \operatorname{coker}(d^2\colon A_2 \to A_0)$ must vanish. $A_1 = 0$ (simply connected), and $d^2\colon A_2 \to A_0 = \mathbb{Z}$ must be onto. Total degree 2: $E^\infty_{2,0} = \ker(d^2|A_2)$ and $E^\infty_{1,1} = \operatorname{coker}(A_3 \to A_1) = 0$; so $d^2\colon A_2 \to \mathbb{Z}$ is injective, hence an isomorphism, $A_2 \cong \mathbb{Z}$. Inductively, for $2 \leq k \leq 2n$, total degree $k-1$ forces $\operatorname{coker}(d^2\colon A_k \to A_{k-2}) = 0$ and total degree k forces $\ker(d^2\colon A_k \to A_{k-2}) = 0$; so $A_k \cong A_{k-2}$. Starting from $A_0 = \mathbb{Z}$, $A_1 = 0$ we get $A_k = \mathbb{Z}$ for k even $\leq 2n$ and 0 for k odd. In total degree $2n+1$ the surviving class is $E^\infty_{2n,1} = \operatorname{coker}\left(d^2\colon A_{2n+2} \to A_{2n}\right) = A_{2n} = \mathbb{Z}$, since $A_{2n+2} = 0$: it gives $H_{2n+1}(S^{2n+1}) = \mathbb{Z}$, as it must. So $H_*(\mathbb{CP}^n) = \mathbb{Z}$ in even degrees $\leq 2n$ and 0 otherwise.

Exercise 15.22 (medium)**.** Let $F \to E \to S^n$ be a fibration with $n \geq 2$. Derive from the Serre spectral sequence a long exact sequence

$$\cdots \to H_k(F) \xrightarrow{i_*} H_k(E) \to H_{k-n}(F) \xrightarrow{\tau} H_{k-1}(F) \xrightarrow{i_*} H_{k-1}(E) \to \cdots$$

(the homology Wang sequence for a fibration over a sphere). What changes for $n = 1$?

Source: MIT OCW 18.906 (Spring 2020), Problem Set V, Problem 24(a), reproduced in substance

Hint. *For $n \geq 2$ the base is simply connected; the only nonzero columns are $p = 0$ and $p = n$, both equal to $H_*(F)$, and the only possible differential is d^n.*

Solution. For $n \geq 2$, S^n is simply connected, so $E^2_{p,q} = H_p(S^n; H_q(F))$, which is $H_q(F)$ for $p = 0, n$ and 0 otherwise (UCT, $H_*(S^n)$ being free). The only differential that can be nonzero is $\tau = d^n \colon E^n_{n,q} = H_q(F) \to E^n_{0,q+n-1} = H_{q+n-1}(F)$, and $E^{n+1} = E^\infty$, with $E^\infty_{0,k} = \operatorname{coker}\big(\tau \colon H_{k-n+1}(F) \to H_k(F)\big)$ and $E^\infty_{n,k-n} = \ker\big(\tau \colon H_{k-n}(F) \to H_{k-1}(F)\big)$. The filtration of $H_k(E)$ has two steps: $0 \to E^\infty_{0,k} \to H_k(E) \to E^\infty_{n,k-n} \to 0$, the first map being induced by i (the bottom of the filtration is the image of the fibre). Splicing these short exact sequences with the exact sequences $0 \to \ker\tau \to H_{k-n}(F) \xrightarrow{\tau} H_{k-1}(F) \to \operatorname{coker}\tau \to 0$ gives the stated long exact sequence (as in the derivation of the Gysin sequence, Exercise 16.12). For $n = 1$ the base is not simply connected and the E^2-page involves local coefficients; there are no differentials, and the result is the Wang sequence with $1 - t$ in place of τ (Exercise 15.19).

Exercise 15.23 (hard)**.** Let $f \colon \mathbb{CP}^\infty \to \mathbb{CP}^\infty$ induce multiplication by 2 on H^2, and let $\mathbb{RP}^\infty \to E \xrightarrow{p} \mathbb{CP}^\infty$ be the fibration obtained by turning f into a fibration (its fibre is $\mathbb{RP}^\infty$ by Exercise 14.7). Determine the behaviour of the homology Serre spectral sequence of p: the E^2-page, all nonzero differentials, and E^∞.

Source: MIT OCW 18.906 (Spring 2020), Problem Set V, Problem 24(c), reproduced in substance

Hint. *$E \simeq \mathbb{CP}^\infty$ and p corresponds to f, which induces multiplication by 2^k on H_{2k}. The edge map $H_{2k}(E) \to E^\infty_{2k,0} \subseteq E^2_{2k,0} = H_{2k}(\mathbb{CP}^\infty)$ is p_*.*

Solution. E^2. The base is simply connected, so $E^2_{p,q} = H_p(\mathbb{CP}^\infty; H_q(\mathbb{RP}^\infty))$. With $H_q(\mathbb{RP}^\infty) = \mathbb{Z},\ \mathbb{Z}/2,\ 0$ for $q = 0$, q odd, $q > 0$ even, and $H_p(\mathbb{CP}^\infty; G) = G$ for p even, 0 for p odd (UCT; $H_*(\mathbb{CP}^\infty)$ is free), the nonzero entries are $E^2_{2j,0} = \mathbb{Z}$ and $E^2_{2j,q} = \mathbb{Z}/2$ for q odd.

The answer. $E \simeq \mathbb{CP}^\infty$, so $H_n(E) = \mathbb{Z}$ for n even, 0 for n odd. In odd total degree $2k - 1$ the page contains the k groups $E^2_{2j,2k-1-2j} = \mathbb{Z}/2$, $0 \leq j \leq k-1$, all of which must die. In even total degree $2k$ the only entry is $E^2_{2k,0} = \mathbb{Z}$, since $2j + q$ with q odd is odd.

Differentials. A $\mathbb{Z}/2$ at $(2j, q)$ (q odd) cannot support a nonzero differential: d^r maps it to $(2j - r, q + r - 1)$, in even total degree $2j + q - 1$ and positive row, where everything is zero. So each $\mathbb{Z}/2$ must be hit, and the only entries in total degree $2k$ are in the bottom row: the $\mathbb{Z}/2$ at $(2j, 2k - 1 - 2j)$ must be hit by $d^{2k-2j} \colon E^{2k-2j}_{2k,0} \to E^{2k-2j}_{2j,2k-2j-1}$. So the bottom class at $(2k, 0)$ supports exactly k nonzero differentials, $d^2, d^4, \ldots, d^{2k}$, each surjective onto a $\mathbb{Z}/2$; each replaces the current subgroup of $E^2_{2k,0} = \mathbb{Z}$ by its kernel, of index 2. Hence $E^\infty_{2k,0} = 2^k\mathbb{Z} \subseteq \mathbb{Z}$, all $\mathbb{Z}/2$'s die, and E^∞ is $2^k\mathbb{Z}$ at $(2k, 0)$ and 0 elsewhere.

Consistency. The edge map $H_{2k}(E) \to E^\infty_{2k,0} \subseteq H_{2k}(\mathbb{CP}^\infty)$ is $p_* = f_*$. Since $f^*(c) = 2c$ and f^* is a ring map, $f^*(c^k) = 2^k c^k$, so f_* on $H_{2k} = \operatorname{Hom}(H^{2k}, \mathbb{Z})$ is multiplication by 2^k: its image is $2^k\mathbb{Z}$, exactly $E^\infty_{2k,0}$.

15.5 Serre classes

Often one wants a conclusion only up to some class of groups that is being ignored — finite groups, torsion groups, finitely generated groups. The axioms that make such ignoring legitimate are these.

Definition 15.24. Call a class $\mathcal{C}$ of abelian groups a *Serre class* if it contains the zero group and is closed under subgroups, quotients and extensions — all three at once, which is to say that in any exact sequence $0 \to A \to B \to C \to 0$ the middle group belongs to $\mathcal{C}$ exactly when the two outer ones do.

Exercise 15.25 (easy)**.** Show that the finitely generated abelian groups form a Serre class, and that the free abelian groups do not.

Solution. Let $0 \to A \to B \to C \to 0$ be exact. If B is finitely generated then so is C, being a quotient, and so is A, being a subgroup of a finitely generated abelian group. Conversely if A and C are finitely generated, lift generators of C and adjoin generators of A to generate B. So the class is a Serre class.

Free abelian groups do not form a Serre class: in $0 \to \mathbb{Z} \xrightarrow{2} \mathbb{Z} \to \mathbb{Z}/2 \to 0$ the middle term $\mathbb{Z}$ is free but its quotient $\mathbb{Z}/2$ is not, so the class is not closed under quotients (the "only if" direction of Definition 15.24 fails).

Exercise 15.26 (easy)**.** Show that each of the following is a Serre class: (a) finite abelian groups; (b) torsion abelian groups; (c) for a prime p, torsion abelian groups in which every element has order prime to p.

SOURCE: original

Solution. Let $0 \to A \to B \to C \to 0$ be exact.

(a) If B is finite, so are $A \subseteq B$ and $C = B/A$; if A and C are finite, $|B| = |A||C|$ is finite.

(b) A subgroup or quotient of a torsion group is torsion. If A and C are torsion and $b \in B$, the image of b in C is killed by some m, so $mb \in A$ is killed by some n, and $nmb = 0$.

(c) The same argument: the orders m and n above are prime to p, hence so is nm; subgroups and quotients only lower orders (the order of an image divides the order of the element).

Remark 15.27 (What they are for)*.* The mod-$\mathcal{C}$ Hurewicz theorem says that if X is simply connected and $\pi_i(X) \in \mathcal{C}$ for $i < n$, then $H_i(X) \in \mathcal{C}$ for $i < n$ and the Hurewicz map $\pi_n(X) \to H_n(X)$ is an isomorphism modulo $\mathcal{C}$. With $\mathcal{C}$ the finitely generated groups this yields, via the Serre spectral sequence applied to Postnikov towers, that the homotopy groups of a simply connected finite complex are all finitely generated — a theorem with no elementary proof, and one that Chapter 14 could not have reached.

Verifying the axioms for a proposed class is routine and worth doing once for each class one intends to use; the only trap is the two-sidedness noted in Exercise 15.25.

15.6 What to carry forward

- $E^2_{p,q} = H_p(B; H_qF) \Rightarrow H_{p+q}(E)$ (Theorem 15.5); two nonzero columns force everything (Example 15.7).
- Read differentials backwards from what must vanish (Warning 15.8); do not expect collapse (Example 15.9).
- Twisted coefficients are not optional (Example 15.12); the monodromy of a mapping torus is f_* (Exercise 15.15).
- E^∞ gives the associated graded, not the group (Warning 15.18); order counting (Exercise 15.17) is the standard way to force differentials.
- The Wang sequence is the case of a circle base (Exercise 15.19).

- Serre classes let whole families of groups be ignored (Definition 15.24 and Remark 15.27).

Chapter 16

Multiplicative structure: Leray–Hirsch, Gysin, Thom

The cohomology Serre spectral sequence carries a multiplication, and that changes what it can do. Degree bookkeeping now constrains not only which groups appear but which products are possible, and a single differential propagates through a whole ring by the Leibniz rule. Most of the classical computations in this subject are one differential plus multiplicativity.

The chapter then extracts three named tools from the machine — Leray–Hirsch, Gysin, Thom — and ends with the invariant that measures how far a map $S^{2n-1} \to S^n$ is from being inessential.

16.1 Products

Theorem 16.1 (Multiplicativity; recalled)**.** *For a fibration $F \to E \to B$ with B path-connected and untwisted coefficients, the cohomology Serre spectral sequence is multiplicative: each E_r is a bigraded ring, each differential is a derivation,*

$$d_r(xy) = d_r(x)\, y + (-1)^{|x|} x\, d_r(y),$$

the product on E_{r+1} is induced from that on E_r, and the product on $E_2 = H^*(B; H^*(F))$ is the cup product on the base tensored with that on the fibre. The filtration of $H^*(E)$ is multiplicative, and E_∞ is its associated graded ring.

The last sentence contains the standing caveat: E_∞ gives the associated graded *ring*, so products landing in higher filtration are invisible.

Example 16.2 (The cohomology of $K(\mathbb{Z}, 2)$)**.** Run the machine backwards: base unknown, fibre and total space known. Take the path–loop fibration

$$K(\mathbb{Z}, 1) \longrightarrow PK(\mathbb{Z}, 2) \longrightarrow K(\mathbb{Z}, 2),$$

whose fibre is S^1 and whose total space is contractible. Write $B = K(\mathbb{Z}, 2)$, which is simply connected, so coefficients are untwisted. The E_2-page has two rows, $q = 0$ and $q = 1$, with $E_2^{p,q} = H^p(B) \otimes H^q(S^1)$.

Everything except $E_\infty^{0,0} = \mathbb{Z}$ must die. The only differential available is $d_2 \colon E_2^{p,1} \to E_2^{p+2,0}$. Let x generate $H^1(S^1) = \mathbb{Z}$, sitting in $E_2^{0,1}$, and put

$$c = d_2(x) \in H^2(B).$$

This d_2 must be injective on $E_2^{0,1}$ and onto $E_2^{2,0}$, so c generates $H^2(B) \cong \mathbb{Z}$.

Now multiplicativity does the rest. Elements of the top row are $x \cdot b$ with $b \in H^p(B)$, and since $d_2 b = 0$,

$$d_2(x\,b) = d_2(x)\,b = c\,b.$$

So $d_2 \colon E_2^{p,1} \to E_2^{p+2,0}$ is multiplication by c, and it must be an isomorphism for every $p \geq 0$ — injective so the top row dies, surjective so the bottom row dies above degree 0. Hence multiplication by c is an isomorphism $H^p(B) \to H^{p+2}(B)$ for all $p \geq 0$, and starting from $H^0(B) = \mathbb{Z}$, $H^1(B) = 0$ we get

$$H^*\big(K(\mathbb{Z},2);\mathbb{Z}\big) = \mathbb{Z}[c], \qquad |c| = 2.$$

Combined with Exercise 14.6, this is the computation of $H^*(\mathbb{CP}^\infty)$ — now derived rather than quoted.

Exercise 16.3 (medium)**.** Compute $H^*(\Omega S^{2n+1};\mathbb{Q})$ as a ring, and say what changes over $\mathbb{Z}$.

Hint. *Use the path–loop fibration over S^{2n+1}. Only one differential can act; call the transgressing class x and apply the Leibniz rule to x^k.*

Solution. The fibration $\Omega S^{2n+1} \to P \to S^{2n+1}$ has contractible total space and simply connected base. The base has cohomology only in degrees 0 and $2n{+}1$, so only d_{2n+1} can be nonzero. By Example 15.7 the fibre has homology $\mathbb{Q}$ in degrees $2nk$; let x generate $H^{2n}(\Omega S^{2n+1};\mathbb{Q})$, and let ι generate $H^{2n+1}(S^{2n+1};\mathbb{Q})$, with $d_{2n+1}(x) = \iota$.

By the Leibniz rule, $d_{2n+1}(x^k) = k\,x^{k-1}\iota$. Over $\mathbb{Q}$ this is an isomorphism from the class x^k in column 0 onto $x^{k-1}\iota$ in column $2n+1$ for every $k \geq 1$, so everything above total degree 0 cancels, as it must. Hence $x^k \neq 0$ for all k, and

$$H^*\big(\Omega S^{2n+1};\mathbb{Q}\big) = \mathbb{Q}[x], \qquad |x| = 2n.$$

Over $\mathbb{Z}$ the factor k is an obstruction: the differential is multiplication by k, so cancellation forces x^k to be $k!$ times a generator. The integral cohomology is the divided power algebra on x, additively $\mathbb{Z}$ in each degree $2nk$ but with $x^k = k!\,\gamma_k$.

16.2 Leray–Hirsch

Theorem 16.4 (Leray–Hirsch; recalled)**.** *Let $F \to E \xrightarrow{p} B$ be a fibration with B path-connected, and suppose each $H^k(F;R)$ is a finitely generated free R-module. If there are classes $c_1,\dots,c_m \in H^*(E;R)$ whose restrictions i^*c_j form a basis of $H^*(F;R)$, then $H^*(E;R)$ is a free module over $H^*(B;R)$ with basis $c_1,\dots,c_m$; equivalently*

$$H^*(F;R) \otimes_R H^*(B;R) \longrightarrow H^*(E;R), \qquad i^*c_j \otimes b \longmapsto c_j \smile p^*b,$$

is an isomorphism of $H^*(B;R)$-modules.

Warning 16.5. This is an isomorphism of *modules*, not of rings. It says nothing about the products $c_i c_j$, which have to be computed separately and which are exactly where the interesting information sits. A bundle whose Leray–Hirsch decomposition looks like a product can have a ring structure that is not the tensor product, and comparing two bundles with the same module structure and different rings is a standard way to produce spaces with identical cohomology groups and different homotopy type.

Exercise 16.6 (medium)**.** Let $S^1 \to S^3 \xrightarrow{h} S^2$ be the Hopf fibration and let $S^1 \to E_k \xrightarrow{p_k} S^2$ be its pullback along a map $g\colon S^2 \to S^2$ of degree $k \geq 0$. Use the cohomology Serre spectral sequence to show that $H^*(E_k;\mathbb{Z})$ is $\mathbb{Z}, 0, \mathbb{Z}/k, \mathbb{Z}$ in degrees $0,1,2,3$ for $k \geq 1$, and $\mathbb{Z},\mathbb{Z},\mathbb{Z},\mathbb{Z}$ for $k = 0$. Conclude that the E_k are pairwise non-homotopy-equivalent, although they all have the same fibre and base and the same E_2-page.

SOURCE: original (replaces an exercise of the first draft of this edition that reproduced a webapp question)

Hint. *The bundle map $F\colon E_k \to S^3$ covering g is the identity on fibres, so it induces a map of spectral sequences which is an isomorphism on the fibre column; compare d_2 on $E_2^{0,1}$.*

Solution. For any oriented circle bundle over S^2 the base is simply connected, so $E_2^{p,q} = H^p(S^2) \otimes H^q(S^1)$, nonzero only for $p \in \{0,2\}$, $q \in \{0,1\}$; the only possible differential is $d_2\colon E_2^{0,1} = \mathbb{Z}e \to E_2^{2,0} = \mathbb{Z}u$. For the Hopf fibration $H^1(S^3) = H^2(S^3) = 0$, so this d_2 is an isomorphism: $d_2 e = \pm u$. The map of spectral sequences induced by F is the identity on $E_2^{0,1}$ (the fibres are identified) and g^* on the base, so by naturality $d_2(F^*e) = g^*(d_2 e) = \pm g^* u = \pm k\,u$. Hence for E_k, d_2 is multiplication by $\pm k$ on $\mathbb{Z}$. If $k \geq 1$: $E_\infty^{0,1} = \ker d_2 = 0$, $E_\infty^{2,0} = \mathbb{Z}/k$, and $E_\infty^{2,1} = E_2^{2,1} = \mathbb{Z}$ (no differential can enter or leave it), giving $\mathbb{Z}, 0, \mathbb{Z}/k, \mathbb{Z}$; there are no extension problems since each total degree has a single nonzero term. If $k = 0$ the map g may be taken constant, the pullback is the product $S^2 \times S^1$, and $d_2 = 0$. The groups $H^2(E_k) = \mathbb{Z}/k$ ($k \geq 1$) are pairwise non-isomorphic, so the E_k are pairwise non-homotopy-equivalent. (In fact E_k is the lens space $L(k,1)$; this is not needed.)

Exercise 16.7 (medium)**.** Apply Theorem 16.4 to the bundle $S^1 \to S^\infty/\mathbb{Z}_p \to \mathbb{CP}^\infty$ (where $\mathbb{Z}_p \subseteq S^1$ acts on $S^\infty \subseteq \mathbb{C}^\infty$ by scalars) to compute the ring $H^*(K(\mathbb{Z}/p,1);\mathbb{F}_p)$ for a prime p.

SOURCE: [1, §4.D, Ex. 2], reproduced in substance

Hint. *The fibre is $S^1/\mathbb{Z}_p \cong S^1$. Show that restriction $H^1(S^\infty/\mathbb{Z}_p;\mathbb{F}_p) \to H^1(S^1;\mathbb{F}_p)$ is an isomorphism using π_1.*

Solution. $L = S^\infty/\mathbb{Z}_p$ is a $K(\mathbb{Z}/p,1)$ (S^∞ contractible, free action), and the map $L \to S^\infty/S^1 = \mathbb{CP}^\infty$ is a bundle with fibre $S^1/\mathbb{Z}_p \cong S^1$. The long exact sequence $\pi_1(S^1) \to \pi_1(L) \to \pi_1(\mathbb{CP}^\infty) = 0$ shows that the fibre inclusion is onto $\pi_1(L) = \mathbb{Z}/p$. Hence on $H^1(-;\mathbb{F}_p) = \mathrm{Hom}(\pi_1,\mathbb{F}_p)$ the restriction $\mathrm{Hom}(\mathbb{Z}/p,\mathbb{F}_p) \to \mathrm{Hom}(\mathbb{Z},\mathbb{F}_p)$ is injective between groups of order p, hence an isomorphism. Let $x \in H^1(L;\mathbb{F}_p)$ restrict to a generator. By Theorem 16.4, $H^*(L;\mathbb{F}_p)$ is free over $H^*(\mathbb{CP}^\infty;\mathbb{F}_p) = \mathbb{F}_p[c]$ on $1, x$: with $y = p^*c$, the elements y^k and xy^k form a basis, one in each degree.

Products: y^k and xy^k are what they say. For x^2: if p is odd, graded commutativity gives $x^2 = -x^2$, so $2x^2 = 0$ and $x^2 = 0$. Hence $H^*(K(\mathbb{Z}/p,1);\mathbb{F}_p) = \Lambda(x) \otimes \mathbb{F}_p[y]$, $|x| = 1$, $|y| = 2$, for p odd. For $p = 2$, $L = \mathbb{RP}^\infty$ and $x^2 = y$ (Example 9.20: $x^2 = \beta x \neq 0$ spans H^2), giving $\mathbb{F}_2[x]$. (For p odd one also has $y = \beta x$ up to a unit, by the argument of Example 9.19 applied to the 2-skeleton, a Moore space $M(\mathbb{Z}/p,1)$.)

Exercise 16.8 (easy)**.** Show that Theorem 16.4 does not apply to the Hopf fibration $S^1 \to S^3 \to S^2$, and reconcile this with Example 15.9.

Solution. One would need some $c \in H^1(S^3)$ whose restriction generates $H^1(S^1) = \mathbb{Z}$. Since $H^1(S^3) = 0$, there is no candidate at all. Consistently, Example 15.9 found $d^2 \neq 0$: in the language of the spectral sequence, the generator of $E_2^{0,1}$ is not a permanent cycle, and Leray–Hirsch is exactly the hypothesis that makes the whole fibre row consist of permanent cycles.

Example 16.9 (The projective bundle formula)**.** Let $E \to B$ be a complex vector bundle of rank n and let $\mathbb{P}(E) \to B$ be its projectivisation, with fibre $\mathbb{CP}^{n-1}$. The tautological line bundle on $\mathbb{P}(E)$ has first Chern class $x \in H^2(\mathbb{P}(E);\mathbb{Z})$ restricting on each fibre to the generator of $H^2(\mathbb{CP}^{n-1})$. Then $1, x, \dots, x^{n-1}$ restrict to a basis of $H^*(\mathbb{CP}^{n-1};\mathbb{Z})$, so by Theorem 16.4

$$H^*\big(\mathbb{P}(E);\mathbb{Z}\big) \text{ is free over } H^*(B;\mathbb{Z}) \text{ on } 1, x, \dots, x^{n-1}.$$

The ring structure is then determined by the single relation expressing x^n in terms of the lower powers, and its coefficients are by definition the Chern classes of E. Chapter 26 takes that as the definition.

Exercise 16.10 (easy)**.** Check Example 16.9 in the case $B = \mathrm{pt}$.

Solution. With B a point, $E = \mathbb{C}^n$ and $\mathbb{P}(E) = \mathbb{CP}^{n-1}$, and $H^*(B) = \mathbb{Z}$ concentrated in degree 0. The theorem says $H^*(\mathbb{CP}^{n-1})$ is free on $1, x, \dots, x^{n-1}$ over $\mathbb{Z}$, which together with $x^n = 0$ — there being nothing in $H^{2n}(\mathbb{CP}^{n-1})$ — gives $\mathbb{Z}[x]/(x^n)$.

16.3 Gysin and Thom

A sphere bundle has a fibre with cohomology in only two degrees, so its spectral sequence has two rows and collapses into a long exact sequence.

Theorem 16.11 (Gysin sequence)**.** *Let $E \to B$ be an oriented real vector bundle of rank k over a path-connected base, with sphere bundle $S(E) \xrightarrow{p} B$ and Euler class $e \in H^k(B;\mathbb{Z})$. There is a long exact sequence*

$$\cdots \to H^{n-k}(B) \xrightarrow{\smile e} H^n(B) \xrightarrow{p^*} H^n\big(S(E)\big) \longrightarrow H^{n-k+1}(B) \xrightarrow{\smile e} H^{n+1}(B) \to \cdots$$

Exercise 16.12 (medium)**.** Derive Theorem 16.11 from the Serre spectral sequence of the sphere bundle, and identify the Euler class in spectral-sequence terms.

Hint. *The fibre is S^{k-1}, so only the rows $q = 0$ and $q = k-1$ are nonzero, and only d_k can act. Write out the two-row page and see what the exactness of a two-step filtration says.*

Solution. With fibre S^{k-1} the E_2-page has $E_2^{p,0} = H^p(B)$ and $E_2^{p,k-1} = H^p(B)$, all other rows zero. The only possible differential is

$$d_k \colon E_k^{p,k-1} \longrightarrow E_k^{p+k,0},$$

and by multiplicativity it is determined by its value on the fibre generator $u \in E_2^{0,k-1}$: writing $e := d_k(u) \in H^k(B)$, the Leibniz rule gives $d_k(u\,b) = e\,b$, so d_k is cup product with e.

Then $E_\infty^{p,0} = \operatorname{coker}(\smile e)$ and $E_\infty^{p,k-1} = \ker(\smile e)$, and the two-step filtration of $H^n(S(E))$ gives short exact sequences

$$0 \to \operatorname{coker}\Big(H^{n-k}(B) \xrightarrow{\smile e} H^n(B)\Big) \to H^n\big(S(E)\big) \to \ker\Big(H^{n-k+1}(B) \xrightarrow{\smile e} H^{n+1}(B)\Big) \to 0,$$

which splice into the displayed long exact sequence. The Euler class is the transgression of the fibre's fundamental class.

Example 16.13 (Projective space by Gysin)**.** Apply Theorem 16.11 to $S^1 \to S^{2n+1} \to \mathbb{CP}^n$, the sphere bundle of the tautological complex line bundle, so $k = 2$ and $e = c \in H^2(\mathbb{CP}^n)$. Since $H^m(S^{2n+1}) = 0$ for $0 < m < 2n+1$, exactness forces

$$\smile c\colon H^{m-2}(\mathbb{CP}^n) \longrightarrow H^m(\mathbb{CP}^n)$$

to be an isomorphism for $0 < m \leq 2n$. Starting from $H^0 = \mathbb{Z}$ gives $H^*(\mathbb{CP}^n;\mathbb{Z}) = \mathbb{Z}[c]/(c^{n+1})$, and the truncation appears exactly where the sphere's top class interrupts the sequence.

Exercise 16.14 (medium)**.** Let M be a closed $2n$-manifold admitting a fibre bundle $S^1 \to S^{2n+1} \to M$. Show that M is simply connected and that $H^*(M;\mathbb{Z}) \cong H^*(\mathbb{CP}^n;\mathbb{Z})$ as rings.

Source: [1, §4.D, Ex. 6], first half, reproduced in substance

Hint. *Use the long exact homotopy sequence for π_1, then argue exactly as in Example 16.13, with the Euler class of the associated (oriented) plane bundle.*

Solution. From $\pi_1(S^{2n+1}) = 0 \to \pi_1(M) \to \pi_0(S^1) = 0$ (exact as pointed sets, $n \geq 1$) M is simply connected, and $H^k(M) = 0$ for $k > 2n$ (a closed $2n$-manifold has the homotopy type of a $2n$-dimensional CW complex; or use duality). Since $\pi_1(M) = 0$, the monodromy on $H^*(S^1)$ is trivial and the argument of Exercise 16.12 applies verbatim to this fibration: the cohomology spectral sequence has two rows, the only differential is $d_2 = \smile e$ for the transgression $e = d_2(u) \in H^2(M)$ of the fibre generator, and one obtains the Gysin sequence of Theorem 16.11 with $k = 2$. Now use $H^m(S^{2n+1}) = 0$ for $0 < m < 2n+1$. Exactness of $H^{m-2}(M) \xrightarrow{\smile e} H^m(M) \to H^m(S^{2n+1})$ for $1 \leq m \leq 2n$ shows that $\smile e$ is onto in that range, and exactness of $H^{m-1}(S^{2n+1}) \to H^{m-2}(M) \xrightarrow{\smile e} H^m(M)$ for $2 \leq m \leq 2n+1$ shows that it is injective. Starting from $H^0(M) = \mathbb{Z}$ and $H^{-1} = 0$: $H^{2k}(M) = \mathbb{Z}e^k$ and $H^{2k+1}(M) = 0$ for $2k \leq 2n$, while $e^{n+1} \in H^{2n+2}(M) = 0$. So $H^*(M) = \mathbb{Z}[e]/(e^{n+1}) \cong H^*(\mathbb{CP}^n)$ as rings.

Theorem 16.15 (Thom isomorphism; recalled)**.** *For an oriented rank-k vector bundle $E \to B$ there is a class $u \in H^k\big(\mathrm{Th}(E)\big)$, the Thom class, such that*

$$H^n(B) \xrightarrow{\cong} \widetilde{H}^{\,n+k}\big(\mathrm{Th}(E)\big), \qquad b \longmapsto p^*b \smile u,$$

is an isomorphism for all n. The Euler class is the restriction of u to the zero section.

Thom spaces. For a real vector bundle $E \to B$ of rank k with a metric, the *Thom space* is $\mathrm{Th}(E) = D(E)/S(E)$, the unit disc bundle with the unit sphere bundle collapsed to a point. For compact B it is the one-point compactification of the total space. Over a point it is S^k, and in general each fibre contributes a sphere S^k, all of them sharing the basepoint. A *Thom class* is $u \in \widetilde{H}^k(\mathrm{Th}(E)) = H^k(D(E), S(E))$ restricting on each fibre to a generator of $H^k(D^k, S^{k-1})$; it exists over $\mathbb{Z}$ iff the bundle is orientable, and always over $\mathbb{F}_2$. The Euler class is the image of u under $H^k(D(E), S(E)) \to H^k(D(E)) \cong H^k(B)$, which is why it measures the failure of the zero section to be pushed off itself: a nowhere-zero section makes the zero section disjoint from a nearby section, and the Euler class vanishes.

Exercise 16.16 (easy)**.** Show that for the product bundle $B \times \mathbb{R}^n$ over a CW complex B the Thom space is the n-fold reduced suspension $\Sigma^n(B_+)$, where B_+ is B with a disjoint basepoint, and that the Thom isomorphism is the suspension isomorphism $H^i(B) = \widetilde{H}^i(B_+) \cong \widetilde{H}^{n+i}(\Sigma^n B_+)$.

Source: [1, §4.D, Ex. 8], reproduced in substance

Solution. $D(E) = B \times D^n$ and $S(E) = B \times S^{n-1}$, so $\mathrm{Th}(E) = B \times D^n / B \times S^{n-1}$. Since $D^n/S^{n-1} = S^n$ with basepoint the image of S^{n-1}, this is $(B \times S^n)/(B \times \{*\})$, which equals $B_+ \wedge S^n = \Sigma^n(B_+)$ (in $B_+ \wedge S^n = (B_+ \times S^n)/(B_+ \vee S^n)$ the extra point of B_+ times S^n and $B \times \{*\}$ are collapsed together). A Thom class is the pullback $u = \mathrm{pr}^*\sigma$ of the generator σ of $\widetilde{H}^n(S^n)$, and $b \mapsto p^*b \smile u$ is the cross product with σ, $b \mapsto b \times \sigma$, which is the suspension isomorphism $\widetilde{H}^i(B_+) \to \widetilde{H}^{i+n}(B_+ \wedge S^n)$ (the reduced cross product with the generator of $\widetilde{H}^n(S^n)$).

16.4 The Hopf invariant

Definition 16.17. Let $f\colon S^{2n-1} \to S^n$ with $n \geq 2$, and let $C_f = S^n \cup_f e^{2n}$ be its mapping cone. Then $H^*(C_f;\mathbb{Z})$ is $\mathbb{Z}$ in degrees $0, n, 2n$; choose generators $\alpha \in H^n$ and $\beta \in H^{2n}$. The *Hopf invariant* $H(f)$ is the integer with

$$\alpha \smile \alpha = H(f)\,\beta.$$

Example 16.18 (The Hopf map)**.** For the Hopf map $p\colon S^3 \to S^2$ the mapping cone is $\mathbb{CP}^2$, since attaching a 4-cell to $S^2 = \mathbb{CP}^1$ along p is precisely how $\mathbb{CP}^2$ is built. By Example 16.13, $H^*(\mathbb{CP}^2) = \mathbb{Z}[c]/(c^3)$ and c^2 generates H^4, so $H(p) = \pm 1$.

Exercise 16.19 (medium)**.** Show that $H(\Sigma g) = 0$ for every $g\colon S^{2n-2} \to S^{n-1}$; that is, the Hopf invariant vanishes on suspensions.

Hint. *What is the mapping cone of a suspension, and what do cup products of positive-degree classes do on a suspension?*

Solution. The mapping cone of Σg is ΣC_g. On a suspension every product of positive-degree reduced classes vanishes: the reduced diagonal $\Sigma Z \to \Sigma Z \wedge \Sigma Z$ is nullhomotopic, so the cup product of two reduced classes is zero. Hence $\alpha^2 = 0$ in $H^*(\Sigma C_g)$ and $H(\Sigma g) = 0$.

Exercise 16.20 (medium)**.** Show that $H(f) = 0$ whenever n is odd.

Solution. Graded commutativity gives $\alpha \smile \alpha = (-1)^{n\cdot n}\alpha \smile \alpha$, which for n odd reads $\alpha^2 = -\alpha^2$, so $2\alpha^2 = 0$. But α^2 lies in $H^{2n}(C_f) \cong \mathbb{Z}$, which is torsion-free, so $\alpha^2 = 0$ and $H(f) = 0$.

Exercise 16.21 (medium)**.** For maps $S^{2n-1} \xrightarrow{f} S^{2n-1} \xrightarrow{g} S^n$ show $H(g \circ f) = (\deg f)\, H(g)$, and for $S^{2n-1} \xrightarrow{f} S^n \xrightarrow{g} S^n$ show $H(g \circ f) = (\deg g)^2\, H(f)$. Deduce that if some $f_0\colon S^{2n-1} \to S^n$ has $H(f_0) \neq 0$, then every multiple of $H(f_0)$ is a Hopf invariant and $\pi_{2n-1}(S^n)$ is infinite.

Source: [1, §4.B, Ex. 1] (first two statements), reproduced in substance; the deduction is original

Hint. *Build maps of mapping cones that are the identity on one cell and have the relevant degree on the other.*

Solution. Write $C_h = S^n \cup_h e^{2n}$ with generators $\alpha_h \in H^n$, $\beta_h \in H^{2n}$ and $\alpha_h^2 = H(h)\beta_h$.

First formula. Define $\phi\colon C_{gf} \to C_g$ as the identity on S^n and, on the $2n$-cell, as the cone on f: $D^{2n} = C(S^{2n-1}) \xrightarrow{Cf} C(S^{2n-1}) = D^{2n}$ followed by the characteristic map of C_g. This is compatible with the attaching maps ($g \circ f$ on the boundary). On cellular cochains ϕ is the identity in degree n and multiplication by $\deg f$ in degree $2n$, so $\phi^*\alpha_g = \alpha_{gf}$ and $\phi^*\beta_g = (\deg f)\beta_{gf}$. Hence $H(gf)\beta_{gf} = \alpha_{gf}^2 = \phi^*(\alpha_g^2) = H(g)\phi^*\beta_g = (\deg f)H(g)\beta_{gf}$.

Second formula. Define $\psi\colon C_f \to C_{gf}$ as g on S^n and the identity on the $2n$-cell; compatible since the attaching map of C_{gf} is $g \circ f$. Then $\psi^*\alpha_{gf} = (\deg g)\alpha_f$ and $\psi^*\beta_{gf} = \beta_f$. So $H(gf)\beta_f = \psi^*(\alpha_{gf}^2) = (\deg g)^2\alpha_f^2 = (\deg g)^2 H(f)\beta_f$.

Deduction. With f_0 fixed and $f = f_0 \circ k$, $k\colon S^{2n-1} \to S^{2n-1}$ of degree m (Exercise 4.25), $H(f) = mH(f_0)$. If $H(f_0) \neq 0$ these are infinitely many distinct values, and the Hopf invariant is a homotopy invariant, so $\pi_{2n-1}(S^n)$ contains infinitely many elements; by Exercise 16.20 this can happen only for n even. (That H is a homomorphism on $\pi_{2n-1}(S^n)$, so that the image is a subgroup, is also true, [1, §4.B].)

Remark 16.22 (What the three tools do together). The tools of this chapter interlock in a way worth naming, because the combination is the standard route to the hardest classical results.

Suppose one is given a fibre bundle whose fibre, total space and base are all spheres. Theorem 16.11 constrains the dimensions immediately: the sequence relates the cohomology of the base to that of the total space through cup product with the Euler class, and a sphere has cohomology in only two degrees, so most of the maps are forced to be zero or isomorphisms and the possible dimensions are pinned down by arithmetic alone.

Then Theorem 16.15 converts the geometry into the Hopf invariant: the mapping cone of the bundle projection is the Thom space of the associated vector bundle, so the Thom class and the Euler class identify the cup square of Definition 16.17 with a characteristic class of the bundle. That identification turns "which sphere bundles over spheres exist" into "which Hopf invariants occur".

At which point the answer is not elementary: Adams proved that $H(f) = \pm 1$ is possible only for $n = 1, 2, 4, 8$. The book stops there and says so. What it does supply is everything up to that boundary — and Example 16.18 shows the boundary is attained.

16.5 What to carry forward

- The spectral sequence is a ring and its differentials are derivations (Theorem 16.1); one differential plus Leibniz computes a whole ring (Example 16.2 and Exercise 16.3).
- Leray–Hirsch (Theorem 16.4) gives a module isomorphism and never a ring isomorphism (Warning 16.5); what it assumes is that the fibre row consists of permanent cycles (Exercise 16.8).
- The projective bundle formula (Example 16.9) is where Chern classes come from.
- Gysin is the two-row spectral sequence, with the Euler class as the transgression (Exercise 16.12); Thom converts bundles into cones (Theorem 16.15).
- The Hopf invariant (Definition 16.17) is unstable (Exercise 16.19) and vanishes in odd degrees (Exercise 16.20); it equals ± 1 for the Hopf map (Example 16.18).

Chapter 17

Obstruction theory

17.1 The problem

Three questions recur throughout topology.

- *Extension.* Given a CW pair (W, A) and a map $f\colon A \to Y$, does f extend to W?
- *Classification.* Given two extensions, are they homotopic rel A? More ambitiously: what is the set $[W, Y]$ of homotopy classes of maps?
- *Lifting.* Given a fibration $p\colon E \to B$ and a map $g\colon W \to B$, is there a lift $\tilde{g}\colon W \to E$ with $p\tilde{g} = g$? The special case $g = \mathrm{id}_B$ asks for a section of p; for instance, a nowhere-zero vector field is a section of the unit tangent bundle.

The extension lemma (Theorem 11.15) answered the first question when every relevant homotopy group of Y vanishes. Obstruction theory keeps track of what happens when they do not: it attaches to each stage of a cell-by-cell construction a *cohomology class* whose vanishing is exactly the condition for the construction to continue, after possibly changing what was built at the previous stage. Its two main outputs in this book are the classification theorem $[X, K(G, n)] \cong H^n(X; G)$, on which the theory of cohomology operations (Chapter 18) rests, and the interpretation of characteristic classes as obstructions (Chapter 26).

Prerequisites and notation. CW complexes and cellular (co)homology with coefficients (Chapters 4 and 6); relative CW complexes; homotopy groups, the action of π_1 on π_n and the Hurewicz theorem (Chapter 13, Theorem 13.5); the extension lemma and cellular approximation (Chapter 11); fibrations and their long exact sequence (Chapter 14); Postnikov towers (Theorem 13.16). For a relative CW complex (W, A) we write $X_n = W^{(n)} = A \cup W^n$ for the relative n-skeleton, and $C^k(W, A; G)$ for the cellular cochains: functions from the oriented relative k-cells to G.

Definition 17.1. A path-connected space Y is *n-simple* if $\pi_1(Y)$ acts trivially on $\pi_n(Y)$ (for $n = 1$: $\pi_1(Y)$ is abelian), and *simple* if it is n-simple for all $n \geq 1$.

Simply connected spaces are simple, and so are H-spaces (by the argument of Exercise 1.30, extended to π_n), in particular topological groups and Eilenberg–MacLane spaces $K(G, n)$ with G abelian. The point of the hypothesis is the following.

Lemma 17.2. *If Y is n-simple, the map $\pi_n(Y, y_0) \to [S^n, Y]$ forgetting basepoints is a bijection, and for any two points y_0, y_1 the isomorphism $\beta_\gamma \colon \pi_n(Y, y_1) \to \pi_n(Y, y_0)$ is independent of the path γ. Hence "the homotopy class of a map $S^n \to Y$" is a well-defined element of a single group $\pi_n(Y)$.*

Proof. As for $n = 1$ (Chapter 1), two based maps are freely homotopic iff they differ by the action of an element of $\pi_1(Y, y_0)$, and every free homotopy class contains based maps (the higher-dimensional version of Lemma 1.16; [1, §4.A]). Under n-simplicity the action is trivial, giving the bijection; two paths from y_0 to y_1 differ by a loop, which acts trivially. □

Throughout this chapter Y is path-connected and, when the obstruction lives in $\pi_n(Y)$, n-simple. Without this hypothesis everything remains true with *local coefficients* (Remark 17.27).

17.2 The obstruction cocycle

Construction 17.3 (The obstruction cochain)**.** Let (W, A) be a relative CW complex and $f \colon X_n \to Y$ a map, $n \geq 1$. For each relative $(n+1)$-cell e with characteristic map $\Phi_e \colon D^{n+1} \to W$ and attaching map $\varphi_e = \Phi_e|_{S^n} \colon S^n \to X_n$, put

$$c(f)(e) = \big[f \circ \varphi_e\big] \in \pi_n(Y),$$

well defined by Lemma 17.2. Reversing the orientation of e precomposes φ_e with a reflection and negates the class, so $c(f)$ is a cellular cochain $c(f) \in C^{n+1}(W, A; \pi_n Y)$, the *obstruction cochain* of f.

Lemma 17.4. *$c(f) = 0$ if and only if f extends to a map $X_{n+1} \to Y$.*

Proof. A map extends over a single cell attached along φ_e iff $f \circ \varphi_e \colon S^n \to Y$ extends over D^{n+1}, iff it is freely nullhomotopic (a map out of the cone on S^n is a nullhomotopy), iff its class in $\pi_n(Y)$ is zero (Lemma 17.2). Extensions over different cells are independent, and the CW topology makes the result continuous. □

The cochain is the naive answer to "does f extend?". The theory begins with the observation that it has cohomological structure.

Proposition 17.5. *$c(f)$ is a cocycle: $\delta c(f) = 0$ in $C^{n+2}(W, A; \pi_n Y)$.*

Proof. We express $c(f)$ through homotopy groups. The pair (X_{n+1}, X_n) is n-connected [1, Cor. 4.12], so by the relative Hurewicz theorem [1, Thm. 4.37] the Hurewicz map

$$h \colon \pi_{n+1}(X_{n+1}, X_n) \longrightarrow H_{n+1}(X_{n+1}, X_n) = C_{n+1}(W, A)$$

is surjective, and becomes an isomorphism after dividing out the action of $\pi_1(X_n)$ (the relative Hurewicz theorem in the form $\pi'_{n+1} \cong H_{n+1}$). The characteristic map Φ_e defines an element of $\pi_{n+1}(X_{n+1}, X_n)$ with $h(\Phi_e) = e$, whose image under $\partial \colon \pi_{n+1}(X_{n+1}, X_n) \to \pi_n(X_n)$ is $[\varphi_e]$. Hence

$$c(f)\big(h(x)\big) = f_*\partial(x) \qquad \text{for all } x \in \pi_{n+1}(X_{n+1}, X_n) : \tag{$*$}$$

both sides are additive and invariant under the action of $\pi_1(X_n)$ — the right side because Y is n-simple, $f_*\partial(\gamma \cdot x) = f_*(\gamma) \cdot f_*\partial x = f_*\partial x$ — and they agree on the generators Φ_e.

Let E be a relative $(n+2)$-cell, $\Phi_E \in \pi_{n+2}(X_{n+2}, X_{n+1})$. Its cellular boundary is $dE = h\big(j(\partial\Phi_E)\big)$, where $\partial\Phi_E \in \pi_{n+1}(X_{n+1})$ and $j\colon \pi_{n+1}(X_{n+1}) \to \pi_{n+1}(X_{n+1}, X_n)$: the cellular boundary map is the composite $H_{n+2}(X_{n+2}, X_{n+1}) \to H_{n+1}(X_{n+1}) \to H_{n+1}(X_{n+1}, X_n)$, and the Hurewicz map is natural and compatible with boundary maps. By $(*)$,

$$(\delta c(f))(E) = c(f)(dE) = f_*\,\partial\, j\,(\partial\Phi_E) = 0,$$

because $\partial \circ j$ is the composite of two consecutive maps in the long exact sequence of the pair (X_{n+1}, X_n). □

Remark 17.6 (The geometric content). The boundary of an $(n+2)$-cell is a sphere S^{n+1} mapped into X_{n+1}; the $(n+1)$-cells it crosses, with multiplicity, form its cellular boundary; and the obstructions on those cells add up to the class of a map $S^n \to Y$ that bounds. A sum of obstructions around the boundary of a higher cell is always zero, just as the integral of a gradient around a closed loop is zero. That is what makes $c(f)$ closed.

17.3 The difference cochain

To pass from the cocycle to its class we compare two maps agreeing on the $(n-1)$-skeleton.

Construction 17.7 (The difference cochain)**.** Let $f, g\colon X_n \to Y$ agree on X_{n-1}. For a relative n-cell e the maps $f \circ \Phi_e$, $g \circ \Phi_e\colon D^n \to Y$ agree on S^{n-1}, so they glue to a map $S^n = D^n_+ \cup_{S^{n-1}} D^n_- \to Y$ (f on the upper, g on the lower hemisphere). Its class is $d(f,g)(e) \in \pi_n(Y)$; $d(f,g) \in C^n(W, A; \pi_n Y)$ is the *difference cochain.*

Equivalently: give $W \times I$ the product cell structure and let F be defined on $X_n \times \{0\} \cup X_n \times \{1\} \cup X_{n-1} \times I$ by f, g and the stationary homotopy. The cell $e \times (0,1)$, for e an n-cell, is an $(n+1)$-cell of $W \times I$, and the obstruction cochain of F on it is $d(f,g)(e)$: the boundary of $D^n \times I$, with its side collapsed by the stationary homotopy, is exactly the sphere of Construction 17.7.

Proposition 17.8. *Let $f, g\colon X_n \to Y$ agree on X_{n-1}.*

(a) $\delta d(f,g) = \pm\big(c(f) - c(g)\big)$, with a sign depending only on n and the orientation conventions.

(b) $d(f,g) = 0$ if and only if $f \simeq g$ rel X_{n-1}.

(c) $d(f,g) + d(g,h) = d(f,h)$ for a third map h agreeing with them on X_{n-1}.

(d) Given f and any cochain $\eta \in C^n(W, A; \pi_n Y)$ there is $g\colon X_n \to Y$ agreeing with f on X_{n-1} with $d(f,g) = \eta$.

Proof. (a) Apply Proposition 17.5 to F on the relative CW complex $(W \times I, A \times I)$; F is defined on its relative n-skeleton. Its obstruction cocycle takes the value $c(f)(e)$ on $e \times 0$, $c(g)(e)$ on $e \times 1$ (e an $(n+1)$-cell), and $d(f,g)(e')$ on $e' \times (0,1)$ (e' an n-cell). In the product, $d(e \times (0,1)) = de \times (0,1) + (-1)^{\dim e}(e \times 1 - e \times 0)$. Evaluating $\delta c(F) = 0$ on $e \times (0,1)$, e an $(n+1)$-cell, gives $d(f,g)(de) + (-1)^{n+1}\big(c(g)(e) - c(f)(e)\big) = 0$.

(b) A homotopy rel X_{n-1} is an extension of F over the cells $e \times (0,1)$; by Lemma 17.4 on the cylinder it exists iff $d(f,g) = 0$.

(c) On a cell e, $d(f,g)(e)+d(g,h)(e)$ is represented by gluing the two spheres along their common hemisphere $g\circ\Phi_e$; this is the sum in π_n (the pinch map), and cancelling the common hemisphere gives $d(f,h)(e)$.

(d) For each n-cell e choose $\alpha_e\colon S^n\to Y$ representing $\eta(e)$, and change f inside e: use $f\circ\Phi_e$ rescaled on the disc of radius $\frac12$, and on the annulus $\frac12\le|x|\le1$ the map collapsing the annulus onto a sphere ($S^n=$ annulus with both boundary spheres identified to a point) followed by α_e (precomposed with a path to the correct basepoint, harmless by Lemma 17.2). Then g agrees with f on X_{n-1} and $d(f,g)(e)=\pm[\alpha_e]$, with a sign fixed by conventions; replacing α_e by $-\alpha_e$ if necessary gives $d(f,g)=\eta$. □

17.4 The extension theorem

Theorem 17.9 (Obstruction theory for extensions)**.** *Let (W,A) be a relative CW complex, Y path-connected and n-simple, $n\ge1$, and $f\colon X_n\to Y$. Then:*

(a) $c(f)\in C^{n+1}(W,A;\pi_nY)$ is a cocycle, and it vanishes iff f extends over X_{n+1};

(b) its class $\mathfrak{o}_{n+1}(f)=[c(f)]\in H^{n+1}(W,A;\pi_nY)$ vanishes iff the restriction of f to X_{n-1} extends over X_{n+1};

(c) the class depends only on the homotopy class of $f|_{X_{n-1}}$ rel A.

Proof. (a) is Lemma 17.4 and Proposition 17.5.

(b) If $g\colon X_{n+1}\to Y$ extends $f|_{X_{n-1}}$, then $c(g|_{X_n})=0$ and by Proposition 17.8(a) $c(f)=c(f)-c(g)=\pm\delta d(f,g)$ is a coboundary. Conversely if $c(f)=\delta\eta$, choose by Proposition 17.8(d) a map g agreeing with f on X_{n-1} with $d(f,g)=\pm\eta$, the sign chosen so that $c(f)-c(g)=\delta\eta$; then $c(g)=0$ and g extends over X_{n+1}.

(c) If $f'|_{X_{n-1}}\simeq f|_{X_{n-1}}$ rel A, the homotopy extension property lets us replace f' by a homotopic map agreeing with f on X_{n-1}; homotopic maps on X_n have equal obstruction cochains (the attaching maps, composed with homotopic maps, are homotopic), and maps agreeing on X_{n-1} have cohomologous ones by Proposition 17.8(a). □

Warning 17.10 (Cochain versus class). The two statements (a) and (b) differ, and the difference is the whole point. The cochain $c(f)$ answers "does *this* f extend?"; its class answers "does *some* map agreeing with f away from the n-cells extend?". A reader who computes $c(f)\neq0$ and concludes that no extension exists has proved nothing. And a reader who finds $\mathfrak{o}_{n+1}(f)\neq0$ has proved only that no map agreeing with f on X_{n-1} extends: going back further, to X_{n-2}, may change the answer. This is the phenomenon of *secondary obstructions* (Remark 17.33).

Corollary 17.11 (The extension lemma, recovered)**.** *Let Y be simple. If $H^{n+1}(W,A;\pi_nY)=0$ for all $n\ge1$ — for instance, if $\pi_n(Y)=0$ whenever $W\setminus A$ has cells of dimension $n+1$ — then every map $A\to Y$ extends over W.*

Proof. Extend over X_1 using path-connectivity of Y. Inductively, given an extension f_n on X_n, $\mathfrak{o}_{n+1}(f_n)=0$, so by Theorem 17.9(b) $f_n|_{X_{n-1}}$ extends to some f_{n+1} on X_{n+1}. The maps f_n and f_{n+1} agree on X_{n-1}, so the sequence stabilises on each skeleton and defines a map on W. □

Example 17.12 (Extending over a disc)**.** If $W=A\cup e^{n+1}$ is a single cell attached along φ, then $H^{n+1}(W,A;\pi_nY)\cong\pi_n(Y)$ and the obstruction is literally $[f\circ\varphi]$: a

nullhomotopy of the composite *is* an extension over the cell. For $W = D^{n+1}$, $A = S^n$, $Y = S^n$, the obstruction to extending $f\colon S^n \to S^n$ is its class in $\pi_n(S^n) \cong \mathbb{Z}$, i.e. its degree.

Example 17.13 (A nonzero cochain with zero class)**.** Let W be the closed disc D^2 with the CW structure $e^0 \cup e^1 \cup e^2$ (boundary circle $e^0 \cup e^1$) and $A = e^0$. Take $Y = S^1$ and let f be defined on the 1-skeleton ∂D^2 by a map of degree $k \neq 0$. Then $c(f)(e^2) = k \neq 0$, so this f does not extend. But $H^2(D^2, e^0; \mathbb{Z}) = 0$, so $\mathfrak{o}_2(f) = 0$: changing f on the 1-cell (to degree 0) removes the obstruction. Both statements of Theorem 17.9 are visible: f itself does not extend, but $f|_{X_0}$ (a point) does.

17.5 Homotopies and the classification of maps

Applying Theorem 17.9 to the pair $(W \times I,\ W \times \partial I \cup A \times I)$ turns existence into uniqueness, with the degree shifted down by one.

Proposition 17.14. *The relative cells of $(W \times I,\ W \times \partial I \cup A \times I)$ are the products $e \times (0,1)$ with e a cell of $W \setminus A$; a k-cell of $W \setminus A$ gives a relative $(k+1)$-cell. For every abelian group G,*

$$H^{k+1}\big(W \times I,\ W \times \partial I \cup A \times I; G\big) \cong H^k(W, A; G).$$

Proof. The first statement is the product cell structure. On cellular cochains, $\phi \mapsto \big(e \times (0,1) \mapsto \phi(e)\big)$ is an isomorphism from $C^k(W, A; G)$ to the relative cochains of the cylinder in degree $k+1$, commuting with δ up to sign, because $d(e \times (0,1)) \equiv de \times (0,1)$ modulo cells of the subcomplex. □

Theorem 17.15 (Obstructions to homotopy)**.** *Let $f_0, f_1\colon W \to Y$ agree on A, Y simple, and suppose $f_0 = f_1$ on X_{n-1}. Then $\mathfrak{d}_n(f_0, f_1) = [d(f_0|_{X_n}, f_1|_{X_n})] \in H^n(W, A; \pi_n Y)$ vanishes iff, after changing the stationary homotopy on $X_{n-1} \times I$ rel $X_{n-2} \times I$, the maps $f_0|_{X_n}$ and $f_1|_{X_n}$ become homotopic rel A.*

Proof. Apply Theorem 17.9(b) to the cylinder pair, using Proposition 17.14; the relevant obstruction cochain is $d(f_0, f_1)$ by the remark after Construction 17.7. □

The cleanest consequence is when only one obstruction group can be nonzero.

Theorem 17.16 (Hopf–Whitney)**.** *Let X be a CW complex of dimension at most n and Y an $(n-1)$-connected space, $n \geq 1$ (for $n = 1$ assume $\pi_1(Y)$ abelian). Then*

$$[X, Y] \xrightarrow{\ \cong\ } H^n\big(X; \pi_n(Y)\big), \qquad [f] \mapsto f^*(\iota),$$

where $\iota \in H^n(Y; \pi_n Y) \cong \operatorname{Hom}(H_n(Y), \pi_n(Y))$ corresponds to the inverse of the Hurewicz isomorphism.

Proof. Normalisation. Every map $X \to Y$ is homotopic to one sending X^{n-1} to the base point: the extension lemma on $X^{n-1} \times I$ rel $X^{n-1} \times \{0\}$, using $\pi_k(Y) = 0$ for $k \leq n-1$, deforms $f|_{X^{n-1}}$ to a constant, and the HEP extends the deformation. Two normalised maps f, g agree on X^{n-1}, so $d(f, g) \in C^n(X; \pi_n Y)$ is defined; it is a cocycle because $C^{n+1} = 0$.

Surjectivity. Given a cocycle z, Proposition 17.8(d) produces g with $d(*, g) = z$, where $*$ is the constant map.

Injectivity. If $d(*,f)$ and $d(*,g)$ are cohomologous then $d(f,g) = d(*,g) - d(*,f)$ (Proposition 17.8(c)) is a coboundary, so by Theorem 17.15 f and g become homotopic on $X^n = X$ after a change of the homotopy on $X^{n-1} \times I$; such a change is possible because the further obstructions to homotopies on $X^k \times I$, $k < n-1$, lie in groups with coefficients $\pi_k(Y) = 0$. Conversely homotopic maps have cohomologous difference cochains. So $[f] \mapsto [d(*,f)]$ is a bijection $[X,Y] \to H^n(X;\pi_n Y)$.

Identification with $f^*\iota$. For normalised f and an n-cell e, $d(*,f)(e)$ is the class of $f \circ \Phi_e$ with ∂e collapsed, i.e. the preimage under Hurewicz of f_* of the cellular chain e; this is the cellular cochain $\iota \circ f_\#$, which represents $f^*\iota$. □

Corollary 17.17 (Hopf)**.** *For a CW complex X of dimension $\le n$, $[X, S^n] \cong H^n(X;\mathbb{Z})$. In particular, for a closed connected n-manifold M, homotopy classes of maps $M \to S^n$ are classified by their degree in $\mathbb{Z}$ if M is orientable, and by their degree mod 2 if not.*

Proof. $\pi_n(S^n) = \mathbb{Z}$, and $H^n(M;\mathbb{Z})$ is $\mathbb{Z}$ or $\mathbb{Z}/2$ according to orientability (Theorems 6.1 and 7.11); the class $f^*\iota$ is the degree, resp. its reduction mod 2, by the identification in the proof. □

Example 17.18. $[\mathbb{RP}^2, S^2] \cong H^2(\mathbb{RP}^2;\mathbb{Z}) \cong \mathbb{Z}/2$: there are exactly two homotopy classes of maps $\mathbb{RP}^2 \to S^2$, the constant map and the collapse $\mathbb{RP}^2 \to \mathbb{RP}^2/\mathbb{RP}^1 \cong S^2$. The collapse is essential although it induces zero on all homology groups (Example 6.13): obstruction theory sees the class in $H^2(\mathbb{RP}^2;\mathbb{Z}) = \operatorname{Ext}(H_1,\mathbb{Z})$ that homology misses.

17.6 Eilenberg–MacLane spaces represent cohomology

Theorem 17.19. *Let G be abelian and $n \ge 1$. For every CW complex X,*

$$T\colon [X, K(G,n)] \xrightarrow{\cong} H^n(X;G), \qquad T[f] = f^*(\iota_n),$$

where $\iota_n \in H^n(K(G,n);G) \cong \operatorname{Hom}(G,G)$ is the fundamental class *corresponding to the identity. The bijection is natural in X, and the same holds for based maps when X is connected.*

Proof. $K = K(G,n)$ is simple (simply connected if $n \ge 2$; for $n = 1$, $\pi_1 = G$ is abelian and there are no higher groups). The proof of Theorem 17.16 applies verbatim without the dimension bound, because every obstruction group other than those with coefficients $\pi_n(K)$ vanishes: obstructions to homotopies on the $(k+1)$-cells of the cylinder lie in $H^k(X;\pi_k K)$, zero for $k \ne n$; and the existence part needs no obstruction theory, since $d(*,g)$ can be prescribed. Naturality: $T[f \circ g] = g^* f^* \iota_n = g^* T[f]$. □

Remark 17.20. Theorem 17.19 is the fact used in Remark 10.16 of Chapter 10: the H-space structure of $K(G,n)$ is the map $K \times K \to K$ representing $\mathrm{pr}_1^*\iota_n + \mathrm{pr}_2^*\iota_n$, and under T the pointwise product of maps corresponds to addition in $H^n(X;G)$. Hatcher proves the theorem by a different route, through Postnikov towers [1, Thm. 4.57].

Corollary 17.21. *A map $A \to K(G,n)$ extends over a CW complex $W \supseteq A$ iff its class in $H^n(A;G)$ lies in the image of the restriction $H^n(W;G) \to H^n(A;G)$.*

Proof. By naturality of T, an extension up to homotopy is a class on W restricting to the given one; the HEP converts extensions up to homotopy into extensions. □

17.7 Lifting through fibrations, and sections

Theorem 17.22 (Obstructions to lifting)**.** *Let $F \to E \xrightarrow{p} B$ be a fibration with B and F path-connected, F n-simple, and $\pi_1(B)$ acting trivially on $\pi_n(F)$. Let (W, A) be a relative CW complex, $g\colon W \to B$, and $\tilde g\colon X_n \to E$ a lift of $g|_{X_n}$. There is an obstruction cocycle $c(\tilde g) \in C^{n+1}(W, A; \pi_n F)$, defined cell by cell as in Construction 17.3, such that:*

(a) $c(\tilde g) = 0$ iff $\tilde g$ extends to a lift on X_{n+1};

(b) $[c(\tilde g)] \in H^{n+1}(W, A; \pi_n F)$ vanishes iff $\tilde g|_{X_{n-1}}$ extends to a lift on X_{n+1}.

Without the triviality hypotheses the same holds with $\pi_n(F)$ replaced by the local coefficient system it forms over B, pulled back to W.

Proof sketch. Over a cell e with characteristic map $\Phi_e\colon D^{n+1} \to W$, the pulled-back fibration $\Phi_e^* E \to D^{n+1}$ is fibre homotopy equivalent to the product $D^{n+1} \times F$, since D^{n+1} is contractible [1, Prop. 4.62]. The given lift on ∂e becomes, after projecting to F, a map $S^n \to F$, whose class in $\pi_n(F)$ is $c(\tilde g)(e)$; changing the trivialisation changes it by the action of $\pi_1(B)$, trivial by hypothesis. A lift over e exists iff this class vanishes. The rest of the proof of Theorem 17.9 — the cocycle property, difference cochains, the class — goes through with lifts in place of maps and lifts of $g \circ \mathrm{pr}\colon W \times I \to B$ in place of homotopies. See [10, §12] for sphere bundles and [11, Part III] for the general case. □

Example 17.23 (Sections in low dimensions)**.** Take $A = \varnothing$, $g = \mathrm{id}_B$. If $\dim B \le n$ and $\pi_i(F) = 0$ for $i < n$, every obstruction group $H^{i+1}(B; \pi_i F)$ vanishes — for $i < n$ the coefficients, for $i \ge n$ the cohomology of B — and a section exists. For example, the space of orthonormal k-frames in $\mathbb{R}^N$ is $(N-k-1)$-connected [1, §4.2], so an N-dimensional vector bundle over a complex of dimension $\le N - k$ admits k linearly independent sections.

Theorem 17.24 (The Euler class as an obstruction)**.** *Let ξ be an oriented real vector bundle of rank $k \ge 2$ over a CW complex B, and $S(\xi) \to B$ its unit sphere bundle, with fibre S^{k-1}. A section of $S(\xi)$ exists over the $(k-1)$-skeleton, and the obstruction class to extending one over the k-skeleton is the Euler class $e(\xi) \in H^k(B; \mathbb{Z})$, up to a sign fixed by conventions. In particular, if $\dim B = k$ then ξ has a nowhere-zero section iff $e(\xi) = 0$.*

Proof sketch. The fibre S^{k-1} is $(k-2)$-connected, so by Theorem 17.22 the first possibly nonzero obstruction lies in $H^k(B; \pi_{k-1} S^{k-1}) = H^k(B; \mathbb{Z})$, with untwisted coefficients because the orientation makes the monodromy act trivially on $\pi_{k-1}(S^{k-1}) \cong H_{k-1}(S^{k-1})$. Moreover the primary obstruction does not depend on the choice of section over the $(k-1)$-skeleton, since sections over that skeleton are unique up to homotopy over the $(k-2)$-skeleton (the analogue of Theorem 17.16). So it is a natural characteristic class of oriented bundles, and that it equals the Euler class defined through the Thom class (Theorem 16.15) is proved in [10, Thm. 12.5] by comparing both on the universal bundle. If $\dim B = k$ there are no further obstructions. □

Example 17.25 (Vector fields on surfaces)**.** For a closed oriented surface M_g, $\langle e(TM_g), [M_g] \rangle = \chi(M_g) = 2 - 2g$ [10, §11]. By Theorem 17.24, M_g has a nowhere-zero vector field iff $g = 1$. For S^2 the obstruction is 2, which recovers the hairy ball theorem; compare Exercise 12.13, where the mod-2 invariant $w_2(TS^2) = 0$ could not see this.

Remark 17.26 (Stiefel–Whitney classes as obstructions). For a real bundle ξ of rank k, the obstruction to finding $k-i+1$ linearly independent sections over the i-skeleton lies in $H^i(B;\pi_{i-1}V_{k-i+1}(\mathbb{R}^k))$ (with local coefficients in general), where $V_j(\mathbb{R}^k)$ is the Stiefel manifold of orthonormal j-frames. The coefficient group is $\mathbb{Z}$ or $\mathbb{Z}/2$, and the primary obstruction reduced mod 2 is $w_i(\xi)$ [10, §12]. This was the original definition of the Stiefel–Whitney classes: $w_i(\xi)\neq 0$ obstructs $k-i+1$ independent sections over the i-skeleton.

Remark 17.27 (Local coefficients). If Y is not n-simple, a map $S^n\to Y$ only has a class in $\pi_n(Y)$ modulo the action of $\pi_1(Y)$, and the obstruction cochain takes values in the local system $f^*\pi_n(Y)$ on W; all statements above hold with cohomology with local coefficients (Definition 15.10). This matters in Part VI, where configuration spaces of surfaces are $K(\pi,1)$'s with highly nonabelian π_1.

17.8 Postnikov towers: obstructions as k-invariants

When the target has a Postnikov tower of principal fibrations, the obstructions have a second description, which is the one that computes.

Definition 17.28. A fibration $E\to B$ is *principal* if it is, up to fibre homotopy equivalence, the pullback of the path fibration $PC\to C$ along a map $k\colon B\to C$; its fibre is then ΩC.

Example 17.29. For a connected CW complex X on whose higher homotopy groups $\pi_1(X)$ acts trivially — in particular for simple X — the Postnikov tower can be chosen with every stage $X^n\to X^{n-1}$ the pullback of the path fibration over $K(\pi_nX,n+1)$ along a k-invariant $k^n\colon X^{n-1}\to K(\pi_nX,n+1)$ [1, Thm. 4.69]. The fibre is $K(\pi_nX,n)=\Omega K(\pi_nX,n+1)$.

Warning 17.30. Principal does not mean trivial: a principal fibration is a pullback of a *universal* fibration, and it is trivial exactly when the classifying map is nullhomotopic (Exercise 17.44).

Proposition 17.31 (Lifting through a principal fibration)**.** *Let $E\to B$ be the pullback of $PK(G,m)\to K(G,m)$ along $k\colon B\to K(G,m)$, with $\kappa=T[k]\in H^m(B;G)$. A map $f\colon X\to B$ from a CW complex lifts to E iff $f^*\kappa=0\in H^m(X;G)$. If a lift exists, the group $H^{m-1}(X;G)$ acts transitively on the set of homotopy classes of lifts of f (lifts up to vertical homotopy).*

Proof. A lift of f to the pullback is a pair consisting of f and a path in the mapping space from $k\circ f$ to the constant map, i.e. a nullhomotopy of $k\circ f$. So a lift exists iff $k\circ f\simeq *$, iff $f^*\kappa=T[k\circ f]=0$ (Theorem 17.19). Two nullhomotopies differ by a map $X\to\Omega K(G,m)\simeq K(G,m-1)$ (glue one to the reverse of the other), and $[X,K(G,m-1)]=H^{m-1}(X;G)$ acts by concatenating a nullhomotopy with a loop of maps; any two nullhomotopies are related this way. □

Example 17.32 (Maps into the 3-type of S^2)**.** Let P be the third Postnikov stage of S^2, with $\pi_2(P)=\pi_3(P)=\mathbb{Z}$, sitting in a principal fibration $K(\mathbb{Z},3)\to P\to\mathbb{CP}^\infty$ with k-invariant $\kappa\in H^4(\mathbb{CP}^\infty;\mathbb{Z})=\mathbb{Z}\iota^2$. We claim $\kappa=\pm\iota^2$.

The map $S^2\to P$ is an isomorphism on π_i for $i\le 3$ and onto on π_4, so by the relative Hurewicz theorem its mapping cylinder pair is 4-connected and the map is

an isomorphism on H_i for $i \le 3$ and onto on H_4. Hence $H_3(P) = H_4(P) = 0$ and $H^4(P;\mathbb{Z}) = 0$ by Theorem 6.1. Now run the cohomology Serre spectral sequence of $K(\mathbb{Z},3) \to P \to \mathbb{CP}^\infty$, using $\widetilde{H}^i(K(\mathbb{Z},3);\mathbb{Z}) = 0$ for $i \le 4$ except $H^3 = \mathbb{Z}$ (Hurewicz gives $H_3 = \mathbb{Z}$, and $H_4(K(\mathbb{Z},3)) = 0$ by [1, §4.2, Ex. 22]). In total degree 4 the only nonzero entry is $E_2^{4,0} = \mathbb{Z}\iota^2$, and the only differential reaching it is the transgression $d_4 \colon E_4^{0,3} = \mathbb{Z} \to E_4^{4,0}$, which sends the generator to κ (this is the defining property of the k-invariant as the transgression of the fundamental class of the fibre). So $H^4(P) \cong \mathbb{Z}\iota^2/\mathbb{Z}\kappa$, and $H^4(P) = 0$ forces $\kappa = \pm\iota^2$.

Consequently, by Proposition 17.31, a map $X \to \mathbb{CP}^\infty$ with class $x \in H^2(X;\mathbb{Z})$ lifts to P iff $x^2 = 0$. Since $S^2 \to P$ is 4-connected, the compression lemma [1, Lemma 4.6] applied to its mapping cylinder shows that $[X, S^2] \to [X, P]$ is bijective for $\dim X \le 3$ and surjective for $\dim X = 4$. Hence for $\dim X \le 4$, a class $x \in H^2(X;\mathbb{Z})$ is f^* of the generator of $H^2(S^2)$ for some $f \colon X \to S^2$ iff $x^2 = 0$. For instance every map $\mathbb{CP}^2 \to S^2$ induces zero on H^2, since nonzero classes of $H^2(\mathbb{CP}^2)$ have nonzero square — the obstruction is the cup square, a *quadratic* function of x.

Remark 17.33 (Secondary obstructions). When $\mathfrak{o}_{n+1}(f) \neq 0$ one may try to change f on X_{n-1} instead. The set of obstruction classes $\mathfrak{o}_{n+1}$ obtained from maps agreeing with a fixed map on X_{n-2} is a coset of a subgroup of $H^{n+1}(W, A; \pi_n Y)$ — the *secondary obstruction* with its indeterminacy. For a target with a two-stage Postnikov tower, the k-invariant is a class $\kappa \in H^m(K(\pi_p, p); \pi_{m-1})$, i.e. by Theorem 18.2 a cohomology operation, and the obstruction to lifting $x \in H^p(X;\pi_p)$ is that operation applied to x: the cup square in Example 17.32, a Steenrod square for many stable two-stage spaces. This is why cohomology operations are the subject of the next chapter.

17.9 Looping a tower

Lemma 17.34. *The loop functor preserves pullbacks:* $\Omega(B \times_C D) \cong \Omega B \times_{\Omega C} \Omega D$. *And* $\Omega K(G, n+1) \simeq K(G, n)$ *for* $n \ge 0$.

Proof. $\Omega = \mathrm{Map}_*(S^1, -)$ is right adjoint to $S^1 \wedge -$, and right adjoints preserve limits; concretely, a loop in $B \times_C D$ is a pair of loops agreeing in C. For the second statement, $\pi_i(\Omega Y) \cong \pi_{i+1}(Y)$ (Example 14.5), so $\Omega K(G, n+1)$ has a single nonzero homotopy group G, in degree n; it has the homotopy type of a CW complex by Milnor's theorem, and CW $K(G,n)$'s are unique up to homotopy [1, Prop. 4.30]. □

Corollary 17.35. *Looping the Postnikov tower of a simple space X gives a Postnikov tower of (the base point component of) ΩX, whose k-invariants are the loopings $\Omega k^n \colon \Omega X^{n-1} \to K(\pi_n X, n)$. In particular the k-invariants of a loop space are loop maps, and in cohomology they are* primitive *classes (Definition 10.9).*

Proof. Apply Ω to the pullback squares of Example 17.29 and use Lemma 17.34. A loop map Ωk commutes up to homotopy with the loop multiplications, so with $\kappa = (\Omega k)^* \iota$ we get $\mu^*\kappa = (\Omega k \times \Omega k)^* \mu^* \iota$, and $\mu^*\iota = \iota \otimes 1 + 1 \otimes \iota$ because the H-structure of $K(G,n)$ represents addition (Remark 17.20). So κ is primitive. □

Remark 17.36. Corollary 17.35 is a strong constraint. The cup square $\iota^2 \in H^4(K(\mathbb{Z},2))$ is not primitive ($\mu^*\iota^2$ contains $2\,\iota \otimes \iota$, Example 10.3), so it cannot be a k-invariant of a loop space; consistently, $\Omega S^2 \simeq S^1 \times \Omega S^3$, and the Hopf fibration loops to a trivial fibration. k-invariants surviving infinitely many loopings are *stable* cohomology operations, i.e. elements of the Steenrod algebra (Chapter 18).

17.10 Exercises

Exercise 17.37 (easy)**.** Deduce Theorem 11.15 from Theorem 17.9 in the case of a simple target, and say which hypothesis of the extension lemma corresponds to which obstruction group vanishing.

SOURCE: orig-book (retained)

Solution. The hypothesis of Theorem 11.15 is $\pi_{k-1}(Y) = 0$ whenever $W \setminus A$ has a k-cell. The class $\mathfrak{o}_{n+1}$ lies in $H^{n+1}(W, A; \pi_n Y)$, computed from cochains on the relative $(n+1)$-cells: if there are none the cochain group is zero; if there are some, $\pi_n(Y) = 0$ by hypothesis ($k = n+1$). Either way the obstruction group vanishes for every n, and Corollary 17.11 gives the extension. (Simplicity is not needed for the extension lemma itself, whose proof uses only that the obstructions vanish as elements.)

Exercise 17.38 (medium)**.** Let (W, A) be a CW pair. Show that a map $A \to K(G, n)$ with class $a \in H^n(A; G)$ extends over W iff $\delta a = 0$ in $H^{n+1}(W, A; G)$, and identify δa with the primary obstruction $\mathfrak{o}_{n+1}$.

SOURCE: original (refines Corollary 17.21)

Solution. By Corollary 17.21 the map extends iff a is a restriction, iff $\delta a = 0$ by exactness of $H^n(W; G) \to H^n(A; G) \xrightarrow{\delta} H^{n+1}(W, A; G)$. For the identification, normalise the map f on A to be constant on A^{n-1} and extend it to $X_n = A \cup W^n$ by the constant map on the new cells. For a relative $(n+1)$-cell e, the obstruction $c(f)(e)$ is the class of $f \circ \varphi_e$, which only sees the part of ∂e lying in A, where f is represented by a cocycle z for a; so $c(f)(e)$ is the value on e of the cochain "extend z by zero to W, then apply δ". By Lemma 5.1 that is a representative of δa.

Exercise 17.39 (medium)**.** Compute $[M_g, S^2]$, $[N_h, S^2]$ (with N_h the non-orientable surface of Exercise 5.18) and $[T^3, S^3]$, and give a representative of each class.

SOURCE: original

Solution. By Corollary 17.17: $[M_g, S^2] \cong H^2(M_g) = \mathbb{Z}$, classified by degree; a map of degree d is the collapse $M_g \to M_g/M_g^1 \cong S^2$ (degree ± 1) followed by a self-map of S^2 of degree $\pm d$. $[N_h, S^2] \cong H^2(N_h; \mathbb{Z}) = \mathbb{Z}/2$ (by Exercise 5.18 and Theorem 6.1, $H^2 = \mathrm{Ext}(\mathbb{Z}^{h-1} \oplus \mathbb{Z}/2, \mathbb{Z}) = \mathbb{Z}/2$); the two classes are the constant map and the collapse of the 1-skeleton, and the collapse followed by a degree-2 map of S^2 is nullhomotopic. $[T^3, S^3] \cong H^3(T^3) = \mathbb{Z}$, by degree; the collapse of the 2-skeleton has degree ± 1.

Exercise 17.40 (medium)**.** Show that $[X, \mathbb{CP}^\infty] \cong H^2(X; \mathbb{Z})$ for every CW complex X, and deduce that complex line bundles over a CW complex are classified by $c_1 \in H^2(X; \mathbb{Z})$. Which line bundle over S^2 corresponds to $d \in \mathbb{Z} = H^2(S^2)$?

SOURCE: original

Solution. $\mathbb{CP}^\infty = K(\mathbb{Z}, 2)$ (Exercise 14.6), so Theorem 17.19 gives $[X, \mathbb{CP}^\infty] \cong H^2(X; \mathbb{Z})$, $[f] \mapsto f^*\iota$. Complex line bundles over a paracompact space are classified by $[X, \mathbb{CP}^\infty]$, the universal bundle being the tautological γ; taking $\iota = c_1(\gamma)$, naturality gives $f^*\iota = c_1(f^*\gamma)$. So $L \mapsto c_1(L)$ is a bijection onto $H^2(X; \mathbb{Z})$. Over $S^2 = \mathbb{CP}^1$ the tautological bundle has c_1 a generator, and the bundle with $c_1 = d$ is the pullback of the tautological bundle along a map $S^2 \to S^2$ of degree d (up to the sign convention for the generator), equivalently a tensor power of the tautological bundle or of its dual, using additivity of c_1 under $\otimes$ (Exercise 26.9).

Exercise 17.41 (easy)**.** Show that $[X, S^1] \cong H^1(X;\mathbb{Z})$ for every CW complex X. In particular two maps $T^2 \to S^1$ are homotopic iff they induce the same map on H^1.

SOURCE: original

Solution. S^1 is a $K(\mathbb{Z},1)$ (its universal cover $\mathbb{R}$ is contractible) and a topological group, hence simple. Theorem 17.19 with $n=1$ gives $[X,S^1] \cong H^1(X;\mathbb{Z})$, $f \mapsto f^*\iota$, and $f^*\iota$ determines and is determined by $f^*\colon H^1(S^1) \to H^1(X)$. For $X = T^2$ this says two maps are homotopic iff they agree on $H^1(-;\mathbb{Z})$.

Exercise 17.42 (medium)**.** Let $p\colon E \to B$ be a fibration with $(n-1)$-connected fibre F, $n \geq 2$, and let B be a CW complex of dimension $\leq n$ with $\pi_1(B)$ acting trivially on $\pi_n(F)$. Show that p has a section; that any two sections are homotopic through sections if $\dim B \leq n-1$; and that when $\dim B = n$ the homotopy classes of sections are in bijection with $H^n(B;\pi_n F)$ (non-canonically).

SOURCE: original

Solution. *Existence.* The obstructions lie in $H^{i+1}(B;\pi_i F)$: zero for $i \leq n-1$ ($\pi_i F = 0$) and for $i \geq n$ ($H^{i+1}(B) = 0$). So a section exists (Theorem 17.22).

Uniqueness. A homotopy through sections from s_0 to s_1 is a lift of $\mathrm{pr}\colon B \times I \to B$ extending $s_0 \sqcup s_1$. By Proposition 17.14 the obstructions lie in $H^{i+1}(B \times I, B \times \partial I; \pi_i F) \cong H^i(B;\pi_i F)$, zero for $i \leq n-1$ and for $i > \dim B$. If $\dim B \leq n-1$ all vanish. If $\dim B = n$ only the difference class in $H^n(B;\pi_n F)$ remains; by the analogue of Proposition 17.8(c),(d) for sections, the classes $\mathfrak{d}(s_0, s)$, s a section, realise every element of $H^n(B;\pi_n F)$ and determine s up to homotopy. This gives a bijection depending on the choice of s_0.

Exercise 17.43 (medium)**.** Let M be a closed oriented surface and UM its unit tangent bundle, with fibre S^1. Show that $UM \to M$ has a section iff M is a torus, and that the homotopy classes of nowhere-zero vector fields on T^2 (up to homotopy through nowhere-zero vector fields) are in bijection with $H^1(T^2;\mathbb{Z}) \cong \mathbb{Z}^2$.

SOURCE: original

Solution. Existence: by Example 17.25 the only obstruction ($\dim M = 2$) is $e(TM)$, with $\langle e, [M]\rangle = \chi(M)$, which vanishes iff $M = T^2$.

Classification on T^2: T^2 is a Lie group, so its tangent bundle is trivial, $UT^2 \cong T^2 \times S^1$, and sections are graphs of maps $T^2 \to S^1$; homotopies through sections are homotopies of such maps. By Exercise 17.41, $[T^2, S^1] \cong H^1(T^2;\mathbb{Z}) \cong \mathbb{Z}^2$. In obstruction-theoretic terms: the fibre is $S^1 = K(\mathbb{Z},1)$, so homotopies between sections meet obstructions only in $H^1(T^2;\pi_1 S^1)$, and the sections form a torsor under that group.

Exercise 17.44 (medium)**.** Show that a principal fibration $E \to B$ classified by $k\colon B \to K(G,m)$ is fibre homotopy equivalent to the product $B \times K(G, m-1)$ iff k is nullhomotopic. Deduce that a two-stage Postnikov tower with vanishing k-invariant is a product of Eilenberg–MacLane spaces.

SOURCE: orig-book (retained; "only if" direction added)

Solution. If $k \simeq *$, the pullback along k is fibre homotopy equivalent to the pullback along the constant map [1, Prop. 4.62], which is $B \times \Omega K(G,m) = B \times K(G, m-1)$. Conversely a product fibration has a section, i.e. a lift of id_B, which by Proposition 17.31 exists iff $\kappa = \mathrm{id}^*\kappa = 0$, i.e. $k \simeq *$. For a two-stage tower, $X^n \to X^{n-1} = K(\pi_{n-1}, n-1)$ with $k^n \simeq *$ gives $X^n \simeq K(\pi_{n-1}, n-1) \times K(\pi_n, n)$.

17.11 Connections and further reading

Obstruction theory converts the cell-by-cell construction of maps into a sequence of cohomology classes. Its two most important consequences for this book are the representability theorem $[X, K(G,n)] \cong H^n(X;G)$, which makes cohomology operations computable (Chapter 18), and the identification of characteristic classes with obstructions (Chapter 26). In Chapter 23 the same skeleton-by-skeleton pattern reappears in combinatorial form, as horn filling, where the obstruction groups are not computed but abolished by hypothesis.

The classical references are Steenrod's *Topology of Fibre Bundles* [11, Part III], which develops the cochain approach followed here, and Whitehead's *Elements of Homotopy Theory* [12]. Hatcher [1, §4.3] derives the same results from Postnikov towers; Milnor–Stasheff [10, §12] treats characteristic classes as obstructions.

Chapter 18

Cohomology operations: Steenrod squares and reduced powers

18.1 The problem

Three times the book has stopped at the same wall. Chapter 4 found that homology cannot separate certain spaces; Chapter 7 that the cohomology ring is stronger but still incomplete (Remark 7.21); Chapter 13 that even all homotopy groups together can fail. And Chapter 17 ended with the observation that the obstructions met when classifying maps into two-stage Postnikov towers are *natural transformations of cohomology* (Remark 17.33). This chapter constructs the most important such transformations, the Steenrod squares Sq^i and, for odd primes, the reduced powers P^i, and uses them to answer questions that rings cannot: whether a space is a suspension of another, whether a map between spheres is stably essential, which dimensions can carry a Hopf invariant one map, and how the characteristic classes of a manifold depend on its cohomology.

Prerequisites. Cohomology with coefficients and cup products (Chapters 6 and 7); the Bockstein (Chapter 9); Eilenberg–MacLane spaces and the representability theorem $[X, K(G,n)] \cong H^n(X;G)$ (Theorem 17.19); Thom spaces and the Thom isomorphism (Chapter 16); Stiefel–Whitney classes (Chapter 12).

18.2 Cohomology operations

Definition 18.1. A *cohomology operation* of type $(m, \pi; n, G)$ is a family of functions $\theta_X \colon H^m(X;\pi) \to H^n(X;G)$, one for each space X, natural in X: $\theta_X(f^*\alpha) = f^*\theta_Y(\alpha)$ for every $f \colon X \to Y$.

Nothing is assumed about additivity (Exercise 18.5).

Theorem 18.2 (Operations are classes on Eilenberg–MacLane spaces)**.** *For CW complexes, cohomology operations of type $(m, \pi; n, G)$ are in natural bijection with $H^n(K(\pi,m);G)$, via $\theta \mapsto \theta(\iota_m)$.*

Proof. Write $\Phi(\theta) = \theta(\iota_m)$. Given $u \in H^n(K(\pi,m);G)$, define $\Psi(u)$ on $\alpha \in H^m(X;\pi)$ by $\Psi(u)(\alpha) = f_\alpha^* u$, where $f_\alpha \colon X \to K(\pi,m)$ is the map, unique up to homotopy, with

$f_\alpha^*\iota_m = \alpha$ (Theorem 17.19). Naturality of $\Psi(u)$: for $g\colon X' \to X$, $f_{g^*\alpha} \simeq f_\alpha \circ g$ by uniqueness, so $\Psi(u)(g^*\alpha) = g^* f_\alpha^* u = g^*\Psi(u)(\alpha)$. Then $\Phi\Psi(u) = \mathrm{id}^* u = u$, and for any θ, $\theta(\alpha) = \theta(f_\alpha^*\iota_m) = f_\alpha^*\theta(\iota_m) = \Psi\Phi(\theta)(\alpha)$. □

Exercise 18.3 (medium)**.** Show that an operation θ is additive, $\theta(\alpha+\beta) = \theta(\alpha) + \theta(\beta)$, if and only if $\theta(\iota_m)$ is *primitive* with respect to the H-space structure of $K(\pi, m)$: $\mu^*\theta(\iota_m) = \theta(\iota_m)\otimes 1 + 1 \otimes \theta(\iota_m)$ (interpreting the right side as $\mathrm{pr}_1^* + \mathrm{pr}_2^*$ when there is no Künneth isomorphism).

SOURCE: original (makes precise a remark of the original edition)

Solution. The multiplication $\mu\colon K\times K \to K$ represents $\mathrm{pr}_1^*\iota + \mathrm{pr}_2^*\iota$ (Remark 17.20). For classes α, β on X, with $f = (f_\alpha, f_\beta)\colon X \to K\times K$, we have $f_{\alpha+\beta} \simeq \mu\circ f$. So $\theta(\alpha+\beta) = f^*\mu^*\theta(\iota)$, while $\theta(\alpha)+\theta(\beta) = f^*(\mathrm{pr}_1^*\theta(\iota) + \mathrm{pr}_2^*\theta(\iota))$. If $\mu^*\theta(\iota) = \mathrm{pr}_1^*\theta(\iota) + \mathrm{pr}_2^*\theta(\iota)$, θ is additive. Conversely, take $X = K\times K$ and $\alpha = \mathrm{pr}_1^*\iota$, $\beta = \mathrm{pr}_2^*\iota$, so $f = \mathrm{id}$: additivity gives $\mu^*\theta(\iota) = \theta(\mathrm{pr}_1^*\iota) + \theta(\mathrm{pr}_2^*\iota) = \mathrm{pr}_1^*\theta(\iota) + \mathrm{pr}_2^*\theta(\iota)$ by naturality.

Example 18.4 (Operations that preserve the degree)**.** Take $\pi = G = \mathbb{Z}$, $n = m \geq 1$. $H^m(K(\mathbb{Z}, m); \mathbb{Z}) \cong \mathrm{Hom}(H_m, \mathbb{Z}) \oplus \mathrm{Ext}(H_{m-1}, \mathbb{Z}) = \mathrm{Hom}(\mathbb{Z}, \mathbb{Z}) = \mathbb{Z}\iota_m$ (Hurewicz). So the only operations $H^m(-;\mathbb{Z}) \to H^m(-;\mathbb{Z})$ are $\alpha \mapsto k\alpha$. Everything of substance raises degree.

Exercise 18.5 (easy)**.** Exhibit a cohomology operation that is not a homomorphism.

SOURCE: orig-book

Solution. $\theta(\alpha) = \alpha \smile \alpha$ on $H^m(X;\mathbb{Z})$ is natural since f^* is a ring map. It is not additive: $(\alpha+\beta)^2 = \alpha^2 + \beta^2 + \alpha\beta + \beta\alpha$, and for m even $\alpha\beta + \beta\alpha = 2\alpha\beta$, which is nonzero for α, β the two degree-2 generators of $H^*(\mathbb{CP}^\infty \times \mathbb{CP}^\infty; \mathbb{Z})$. Its class is $\iota_m^2 \in H^{2m}(K(\mathbb{Z}, m); \mathbb{Z})$, which for $m = 2$ is not primitive, in accordance with Exercise 18.3.

Definition 18.6. A *stable operation* of degree k with coefficients G is a family of additive operations $\theta^{(m)}\colon H^m(-;G) \to H^{m+k}(-;G)$, $m \geq 0$, commuting with the suspension isomorphisms

$$\sigma\colon \widetilde{H}^m(X;G) \cong \widetilde{H}^{m+1}(\Sigma X; G).$$

Equivalently, a stable operation is a sequence of classes $\theta_m \in H^{m+k}(K(G,m);G)$ which correspond to each other under the maps $\Sigma K(G,m) \to K(G, m+1)$ adjoint to the equivalences $K(G,m) \simeq \Omega K(G, m+1)$. The Bockstein β of Definition 9.14 is a stable operation of degree 1 with $\mathbb{F}_p$ coefficients: it is the connecting homomorphism of a coefficient sequence, and connecting homomorphisms of coefficient sequences commute with suspension (up to a sign convention).

18.3 The Steenrod squares

18.3.1 What they are

The cup product is graded commutative in cohomology, but on cochains it is commutative only up to homotopy (Chapter 7). Steenrod's discovery was that this homotopy, and the higher homotopies between homotopies, carry information. For a mod-2 cocycle u of degree n, $u \smile u$ and its "transpose" agree up to a specified coboundary; the

specified homotopies, called *cup-i products* $u \smile_i u$, are again cocycles, of degree $2n-i$, and their classes define

$$\mathrm{Sq}^{n-i}[u] = [u \smile_i u] \in H^{2n-i}(X;\mathbb{F}_2).$$

A cleaner formulation, due to Steenrod and used in [1, §4.L], works with the space $S^\infty \times_{\mathbb{Z}/2} (X \times X)$, where $\mathbb{Z}/2$ swaps the factors and acts antipodally on S^∞. A class $u \in H^n(X;\mathbb{F}_2)$ has an "external square" $P(u) \in H^{2n}(S^\infty \times_{\mathbb{Z}/2} (X \times X);\mathbb{F}_2)$ extending $u \times u$, and its restriction along the diagonal $\mathbb{RP}^\infty \times X \to S^\infty \times_{\mathbb{Z}/2} (X \times X)$ can be written uniquely as

$$\Delta^* P(u) = \sum_{i=0}^{n} w^{n-i} \times \mathrm{Sq}^i(u), \qquad w \in H^1(\mathbb{RP}^\infty;\mathbb{F}_2) \text{ the generator},$$

which defines $\mathrm{Sq}^i(u) \in H^{n+i}(X;\mathbb{F}_2)$. All the properties below are proved from this definition in [1, §4.L] and in Steenrod–Epstein [14]; Mosher–Tangora [15] is the standard place to learn to compute with them.

Theorem 18.7 (Properties of the Steenrod squares)**.** *There are operations* $\mathrm{Sq}^i\colon H^n(X,A;\mathbb{F}_2) \to H^{n+i}(X,A;\mathbb{F}_2)$*, $i \geq 0$, for all pairs (X,A), with the following properties.*

(1) Naturality*:* $f^* \mathrm{Sq}^i = \mathrm{Sq}^i f^*$.

(2) Additivity*:* $\mathrm{Sq}^i(x+y) = \mathrm{Sq}^i(x) + \mathrm{Sq}^i(y)$.

(3) Cartan formula*:* $\mathrm{Sq}^k(xy) = \sum_{i+j=k} \mathrm{Sq}^i(x)\,\mathrm{Sq}^j(y)$.

(4) Stability*:* $\mathrm{Sq}^i \sigma = \sigma\, \mathrm{Sq}^i$ *for the suspension isomorphism* $\sigma\colon \widetilde{H}^n(X) \to \widetilde{H}^{n+1}(\Sigma X)$*; more generally* Sq^i *commutes with connecting homomorphisms* $\delta\colon H^n(A) \to H^{n+1}(X,A)$.

(5) Unit*:* $\mathrm{Sq}^0 = \mathrm{id}$.

(6) Instability*:* $\mathrm{Sq}^n(x) = x^2$ *if* $|x| = n$*, and* $\mathrm{Sq}^i(x) = 0$ *if* $i > |x|$.

(7) Bockstein*:* Sq^1 *is the Bockstein* β *associated with* $0 \to \mathbb{Z}/2 \to \mathbb{Z}/4 \to \mathbb{Z}/2 \to 0$.

Moreover, (1)–(6) characterise the Sq^i *uniquely, and they satisfy the* Adem relations *(Theorem 18.17).*

Property (7) is not independent: we prove it from (1)–(6) in Proposition 18.14. Properties (1)–(6) are [1, Thm. 4L.12]; uniqueness is [14].

Remark 18.8 (Reading the axioms)*.* Instability (6) is the bridge to the ring: in the top degree the operation *is* the cup square, and below the top degree it is a genuinely new invariant. Stability (4) says that the operation survives suspension, although on a suspension all cup products of positive-degree classes vanish (Exercise 16.19). Taken together: *the Steenrod squares are the part of the multiplicative structure that survives suspension.* That sentence explains most of their applications.

Example 18.9 (The smallest computation)**.** In $H^*(\mathbb{CP}^2;\mathbb{F}_2) = \mathbb{F}_2[x]/(x^3)$, $|x| = 2$, axiom (6) gives $\mathrm{Sq}^2 x = x^2 \neq 0$, while $\mathrm{Sq}^1 x \in H^3 = 0$ and $\mathrm{Sq}^i x = 0$ for $i > 2$.

18.3.2 The total square and standard spaces

Definition 18.10. The *total square* of $x \in H^*(X;\mathbb{F}_2)$ is $\mathrm{Sq}(x) = \sum_{i\geq 0} \mathrm{Sq}^i(x)$, a finite sum by (6).

The Cartan formula says exactly that $\mathrm{Sq}(xy) = \mathrm{Sq}(x)\,\mathrm{Sq}(y)$: Sq is a ring homomorphism (collect terms of the Cartan formula by total degree). This is the form in which the squares are computed.

Proposition 18.11. *(a) In* $H^*(\mathbb{RP}^\infty;\mathbb{F}_2) = \mathbb{F}_2[w]$, $\mathrm{Sq}^i(w^k) = \binom{k}{i} w^{k+i}$.

(b) In $H^*(\mathbb{CP}^\infty;\mathbb{F}_2) = \mathbb{F}_2[c]$, $\mathrm{Sq}^{2i}(c^k) = \binom{k}{i} c^{k+i}$ *and the odd squares vanish.*

(c) The same formulas hold in $\mathbb{RP}^n$ *and* $\mathbb{CP}^n$*, reading* $w^{n+1} = 0$*, resp.* $c^{n+1} = 0$.

Proof. (a) By (6), $\mathrm{Sq}(w) = w + w^2 = w(1+w)$. Since Sq is a ring map, $\mathrm{Sq}(w^k) = w^k(1+w)^k = \sum_i \binom{k}{i} w^{k+i}$; compare degrees. (b) $\mathrm{Sq}(c) = c + c^2$ by (6) and since $\mathrm{Sq}^1 c \in H^3 = 0$; then $\mathrm{Sq}(c^k) = c^k(1+c)^k$. (c) Naturality along the inclusions, which are onto in cohomology. □

Example 18.12. In $H^*(\mathbb{RP}^\infty)$: $\mathrm{Sq}^1(w^k) = kw^{k+1}$, so Sq^1 is nonzero exactly on odd powers; $\mathrm{Sq}^2(w^k) = \binom{k}{2} w^{k+2}$, nonzero iff $k \equiv 2, 3 \pmod 4$; and $\mathrm{Sq}^{2^j}(w^k) \neq 0$ iff the 2^j-digit of k in binary is 1 (Lucas). In particular $\mathrm{Sq}^i(w^{2^j}) = 0$ for $0 < i < 2^j$: the powers w^{2^j} are reached from w by squaring but are invisible to all lower squares.

18.3.3 Sq^1 is the Bockstein

Lemma 18.13. *For* $n \geq 1$, $H^{n+1}(K(\mathbb{Z}/2, n);\mathbb{F}_2) \cong \mathbb{F}_2$. *Consequently there is exactly one nonzero operation* $H^n(-;\mathbb{F}_2) \to H^{n+1}(-;\mathbb{F}_2)$.

Proof. Build $K = K(\mathbb{Z}/2, n)$ from the Moore space $M = M(\mathbb{Z}/2, n) = S^n \cup_2 e^{n+1}$ (Proposition 6.9) by attaching cells of dimension $\geq n+2$ to kill the higher homotopy groups. Then $H^{n+1}(K;\mathbb{F}_2) \to H^{n+1}(M;\mathbb{F}_2) = \mathbb{F}_2$ is injective (the new cells have dimension $\geq n+2$, so the restriction on $(n+1)$-cochains is an isomorphism and only coboundaries from new $(n+2)$-cells could change cocycles). It is onto: the dual cochain e^{n+1*} is a mod-2 cocycle on K, because the cellular boundary of each new $(n+2)$-cell is an integral cycle of M in degree $n+1$, and the only such cycle is 0 ($d(e^{n+1}) = 2e^n \neq 0$); so $\delta(e^{n+1*})$ vanishes on the new cells. And it is not a coboundary, since the only cochain e^{n*} in degree n has $\delta e^{n*} = 2e^{n+1*} = 0 \bmod 2$. The last statement follows from Theorem 18.2. □

Proposition 18.14. $\mathrm{Sq}^1 = \beta$*, the Bockstein of* $0 \to \mathbb{Z}/2 \to \mathbb{Z}/4 \to \mathbb{Z}/2 \to 0$.

Proof. Both are operations $H^n(-;\mathbb{F}_2) \to H^{n+1}(-;\mathbb{F}_2)$, so by Lemma 18.13 it suffices to show both are nonzero for each $n \geq 1$. For β: on $M(\mathbb{Z}/2, n)$, β is an isomorphism $H^n \to H^{n+1}$ (Example 9.19). For Sq^1: on $\mathbb{RP}^\infty$, $\mathrm{Sq}^1 w = w^2 \neq 0$ by (6); by stability, $\mathrm{Sq}^1(\sigma^{n-1}w) = \sigma^{n-1}(w^2) \neq 0$ on $\Sigma^{n-1}\mathbb{RP}^\infty$, so $\mathrm{Sq}^1 \neq 0$ in every degree $n \geq 1$. (The case $n = 0$: $H^0 \to H^1$, both operations are zero, since a class of $H^0(X;\mathbb{F}_2)$ is locally constant and lifts to $\mathbb{Z}/4$.) □

Warning 18.15. The argument needs Lemma 18.13. Checking that Sq^1 and β agree on one space proves nothing without knowing that there is only one nonzero operation of this type: operations that agree on $\mathbb{RP}^\infty$ in degree 1 could a priori differ elsewhere.

Corollary 18.16. $\mathrm{Sq}^1 = \rho \circ \tilde{\beta}$, *where* $\tilde{\beta}\colon H^n(X;\mathbb{F}_2) \to H^{n+1}(X;\mathbb{Z})$ *is the integral Bockstein and* ρ *is reduction mod* 2 *(Exercise 9.26). In particular, if* $H^*(X;\mathbb{Z})$ *has no* 2*-torsion then* $\mathrm{Sq}^1 = 0$ *on* $H^*(X;\mathbb{F}_2)$*, and* Sq^1 *computes the order-*2 *part of the integral torsion via Proposition 9.17.*

Proof. The first statement is Exercise 9.26. The image of $\tilde{\beta}$ consists of classes killed by 2; if there is no 2-torsion, $\tilde{\beta} = 0$. □

18.4 The Adem relations and the Steenrod algebra

Theorem 18.17 (Adem relations)**.** *For* $0 < a < 2b$*,*

$$\mathrm{Sq}^a \mathrm{Sq}^b = \sum_{c=0}^{\lfloor a/2 \rfloor} \binom{b-c-1}{a-2c} \mathrm{Sq}^{a+b-c} \mathrm{Sq}^c,$$

binomial coefficients being read mod 2*.*

This is [1, Thm. 4L.13]; the proof there evaluates both sides on a product of copies of $\mathbb{RP}^\infty$, where everything is computable by Proposition 18.11, and uses that $H^*(K(\mathbb{Z}/2,n))$ is detected on such products in the relevant degrees.

Definition 18.18. The *(mod* 2*) Steenrod algebra* $\mathcal{A}$ is the graded $\mathbb{F}_2$-algebra generated by symbols Sq^i, $i \geq 1$ (with $\mathrm{Sq}^0 = 1$), modulo the Adem relations. It acts on $H^*(X;\mathbb{F}_2)$ for every X, making it an *unstable* $\mathcal{A}$-module: $\mathrm{Sq}^i x = 0$ for $i > |x|$.

By Theorem 18.2, $\mathcal{A}$ is exactly the algebra of stable mod-2 operations: $\mathcal{A}^k = \varprojlim_m H^{m+k}(K(\mathbb{Z}/2,m);\mathbb{F}_2)$, a theorem of Serre and Cartan which we do not prove; the algebraic structure of $\mathcal{A}$ — the admissible basis, the coproduct, Milnor's dual description — is Chapter 27.

Example 18.19. $a = b = 1$: only $c = 0$ contributes, with coefficient $\binom{0}{1} = 0$. So $\mathrm{Sq}^1 \mathrm{Sq}^1 = 0$, which by Proposition 18.14 is $\beta^2 = 0$ again (Exercise 9.15). $a = 1$, $b = 2$: $\mathrm{Sq}^1 \mathrm{Sq}^2 = \binom{1}{1} \mathrm{Sq}^3 = \mathrm{Sq}^3$. $a = b = 2$: $\mathrm{Sq}^2 \mathrm{Sq}^2 = \binom{1}{2} \mathrm{Sq}^4 + \binom{0}{0} \mathrm{Sq}^3 \mathrm{Sq}^1 = \mathrm{Sq}^3 \mathrm{Sq}^1$.

Proposition 18.20 (Decomposability)**.** *If* n *is not a power of* 2*, then* $\mathrm{Sq}^n = \sum_{0<c<n} a_c \mathrm{Sq}^{n-c} \mathrm{Sq}^c$ *for suitable* $a_c \in \mathbb{F}_2$*. Hence* $\mathcal{A}$ *is generated as an algebra by the* Sq^{2^j}*.*

Proof. Let 2^j be the largest power of 2 less than n, and put $b = 2^j$, $a = n - b$. Since n is not a power of 2, $0 < a < b < 2b$. In the Adem relation for $\mathrm{Sq}^a \mathrm{Sq}^b$, the term $c = 0$ has coefficient $\binom{2^j - 1}{a}$, which is odd by Lucas' theorem: $2^j - 1$ has all binary digits 1 below position j, and $a < 2^j$. So $\mathrm{Sq}^a \mathrm{Sq}^b = \mathrm{Sq}^n + \sum_{c \geq 1} \binom{b-c-1}{a-2c} \mathrm{Sq}^{n-c} \mathrm{Sq}^c$, and solving for Sq^n gives the claim (the term $\mathrm{Sq}^a \mathrm{Sq}^b$ has $0 < b < n$). Induction on n gives generation by the Sq^{2^j}. □

18.5 Applications

18.5.1 Where the ring is not enough

Theorem 18.21. *Over every commutative coefficient ring,* $\Sigma\mathbb{CP}^2$ *and* $S^3 \vee S^5$ *have isomorphic cohomology rings. They are nevertheless not homotopy equivalent.*

Proof. The rings agree. $\widetilde{H}^*(\Sigma\mathbb{CP}^2;R)$ and $\widetilde{H}^*(S^3\vee S^5;R)$ are both R in degrees $3,5$, and in both every product of positive-degree classes vanishes: on $\Sigma\mathbb{CP}^2$ because it is a suspension (Exercise 16.19), on the wedge by Exercise 7.4 and degree reasons.

The spaces differ. Let x generate $H^2(\mathbb{CP}^2;\mathbb{F}_2)$. By stability and Example 18.9, $\mathrm{Sq}^2(\sigma x)=\sigma(x^2)\neq 0$. On $S^3\vee S^5$, the generator a of H^3 restricts to the generator on S^3, where Sq^2 lands in $H^5(S^3)=0$, and to 0 on S^5; restriction to the summands is injective, so $\mathrm{Sq}^2 a=0$. A homotopy equivalence would commute with Sq^2 and carry generator to generator. □

Remark 18.22 (The debt, paid)*.* This is the pair promised in Remark 7.21: isomorphic rings over every coefficient ring, different homotopy types. The invariant separating them is not induced by any product — both products vanish — and it is natural, hence a homotopy invariant. It remembers the cup square x^2 of $\mathbb{CP}^2$ after suspension has destroyed it (Remark 18.8).

Exercise 18.23 (medium)**.** Let X be any space with $\widetilde{H}^*(X;\mathbb{F}_2)$ concentrated in degrees n and $2n$, the top class being the square of the bottom one. Show that $\Sigma^k X$ is not homotopy equivalent to $S^{n+k}\vee S^{2n+k}$ for any $k\geq 0$.

SOURCE: orig-book (generalised to iterated suspensions)

Solution. With x the generator of $H^n(X;\mathbb{F}_2)$, $\mathrm{Sq}^n x=x^2\neq 0$ by (6), so by stability $\mathrm{Sq}^n(\sigma^k x)=\sigma^k(x^2)\neq 0$ in $H^{2n+k}(\Sigma^k X;\mathbb{F}_2)$. On the wedge, the generator a of H^{n+k} restricts to the generator on S^{n+k}, where $\mathrm{Sq}^n a$ lands in $H^{2n+k}(S^{n+k})=0$ (as $n\geq 1$), and to 0 on S^{2n+k}; restriction being injective on $\widetilde{H}^{2n+k}$ of a wedge, $\mathrm{Sq}^n a=0$. A homotopy equivalence would commute with Sq^n and preserve the generators of the one-dimensional groups.

18.5.2 Hopf invariant one and stable maps

Theorem 18.24. *If $f\colon S^{2n-1}\to S^n$ has odd Hopf invariant, then n is a power of 2.*

Proof. In $H^*(C_f;\mathbb{F}_2)$, with $C_f=S^n\cup_f e^{2n}$ and generators $\alpha\in H^n$, $\beta\in H^{2n}$, the Hopf invariant being odd means $\alpha^2=\beta\neq 0$, so $\mathrm{Sq}^n\alpha=\alpha^2\neq 0$. If n were not a power of 2, Proposition 18.20 would give $\mathrm{Sq}^n\alpha=\sum_{0<c<n}a_c\,\mathrm{Sq}^{n-c}(\mathrm{Sq}^c\alpha)$, and each $\mathrm{Sq}^c\alpha$ lies in $H^{n+c}(C_f;\mathbb{F}_2)=0$ for $0<c<n$. Contradiction. □

The Hopf maps realise $n=1,2,4,8$; Adams proved these are the only cases [13]. The same argument with stability gives more:

Theorem 18.25. *If $f\colon S^{2n-1}\to S^n$ has odd Hopf invariant, then all its suspensions $\Sigma^k f\colon S^{2n-1+k}\to S^{n+k}$ are essential. In particular the Hopf map $\eta\colon S^3\to S^2$ represents a nonzero element of the stable stem π_1^s.*

Proof. $C_{\Sigma^k f}\cong\Sigma^k C_f$, and by stability Sq^n is nonzero from H^{n+k} to H^{2n+k} of $\Sigma^k C_f$. If $\Sigma^k f$ were nullhomotopic, $C_{\Sigma^k f}$ would be homotopy equivalent to $S^{n+k}\vee S^{2n+k}$, on which Sq^n vanishes (Exercise 18.23). □

Remark 18.26*.* Exercise 14.17 shows that $\pi_{n+1}(S^n)$, $n\geq 3$, is a cyclic quotient of $\pi_3(S^2)=\mathbb{Z}$, generated by a suspension of η, and Theorem 18.25 shows it is nonzero. In fact $\pi_{n+1}(S^n)\cong\mathbb{Z}/2$ for $n\geq 3$ [1, Cor. 4J.4], so $\pi_1^s\cong\mathbb{Z}/2$ generated by η; that $2\eta=0$ requires an argument beyond primary operations.

18.5.3 Thom spaces, Stiefel–Whitney classes and Wu's formula

Theorem 18.27 (Thom)**.** *Let ξ be a real vector bundle of rank k over B, with mod-2 Thom class $u \in \widetilde{H}^k(\mathrm{Th}(\xi);\mathbb{F}_2)$ and Thom isomorphism $\Phi(b) = p^*b \smile u$. Then*

$$w_i(\xi) = \Phi^{-1}\big(\mathrm{Sq}^i u\big).$$

These classes satisfy the axioms of Theorem 12.4, so this formula may be taken as the definition *of the Stiefel–Whitney classes.*

Proof sketch. Naturality is clear. Normalisation: for the tautological line bundle γ over $\mathbb{RP}^n$ the Thom space is $\mathbb{RP}^{n+1}$ (the disc bundle is $\mathbb{RP}^{n+1}$ minus a point, with the sphere bundle, a copy of S^n double covering $\mathbb{RP}^n$, collapsed), with Thom class $u = w$, and $\mathrm{Sq}^1 u = w^2 = \Phi(w)$, so $w_1(\gamma) = w$. Whitney formula: the Thom space of $\xi \oplus \eta$ is (a quotient of) the smash of the Thom spaces, with Thom class $u_\xi \times u_\eta$, and the Cartan formula gives $\mathrm{Sq}(u_\xi \times u_\eta) = \mathrm{Sq}\,u_\xi \times \mathrm{Sq}\,u_\eta$, which translates into $w(\xi\oplus\eta) = w(\xi)w(\eta)$. Vanishing above the rank is instability: $\mathrm{Sq}^i u = 0$ for $i > k$. Full details: [10, §8]. □

Remark 18.28. Theorem 18.27 explains why Stiefel–Whitney classes are "stable": they depend on ξ only through the $\mathcal{A}$-module structure of its Thom space, which is unchanged when a trivial summand is added (the Thom space is suspended).

For a closed n-manifold M the squares landing in the top degree are linear functionals on $H^{n-k}(M;\mathbb{F}_2)$, so by Poincaré duality they are cup products with fixed classes.

Definition 18.29. The *Wu class* $v_k \in H^k(M;\mathbb{F}_2)$ of a closed n-manifold is the unique class with $\langle v_k \smile x, [M]\rangle = \langle \mathrm{Sq}^k x, [M]\rangle$ for all $x \in H^{n-k}(M;\mathbb{F}_2)$; $v = 1 + v_1 + v_2 + \cdots$.

Existence and uniqueness follow from the nondegeneracy of the mod-2 cup pairing (Theorem 7.11(b)). By instability $\mathrm{Sq}^k x = 0$ for $k > n-k$, so $v_k = 0$ for $k > n/2$.

Theorem 18.30 (Wu's formula)**.** *For a closed manifold M, $w(TM) = \mathrm{Sq}(v)$, that is, $w_k(TM) = \sum_i \mathrm{Sq}^{k-i} v_i$.*

This is [10, Thm. 11.14]. Its significance: the Stiefel–Whitney classes of the tangent bundle of a closed manifold are determined by the cohomology ring and the action of $\mathcal{A}$, hence are *homotopy invariants* of M — a fact with no evident reason from the definition via the tangent bundle.

Example 18.31. $M = \mathbb{CP}^2$, $H^* = \mathbb{F}_2[x]/(x^3)$. $v_1 = 0$, since v_1 lies in $H^1 = 0$; and v_2 satisfies $v_2 x = \mathrm{Sq}^2 x = x^2$, so $v_2 = x$. Hence $w(\mathbb{CP}^2) = \mathrm{Sq}(1+x) = 1 + x + x^2$: $w_2(\mathbb{CP}^2) = x \neq 0$, so $\mathbb{CP}^2$ does not admit a spin structure, and $w_4 = x^2$ evaluates to $1 = \chi(\mathbb{CP}^2) \bmod 2$. Compare $w(\mathbb{CP}^2) = (1+x)^3$ from the tangent bundle formula $T\mathbb{CP}^n \oplus \mathbb{C} = (n+1)\gamma^*$ [10, §14]: $(1+x)^3 = 1 + 3x + 3x^2 = 1 + x + x^2 \bmod 2$.

18.5.4 Reduced powers

For an odd prime p there is a parallel family.

Theorem 18.32 (Reduced powers; recalled)**.** *There are stable operations*

$$P^i \colon H^n(X;\mathbb{F}_p) \to H^{n+2i(p-1)}(X;\mathbb{F}_p), \qquad i \geq 0,$$

natural, additive, satisfying the Cartan formula $P^k(xy) = \sum_{i+j=k} P^i(x)P^j(y)$, $P^0 = \mathrm{id}$, $P^i(x) = x^p$ if $|x| = 2i$, and $P^i(x) = 0$ if $2i > |x|$. Together with the Bockstein β they generate the mod-p Steenrod algebra $\mathcal{A}_p$, subject to Adem relations; P^i is decomposable unless i is a power of p.

See [1, §4.L, Thm. 4L.16 and Prop. 4L.8]. On $H^*(\mathbb{CP}^\infty;\mathbb{F}_p) = \mathbb{F}_p[c]$ one gets, as in Proposition 18.11, $P(c) = c + c^p$ and $P^i(c^k) = \binom{k}{i} c^{k+i(p-1)}$.

Corollary 18.33. *If $H^*(X;\mathbb{Z})$ is a polynomial algebra $\mathbb{Z}[\alpha]$, possibly truncated by $\alpha^m = 0$ with $m > 3$, then $|\alpha|$ is 2 or 4.*

This is [1, Cor. 4L.10]: mod 2, $\mathrm{Sq}^{|\alpha|}\,\alpha = \alpha^2 \neq 0$ forces $|\alpha|$ to be a power of 2; mod 3, $|\alpha|$ even and $P^{|\alpha|/2}\alpha = \alpha^3 \neq 0$ decomposes into P^{3^j}'s, which forces $|\alpha|$ to be a power of 3 times a divisor of 4. So $\mathbb{CP}^\infty$ and $\mathbb{HP}^\infty$ exhaust the possibilities: there is no "octonionic projective space" of dimension > 16.

18.6 Further exercises

Exercise 18.34 (easy)**.** Show that $\mathrm{Sq}^1 = 0$ on $H^*(\mathbb{CP}^n;\mathbb{F}_2)$, on $H^*(S^p \times S^q;\mathbb{F}_2)$ and on $H^*(L(p,q);\mathbb{F}_2)$ for p odd, and compute Sq^1 on $H^*(\mathbb{RP}^2 \times \mathbb{RP}^2;\mathbb{F}_2)$.

Source: original

Solution. In the first three cases the integral cohomology has no 2-torsion (for $L(p,q)$ with p odd the only torsion is $\mathbb{Z}/p$, Exercise 6.19), so $\mathrm{Sq}^1 = 0$ by Corollary 18.16. (For $L(p,q)$, p odd, $H^*(L;\mathbb{F}_2) = H^*(S^3;\mathbb{F}_2)$ anyway.) For $\mathbb{RP}^2 \times \mathbb{RP}^2$, $H^* = \mathbb{F}_2[a,b]/(a^3,b^3)$ with $\mathrm{Sq}^1 a = a^2$, $\mathrm{Sq}^1 b = b^2$ (Proposition 18.11(c)). By Cartan, $\mathrm{Sq}^1(ab) = a^2b + ab^2$, $\mathrm{Sq}^1(a^2) = 2a^3 = 0$, $\mathrm{Sq}^1(a^2b) = a^2b^2$, $\mathrm{Sq}^1(ab^2) = a^2b^2$, $\mathrm{Sq}^1(a^2b^2) = 0$. In particular $\ker \mathrm{Sq}^1 / \operatorname{im} \mathrm{Sq}^1$ is $\mathbb{F}_2$ in degree 0 only (check: it is the tensor square of the Bockstein homology of $\mathbb{RP}^2$, which is $\mathbb{F}_2$ in degree 0), so all 2-torsion in $H^*(\mathbb{RP}^2 \times \mathbb{RP}^2;\mathbb{Z})$ has order 2 and the free part is in degree 0 (Proposition 9.17), consistent with Example 8.22.

Exercise 18.35 (medium)**.** Compute the Wu classes of $\mathbb{RP}^4$ and use Theorem 18.30 to recover $w(\mathbb{RP}^4)$; compare with $(1+a)^5$ from Example 12.6.

Source: original

Solution. $H^*(\mathbb{RP}^4;\mathbb{F}_2) = \mathbb{F}_2[a]/(a^5)$. $v_k = 0$ for $k > 2$. v_1: for $x = a^3$, $\mathrm{Sq}^1 a^3 = 3a^4 = a^4$, so $v_1 a^3 = a^4$ and $v_1 = a$. v_2: for $x = a^2$, $\mathrm{Sq}^2 a^2 = a^4$, so $v_2 = a^2$. Thus $v = 1 + a + a^2$. Then $\mathrm{Sq}(v) = 1 + \mathrm{Sq}(a) + \mathrm{Sq}(a^2) = 1 + (a + a^2) + (a^2 + a^4)$, using $\mathrm{Sq}(a^2) = \mathrm{Sq}(a)^2 = a^2(1+a)^2 = a^2 + a^4$. So $w(\mathbb{RP}^4) = 1 + a + a^4$. And $(1+a)^5 = (1+a)^4(1+a) = (1+a^4)(1+a) = 1 + a + a^4$. They agree.

Exercise 18.36 (medium)**.** Show that all Stiefel–Whitney classes of a closed orientable 3-manifold vanish.

Source: original

Solution. $v_k = 0$ for $k > 3/2$, so $v = 1 + v_1$. For $x \in H^2$, $v_1 x = \mathrm{Sq}^1 x$; and $w_1 = v_1$ by Theorem 18.30 ($w_1 = \mathrm{Sq}^0 v_1 + \mathrm{Sq}^1 v_0 = v_1$). Orientable means $w_1 = 0$, so $v = 1$ and $w = \mathrm{Sq}(1) = 1$.

Exercise 18.37 (medium)**.** Let $\eta\colon S^{n+1} \to S^n$ be a suspension of the Hopf map ($n \geq 3$). Show, using Sq^2 on the mapping cone of the composite $\eta \circ \Sigma\eta\colon S^{n+2} \to S^n$, that this is not by itself enough to decide whether η^2 is essential, and explain what extra structure is needed.

SOURCE: original (contrast with [1, Prop. 4L.11], which proves $\eta^2 \neq 0$ by a finer argument)

Solution. $C = S^n \cup_{\eta^2} e^{n+3}$ has $\widetilde{H}^*(C;\mathbb{F}_2)$ in degrees n and $n+3$. A primary operation detecting η^2 would have to be a nonzero element of $\mathcal{A}^3$ acting from degree n to $n+3$; $\mathcal{A}^3$ is spanned by $\mathrm{Sq}^3 = \mathrm{Sq}^1\mathrm{Sq}^2$ and $\mathrm{Sq}^2\mathrm{Sq}^1$, and both factor through degree $n+1$ or $n+2$, where $H^*(C) = 0$. So every primary operation vanishes on C, exactly as on $S^n \vee S^{n+3}$: primary operations cannot detect η^2. One needs either a *secondary* operation (built from the relation $\mathrm{Sq}^2\mathrm{Sq}^2 + \mathrm{Sq}^3\mathrm{Sq}^1 = 0$), or the argument of [1, Prop. 4L.11], which uses the three-cell complex $S^n \cup_\eta e^{n+2} \cup e^{n+4}$ and Sq^2 twice.

18.7 Connections and further reading

The Steenrod operations are the first invariants of the book that are not induced by a product. They resolve the "ring is not enough" question, decide the Hopf invariant one problem up to powers of 2, show that η is stably essential, and make the Stiefel–Whitney classes of manifolds homotopy invariant. Chapter 27 studies the algebra $\mathcal{A}$ itself; Chapter 26 uses $\mathrm{Sq}^1 = \rho\tilde{\beta}$ to identify integral Euler classes; Part VI uses the action of $\mathcal{A}$ on the mod-2 homology of configuration spaces, which is the classical route (Cohen–Lada–May) to their structure.

References: Hatcher [1, §4.L] for constructions and proofs; Steenrod–Epstein [14] for the original treatment; Mosher–Tangora [15] for applications and computations; Milnor–Stasheff [10, §8, §11] for Thom's definition and Wu's formula.

Chapter 19

Rational homotopy: the first computations

19.1 The problem

Homotopy groups of spheres are wild: $\pi_{n+k}(S^n)$ is known only for small k and displays no simple pattern. Yet after tensoring with $\mathbb{Q}$ the picture is completely regular: $\pi_*(S^n)\otimes\mathbb{Q}$ is $\mathbb{Q}$ in degree n, and in degree $2n-1$ when n is even, and zero otherwise. This chapter proves that theorem of Serre, and in doing so sets up the three tools that make rational homotopy theory work: Serre's "modulo a class of groups" version of the Hurewicz theorem, the rational cohomology of Eilenberg–MacLane spaces, and the transgression in the Serre spectral sequence. The parity phenomenon — even spheres carry an extra rational class, detected by a Whitehead product — is the seed of the algebraic models of Chapter 24, and through them of the rational models of configuration spaces in Part VI.

Prerequisites. The Serre spectral sequence with its multiplicative structure (Chapters 15 and 16); Serre classes (Definition 15.24); the Hurewicz and Whitehead theorems (Chapter 13); Hopf invariant (Definition 16.17).

19.2 Homotopy modulo a class of groups

Let $\mathcal{C}$ be one of the following Serre classes of abelian groups (Definition 15.24, Exercises 15.25 and 15.26): the finitely generated groups; the torsion groups (or those whose orders involve only primes from a set P); the finite groups (or those of P-power order). A homomorphism is an *isomorphism mod $\mathcal{C}$* if its kernel and cokernel lie in $\mathcal{C}$.

Theorem 19.1 (Serre)**.** *Let X be simply connected (more generally, path-connected with π_1 acting trivially on all π_n).*

(a) $\pi_n(X)\in\mathcal{C}$ for all n iff $H_n(X;\mathbb{Z})\in\mathcal{C}$ for all $n>0$.

(b) (Hurewicz mod $\mathcal{C}$) If $\pi_i(X)\in\mathcal{C}$ for $i<n$, then $H_i(X)\in\mathcal{C}$ for $0<i<n$ and the Hurewicz map $\pi_n(X)\to H_n(X)$ is an isomorphism mod $\mathcal{C}$.

(c) If $F\to E\to B$ is a fibration of path-connected spaces with $\pi_1(B)$ acting trivially on $H_*(F)$, and two of F,E,B have $\widetilde{H}_*\in\mathcal{C}$, so does the third.

(d) If $\pi \in \mathcal{C}$ (for the classes of finite or torsion groups), then $\widetilde{H}_*(K(\pi,n);\mathbb{Z}) \in \mathcal{C}$ for all $n \geq 1$.

This is [16, Thm. 1.7, Thm. 1.8, Lemma 1.9, Lemma 1.10]. The engine of the proofs is the Serre spectral sequence: in (c), every E^2-term $H_p(B;H_q(F))$ is built from groups in $\mathcal{C}$ by the operations under which $\mathcal{C}$ is closed (tensor products and Tor with finitely generated groups, subquotients, extensions), and so is the abutment; (b) is proved by induction on n using the path–loop fibration and (c).

Corollary 19.2. *A simply connected CW complex with finitely generated homology groups (for instance a finite simply connected complex) has finitely generated homotopy groups.*

For rational homotopy we use $\mathcal{C}$ = torsion groups: a group lies in $\mathcal{C}$ iff it vanishes after $\otimes\mathbb{Q}$, and an isomorphism mod $\mathcal{C}$ is a homomorphism that becomes an isomorphism after $\otimes\mathbb{Q}$ ($\mathbb{Q}$ is flat). So (b) says: if $\pi_i(X)\otimes\mathbb{Q}=0$ for $i<n$ then $\widetilde{H}_i(X;\mathbb{Q})=0$ for $i<n$ and $\pi_n(X)\otimes\mathbb{Q}\cong H_n(X;\mathbb{Q})$ — the *rational Hurewicz theorem* in its first nontrivial degree.

19.3 Eilenberg–MacLane spaces over $\mathbb{Q}$

Theorem 19.3. *For $n \geq 1$,*

$$H^*\big(K(\mathbb{Z},n);\mathbb{Q}\big) = \begin{cases} \mathbb{Q}[x_n], & n \text{ even},\\ \Lambda(x_n), & n \text{ odd}, \end{cases} \qquad |x_n| = n,$$

and $\widetilde{H}^*(K(\pi,n);\mathbb{Q})=0$ for every finite group π (for $n=1$, any finite group; for $n\geq 2$, finite abelian).

Proof. The finite case is Theorem 19.1(d) with $\mathcal{C}$ the finite groups ($\widetilde{H}_*$ finite, hence zero rationally, and then $\widetilde{H}^*(-;\mathbb{Q})=0$ by universal coefficients). For $\mathbb{Z}$ we induct on n; $K(\mathbb{Z},1)=S^1$ has $H^*=\Lambda(x_1)$. Assume the result for $n-1$ and consider the path–loop fibration $K(\mathbb{Z},n-1)\to P\to B=K(\mathbb{Z},n)$, $n\geq 2$, with P contractible and B simply connected, so the rational cohomology spectral sequence has $E_2=H^*(B;\mathbb{Q})\otimes H^*(K(\mathbb{Z},n-1);\mathbb{Q})$ and $E_\infty=\mathbb{Q}$ in bidegree $(0,0)$. Let y generate $H^{n-1}(K(\mathbb{Z},n-1);\mathbb{Q})$, in $E_2^{0,n-1}$. Since B is $(n-1)$-connected, $E_2^{p,q}=0$ for $0<p<n$, so the only differential that can act on y is $d_n\colon E_n^{0,n-1}\to E_n^{n,0}=H^n(B;\mathbb{Q})$, and y must die: $x:=d_n(y)\neq 0$.

n even (y odd, fibre $\Lambda(y)$). The page has two rows, $q=0$ and $q=n-1$, each a copy of $H^*(B;\mathbb{Q})$, and by the Leibniz rule $d_n(yb)=xb$. Everything above $(0,0)$ must die, so $d_n\colon E^{p,n-1}=y\,H^p(B)\to H^{p+n}(B)$ is an isomorphism for all $p\geq 0$, and $H^p(B)=0$ for $0<p<n$. Hence multiplication by x is an isomorphism $H^p(B)\to H^{p+n}(B)$ for all $p\geq 0$, and $H^*(B;\mathbb{Q})=\mathbb{Q}[x]$.

n odd (y even, fibre $\mathbb{Q}[y]$). Now $|x|=n$ is odd, so $x^2=0$, and $d_n(y^k)=ky^{k-1}x$, an isomorphism $\mathbb{Q}y^k\to\mathbb{Q}y^{k-1}x$ over $\mathbb{Q}$. So the subquotient $\mathbb{Q}[y]\otimes\Lambda(x)$ of E_n is acyclic for d_n above degree 0. It remains to show that $H^p(B;\mathbb{Q})=0$ for $p\neq 0,n$. Suppose not, and let p be minimal with $p\neq 0,n$ and $H^p(B)\neq 0$; take $0\neq b\in E_2^{p,0}$. Classes in the bottom row support no differentials, so b survives unless it is hit by some d_r from $E_r^{p-r,r-1}$. By minimality $E_2^{p-r,r-1}\neq 0$ only if $p-r\in\{0,n\}$ and $r-1$ is a multiple of $n-1$. The classes there are y^k (if $p-r=0$) or xy^k (if $p-r=n$); but by the computation just made every y^k with $k\geq 1$ supports a nonzero d_n and every xy^k is hit

by d_n, so none of them survives to E_r with $r > n$, and for $r \le n$ the degrees do not match ($r = n$ would need $p = n$ or $p = 2n$ with target in row $n-1$, not row 0). So b survives to E_∞, contradicting contractibility of P. Hence $H^*(B;\mathbb{Q}) = \Lambda(x)$. □

Example 19.4 (Degree two)**.** $K(\mathbb{Z},2) \simeq \mathbb{CP}^\infty$ (Exercise 14.6) and $H^*(\mathbb{CP}^\infty;\mathbb{Z}) = \mathbb{Z}[c]$ (Example 16.2); rationally $\mathbb{Q}[c]$, the even case of Theorem 19.3.

Exercise 19.5 (medium)**.** Compute $H^*(K(\mathbb{Z},3);\mathbb{Q})$ directly from $H^*(K(\mathbb{Z},2);\mathbb{Q}) = \mathbb{Q}[c]$ by the argument of Theorem 19.3, and explain why the same computation over $\mathbb{Z}$ breaks down.

SOURCE: orig-book (expanded)

Solution. Fibration $K(\mathbb{Z},2) \to P \to K(\mathbb{Z},3)$; with $x = d_3(c)$, the Leibniz rule gives $d_3(c^k) = kc^{k-1}x$, an isomorphism over $\mathbb{Q}$. As in the proof, every possible additional class in the base would survive, so $H^*(K(\mathbb{Z},3);\mathbb{Q}) = \Lambda(x)$, $|x| = 3$. Over $\mathbb{Z}$, $d_3(c^k) = kc^{k-1}x$ is *not* onto $\mathbb{Z}c^{k-1}x$ for $k \ge 2$: the cokernel $\mathbb{Z}/k$ must be killed by something else, so $H^*(K(\mathbb{Z},3);\mathbb{Z})$ has torsion in infinitely many degrees — the first being $H^6(K(\mathbb{Z},3);\mathbb{Z}) = \mathbb{Z}/2$, coming from $k = 2$ (the class cx is twice $d_3(c^2/2)$, which does not exist integrally). This is the integral shadow of the operation Sq^2 on ι_3.

19.4 Serre's theorem on the rational homotopy of spheres

Theorem 19.6 (Serre)**.** *For $n \ge 2$, $\pi_k(S^n)$ is finite for $k \ne n$, except that for n even $\pi_{2n-1}(S^n)$ is $\mathbb{Z}$ plus a finite group. Hence*

$$\pi_k(S^n) \otimes \mathbb{Q} = \begin{cases} \mathbb{Q}, & k = n, \\ \mathbb{Q}, & k = 2n-1 \text{ and } n \text{ even}, \\ 0, & \text{otherwise}. \end{cases}$$

Proof. All groups are finitely generated (Corollary 19.2), so "finite" is equivalent to "zero after $\otimes\mathbb{Q}$", and we may work mod the class $\mathcal{C}$ of finite groups. Let $S^n \to K(\mathbb{Z},n)$ induce an isomorphism on π_n and turn it into a fibration with fibre F. From the long exact sequence, F is n-connected and $\pi_k(F) \cong \pi_k(S^n)$ for $k > n$. Turning $F \to S^n$ into a fibration in turn gives a fibration $K(\mathbb{Z},n-1) \to X \to S^n$ with $X \simeq F$ (the homotopy fibre of $F \to S^n$ is $\Omega K(\mathbb{Z},n) = K(\mathbb{Z},n-1)$). Run the rational cohomology spectral sequence: $E_2 = H^*(S^n;\mathbb{Q}) \otimes H^*(K(\mathbb{Z},n-1);\mathbb{Q})$, two columns $p = 0, n$, and since X is n-connected, the transgression d_n of the generator $y \in H^{n-1}(K(\mathbb{Z},n-1);\mathbb{Q})$ is a nonzero multiple of the generator ι of $H^n(S^n;\mathbb{Q})$.

n odd. Then $|y| = n-1$ is even, the fibre is $\mathbb{Q}[y]$, and $d_n(y^k) = ky^{k-1}\iota$ (up to a nonzero scalar) is an isomorphism for all $k \ge 1$: everything cancels and $\widetilde{H}^*(X;\mathbb{Q}) = 0$. By Theorem 19.1(a) for $\mathcal{C}$ = finite groups (applied to the simply connected X with finitely generated homology, whose rational homology vanishes, so its reduced homology is finite), all $\pi_k(X)$ are finite, hence $\pi_k(S^n)$ is finite for $k > n$.

n even. Then the fibre is $\Lambda(y)$, $d_n(y) = \iota$ (up to scalar) kills y and ι, and $y\iota$ survives (it is a cycle: $d_n(y\iota) = \pm\iota^2 = 0$). So $H^*(X;\mathbb{Q}) \cong H^*(S^{2n-1};\mathbb{Q})$. By Theorem 19.1(b), $\pi_k(X)$ is finite for $k < 2n-1$ and $\pi_{2n-1}(X) \cong H_{2n-1}(X)$ mod finite groups, i.e. $\mathbb{Z}$ plus a finite group. For $k > 2n-1$, let Y be the $(2n-2)$-nd Postnikov section of X; its homotopy groups are $\pi_k(X)$ for $k \le 2n-2$, all finite, so $\widetilde{H}_*(Y)$ is finite by Theorem 19.1(a). Turn $X \to Y$ into a fibration with fibre Z. Then Z is $(2n-2)$-connected and $\pi_k(Z) \cong \pi_k(X)$

for $k \geq 2n-1$. In the rational homology spectral sequence of $Z \to X \to Y$ (Y simply connected) every column $p > 0$ is $H_p(Y; H_q(Z;\mathbb{Q})) = 0$, since $\widetilde{H}_*(Y;\mathbb{Q}) = 0$; so the edge map is an isomorphism $H_*(Z;\mathbb{Q}) \cong H_*(X;\mathbb{Q}) \cong H_*(S^{2n-1};\mathbb{Q})$. Now repeat the odd case with Z in place of S^{2n-1}: a map $Z \to K(\mathbb{Z}, 2n-1)$ inducing an isomorphism on π_{2n-1} modulo torsion is a rational cohomology isomorphism (both are Λ(generator) rationally), so its fibre has finite reduced homology and finite homotopy groups. Hence $\pi_k(Z)$, and with it $\pi_k(S^n)$, is finite for $k > 2n-1$. □

Remark 19.7. The proof illustrates the standard strategy: compare X with an Eilenberg–MacLane space by a map inducing an isomorphism on the first homotopy group, compute the fibre's rational cohomology with the spectral sequence, and iterate. For n even the iteration stops after one step because $H^*(S^n;\mathbb{Q})$ differs from $H^*(K(\mathbb{Z},n);\mathbb{Q}) = \mathbb{Q}[x]$ exactly by the relation $x^2 = 0$, and killing that relation costs one new class, in degree $2n-1$. This is precisely the generator y with $dy = x^2$ of the Sullivan model of S^n (Example 24.7).

19.5 Rational equivalences

Definition 19.8. A map $f\colon X \to Y$ of simply connected spaces is a *rational equivalence* if it induces an isomorphism $\pi_*(X) \otimes \mathbb{Q} \to \pi_*(Y) \otimes \mathbb{Q}$.

Theorem 19.9 (Rational Whitehead theorem)**.** *For a map $f\colon X \to Y$ of simply connected spaces, the following are equivalent: f induces an isomorphism on $\pi_* \otimes \mathbb{Q}$; on $H_*(-;\mathbb{Q})$; on $H^*(-;\mathbb{Q})$.*

This is the Serre–Whitehead theorem; see [17, Thm. 8.6]. The proof applies Theorem 19.1 with $\mathcal{C}$ the torsion groups to the homotopy fibre of f (the equivalence of homology and cohomology is universal coefficients over $\mathbb{Q}$).

Example 19.10. For n odd the map $S^n \to K(\mathbb{Z},n)$ is a rational equivalence: both have rational cohomology $\Lambda(x_n)$. Rationally, an odd sphere is an Eilenberg–MacLane space. For n even this fails, and the defect is Theorem 19.6.

Rationalisation. Every simply connected space X has a *rationalisation* $X \to X_{\mathbb{Q}}$, a map to a space with rational vector spaces as homotopy groups which is a rational equivalence; it is unique up to homotopy and functorial [16, Thm. 1.23]. Two simply connected spaces have the *same rational homotopy type* if their rationalisations are homotopy equivalent, equivalently if they are connected by a zig-zag of rational equivalences. Rational homotopy theory is the study of these types, and its central result (Chapter 24) is that, for spaces of finite type, they are classified by purely algebraic objects.

19.6 Loop spaces and Whitehead products

Exercise 19.11 (hard)**.** Compute $H^*(\Omega S^{2m};\mathbb{Q})$ as a ring.

SOURCE: orig-book

Hint. *Use the path–loop fibration over S^{2m}. After $d_{2m}(x) = \iota$, the class $x\iota$ is left over; a class killing it would sit in degree $4m-2$.*

Solution. By Example 15.7, $H_*(\Omega S^{2m};\mathbb{Q})$ is $\mathbb{Q}$ in each degree $(2m-1)k$. Let x generate degree $2m-1$; $x^2=0$ since x is odd. In the spectral sequence of $\Omega S^{2m}\to P\to S^{2m}$ only d_{2m} acts and $d_{2m}(x)=\iota$; then $d_{2m}(x\iota)=\pm\iota^2=0$, so $x\iota\in E^{2m,2m-1}$ must be hit from $E^{0,4m-2}$: there is $y\in H^{4m-2}$ with $d_{2m}(y)=x\iota$. Then $d_{2m}(y^k)=ky^{k-1}x\iota$ and $d_{2m}(xy^k)=\iota y^k$ (up to sign), isomorphisms over $\mathbb{Q}$, so everything cancels. The algebra $\Lambda(x)\otimes\mathbb{Q}[y]$ has exactly one basis element (y^j or xy^j) in each degree $(2m-1)k$, matching the additive answer, so $H^*(\Omega S^{2m};\mathbb{Q})=\Lambda(x)\otimes\mathbb{Q}[y]$, $|x|=2m-1$, $|y|=4m-2$.

Warning 19.12. For an odd sphere $H^*(\Omega S^{2m+1};\mathbb{Q})=\mathbb{Q}[x]$ (Exercise 16.3); for an even sphere one needs an extra polynomial generator. By the Milnor–Moore theorem (Chapter 25) this reflects the extra rational homotopy class of Theorem 19.6: $\pi_*(\Omega S^{2m})\otimes\mathbb{Q}$ is $\mathbb{Q}$ in degrees $2m-1$ and $4m-2$, the second being the loop of the class in $\pi_{4m-1}(S^{2m})$.

Definition 19.13. Let $\alpha\in\pi_m(X)$ and $\beta\in\pi_n(X)$. The product $S^m\times S^n$ is $S^m\vee S^n$ with one $(m+n)$-cell attached along a map $w\colon S^{m+n-1}\to S^m\vee S^n$. The *Whitehead product* is $[\alpha,\beta]=(\alpha\vee\beta)_*[w]\in\pi_{m+n-1}(X)$.

Exercise 19.14 (medium)**.** Verify that $S^m\times S^n$ is obtained from $S^m\vee S^n$ by attaching a single $(m+n)$-cell.

SOURCE: orig-book

Solution. With $S^m=e^0\cup e^m$ and $S^n=e^0\cup e^n$, the product CW structure has cells $e^0\times e^0$, $e^m\times e^0$, $e^0\times e^n$, $e^m\times e^n$; the first three form $S^m\vee S^n$, so the product is that wedge with one $(m+n)$-cell, whose characteristic map is $D^m\times D^n\cong D^{m+n}\to S^m\times S^n$ and whose attaching map is its restriction to $\partial(D^m\times D^n)=\partial D^m\times D^n\cup D^m\times\partial D^n$.

Exercise 19.15 (easy)**.** In $H^*(S^n\times S^n;\mathbb{Z})$ let a,b be the pullbacks of a generator along the two projections. Show $(a+b)^2=(1+(-1)^n)\,ab$.

SOURCE: orig-book

Solution. $a^2=b^2=0$ (pulled back from $H^{2n}(S^n)=0$) and $ba=(-1)^{n^2}ab=(-1)^nab$. So $(a+b)^2=ab+ba=(1+(-1)^n)ab$.

Proposition 19.16 (The Whitehead square)**.** *Let $n\geq 2$ and $\iota\in\pi_n(S^n)$ the identity class. The Whitehead square $[\iota,\iota]\in\pi_{2n-1}(S^n)$ has Hopf invariant $H([\iota,\iota])=\pm2$ for n even, and 0 for n odd. In particular, for n even $[\iota,\iota]$ has infinite order and represents a generator of $\pi_{2n-1}(S^n)\otimes\mathbb{Q}$.*

Proof. Let $C=S^n\cup_{[\iota,\iota]}e^{2n}$. The fold map $\nabla\colon S^n\vee S^n\to S^n$ satisfies $\nabla\circ w\simeq[\iota,\iota]$ by definition, so ∇ extends to a map $\Phi\colon S^n\times S^n\to C$ which is ∇ on the wedge and maps the top cell to the top cell by a homeomorphism on the interior (the characteristic map of the top cell of $S^n\times S^n$, followed by the extension). On cellular cochains, Φ^* sends the generator $\alpha\in H^n(C)$ to $a+b$ (the fold map is the identity on each summand) and the generator $\beta\in H^{2n}(C)$ to $\pm ab$ (degree one on the top cell). By Exercise 19.15, $\Phi^*(\alpha^2)=(a+b)^2=(1+(-1)^n)ab$, while $\Phi^*(H\beta)=\pm H\,ab$. Since Φ^* is injective on H^{2n} (an isomorphism $\mathbb{Z}\to\mathbb{Z}$), $H([\iota,\iota])=\pm(1+(-1)^n)$. For n even this is $\pm2\neq0$; the Hopf invariant is a homomorphism $\pi_{2n-1}(S^n)\to\mathbb{Z}$ [1, §4.B], so $[\iota,\iota]$ has infinite order, and by Theorem 19.6 it spans $\pi_{2n-1}(S^n)\otimes\mathbb{Q}$. □

Remark 19.17 (Where the extra class comes from). Three descriptions of one phenomenon: the extra generator y of $H^*(\Omega S^{2m};\mathbb{Q})$ (Exercise 19.11); the nonvanishing

of $(a+b)^2$ in $S^n \times S^n$ for n even (Exercise 19.15); and the Whitehead square of infinite order (Proposition 19.16). The Hopf invariant 2, never 1, shows that the Whitehead square does not solve the Hopf invariant one problem. In Chapter 24 the same class appears as the generator y, $dy = x^2$, of the minimal model of S^{2m}, and in Chapter 25 as the bracket $[x, x]$ in the homotopy Lie algebra.

19.7 Stable rational homotopy

Theorem 19.18. *For every CW complex X, the stable homotopy groups satisfy $\pi_k^s(X) \otimes \mathbb{Q} \cong \widetilde{H}_k(X; \mathbb{Q})$.*

Proof sketch. $X \mapsto \pi_*^s(X) \otimes \mathbb{Q}$ is a reduced homology theory ($\otimes\mathbb{Q}$ preserves exactness), and its coefficients are $\pi_k^s \otimes \mathbb{Q}$, which is $\mathbb{Q}$ for $k = 0$ and 0 otherwise by Theorem 19.6 (for $k > 0$ and n large, $\pi_{n+k}(S^n)$ is finite). The stable Hurewicz map is a natural transformation to $\widetilde{H}_*(-; \mathbb{Q})$ which is an isomorphism on coefficients, hence on all CW complexes by the uniqueness of homology theories with given coefficients [1, §4.F], [16, §1.2]. □

Unstably the situation is richer: the rational homotopy of a finite complex can be infinite-dimensional and grows exponentially for wedges of spheres (Chapter 25).

19.8 Exercises

Exercise 19.19 (medium)**.** Compute $\pi_*(\mathbb{CP}^n) \otimes \mathbb{Q}$.

SOURCE: original

Solution. From $S^1 \to S^{2n+1} \to \mathbb{CP}^n$: $\pi_k(\mathbb{CP}^n) \cong \pi_k(S^{2n+1})$ for $k \geq 3$, $\pi_2(\mathbb{CP}^n) = \mathbb{Z}$, $\pi_1 = 0$. By Theorem 19.6 for the odd sphere S^{2n+1}, $\pi_*(\mathbb{CP}^n) \otimes \mathbb{Q}$ has rank one in degrees 2 and $2n+1$ and vanishes in all other degrees.

Exercise 19.20 (medium)**.** Using the Hopf fibration $S^3 \to S^7 \to S^4$, compute $\pi_*(S^4) \otimes \mathbb{Q}$ in a second way and compare with Theorem 19.6. Which element generates $\pi_7(S^4) \otimes \mathbb{Q}$?

SOURCE: original

Solution. The long exact sequence, tensored with $\mathbb{Q}$ (exact), gives $\pi_k(S^3) \otimes \mathbb{Q} \to \pi_k(S^7) \otimes \mathbb{Q} \to \pi_k(S^4) \otimes \mathbb{Q} \to \pi_{k-1}(S^3) \otimes \mathbb{Q} \to \pi_{k-1}(S^7) \otimes \mathbb{Q}$. By the odd case of Theorem 19.6, $\pi_*(S^3) \otimes \mathbb{Q}$ is $\mathbb{Q}$ in degree 3 and $\pi_*(S^7) \otimes \mathbb{Q}$ is $\mathbb{Q}$ in degree 7. The connecting map $\pi_4(S^4) \otimes \mathbb{Q} \to \pi_3(S^3) \otimes \mathbb{Q}$ is an isomorphism (it is the isomorphism of the classical computation $\pi_4(S^4) \cong \pi_3(S^3)$ from the bundle, the total space S^7 being 6-connected). So $\pi_*(S^4) \otimes \mathbb{Q}$ is $\mathbb{Q}$ in degree 4 and, via p_*, $\mathbb{Q}$ in degree 7, generated by the Hopf map $\nu = p\colon S^7 \to S^4$, which has Hopf invariant ± 1. The Whitehead square $[\iota_4, \iota_4]$ has Hopf invariant ± 2, so rationally $[\iota_4, \iota_4] = \pm 2\nu$ (the Hopf invariant $\pi_7(S^4) \to \mathbb{Z}$ is injective after $\otimes\mathbb{Q}$, both groups being $\mathbb{Q}$).

Exercise 19.21 (easy)**.** Compute $\pi_*(S^2 \times S^3) \otimes \mathbb{Q}$ and $\pi_*(S^2 \vee S^3) \otimes \mathbb{Q}$ in degrees ≤ 4, and decide whether $S^2 \times S^3$ and $S^2 \vee S^3 \cup e^5$ (any 5-cell) can have the same rational homotopy groups in degrees ≤ 3.

SOURCE: original

Solution. $\pi_k(X \times Y) = \pi_k(X) \oplus \pi_k(Y)$, so $\pi_*(S^2 \times S^3) \otimes \mathbb{Q}$ is $\mathbb{Q}$ in degrees 2, 3 (from $\pi_3(S^2) \otimes \mathbb{Q}$, spanned by the Hopf map, by Theorem 19.6) and 3 again (from S^3): dimensions $1, 2, 0$ in degrees

$2, 3, 4$. For the wedge, the inclusion $S^2 \vee S^3 \to S^2 \times S^3$ is the 4-skeleton (the product has one more cell, in dimension 5), hence an isomorphism on π_k for $k \leq 3$ and onto on π_4 (Exercise 11.12). So in degrees ≤ 3 both have $\pi_2 \otimes \mathbb{Q} = \mathbb{Q}$, $\pi_3 \otimes \mathbb{Q} = \mathbb{Q}^2$, and so does $S^2 \vee S^3 \cup e^5$ for any attaching map (adding a 5-cell does not change π_k, $k \leq 3$). In degree 4 the wedge has more: $\pi_4(S^2 \vee S^3) \otimes \mathbb{Q} \supseteq$ the Whitehead product $[\iota_2, \iota_3]$, which is killed in the product by the top cell (Definition 19.13).

19.9 Connections and further reading

The rational homotopy groups of spheres are determined by the parity of the dimension, and the even case is governed by the cup square, the Whitehead square and the Hopf invariant. Chapter 24 reorganises everything here into algebra: the minimal Sullivan model encodes exactly the rational homotopy groups (as generators) and the Whitehead products (as the quadratic part of the differential).

References: Hatcher's spectral sequences notes [16, §1.1–1.3] for the mod-$\mathcal{C}$ theory and Serre's theorem; Serre's original paper [18]; Félix–Halperin–Thomas [17] for rational homotopy theory in general.

Part IV

Abstract homotopy theory

Chapter 20

Adjunctions, Kan extensions, and nerve–realisation

20.1 The problem

Geometric realisation of simplicial sets, the nerve of a category, the classifying space of a group, the cellular chain complex of a CW complex: each is defined by a formula, and for each one proves properties — preserves colimits, has a right adjoint, is homotopy invariant — by a separate computation. This chapter isolates the single construction of which they are all instances, and proves its basic property once. The payoff is not elegance but transferability: in Chapters 22 and 23 the same construction produces the comparison between spaces and simplicial sets, and in Part VI the operadic constructions (factorisation homology, the realisation of the little-disks operad) are of exactly this form.

Prerequisites. Categories, functors, natural transformations, limits and colimits at the level of a first course in category theory [19].

20.2 Adjunctions

Definition 20.1. Functors $F\colon \mathcal{C} \to \mathcal{D}$ and $G\colon \mathcal{D} \to \mathcal{C}$ are *adjoint*, $F \dashv G$, if there are bijections $\phi_{c,d}\colon \mathcal{D}(Fc, d) \cong \mathcal{C}(c, Gd)$ natural in c and d.

Proposition 20.2. *An adjunction is equivalently given by natural transformations $\eta\colon \mathrm{id} \Rightarrow GF$ (unit) and $\varepsilon\colon FG \Rightarrow \mathrm{id}$ (counit) satisfying the triangle identities $\varepsilon_{Fc} \circ F\eta_c = \mathrm{id}_{Fc}$ and $G\varepsilon_d \circ \eta_{Gd} = \mathrm{id}_{Gd}$.*

Proof. Given ϕ, put $\eta_c = \phi(\mathrm{id}_{Fc})$ and $\varepsilon_d = \phi^{-1}(\mathrm{id}_{Gd})$. Naturality of ϕ gives $\phi(g) = Gg \circ \eta_c$ for $g\colon Fc \to d$ and $\phi^{-1}(h) = \varepsilon_d \circ Fh$ for $h\colon c \to Gd$. Then $\mathrm{id} = \phi^{-1}\phi(\mathrm{id}_{Fc}) = \varepsilon_{Fc} \circ F\eta_c$, and similarly for the other identity. Conversely, given η, ε, define $\phi(g) = Gg \circ \eta_c$ and $\psi(h) = \varepsilon_d \circ Fh$; the triangle identities and naturality show $\psi\phi = \mathrm{id}$ and $\phi\psi = \mathrm{id}$: e.g. $\psi\phi(g) = \varepsilon_d \circ FGg \circ F\eta_c = g \circ \varepsilon_{Fc} \circ F\eta_c = g$. □

Exercise 20.3 (easy)**.** Show that a left adjoint preserves all colimits that exist, and a right adjoint all limits.

Source: orig-book

Solution. Let $D\colon J \to \mathcal{C}$ have colimit $\operatorname{colim} D$. For every $d \in \mathcal{D}$, naturally in d, $\mathcal{D}(F\operatorname{colim} D, d) \cong \mathcal{C}(\operatorname{colim} D, Gd) \cong \lim_J \mathcal{C}(Dj, Gd) \cong \lim_J \mathcal{D}(FDj, d)$, the middle step being the universal property of the colimit and the last the adjunction in each variable. So $F\operatorname{colim} D$ represents the functor whose representing object, if it exists, is $\operatorname{colim} FD$; by Yoneda (Lemma 20.6) $F\operatorname{colim} D$ with the induced cocone is a colimit of FD. The dual argument handles limits.

Exercise 20.4 (medium)**.** Suppose F and F' are both left adjoint to $G\colon \mathcal{D} \to \mathcal{C}$. Show that there is a unique natural isomorphism $F \cong F'$ compatible with the adjunctions.

Source: MIT OCW 18.906 (Spring 2020), Problem Set I, Problem 1(c), reproduced in substance

Solution. For each c, $\mathcal{D}(Fc, -) \cong \mathcal{C}(c, G-) \cong \mathcal{D}(F'c, -)$ naturally, so by Yoneda (applied in $\mathcal{D}^{\mathrm{op}}$) there is a unique isomorphism $\alpha_c\colon F'c \to Fc$ inducing this composite; explicitly, α_c is the morphism corresponding under the adjunction $F' \dashv G$ to the unit $\eta_c\colon c \to GFc$ of $F \dashv G$. Naturality in c follows from naturality of the bijections, and compatibility with the adjunctions means precisely that α induces the composite bijection above, which determines it uniquely.

Exercise 20.5 (medium)**.** Show that every limit can be expressed as an equaliser of two maps between products: $\lim_J D$ is the equaliser of the two maps $\prod_{j\in J} Dj \rightrightarrows \prod_{u\colon j\to j'} Dj'$.

Source: MIT OCW 18.906 (Spring 2020), Problem Set I, Problem 1(a), reproduced in substance

Solution. Define $s, t\colon \prod_j Dj \to \prod_{u\colon j\to j'} Dj'$ by $s(x)_u = D(u)(x_j)$ and $t(x)_u = x_{j'}$. A map $f\colon c \to \prod_j Dj$, i.e. a family $f_j\colon c \to Dj$, equalises s and t iff $D(u)f_j = f_{j'}$ for all $u\colon j \to j'$, i.e. iff the f_j form a cone over D. So cones over D with vertex c are the same as maps from c into the equaliser, which is therefore a limit.

20.3 The Yoneda lemma and representable functors

Lemma 20.6 (Yoneda)**.** *For a functor $X\colon \mathcal{C}^{\mathrm{op}} \to \mathbf{Set}$ and $c \in \mathcal{C}$, the map $\mathrm{Nat}\big(\mathcal{C}(-,c), X\big) \to X(c)$, $\theta \mapsto \theta_c(\mathrm{id}_c)$, is a bijection, natural in c and X. In particular the* Yoneda embedding *$\mathcal{Y}\colon \mathcal{C} \to [\mathcal{C}^{\mathrm{op}}, \mathbf{Set}]$, $c \mapsto \mathcal{C}(-,c)$, is fully faithful.*

Proof. A natural θ is determined by $x = \theta_c(\mathrm{id}_c)$: for $f\colon b \to c$, $\theta_b(f) = \theta_b(\mathcal{C}(f,c)\mathrm{id}_c) = X(f)\theta_c(\mathrm{id}_c) = X(f)(x)$. Conversely $\theta_b(f) = X(f)(x)$ defines a natural transformation for every $x \in X(c)$. □

We have used the lemma repeatedly already: the classification of cohomology operations (Theorem 18.2) is the Yoneda lemma for the functor $H^n(-;G)$ on the homotopy category, represented by $K(\pi, m)$.

20.4 Kan extensions

Definition 20.7. Let $K\colon \mathcal{D} \to \mathcal{E}$ and $F\colon \mathcal{D} \to \mathcal{C}$. A *left Kan extension* of F along K is a functor $\mathrm{Lan}_K F\colon \mathcal{E} \to \mathcal{C}$ with a natural transformation $\eta\colon F \Rightarrow (\mathrm{Lan}_K F)K$ that is universal: every $\gamma\colon F \Rightarrow GK$ factors uniquely as $(\delta K)\circ\eta$ with $\delta\colon \mathrm{Lan}_K F \Rightarrow G$.

Equivalently, Lan_K is left adjoint to restriction $K^*\colon [\mathcal{E},\mathcal{C}] \to [\mathcal{D},\mathcal{C}]$, when it exists for all F.

Proposition 20.8 (Pointwise formula)**.** *If $\mathcal{D}$ is small and $\mathcal{C}$ cocomplete, then* $\operatorname{Lan}_K F$ *exists and*

$$(\operatorname{Lan}_K F)(e) = \operatorname*{colim}_{(K\downarrow e)} F,$$

the colimit over the comma category of pairs $(d,\ Kd \to e)$, of the functor $(d, Kd \to e) \mapsto Fd$.

Proof. A map $e \to e'$ induces a functor $(K \downarrow e) \to (K \downarrow e')$ (compose), hence a map of colimits: the formula defines a functor L. The component $\eta_d \colon Fd \to LKd$ is the coprojection at the object (d, id_{Kd}). Given $\gamma \colon F \Rightarrow GK$, define $\delta_e \colon Le \to Ge$ on the coprojection at $(d, f \colon Kd \to e)$ by $G(f) \circ \gamma_d \colon Fd \to GKd \to Ge$; these are compatible over the comma category by naturality of γ, so define δ_e, and $\delta K \circ \eta = \gamma$. Uniqueness: any such δ is determined on each coprojection, because (d, f) is the image of (d, id) under the functor induced by f, so the coprojection at (d, f) is $L(f) \circ \eta_d$, on which δ_e must be $G(f)\delta_{Kd}\eta_d = G(f)\gamma_d$. □

Example 20.9 (Colimits are Kan extensions)**.** With $\mathcal{E} = \mathbf{1}$, $(K \downarrow *) = \mathcal{D}$ and $\operatorname{Lan}_K F = \operatorname{colim}_{\mathcal{D}} F$.

Definition 20.10. A left Kan extension is *absolute* if it is preserved by every functor out of $\mathcal{C}$: for every $H \colon \mathcal{C} \to \mathcal{C}'$, $(H \operatorname{Lan}_K F, H\eta)$ is a left Kan extension of HF along K.

Absolute Kan extensions are exactly what one needs to transport derived functors (Chapter 22): a total left derived functor is an absolute Kan extension, and it is therefore preserved by composition with any functor.

20.5 Density

Theorem 20.11 (Density)**.** *Let $\mathcal{D}$ be small. Every presheaf $X \colon \mathcal{D}^{\mathrm{op}} \to \mathbf{Set}$ is canonically the colimit of representables, $X \cong \operatorname{colim}_{\int X} \mathcal{Y}$, where $\int X$ is the category of elements: objects (d, x) with $x \in X(d)$, morphisms $f \colon (d, x) \to (d', x')$ with $X(f)(x') = x$.*

Proof. For any presheaf Z, $\operatorname{Nat}(\operatorname{colim}_{\int X} \mathcal{Y}, Z) \cong \lim_{(d,x)\in \int X} \operatorname{Nat}(\mathcal{Y}d, Z) \cong \lim_{(d,x)} Z(d)$ by Lemma 20.6. An element of this limit assigns to each $x \in X(d)$ an element of $Z(d)$, compatibly with the maps of $\int X$ — exactly a natural transformation $X \Rightarrow Z$. So $\operatorname{colim} \mathcal{Y}$ and X represent the same functor, and Yoneda identifies them. □

Remark 20.12 (Coends). The colimit is written $X \cong \int^d X(d) \cdot \mathcal{Y}d$, where $S \cdot c$ is the coproduct of S copies of c; in general $\int^d F(d,d)$ is the coequaliser of $\coprod_{d\to d'} F(d', d) \rightrightarrows \coprod_d F(d,d)$.

20.6 Nerve and realisation

Construction 20.13. Let $\mathcal{D}$ be small, $\mathcal{C}$ cocomplete, $F \colon \mathcal{D} \to \mathcal{C}$. Define

$$\operatorname{Sing}_F \colon \mathcal{C} \to [\mathcal{D}^{\mathrm{op}}, \mathbf{Set}], \quad \operatorname{Sing}_F(c)(d) = \mathcal{C}(Fd, c),$$

$$|X|_F = \operatorname{Lan}_{\mathcal{Y}} F(X) = \int^d X(d) \cdot Fd.$$

Theorem 20.14 (Nerve–realisation adjunction)**.** $|\cdot|_F \dashv \operatorname{Sing}_F$*: there are natural bijections $\mathcal{C}(|X|_F, c) \cong \operatorname{Nat}(X, \operatorname{Sing}_F(c))$. Moreover $|\mathcal{Y}d|_F \cong Fd$.*

Proof. By the coend formula and the universal property of colimits,

$$\mathcal{C}\Big(\int^{d} X(d)\cdot Fd,\ c\Big) \cong \int_{d} \mathcal{C}\big(X(d)\cdot Fd, c\big) \cong \int_{d} \mathbf{Set}\big(X(d), \mathcal{C}(Fd,c)\big),$$

where $\int_d$ is the end (the limit dual to the coend). An element of the last end is a family of functions $X(d) \to \mathcal{C}(Fd,c) = \mathrm{Sing}_F(c)(d)$, natural in d: a natural transformation $X \Rightarrow \mathrm{Sing}_F(c)$. Naturality in X and c is evident. For $X = \mathcal{Y}d$, $\mathrm{Nat}(\mathcal{Y}d, \mathrm{Sing}_F\, c) \cong \mathrm{Sing}_F(c)(d) = \mathcal{C}(Fd,c)$ by Yoneda, so $|\mathcal{Y}d|_F$ and Fd represent the same functor. □

Example 20.15 (Geometric realisation)**.** $\mathcal{D} = \Delta$, $\mathcal{C} = \mathbf{Top}$, $F[n] = |\Delta^n|$. Then Sing_F is the singular simplicial set and $|X| = \big(\coprod_n X_n \times |\Delta^n|\big)/\sim$ is the usual realisation. Theorem 20.14 gives $|\cdot| \dashv \mathrm{Sing}$, and therefore realisation preserves colimits and Sing preserves limits (Exercise 20.3).

Example 20.16 (The nerve of a category)**.** $\mathcal{C} = \mathbf{Cat}$, $F[n] = [n]$, the poset $0 < \cdots < n$. Then $\mathrm{Sing}_F = N$ is the nerve, whose n-simplices are strings of n composable arrows, and $|\cdot|_F$ is its left adjoint τ_1, the "fundamental category" of a simplicial set (Chapter 23).

Exercise 20.17 (medium)**.** Let G be a discrete group, viewed as a one-object category. Show that $|NG|$ is a $K(G,1)$.

Source: orig-book

Solution. $(NG)_n = G^n$, with faces multiplying adjacent entries or dropping an outer one. Let EG be the simplicial set with $(EG)_n = G^{n+1}$ and d_i deleting the i-th entry; G acts freely by right multiplication on all entries, and $EG/G \cong NG$ via $(g_0,\dots,g_n) \mapsto (g_0 g_1^{-1}, \dots, g_{n-1}g_n^{-1})$. EG is the nerve of the category with objects G and exactly one morphism between any two objects; that category has a terminal object, so its nerve is contractible (a natural transformation from the identity functor to the constant functor at the terminal object realises to a homotopy). The free G-action realises to a covering space action (free actions of discrete groups on CW complexes permuting cells freely), so $|EG| \to |NG|$ is a covering with contractible total space; by Proposition 3.16 and Exercise 13.10, $\pi_1|NG| \cong G$ and higher homotopy vanishes.

Exercise 20.18 (medium)**.** Construct left and right adjoints to the forgetful functor $u\colon \mathbf{Top} \to \mathbf{Set}$, and deduce that the limit and the colimit in **Top** of any diagram are the limit and colimit of underlying sets with suitable topologies.

Source: MIT OCW 18.906 (Spring 2020), Problem Set I, Problem 2(c), reproduced in substance

Solution. Left adjoint: the discrete topology δS; a function $S \to uX$ is the same as a continuous map $\delta S \to X$. Right adjoint: the indiscrete topology ιS; every function $uX \to S$ is continuous $X \to \iota S$. Hence u preserves limits and colimits (Exercise 20.3): the underlying set of $\lim D$ is $\lim uD$, and of $\operatorname{colim} D$ is $\operatorname{colim} uD$. The topologies are forced by the universal properties: the initial topology for the limit (coarsest making the projections continuous) and the final topology for the colimit (finest making the coprojections continuous); one checks directly that these satisfy the universal properties.

20.7 Connections and further reading

Nerve–realisation is the template for every "build an object out of simplices" construction in the book: the bar construction, homotopy colimits (Chapter 21), Sullivan's spatial realisation (Chapter 24), and the realisation of operads (Part VI). References: Mac Lane [19] for adjunctions, Kan extensions and coends; Riehl [20] for the homotopical uses; Haynes Miller's MIT notes [21] for the topological setting.

Chapter 21

Homotopy limits and colimits

21.1 The problem

Strict limits and colimits are not homotopy invariant: the pushout of $* \leftarrow S^1 \to *$ is a point, while that of the homotopy equivalent diagram $D^2 \leftarrow S^1 \to D^2$ is S^2. Every construction of Parts I–III was only ever meaningful up to homotopy, so we need versions of limits and colimits that are invariant by design. This chapter constructs them, identifies the familiar constructions (mapping cylinder, cone, suspension, homotopy fibre, loop space) as special cases, proves the exactness properties that make them computable, and introduces the *homotopy orbit space* X_{hG}, which in Part VI identifies an unordered configuration space with a construction to which spectral sequences apply.

Prerequisites. Cofibrations and the HEP (Chapter 11), fibrations (Chapter 14), and Chapter 20 for the general formula.

21.2 Homotopy pushouts

Definition 21.1. The *homotopy pushout* of $Y \xleftarrow{f} X \xrightarrow{g} Z$ is the double mapping cylinder $M(f,g) = Y \sqcup (X \times I) \sqcup Z / \big((x,0) \sim f(x),\ (x,1) \sim g(x)\big)$.

Example 21.2. The homotopy pushout of $* \leftarrow X \to *$ is the unreduced suspension SX; of $Y \xleftarrow{f} X \xrightarrow{\mathrm{id}} X$ the mapping cylinder $M_f \simeq Y$; of $Y \xleftarrow{f} X \to *$ the mapping cone C_f. So for $* \leftarrow S^1 \to *$ it is S^2, the same as the strict pushout of $D^2 \leftarrow S^1 \to D^2$.

Lemma 21.3 (Gluing lemma)**.** *Consider a map of pushout diagrams $(Y \leftarrow X \to Z) \to (Y' \leftarrow X' \to Z')$ in which $X \to Y$ and $X' \to Y'$ are cofibrations and the three component maps are homotopy equivalences. Then the induced map of strict pushouts $Y \cup_X Z \to Y' \cup_{X'} Z'$ is a homotopy equivalence.*

This is [1, Prop. 0.18 and its proof] in the CW case and [4, Ch. 6] in general; the proof constructs homotopy inverses piece by piece, using the HEP to make them compatible on X.

Theorem 21.4 (Homotopy invariance)**.** *A map of diagrams $(Y \leftarrow X \to Z) \to (Y' \leftarrow X' \to Z')$ which is a homotopy equivalence on each object induces a homotopy equivalence of homotopy pushouts.*

Proof. The double mapping cylinder is the strict pushout of $M_f \hookleftarrow X \times \{0\} \sqcup X \times \{1\} \hookrightarrow \dots$; more simply, it is the strict pushout of $M_f \hookleftarrow X \to Z$, where $X \hookrightarrow M_f$ is the inclusion of the top of the mapping cylinder, a cofibration. The given map induces a map of these diagrams which is a homotopy equivalence on each piece ($M_f \simeq Y \to Y' \simeq M_{f'}$). Apply Lemma 21.3. □

Exercise 21.5 (medium)**.** Show that if $f\colon X \to Y$ is a cofibration then the natural map from the homotopy pushout $M(f,g)$ to the strict pushout $Y \cup_X Z$ is a homotopy equivalence.

SOURCE: orig-book (solution rewritten using Lemma 21.3)

Solution. $M(f,g)$ is the strict pushout of $M_f \hookleftarrow X \xrightarrow{g} Z$ (top inclusion, a cofibration), and $Y \cup_X Z$ is the strict pushout of $Y \hookleftarrow X \xrightarrow{g} Z$. The collapse $M_f \to Y$ is a homotopy equivalence compatible with the maps from X, and both $X \to M_f$ and $X \to Y$ are cofibrations. By Lemma 21.3, the induced map of pushouts — which is the collapse $M(f,g) \to Y \cup_X Z$ — is a homotopy equivalence.

21.3 Homotopy pullbacks and homotopy fibres

Definition 21.6. The *homotopy pullback* of $Y \xrightarrow{p} X \xleftarrow{q} Z$ is

$$Y \times_X^h Z = \{(y, \gamma, z) \in Y \times X^I \times Z : \ \gamma(0) = p(y),\ \gamma(1) = q(z)\}.$$

The *homotopy fibre* of $f\colon Y \to X$ over x_0 is the special case $Z = \{x_0\}$: $F_f = \{(y,\gamma) : \gamma(0) = f(y),\ \gamma(1) = x_0\}$.

Dually to Theorem 21.4, the homotopy pullback is invariant under objectwise homotopy equivalences, and the strict pullback is homotopy equivalent to it when p or q is a fibration [1, §4.H]. In particular the strict fibre of a fibration is equivalent to its homotopy fibre, and $F_f \to Y$, $(y,\gamma) \mapsto y$, is a fibration with fibre ΩX.

Exercise 21.7 (easy)**.** Give a strict pullback that is not a homotopy pullback, and verify that the homotopy pullback of $* \to S^1 \leftarrow *$ is $\Omega S^1 \simeq \mathbb{Z}$.

SOURCE: orig-book

Solution. The strict pullback of $* \to S^1 \leftarrow *$ (both maps to the base point) is a point. The homotopy pullback is the space of paths from the base point to itself, ΩS^1, which has $\pi_0 = \pi_1(S^1) = \mathbb{Z}$ and contractible components (Example 14.5, $\pi_k(\Omega S^1) = \pi_{k+1}(S^1) = 0$ for $k \geq 1$), so $\Omega S^1 \simeq \mathbb{Z}$. Neither map is a fibration; replacing one by the path fibration $PS^1 \to S^1$ makes the strict pullback the homotopy pullback.

Exercise 21.8 (medium)**.** Let $f\colon X \to Y$ with Y path-connected. The loop space ΩY acts on the right on F_f by concatenating the path coordinate with a loop. Show that, on path components, this gives a right action of $\pi_1(Y)$ on $\pi_0(F_f)$, that two elements of $\pi_0(F_f)$ have the same image in $\pi_0(X)$ iff they lie in the same orbit, and that the stabiliser of the component of (x,ω) is conjugate (via ω) to $f_*\pi_1(X,x)$.

SOURCE: MIT OCW 18.906 (Spring 2020), Problem Set II, Problem 8(a,b), reproduced in substance

Solution. Use the convention $\gamma(0) = f(x)$, $\gamma(1) = y_0$, so a loop ω at y_0 acts by $(x,\gamma)\cdot\omega = (x, \gamma \cdot \omega)$. Homotopic loops give homotopic actions, so $\pi_1(Y, y_0)$ acts on $\pi_0(F_f)$. If (x,γ) and

(x', γ') have x, x' in the same component, choose a path λ from x to x'; then (x', γ') is connected to $(x, (f\lambda)\cdot\gamma')$ (move along λ), and $(f\lambda)\cdot\gamma' = \gamma\cdot(\bar\gamma\cdot f\lambda\cdot\gamma')$, with $\bar\gamma\cdot f\lambda\cdot\gamma'$ a loop at y_0. So they are in the same orbit. Conversely the action does not change the X-coordinate, so orbits lie in fibres of $\pi_0(F_f) \to \pi_0(X)$. For the stabiliser: a path in F_f from (x,γ) to $(x, \gamma\cdot\omega)$ is a loop λ at x together with a family of paths γ_s from $f(\lambda(s))$ to y_0, with $\gamma_0 = \gamma$ and $\gamma_1 = \gamma\cdot\omega$. The map $(s,t) \mapsto \gamma_s(t)$ on the square shows (reading its boundary, as in Lemma 1.16) that $\gamma\cdot\omega \simeq \overline{f\lambda}\cdot\gamma$ rel endpoints, i.e. $\omega \simeq \bar\gamma\cdot\overline{f\lambda}\cdot\gamma$; conversely such a homotopy produces the family. Hence the stabiliser is $\{[\bar\gamma\cdot f(\mu)\cdot\gamma] : \mu \text{ a loop at } x\}$, the subgroup $f_*\pi_1(X,x)$ transported to y_0 along γ.

21.4 Exactness

Proposition 21.9. *Let $F \to E \xrightarrow{p} B$ be a fibration of based spaces and Z a based CW complex. Then $[Z,F]_* \to [Z,E]_* \xrightarrow{p_*} [Z,B]_*$ is exact as pointed sets.*

Proof. A map $Z \to F \to E \to B$ is constant. Conversely if $p\circ g$ is nullhomotopic by H, the homotopy lifting property lifts H to $\widetilde H$ starting at g, and $\widetilde H_1$ lands in $p^{-1}(b_0) = F$, with $g \simeq \widetilde H_1$. □

Theorem 21.10 (Puppe sequences)**.** *For a based map $f\colon X \to Y$ of CW complexes there is a sequence*

$$X \xrightarrow{f} Y \xrightarrow{i} C_f \xrightarrow{q} \Sigma X \xrightarrow{-\Sigma f} \Sigma Y \xrightarrow{-\Sigma i} \Sigma C_f \to \cdots$$

such that applying $[-,Z]_*$ gives an exact sequence of pointed sets, which are groups from ΣX on and abelian from $\Sigma^2 X$ on. Dually $\cdots \to \Omega F_f \to \Omega X \to \Omega Y \to F_f \to X \to Y$ gives exact sequences after $[W,-]_*$.

Proof. Exactness at $[Y,Z]$: a map $g\colon Y \to Z$ with $g\circ f \simeq *$ extends over the cone on X, i.e. over $C_f = Y \cup_f CX$, because a nullhomotopy of gf is precisely a map $CX \to Z$ compatible with g; conversely $i\circ f$ is nullhomotopic. The next term: $C_i = C_f \cup CY \simeq C_f/CY \cong \Sigma X$ (Exercise 11.10), so exactness at $[C_f, Z]$ is the same statement for i, and so on; the identification of the third map with Σf up to sign is a direct inspection of the double cone. The group structures come from the co-H-structure of suspensions (pinch map). See [1, §4.3] or [4, Ch. 8]. □

Example 21.11. Applying Theorem 21.10 with $Z = K(G,n)$ and Theorem 17.19 recovers the long exact cohomology sequence of the pair (M_f, X), i.e. of the cofibre sequence: the Puppe sequence is the homotopy-theoretic origin of all long exact sequences in cohomology.

21.5 Homotopy colimits in general

Definition 21.12 (Bousfield–Kan)**.** For a diagram $D\colon J \to \mathbf{Top}$ of spaces indexed by a small category, the *homotopy colimit* is the coend

$$\operatorname*{hocolim}_J D = \int^{j\in J} D(j) \times |N(J\downarrow j)|^{\mathrm{op}} \;\cong\; \Big|[n] \mapsto \coprod_{j_0\to\cdots\to j_n} D(j_0)\Big|,$$

the realisation of the *simplicial replacement*. Dually, the homotopy limit is the end $\int_j D(j)^{|N(j\downarrow J)|}$.

Theorem 21.13. *A natural transformation $D \to D'$ which is objectwise a weak equivalence induces a weak equivalence* $\operatorname{hocolim} D \to \operatorname{hocolim} D'$. *There is a natural map* $\operatorname{hocolim} D \to \operatorname{colim} D$, *which is a weak equivalence when D is "cofibrant" (e.g. for pushouts along cofibrations, Exercise 21.5).*

See [22, Ch. XII], [20]. For $J = \bullet \leftarrow \bullet \to \bullet$ the formula gives the double mapping cylinder of Definition 21.1, and for $J = \mathbb{N}$ (a sequence $X_0 \to X_1 \to \cdots$) the *mapping telescope*, used in Exercise 9.12.

21.6 Homotopy orbits

Definition 21.14. Let a discrete group G act on a space X, and let EG be a contractible space with a free G-action (e.g. $|EG|$ of Exercise 20.17). The *homotopy orbit space* or *Borel construction* is $X_{hG} = EG \times_G X = (EG \times X)/G$, with G acting diagonally. It is the homotopy colimit of X viewed as a functor from the one-object category G.

Proposition 21.15. *(a) $X \to X_{hG} \to BG$ is a fibration (fibre bundle), with $BG = EG/G$.*

(b) If G acts freely and properly discontinuously (e.g. G finite acting freely on a Hausdorff space), the projection $X_{hG} \to X/G$ is a homotopy equivalence (indeed a fibre bundle with contractible fibre EG).

Proof. (a) $EG \times X \to EG$ is G-equivariant, so it descends to $EG \times_G X \to EG/G = BG$; since $EG \to BG$ is a covering (principal G-bundle), $EG \times_G X \to BG$ is the associated bundle with fibre X. (b) Symmetrically, $EG \times_G X \to X/G$ is the bundle associated with the covering $X \to X/G$, with fibre EG, contractible; a fibration with contractible fibre over a CW-type base is a homotopy equivalence (long exact sequence and Whitehead). □

Example 21.16 (Unordered configurations)**.** For a manifold M, the symmetric group $\mathfrak{S}_k$ acts freely on the ordered configuration space $F_k(M)$ (Exercise 3.20, Exercise 3.19). By Proposition 21.15, $B_k(M) = F_k(M)/\mathfrak{S}_k \simeq F_k(M)_{h\mathfrak{S}_k}$, which fibres over $B\mathfrak{S}_k$ with fibre $F_k(M)$. The Serre spectral sequence of this fibration is the Cartan–Leray spectral sequence $H_p(\mathfrak{S}_k; H_q(F_k(M))) \Rightarrow H_{p+q}(B_k(M))$ of Part VI, with local coefficients given by the $\mathfrak{S}_k$-action on $H_*(F_k(M))$.

Exercise 21.17 (medium)**.** Show that if G acts trivially on X then $X_{hG} \simeq BG \times X$, and that if G is a finite group acting freely then $H^*(X/G; \mathbb{Q}) \cong H^*(X; \mathbb{Q})^G$, the invariants.

SOURCE: original

Solution. For a trivial action, $EG \times_G X = (EG/G) \times X = BG \times X$ on the nose.

For a free action of a finite group, $X \to X/G$ is a $|G|$-sheeted covering (Exercise 3.19). The *transfer* $\tau\colon H^*(X;\mathbb{Q}) \to H^*(X/G;\mathbb{Q})$, induced on cochains by summing a cochain over the $|G|$ lifts of each simplex, satisfies $\tau \circ p^* = |G| \cdot \mathrm{id}$ and $p^* \circ \tau = \sum_{g \in G} g^*$. Over $\mathbb{Q}$, $|G|$ is invertible, so p^* is injective, and its image is contained in the invariants (as $p \circ g = p$); conversely an invariant class x equals $p^*\big(\tau(x)/|G|\big)$ since $p^*\tau(x) = \sum g^* x = |G|x$. So $p^*\colon H^*(X/G;\mathbb{Q}) \to H^*(X;\mathbb{Q})^G$ is an isomorphism. Integrally the argument fails because $|G|$ is not invertible; e.g. $X = S^\infty$, $G = \mathbb{Z}/2$: $H^*(S^\infty;\mathbb{Z})^G = \mathbb{Z}$ in degree 0, but $H^*(\mathbb{RP}^\infty;\mathbb{Z})$ has $\mathbb{Z}/2$ in every even positive degree.

21.7 Connections and further reading

Homotopy colimits replace gluing by gluing along cylinders, homotopy limits replace equations by paths; the Puppe sequences are the source of all long exact sequences; and homotopy orbits turn quotients by group actions into fibrations over classifying spaces. In Chapter 22 all of this becomes "derived functors of colim and lim". References: Hatcher [1, §4.G, §4.H]; May [4, Ch. 6–8]; Bousfield–Kan [22]; Riehl [20].

Chapter 22

Model categories and derived functors

22.1 The problem

Chapter 21 repaired colimits one diagram shape at a time. A model structure repairs everything at once: from three classes of maps — weak equivalences, cofibrations, fibrations — one extracts the homotopy category, derived functors (homotopy colimits among them), and a criterion for when two homotopy theories are the same. We use the language in three places: to compare spaces with simplicial sets (Chapter 23), to set up Sullivan's rational models (Chapter 24), and to make sense of "algebras up to homotopy" over operads (Chapter 30, Part VI).

22.2 Lifting and factorisation

Definition 22.1. A map $i\colon A \to B$ has the *left lifting property* (LLP) against $p\colon X \to Y$, and p the RLP against i, if every commutative square with i on the left and p on the right has a diagonal filler $B \to X$ making both triangles commute. A *weak factorisation system* $(\mathcal{L}, \mathcal{R})$ is a pair of classes, each consisting exactly of the maps with the LLP (resp. RLP) against the other, such that every map factors as a map of $\mathcal{L}$ followed by a map of $\mathcal{R}$.

Exercise 22.2 (medium)**.** (Retract argument.) If $f = p \circ i$ with $p \in \mathcal{R}$, and f has the LLP against p, show that f is a retract of i in the arrow category.

Source: orig-book

Solution. Write $f\colon A \to B$ as $A \xrightarrow{i} C \xrightarrow{p} B$. The square with top i, left f, right p, bottom id_B commutes, so there is $r\colon B \to C$ with $rf = i$ and $pr = \mathrm{id}_B$. Then $(\mathrm{id}_A, r)\colon f \to i$ and $(\mathrm{id}_A, p)\colon i \to f$ are maps of arrows (commuting squares by $rf = i$ and $pi = f$) composing to the identity of f.

22.3 Model categories

Definition 22.3. A *model structure* on a complete and cocomplete category $\mathcal{M}$ consists of classes $\mathcal{W}$ (weak equivalences), $\mathcal{C}$ (cofibrations) and $\mathcal{F}$ (fibrations) such that: $\mathcal{W}$ satisfies two-out-of-three and contains isomorphisms; all three classes are closed under

retracts; and $(\mathcal{C} \cap \mathcal{W}, \mathcal{F})$ and $(\mathcal{C}, \mathcal{F} \cap \mathcal{W})$ are weak factorisation systems. An object is *cofibrant* if $\varnothing \to X$ is a cofibration, *fibrant* if $X \to *$ is a fibration.

Example 22.4 (The main examples)**.** (a) *Spaces, Quillen (Serre) structure*: weak equivalences the weak homotopy equivalences, fibrations the Serre fibrations, cofibrations the retracts of relative cell complexes. All spaces are fibrant; cofibrant spaces are retracts of CW complexes.

(b) *Spaces, Strøm (Hurewicz) structure*: homotopy equivalences, Hurewicz fibrations, closed Hurewicz cofibrations. Every space is fibrant and cofibrant.

(c) *Simplicial sets, Kan–Quillen structure*: weak equivalences the maps whose realisations are weak equivalences, cofibrations the monomorphisms, fibrations the Kan fibrations (Chapter 23). Every object is cofibrant; the fibrant ones are the Kan complexes.

(d) *Nonnegatively graded chain complexes over a ring R, projective structure*: quasi-isomorphisms, degreewise surjections in positive degrees, and degreewise monomorphisms with projective cokernels. Cofibrant objects are complexes of projectives, so cofibrant replacement is a projective resolution.

(e) *Commutative dg algebras over $\mathbb{Q}$* concentrated in nonnegative degrees: quasi-isomorphisms, surjections, and retracts of relative Sullivan algebras [17, §14], [28]. Cofibrant replacement is a Sullivan model (Chapter 24).

Example 22.5 (Weak equivalence is weaker than homotopy equivalence)**.** The Warsaw circle — the closed topologist's sine curve with its ends joined by an arc — has all homotopy groups trivial, so the map to a point is a weak equivalence, but it is not contractible. Whitehead's theorem says that for cofibrant (CW) spaces the two notions agree; this is the prototype of Theorem 22.9(c).

Exercise 22.6 (easy)**.** Show that weak homotopy equivalences of spaces satisfy two-out-of-three.

Source: MIT OCW 18.906 (Spring 2020), Problem Set II, Problem 6(a), reproduced in substance

Solution. Let $X \xrightarrow{f} Y \xrightarrow{g} Z$. A map is a weak equivalence iff it induces a bijection on π_0 and isomorphisms on π_n at every base point. If two of f, g, gf do, then by functoriality $(gf)_* = g_* f_*$ and the third does too, provided the base points match: for gf and g weak equivalences, $f_* \colon \pi_n(X, x) \to \pi_n(Y, f(x))$ equals $g_*^{-1}(gf)_*$ with g_* taken at the base point $f(x)$ — legitimate because g is a weak equivalence at *every* base point. This is why the definition quantifies over all base points; otherwise the case "g and gf" fails.

Exercise 22.7 (medium)**.** Show that in a model category cofibrations are stable under cobase change and fibrations under base change.

Source: orig-book

Solution. Let $i \colon A \to B$ be a cofibration, $A \to A'$ any map, $i' \colon A' \to B' = A' \cup_A B$. Given a lifting problem for i' against a trivial fibration p, compose with $B \to B'$ to get one for i, solve it, and use the universal property of the pushout to assemble the solution with the given map on A'; this solves the original problem. So i' has the LLP against trivial fibrations, i.e. is a cofibration. Dually for fibrations.

22.4 The homotopy category

Definition 22.8. A *cylinder* for A is a factorisation $A \sqcup A \xrightarrow{(i_0,i_1)} \mathrm{Cyl}(A) \xrightarrow{\sim} A$ of the fold map with the first map a cofibration and the second a weak equivalence. Two maps $f, g\colon A \to X$ are *left homotopic* if (f, g) extends over some cylinder. Dually, *path objects* $X \xrightarrow{\sim} \mathrm{Path}(X) \twoheadrightarrow X \times X$ define *right homotopy*.

Theorem 22.9 (Fundamental theorem of model categories)**.** *(a) The localisation* $\mathrm{Ho}(\mathcal{M}) = \mathcal{M}[\mathcal{W}^{-1}]$ *exists.*

(b) For A cofibrant and X fibrant, left and right homotopy coincide and are equivalence relations on $\mathcal{M}(A, X)$, and $\mathrm{Ho}(\mathcal{M})(A, X) = \mathcal{M}(A, X)/\simeq$.

(c) A weak equivalence between fibrant-cofibrant objects is a homotopy equivalence.

(d) $\mathrm{Ho}(\mathcal{M})$ *is equivalent to the category of fibrant-cofibrant objects and homotopy classes of maps; every object is isomorphic in* $\mathrm{Ho}(\mathcal{M})$ *to a fibrant-cofibrant replacement RQX.*

This is Quillen's theorem [23]; see [24, §1.2] or [26, §4–5] for complete proofs. For spaces it says: homotopy classes of maps from a CW complex to any space compute maps in the homotopy category, and Example 22.5 shows why the source must be cofibrant.

22.5 Derived functors

Theorem 22.10 (Ken Brown's lemma)**.** *Let $F\colon \mathcal{M} \to \mathcal{N}$ be a functor from a model category to a category with weak equivalences satisfying two-out-of-three. If F sends trivial cofibrations between cofibrant objects to weak equivalences, then it sends all weak equivalences between cofibrant objects to weak equivalences.*

Proof. Let $f\colon A \to B$ be a weak equivalence of cofibrant objects. Factor $(f, \mathrm{id})\colon A \sqcup B \to B$ as a cofibration j followed by a trivial fibration q. The inclusions $A \to A \sqcup B$, $B \to A \sqcup B$ are cofibrations (cobase changes of $\varnothing \to B$, $\varnothing \to A$), so j_A, j_B are cofibrations, and by two-out-of-three ($qj_A = f$, $qj_B = \mathrm{id}$) trivial cofibrations. So $F(j_A)$, $F(j_B)$ are weak equivalences; $F(q)F(j_B) = \mathrm{id}$ makes $F(q)$ one; hence $F(f) = F(q)F(j_A)$ is one. □

Definition 22.11. The *total left derived functor* $\mathbb{L}F$ of $F\colon \mathcal{M} \to \mathcal{N}$ is obtained by right Kan extension of $\mathcal{M} \to \mathcal{N} \to \mathrm{Ho}(\mathcal{N})$ along $\mathcal{M} \to \mathrm{Ho}(\mathcal{M})$ (it is the best approximation *from the left*); $\mathbb{R}G$ is defined dually.

Definition 22.12. An adjunction $F\colon \mathcal{M} \rightleftarrows \mathcal{N}\colon U$ between model categories is a *Quillen adjunction* if F preserves cofibrations and trivial cofibrations (equivalently, U preserves fibrations and trivial fibrations).

Theorem 22.13. *For a Quillen adjunction, $\mathbb{L}F(X) = F(QX)$ (with $QX \to X$ a cofibrant replacement) and $\mathbb{R}U(Y) = U(RY)$ exist, are absolute Kan extensions (Definition 20.10), and form an adjunction $\mathbb{L}F \dashv \mathbb{R}U$ between homotopy categories.*

Proof sketch. By Theorem 22.10, F sends weak equivalences between cofibrant objects to weak equivalences, so $X \mapsto F(QX)$ is well defined on $\mathrm{Ho}(\mathcal{M})$; the universal property follows because every object is weakly equivalent to a cofibrant one. For the adjunction: for X cofibrant and Y fibrant, left homotopies $FX \times \dots$ correspond to right homotopies under the adjunction (a cylinder on X is sent by F to a cylinder on FX, F being left Quillen), so $\mathrm{Ho}(\mathcal{N})(FQX, RY) = \mathcal{N}(FQX, RY)/\simeq \;\cong\; \mathcal{M}(QX, URY)/\simeq = \mathrm{Ho}(\mathcal{M})(X, URY)$. See [24, §1.3]. □

Warning 22.14. Definition 22.11 defines $\mathbb{L}F$ by a universal property; cofibrant replacement is one way to compute it. Any other construction satisfying the universal property — a bar resolution, a different replacement — gives the same functor, with no further comparison needed.

Example 22.15. On $\mathrm{Ch}_{\geq 0}(R)$ with the projective structure, the functor $- \otimes_R M$ sends quasi-isomorphisms between complexes of projectives to quasi-isomorphisms, so (by the argument of Theorem 22.13) it has a total left derived functor computed by cofibrant, i.e. projective, replacement; for a module A in degree 0, $\mathbb{L}(A \otimes_R M)$ has homology $\mathrm{Tor}^R_*(A, M)$: derived functors in the sense of Chapter 8 are total derived functors in the present sense. Homotopy colimits are $\mathbb{L}\,\mathrm{colim}$ for the projective model structure on diagram categories.

22.6 Quillen equivalences

Definition 22.16. A Quillen adjunction is a *Quillen equivalence* if $\mathbb{L}F \dashv \mathbb{R}U$ is an equivalence of categories.

Theorem 22.17 (Pointwise criterion)**.** *A Quillen adjunction $F \dashv U$ is a Quillen equivalence iff, whenever $X \in \mathcal{M}$ is cofibrant and $Y \in \mathcal{N}$ is fibrant, a morphism $FX \to Y$ is a weak equivalence exactly when its adjoint $X \to UY$ is.*

Proof. Assume the criterion. The derived unit at a cofibrant X is $X \to URFX$, adjoint to the fibrant replacement $FX \to RFX$, a weak equivalence; so the derived unit is an isomorphism in $\mathrm{Ho}(\mathcal{M})$. Dually the derived counit $FQUY \to Y$ at a fibrant Y is adjoint to $QUY \to UY$, a weak equivalence; so it is an isomorphism. Hence $\mathbb{L}F \dashv \mathbb{R}U$ is an equivalence.

Conversely, suppose the derived adjunction is an equivalence, and let $g\colon FX \to Y$, with adjoint $g^\flat\colon X \to UY$, X cofibrant, Y fibrant. In the homotopy category, $g^\flat$ factors as $X \xrightarrow{\eta} \mathbb{R}U\mathbb{L}F(X) \xrightarrow{\mathbb{R}U(g)} \mathbb{R}U(Y)$ (naturality of the derived adjunction; for cofibrant X and fibrant Y the derived functors are computed by F and U up to the replacements). The unit η is an isomorphism, so $g^\flat$ is an isomorphism in $\mathrm{Ho}(\mathcal{M})$ iff $\mathbb{R}U(g)$ is, iff g is (an equivalence of categories reflects isomorphisms). Finally, a map is a weak equivalence iff it becomes an isomorphism in the homotopy category (this saturation property holds in every model category, [24, §1.2]). □

Example 22.18. $|\cdot|\colon \mathbf{sSet} \rightleftarrows \mathbf{Top}\colon \mathrm{Sing}$ (Theorem 20.14) is a Quillen equivalence between the Kan–Quillen and the Quillen structures [27, Ch. I]. This is the precise sense in which "simplicial sets model spaces".

Example 22.19. Sullivan's polynomial forms give a Quillen adjunction $A_{\mathrm{PL}}\colon \mathbf{sSet} \rightleftarrows \mathbf{CDGA}_{\mathbb{Q}}^{\mathrm{op}}\colon \langle - \rangle$ which, restricted to simply connected spaces of finite rational type and the corresponding cdgas, induces an equivalence between the rational homotopy category and a homotopy category of cdgas [28], [17, §17]. This is the foundation of Chapter 24.

22.7 Exercises

Exercise 22.20 (medium)**.** In the projective model structure on $\mathrm{Ch}_{\geq 0}(\mathbb{Z})$, show that every object is fibrant, identify a cofibrant replacement of $\mathbb{Z}/n$ (in degree 0), and use it to show that the derived tensor product $\mathbb{Z}/n \otimes^{\mathbb{L}} \mathbb{Z}/m$ has homology $\mathbb{Z}/d$ in degrees 0 and 1, $d = \gcd(m, n)$.

Source: original

Solution. Fibrations are maps surjective in positive degrees; $X \to 0$ is one, so every object is fibrant. The complex $Q = (\mathbb{Z} \xrightarrow{n} \mathbb{Z})$ in degrees $1, 0$ is a complex of projectives (cofibrant) and $Q \to \mathbb{Z}/n$ is a quasi-isomorphism. Then $\mathbb{Z}/n \otimes^{\mathbb{L}} \mathbb{Z}/m = Q \otimes \mathbb{Z}/m = (\mathbb{Z}/m \xrightarrow{n} \mathbb{Z}/m)$, with $H_0 = \mathbb{Z}/m/n\mathbb{Z}/m = \mathbb{Z}/d$ and $H_1 = \ker(n) = \mathbb{Z}/d$, i.e. Tor_0 and Tor_1 (Example 8.5).

Exercise 22.21 (medium)**.** Show that the identity functor from spaces with the Strøm structure to spaces with the Quillen structure is right Quillen, but not a Quillen equivalence.

Source: original

Solution. Every Hurewicz fibration is a Serre fibration, and every homotopy equivalence is a weak homotopy equivalence, so the identity $\mathrm{id}\colon \mathbf{Top}_{\text{Strøm}} \to \mathbf{Top}_{\text{Quillen}}$ preserves fibrations and trivial fibrations: it is right Quillen, with left adjoint the identity. It is not a Quillen equivalence: in the criterion take $X = *$ (cofibrant in the Quillen structure) and Y the Warsaw circle W (fibrant in the Strøm structure); the map $F(*) = * \to W$ is not a weak equivalence in the Strøm structure (not a homotopy equivalence), while its adjoint $* \to W$ is a weak equivalence in the Quillen structure. The two homotopy categories differ: in one, W is isomorphic to a point; in the other not.

22.8 Connections and further reading

The language of model categories is used in the rest of the book mainly to make three statements precise: simplicial sets and spaces have the same homotopy theory; rational homotopy types of simply connected spaces are cdgas up to quasi-isomorphism; and algebras over a cofibrant replacement of an operad are the "up to homotopy" algebras. References: Dwyer–Spaliński [26] (the best introduction), Hovey [24], Hirschhorn [25], Goerss–Jardine [27], Quillen's original [23].

Chapter 23

Simplicial sets and quasi-categories

23.1 The problem

Singular homology is defined from simplices, and so is the nerve of a category; both are *simplicial sets*. If homotopy theory could be done directly with such combinatorial objects, many constructions — classifying spaces, bar constructions, homotopy colimits, the passage from algebra to topology in Chapter 24 — would become explicit. This chapter shows that it can: Kan complexes behave like spaces (Example 22.18), and weakening the Kan condition gives *quasi-categories*, a model for categories in which composition is defined only up to coherent homotopy — the setting of much of modern homotopy theory, including the ∞-operads that appear in research on configuration spaces.

Prerequisites. Chapter 20 (nerve and realisation), Chapter 22.

23.2 Simplicial sets

Definition 23.1. Let Δ be the category of the ordered sets $[n] = \{0 < 1 < \cdots < n\}$ and order-preserving maps. A *simplicial set* is a functor $X\colon \Delta^{\mathrm{op}} \to \mathbf{Set}$; write $X_n = X[n]$, $d_i\colon X_n \to X_{n-1}$ (faces) and $s_i\colon X_n \to X_{n+1}$ (degeneracies) for the maps induced by the coface δ^i (skip i) and codegeneracy σ^i (repeat i). They satisfy the *simplicial identities*, e.g. $d_i d_j = d_{j-1} d_i$ for $i < j$.

By Yoneda, n-simplices of X are maps $\Delta^n = \Delta(-,[n]) \to X$. The *boundary* $\partial\Delta^n$ is the union of the faces $d_i\Delta^{n-1}$, and the *horn* Λ^n_k the union of all faces except the k-th. A simplex is *degenerate* if it is in the image of some s_i.

Example 23.2. $\mathrm{Sing}(Y)_n = \mathbf{Top}(|\Delta^n|, Y)$; the nerve $N(\mathcal{C})_n =$ strings of n composable arrows; for a group G, NG with $NG_n = G^n$ (Exercise 20.17); and the standard simplices Δ^n themselves, with $(\Delta^n)_m$ the order-preserving maps $[m] \to [n]$.

23.3 Kan complexes

Definition 23.3. A map $p\colon X \to Y$ is a *Kan fibration* if it has the RLP against all horn inclusions $\Lambda^n_k \hookrightarrow \Delta^n$, $0 \le k \le n$; X is a *Kan complex* if $X \to *$ is a Kan fibration,

i.e. every horn in X has a filler.

Proposition 23.4. $\mathrm{Sing}(Y)$ *is a Kan complex for every space* Y.

Proof. By Theorem 20.14, a horn $\Lambda^n_k \to \mathrm{Sing}(Y)$ is a map $|\Lambda^n_k| \to Y$ and a filler is an extension over $|\Delta^n|$. The horn is a retract of the simplex: project from a point outside $|\Delta^n|$ near the barycentre of the missing face (the same picture as the retraction $D^n \times I \to D^n \times 0 \cup \partial D^n \times I$ of Exercise 11.3). □

Definition 23.5. For a Kan complex X and a vertex x, $\pi_n(X, x)$ is the set of n-simplices all of whose faces are the degenerate simplex at x, modulo the relation $a \sim b$ iff there is an $(n+1)$-simplex with faces $(x, \dots, x, a, b)$ (in the last two positions). The Kan condition makes this an equivalence relation and defines the group structure (fill a horn).

Theorem 23.6. *For a Kan complex* X, $\pi_n(X, x) \cong \pi_n(|X|, x)$. *A map of Kan complexes is a weak equivalence iff it induces isomorphisms on all* π_n *at all vertices, and a weak equivalence between Kan complexes is a simplicial homotopy equivalence.*

See [27, Ch. I], [29, Ch. I].

Theorem 23.7 (Moore)**.** *Every simplicial group is a Kan complex.*

Proof sketch. Let $(y_0, \dots, \widehat{y_k}, \dots, y_n)$ be a horn in G_{n-1}. One constructs a filler $w \in G_n$ by successive corrections: start with $w_0 = s_0 y_0$ (which has the correct 0-th face when $k \neq 0$), and at each step multiply by $s_i w'$ for a suitable w' built from the current error $d_i w \cdot y_i^{-1}$, which by the simplicial identities does not disturb the faces already corrected. Group multiplication and inverses are what make the correction possible; for details see [27, §I.3]. □

Theorem 23.8 (Dold–Kan)**.** *The normalised chain complex functor* N *(with* $N_n(A) = \bigcap_{i>0} \ker d_i$ *and differential* d_0*) is an equivalence between simplicial abelian groups and nonnegatively graded chain complexes, with* $\pi_n(A) \cong H_n(NA)$.

See [27, §III.2]. Consequently every chain complex is "a space": the realisation of the simplicial abelian group corresponding to C_* has homotopy groups $H_*(C)$; for $C = G[n]$ this is a $K(G, n)$ with a strictly commutative multiplication — a concrete model of the H-space structure used in Chapter 17.

Proposition 23.9. *The nerve* $N\mathcal{C}$ *of a category is a Kan complex iff* $\mathcal{C}$ *is a groupoid.*

Proof. A horn $\Lambda^2_0 \to N\mathcal{C}$ is a pair of arrows $f\colon a \to b$, $g\colon a \to c$; a filler is $h\colon b \to c$ with $hf = g$. If every f is invertible take $h = gf^{-1}$; conversely filling the horn (f, id_a) gives a left inverse of f, and Λ^2_2-horns give right inverses. For $n \geq 3$ every horn in a nerve has a (unique) filler, since a simplex of a nerve is determined by its 1-skeleton (Exercise 23.18) and the relevant composites exist. So $N\mathcal{C}$ is Kan iff all Λ^2_0 and Λ^2_2 horns fill, iff $\mathcal{C}$ is a groupoid. □

23.4 Quasi-categories

Definition 23.10. A simplicial set X is a *quasi-category* (an ∞-category, a weak Kan complex) if every *inner* horn $\Lambda^n_k \to X$, $0 < k < n$, has a filler.

Theorem 23.11 (Nerves are the quasi-categories with unique fillers)**.** *A simplicial set is isomorphic to the nerve of a category iff every inner horn has a* unique *filler.*

Proof. For a nerve, an inner horn Λ^n_k is a diagram of composable arrows in which one composite is missing; it exists and is unique. Conversely, given unique inner fillers, define a category with objects X_0, morphisms X_1, identities s_0, and $g \circ f$ the 1-face of the unique filler of the Λ^2_1-horn $(g, \cdot, f)$; associativity comes from the unique filler of the Λ^3_1-horn built from the three relevant 2-simplices, and one checks that the higher simplices of X are determined by their 1-skeleta (unique fillers of inner horns in all dimensions), giving $X \cong N\mathcal{C}$. See [31, Prop. 1.1.2.2]. □

Definition 23.12. Let X be a quasi-category. Two 1-simplices $f, g\colon x \to y$ are *homotopic* if there is a 2-simplex σ with $d_2\sigma = f$, $d_1\sigma = g$ and $d_0\sigma = s_0(y)$ (the identity of y): a triangle exhibiting g as "$\mathrm{id}_y \circ f$". The *homotopy category* $\mathrm{h}X$ has objects X_0, morphisms homotopy classes of 1-simplices, and composition $[g][f] = [h]$ whenever there is a 2-simplex with $d_2 = f$, $d_0 = g$, $d_1 = h$.

Theorem 23.13 (Boardman–Vogt, Joyal)**.** *For a quasi-category X, homotopy of* 1*-simplices is an equivalence relation, composition in* $\mathrm{h}X$ *is well defined and associative, and* $\mathrm{h} \dashv N$*: functors* $\mathrm{h}X \to \mathcal{C}$ *correspond to simplicial maps* $X \to N\mathcal{C}$*.*

Proof sketch. Each property is a horn filling in dimension 3. For instance, given 2-simplices witnessing $[g][f] = [h]$ and $[g][f] = [h']$, the Λ^3_1-horn whose faces are these two simplices and the degenerate simplex $s_0(g)$ has a filler (an inner horn), whose remaining face witnesses $h \simeq h'$; associativity uses a Λ^3_1 or Λ^3_2 horn built from the composition simplices of (f, g), (g, k) and (gf, k). The adjunction follows because $N\mathcal{C}$ is 2-coskeletal (Exercise 23.18), so a map into it is determined by vertices, edges and triangles. See [31, §1.2.3], [32]. □

Theorem 23.14 (Joyal)**.** *A quasi-category X is a Kan complex iff* $\mathrm{h}X$ *is a groupoid. There is a model structure on simplicial sets (the Joyal structure) whose fibrant objects are the quasi-categories and whose weak equivalences are the categorical equivalences.*

See [30], [31, Thm. 2.2.5.1]. Quasi-categories are one model of $(\infty, 1)$-categories; others (simplicial categories, complete Segal spaces) are related by Quillen equivalences, one of which is the rigidification $\mathfrak{C}$ of Chapter 29.

23.5 Filling is obstruction theory

Remark 23.15. In Chapter 17 a map was built over a CW complex skeleton by skeleton; at each stage the obstruction lived in a homotopy group and was recorded, when nonzero, as a cohomology class. Here a map is built over a simplicial set simplex by simplex, and the question at each stage is whether a horn extends. The Kan and quasi-category conditions simply demand that the extensions exist. Obstruction theory measures the obstruction; fibrancy conditions abolish it by hypothesis, at the price of restricting to fibrant objects — which is why fibrant replacement is unavoidable in Chapter 22.

23.6 Left and right fibrations

Definition 23.16. $p\colon X \to Y$ is a *left fibration* if it has the RLP against $\Lambda^n_k \hookrightarrow \Delta^n$ for $0 \le k < n$, and a *right fibration* for $0 < k \le n$.

Warning 23.17 (Handedness is not a convention). Reversing simplices exchanges left and right fibrations but also reverses all arrows, so no statement transfers by symmetry within one quasi-category. Left fibrations over Y model functors $Y \to$ (spaces); right fibrations model presheaves $Y^{\mathrm{op}} \to$ (spaces). In any argument record which horns are available.

23.7 Exercises

Exercise 23.18 (easy)**.** Show that $N\colon \mathbf{Cat} \to \mathbf{sSet}$ is fully faithful and that $N\mathcal{C}$ is 2-coskeletal: for $n \geq 3$ every map $\partial\Delta^n \to N\mathcal{C}$ extends uniquely to Δ^n.

Source: orig-book

Solution. A simplicial map $N\mathcal{C} \to N\mathcal{D}$ is determined on 0- and 1-simplices (every simplex of a nerve is determined by its vertices and edges: the string $c_0 \to \cdots \to c_n$), and compatibility with 2-simplices says composites are preserved; so maps of nerves are functors. For $n \geq 3$, a map $\partial\Delta^n \to N\mathcal{C}$ contains all edges $i \to i+1$ (each lies in some face) and all triangles, which say that composites agree; the string of edges $i \to i+1$ is the unique n-simplex with this boundary.

Exercise 23.19 (easy)**.** Show that $\mathrm{h}N\mathcal{C} \cong \mathcal{C}$.

Source: orig-book

Solution. Objects and morphisms agree; two morphisms are homotopic iff a 2-simplex exhibits one as the other composed with an identity, i.e. iff they are equal; and composition is given by 2-simplices, i.e. commuting triangles, which is composition in $\mathcal{C}$.

Exercise 23.20 (medium)**.** Let $X = \partial\Delta^2$, the boundary of the 2-simplex (three vertices, three edges). It is not a Kan complex. Compute $\pi_1|X|$, and find a horn in X without a filler.

Source: original

Solution. $|X|$ is a triangle, a circle: $\pi_1 = \mathbb{Z}$. The horn $\Lambda^2_1 \to X$ given by the edges $[0,1]$ and $[1,2]$ has no filler, since X has no nondegenerate 2-simplex and a degenerate 2-simplex has two equal vertices. (Also Λ^2_0 given by $[0,1], [0,2]$ cannot be filled.) So X is not even a quasi-category; its fibrant replacement $\mathrm{Ex}^\infty X$ or $\mathrm{Sing}\,|X|$ has $\pi_1 = \mathbb{Z}$ computed simplicially.

Exercise 23.21 (medium)**.** Using Theorem 23.8, describe a simplicial abelian group $K(A,n)$ whose only nonzero homotopy group is A in degree n, and identify its m-simplices for $m \leq n$.

Source: original

Solution. Take the chain complex $A[n]$, A in degree n and 0 elsewhere, and let $K(A,n) = \Gamma(A[n])$ be the simplicial abelian group corresponding to it under Dold–Kan. Then $\pi_m K(A,n) = H_m(A[n])$, which is A for $m = n$ and 0 otherwise. Its m-simplices are $\Gamma(C)_m = \bigoplus_{[m]\twoheadrightarrow[k]} C_k$, the sum over surjections; for $m < n$ this is 0 (one point), and for $m = n$ it is A (only the identity surjection $[n] \to [n]$). (A useful description: $K(A,n)_m = Z^n(\Delta^m; A)$, the normalised n-cocycles on the simplex Δ^m.)

23.8 Connections and further reading

Simplicial sets give combinatorial models of spaces (Kan complexes) and of $(\infty, 1)$-categories (quasi-categories); the obstruction-theoretic pattern of Chapter 17 reappears as horn filling. References: Goerss–Jardine [27], May [29], Riehl [20], Lurie [31], Joyal [30], Riehl–Verity [32].

Chapter 24

Sullivan models

24.1 The problem

Chapter 19 computed rational homotopy groups one space at a time with spectral sequences. Sullivan's theory replaces the space by a commutative differential graded algebra and makes the computation algebraic: for simply connected spaces of finite type, the rational homotopy type is encoded by a *minimal Sullivan model* $(\Lambda V, d)$, whose generators V are dual to the rational homotopy groups and whose differential encodes the Whitehead products and the k-invariants. This is the language in which the rational homotopy of configuration spaces is studied (Kriz, Totaro, Lambrechts–Stanley, Idrissi; Part VI).

Prerequisites. Chapter 19; model categories (Chapter 22) for the conceptual framework; simplicial sets (Chapter 23) for the construction of A_{PL}.

24.2 Commutative differential graded algebras

Definition 24.1. A *commutative differential graded algebra* (cdga) over $\mathbb{Q}$ is a graded algebra $A = \bigoplus_{n \geq 0} A^n$ with $ab = (-1)^{|a||b|} ba$, together with a differential d of degree $+1$, $d^2 = 0$, satisfying the Leibniz rule $d(ab) = (da)b + (-1)^{|a|} a\, db$. Its cohomology $H(A)$ is a graded commutative algebra. A morphism of cdgas is a *quasi-isomorphism* if it induces an isomorphism on cohomology. A is *connected* if $H^0(A) = \mathbb{Q}$, *simply connected* if moreover $H^1(A) = 0$, and of *finite type* if each $H^n(A)$ is finite-dimensional.

Example 24.2. (a) $H^*(X; \mathbb{Q})$ with $d = 0$. (b) For a smooth manifold, the de Rham algebra $\Omega^*_{dR}(M)$ (over $\mathbb{R}$). (c) The free graded commutative algebra ΛV on a graded vector space V: the tensor product of the polynomial algebra on V^{even} and the exterior algebra on V^{odd}. A differential on ΛV is determined by its values on V, which can be prescribed arbitrarily (in degree $+1$) subject to $d^2 = 0$ on generators.

The singular cochain algebra $C^*(X; \mathbb{Q})$ is associative but not commutative, so it is not a cdga; over $\mathbb{Q}$ one can replace it by a commutative model.

Theorem 24.3 (Polynomial de Rham theory)**.** *Let*

$$\nabla(n) = \Lambda(t_0, \dots, t_n, dt_0, \dots, dt_n)/(\textstyle\sum t_i - 1,\ \sum dt_i), \qquad |t_i| = 0,$$

the polynomial differential forms on the n-simplex; these form a simplicial cdga $\nabla(\bullet)$. For a space X put $A_{\mathrm{PL}}(X) = \mathbf{sSet}(\operatorname{Sing} X, \nabla(\bullet))$, the cdga of "polynomial forms on singular simplices compatible with faces". Then there is a natural isomorphism of graded algebras $H(A_{\mathrm{PL}}(X)) \cong H^*(X;\mathbb{Q})$, and $A_{\mathrm{PL}}(X)$ is connected to $C^*(X;\mathbb{Q})$ by natural quasi-isomorphisms of differential graded algebras. For a smooth manifold M, $A_{\mathrm{PL}}(M)\otimes \mathbb{R}$ is quasi-isomorphic to $\Omega^*_{dR}(M)$.

See [17, §10–11]. The rational homotopy type of X is encoded by $A_{\mathrm{PL}}(X)$ up to quasi-isomorphism; Sullivan models are the cofibrant (Example 22.4(e)) and minimal representatives.

24.3 Sullivan algebras and minimal models

Definition 24.4. A *Sullivan algebra* is a cdga $(\Lambda V, d)$, $V = V^{\geq 1}$, such that V has an increasing filtration $V(0) \subseteq V(1) \subseteq \cdots$, $\bigcup V(k) = V$, with $d(V(0)) = 0$ and $d(V(k)) \subseteq \Lambda V(k-1)$. It is *minimal* if $d(V) \subseteq \Lambda^{\geq 2} V$: differentials of generators have no linear part. A *Sullivan model* of a cdga A is a quasi-isomorphism $(\Lambda V, d) \to A$; of a space X, a Sullivan model of $A_{\mathrm{PL}}(X)$.

Warning 24.5 (Free is not the condition). Freeness of ΛV is half of the definition; the filtration is the other half, and every inductive argument runs over it. For simply connected models ($V^1 = 0$) the filtration is automatic: filter by degree, since d of a generator of degree n is a polynomial in generators of degree $< n$ (a product of at least two generators of degree ≥ 2 has degree ≥ 4; in general $d(V^n) \subseteq \Lambda(V^{<n})$). In degree 1 it is a real condition: $\Lambda(x,y,z)$, $|x| = |y| = |z| = 1$, with $dx = yz$, $dy = zx$, $dz = xy$ is free with $d^2 = 0$ but has no closed generator, hence is not Sullivan (it is the Chevalley–Eilenberg algebra of $\mathfrak{so}(3)$).

Theorem 24.6 (Existence and uniqueness)**.** *Every cdga A with $H^0(A) = \mathbb{Q}$ and $H^1(A) = 0$ has a minimal Sullivan model $m\colon (\Lambda V, d) \xrightarrow{\simeq} A$ with $V^1 = 0$, unique up to isomorphism. If $H(A)$ is of finite type, so is V.*

Proof sketch. Construct V degree by degree. Choose $V^2 \cong H^2(A)$, with $d = 0$ and m sending a basis to representative cocycles. Suppose $m_n\colon \Lambda V^{\leq n} \to A$ is an isomorphism on $H^{\leq n}$ and injective on H^{n+1}. In degree $n+1$ adjoin: (i) generators v with $dv = 0$ mapping to cocycles that complete the image to all of $H^{n+1}(A)$; (ii) generators w of degree $n+1$ with $dw = z$, one for each element z of a basis of $\ker\big(H^{n+2}(\Lambda V^{\leq n}) \to H^{n+2}(A)\big)$, choosing $m(w)$ with $d\,m(w) = m(z)$. Since z is decomposable when it comes from $\Lambda^{\geq 2}$ (simple connectivity forces $H^{n+2}(\Lambda V^{\leq n})$ to be spanned by products), the result is minimal; after adjoining, the map is an isomorphism on $H^{\leq n+1}$ and injective on H^{n+2}. Uniqueness: a quasi-isomorphism between minimal Sullivan algebras is an isomorphism (it induces an isomorphism on indecomposables V, by an inductive argument using minimality), and any two minimal models are connected by a quasi-isomorphism because Sullivan algebras are cofibrant. See [17, §12, §14]. □

24.4 First examples

Example 24.7 (Spheres)**.** For n odd, $(\Lambda(x), 0)$, $|x| = n$, is a minimal model of S^n: $\Lambda(x) = \mathbb{Q} \oplus \mathbb{Q}x$ since $x^2 = 0$. For n even, $(\Lambda(x,y),\ dx = 0,\ dy = x^2)$, $|x| = n$,

$|y| = 2n-1$: the generator y kills x^2, which is not zero in the polynomial algebra $\mathbb{Q}[x]$ but is zero in $H^*(S^n)$.

Exercise 24.8 (easy)**.** Verify that $(\Lambda(x,y), dy = x^2)$, $|x|$ even, has the cohomology of $S^{|x|}$ and is minimal, and that a quasi-isomorphism to $A_{\mathrm{PL}}(S^n)$ exists.

SOURCE: orig-book (last clause added)

Solution. Elements are $p(x) + q(x)y$ $(y^2 = 0)$. $d(x^k) = 0$, $d(x^k y) = x^{k+2}$. Cocycles: $\mathbb{Q}[x]$; coboundaries: $x^2\mathbb{Q}[x]$. So $H = \mathbb{Q}\{1, x\}$. Minimal: $dy = x^2 \in \Lambda^{\geq 2}$. For the map: choose a cocycle $\omega \in A^n_{\mathrm{PL}}(S^n)$ representing the generator; ω^2 is a cocycle of degree $2n$ in a cdga with $H^{2n} = 0$, so $\omega^2 = d\eta$; send $x \mapsto \omega$, $y \mapsto \eta$. This is a cdga map (it respects d on generators) and induces an isomorphism on cohomology.

Example 24.9 (Projective spaces)**.** $H^*(\mathbb{CP}^n; \mathbb{Q}) = \mathbb{Q}[x]/(x^{n+1})$, $|x| = 2$. The minimal model is $(\Lambda(x,y), dy = x^{n+1})$, $|y| = 2n+1$: as in the sphere case, cocycles are $\mathbb{Q}[x]$, coboundaries $x^{n+1}\mathbb{Q}[x]$. For $\mathbb{CP}^\infty$, $(\Lambda(x), 0)$.

Example 24.10 (Products and wedges)**.** A model of $X \times Y$ is the tensor product of models. For the wedge $S^2 \vee S^2$ the minimal model must kill the products of the two degree-2 generators: $V^2 = \langle a, b\rangle$, then in degree 3 generators u, v, w with $du = a^2$, $dv = ab$, $dw = b^2$; then in degree 4 new generators killing the cocycles $av - bu$, $aw - bv$ (of degree 5; $d(av - bu) = a\cdot ab - b\cdot a^2 = 0$), and so on. The model is infinitely generated, reflecting the infinitely many rational homotopy groups of $S^2 \vee S^2$ (Chapter 25).

Example 24.11 (Lie groups and H-spaces)**.** For a connected Lie group, or any simply connected H-space of finite type, $H^*(G;\mathbb{Q}) = \Lambda(x_1, \ldots, x_r)$ with $|x_i|$ odd (Theorem 10.4), and $(\Lambda(x_1, \ldots, x_r), 0)$ is a minimal model. Hence $\pi_*(G) \otimes \mathbb{Q}$ has dimension r, concentrated in the odd degrees $|x_i|$; e.g. $SU(n)$: degrees $3, 5, \ldots, 2n-1$.

24.5 Homotopy groups from generators

Theorem 24.12 (Sullivan)**.** *Let X be simply connected with $H^*(X;\mathbb{Q})$ of finite type and minimal model $(\Lambda V, d)$. Then $\pi_k(X) \otimes \mathbb{Q} \cong \mathrm{Hom}_{\mathbb{Q}}(V^k, \mathbb{Q})$ for all $k \geq 2$. Moreover, writing $d_1 \colon V \to \Lambda^2 V$ for the quadratic part of d, the dual of d_1 is the Whitehead product $\pi_p \otimes \pi_q \to \pi_{p+q-1}$ (rationally, up to sign).*

See [17, Thm. 15.11 and §13]. The proof compares the minimal model of X with that of its Postnikov tower: adjoining a generator of degree k with differential z models the principal fibration $K(\mathbb{Q}, k) \to X_k \to X_{k-1}$ with k-invariant represented by z (Example 17.29).

Remark 24.13 (The parity explained)*.* Compare Example 24.7 with Theorem 19.6: one generator in degree n for n odd, two generators (degrees n, $2n-1$) for n even. And $dy = x^2$ is quadratic: by Theorem 24.12 the Whitehead square $[\iota, \iota]$ is rationally nonzero, its dual being y — Proposition 19.16 recovered algebraically.

Example 24.14 (Spheres, one last time)**.** For S^n the theorem gives $\pi_*(S^n) \otimes \mathbb{Q}$ in degrees n (and $2n-1$ for n even), and for $\mathbb{CP}^n$ in degrees 2 and $2n+1$, matching Exercise 19.19.

24.6 Relative models and fibrations

Definition 24.15. A *relative Sullivan algebra* over a cdga B is $(B \otimes \Lambda V, d)$ with V filtered and $d(V(k)) \subseteq B \otimes \Lambda V(k-1)$, $d|_B = d_B$.

Theorem 24.16 (Models of fibrations)**.** *Let $F \to E \to B$ be a fibration with B simply connected, F connected, all of finite rational type. If $(\Lambda W, d)$ models B, there is a relative Sullivan algebra $(\Lambda W \otimes \Lambda V, D)$ modelling $E \to B$, and its quotient $(\Lambda V, \bar{D}) = \mathbb{Q} \otimes_{\Lambda W} (\Lambda W \otimes \Lambda V)$ is a Sullivan model of F.*

See [17, §15].

Warning 24.17 (Cofibrancy is what makes the quotient correct)*.* The fibre is modelled by the quotient only because the model is a relative *Sullivan* algebra, i.e. a cofibration; for a non-cofibrant map the quotient is a non-derived tensor product and gives the wrong answer, exactly as strict pushouts did in Chapter 21.

Exercise 24.18 (medium)**.** Let B be a cdga with zero differential and $f \in B$ a non-zero-divisor of even degree. Show that $(B \otimes \Lambda(w), dw = f)$, $|w| = |f| - 1$, is quasi-isomorphic to $B/(f)$.

Source: orig-book

Solution. w has odd degree, so elements are $b_0 + b_1 w$ and $d(b_0 + b_1 w) = \pm b_1 f$. Cocycles: $b_1 f = 0$, i.e. $b_1 = 0$ (f a non-zero-divisor), so cocycles $= B$; coboundaries $= (f)$. Hence $H = B/(f)$, and the map $B \otimes \Lambda(w) \to B/(f)$, $w \mapsto 0$, is a quasi-isomorphism.

Example 24.19 (The quaternionic Hopf fibration)**.** $S^3 \to S^7 \to S^4$. The base has model $(\Lambda(x,y), dy = x^2)$, $|x| = 4$, $|y| = 7$. A relative model is $(\Lambda(x,y) \otimes \Lambda(z),\ Dz = x)$, $|z| = 3$; its quotient by (x,y) is $(\Lambda(z), 0)$, the model of S^3. The total algebra $\Lambda(x,y,z)$ with $dz = x$, $dy = x^2$ is not minimal, but the change of generators $y' = y - xz$ gives $dy' = x^2 - x \cdot x = 0$, and $\Lambda(x,z)$ with $dz = x$ is acyclic (contractible); so the total space has model $(\Lambda(y'), 0)$, $|y'| = 7$: rationally S^7. The differential $Dz = x$ is the transgression of the fibre class to the Euler class of the bundle.

24.7 Spatial realisation

Definition 24.20. For a cdga A the *spatial realisation* is the simplicial set $\langle A \rangle_n = \mathrm{CDGA}(A, \nabla(n))$.

Theorem 24.21 (Sullivan)**.** *$\langle A \rangle$ is a Kan complex. For a simply connected minimal Sullivan algebra of finite type $(\Lambda V, d)$, $|\langle \Lambda V \rangle|$ is a simply connected space with minimal model $(\Lambda V, d)$ and $\pi_k \cong \mathrm{Hom}(V^k, \mathbb{Q})$; and for a simply connected X of finite rational type, $X \to \langle \Lambda V \rangle$ is a rationalisation. Every such algebra is the model of a space.*

See [17, §17]. Together: rational homotopy types of simply connected spaces of finite type correspond bijectively to isomorphism classes of simply connected minimal Sullivan algebras of finite type [34].

24.8 Formality

Definition 24.22. A connected cdga A (or a space X) is *formal* if it is connected to its cohomology $(H(A), 0)$ by a zig-zag of quasi-isomorphisms; equivalently its minimal model is the minimal model of $(H(A), 0)$. A formal space's rational homotopy type is determined by its rational cohomology ring.

Example 24.23. Spheres, projective spaces, H-spaces, products and wedges of formal spaces are formal (in each case the models above admit a quasi-isomorphism to the cohomology: for S^n, n even, send $x \mapsto x$, $y \mapsto 0$). Compact Kähler manifolds are formal (Deligne–Griffiths–Morgan–Sullivan [33]). The ordered configuration spaces $F_k(\mathbb{R}^n)$ are formal (Arnold for $n = 2$, Kontsevich and Lambrechts–Volić in general; Part VI).

Example 24.24 (A non-formal simply connected space)**.** Let $A = (\Lambda(a, b, c),\ da = db = 0,\ dc = ab)$ with $|a| = |b| = 3$, $|c| = 5$; by Theorem 24.21 it is the minimal model of a simply connected space X. Its cohomology in low degrees: $H^3 = \langle a, b\rangle$; $H^6 = 0$ since $ab = dc$; $H^8 = \langle ac, bc\rangle$ ($d(ac) = -a\cdot ab = 0$ because $a^2 = 0$, and there is nothing in degree 7 to bound). The classes $[ac]$, $[bc]$ are not products of lower-degree cohomology classes ($H^5 = 0$), yet they are built from the degree-3 classes by the relation $ab = dc$: they are *Massey products*. Precisely, the triple Massey product $\langle a, a, b\rangle$ contains $\pm[ac] \neq 0$ and has zero indeterminacy (Example 31.10 in Chapter 31), while in a formal cdga every Massey product contains 0 (Proposition 31.9). So X is not formal.

24.9 Exercises

Exercise 24.25 (medium)**.** Compute the minimal models of $S^2 \times S^2$ and of $\mathbb{CP}^2 \# \mathbb{CP}^2$, and decide whether they are isomorphic. (Recall $H^*(\mathbb{CP}^2 \# \mathbb{CP}^2; \mathbb{Q}) = \mathbb{Q}[a, b]/(ab,\ a^2 - b^2)$, $|a| = |b| = 2$.)

SOURCE: original

Solution. $S^2 \times S^2$: tensor product of two sphere models, $\Lambda(x_1, y_1, x_2, y_2)$, $dy_i = x_i^2$, $|x_i| = 2$, $|y_i| = 3$. $\mathbb{CP}^2 \# \mathbb{CP}^2$ is formal (it is a simply connected 4-manifold; all simply connected closed manifolds of dimension ≤ 6 are formal, [37], but we can build the model directly): $V^2 = \langle a, b\rangle$; the degree-4 products a^2, ab, b^2 span a 3-dimensional space mapping onto $H^4 = \mathbb{Q}$ with kernel spanned by ab and $a^2 - b^2$, so add u, v in degree 3 with $du = ab$, $dv = a^2 - b^2$; for $S^2 \times S^2$ the kernel is spanned by a^2, b^2 (with $a = x_1$, $b = x_2$, ab generating H^4). The two algebras $\Lambda(a, b, u, v)$ (through degree 3) are isomorphic iff there is a linear change of $\langle a, b\rangle$ carrying the plane of relations $\langle a^2, b^2\rangle$ to $\langle ab, a^2 - b^2\rangle$, i.e. iff the quadratic forms $Q_1(\alpha a + \beta b) =$ coefficient of the top class are equivalent over $\mathbb{Q}$: for $S^2 \times S^2$ the cup square of $\alpha a + \beta b$ is $2\alpha\beta \cdot ab$ (the hyperbolic form $2\alpha\beta$); for $\mathbb{CP}^2 \# \mathbb{CP}^2$ it is $(\alpha^2 + \beta^2)\cdot a^2$ (the form $\alpha^2 + \beta^2$). The first is indefinite, the second definite, so they are not equivalent over $\mathbb{Q}$ (or $\mathbb{R}$), and the minimal models are not isomorphic: the spaces are not rationally equivalent.

Exercise 24.26 (easy)**.** Write down minimal models of $K(\mathbb{Z}, n)$ and of $K(\mathbb{Z}, 2)\times K(\mathbb{Z}, 3)$, and of a space X with $\pi_*(X) \otimes \mathbb{Q}$ one-dimensional in degrees 2 and 3 and a nontrivial Whitehead product $\pi_2 \otimes \pi_2 \to \pi_3$. Identify X rationally.

SOURCE: original

Solution. $K(\mathbb{Z}, n)$: $(\Lambda(x_n), 0)$ (Theorem 19.3). The product: $(\Lambda(x_2, x_3), 0)$. A space with generators $x \in V^2$, $y \in V^3$ and nontrivial quadratic differential must have $dy = \lambda x^2$, $\lambda \neq 0$

(Theorem 24.12); rescaling y gives $dy = x^2$, the model of S^2. So $X \simeq_{\mathbb{Q}} S^2$, and $\pi_*(S^2) \otimes \mathbb{Q}$ is the only possibility: rationally S^2 *is* the two-stage Postnikov tower with k-invariant ι^2 (Example 17.32).

Exercise 24.27 (medium)**.** Use Theorem 24.16 for the fibration $S^1 \to S^{2n+1} \to \mathbb{CP}^n$, with base model $(\Lambda(x,y), dy = x^{n+1})$, to show that the relative model is $\Lambda(x,y,z)$ with $Dz = x$, $|z| = 1$, and deduce that the total space is rationally S^{2n+1}. Where does the fibre's non-simple-connectivity enter?

SOURCE: original

Solution. The fibre S^1 has model $(\Lambda(z), 0)$, $|z| = 1$; a relative model has $Dz \in (\Lambda(x,y))^2 = \mathbb{Q}x$, and since the total space has $H^2 = 0$, $Dz = \lambda x$ with $\lambda \neq 0$; rescale to $Dz = x$. As in Example 24.19, put $y' = y - x^n z$: $Dy' = x^{n+1} - x^n \cdot x = 0$, and $\Lambda(x,z)$ with $Dz = x$ is acyclic, so the total algebra is quasi-isomorphic to $(\Lambda(y'), 0)$ with $|y'| = 2n+1$. The fibre S^1 is not simply connected, but Theorem 24.16 only needs the base simply connected (and the fibre nilpotent, which S^1 is); the degree-1 generator is harmless because Dz lies in the base.

24.10 Connections and further reading

Minimal models turn rational homotopy theory into commutative algebra: generators are homotopy groups, the quadratic differential is the Whitehead product, higher terms are higher k-invariants, and formality is the statement that the cohomology ring determines everything. Chapter 31 adds the dual (Lie) picture and Massey products; Part VI applies all of it to configuration spaces. References: Félix–Halperin–Thomas [17], Sullivan [34], Griffiths–Morgan [35], Félix–Oprea–Tanré [36] (with many geometric examples).

Chapter 25

Operads and homotopy Lie algebras

25.1 The problem

Two questions motivate this chapter. First: what is the structure on the rational homotopy groups of a space? The Whitehead product makes $\pi_*(\Omega X)\otimes\mathbb{Q}$ a graded Lie algebra, and the Milnor–Moore theorem says that this Lie algebra determines the rational homology of the loop space; counting dimensions then shows that the rational homotopy of a finite complex is either finite or grows exponentially. Second: how does one describe algebraic structures — Lie, commutative, associative, and their "up to homotopy" versions — uniformly? Operads answer the second question, and through the *little disks operads*, built out of configuration spaces, they connect the two: n-fold loop spaces are exactly the algebras over the little n-disks, and the homology of configuration spaces of points in $\mathbb{R}^n$ is the operad of n-Poisson algebras.

Prerequisites. Hopf algebras and primitives (Chapter 10); Whitehead products and Chapter 19; Sullivan models (Chapter 24).

25.2 Operads

Definition 25.1. A (symmetric) *operad* $\mathcal{P}$ in a symmetric monoidal category (**Set**, **Top**, graded vector spaces, chain complexes) consists of objects $\mathcal{P}(n)$, $n\geq 0$, with a right action of the symmetric group $\mathfrak{S}_n$, a unit $\mathrm{id}\in\mathcal{P}(1)$, and composition maps

$$\gamma\colon \mathcal{P}(k)\otimes\mathcal{P}(n_1)\otimes\cdots\otimes\mathcal{P}(n_k)\to\mathcal{P}(n_1+\cdots+n_k)$$

which are associative, unital and equivariant. Equivalently (for unital operads) one gives partial compositions $\circ_i\colon \mathcal{P}(m)\otimes\mathcal{P}(n)\to\mathcal{P}(m+n-1)$, "plug the second operation into the i-th input of the first", satisfying the sequential and parallel associativity axioms and equivariance.

Definition 25.2. For an object A, the *endomorphism operad* has $\mathrm{End}_A(n)=\mathrm{Hom}(A^{\otimes n},A)$, with $\mathfrak{S}_n$ permuting the inputs and γ given by composition of multilinear maps. An *algebra over* $\mathcal{P}$ is an operad morphism $\mathcal{P}\to\mathrm{End}_A$; equivalently, maps $\mathcal{P}(n)\otimes A^{\otimes n}\to A$ compatible with the compositions and the symmetric group actions.

Example 25.3 (Ass, Com, Lie)**.** $\mathrm{Ass}(n) = k[\mathfrak{S}_n]$, the regular representation: an Ass-algebra is an associative algebra. $\mathrm{Com}(n) = k$, the trivial representation (for $n \geq 1$): commutative associative algebras. $\mathrm{Lie}(n)$ is the subspace of the free Lie algebra on $x_1, \dots, x_n$ spanned by brackets containing each x_i exactly once; $\dim \mathrm{Lie}(n) = (n-1)!$, with basis the left-normed brackets $[x_{\sigma(1)}, [x_{\sigma(2)}, [\dots, x_n]]]$, $\sigma \in \mathfrak{S}_{n-1}$ permuting $1, \dots, n-1$: a Lie-algebra is a Lie algebra. Morphisms of operads $\mathrm{Lie} \to \mathrm{Ass} \to \mathrm{Com}$ send the bracket to the commutator and every permutation to 1; they correspond to the facts that an associative algebra is a Lie algebra under the commutator and a commutative algebra is associative.

Warning 25.4 (Arity-wise maps need not be operad maps)*.* The $\mathfrak{S}_n$-equivariant inclusion $\mathrm{Com}(n) \to \mathrm{Ass}(n)$, $1 \mapsto \sum_{\sigma \in \mathfrak{S}_n} \sigma$ (symmetrisation), is not compatible with composition: composing $\mu_2 = e + (12)$ with itself in the first slot gives a sum of 4 ternary operations, while the symmetrisation in arity 3 has 6 terms. So it is not an operad morphism, although each arity component is equivariant.

Example 25.5 (Topological operads)**.** An operad in spaces induces operads $H_*(\mathcal{P}; k)$ in graded vector spaces (by Künneth over a field). The basic example: the *little n-disks operad* D_n, where $\mathsf{D}_n(k)$ is the space of k-tuples of embeddings $D^n \to D^n$ of the form $x \mapsto rx + c$ ($r > 0$), with pairwise disjoint images; $\mathfrak{S}_k$ permutes the disks and γ inserts configurations into disks. Sending a configuration of disks to their centres gives a $\mathfrak{S}_k$-equivariant homotopy equivalence $\mathsf{D}_n(k) \simeq F_k(\mathbb{R}^n)$, the ordered configuration space of Part VI.

Theorem 25.6 (Boardman–Vogt, May)**.** *Every n-fold loop space $\Omega^n Y$ is a D_n-algebra in spaces, and conversely every D_n-algebra whose π_0 is a group is weakly equivalent to an n-fold loop space.*

See [38], [39]. The action of $\mathsf{D}_n(k)$ on $\Omega^n Y$ (maps $(D^n, \partial D^n) \to (Y, *)$) puts the k maps into the k little disks and extends by the base point.

Theorem 25.7 (F. Cohen)**.** *For $n \geq 2$, $H_*(\mathsf{D}_n; \mathbb{Q})$, equivalently the operad formed by $H_*(F_k(\mathbb{R}^n); \mathbb{Q})$, is the operad $\mathsf{e}_n = \mathrm{Pois}_n$ of n-*Poisson algebras*: graded commutative algebras with a Lie bracket of degree $n-1$ which is a derivation in each variable. For $n = 2$ these are* Gerstenhaber algebras*. In particular $H_*(\mathsf{D}_n(2)) = H_*(S^{n-1})$ carries the product (degree 0) and the bracket (degree $n-1$).*

See [40] (the computation is over $\mathbb{Z}$ and includes the action of Dyer–Lashof operations in characteristic p); Part VI derives the additive structure from the Fadell–Neuwirth fibrations. The little disks operads are the meeting point of iterated loop spaces, configuration spaces and deformation quantisation (Kontsevich's formality of D_n).

25.3 The homotopy Lie algebra

Definition 25.8. For a simply connected space X, the graded vector space $L_X = \pi_*(\Omega X) \otimes \mathbb{Q}$, with $(L_X)_k = \pi_{k+1}(X) \otimes \mathbb{Q}$, carries the *Samelson product*: for $\alpha\colon S^p \to \Omega X$, $\beta\colon S^q \to \Omega X$, the commutator map $S^p \times S^q \to \Omega X$, $(s,t) \mapsto \alpha(s)\beta(t)\alpha(s)^{-1}\beta(t)^{-1}$, factors through $S^p \wedge S^q = S^{p+q}$. Under $\pi_k(\Omega X) \cong \pi_{k+1}(X)$ it agrees up to sign with the Whitehead product, and it makes L_X a graded Lie algebra: $[a,b] = -(-1)^{|a||b|}[b,a]$ and the graded Jacobi identity hold. L_X is the *homotopy Lie algebra* of X.

Example 25.9. For S^n, n odd, L is one-dimensional, spanned by a in degree $n-1$ (even) with $[a,a]=0$ (forced by antisymmetry in even degree). For n even, L has a in degree $n-1$ (odd) and b in degree $2n-2$, with $[a,a]=\pm 2b$ — the Whitehead square, Proposition 19.16. In the Sullivan model $(\Lambda(x,y), dy=x^2)$ this is visible as the quadratic differential (Theorem 24.12).

Theorem 25.10 (Milnor–Moore)**.** *For X simply connected, the Hurewicz map induces an isomorphism of Hopf algebras $U(L_X)\cong H_*(\Omega X;\mathbb{Q})$, where U is the universal enveloping algebra and $H_*(\Omega X;\mathbb{Q})$ has the Pontryagin product. The primitive elements of $H_*(\Omega X;\mathbb{Q})$ are exactly the image of L_X.*

See [41], [17, §21].

Proposition 25.11. *For a graded vector space V there is a free graded Lie algebra $\mathbb{L}(V)$, and $U\mathbb{L}(V)\cong T(V)$, the tensor algebra.*

Proof. Both $U\mathbb{L}$ and T are left adjoint to the forgetful functor from associative algebras to graded vector spaces: $\mathrm{Alg}(U\mathbb{L}V,A)\cong \mathrm{Lie}(\mathbb{L}V,A_{\mathrm{Lie}})\cong \mathrm{Vect}(V,A)\cong \mathrm{Alg}(TV,A)$; adjoints are unique (Exercise 20.4). □

Theorem 25.12 (Graded Poincaré–Birkhoff–Witt)**.** *Over a field of characteristic zero, $U(L)\cong\Lambda(L)$ as graded vector spaces (indeed as graded coalgebras), where $\Lambda(L)$ is the free graded commutative algebra (polynomial on even, exterior on odd elements). Hence, with $\ell_k=\dim L_k$,*

$$\sum_n \dim U(L)_n t^n=\prod_{k\ \text{even}}(1-t^k)^{-\ell_k}\prod_{k\ \text{odd}}(1+t^k)^{\ell_k}.$$

Example 25.13 (The rational homotopy of $S^3\vee S^3$)**.** $\Omega(S^3\vee S^3)$ has $H_*=T(u,v)$ with $|u|=|v|=2$ (Bott–Samelson), and L is free on u,v (both even). By Theorem 25.12, $\prod_{m\ge1}(1-t^{2m})^{-\ell_{2m}}=\frac{1}{1-2t^2}$. Applying $s\frac{d}{ds}\log$ with $s=t^2$ to both sides and comparing coefficients of s^n gives $\sum_{d\mid n} d\,\ell_{2d}=2^n$, and Möbius inversion yields the *Witt formula*

$$\ell_{2n}=\frac{1}{n}\sum_{d\mid n}\mu(d)\,2^{n/d}:\qquad \ell_2=2,\ \ell_4=1,\ \ell_6=2,\ \ell_8=3,\ \ell_{10}=6,\ \ell_{12}=9,\dots$$

So $\dim\pi_{2n+1}(S^3\vee S^3)\otimes\mathbb{Q}=\ell_{2n}$ grows like $2^n/n$: exponential growth.

Theorem 25.14 (Dichotomy; Félix–Halperin–Thomas)**.** *Let X be a simply connected finite CW complex. Either $\pi_*(X)\otimes\mathbb{Q}$ is finite-dimensional (X is* elliptic*), or the sequence $\sum_{k\le r}\dim\pi_k(X)\otimes\mathbb{Q}$ grows exponentially in r (X is* hyperbolic*).*

See [17, §33]. Elliptic spaces satisfy strong constraints: their rational cohomology satisfies Poincaré duality, $\chi(X)\ge 0$, and $\dim\pi_{\mathrm{even}}\otimes\mathbb{Q}\le\dim\pi_{\mathrm{odd}}\otimes\mathbb{Q}$ [17, §32]. Spheres, Lie groups and homogeneous spaces are elliptic; wedges of at least two spheres are hyperbolic (Example 25.13).

25.4 Exercises

Exercise 25.15 (medium)**.** Verify the Witt formula numbers $\ell_2,\dots,\ell_{10}$ in Example 25.13 directly from $\prod(1-t^{2m})^{-\ell_{2m}}=(1-2t^2)^{-1}$, and identify explicit brackets spanning L_4 and L_6.

SOURCE: original

Solution. Write $s = t^2$. Expanding $\prod_m (1-s^m)^{-\ell_m} = 1 + 2s + 4s^2 + 8s^3 + 16s^4 + 32s^5 + \cdots$ (here ℓ_m means ℓ_{2m}): the coefficient of s gives $\ell_1 = 2$. Coefficient of s^2: $\binom{\ell_1+1}{2} + \ell_2 = 3 + \ell_2 = 4$, so $\ell_2 = 1$. Coefficient of s^3: from ℓ_1 alone $\binom{4}{3} = 4$, from $\ell_1\ell_2$: $2 \cdot 1 = 2$, plus ℓ_3: $6 + \ell_3 = 8$, $\ell_3 = 2$. Using $\sum_{d|n} d\ell_d = 2^n$ is quicker: $n = 4$: $\ell_1 + 2\ell_2 + 4\ell_4 = 2 + 2 + 4\ell_4 = 16$, $\ell_4 = 3$; $n = 5$: $2 + 5\ell_5 = 32$, $\ell_5 = 6$. (This identity follows by taking $s\frac{d}{ds}$ log of both sides: $\sum_m \ell_m \frac{ms^m}{1-s^m} = \frac{2s}{1-2s}$, and comparing coefficients of s^n.) Brackets: L_4 (degree 4, i.e. $n = 2$) is spanned by $[u, v]$; L_6 by $[u, [u, v]]$ and $[v, [u, v]]$.

Exercise 25.16 (medium)**.** Show that $S^2 \vee S^2$ is hyperbolic, by exhibiting infinitely many linearly independent iterated Whitehead products in $\pi_*(S^2 \vee S^2) \otimes \mathbb{Q}$.

SOURCE: original

Solution. $H_*(\Omega(S^2 \vee S^2); \mathbb{Q}) = T(a, b)$ with $|a| = |b| = 1$ (Bott–Samelson), so by Theorem 25.10 and Proposition 25.11 $L = \mathbb{L}(a, b)$ is the free graded Lie algebra on two odd generators. It contains the elements $[a, [a, \ldots, [a, b]]]$ (k copies of a), which are nonzero for all k: their images in $T(a, b)$ are iterated graded commutators $\mathrm{ad}_a^k(b)$, and for odd a, $\mathrm{ad}_a^2 = \frac{1}{2}\mathrm{ad}_{[a,a]} = \mathrm{ad}_{a^2}$ (as $[a, a] = 2a^2$ in T), so $\mathrm{ad}_a^{2j}(b) = \mathrm{ad}_{a^2}^j(b)$, whose expansion in $T(a, b)$ contains the monomial $a^{2j}b$ with coefficient 1: nonzero. These lie in distinct degrees, giving infinitely many nonzero rational homotopy groups (indeed exponential growth by PBW).

Exercise 25.17 (easy)**.** Verify that Ass $\to$ Com, sending every permutation to 1, is a morphism of operads, and that an operad morphism $\mathcal{P} \to \mathcal{Q}$ induces a functor from $\mathcal{Q}$-algebras to $\mathcal{P}$-algebras.

SOURCE: orig-book (made into an exercise with solution)

Solution. γ in Ass substitutes words (permutations) into a word; applying "every permutation $\mapsto 1$" before or after substitution gives 1, and γ in Com is multiplication $k \otimes \cdots \otimes k \to k$; units and equivariance are respected since the trivial representation is a quotient of the regular one. For the second statement compose: $\mathcal{P} \to \mathcal{Q} \to \mathrm{End}_A$.

25.5 Connections and further reading

Operads encode algebraic structures and their homotopy-invariant versions; the little disks operads, built from configuration spaces, encode iterated loop spaces; and the homotopy Lie algebra encodes rational homotopy, with Milnor–Moore and PBW converting its dimensions into the elliptic–hyperbolic dichotomy. References: Loday–Vallette [42], Fresse [43] (operads and the little disks), May [38], Cohen–Lada–May [40], Félix–Halperin–Thomas [17, Part IV–V].

Part V

Research preparation

Chapter 26

Integral characteristic classes

26.1 The problem

Stiefel–Whitney classes (Chapter 12) see only mod-2 information. Over $\mathbb{Z}$ there are three families: the Euler class of oriented bundles, the Chern classes of complex bundles, and the Pontryagin classes of real bundles. They detect signs and odd torsion, they are obstructions (Theorem 17.24), and their characteristic numbers carry geometric content (the Euler characteristic, the signature). The chapter also closes a triangle opened in Chapter 12: an odd-rank Euler class is the integral Bockstein of a Stiefel–Whitney class, which explains why it is 2-torsion.

Prerequisites. Thom isomorphism and Gysin sequence (Chapter 16), Leray–Hirsch (Theorem 16.4), Steenrod squares and Wu's formula (Chapter 18), the Bockstein (Chapter 9).

26.2 The Euler class

Definition 26.1. Let $\xi\colon E \to B$ be an oriented real vector bundle of rank k with Thom class $u \in H^k(D(\xi), S(\xi); \mathbb{Z})$ (Theorem 16.15). Its *Euler class* $e(\xi) \in H^k(B; \mathbb{Z})$ is the image of u under $H^k(D, S) \to H^k(D) \cong H^k(B)$.

Proposition 26.2. *(a) Naturality:* $e(f^*\xi) = f^*e(\xi)$.

(b) Reversing the orientation negates e.

(c) Whitney: $e(\xi \oplus \eta) = e(\xi) \smile e(\eta)$.

(d) If ξ *has a nowhere-zero section,* $e(\xi) = 0$.

(e) The mod-2 reduction of $e(\xi)$ *is* $w_k(\xi)$.

(f) $e(TM)$ *evaluated on* $[M]$ *is* $\chi(M)$ *for a closed oriented manifold* M.

Proof. (a), (b): the Thom class is natural and changes sign with the orientation. (c): the Thom class of $\xi \oplus \eta$ is the cross product of the Thom classes (up to the identification of the Thom space of the sum with a quotient of the smash of the Thom spaces), and restricting to zero sections gives the cup product. (d): a nowhere-zero section s gives a homotopy of the zero section into the sphere bundle, so the composite $B \to D(\xi)$ factors

up to homotopy through $S(\xi)$, where u restricts to zero. (e): the mod-2 Thom class is the reduction of u, and $w_k = \Phi^{-1}\operatorname{Sq}^k u = \Phi^{-1}(u^2)$ (Theorem 18.27 and instability), while $u \smile u = (p^*e) \smile u = \Phi(e)$ because u restricts to p^*e on $D(\xi)$; reduce mod 2. (f) is [10, §11]. □

Exercise 26.3 (medium)**.** Let ξ be oriented of odd rank. Show $2e(\xi) = 0$.

SOURCE: orig-book; also MIT OCW 18.906 (Spring 2020), PSet VI #31(b), first part

Solution. Fibrewise multiplication by -1 is a bundle automorphism covering the identity, with determinant $(-1)^{\operatorname{rank}} = -1$ on each fibre; it carries the orientation to the opposite one. Being an isomorphism of bundles, it preserves e (naturality), while reversing orientation negates e. So $e = -e$.

26.3 Chern classes

Definition 26.4. For a complex line bundle L put $c_1(L) = e(L_{\mathbb{R}})$, the Euler class of the underlying real plane bundle with its complex orientation. Let ξ be a complex vector bundle of rank n over B, $\mathbb{P}(\xi) \xrightarrow{\pi} B$ its projectivisation, $\lambda \subseteq \pi^*\xi$ the tautological line subbundle, and $x = c_1(\lambda^*) \in H^2(\mathbb{P}(\xi);\mathbb{Z})$. By Leray–Hirsch (Example 16.9) $H^*(\mathbb{P}(\xi))$ is free over $H^*(B)$ on $1, x, \dots, x^{n-1}$, so there are unique classes $c_i(\xi) \in H^{2i}(B;\mathbb{Z})$ with

$$x^n + c_1(\xi)x^{n-1} + \cdots + c_n(\xi) = 0.$$

These are the *Chern classes*; $c = 1 + c_1 + \cdots + c_n$.

The sign conventions vary between authors; with this one c_1 of the tautological line bundle on $\mathbb{CP}^\infty$ is minus the generator dual to $\mathbb{CP}^1$, and c_1 of its dual $\mathcal{O}(1)$ is the generator.

Theorem 26.5 (Properties of Chern classes)**.** *Chern classes are natural, $c(\xi \oplus \eta) = c(\xi)c(\eta)$, $c_n(\xi) = e(\xi_{\mathbb{R}})$ for the underlying oriented real bundle, and for a line bundle $c_1(L)$ is its classifying class in $H^2(B;\mathbb{Z})$ (Exercise 17.40). They are uniquely determined by these properties and normalisation.*

The proof rests on:

Theorem 26.6 (Splitting principle)**.** *For a complex bundle ξ over B there is a space $F(\xi) \xrightarrow{p} B$ (the flag bundle) such that $p^*\colon H^*(B;\mathbb{Z}) \to H^*(F(\xi);\mathbb{Z})$ is injective and $p^*\xi$ splits as a sum of line bundles.*

Proof. Iterate projectivisation: on $\mathbb{P}(\xi)$, $\pi^*\xi = \lambda \oplus \lambda^\perp$, and π^* is injective by Leray–Hirsch (the basis contains 1); repeat with $\lambda^\perp$. □

So identities among Chern classes may be proved for sums of line bundles $L_1 \oplus \cdots \oplus L_n$, where $c = \prod(1 + x_i)$, $x_i = c_1(L_i)$: the c_j are the elementary symmetric polynomials in the *Chern roots* x_i.

Example 26.7. $T\mathbb{CP}^n \oplus \mathbb{C} \cong (n+1)\mathcal{O}(1)$ (the complex analogue of Proposition 12.3), so $c(T\mathbb{CP}^n) = (1+h)^{n+1}$, h the positive generator. Then $c_n(T\mathbb{CP}^n) = (n+1)h^n$ and $\chi(\mathbb{CP}^n) = n+1$, consistent with Proposition 26.2(f).

Exercise 26.8 (medium)**.** Let ξ be a complex n-plane bundle and λ a complex line bundle with $c_1(\lambda) = y$. Express $c(\lambda \otimes \xi)$ in terms of $c(\xi)$ and y.

SOURCE: MIT OCW 18.906 (Spring 2020), Problem Set VI, Problem 31(a), reproduced in substance

Solution. By the splitting principle assume $\xi = L_1 \oplus \cdots \oplus L_n$ with roots x_i. Then $\lambda \otimes \xi = \bigoplus \lambda \otimes L_i$ and $c_1(\lambda \otimes L_i) = x_i + y$ (Exercise 26.9). So $c(\lambda \otimes \xi) = \prod_i (1 + y + x_i) = \sum_{j=0}^{n} c_j(\xi)(1+y)^{n-j}$, i.e. $c_k(\lambda \otimes \xi) = \sum_{j=0}^{k} \binom{n-j}{k-j} c_j(\xi) y^{k-j}$.

Exercise 26.9 (easy)**.** Show $c_1(L \otimes L') = c_1(L) + c_1(L')$ for complex line bundles.

SOURCE: orig-book

Solution. Line bundles are classified by maps to $\mathbb{CP}^\infty = K(\mathbb{Z}, 2)$ and c_1 is the pulled-back fundamental class (Exercise 17.40). The tensor product of the tautological bundles over $\mathbb{CP}^\infty \times \mathbb{CP}^\infty$ is classified by a map $\mu\colon \mathbb{CP}^\infty \times \mathbb{CP}^\infty \to \mathbb{CP}^\infty$ with $\mu^* \iota = a\iota \times 1 + b\, 1 \times \iota$; restricting to $\mathbb{CP}^\infty \times *$ and $* \times \mathbb{CP}^\infty$ (where the tensor product with the trivial bundle is the bundle itself) gives $a = b = 1$. Pull back along the classifying map of (L, L').

26.4 Pontryagin classes

Definition 26.10. For a real bundle ξ, $p_i(\xi) = (-1)^i c_{2i}(\xi \otimes \mathbb{C}) \in H^{4i}(B; \mathbb{Z})$.

The odd Chern classes of a complexification are 2-torsion ($\xi \otimes \mathbb{C} \cong \overline{\xi \otimes \mathbb{C}}$ and $c_i(\bar\eta) = (-1)^i c_i(\eta)$), which is why only even ones are used. Whitney: $p(\xi \oplus \eta) = p(\xi)p(\eta)$ modulo 2-torsion. For an oriented rank-$2k$ bundle, $p_k = e^2$ [10, §15].

Example 26.11. $T\mathbb{CP}^2 \otimes \mathbb{C} = T\mathbb{CP}^2 \oplus \overline{T\mathbb{CP}^2}$, so $c(T\mathbb{CP}^2 \otimes \mathbb{C}) = (1+h)^3(1-h)^3 = (1-h^2)^3$, and $p_1(\mathbb{CP}^2) = 3h^2$. Hirzebruch's signature theorem [10, §19] gives $\sigma(M^4) = \frac{1}{3}\langle p_1, [M]\rangle$; for $\mathbb{CP}^2$, $\sigma = 1$. Since the signature is a homotopy invariant (it is defined by the cup product form) while p_1 is a priori a bundle invariant, the theorem shows p_1 of a 4-manifold is a homotopy invariant — in contrast to higher Pontryagin classes, which are only rational topological invariants (Novikov).

26.5 The triangle: Stiefel–Whitney, Bockstein, Euler

Theorem 26.12. *For an oriented real bundle ξ of rank $2k+1$, $e(\xi) = \tilde\beta\big(w_{2k}(\xi)\big)$, where $\tilde\beta\colon H^{2k}(B; \mathbb{F}_2) \to H^{2k+1}(B; \mathbb{Z})$ is the integral Bockstein.*

Proof. By naturality it suffices to treat the universal bundle over $BSO(2k+1)$. There:

(i) $\rho e = w_{2k+1}$ (Proposition 26.2(e));

(ii) Wu's formula for the squares of Stiefel–Whitney classes, $\mathrm{Sq}^1 w_i = w_1 w_i + (i+1) w_{i+1}$ [10, §8] (verified for $\mathbb{RP}^n$ in Exercise 26.14), gives $\mathrm{Sq}^1 w_{2k} = w_{2k+1}$ because $w_1 = 0$;

(iii) $\mathrm{Sq}^1 = \rho\tilde\beta$ (Corollary 18.16).

So $\rho(e - \tilde\beta w_{2k}) = 0$, and $z = e - \tilde\beta w_{2k}$ lies in $2H^{2k+1}(BSO(2k+1); \mathbb{Z})$. Both e (Exercise 26.3) and $\tilde\beta w_{2k}$ (image of $\tilde\beta$) are killed by 2, so z is torsion. By a theorem of E. Brown [44], all torsion in $H^*(BSO(n); \mathbb{Z})$ has order 2. Write $z = 2y$; then $4y = 2z = 0$, so y is torsion, hence $2y = 0$, i.e. $z = 0$. □

Exercise 26.13 (medium)**.** Over $B = \mathbb{RP}^\infty \times \mathbb{RP}^\infty$ with $H^*(B; \mathbb{F}_2) = \mathbb{F}_2[a, b]$, let γ_1, γ_2 be the pulled-back tautological line bundles and $\xi = \gamma_1 \oplus \gamma_2 \oplus (\gamma_1 \otimes \gamma_2)$. Show that ξ is orientable and that $e(\xi) \neq 0$, although $2e(\xi) = 0$.

SOURCE: original solution of MIT OCW 18.906 (Spring 2020), PSet VI #31(b), second part ("give an example in which e is nevertheless nonzero")

Solution. $w(\gamma_1) = 1+a$, $w(\gamma_2) = 1+b$, $w(\gamma_1 \otimes \gamma_2) = 1+a+b$ (for real line bundles w_1 is additive under $\otimes$, as c_1 is in Exercise 26.9). So $w(\xi) = (1+a)(1+b)(1+a+b)$, giving $w_1 = 2(a+b) = 0$ (orientable), $w_2 = ab + a(a+b) + b(a+b) = a^2 + ab + b^2$, $w_3 = ab(a+b) = a^2b + ab^2 \neq 0$. By Proposition 26.2(e), $\rho e(\xi) = w_3 \neq 0$, so $e(\xi) \neq 0$; and $2e = 0$ by Exercise 26.3. (By Theorem 26.12, $e = \tilde{\beta}(a^2 + ab + b^2)$.)

Exercise 26.14 (medium)**.** Verify $\mathrm{Sq}^1 w_i = w_1 w_i + (i+1) w_{i+1}$ for the tangent bundle of $\mathbb{RP}^n$.

SOURCE: orig-book

Solution. $w_i = \binom{n+1}{i} a^i$ and $\mathrm{Sq}^1 a^i = i a^{i+1}$ (Proposition 18.11). The identity becomes $i\binom{n+1}{i} \equiv (n+1)\binom{n+1}{i} + (i+1)\binom{n+1}{i+1} \pmod 2$; since $(i+1)\binom{n+1}{i+1} = (n+1-i)\binom{n+1}{i}$, the right side is $(2n+2-i)\binom{n+1}{i} \equiv i\binom{n+1}{i}$.

26.6 Which classes are characteristic classes?

Every class in $H^2(B;\mathbb{Z})$ is c_1 of a line bundle (Exercise 17.40). In higher degrees there are constraints. Bott proved, using his periodicity theorem, that a class in $H^{2n}(S^{2n};\mathbb{Z})$ is c_n of a complex bundle over S^{2n} iff it is divisible by $(n-1)!$ [9, Cor. 4.4]. So c_2 of a bundle over S^4 can be any integer, but c_3 over S^6 must be even, and c_4 over S^8 divisible by 6.

26.7 Connections and further reading

Euler, Chern and Pontryagin classes refine Stiefel–Whitney classes integrally; the odd Euler class is a Bockstein, closing the triangle of Chapters 9, 12 and 18. In Part VI the Euler class of the tangent bundle of a closed manifold reappears as the class of the diagonal, which controls the cohomology of $F_2(M)$. References: Milnor–Stasheff [10], Bott–Tu [6], Hatcher [9].

Chapter 27

The Steenrod algebra in earnest

27.1 The problem

Chapter 18 used individual squares. This chapter studies the algebra $\mathcal{A}$ they generate: its basis, its coproduct and Milnor's dual description, and its role as the algebra of all stable mod-2 operations. These structures are what one computes with in the mod-2 homology of configuration spaces and iterated loop spaces (Part VI), and they are the input of the Adams spectral sequence.

Prerequisites. Chapter 18; Hopf algebras (Chapter 10).

27.2 Admissible monomials

Definition 27.1. For $I = (i_1, \dots, i_k)$ write $\mathrm{Sq}^I = \mathrm{Sq}^{i_1} \cdots \mathrm{Sq}^{i_k}$. I is *admissible* if $i_j \geq 2i_{j+1}$ for all j; its *excess* is $e(I) = \sum_j (i_j - 2i_{j+1}) = i_1 - i_2 - \cdots - i_k$; its degree is $\sum i_j$.

Theorem 27.2 (Serre, Cartan)**.** *The admissible monomials form a basis of $\mathcal{A}$. Moreover, for $|x| = n$: $\mathrm{Sq}^I x = 0$ if $e(I) > n$, and $\mathrm{Sq}^I x = (\mathrm{Sq}^{I'} x)^2$ if $e(I) = n$, where $I' = (i_2, \dots, i_k)$.*

Proof. Spanning. Define the moment of I as $\sum_j j\, i_j$. If I is not admissible, some $i_j < 2i_{j+1}$, and the Adem relation (Theorem 18.17) rewrites $\mathrm{Sq}^{i_j}\,\mathrm{Sq}^{i_{j+1}}$ as a sum of terms $\mathrm{Sq}^{i_j + i_{j+1} - c}\,\mathrm{Sq}^c$ with $c \leq i_j/2 < i_{j+1}$, which moves weight to the left and strictly lowers the moment. The moment is a nonnegative integer, so repeated rewriting terminates in admissible monomials.

Independence is proved by evaluating on $w_1 w_2 \cdots w_N \in H^N((\mathbb{RP}^\infty)^N)$ for large N, where distinct admissible monomials give linearly independent results [1, §4.L], or via Serre's computation Theorem 27.6 [16, Cor. 1.38].

Excess. Write $\mathrm{Sq}^I x = \mathrm{Sq}^{i_1}(\mathrm{Sq}^{I'} x)$ with $|\mathrm{Sq}^{I'} x| = n + i_2 + \cdots + i_k$; if $i_1 > n + i_2 + \cdots + i_k$, i.e. $e(I) > n$, instability gives 0, and equality gives the square. □

Exercise 27.3 (easy)**.** Express $\mathrm{Sq}^2\,\mathrm{Sq}^2$, $\mathrm{Sq}^1\,\mathrm{Sq}^2$, $\mathrm{Sq}^2\,\mathrm{Sq}^3$ and $\mathrm{Sq}^3\,\mathrm{Sq}^2$ in the admissible basis.

SOURCE: orig-book (first case); remaining cases original

Solution. Adem with (a,b): $\mathrm{Sq}^a\mathrm{Sq}^b = \sum_c \binom{b-c-1}{a-2c}\mathrm{Sq}^{a+b-c}\mathrm{Sq}^c$. $(2,2)$: $c=0$: $\binom{1}{2}=0$; $c=1$: $\binom{0}{0}=1$: $\mathrm{Sq}^2\mathrm{Sq}^2 = \mathrm{Sq}^3\mathrm{Sq}^1$. $(1,2)$: $c=0$: $\binom{1}{1}=1$: $\mathrm{Sq}^1\mathrm{Sq}^2=\mathrm{Sq}^3$. $(2,3)$: $c=0$: $\binom{2}{2}=1$; $c=1$: $\binom{1}{0}=1$: $\mathrm{Sq}^2\mathrm{Sq}^3 = \mathrm{Sq}^5+\mathrm{Sq}^4\mathrm{Sq}^1$. $(3,2)$: $c=0$: $\binom{1}{3}=0$; $c=1$: $\binom{0}{1}=0$: $\mathrm{Sq}^3\mathrm{Sq}^2=0$.

Exercise 27.4 (medium)**.** List the admissible monomials of degree ≤ 7, and compare their number in each degree with the number of sequences $(r_1, r_2, \dots)$ of nonnegative integers with $\sum r_i(2^i-1) = d$.

SOURCE: original

Solution. Degree 0: 1. 1: Sq^1. 2: Sq^2. 3: $\mathrm{Sq}^3, \mathrm{Sq}^2\mathrm{Sq}^1$. 4: $\mathrm{Sq}^4, \mathrm{Sq}^3\mathrm{Sq}^1$. 5: $\mathrm{Sq}^5, \mathrm{Sq}^4\mathrm{Sq}^1$. 6: $\mathrm{Sq}^6, \mathrm{Sq}^5\mathrm{Sq}^1, \mathrm{Sq}^4\mathrm{Sq}^2$. 7: $\mathrm{Sq}^7, \mathrm{Sq}^6\mathrm{Sq}^1, \mathrm{Sq}^5\mathrm{Sq}^2, \mathrm{Sq}^4\mathrm{Sq}^2\mathrm{Sq}^1$. Dimensions $1,1,1,2,2,2,3,4$. Sequences with $r_1+3r_2+7r_3 = d$: $d=3$: $(3),(0,1)$; $d=6$: $(6),(3,1),(0,2)$; $d=7$: $(7),(4,1),(1,2),(0,0,1)$; similarly the others. The counts agree.

27.3 Unstable modules and the cohomology of $K(\mathbb{Z}/2, n)$

Definition 27.5. An $\mathcal{A}$-module M is *unstable* if $\mathrm{Sq}^i x = 0$ for $i > |x|$; an *unstable algebra* if moreover $\mathrm{Sq}^{|x|}x = x^2$ and the Cartan formula holds.

$H^*(X;\mathbb{F}_2)$ is an unstable algebra for every space. The free unstable module on a class of degree n has basis the admissible $\mathrm{Sq}^I\iota_n$ with $e(I)\le n$, and the free unstable algebra is the polynomial algebra on those with $e(I)<n$ (the case $e(I)=n$ being a square):

Theorem 27.6 (Serre)**.** $H^*(K(\mathbb{Z}/2,n);\mathbb{F}_2) = \mathbb{F}_2[\mathrm{Sq}^I\iota_n :\ I \text{ admissible},\ e(I)<n]$.

See [16, Thm. 1.32], proved with the Serre spectral sequence and Borel's transgression theorem. Combined with Theorem 18.2, it shows that $\mathcal{A}$ is exactly the algebra of stable mod-2 operations: in degrees $<n$, $H^{n+*}(K(\mathbb{Z}/2,n))$ is spanned by the admissible $\mathrm{Sq}^I\iota_n$, all of excess $\le$ degree $<n$ [16, Cor. 1.38].

Example 27.7. $n=1$: admissible I with $e(I)<1$, i.e. $e(I)=0$: $I=(2^{k-1},\dots,2,1)$, and $\mathrm{Sq}^I\iota_1 = \iota_1^{2^k}$; so $H^*(K(\mathbb{Z}/2,1)) = \mathbb{F}_2[\iota_1]$, as $\mathbb{RP}^\infty$ requires. $n=2$: generators ι_2, $\mathrm{Sq}^1\iota_2$, $\mathrm{Sq}^2\mathrm{Sq}^1\iota_2$, $\mathrm{Sq}^4\mathrm{Sq}^2\mathrm{Sq}^1\iota_2,\dots$ in degrees $2,3,5,9,\dots,2^k+1$.

27.4 The Hopf algebra structure and Milnor's dual

Proposition 27.8. *There is a unique algebra map $\psi\colon \mathcal{A}\to\mathcal{A}\otimes\mathcal{A}$ with $\psi(\mathrm{Sq}^k) = \sum_{i+j=k}\mathrm{Sq}^i\otimes\mathrm{Sq}^j$, coassociative and cocommutative, making $\mathcal{A}$ a Hopf algebra; for $\theta\in\mathcal{A}$ with $\psi(\theta)=\sum\theta'\otimes\theta''$, $\theta(xy)=\sum\theta'(x)\theta''(y)$.*

Proof sketch. Define ψ by the last formula: an operation applied to a cross product $x\times y$ on $X\times Y$ is determined, by Serre's theorem on $K(\mathbb{Z}/2,m)\times K(\mathbb{Z}/2,n)$, as a sum $\sum\theta'(x)\times\theta''(y)$; this defines $\psi(\theta)$ uniquely, and it is multiplicative because composition of operations corresponds to composition. On generators the Cartan formula gives the stated value. See [45]. □

Theorem 27.9 (Milnor)**.** *The dual Hopf algebra $\mathcal{A}_* = \mathrm{Hom}(\mathcal{A},\mathbb{F}_2)$ is a polynomial algebra $\mathbb{F}_2[\xi_1,\xi_2,\dots]$, $|\xi_i| = 2^i-1$, with coproduct $\Delta(\xi_k) = \sum_{i=0}^k \xi_{k-i}^{2^i}\otimes\xi_i$ ($\xi_0=1$). Here ξ_i is dual to $\mathrm{Sq}^{2^{i-1}}\cdots\mathrm{Sq}^2\mathrm{Sq}^1$ in the admissible basis.*

This is [45]. The dual basis to the monomials $\xi_1^{r_1}\xi_2^{r_2}\cdots$ is the *Milnor basis* $\mathrm{Sq}(r_1, r_2, \dots)$, with $\mathrm{Sq}(r, 0, \dots) = \mathrm{Sq}^r$; Exercise 27.4 checks the dimension count. The elements P_t^0 dual to ξ_t satisfy $(P_t^0)^2 = 0$ and generate exterior subalgebras used in computations of Margolis homology.

Exercise 27.10 (medium)**.** Show that the subalgebra $\mathcal{A}(1) \subseteq \mathcal{A}$ generated by Sq^1 and Sq^2 has dimension 8, with basis

$$1,\ \mathrm{Sq}^1,\ \mathrm{Sq}^2,\ \mathrm{Sq}^1\mathrm{Sq}^2,\ \mathrm{Sq}^2\mathrm{Sq}^1,\ \mathrm{Sq}^1\mathrm{Sq}^2\mathrm{Sq}^1,\ \mathrm{Sq}^2\mathrm{Sq}^2,\ \mathrm{Sq}^2\mathrm{Sq}^1\mathrm{Sq}^2$$

(one may also write $\mathrm{Sq}^3 = \mathrm{Sq}^1\mathrm{Sq}^2$, $\mathrm{Sq}^2\mathrm{Sq}^2 = \mathrm{Sq}^3\mathrm{Sq}^1$, …), top element in degree 6.

Source: [1, §4.L, Ex. 5], reproduced in substance

Solution. The Adem relations needed: $\mathrm{Sq}^1\mathrm{Sq}^1 = 0$, $\mathrm{Sq}^1\mathrm{Sq}^2 = \mathrm{Sq}^3$, $\mathrm{Sq}^2\mathrm{Sq}^2 = \mathrm{Sq}^3\mathrm{Sq}^1$, $\mathrm{Sq}^2\mathrm{Sq}^3 = \mathrm{Sq}^5 + \mathrm{Sq}^4\mathrm{Sq}^1$, $\mathrm{Sq}^3\mathrm{Sq}^3 = \mathrm{Sq}^5\mathrm{Sq}^1$, $\mathrm{Sq}^1\mathrm{Sq}^5 = 0$, $\mathrm{Sq}^2\mathrm{Sq}^5 = \mathrm{Sq}^6\mathrm{Sq}^1$ (from Theorem 18.17; e.g. for $(3,3)$ only $c = 1$ contributes, with $\binom{1}{1} = 1$; for $(1,5)$, $\binom{4}{1} = 4 \equiv 0$; for $(2,5)$, $\binom{4}{2} \equiv 0$ and $\binom{3}{0} = 1$).

Spanning. In a word in $\mathrm{Sq}^1, \mathrm{Sq}^2$, a factor $\mathrm{Sq}^1\mathrm{Sq}^1$ kills the word and a factor $\mathrm{Sq}^2\mathrm{Sq}^2$ may be replaced by $\mathrm{Sq}^3\mathrm{Sq}^1 = \mathrm{Sq}^1\mathrm{Sq}^2\mathrm{Sq}^1$, which does not change the degree and shortens nothing but removes the repeated Sq^2; repeating, every word becomes a sum of alternating words. Alternating words of length ≤ 3 are the first seven listed elements. In length 4: $\mathrm{Sq}^1\mathrm{Sq}^2\mathrm{Sq}^1\mathrm{Sq}^2 = \mathrm{Sq}^3\mathrm{Sq}^3 = \mathrm{Sq}^5\mathrm{Sq}^1$ and $\mathrm{Sq}^2\mathrm{Sq}^1\mathrm{Sq}^2\mathrm{Sq}^1 = \mathrm{Sq}^2\mathrm{Sq}^3\mathrm{Sq}^1 = (\mathrm{Sq}^5 + \mathrm{Sq}^4\mathrm{Sq}^1)\mathrm{Sq}^1 = \mathrm{Sq}^5\mathrm{Sq}^1$; also $\mathrm{Sq}^2\mathrm{Sq}^2\mathrm{Sq}^2 = \mathrm{Sq}^2\mathrm{Sq}^3\mathrm{Sq}^1 = \mathrm{Sq}^5\mathrm{Sq}^1$. Call this element t. In length 5, every alternating word is $\mathrm{Sq}^1 t$ or $\mathrm{Sq}^2 t$ (or t followed by Sq^1 or Sq^2, which is the same word read differently): $\mathrm{Sq}^1 t = \mathrm{Sq}^1\mathrm{Sq}^5\mathrm{Sq}^1 = 0$ and $\mathrm{Sq}^2 t = \mathrm{Sq}^2\mathrm{Sq}^5\mathrm{Sq}^1 = \mathrm{Sq}^6\mathrm{Sq}^1\mathrm{Sq}^1 = 0$. So $\mathcal{A}(1)$ is spanned by the eight listed elements.

Independence. Their expansions in the admissible basis are

$$1,\ \mathrm{Sq}^1,\ \mathrm{Sq}^2,\ \mathrm{Sq}^3,\ \mathrm{Sq}^2\mathrm{Sq}^1,\ \mathrm{Sq}^3\mathrm{Sq}^1,\ \mathrm{Sq}^5 + \mathrm{Sq}^4\mathrm{Sq}^1,\ \mathrm{Sq}^5\mathrm{Sq}^1,$$

in degrees $0, 1, 2, 3, 3, 4, 5, 6$; within each degree they are distinct admissible monomials (or, in degree 5, a sum of two), hence linearly independent by Theorem 27.2. So $\dim \mathcal{A}(1) = 8$, with top element $t = \mathrm{Sq}^5\mathrm{Sq}^1$ in degree 6.

27.5 The Adams spectral sequence

The Steenrod algebra controls stable homotopy through the *Adams spectral sequence*

$$E_2^{s,t} = \mathrm{Ext}_{\mathcal{A}}^{s,t}(\mathbb{F}_2, \mathbb{F}_2) \Longrightarrow \pi_{t-s}^s \otimes \mathbb{Z}_2^\wedge,$$

converging to the 2-completed stable stems [46], [8, Ch. 9]. Filtration $s = 1$ is spanned by classes h_i dual to the indecomposables Sq^{2^i}; a stable map of Hopf invariant one in stem $2^i - 1$ would be detected by h_i, and Adams' theorem is the statement that h_i supports a nonzero differential for $i \ge 4$ [13]. The decomposability of Sq^n for n not a power of 2 (Proposition 18.20) is the algebraic reason that filtration one contains only the h_i.

27.6 Connections and further reading

The Steenrod algebra is the algebra of stable mod-2 operations; its admissible and Milnor bases make it computable, Serre's theorem identifies it with the cohomology of Eilenberg–MacLane spaces, and its Ext groups are the E_2-page of the Adams spectral sequence. References: Steenrod–Epstein [14], Milnor [45], Mosher–Tangora [15], Hatcher [1, §4.L], McCleary [8, Ch. 9], Ravenel [47].

Chapter 28

Spectra: stable homotopy and where the signs come from

28.1 The problem

Freudenthal's theorem (Theorem 14.15) says that $\pi_{n+k}(S^n)$ stabilises; the stable homotopy category is the place where suspension is invertible and these stable groups are the homotopy groups of an object, the sphere spectrum. Spectra also represent generalised cohomology theories, and they are where the Koszul sign rule finally gets an explanation. For configuration spaces, spectra enter through stable splittings (Snaith, Cohen–May–Taylor) and through scanning maps to section spaces (Part VI).

Prerequisites. Chapters 14, 18 and 21.

28.2 Spectra and their homotopy groups

Definition 28.1. A *spectrum* X is a sequence of based spaces X_n with structure maps $\sigma_n\colon \Sigma X_n \to X_{n+1}$. Its homotopy groups are $\pi_k(X) = \operatorname{colim}_n \pi_{k+n}(X_n)$, $k \in \mathbb{Z}$. An *Ω-spectrum* is one whose adjoint structure maps $X_n \to \Omega X_{n+1}$ are weak equivalences; then $\pi_k(X) = \pi_{k+n}(X_n)$ for $n \geq \max(0, -k)$.

Example 28.2. The *sphere spectrum* $\mathbb{S}$, $\mathbb{S}_n = S^n$: $\pi_k(\mathbb{S}) = \pi_k^s$, the stable stems (Example 14.16). The *suspension spectrum* $\Sigma^\infty Y$, $(\Sigma^\infty Y)_n = \Sigma^n Y$: $\pi_k = \pi_k^s(Y)$. The *Eilenberg–MacLane spectrum* HG, $(HG)_n = K(G,n)$ with structure maps adjoint to $K(G,n) \simeq \Omega K(G,n+1)$ (Lemma 17.34): an Ω-spectrum with $\pi_0 = G$ and no other homotopy.

Theorem 28.3 (Spectra and cohomology theories)**.** *An Ω-spectrum E defines a reduced cohomology theory on CW complexes, $\widetilde{E}^n(X) = [X, E_n]_*$ (for $n \geq 0$), satisfying exactness, suspension and the wedge axiom; for $E = HG$ this is ordinary cohomology by Theorem 17.19. Conversely (Brown representability) every reduced cohomology theory on based CW complexes satisfying the wedge axiom is represented by an Ω-spectrum, unique up to equivalence.*

See [1, §4.E–4.F], [48]. Exactness comes from the Puppe sequence (Theorem 21.10); the group structure from $E_n \simeq \Omega E_{n+1}$.

Remark 28.4 (The stable homotopy category). There is a homotopy category of spectra, the *stable homotopy category*, in which Σ is an equivalence, cofibre sequences are also fibre sequences, finite wedges are products, and there is a smash product $\wedge$, associative and commutative *up to coherent isomorphism* [48]. Defining a smash product that is associative and commutative *on the nose* at the level of spectra (needed to talk about ring spectra and module spectra as honest algebraic objects) requires more structure: symmetric spectra, orthogonal spectra or S-modules [49, 50].

28.3 Where the signs come from

Lemma 28.5. *Let $\tau \in \mathfrak{S}_k$ act on $S^k = (\mathbb{R}^k)^+$ by permuting coordinates. Then $\deg \tau = \operatorname{sgn}(\tau)$. In particular the shuffle exchanging a block of n coordinates with a block of m coordinates has degree $(-1)^{nm}$.*

Proof. τ is the one-point extension of a linear map with determinant $\operatorname{sgn}\tau$. A linear automorphism of $\mathbb{R}^k$ extends to S^k with degree the sign of its determinant: $GL_k(\mathbb{R})$ has two path components, and the linear maps in one component are homotopic through linear automorphisms (hence through maps of S^k) to the identity, in the other to a reflection, of degree -1 (Definition 4.24). The block shuffle is a product of nm transpositions. □

Remark 28.6 (The sign rule is a theorem). In the stable category an element of degree n is a map out of S^n, and exchanging two such elements in a smash product exchanges sphere coordinates; by Lemma 28.5 this costs $(-1)^{nm}$. This is the origin of graded commutativity of cup products (Chapter 7), of the Koszul sign rule in tensor products of graded objects, and of the signs in the graded Lie algebras of Chapter 25.

28.4 Symmetric spectra

Definition 28.7. A *symmetric spectrum* is a spectrum with a based $\mathfrak{S}_n$-action on each X_n such that the iterated structure maps $X_n \wedge S^m \to X_{n+m}$ are $\mathfrak{S}_n \times \mathfrak{S}_m$-equivariant, $\mathfrak{S}_m$ permuting the coordinates of $S^m = (\mathbb{R}^m)^+$ and acting on X_{n+m} through $\mathfrak{S}_m \subseteq \mathfrak{S}_{n+m}$ (the last m letters).

Example 28.8. $\mathbb{S}$ is symmetric: $\mathfrak{S}_n$ permutes the coordinates of S^n. So is HG with $(HG)_n = \widetilde{G}[S^n]$, the reduced free G-module on the simplicial sphere $S^n = (S^1)^{\wedge n}$, with permutation of smash factors.

Symmetric spectra carry a smash product which is associative, commutative and unital up to isomorphism, so commutative ring spectra are commutative monoids [49]. The price is that the naive homotopy groups of Definition 28.1 are not invariant under the correct weak equivalences (the stable equivalences): there are stable equivalences of symmetric spectra which are not isomorphisms on naive homotopy groups. Spectra for which naive homotopy groups behave correctly are called *semistable*; suspension spectra and Ω-spectra are semistable [49], [51]. Two elementary facts about colimits are behind such phenomena:

Exercise 28.9 (easy)**.** If in a sequence $A_0 \to A_1 \to \cdots$ of abelian groups infinitely many maps are zero, then $\operatorname{colim} A_n = 0$.

SOURCE: orig-book

Solution. An element of the colimit is represented by some $a \in A_n$; choose $m \geq n$ with $A_m \to A_{m+1}$ zero; the image of a in A_{m+1} is 0.

Exercise 28.10 (easy)**.** With $\mathbb{Z}^n \hookrightarrow \mathbb{Z}^{n+1}$ the inclusion of the first n coordinates, $\operatorname{colim}_n \mathbb{Z}^n = \bigoplus_{n\geq 1} \mathbb{Z}$, which differs from $\prod_{n\geq 1} \mathbb{Z}$. Deduce that a colimit of finite products need not be a product.

SOURCE: orig-book

Solution. Every element of the colimit comes from some $\mathbb{Z}^n$, hence has finite support, and every finitely supported sequence occurs; $\bigoplus \mathbb{Z}$ is countable while $\prod \mathbb{Z}$ is not (Example 5.12). So naive homotopy groups, being colimits, do not obviously commute with infinite products.

28.5 Moore spectra

Definition 28.11. $\mathbb{S}/m$ is the cofibre of $m\colon \mathbb{S} \to \mathbb{S}$: $\mathbb{S} \xrightarrow{m} \mathbb{S} \xrightarrow{i} \mathbb{S}/m \xrightarrow{p} \Sigma\mathbb{S}$.

Exercise 28.12 (easy)**.** Show that $H^*(\mathbb{S}/2;\mathbb{F}_2)$ is $\mathbb{F}_2$ in degrees 0 and 1, with Sq^1 an isomorphism between them.

SOURCE: orig-book

Solution. Mod 2, $2\colon \mathbb{S} \to \mathbb{S}$ is zero on cohomology, so the long exact sequence gives $\mathbb{F}_2$ in degrees 0 and 1. $\mathbb{S}/2 = \Sigma^\infty\Sigma^{-1}\mathbb{RP}^2$, up to a shift: the Moore space $M(\mathbb{Z}/2,1) = \mathbb{RP}^2$ has Sq^1 an isomorphism $H^1 \to H^2$ (Example 18.12), and stability transports this to the spectrum.

Exercise 28.13 (easy)**.** Show that $H^*(X \vee Y;\mathbb{F}_2) \cong H^*(X;\mathbb{F}_2) \oplus H^*(Y;\mathbb{F}_2)$ as $\mathcal{A}$-modules.

SOURCE: orig-book

Solution. The inclusions and projections of a wedge induce a splitting as in Exercise 7.4; both are induced by maps, so the splitting is one of $\mathcal{A}$-modules.

Theorem 28.14. *$\mathbb{S}/2$ admits no unital multiplication $\mu\colon \mathbb{S}/2 \wedge \mathbb{S}/2 \to \mathbb{S}/2$ (i.e. with $\mu \circ (i \wedge 1) \simeq \mathrm{id}$, $i\colon \mathbb{S} \to \mathbb{S}/2$ the unit).*

Proof. Cohomology of the smash square. By Künneth, $H^*(\mathbb{S}/2 \wedge \mathbb{S}/2;\mathbb{F}_2)$ has basis $x_0y_0, x_1y_0, x_0y_1, x_1y_1$ in degrees $0,1,1,2$, where x_j, y_j are the classes of the two factors, and the Cartan formula (valid for smash products of spectra) gives

$$\mathrm{Sq}^2(x_0y_0) = \mathrm{Sq}^2 x_0\, y_0 + \mathrm{Sq}^1 x_0\ \mathrm{Sq}^1 y_0 + x_0\ \mathrm{Sq}^2 y_0 = x_1y_1 \neq 0,$$

since $\mathrm{Sq}^1 x_0 = x_1$, $\mathrm{Sq}^1 y_0 = y_1$ and Sq^2 vanishes on the degree-0 classes of $\mathbb{S}/2$ (there is nothing in degree 2).

A unital multiplication splits the smash square. Smashing the cofibre sequence $\mathbb{S} \xrightarrow{2} \mathbb{S} \xrightarrow{i} \mathbb{S}/2$ with $\mathbb{S}/2$ gives a cofibre sequence $\mathbb{S}/2 \xrightarrow{i\wedge 1} \mathbb{S}/2 \wedge \mathbb{S}/2 \to \Sigma\mathbb{S}/2$ (preceded by $2 \wedge 1$). If $\mu(i \wedge 1) \simeq \mathrm{id}$, then $i \wedge 1$ has a retraction, and in the stable homotopy category (a triangulated category) a cofibre sequence whose first map has a retraction splits: $\mathbb{S}/2 \wedge \mathbb{S}/2 \simeq \mathbb{S}/2 \vee \Sigma\mathbb{S}/2$.

Contradiction. By Exercise 28.13 the cohomology of $\mathbb{S}/2 \vee \Sigma\mathbb{S}/2$ is the sum of $\mathcal{A}$-modules concentrated in degrees $\{0,1\}$ and $\{1,2\}$, on which Sq^2 is identically zero (it would have to map degree 0 to degree 2 within one summand). But $\mathrm{Sq}^2 \neq 0$ on the smash square. □

Remark 28.15. For odd p, $\mathbb{S}/p$ does admit a unital multiplication (homotopy commutative and associative for $p \geq 5$). The failure at $p = 2$ is equivalent to the fact that $2 \cdot \mathrm{id}_{\mathbb{S}/2} \neq 0$, which is detected by $\eta \in \pi_1^s$: $2 \cdot \mathrm{id} = i\eta p$. The mod-2 Moore spectrum is the basic counterexample to naive expectations in stable homotopy theory.

28.6 Connections and further reading

Spectra make suspension invertible, represent cohomology theories, and explain the Koszul sign rule; symmetric spectra provide a point-set smash product at the cost of subtle homotopy groups. References: Adams [48], Hatcher [1, §4.F], Hovey–Shipley–Smith [49], Schwede [51], Elmendorf–Kriz–Mandell–May [50].

Chapter 29

Quasi-category technology

29.1 The problem

Chapter 23 defined quasi-categories. To use them one needs three tools that research papers take for granted: lifting arguments that work against all monomorphisms, the comparison with categories enriched in simplicial sets ("rigidification"), and criteria replacing explicit contracting homotopies. This short chapter supplies them.

29.2 Lifting against monomorphisms

Definition 29.1. A map of simplicial sets is a *trivial fibration* if it has the RLP against all $\partial\Delta^n \hookrightarrow \Delta^n$, $n \geq 0$.

Proposition 29.2. *A trivial fibration has the RLP against every monomorphism $A \hookrightarrow B$.*

Proof. Let $B^{(n)} = A \cup \mathrm{sk}_n B$. Then $B^{(n)}$ is obtained from $B^{(n-1)}$ as a pushout of $\coprod \partial\Delta^n \to \coprod \Delta^n$ over the nondegenerate n-simplices of B not in A. The class of maps with the LLP against a fixed map is closed under coproducts, cobase change and (transfinite) composition (Exercise 22.7), so $A \hookrightarrow B = \bigcup_n B^{(n)}$ has the LLP against every trivial fibration. □

Remark 29.3. This is the form in which lifting properties are used: to construct sections of trivial fibrations ($\varnothing \to Y$), or homotopies ($A \times \Delta^1 \cup B \times \partial\Delta^1 \hookrightarrow B \times \Delta^1$), neither of which is a boundary inclusion. In particular a trivial fibration between quasi-categories has a section and is an equivalence of quasi-categories [31, §2.2.5].

29.3 Rigidification

Construction 29.4. Let $\mathfrak{C}[\Delta^n]$ be the simplicial category with objects $0, \dots, n$ and $\mathfrak{C}[\Delta^n](i,j) = N(P_{i,j})$, where $P_{i,j}$ is the poset of subsets $S \subseteq \{i, \dots, j\}$ containing i and j, ordered by inclusion (empty if $i > j$); composition is union. This is a functor $\Delta \to \mathbf{sCat}$, so by Theorem 20.14 it induces an adjunction $\mathfrak{C}\colon \mathbf{sSet} \rightleftarrows \mathbf{sCat} \colon N_{\mathrm{hc}}$, where the *homotopy coherent nerve* is $N_{\mathrm{hc}}(\mathcal{C})_n = \mathbf{sCat}(\mathfrak{C}[\Delta^n], \mathcal{C})$.

Proposition 29.5 (Hom-cubes)**.** *$\mathfrak{C}[\Delta^n](i,j)$ is isomorphic to the cube $(\Delta^1)^{j-i-1}$ for $i < j$.*

Proof. A subset S containing i, j is determined by which of the $j-i-1$ intermediate elements it contains; inclusion is the product order on $\{0<1\}^{j-i-1}$, whose nerve is $(\Delta^1)^{j-i-1}$. □

Example 29.6. $\mathfrak{C}[\Delta^1](0,1) = \Delta^0$; $\mathfrak{C}[\Delta^2](0,2) = \Delta^1$, an edge from $\{0,2\}$ (the direct morphism) to $\{0,1,2\}$ (the composite through 1): the composite and the direct arrow are not equal but connected by a specified homotopy. $\mathfrak{C}[\Delta^3](0,3)$ is a square, whose two edges from $\{0,3\}$ to $\{0,1,2,3\}$ are the two ways of bracketing.

Theorem 29.7 (Comparison)**.** *$\mathfrak{C} \dashv N_{\mathrm{hc}}$ is a Quillen equivalence between the Joyal model structure on simplicial sets and the Bergner model structure on simplicial categories. For a simplicial category whose hom-spaces are Kan complexes, $N_{\mathrm{hc}}(\mathcal{C})$ is a quasi-category.*

See [31, Thm. 2.2.5.1]. In practice: every quasi-category is equivalent to the homotopy coherent nerve of a simplicial category, so $(\infty,1)$-categories may be presented either way.

Example 29.8 ($\mathfrak{C}$ does not preserve products)**.** $\mathfrak{C}[\Delta^1]\times\mathfrak{C}[\Delta^1]$ is the poset $[1]\times[1]$ with discrete homs, so its hom from $(0,0)$ to $(1,1)$ is a point. In $\mathfrak{C}[\Delta^1\times\Delta^1]$ (the square is a union of two 2-simplices glued along the diagonal) the hom from $(0,0)$ to $(1,1)$ is the union of two intervals, $\mathfrak{C}[\Delta^2](0,2)$ for each triangle, glued at the common vertex given by the diagonal: three vertices (the diagonal and the two paths along the edges), two edges. It is contractible but not a point. So $\mathfrak{C}$, a left adjoint, preserves colimits but not products — up to weak equivalence it does, which is all homotopy theory requires.

29.4 Extra degeneracies and cofinality

Proposition 29.9. *Let $A_\bullet \to A_{-1}$ be an augmented simplicial abelian group with an* extra degeneracy*: maps $s_{-1}\colon A_n \to A_{n+1}$ ($n \geq -1$) with $d_0 s_{-1} = \mathrm{id}$ and $d_{i+1}s_{-1} = s_{-1}d_i$ for $i \geq 0$ (where $d_0\colon A_0 \to A_{-1}$ is the augmentation). Then the augmented chain complex $\cdots \to A_1 \to A_0 \to A_{-1} \to 0$ is contractible; in particular $H_n(A) = 0$ for $n > 0$ and $H_0(A) \cong A_{-1}$.*

Proof. Let $h = s_{-1}$ and $\partial = \sum_i (-1)^i d_i$. On A_n,

$$\partial h = \sum_{i=0}^{n+1}(-1)^i d_i s_{-1} = d_0 s_{-1} + \sum_{i=1}^{n+1}(-1)^i s_{-1} d_{i-1} = \mathrm{id} - s_{-1}\sum_{j=0}^{n}(-1)^j d_j = \mathrm{id} - h\partial.$$

So $\partial h + h\partial = \mathrm{id}$: h is a contracting homotopy of the augmented complex. □

Example 29.10 (The bar resolution)**.** For a group G, $E_nG = G^{n+1}$ augmented over a point has the extra degeneracy $(g_0,\dots,g_n) \mapsto (e, g_0, \dots, g_n)$. Applying Proposition 29.9 to $\mathbb{Z}[E_\bullet G]$ shows that the bar complex $\mathbb{Z}[G^{\bullet+1}]$ is a free resolution of $\mathbb{Z}$ over $\mathbb{Z}[G]$: this is the resolution computing group cohomology in Part VI. Topologically the same extra degeneracy contracts $|EG|$ (Exercise 20.17).

Definition 29.11. A functor $u\colon J' \to J$ is *homotopy final* (resp. *initial*) if for every $j \in J$ the nerve of the comma category $(j \downarrow u)$ (resp. $(u \downarrow j)$) is contractible. Then for every diagram D on J, $\mathrm{hocolim}_{J'} Du \to \mathrm{hocolim}_J D$ is a weak equivalence (resp. dually for holim).

See [20]. An augmented simplicial object with extra degeneracy is a functor on the category $\Delta_{-\infty}^{\mathrm{op}}$, where $\Delta_{-\infty} \supseteq \Delta$ has an extra object $[-1]$ and the extra degeneracies are adjoined as morphisms; the inclusion $\Delta^{\mathrm{op}} \hookrightarrow \Delta_{-\infty}^{\mathrm{op}}$ is homotopy final, so the homotopy colimit (the realisation) is equivalent to the value at the adjoined object — Proposition 29.9 in categorical form.

29.5 Exercises

Exercise 29.12 (medium)**.** Describe the composition $\mathfrak{C}[\Delta^3](1,3) \times \mathfrak{C}[\Delta^3](0,1) \to \mathfrak{C}[\Delta^3](0,3)$ in terms of cubes, and identify its image.

Source: original

Solution. $\mathfrak{C}[\Delta^3](0,1)$ is a point $\{0,1\}$; $\mathfrak{C}[\Delta^3](1,3) = \Delta^1$, from $\{1,3\}$ to $\{1,2,3\}$. Composition is union: $\{0,1\} \cup S = \{0\} \cup S$. The image is the edge of the square $\mathfrak{C}[\Delta^3](0,3)$ (coordinates: contains 1? contains 2?) consisting of subsets containing 1: from $\{0,1,3\}$ to $\{0,1,2,3\}$. Geometrically, the face of the cube where the first coordinate is 1.

29.6 Connections and further reading

These three tools — lifting against monomorphisms, rigidification, and cofinality — are the working machinery of quasi-category theory, and of ∞-operads in modern configuration-space research (factorisation homology, Part VI). References: Lurie [31], Riehl [20], Riehl–Verity [32].

Chapter 30

Operads, Koszul duality, and twisting morphisms

30.1 The problem

"Algebra up to homotopy" — an associative product associative only up to a specified homotopy, a Lie bracket satisfying Jacobi up to homotopy — appears whenever algebraic structures are transported along homotopy equivalences: on the cohomology of a space (Chapter 31), on deformation complexes, and in the models of configuration spaces and of the little disks operads (Part VI). Koszul duality produces, for each quadratic operad $\mathcal{P}$, an explicit cofibrant resolution $\mathcal{P}_\infty$ whose algebras are the homotopy $\mathcal{P}$-algebras; its structure equations are a single Maurer–Cartan equation.

Prerequisites. Operads (Chapter 25); chain complexes and the Koszul sign rule (Remark 28.6).

30.2 $\mathfrak{S}$-modules, Schur functors, composition

Definition 30.1. An $\mathfrak{S}$-*module* is a sequence $E = (E(n))_{n\geq 0}$ of (dg) representations of the $\mathfrak{S}_n$. Its *Schur functor* is $\widetilde{E}(V) = \bigoplus_n E(n) \otimes_{\mathfrak{S}_n} V^{\otimes n}$.

Example 30.2. Trivial representations give $\bigoplus \mathrm{Sym}^n V$; sign representations $\bigoplus \Lambda^n V$; regular representations $T(V)$. So the free commutative, exterior and associative algebras are the Schur functors of the $\mathfrak{S}$-modules underlying Com, its signed version, and Ass.

Definition 30.3. The *composition product* of $\mathfrak{S}$-modules is

$$(E \circ F)(n) = \bigoplus_k E(k) \otimes_{\mathfrak{S}_k} \Big(\bigoplus_{n_1+\cdots+n_k=n} \mathrm{Ind}_{\mathfrak{S}_{n_1}\times\cdots\times\mathfrak{S}_{n_k}}^{\mathfrak{S}_n} F(n_1)\otimes\cdots\otimes F(n_k) \Big),$$

so that $\widetilde{E\circ F} = \widetilde{E}\circ\widetilde{F}$. An operad is a monoid $(\mathcal{P}, \gamma\colon \mathcal{P}\circ\mathcal{P}\to\mathcal{P}, \eta\colon I\to\mathcal{P})$ for $\circ$, I being the unit $\mathfrak{S}$-module (k in arity 1).

The symmetric group $\mathfrak{S}_k$ acts on the k blocks and the $\mathfrak{S}_{n_i}$ inside them: the relevant symmetry is a wreath product, not a product of symmetric groups — the classical *plethysm* of representations.

30.3 Quadratic operads and Koszul duality

Definition 30.4. The *free operad* $\mathcal{T}(E)$ on an $\mathfrak{S}$-module E is spanned by rooted trees with vertices labelled by elements of E; it is graded by *weight*, the number of vertices. A *quadratic operad* is $\mathcal{P} = \mathcal{T}(E)/(R)$ with E concentrated in arity 2 and $R \subseteq \mathcal{T}(E)^{(2)}(3)$, a space of relations among composites of two generators.

Example 30.5. Ass: $E = k[\mathfrak{S}_2]$ (spanned by μ and $\mu^{(12)}$), $R =$ associativity. Com: E trivial, $R =$ associativity. Lie: $E =$ sign representation (bracket), $R =$ Jacobi identity.

Definition 30.6. The *Koszul dual cooperad* $\mathcal{P}^{\text{¡}} = \mathcal{T}^c(sE, s^2R)$ is the sub-cooperad of the cofree cooperad on the suspension sE cogenerated by s^2R; its weight-w part sits in homological degree w (up to the suspension convention). The *Koszul dual operad* $\mathcal{P}^!$ is its arity-wise linear dual twisted by the sign representation and desuspended, presented by $E^\vee \otimes \mathrm{sgn}$ with relations $R^\perp$.

Theorem 30.7. $\mathrm{Ass}^! = \mathrm{Ass}$, $\mathrm{Com}^! = \mathrm{Lie}$, $\mathrm{Lie}^! = \mathrm{Com}$*; these operads are* Koszul*: the cobar construction* $\Omega\mathcal{P}^{\text{¡}}$ *is a quasi-free resolution* $\Omega\mathcal{P}^{\text{¡}} \xrightarrow{\sim} \mathcal{P}$.

See [42, Ch. 7] and the original [52]. The resulting algebras are $\mathrm{Ass}_\infty = A_\infty$, $\mathrm{Com}_\infty = C_\infty$ and $\mathrm{Lie}_\infty = L_\infty$.

Proposition 30.8 (Generating series test)**.** *For an operad with finite-dimensional components let* $f_{\mathcal{P}}(x) = \sum_{n\geq 1} \dim \mathcal{P}(n)\, x^n/n!$. *If* $\mathcal{P}$ *is Koszul then* $f_{\mathcal{P}}\big(-f_{\mathcal{P}^!}(-x)\big) = x$.

See [42, Ch. 7]. It is a necessary condition, a quick check on proposed Koszul dualities (Exercise 30.15).

30.4 A_∞-algebras

Definition 30.9. An *A_∞-algebra* is a graded vector space A with maps $m_k \colon A^{\otimes k} \to A$ of degree $k-2$ (homological grading), $k \geq 1$, such that for every $n \geq 1$

$$\sum_{r+s+t=n} (-1)^{r+st}\, m_{r+1+t}\big(1^{\otimes r} \otimes m_s \otimes 1^{\otimes t}\big) = 0.$$

Proposition 30.10. *An A_∞-structure on A is equivalently a coderivation D of degree -1 of the tensor coalgebra $T^c(sA) = \bigoplus_{k\geq 1} (sA)^{\otimes k}$ (deconcatenation coproduct) with $D^2 = 0$.*

Proof sketch. A coderivation of a cofree coalgebra is determined by its projection to cogenerators, $\bigoplus_k (sA)^{\otimes k} \to sA$, i.e. by a family of maps $b_k \colon (sA)^{\otimes k} \to sA$ of degree -1, which correspond to the m_k under the suspension isomorphism (the signs in Definition 30.9 are those produced by Lemma 28.5 when s is moved past elements). $D^2 = \frac{1}{2}[D, D]$ is again a coderivation, so $D^2 = 0$ iff its projection to sA vanishes, and that projection in arity n is the left side of the n-th relation. □

Exercise 30.11 (easy)**.** Write the A_∞ relations for $n = 1, 2, 3$ and interpret them.

Source: orig-book

Solution. $n=1$: $m_1 m_1 = 0$, a differential. $n=2$: $m_1 m_2 = m_2(m_1 \otimes 1 + 1 \otimes m_1)$: Leibniz. $n=3$: $m_2(m_2 \otimes 1 - 1 \otimes m_2) = m_1 m_3 + m_3(m_1 \otimes 1 \otimes 1 + 1 \otimes m_1 \otimes 1 + 1 \otimes 1 \otimes m_1)$ (signs as in Definition 30.9): the associator of m_2 is the boundary of m_3 in the complex of maps $A^{\otimes 3} \to A$; m_2 is associative up to the specified homotopy m_3, and on homology m_2 is associative.

30.5 Twisting morphisms and the Maurer–Cartan equation

Definition 30.12. For a dg cooperad $\mathcal{C}$ and dg operad $\mathcal{P}$, $\operatorname{Hom}_{\mathfrak{S}}(\mathcal{C},\mathcal{P}) = \prod_n \operatorname{Hom}_{\mathfrak{S}_n}(\mathcal{C}(n),\mathcal{P}(n))$ is a dg Lie algebra (indeed pre-Lie) under the *convolution product* $f\star g = \gamma_{(1)}\circ(f\otimes g)\circ\Delta_{(1)}$ (decompose infinitesimally, apply f,g, compose). A *twisting morphism* is α of degree -1 with $\partial\alpha+\alpha\star\alpha=0$ (and α vanishing on the counit).

Theorem 30.13. *For $\mathcal{C}=\mathcal{P}^{\text{¡}}$, $\mathcal{P}_\infty$-algebra structures on a chain complex (A,d) are in bijection with twisting morphisms $\mathcal{P}^{\text{¡}}\to\operatorname{End}_A$, i.e. with Maurer–Cartan elements in the convolution dg Lie algebra $\operatorname{Hom}_{\mathfrak{S}}(\mathcal{P}^{\text{¡}},\operatorname{End}_A)$.*

See [42, Ch. 6 and Ch. 10]. For $\mathcal{P}=\mathrm{Ass}$ this recovers Proposition 30.10.

Exercise 30.14 (medium)**.** Let $(\mathfrak{g},d,[\,,])$ be a dg Lie algebra (cohomological grading) and $x\in\mathfrak{g}^1$. Show that $d_x=d+[x,-]$ satisfies $d_x^2=\operatorname{ad}\big(dx+\tfrac12[x,x]\big)$; in particular the Maurer–Cartan equation $dx+\tfrac12[x,x]=0$ implies $d_x^2=0$, and the converse holds when $\operatorname{ad}\colon\mathfrak{g}^2\to\operatorname{End}(\mathfrak{g})$ is injective.

Source: orig-book

Solution. $d_x^2(y)=d[x,y]+[x,dy]+[x,[x,y]]$, and $d[x,y]=[dx,y]-[x,dy]$ ($|x|=1$), so $d_x^2(y)=[dx,y]+[x,[x,y]]$. Graded Jacobi for odd x gives $[x,[x,y]]=\tfrac12[[x,x],y]$. Hence $d_x^2=\operatorname{ad}\big(dx+\tfrac12[x,x]\big)$, as claimed; the two consequences are immediate. (Without injectivity of ad the converse fails: take $\mathfrak{g}$ spanned by x in degree 1 and z in degree 2, $d=0$, $[x,x]=z$, z central. Then $d_x^2=\tfrac12\operatorname{ad}(z)=0$, but $dx+\tfrac12[x,x]=\tfrac12 z\neq0$. This is why the Maurer–Cartan equation, not $d_x^2=0$, is taken as the definition.)

30.6 Exercises

Exercise 30.15 (easy)**.** Verify Proposition 30.8 for $(\mathrm{Com},\mathrm{Lie})$ and for $(\mathrm{Ass},\mathrm{Ass})$, using $\dim\mathrm{Lie}(n)=(n-1)!$.

Source: original

Solution. $f_{\mathrm{Com}}(x)=e^x-1$, $f_{\mathrm{Lie}}(x)=\sum(n-1)!x^n/n!=\sum x^n/n=-\log(1-x)$. Then $-f_{\mathrm{Lie}}(-x)=\log(1+x)$ and $f_{\mathrm{Com}}(\log(1+x))=x$. For Ass, $f(x)=\sum x^n=x/(1-x)$, $-f(-x)=x/(1+x)$, and $f(x/(1+x))=\frac{x/(1+x)}{1-x/(1+x)}=x$.

Exercise 30.16 (medium)**.** Let A be concentrated in a single degree with $m_1=0$. Show that every A_∞-structure on A is an associative algebra structure.

Source: original

Solution. $m_k\colon A^{\otimes k}\to A$ has degree $k-2$; if A is concentrated in degree d, $A^{\otimes k}$ is concentrated in degree kd and the target in degree d, so m_k can be nonzero only if $kd+k-2=d$, i.e. $(k-1)(d+1)=1$, which for $k\geq3$ has no integer solution ($k-1\geq2$). So $m_k=0$ for $k\geq3$ (and m_2 is possible only for $d=0$, since $(k-1)(d+1)=1$ with $k=2$ gives $d=0$). The relation for $n=3$ then says m_2 is associative.

30.7 Connections and further reading

Koszul duality turns "up to homotopy" into explicit algebra: $\mathcal{P}_\infty$-structures are Maurer–Cartan elements in a convolution dg Lie algebra. In Chapter 31 this produces the transferred A_∞-structure on cohomology, whose m_3 is the Massey product; in Part VI, graph complexes and the formality of the little disks operads are formulated in the same language. References: Loday–Vallette [42], Ginzburg–Kapranov [52], Fresse [43].

Chapter 31

Lie models, transferred structures, and Massey products

31.1 The problem

The cohomology ring of a space forgets part of the cochain algebra: two spaces can have isomorphic cohomology rings and different rational homotopy types (Example 24.24). What survives on cohomology is an A_∞-structure, whose first new operation is (essentially) the *triple Massey product*. Massey products are the classical tool for proving non-formality, they appear as differentials in spectral sequences, and they detect linking phenomena invisible to cup products (the Borromean rings). In configuration-space research they are the subject of Félix–Thomas and of the non-formality results for configuration spaces of closed manifolds (Part VI). The chapter also completes the rational picture by describing the dual, Lie-algebraic model and the rational Hurewicz map.

Prerequisites. Sullivan models (Chapter 24), the homotopy Lie algebra (Chapter 25), A_∞-algebras (Chapter 30).

31.2 Lie models and the rational Hurewicz map

Theorem 31.1 (Quillen)**.** *The rational homotopy category of simply connected spaces is equivalent to the homotopy category of connected dg Lie algebras over $\mathbb{Q}$ (reduced: $L = L_{\geq 1}$). The dg Lie algebra $\lambda(X)$ of X satisfies $H_*(\lambda(X)) \cong \pi_*(\Omega X) \otimes \mathbb{Q} = L_X$ as graded Lie algebras.*

See [53], [17, Part IV]. The two models are related by the Chevalley–Eilenberg functor: a dg Lie algebra L of finite type gives the cdga $C^*(L) = (\Lambda(sL)^\vee, d)$, where the quadratic part of d is dual to the bracket; for a minimal Sullivan model $(\Lambda V, d)$ the quadratic part d_1 is dual to the bracket of L_X (Theorem 24.12).

Exercise 31.2 (medium)**.** Let L be a graded Lie algebra. Show that $\Delta(x) = x\otimes 1 + 1\otimes x$ ($x \in L$) extends to an algebra map $U(L) \to U(L)\otimes U(L)$ making $U(L)$ a cocommutative Hopf algebra with L primitive.

SOURCE: orig-book

Solution. By the universal property of U it suffices that $x \mapsto x \otimes 1 + 1 \otimes x$ is a Lie map into $(U(L) \otimes U(L))_{\mathrm{Lie}}$. With $a = x \otimes 1$, $b = 1 \otimes y$: $ab = x \otimes y$ and $ba = (-1)^{|x||y|} x \otimes y$, so

$[a,b]=0$; hence $[\Delta x,\Delta y]=[x,y]\otimes 1+1\otimes[x,y]=\Delta[x,y]$. Coassociativity and counitality hold on generators, hence everywhere (both sides are algebra maps), and cocommutativity likewise.

Theorem 31.3 (Dual Hurewicz)**.** *Let $(\Lambda V,d)$ be the minimal model of a simply connected X of finite type. Every cocycle of positive degree has a linear part in V, and $\zeta\colon H^+(\Lambda V,d)\to V$, $[z]\mapsto$ linear part of z, is well defined (coboundaries are decomposable by minimality). Under $V^k\cong\operatorname{Hom}(\pi_k(X)\otimes\mathbb{Q},\mathbb{Q})$ the map ζ is the dual of the rational Hurewicz map $\pi_k(X)\otimes\mathbb{Q}\to H_k(X;\mathbb{Q})$.*

See [17, §13]. Consequently the image of the rational Hurewicz map is dual to the space of *closed generators*.

Example 31.4. For S^2, $(\Lambda(x,y),dy=x^2)$: H^2 is spanned by $[x]$ with $\zeta[x]=x$, an isomorphism; $H^3=0$, so $\zeta=0$ on $V^3=\mathbb{Q}y$. Dually, the rational Hurewicz map is an isomorphism on π_2 and zero on π_3: the Hopf map is invisible to homology because y is not closed.

31.3 Homotopy transfer

Theorem 31.5 (Homotopy transfer; Kadeishvili)**.** *Let (A,d) be a dg algebra and $(H,0)$ a complex with maps $i\colon H\to A$, $p\colon A\to H$ and a homotopy h on A such that $pi=\mathrm{id}_H$, $ip-\mathrm{id}=dh+hd$ (a* deformation retract*). Then H carries an A_∞-structure (m_k) with $m_1=0$, $m_2=p\mu(i\otimes i)$, and, up to a sign convention,*

$$m_3=p\,\mu\big(h\mu(i\otimes i)\otimes i-i\otimes h\mu(i\otimes i)\big),$$

and in general m_k is a sum over planar binary trees with k leaves, with μ at the vertices, h on the internal edges, i at the leaves and p at the root. The map i extends to an A_∞-quasi-isomorphism. Taking $H=H(A)$ (possible over a field) equips the cohomology of every dg algebra with an A_∞-structure, unique up to A_∞-isomorphism.

See [54], [42, Ch. 9–10]. For a commutative A the transferred structure is C_∞, and A is formal iff the transferred structure on $H(A)$ is A_∞-isomorphic to one with $m_k=0$ for $k\ge 3$.

31.4 Massey products

Definition 31.6 (Triple Massey product)**.** Let A be a dga and $a,b,c\in H(A)$ with $ab=0$ and $bc=0$. Choose cocycle representatives (denoted by the same letters) and cochains u,v with $du=ab$, $dv=bc$. Then

$$z=uc-(-1)^{|a|}av$$

is a cocycle of degree $|a|+|b|+|c|-1$. The *Massey product* $\langle a,b,c\rangle\subseteq H^{|a|+|b|+|c|-1}(A)$ is the set of the classes $[z]$ for all choices.

Verification. $d(uc)=(du)c+(-1)^{|u|}u\,dc=abc$, and $d(av)=(-1)^{|a|}a\,dv=(-1)^{|a|}abc$; so $dz=abc-(-1)^{|a|}(-1)^{|a|}abc=0$. □

Exercise 31.7 (medium)**.** Show that $\langle a,b,c\rangle$ is a coset of the subgroup $a\,H^{|b|+|c|-1}(A)+H^{|a|+|b|-1}(A)\,c$, the *indeterminacy*, and that it does not depend on the choice of representatives of a,b,c.

SOURCE: orig-book (signs made explicit)

Solution. Two choices of u differ by a cocycle $u'-u$ of degree $|a|+|b|-1$, changing $[z]$ by $[u'-u]c \in H\,c$; two choices of v change it by $-(-1)^{|a|}a[v'-v] \in aH$. Conversely every element of $H\,c + aH$ arises this way. Independence of representatives: replace a by $a' = a + de$. Since b is closed, $a'b = d(u+eb)$, so take $u' = u+eb$ and keep v. Then $z'-z = ebc-(-1)^{|a|}(de)v$. Since $d(ev) = (de)v + (-1)^{|a|-1}e\,bc$, we get $(de)v = d(ev) + (-1)^{|a|}ebc$, hence $z'-z = -(-1)^{|a|}d(ev)$, a coboundary. The same computation applies to b and c, so the set $\langle a,b,c\rangle$ depends only on the classes.

Proposition 31.8. *A quasi-isomorphism of dgas $f\colon A \to B$ maps $\langle a,b,c\rangle$ bijectively onto $\langle fa, fb, fc\rangle$. Consequently Massey products are invariants of the quasi-isomorphism type, and for a space X they are invariants of the rational homotopy type when computed in $A_{\mathrm{PL}}(X)$ (or of the homotopy type when computed in singular cochains with any coefficients).*

Proof. Applying f to a defining system (u,v) for $\langle a,b,c\rangle$ gives one for $\langle fa, fb, fc\rangle$, so $f_*\langle a,b,c\rangle \subseteq \langle fa, fb, fc\rangle$. Both sets are cosets of their indeterminacy subgroups, and f_*, an isomorphism of algebras $H(A) \cong H(B)$, maps $aH(A) + H(A)c$ onto $(fa)H(B) + H(B)(fc)$. A coset of a subgroup K mapped by an isomorphism into a coset of $f_*(K)$ is mapped onto it. For a zig-zag of quasi-isomorphisms apply this to each map. □

Proposition 31.9 (Formality kills Massey products)**.** *If A is formal, every triple Massey product $\langle a,b,c\rangle$ of A contains* 0.

Proof. By Proposition 31.8 we may compute in $(H(A),0)$. There $ab = 0$ and $bc = 0$ on the nose, so $u = v = 0$ is a defining system, giving $z = 0$. □

Example 31.10 (A non-formal simply connected space)**.** In $(\Lambda(a,b,c), dc = ab)$ with $|a| = |b| = 3$, $|c| = 5$ (Example 24.24): $a^2 = 0$ on the nose (a is odd), so take $u = 0$; $ab = dc$, so take $v = c$. Then $z = 0\cdot b - (-1)^3 ac = ac$. The indeterminacy is $aH^5 + H^5 b = 0$ since $H^5 = 0$ (degree 5 is spanned by c, which is not closed). And $[ac] \neq 0$: ac is a cocycle, and there are no elements of degree 7. So $\langle a,a,b\rangle = \{[ac]\} \neq \{0\}$ and the space is not formal, although its cohomology ring through degree 8 looks unremarkable.

Exercise 31.11 (medium)**.** Let N be the Heisenberg nilmanifold (the quotient of the group of upper unitriangular real 3×3 matrices by its integer points), with Sullivan model $(\Lambda(x,y,z),\ dx = dy = 0,\ dz = xy)$, $|x| = |y| = |z| = 1$. Compute $H^*(N;\mathbb{Q})$ from the model and show that $\langle x,x,y\rangle$ is a single nonzero class. Deduce that N is not formal (Sullivan models of nilpotent spaces such as N obey the same formalism).

SOURCE: original computation of a standard example (see [36, §3])

Solution. Degree 0: $\mathbb{Q}$. Degree 1: x,y,z with $dx = dy = 0$, $dz = xy$: $H^1 = \langle x,y\rangle$. Degree 2: xy, xz, yz; $d(xz) = -x\,xy = 0$, $d(yz) = -y\,xy = 0$, $d(xy) = 0$; coboundaries: $\langle xy\rangle$. So $H^2 = \langle [xz],[yz]\rangle$. Degree 3: xyz, closed, not a boundary (d of degree-2 elements is zero): $H^3 = \mathbb{Q}$. So the Betti numbers are $1,2,2,1$ (Poincaré duality, as for a closed 3-manifold).

Massey product: $x\cdot x = 0$, take $u = 0$; $x\cdot y = dz$, take $v = z$; then $z_{\mathrm{M}} = 0\cdot y - (-1)^1 xz = xz$. The indeterminacy is $x\,H^1 + H^1\,y = \langle x^2, xy, xy, y^2\rangle = 0$ in H^2 (as $[xy] = 0$). So $\langle x,x,y\rangle = \{[xz]\} \neq 0$, and by Proposition 31.9 N is not formal. (In particular N is not a Kähler manifold, by [33].)

Example 31.12 (The Borromean rings)**.** For the complement X of the Borromean rings in S^3, $H^1(X;\mathbb{Z}) = \mathbb{Z}^3$ with generators u_1, u_2, u_3 dual to meridians, and all products u_iu_j vanish because the pairwise linking numbers are zero. Massey showed that $\langle u_1, u_2, u_3\rangle$ is nonzero modulo its indeterminacy: it detects the triple linking that pairwise products cannot [55]. Cup products see linking numbers; Massey products see Milnor's higher linking invariants.

Remark 31.13 (Higher Massey products and A_∞). An n-fold Massey product $\langle a_1, \dots, a_n\rangle$ is defined using a *defining system*: cochains a_{ij} $(1 \le i \le j \le n$, $(i,j) \neq (1,n))$ with a_{ii} representing a_i and $da_{ij} = \sum_{i\le k<j} \bar{a}_{ik}a_{k+1,j}$ $(\bar{x} = (-1)^{1+|x|}x)$, and the product is the set of classes of $\sum_k \bar{a}_{1k}a_{k+1,n}$ over all defining systems [56], [57]. Three facts link them to Theorem 31.5: (i) if $\langle a,b,c\rangle$ is defined, then up to sign and indeterminacy it contains $m_3(a,b,c)$ for the transferred structure; (ii) more generally, if all lower products vanish appropriately, $m_n(a_1,\dots,a_n) \in \pm\langle a_1,\dots,a_n\rangle$ [58]; (iii) formality forces all Massey products to contain 0, but the converse fails: there are non-formal spaces all of whose Massey products vanish, because the transferred A_∞-structure can be nontrivial in ways not captured by Massey products of single classes.

Remark 31.14 (Massey products and spectral sequences). In the Eilenberg–Moore and Adams spectral sequences, differentials are often computed by Massey products ("d_2 is a Massey product", May's convergence theorem); in the Serre spectral sequence of a fibration, a nontrivial Massey product in the total space can arise from a differential that is invisible to the cup product of the base. This is one of the mechanisms by which configuration spaces of closed manifolds fail to be formal (Part VI).

31.5 Connections and further reading

Rational homotopy has two dual models, commutative (Sullivan) and Lie (Quillen); the Hurewicz map is the projection onto closed generators; the cohomology of a cochain algebra carries a transferred A_∞-structure; and Massey products, its first visible shadow, are the basic obstruction to formality. References: Félix–Halperin–Thomas [17], Quillen [53], Loday–Vallette [42], Kraines [56], May [57], Félix–Oprea–Tanré [36].

Part VI

Configuration spaces

Chapter 32

Group homology, finite covers and the Cartan–Leray spectral sequence

32.1 Why this chapter comes first

The unordered configuration space $B_k(M) = F_k(M)/\mathfrak{S}_k$ is the quotient of a space by a free action of a finite group. Almost every phenomenon of Part VI — rational invariants, torsion, the failure of integral stability — comes from this one fact. For surfaces, moreover, configuration spaces are classifying spaces of groups (braid groups), so their homology *is* group homology. This chapter collects the three tools needed: group (co)homology, the transfer of a finite cover, and the Cartan–Leray spectral sequence of a free action. Everything is a consequence of material already proved: the Serre spectral sequence with local coefficients (Theorem 15.11), the Borel construction (Proposition 21.15) and the bar resolution (Example 29.10).

32.2 Group homology and cohomology

Let G be a group and $\mathbb{Z}G$ its integral group ring. A *G-module* is a left $\mathbb{Z}G$-module. The trivial module $\mathbb{Z}$ has $g \cdot n = n$.

Definition 32.1. For a G-module M,

$$H_n(G;M) = \operatorname{Tor}_n^{\mathbb{Z}G}(\mathbb{Z}, M), \qquad H^n(G;M) = \operatorname{Ext}^n_{\mathbb{Z}G}(\mathbb{Z}, M),$$

computed from any projective resolution $P_\bullet \to \mathbb{Z}$ of the trivial module: $H_n(G;M) = H_n(P_\bullet \otimes_G M)$ and $H^n(G;M) = H^n(\operatorname{Hom}_G(P_\bullet, M))$, where $P \otimes_G M = P \otimes_{\mathbb{Z}G} M$ with P made a right module by $p \cdot g = g^{-1}p$.

Independence of the resolution is the comparison theorem of homological algebra (Chapter 8). In degree 0: $H_0(G;M) = M_G = M/\langle gm - m\rangle$ (the *coinvariants*) and $H^0(G;M) = M^G$ (the *invariants*).

Theorem 32.2 (Topological interpretation)**.** *Let X be a connected CW complex with $\pi_1(X) = G$ and contractible universal cover $\widetilde{X}$ (that is, X is a $K(G,1)$). Then the cellular chain complex $C_\bullet(\widetilde{X})$ is a free resolution of $\mathbb{Z}$ over $\mathbb{Z}G$, and consequently*

$$H_n(G;M) \cong H_n(X;\mathcal{M}), \qquad H^n(G;M) \cong H^n(X;\mathcal{M}),$$

the homology and cohomology of X with local coefficients in the local system $\mathcal{M}$ determined by M (Definition 15.10). In particular $H_n(G;\mathbb{Z}) = H_n(BG;\mathbb{Z})$.

Proof. G acts freely on $\widetilde{X}$ by deck transformations, permuting the cells; choosing one lift of each cell of X gives a $\mathbb{Z}G$-basis of $C_n(\widetilde{X})$, so each $C_n(\widetilde{X})$ is free. Since $\widetilde{X}$ is contractible, $H_n(C_\bullet(\widetilde{X})) = 0$ for $n > 0$ and the augmentation $C_0(\widetilde{X}) \to \mathbb{Z}$ induces $H_0 \cong \mathbb{Z}$; thus $C_\bullet(\widetilde{X}) \to \mathbb{Z}$ is a free resolution. Finally $C_\bullet(\widetilde{X}) \otimes_G M$ is, by definition, the cellular chain complex of X with coefficients in $\mathcal{M}$, and dually for cochains. □

The bar construction of Example 29.10 realises this for every group: $|E_\bullet G|$ is contractible with a free G-action, so $BG = |E_\bullet G|/G$ is a $K(G,1)$.

Example 32.3 (Cyclic groups)**.** Let $G = \mathbb{Z}/m = \langle t \rangle$, $N = 1 + t + \cdots + t^{m-1}$. The sequence

$$\cdots \xrightarrow{N} \mathbb{Z}G \xrightarrow{t-1} \mathbb{Z}G \xrightarrow{N} \mathbb{Z}G \xrightarrow{t-1} \mathbb{Z}G \to \mathbb{Z} \to 0$$

is exact (the kernel of $t-1$ consists of the multiples of N, the kernel of N is spanned by the $t^i - t^{i+1}$), so it is a free resolution; topologically it is the cellular chain complex of S^∞ with the free $\mathbb{Z}/m$-action whose quotient is the infinite lens space (Example 4.10). Tensoring with $\mathbb{Z}$ (trivial action) replaces t by 1 and N by m:

$$H_n(\mathbb{Z}/m;\mathbb{Z}) = \begin{cases} \mathbb{Z} & n = 0, \\ \mathbb{Z}/m & n \text{ odd}, \\ 0 & n > 0 \text{ even}, \end{cases} \qquad H^n(\mathbb{Z}/m;\mathbb{Z}) = \begin{cases} \mathbb{Z} & n = 0, \\ 0 & n \text{ odd}, \\ \mathbb{Z}/m & n > 0 \text{ even}. \end{cases}$$

For the *sign module* $\mathbb{Z}^-$ of $\mathbb{Z}/2$ (t acts by -1), $t-1$ becomes -2 and N becomes 0: $H_n(\mathbb{Z}/2;\mathbb{Z}^-) = \mathbb{Z}/2$ for n even ≥ 0 and 0 for n odd.

Proposition 32.4. $H_1(G;\mathbb{Z}) \cong G^{\mathrm{ab}} = G/[G,G]$.

Proof. For a $K(G,1)$ X, $H_1(X;\mathbb{Z}) \cong \pi_1(X)^{\mathrm{ab}}$ by the Hurewicz theorem (Theorem 13.5). □

Proposition 32.5 (Torsion-freeness)**.** *If a group G has a finite-dimensional $K(G,1)$, then G has no nontrivial elements of finite order.*

Proof. Suppose $g \in G$ has prime order p and let X be a $K(G,1)$ of dimension d. The covering space $\widetilde{X}/\langle g \rangle$ is a $K(\mathbb{Z}/p,1)$ of dimension d, so $H_n(\mathbb{Z}/p;\mathbb{Z}) = 0$ for $n > d$. This contradicts Example 32.3, by which $H_n(\mathbb{Z}/p;\mathbb{Z}) = \mathbb{Z}/p$ for every odd n. An element of finite order > 1 has a power of prime order, so there are none. □

32.3 Finite covers and the transfer

Construction 32.6 (Transfer)**.** Let $p\colon \widetilde{X} \to X$ be a d-sheeted covering of CW complexes (cells of $\widetilde{X}$ are the lifts of cells of X). Each cell e of X has exactly d lifts $\tilde{e}_1, \dots, \tilde{e}_d$. Define $\tau\colon C_n(X) \to C_n(\widetilde{X})$, $\tau(e) = \sum_i \tilde{e}_i$ (orientations lifted). Then τ is a chain map, and $p_\# \tau = d \cdot \mathrm{id}$. If the covering is regular with deck group G, then $\tau p_\# = \sum_{g \in G} g_\#$. The induced map $\mathrm{tr} = \tau_*\colon H_*(X;A) \to H_*(\widetilde{X};A)$ is the *transfer*; similarly $\mathrm{tr}\colon H^*(\widetilde{X};A) \to H^*(X;A)$.

Verification. The boundary of $\tilde{e}_i$ is the lift of ∂e through the lifted characteristic map; summing over all lifts gives the sum over all lifts of the cells in ∂e, so $\partial\tau = \tau\partial$. The identities $p_\#\tau = d$ and $\tau p_\#(\tilde{e}_1) = \tau(e) = \sum_i \tilde{e}_i = \sum_g g\tilde{e}_1$ are immediate (for a regular cover G permutes the lifts simply transitively). □

Theorem 32.7. *Let $p\colon \widetilde{X} \to X$ be a finite regular covering with deck group G of order d.*

(a) If A is a ring in which d is invertible (e.g. $\mathbb{Q}$, or $\mathbb{F}_\ell$ with $\ell \nmid d$), then $p^*\colon H^*(X;A) \to H^*(\widetilde{X};A)^G$ and $p_*\colon H_*(\widetilde{X};A)_G \to H_*(X;A)$ are isomorphisms.

(b) Every element of $\ker\big(\operatorname{tr}\colon H_n(X;\mathbb{Z}) \to H_n(\widetilde{X};\mathbb{Z})\big)$ is annihilated by d. In particular, if $H_n(\widetilde{X};\mathbb{Z})$ is torsion-free, every torsion element of $H_n(X;\mathbb{Z})$ has order dividing d; if $H_n(\widetilde{X};\mathbb{Z}) = 0$, then $d\cdot H_n(X;\mathbb{Z}) = 0$.

Proof. (a) On cohomology $p^*\operatorname{tr} = \sum_g g^*$ and $\operatorname{tr} p^* = d$. The second identity shows p^* injective (after inverting d). Its image lies in the invariants since $pg = p$. If x is invariant, $p^*\operatorname{tr}(x) = \sum_g g^*x = dx$, so $x = p^*(d^{-1}\operatorname{tr} x)$ is in the image. The homology statement is dual.

(b) $d\cdot x = p_*\operatorname{tr}(x)$. If $\operatorname{tr}(x) = 0$ then $dx = 0$. If x is torsion then $\operatorname{tr}(x)$ is torsion, hence zero when $H_n(\widetilde{X};\mathbb{Z})$ is torsion-free. □

Corollary 32.8. *For a finite group G of order d, $d\cdot H_n(G;\mathbb{Z}) = 0$ and $d\cdot H^n(G;\mathbb{Z}) = 0$ for $n > 0$. In particular $H^n(G;\mathbb{Q}) = 0$ for $n > 0$.*

Proof. Apply Theorem 32.7(b) to $EG \to BG$: $H_n(EG) = 0$ for $n > 0$. The cohomological statement follows from the universal coefficient theorem, since the $H_n(G;\mathbb{Z})$ are finitely generated when G is finite (the bar resolution is finitely generated in each degree). □

Caution 32.9. The transfer gives no information on p-torsion when p divides d: for $S^n \to \mathbb{RP}^n$ ($d = 2$), the 2-torsion of $H_*(\mathbb{RP}^n;\mathbb{Z})$ (Example 4.9) is invisible upstairs. This is exactly the situation of $F_k(M) \to B_k(M)$, where $d = k!$ is divisible by every prime $\le k$: all the torsion phenomena of Chapter 36 live at primes $p \le k$.

32.4 The Cartan–Leray spectral sequence

Theorem 32.10 (Cartan–Leray)**.** *Let a discrete group G act freely and properly discontinuously on a path-connected space X (a CW complex with G permuting cells, say). Then there are spectral sequences*

$$E^2_{p,q} = H_p\big(G; H_q(X;\mathbb{Z})\big) \Longrightarrow H_{p+q}(X/G;\mathbb{Z}),$$
$$E_2^{p,q} = H^p\big(G; H^q(X;\mathbb{Z})\big) \Longrightarrow H^{p+q}(X/G;\mathbb{Z}),$$

where G acts on $H_q(X)$ through its action on X. The second is multiplicative. The edge map $H_n(X)_G = E^2_{0,n} \to H_n(X/G)$ is induced by the projection, and the edge map $H_n(X/G) \to E^2_{n,0} = H_n(G)$ by the classifying map $X/G \to BG$ of the covering.

Proof. By Proposition 21.15, $X/G \simeq X_{hG} = EG \times_G X$, and $X \to X_{hG} \to BG$ is a fibration. The fundamental group of BG is G and its action on the homology of the fibre X is induced by the action of G on X (a loop in BG lifts to a path in EG from

e to ge, and transport along it is $x \mapsto gx$). The Serre spectral sequence with local coefficients (Theorem 15.11) has $E^2_{p,q} = H_p(BG; \mathcal{H}_q(X))$, which is $H_p(G; H_q(X))$ by Theorem 32.2. □

Corollary 32.11 (Five-term exact sequence)**.** *In the situation of Theorem 32.10 there is an exact sequence*

$$H_2(X/G) \to H_2(G;\mathbb{Z}) \xrightarrow{d^2} H_1(X)_G \to H_1(X/G) \to H_1(G;\mathbb{Z}) \to 0.$$

Proof. In total degree 1 only $E^2_{1,0}$ and $E^2_{0,1}$ contribute ($E^2_{p,q}$ with $q = 0$ or $p = 0$; $H_0(X) = \mathbb{Z}$). $E^\infty_{1,0} = E^2_{1,0}$ (no differential can hit or leave it: $d^2 \colon E^2_{1,0} \to E^2_{-1,1} = 0$) and $E^\infty_{0,1} = \operatorname{coker}(d^2 \colon E^2_{2,0} \to E^2_{0,1})$. The filtration gives $0 \to E^\infty_{0,1} \to H_1(X/G) \to E^\infty_{1,0} \to 0$. Finally $E^\infty_{2,0} = \ker d^2$ is the image of $H_2(X/G) \to E^2_{2,0}$ (the edge map). Splicing gives the sequence. □

Example 32.12 (Two points in $\mathbb{R}^n$)**.** $F_2(\mathbb{R}^n) \to S^{n-1}$, $(x,y) \mapsto (y-x)/|y-x|$, is a homotopy equivalence (the fibre over v is $\{(x, x+tv) : t > 0\} \cong \mathbb{R}^n \times (0,\infty)$, and the map has a section) and it is $\mathfrak{S}_2$-equivariant for the swap upstairs and the antipodal map downstairs. Hence $B_2(\mathbb{R}^n) \simeq S^{n-1}/\pm = \mathbb{RP}^{n-1}$. Since the antipodal map of S^{n-1} has degree $(-1)^n$, Theorem 32.7(a) gives

$$H^*(B_2(\mathbb{R}^n);\mathbb{Q}) = H^*(S^{n-1};\mathbb{Q})^{\mathfrak{S}_2} = \begin{cases} \mathbb{Q} \oplus \mathbb{Q}[-(n-1)] & n \text{ even}, \\ \mathbb{Q} & n \text{ odd}, \end{cases}$$

whereas integrally $H_*(B_2(\mathbb{R}^n);\mathbb{Z}) = H_*(\mathbb{RP}^{n-1};\mathbb{Z})$ has a $\mathbb{Z}/2$ in every odd degree $< n-1$. In the Cartan–Leray spectral sequence $E^2_{p,q} = H_p(\mathbb{Z}/2; H_q(S^{n-1}))$, the rows $q = 0$ and $q = n-1$ are $H_p(\mathbb{Z}/2;\mathbb{Z})$ and $H_p(\mathbb{Z}/2;\mathbb{Z}^{(-1)^n})$; all classes with $p+q \geq n$ must be killed by the differential $d^n \colon E^n_{p,0} \to E^n_{p-n,n-1}$, which is therefore an isomorphism for $p > n$.

Example 32.13 (The Klein bottle as a quotient of the torus)**.** Let $\mathbb{Z}/2 = \langle t \rangle$ act on $T^2 = \mathbb{R}^2/\mathbb{Z}^2$ by $t(x,y) = (x + \frac{1}{2}, -y)$. The action is free and the quotient is the Klein bottle K. On $H_1(T^2) = \mathbb{Z}a \oplus \mathbb{Z}b$ (a the x-circle, b the y-circle), $t_*a = a$ and $t_*b = -b$. Coinvariants: $H_1(T^2)_G = \mathbb{Z}a \oplus \mathbb{Z}/2\,b$. $H_2(\mathbb{Z}/2;\mathbb{Z}) = 0$, so Corollary 32.11 gives

$$0 \to \mathbb{Z}a \oplus \mathbb{Z}/2\,b \to H_1(K) \to \mathbb{Z}/2 \to 0.$$

This extension is *not* split: the loop s in K given by the image of the path $x \mapsto (x/2, 0)$, $x \in [0,1]$, maps to the generator of $\mathbb{Z}/2$ and $2s = a$ in $H_1(K)$. Hence $H_1(K) \cong \mathbb{Z}s \oplus \mathbb{Z}/2\,b$, in agreement with Example 15.12. The extension problem is solved by geometry, not by the spectral sequence — exactly as in the fibration computation of Example 15.12.

32.5 The Lyndon–Hochschild–Serre spectral sequence

Theorem 32.14 (LHS spectral sequence)**.** *For an extension $1 \to N \to G \to Q \to 1$ and a G-module M there is a spectral sequence $E^2_{p,q} = H_p(Q; H_q(N; M)) \Rightarrow H_{p+q}(G; M)$, and dually in cohomology.*

Proof sketch. G acts freely on EG, so N does, and EG/N is a $K(N,1)$ on which $Q = G/N$ acts freely with quotient BG. Apply Theorem 32.10 (with local coefficients M). For details, see [69, Ch. VII] and [8, Ch. 8bis]. □

Example 32.15. The braid group $\mathrm{Br}_k = \pi_1 B_k(\mathbb{R}^2)$ and the pure braid group $P_k = \pi_1 F_k(\mathbb{R}^2)$ fit in $1 \to P_k \to \mathrm{Br}_k \to \mathfrak{S}_k \to 1$ (the covering $F_k \to B_k$). Since $F_k(\mathbb{R}^2)$ is a $K(P_k, 1)$ (Theorem 33.13), the LHS spectral sequence of this extension is the Cartan–Leray spectral sequence of $F_k(\mathbb{R}^2) \to B_k(\mathbb{R}^2)$, with $E^2_{p,q} = H_p(\mathfrak{S}_k; H_q(P_k))$.

32.6 Exercises

Exercise 32.16 (easy)**.** Show that a free group F_r of rank r has $H_1(F_r; \mathbb{Z}) = \mathbb{Z}^r$ and $H_n(F_r; M) = 0$ for $n \geq 2$ and every M.

Source: original

Solution. A wedge of r circles is a $K(F_r, 1)$ (its universal cover is a tree, which is contractible). It is 1-dimensional, so its (local-coefficient) homology vanishes in degrees ≥ 2; $H_1 = \mathbb{Z}^r$ by cellular homology, or by Proposition 32.4.

Exercise 32.17 (easy)**.** Compute $H_n(\mathbb{Z}^2; \mathbb{Z})$ and $H^n(\mathbb{Z}^2; \mathbb{Z})$.

Source: original

Solution. T^2 is a $K(\mathbb{Z}^2, 1)$ (universal cover $\mathbb{R}^2$). So $H_n = \mathbb{Z}, \mathbb{Z}^2, \mathbb{Z}$ for $n = 0, 1, 2$ and 0 for $n \geq 3$; the same for cohomology (no torsion, so the universal coefficient theorem introduces no Ext terms).

Exercise 32.18 (medium)**.** Verify directly from Definition 32.1 that $H_0(G; M) = M_G$ and $H^0(G; M) = M^G$, using the start $\mathbb{Z}G \xrightarrow{\varepsilon} \mathbb{Z} \to 0$ of any free resolution, whose kernel is the augmentation ideal I generated by the $g - 1$.

Source: original

Solution. $H_0(G; M) = \mathbb{Z} \otimes_{\mathbb{Z}G} M$ by right exactness of $\otimes$. Now $\mathbb{Z} = \mathbb{Z}G/I$, so $\mathbb{Z} \otimes_G M = M/IM = M/\langle gm - m \rangle = M_G$. Dually $H^0(G; M) = \mathrm{Hom}_G(\mathbb{Z}, M)$ by left exactness of Hom, and a G-map $\mathbb{Z} \to M$ is determined by the image m of 1, which must satisfy $gm = m$: this is M^G.

Exercise 32.19 (medium)**.** Use Theorem 32.7(b) to show without computation that $2 \cdot \widetilde{H}_n(\mathbb{RP}^m; \mathbb{Z}) = 0$ for $0 < n < m$, and decide when $H_m(\mathbb{RP}^m; \mathbb{Z})$ is torsion.

Source: original

Solution. $S^m \to \mathbb{RP}^m$ is a 2-sheeted cover and $H_n(S^m) = 0$ for $0 < n < m$, so $2H_n(\mathbb{RP}^m) = 0$. In degree m, $H_m(S^m) = \mathbb{Z}$ is torsion-free, so every torsion element of $H_m(\mathbb{RP}^m)$ is killed by 2; but $H_m(\mathbb{RP}^m)$ is a closed m-manifold's top homology, which is $\mathbb{Z}$ (orientable, m odd) or 0 (m even): it is never torsion.

Exercise 32.20 (medium)**.** In Example 32.13, compute $H^*(T^2; \mathbb{Q})^{\mathbb{Z}/2}$ and check that it equals $H^*(K; \mathbb{Q})$. Then show that the five-term sequence alone does not determine $H_1(K; \mathbb{Z})$: exhibit two non-isomorphic abelian groups fitting in an extension of $\mathbb{Z}/2$ by $\mathbb{Z} \oplus \mathbb{Z}/2$.

Source: original

Solution. t^* fixes a^*, negates b^* and negates the orientation class a^*b^*; the invariants are $\mathbb{Q}$ in degrees $0, 1$ and 0 in degree 2, which is $H^*(K; \mathbb{Q})$ (Example 15.12). Both $\mathbb{Z} \oplus \mathbb{Z}/2 \oplus \mathbb{Z}/2$ (split) and $\mathbb{Z} \oplus \mathbb{Z}/2$ (with $\mathbb{Z} \xrightarrow{2} \mathbb{Z}$) are extensions of $\mathbb{Z}/2$ by $\mathbb{Z} \oplus \mathbb{Z}/2$; so is $\mathbb{Z} \oplus \mathbb{Z}/4$ (with $\mathbb{Z}/2 \hookrightarrow \mathbb{Z}/4$). Only geometry (the relation $2s = a$) decides.

Exercise 32.21 (medium)**.** Show that if a finite group G acts freely on S^{2m} then $|G| \le 2$. (Use the Lefschetz number or the Euler characteristic.)

Source: original; classical

Solution. $S^{2m} \to S^{2m}/G$ is a $|G|$-sheeted covering of finite CW complexes (after a G-CW structure, or using that χ is multiplicative for finite covers of compact manifolds), so $2 = \chi(S^{2m}) = |G|\chi(S^{2m}/G)$. Hence $|G| \in \{1, 2\}$.

Exercise 32.22 (hard)**.** Let $G = \mathbb{Z} \rtimes \mathbb{Z}$ be the Klein bottle group $\langle a, t \mid tat^{-1} = a^{-1}\rangle$. Use the LHS spectral sequence of $1 \to \langle a\rangle \to G \to \langle t\rangle \to 1$ to compute $H_*(G;\mathbb{Z})$.

Source: original

Solution. $Q = \mathbb{Z}$ has a 1-dimensional $K(Q,1)$, so $E^2_{p,q} = 0$ for $p \ge 2$ and the spectral sequence collapses. $H_q(N)$ is $\mathbb{Z}$ (trivial action) for $q = 0$ and $\mathbb{Z}^-$ (t acts by -1) for $q = 1$. For $Q = \mathbb{Z}$: $H_0(\mathbb{Z}; M) = M_t$, $H_1(\mathbb{Z}; M) = M^t$. So $E^2_{0,0} = \mathbb{Z}$, $E^2_{1,0} = \mathbb{Z}$, $E^2_{0,1} = \mathbb{Z}/(2) = \mathbb{Z}/2$, $E^2_{1,1} = (\mathbb{Z}^-)^t = 0$. Hence $H_0 = \mathbb{Z}$, $H_2 = 0$, and $0 \to \mathbb{Z}/2 \to H_1 \to \mathbb{Z} \to 0$, which splits since $\mathbb{Z}$ is free: $H_1(G) = \mathbb{Z} \oplus \mathbb{Z}/2$. This is $H_*(K)$, as it must be since K is a $K(G,1)$.

32.7 Connections and further reading

Brown [69] is the standard reference for group cohomology (resolutions, transfer, the LHS spectral sequence); McCleary [8, Ch. 8bis] treats the Cartan–Leray and LHS spectral sequences, and Hatcher [1, §3.G] the transfer. In Chapter 36 these tools explain where the torsion in $H_*(B_k(M);\mathbb{Z})$ can and cannot occur.

Chapter 33

Configuration spaces and braid groups

33.1 Definitions

Definition 33.1. For a space X and $k \geq 0$, the *ordered configuration space* is

$$F_k(X) = \{(x_1, \dots, x_k) \in X^k : x_i \neq x_j \text{ for } i \neq j\} = X^k \setminus \bigcup_{i<j} \Delta_{ij},$$

where $\Delta_{ij} = \{x_i = x_j\}$. The symmetric group $\mathfrak{S}_k$ acts by permuting coordinates, and the *unordered configuration space* is $B_k(X) = F_k(X)/\mathfrak{S}_k$.

Notation 33.2. The literature uses $F(X,k)$, $\mathrm{Conf}_k(X)$, $\mathrm{PConf}_k(X)$ for $F_k(X)$ and $C_k(X)$, $\mathrm{UConf}_k(X)$, $B(X,k)$ for $B_k(X)$. We use F_k and B_k throughout (as did Example 21.16).

Proposition 33.3. *Let M be a manifold of dimension d (without boundary).*

(a) $F_k(M)$ is an open subset of M^k, hence a kd-manifold; it is orientable if M is.

(b) $\mathfrak{S}_k$ acts freely and properly discontinuously, so $F_k(M) \to B_k(M)$ is a regular $k!$-sheeted covering and $B_k(M)$ is a kd-manifold.

(c) An injective continuous map (e.g. an embedding) $f\colon M \to N$ induces $F_k(f)\colon F_k(M) \to F_k(N)$, $\mathfrak{S}_k$-equivariantly, hence $B_k(f)$. Configuration spaces are functorial for embeddings, not *for arbitrary maps.*

Proof. (a) The diagonals are closed. (b) A point x of $F_k(M)$ has a neighbourhood $U_1 \times \cdots \times U_k$ with the U_i pairwise disjoint; its translates by distinct permutations are disjoint, which gives free and properly discontinuous action and the covering property. (c) Clear. □

Caution 33.4. Because of (c), there is no a priori reason for $F_k(M)$ to depend only on the homotopy type of M: $\mathbb{R}$ and a point are homotopy equivalent, but $F_2(\mathbb{R})$ has two contractible components while $F_2(\mathrm{pt}) = \varnothing$. For closed manifolds the question is subtle and is the subject of Chapter 38.

Example 33.5 (Low-dimensional cases)**.** (a) $F_k(\mathbb{R})$ has $k!$ components, one for each ordering, each a convex open subset of $\mathbb{R}^k$; $B_k(\mathbb{R})$ is contractible.

(b) $F_2(\mathbb{R}^n) \simeq S^{n-1}$ and $B_2(\mathbb{R}^n) \simeq \mathbb{RP}^{n-1}$ (Example 32.12).

(c) $F_k(S^1) \cong S^1 \times F_{k-1}(\mathbb{R})$ via $(z_1, \dots, z_k) \mapsto (z_1, (z_2/z_1, \dots, z_k/z_1))$, using $S^1 \setminus \{1\} \cong \mathbb{R}$. So $F_k(S^1)$ is a disjoint union of $(k-1)!$ circles, and $B_k(S^1) \simeq S^1$ (Exercise 33.23).

Proposition 33.6 (Lie groups)**.** *If G is a topological group with identity e, then $F_k(G) \cong G \times F_{k-1}(G \setminus \{e\})$, $(g_1, \dots, g_k) \mapsto (g_1, (g_2 g_1^{-1}, \dots, g_k g_1^{-1}))$.*

Proof. The inverse is $(g, (h_2, \dots, h_k)) \mapsto (g, h_2 g, \dots, h_k g)$; distinctness of the g_i is equivalent to the h_i being distinct and different from e. □

Example 33.7 ($F_3(\mathbb{C})$)**.** Applying Proposition 33.6 to $(\mathbb{C}, +)$ and then to $(\mathbb{C}^\times, \cdot)$: $F_3(\mathbb{C}) \cong \mathbb{C} \times F_2(\mathbb{C}^\times) \cong \mathbb{C} \times \mathbb{C}^\times \times (\mathbb{C} \setminus \{0, 1\}) \simeq S^1 \times (S^1 \vee S^1)$. Hence $\pi_1 F_3(\mathbb{C}) \cong \mathbb{Z} \times F_2$ and the Poincaré polynomial is $(1+t)(1+2t)$.

33.2 The Fadell–Neuwirth fibrations

Lemma 33.8 (Moving points in a ball)**.** *Let $D \subset \mathbb{R}^d$ be the closed unit ball. For y in the open ball there is a homeomorphism $\varphi_y \colon D \to D$ with $\varphi_y(0) = y$ and $\varphi_y = \mathrm{id}$ on ∂D, depending continuously on y, with $\varphi_0 = \mathrm{id}$: namely $\varphi_y(x) = x + (1 - |x|)y$.*

Proof. For $u \in \partial D$ and $t \in [0, 1]$, $\varphi_y(tu) = (1-t)y + tu$: φ_y maps the radius $[0, u]$ affinely onto the segment $[y, u]$. Since D is strictly convex and y is interior, every point of $D \setminus \{y\}$ lies on exactly one segment $[y, u]$, at a unique parameter; so φ_y is a continuous bijection of a compact Hausdorff space, hence a homeomorphism. Continuity in y is clear from the formula. □

Theorem 33.9 (Fadell–Neuwirth)**.** *Let M be a manifold without boundary, $Q \subset M$ a finite set, and $0 \le l < k$. The projection*

$$\pi \colon F_k(M \setminus Q) \to F_l(M \setminus Q), \qquad (x_1, \dots, x_k) \mapsto (x_1, \dots, x_l),$$

is a locally trivial fibre bundle with fibre $F_{k-l}(M \setminus Q')$, where $Q' = Q \cup \{x_1, \dots, x_l\}$ for any base point $(x_1, \dots, x_l)$.

Proof. Replacing M by $M \setminus Q$ we may assume $Q = \varnothing$. Fix $x = (x_1, \dots, x_l) \in F_l(M)$ and pairwise disjoint closed coordinate balls $D_i \ni x_i$ with x_i at the centre. Let $U = \prod_i \operatorname{int} D_i$, an open neighbourhood of x in $F_l(M)$. For $y \in U$ let $\theta_y \colon M \to M$ be the homeomorphism equal to φ_{y_i} (in the coordinates of D_i) on each D_i and to the identity outside $\bigcup D_i$ (Lemma 33.8). Then $\theta_y(x_i) = y_i$, θ_y depends continuously on y, and $\theta_x = \mathrm{id}$. Define

$$U \times \pi^{-1}(x) \to \pi^{-1}(U),$$
$$\big(y, (x_1, \dots, x_l, z_{l+1}, \dots, z_k)\big) \mapsto \big(y_1, \dots, y_l, \theta_y(z_{l+1}), \dots, \theta_y(z_k)\big).$$

Since θ_y is a bijection taking $\{x_i\}$ to $\{y_i\}$, the image is a configuration; the inverse is $(y, z) \mapsto (y, (x, \theta_y^{-1}(z)))$, continuous because inversion in the homeomorphism group of a compact set, here of each D_i, is continuous. So π is trivial over U, with fibre $\pi^{-1}(x) \cong F_{k-l}(M \setminus \{x_1, \dots, x_l\})$. □

Remark 33.10. The theorem fails for manifolds with boundary if points may lie on the boundary (one cannot move a boundary point to an interior one), and for non-manifolds (in a graph, the fibre changes when a point crosses a vertex). It is the basic reason why configuration spaces of manifolds are tractable.

Corollary 33.11 (Iterated fibrations for $\mathbb{R}^n$)**.** *For $n \geq 2$ the forgetful maps give a tower*

$$\mathbb{R}^n \setminus \{k-1 \text{ points}\} \to F_k(\mathbb{R}^n) \to F_{k-1}(\mathbb{R}^n) \to \cdots \to F_2(\mathbb{R}^n) \to F_1(\mathbb{R}^n) = \mathbb{R}^n,$$

with fibres $\mathbb{R}^n \setminus Q_j \simeq \bigvee_j S^{n-1}$, and each projection has a section.

Proof. The fibres are as in Theorem 33.9. A section of $F_k(\mathbb{R}^n) \to F_{k-1}(\mathbb{R}^n)$ is $(x_1, \dots, x_{k-1}) \mapsto (x_1, \dots, x_{k-1}, x_k)$ with $x_k = (1 + \max_i |x_i|)e_1$, a point outside all of them. □

Corollary 33.12. *For $n \geq 3$, $F_k(\mathbb{R}^n)$ is $(n-2)$-connected, and for every i, $\pi_i F_k(\mathbb{R}^n) \cong \bigoplus_{j=1}^{k-1} \pi_i\big(\bigvee_j S^{n-1}\big)$.*

Proof. The long exact sequences of the fibrations split (sections) and, for $i \geq 2$, give split short exact sequences of abelian groups; for $i = 1$ everything vanishes when $n \geq 3$. Induct on k. □

Theorem 33.13 (Asphericity)**.** *$F_k(\mathbb{R}^2)$ and $B_k(\mathbb{R}^2)$ are $K(\pi, 1)$-spaces. More generally, if Σ is a connected surface other than S^2 and $\mathbb{RP}^2$, then $F_k(\Sigma)$ and $B_k(\Sigma)$ are aspherical for all k.*

Proof. For a connected surface $\Sigma \neq S^2, \mathbb{RP}^2$ and a finite set Q, $\Sigma \setminus Q$ is aspherical: if $Q \neq \varnothing$ or Σ is non-compact, $\Sigma \setminus Q$ deformation retracts to a graph; otherwise Σ is a closed surface with universal cover $\mathbb{R}^2$. Now induct on k using the fibration $\Sigma \setminus \{k-1 \text{ points}\} \to F_k(\Sigma) \to F_{k-1}(\Sigma)$ and its long exact sequence: if base and fibre have $\pi_i = 0$ for $i \geq 2$, so does the total space. Finally B_k is covered by F_k, and coverings induce isomorphisms on π_i, $i \geq 2$. □

33.3 Braid groups

Definition 33.14. The *braid group* on k strands is $\mathrm{Br}_k = \pi_1(B_k(\mathbb{R}^2))$ and the *pure braid group* is $P_k = \pi_1(F_k(\mathbb{R}^2))$. For a surface Σ, the *surface braid groups* are $\mathrm{Br}_k(\Sigma) = \pi_1 B_k(\Sigma)$ and $P_k(\Sigma) = \pi_1 F_k(\Sigma)$.

A loop in $B_k(\mathbb{R}^2)$ based at $\{1, \dots, k\}$ traces k strands in $\mathbb{R}^2 \times [0,1]$ that never meet: a geometric braid. Let σ_i be the braid in which the i-th and $(i+1)$-st points exchange positions by a half turn (counterclockwise), the others staying still.

Theorem 33.15 (Artin)**.** $\mathrm{Br}_k = \big\langle \sigma_1, \dots, \sigma_{k-1} \;\big|\; \sigma_i\sigma_{i+1}\sigma_i = \sigma_{i+1}\sigma_i\sigma_{i+1}\ (1 \leq i \leq k-2),\ \sigma_i\sigma_j = \sigma_j\sigma_i\ (|i-j| \geq 2) \big\rangle$.

For proofs see [68, Ch. 1] (via the Fadell–Neuwirth fibrations, which give a presentation of P_k inductively as an iterated semidirect product of free groups) and [82] (via a cell structure on the one-point compactification, the starting point of Section 36.2).

Corollary 33.16. *$H_1(\mathrm{Br}_k; \mathbb{Z}) \cong \mathbb{Z}$ for $k \geq 2$, generated by the class of any σ_i; the abelianisation is the* exponent sum *(total number of crossings counted with sign).*

Proof. Abelianising, $\sigma_i\sigma_{i+1}\sigma_i = \sigma_{i+1}\sigma_i\sigma_{i+1}$ gives $\sigma_i = \sigma_{i+1}$; the commutation relations become trivial. So $\mathrm{Br}_k^{\mathrm{ab}}$ is free on one generator. □

Corollary 33.17. Br_k *and* P_k *are torsion-free, and their (co)homology vanishes above degree* $2k$*. (In fact above degree* $k-1$*: see Chapter 36.)*

Proof. $B_k(\mathbb{R}^2)$ is a $2k$-dimensional manifold and a $K(\mathrm{Br}_k, 1)$ (Theorem 33.13); a manifold has the homotopy type of a CW complex of dimension at most its dimension, and Proposition 32.5 applies. □

Proposition 33.18 (Polynomials)**.** $B_k(\mathbb{C})$ *is homeomorphic to the space of monic polynomials of degree* k *with* k *distinct roots, i.e. to the complement in* $\mathbb{C}^k$ *of the discriminant hypersurface* $\{\mathrm{disc} = 0\}$*.*

Proof. $\{z_1, \dots, z_k\} \mapsto \prod(t - z_i)$ is a continuous bijection from $B_k(\mathbb{C})$ to monic squarefree polynomials (by the fundamental theorem of algebra, the roots determine and are determined by the polynomial). Its inverse is continuous because the roots of a polynomial depend continuously on the coefficients (the map $\mathbb{C}^k/\mathfrak{S}_k \to \mathbb{C}^k$ given by elementary symmetric functions is a homeomorphism, being a proper continuous bijection). □

This is the identification through which Arnold [61] first studied the cohomology of braid groups.

33.4 The Euler characteristic

Proposition 33.19. *Let* M *be a compact manifold without boundary of dimension* $d \geq 1$ *and* $\chi = \chi(M)$*. Then*

$$\chi(F_k(M)) = \prod_{i=0}^{k-1} \big(\chi - (-1)^d i\big).$$

In particular, if d *is even,* $\chi(B_k(M)) = \binom{\chi}{k}$ *and* $1 + \sum_{k\geq 1} \chi(B_k(M))t^k = (1+t)^{\chi(M)}$*; if* d *is odd,* $\chi(F_k(M)) = \chi(B_k(M)) = 0$ *for* $k \geq 1$*.*

Proof. First, $\chi(M \setminus \{q \text{ points}\}) = \chi - (-1)^d q$. Indeed, for any d-manifold N with finite-dimensional rational homology and a point $x \in N$, Mayer–Vietoris for $N = (N \setminus x) \cup D$, with D an open ball around x and $(N \setminus x) \cap D \simeq S^{d-1}$, gives $\chi(N) = \chi(N \setminus x) + 1 - \chi(S^{d-1}) = \chi(N \setminus x) + (-1)^d$; apply this q times. Second, for a fibration $F \to E \to B$ with B connected and of the homotopy type of a finite CW complex and $H_*(F;\mathbb{Q})$ finite-dimensional, $\chi(E) = \chi(B)\chi(F)$: in the Serre spectral sequence with rational local coefficients, $\chi(E^2) = \sum_q (-1)^q \chi(B; \mathcal{H}_q(F;\mathbb{Q})) = \sum_q (-1)^q \dim H_q(F;\mathbb{Q})\chi(B)$, because cellular chains of B with coefficients in a local system of rank r have rank r times the number of cells; and the Euler characteristic is unchanged by passing to homology, page after page. Applying this to the Fadell–Neuwirth fibrations (the configuration spaces of a compact manifold have the homotopy type of finite CW complexes, as interiors of compact manifolds with boundary — Chapter 40) gives the product formula by induction. For d even, $\chi(F_k(M)) = \chi(\chi-1)\cdots(\chi-k+1)$ and $\chi(B_k) = \chi(F_k)/k!$ (Exercise 4.14). For d odd, $\chi(M) = 0$ and the factor $i = 0$ vanishes. □

Example 33.20. $\chi(B_k(S^2)) = \binom{2}{k}$: $1, 2, 1, 0, 0, \dots$ for $k = 0, 1, 2, 3, \dots$. Indeed $B_2(S^2) \simeq \mathbb{RP}^2$ (Exercise 33.24) has $\chi = 1$, and for $k \geq 3$, $B_k(S^2)$ has rational cohomology $\mathbb{Q}$ in degrees 0 and 3 (Chapter 36), so $\chi = 0$. For the torus, $\chi(B_k(T^2)) = 0$ for all $k \geq 1$.

33.5 Braid groups of closed surfaces

On a closed surface the braid groups have relations coming from the global topology. Two examples will be used repeatedly.

Theorem 33.21 (Fadell–Van Buskirk)**.** $\mathrm{Br}_k(S^2)$ *is the quotient of* Br_k *by the single relation* $\sigma_1\sigma_2\cdots\sigma_{k-2}\sigma_{k-1}^2\sigma_{k-2}\cdots\sigma_1 = 1$.

See [79]. The relation says that the first point, travelling around all others, bounds a disc on the other side of the sphere.

Theorem 33.22 (Bellingeri)**.** *For a closed orientable surface* Σ_g *of genus* $g \geq 1$ *and* $k \geq 1$, $\mathrm{Br}_k(\Sigma_g)$ *has generators* $a_1, b_1, \ldots, a_g, b_g, \sigma_1, \ldots, \sigma_{k-1}$ *and relations: the braid relations among the* σ_i*;* $c\,\sigma_j = \sigma_j\,c$ *for* $j \geq 2$ *and* $c \in \{a_r, b_r\}$*;* $c\,\sigma_1 c\,\sigma_1 = \sigma_1 c\,\sigma_1 c$ *for* $c \in \{a_r, b_r\}$*;* $a_r\sigma_1 b_r = \sigma_1 b_r \sigma_1 a_r \sigma_1$*;* $c\,\sigma_1^{-1} c'\sigma_1 = \sigma_1^{-1} c'\sigma_1 c$ *for* $c \in \{a_s, b_s\}$, $c' \in \{a_r, b_r\}$, $r < s$*; and* $\prod_{r=1}^{g}[a_r^{-1}, b_r] = \sigma_1\sigma_2\cdots\sigma_{k-1}^2\cdots\sigma_2\sigma_1$.

See [64]; we have used the form recorded in [65, Thm. 8].

33.6 Exercises

Exercise 33.23 (easy)**.** Show that $B_k(S^1) \simeq S^1$ and that the covering $F_k(S^1) \to B_k(S^1)$ restricted to one component of $F_k(S^1)$ is the k-fold covering of the circle.

Source: original

Solution. A component of $F_k(S^1)$ consists of configurations in a fixed cyclic order, say counterclockwise $z_1, \ldots, z_k$; by Example 33.5(c) it is homeomorphic to $S^1 \times (\text{convex set}) \simeq S^1$. The subgroup of $\mathfrak{S}_k$ preserving this component is the cyclic group $\mathbb{Z}/k$ of rotations of the labels, and it acts freely; the quotient $B_k(S^1)$ is the quotient of this component by $\mathbb{Z}/k$. Following the loop in which all points rotate by $2\pi/k$ shows that the generator of π_1 of the quotient maps to $1/k$ of a full turn: the covering is the k-fold covering $S^1 \to S^1$, and $B_k(S^1) \simeq S^1$.

Exercise 33.24 (medium)**.** Show that $F_2(S^n) \simeq S^n$ and $B_2(S^n) \simeq \mathbb{RP}^n$. (Hint: for $x \neq y$ on $S^n \subset \mathbb{R}^{n+1}$ put $u = (x-y)/|x-y|$ and deform (x, y) to $(u, -u)$ along straight lines, then normalise.)

Source: original

Solution. Let $x_t = (1-t)x + tu$, $y_t = (1-t)y - tu$. Since $x \cdot u = (1 - x\cdot y)/|x-y| > 0$, $x_t \cdot x > 0$, so $x_t \neq 0$; similarly $y_t \cdot y > 0$. Put $\hat{x}_t = x_t/|x_t|$, $\hat{y}_t = y_t/|y_t|$. They are distinct: $x_t - y_t = ((1-t)|x-y| + 2t)u$, and $x_t \cdot u = (1-t)x\cdot u + t > 0$ while $y_t \cdot u = (1-t)y\cdot u - t < 0$ (as $y \cdot u = (x\cdot y - 1)/|x-y| < 0$), so x_t and y_t are not positive multiples of each other. Swapping x, y replaces u by $-u$ and swaps x_t, y_t: the deformation is $\mathfrak{S}_2$-equivariant, ends in the subspace $A = \{(u, -u)\} \cong S^n$ and fixes A pointwise. So $F_2(S^n)$ equivariantly deformation retracts onto A, on which the swap is the antipodal map. Hence $F_2(S^n) \simeq S^n$ and $B_2(S^n) \simeq S^n/\pm = \mathbb{RP}^n$.

Exercise 33.25 (medium)**.** Deduce from Theorem 33.21 that $H_1(B_k(S^2); \mathbb{Z}) \cong \mathbb{Z}/(2k-2)$ for $k \geq 2$. Conclude that the integral homology of $B_k(S^2)$ does not stabilise as $k \to \infty$.

Source: original; the statement is classical (Fadell–Van Buskirk)

Solution. Abelianising Br_k gives $\mathbb{Z}$ generated by σ (Corollary 33.16); the extra relation becomes $(2(k-2)+2)\sigma = (2k-2)\sigma = 0$. So $H_1 = \mathbb{Z}/(2k-2)$, which is different for each k. For $k = 2$ this is $\mathbb{Z}/2$, consistent with $B_2(S^2) \simeq \mathbb{RP}^2$ (Exercise 33.24).

Exercise 33.26 (medium)**.** Use Theorem 33.22 to show $H_1(B_k(\Sigma_g);\mathbb{Z}) \cong \mathbb{Z}^{2g} \oplus \mathbb{Z}/2$ for $g \geq 1$, $k \geq 2$.

SOURCE: original; the result is in [65, Thm. 1(a)] for $k \geq 3$

Solution. In the abelianisation all σ_i become one element σ (braid relations), and the relations involving $c\,\sigma_1 c\,\sigma_1$ and $c\,\sigma_1^{-1} c'\sigma_1$ and the commutations become trivial. The relation $a_r\sigma_1 b_r = \sigma_1 b_r \sigma_1 a_r \sigma_1$ becomes $a_r+\sigma+b_r = 3\sigma+a_r+b_r$, i.e. $2\sigma = 0$. The last relation becomes $0 = 2(k-1)\sigma$, which follows. So $H_1 = \mathbb{Z}^{2g}$ (free on the a_r, b_r) $\oplus\, \mathbb{Z}/2$ (generated by σ). For $k = 2$ there are no braid relations and the computation is the same. Note the contrast with S^2 (Exercise 33.25) and with $\mathbb{R}^2$, where σ has infinite order.

Exercise 33.27 (medium)**.** Verify Proposition 33.19 for $M = S^2$, $k = 2$ and $k = 3$ directly: $F_2(S^2) \simeq S^2$ (Exercise 33.24) and $F_3(S^2) \cong PGL_2(\mathbb{C}) \simeq SO(3) \cong \mathbb{RP}^3$ (Möbius transformations act simply transitively on ordered triples of distinct points of $\mathbb{CP}^1$).

SOURCE: original

Solution. $\chi = 2$, $d = 2$: the formula gives $\chi(F_2) = 2 \cdot 1 = 2 = \chi(S^2)$ and $\chi(F_3) = 2 \cdot 1 \cdot 0 = 0 = \chi(\mathbb{RP}^3)$. For the identification: $PGL_2(\mathbb{C})$ acts on $\mathbb{CP}^1 = S^2$, and for distinct z_1, z_2, z_3 there is a unique Möbius transformation sending $(0, 1, \infty)$ to (z_1, z_2, z_3); so $F_3(S^2) \cong PGL_2(\mathbb{C})$, which deformation retracts onto its maximal compact subgroup $PU(2) \cong SO(3) \cong \mathbb{RP}^3$.

Exercise 33.28 (medium)**.** Show that the Fadell–Neuwirth fibration $F_k(\mathbb{R}^2) \to F_{k-1}(\mathbb{R}^2)$ gives a split short exact sequence $1 \to F_{k-1} \to P_k \to P_{k-1} \to 1$, with F_{k-1} free of rank $k-1$. Deduce that P_k is an iterated semidirect product of free groups, that $H_1(P_k;\mathbb{Z}) \cong \mathbb{Z}^{\binom{k}{2}}$, and that the centre of $P_3 \cong \mathbb{Z} \times F_2$ is $\mathbb{Z}$.

SOURCE: original

Solution. The long exact sequence of homotopy groups, with $\pi_2 F_{k-1}(\mathbb{R}^2) = 0$ (Theorem 33.13) and π_1 of the fibre $\mathbb{R}^2 \setminus \{k-1 \text{ pts}\} \simeq \bigvee_{k-1} S^1$ free of rank $k-1$, gives the exact sequence; the section of Corollary 33.11 splits it. For H_1: by induction and the Serre (or Wang) spectral sequence, or directly from Chapter 34 ($H^1(F_k(\mathbb{R}^2))$ has basis ω_{ij}, $i < j$), the rank is $(k-1) + \binom{k-1}{2} = \binom{k}{2}$. The centre of $\mathbb{Z} \times F_2$ is $\mathbb{Z} \times Z(F_2) = \mathbb{Z}$ since a free group of rank 2 has trivial centre.

Exercise 33.29 (hard)**.** Let Y be the graph with one vertex of valence 3 (the letter Y). Show that the projection $F_2(Y) \to Y$ is not a fibration, by comparing fibres over the vertex and over an interior point of an edge.

SOURCE: original

Solution. Over the vertex v the fibre is $Y \setminus \{v\}$, three contractible components. Over an interior point x of an edge, the fibre $Y \setminus \{x\}$ has two contractible components. Fibres of a fibration over a path-connected base are homotopy equivalent, so $F_2(Y) \to Y$ is not a fibration. (This is why configuration spaces of graphs need different tools.)

Exercise 33.30 (easy)**.** Using Proposition 33.18, identify $B_2(\mathbb{C})$ with $\{(b, c) \in \mathbb{C}^2 : b^2 \neq 4c\}$ and show directly that it deformation retracts onto a circle.

SOURCE: original

Solution. $t^2 + bt + c$ has distinct roots iff $b^2 - 4c \neq 0$. The map $(b, c) \mapsto (b, b^2 - 4c)$ is a homeomorphism onto $\mathbb{C} \times \mathbb{C}^\times$, which deformation retracts onto $\{0\} \times S^1$.

33.7 Connections and further reading

Fadell–Neuwirth [78] and Fox–Neuwirth [82] are the founding papers; Birman's book [68] and Fadell–Husseini [77] develop braid groups and fibration methods. Kallel's survey [89] is the broadest current overview. The asphericity theorem is due to Fadell and Neuwirth; for S^2 and $\mathbb{RP}^2$ the configuration spaces are not aspherical (Exercise 33.27: $F_3(S^2) \simeq \mathbb{RP}^3$).

Chapter 34

The cohomology of configuration spaces: Arnold, Cohen, Totaro

34.1 The classes ω_{ij}

Throughout, $n \geq 2$ and $[S^{n-1}]^* \in H^{n-1}(S^{n-1};\mathbb{Z})$ is the generator dual to the standard orientation.

Definition 34.1. For $i \neq j$ in $\{1,\dots,k\}$ let $\pi_{ij}\colon F_k(\mathbb{R}^n) \to S^{n-1}$, $\pi_{ij}(x) = (x_j - x_i)/|x_j - x_i|$, and $\omega_{ij} = \pi_{ij}^*[S^{n-1}]^* \in H^{n-1}(F_k(\mathbb{R}^n);\mathbb{Z})$.

Lemma 34.2. *$\omega_{ji} = (-1)^n \omega_{ij}$, $\omega_{ij}^2 = 0$, and for $\sigma \in \mathfrak{S}_k$ acting by $(\sigma x)_i = x_{\sigma^{-1}(i)}$, $\sigma^*\omega_{ij} = \omega_{\sigma^{-1}(i)\sigma^{-1}(j)}$.*

Proof. $\pi_{ji} = a \circ \pi_{ij}$ with a the antipodal map, of degree $(-1)^n$. ω_{ij}^2 is pulled back from $H^{2n-2}(S^{n-1}) = 0$. The last formula is $\pi_{ij}\circ\sigma = \pi_{\sigma^{-1}(i)\sigma^{-1}(j)}$. □

34.2 The Arnold–Cohen theorem

Theorem 34.3 (Arnold, $n = 2$; F. Cohen, all n)**.** *$H^*(F_k(\mathbb{R}^n);\mathbb{Z})$ is the graded-commutative ring generated by the ω_{ij}, $1 \leq i < j \leq k$, of degree $n-1$, subject to the relations $\omega_{ij}^2 = 0$ and, for all distinct i,j,l,*

$$\omega_{ij}\omega_{jl} + \omega_{jl}\omega_{li} + \omega_{li}\omega_{ij} = 0 \qquad \textit{(Arnold relation)}$$

(using $\omega_{ji} = (-1)^n\omega_{ij}$). As an abelian group it is free with basis the monomials

$$\omega_{i_1 j_1}\omega_{i_2 j_2}\cdots\omega_{i_r j_r}, \qquad j_1 < j_2 < \cdots < j_r, \quad i_s < j_s,$$

and its Poincaré polynomial is $\prod_{j=1}^{k-1}(1 + j\,t^{n-1})$.

The proof has three steps: an additive basis from the Fadell–Neuwirth fibrations, the Arnold relation from an explicit computation in $F_3(\mathbb{R}^n)$, and a counting argument.

Step 1: the additive structure. Consider the fibration $p\colon F_k(\mathbb{R}^n) \to F_{k-1}(\mathbb{R}^n)$ of Corollary 33.11 with fibre $Y = \mathbb{R}^n \setminus \{x_1,\dots,x_{k-1}\}$. $H^*(Y;\mathbb{Z})$ is free with basis 1 and the classes $e_1,\dots,e_{k-1}$ in degree $n-1$, e_i dual to a small sphere S_i around x_i. The restriction of ω_{ik} $(i<k)$ to Y is pulled back by $y \mapsto (y-x_i)/|y-x_i|$, which has degree 1

on S_i and degree 0 on S_j, $j \neq i$ (it extends over the ball bounded by S_j, which does not contain x_i). So $\omega_{ik}|_Y = e_i$, and the Leray–Hirsch theorem [1, Thm. 4D.1] gives: $H^*(F_k(\mathbb{R}^n))$ is a free $H^*(F_{k-1}(\mathbb{R}^n))$-module (via p^*) with basis $1, \omega_{1k}, \dots, \omega_{k-1,k}$. By induction on k, $H^*(F_k(\mathbb{R}^n))$ is free with the basis of the theorem (for each j, at most one factor ω_{ij} with $i < j$), and the Poincaré polynomial is multiplied by $1 + (k-1)t^{n-1}$ at each step. □

Step 2: the Arnold relation. By naturality under the forgetful maps $F_k(\mathbb{R}^n) \to F_3(\mathbb{R}^n)$, $x \mapsto (x_i, x_j, x_l)$, it suffices to prove it in $F_3(\mathbb{R}^n)$ for $(i,j,l) = (1,2,3)$. By Step 1, $H^{2n-2}(F_3(\mathbb{R}^n))$ is free with basis $\omega_{12}\omega_{13}$, $\omega_{12}\omega_{23}$. Consider the two maps $S^{n-1} \times S^{n-1} \to F_3(\mathbb{R}^n)$

$$T_1(u,v) = (0,\, u,\, 3v), \qquad T_2(u,v) = (0,\, u,\, u + \tfrac{1}{3}v)$$

("planetary systems": in T_1 the third point orbits the first two; in T_2 it orbits the second). On T_1: $\pi_{12} = u$, $\pi_{13} = v$, and $\pi_{23} = (3v-u)/|3v-u| \simeq v$ via $(3v-su)/|3v-su|$, $s \in [0,1]$. On T_2: $\pi_{12} = u$, $\pi_{23} = v$, and $\pi_{13} \simeq u$ via $u + \frac{s}{3}v$. Writing $\langle c, T\rangle$ for the evaluation of T^*c on the fundamental class, and using that $\langle \mathrm{pr}_1^* a \cup \mathrm{pr}_2^* b, [S^{n-1} \times S^{n-1}]\rangle = 1$ for $a = b = [S^{n-1}]^*$ while classes pulled back from a single factor have zero square:

	$\omega_{12}\omega_{13}$	$\omega_{12}\omega_{23}$	$\omega_{13}\omega_{23}$
T_1	1	1	0
T_2	0	1	1

The first two columns form an invertible integer matrix, so T_1, T_2 give a basis of $H_{2n-2}(F_3(\mathbb{R}^n);\mathbb{Z})$ dual to $\omega_{12}\omega_{13}, \omega_{12}\omega_{23}$ (up to the triangular change of basis). Solving, the third column gives $\omega_{13}\omega_{23} = \omega_{12}\omega_{23} - \omega_{12}\omega_{13}$. On the other hand, by Lemma 34.2 and graded commutativity ($|\omega| = n-1$), $\omega_{23}\omega_{31} = (-1)^n\omega_{23}\omega_{13} = (-1)^n(-1)^{(n-1)^2}\omega_{13}\omega_{23} = -\omega_{13}\omega_{23}$, and similarly $\omega_{31}\omega_{12} = -\omega_{12}\omega_{13}$. So the Arnold relation for $(1,2,3)$ reads $\omega_{12}\omega_{23} - \omega_{13}\omega_{23} - \omega_{12}\omega_{13} = 0$, which is what we found. □

Step 3: no further relations. Let A be the ring presented in the theorem. By Steps 1–2 there is a surjective ring map $A \to H^*(F_k(\mathbb{R}^n))$. We show that the basis monomials span A; since their images are a basis, the map is an isomorphism. Take a monomial containing two factors with the same larger index, $\omega_{ik}\omega_{jk}$ with $i < j < k$. The Arnold relation for (i,j,k) expresses $\omega_{ik}\omega_{jk}$ as $\pm\omega_{ij}\omega_{jk} \pm \omega_{ij}\omega_{ik}$. The sum of the larger indices of the factors decreases (from $2k$ to $j+k$), so repeating terminates, with monomials whose larger indices are distinct (or which vanish because $\omega^2 = 0$). These are $\pm$ the basis monomials. □

Remark 34.4 (Homology and the Poisson operad). Dually, $H_*(F_k(\mathbb{R}^n);\mathbb{Z})$ has a basis of "planetary system" tori like T_1, T_2, and $\bigoplus_k H_*(F_k(\mathbb{R}^n))$ is the operad e_n of n-Poisson algebras (Theorem 25.7): the class of $F_2(\mathbb{R}^n) \simeq S^{n-1}$ in degree $n-1$ is the bracket, and the Arnold relation is dual to the Jacobi identity. Over $\mathbb{C}$, Arnold's original proof uses the forms $\omega_{ij} = \frac{1}{2\pi i}\, d\log(z_j - z_i)$ (Exercise 34.15).

Example 34.5. $F_3(\mathbb{R}^n)$: Poincaré polynomial $(1+t^{n-1})(1+2t^{n-1}) = 1+3t^{n-1}+2t^{2n-2}$. For $n = 2$ this is $(1+t)(1+2t)$, matching Example 33.7. $F_4(\mathbb{R}^2)$: $(1+t)(1+2t)(1+3t) = 1 + 6t + 11t^2 + 6t^3$; the coefficients are the unsigned Stirling numbers of the first kind, and $\sum_r \dim H^{r(n-1)} = k!$.

34.3 The symmetric group action and unordered configurations of $\mathbb{R}^n$

Proposition 34.6. *$H^{n-1}(F_k(\mathbb{R}^n);\mathbb{Q})$, as an $\mathfrak{S}_k$-module, is the permutation module on unordered pairs $\{i,j\}$ if n is even, and the module of "oriented pairs" ($\omega_{ji}=-\omega_{ij}$) if n is odd. Its invariants are 1-dimensional (spanned by $\sum_{i<j}\omega_{ij}$) for n even and 0 for n odd, $k\geq 2$.*

Proof. The first statement is Lemma 34.2. An invariant vector $\sum c_{ij}\omega_{ij}$ has c constant on the $\mathfrak{S}_k$-orbit of pairs, so it is a multiple of $\sum_{i<j}\omega_{ij}$; for n odd the transposition (ij) maps ω_{ij} to $\omega_{ji}=-\omega_{ij}$, and the coefficient of ω_{ij} in an invariant vector must therefore vanish. □

Theorem 34.7. *For $k\geq 2$, $H^*(B_k(\mathbb{R}^n);\mathbb{Q})\cong H^*(F_k(\mathbb{R}^n);\mathbb{Q})^{\mathfrak{S}_k}$ is $\mathbb{Q}$ in degree 0, and in addition $\mathbb{Q}$ in degree $n-1$ if n is even; it vanishes in all other degrees.*

The isomorphism is Theorem 32.7(a); degrees 0 and $n-1$ are Proposition 34.6. The vanishing in degrees $r(n-1)$, $r\geq 2$, is a computation with the $\mathfrak{S}_k$-character of the Arnold algebra; it follows from F. Cohen's computation of $H_*(B_k(\mathbb{R}^n);\mathbb{F})$ [40], and for $n=2$ it is the rational case of Arnold's theorem that $H^i(\mathrm{Br}_k;\mathbb{Z})$ is finite for $i\geq 2$, reproved via the Fuks complex in Section 36.2. In Chapter 41 the invariants are computed by machine for $k\leq 6$, $n=2,3$, confirming the statement in that range.

Caution 34.8. Theorem 34.7 says $B_k(\mathbb{R}^n)$ is rationally almost trivial; integrally it is not. Already $B_2(\mathbb{R}^n)\simeq\mathbb{RP}^{n-1}$ has 2-torsion in every odd degree below $n-1$, and $H^*(\mathrm{Br}_k;\mathbb{Z})$ has p-torsion for every prime p once $k\geq 2p$ (Chapter 36). For $n=\infty$, $B_k(\mathbb{R}^\infty)$ is a $B\mathfrak{S}_k$, whose rational cohomology is trivial and whose integral cohomology is a central object of algebraic topology.

34.4 Two points in a closed manifold

For $k=2$ everything can be done by hand, integrally, for any closed oriented manifold.

Theorem 34.9. *Let M be a closed connected oriented m-manifold with $H^*(M;\mathbb{Z})$ torsion-free, and $\Delta\in H^m(M\times M;\mathbb{Z})$ the class Poincaré dual to the diagonal. Then restriction induces a ring isomorphism*

$$H^*(F_2(M);\mathbb{Z})\cong H^*(M\times M;\mathbb{Z})/(\Delta),$$

the ideal (Δ) is the image of the injective, split map $G\colon H^{*-m}(M)\to H^*(M\times M)$, $G(x)=(x\times 1)\cup\Delta$, and $H^*(F_2(M);\mathbb{Z})$ is torsion-free with Poincaré polynomial $P_M(t)^2-t^mP_M(t)$.

Proof. Let N be a closed tubular neighbourhood of the diagonal $D\cong M$ in $M\times M$; its normal bundle is isomorphic to TM, oriented. By excision and the Thom isomorphism (Chapter 26), $H^*(M\times M,F_2(M))\cong H^*(N,N\setminus D)\cong H^{*-m}(M)$, $x\mapsto\pi^*x\cup u$, where u is the Thom class. The image of u in $H^m(M\times M)$ is the diagonal class Δ [10, §11], and since $\mathrm{pr}_1|_N\simeq\pi$, the composite $H^{j-m}(M)\to H^j(M\times M,F_2(M))\to H^j(M\times M)$ is G. The long exact sequence of the pair becomes

$$\begin{aligned}\cdots\to H^{j-m}(M)\xrightarrow{G}H^j(M\times M)&\to H^j(F_2(M))\\ &\to H^{j-m+1}(M)\xrightarrow{G}H^{j+1}(M\times M)\to\cdots.\end{aligned}$$

G is split injective: by the formula $\Delta = \sum_i (-1)^{|b_i|} b_i \times b_i^{\#}$ for dual bases ($b_i \cup b_i^{\#}$ the orientation class) [10, §11], the slant product with the fundamental class of the second factor satisfies $G(x)/[M] = \pm x$. Hence the sequence splits into short exact sequences $0 \to H^{j-m}(M) \to H^j(M \times M) \to H^j(F_2(M)) \to 0$, and the cokernel of a split injection between free groups is free. The image of G is the ideal generated by Δ: for $z \in H^*(M \times M)$, $z \cup \Delta = z|_N \cup u$ and $z|_N = (\mathrm{pr}_1^* \Delta^* z)|_N$ since $\mathrm{pr}_1 \simeq \Delta \circ \mathrm{pr}_1$ on N; so $z \cup \Delta = G(\Delta^* z)$. Counting ranks gives the Poincaré polynomial. □

Example 34.10. (a) $M = S^m$: $P = 1 + t^m$, so $P^2 - t^m P = 1 + t^m$, matching $F_2(S^m) \simeq S^m$ (Exercise 33.24).

(b) $M = T^2$: $(1+t)^4 - t^2(1+t)^2 = (1+t)^2(1+2t) = 1 + 4t + 5t^2 + 2t^3$, matching $F_2(T^2) \cong T^2 \times (T^2 \setminus \mathrm{pt})$ (Proposition 33.6).

(c) $M = \mathbb{CP}^m$: with $P = 1 + t^2 + \cdots + t^{2m}$, $P^2 - t^{2m} P = P\,(1 + t^2 + \cdots + t^{2m-2})$: Betti numbers $i+1$ in degree $2i < 2m$ and $m - i$ in degree $2m + 2i$. This agrees with the computation from Kriz's model in [89, §5.4.1].

Remark 34.11. Theorem 34.9 shows that the integral cohomology ring of $F_2(M)$ is determined by the cohomology ring of M (with its orientation), hence is a homotopy invariant of closed oriented manifolds with torsion-free cohomology. The *homotopy type* of $F_2(M)$ is not (Chapter 38): the invariant that distinguishes $F_2(L_{7,1})$ from $F_2(L_{7,2})$ is a Massey product on the universal cover.

34.5 The Totaro–Cohen–Taylor spectral sequence and Kriz's model

For $k > 2$ the analogue of Theorem 34.9 is a spectral sequence. Let M be an oriented m-manifold and $j\colon F_k(M) \hookrightarrow M^k$ the inclusion. The Leray spectral sequence of j has $E_2^{p,q} = H^p(M^k; R^q j_* \mathbb{Q})$, where the stalk of the sheaf $R^q j_* \mathbb{Q}$ at a point of M^k is the cohomology of the configuration space of a small neighbourhood — locally, products of $H^*(F_{k_i}(\mathbb{R}^m))$, one factor for each group of coinciding points. This is how the Arnold algebra enters.

Theorem 34.12 (Cohen–Taylor; Totaro)**.** *Let M be an oriented m-manifold and $\mathbb{F}$ a field. The Leray spectral sequence of $F_k(M) \hookrightarrow M^k$ converges to $H^*(F_k(M); \mathbb{F})$ as an algebra. Its first possibly nonzero page E_m (equivalently E_2) is the bigraded algebra generated by $H^*(M; \mathbb{F})^{\otimes k}$ in bidegrees $(*, 0)$ and classes G_{ab}, $a \neq b$, in bidegree $(0, m-1)$, with relations*

$$G_{ab} = (-1)^m G_{ba}, \quad G_{ab}^2 = 0, \quad G_{ab}G_{bc} + G_{bc}G_{ca} + G_{ca}G_{ab} = 0, \quad p_a^*(x)\, G_{ab} = p_b^*(x)\, G_{ab},$$

and the first differential is $d_m(G_{ab}) = p_{ab}^* \Delta$, the pull-back of the diagonal class along the projection to factors a, b. If M is a smooth complex projective variety and $\mathbb{F} = \mathbb{Q}$, d_m is the only nonzero differential.

See [109] (Theorems 1 and 4) and [75]; the formulation above follows the recap in [81]. For $M = \mathbb{R}^m$ there are no diagonal classes, $d_m = 0$, and E_m is the Arnold algebra: the spectral sequence collapses, recovering Theorem 34.3 rationally. For $k = 2$ and M closed, the spectral sequence has two rows and is equivalent to the long exact sequence in the proof of Theorem 34.9.

Theorem 34.13 (Kriz)**.** *If M is a smooth complex projective variety, the differential graded algebra (E_m, d_m) of Theorem 34.12 (with $\mathbb{F} = \mathbb{Q}$) is a rational model of $F_k(M)$: it is quasi-isomorphic to $A_{\mathrm{PL}}(F_k(M))$. In particular the rational homotopy type of $F_k(M)$ depends only on the rational cohomology ring of M (with its orientation class).*

See [92]. Formality of M ([33]) is what makes this possible; for general manifolds the corresponding statement involves a model of M rather than its cohomology (Chapter 37).

Theorem 34.14 (Félix–Thomas)**.** *Let M be a closed, orientable, even-dimensional nilpotent manifold. The rational Betti numbers of $B_k(M)$ are determined by the graded algebra $H^*(M;\mathbb{Q})$ (and not, in general, by the Betti numbers of M alone). If M is a closed odd-dimensional manifold, $H_*(B_k(M);\mathbb{F})$ depends only on the graded vector space $H_*(M;\mathbb{F})$, for every field $\mathbb{F}$; for $\mathbb{F} = \mathbb{F}_2$ the same holds in every dimension (Bödigheimer–Cohen–Taylor, Löffler–Milgram).*

See [80] and [63], as summarised in [89, §5.2]. Schiessl [106] used the Félix–Thomas complex to compute the rational Betti numbers of $B_k(T^2)$ (Theorem 35.4).

34.6 Exercises

Exercise 34.15 (medium)**.** For a, b, c nowhere-vanishing holomorphic functions on an open set of $\mathbb{C}^N$ with $a + b + c = 0$, show $\frac{da}{a} \wedge \frac{db}{b} + \frac{db}{b} \wedge \frac{dc}{c} + \frac{dc}{c} \wedge \frac{da}{a} = 0$. Deduce the Arnold relation in $H^2(F_3(\mathbb{C});\mathbb{C})$ from de Rham's theorem, taking $a = z_2 - z_1$, $b = z_3 - z_2$, $c = z_1 - z_3$.

Source: original; Arnold's argument

Solution. Substitute $dc = -da - db$: the sum is $\frac{da\wedge db}{ab} + \frac{db\wedge(-da)}{bc} + \frac{(-db)\wedge da}{ca} = da\wedge db\big(\frac{1}{ab} + \frac{1}{bc} + \frac{1}{ca}\big) = da \wedge db\,\frac{a+b+c}{abc} = 0$. The closed forms $\frac{1}{2\pi i}\frac{da}{a}$ etc. represent $\omega_{12}, \omega_{23}, \omega_{31}$ in $H^1(F_3(\mathbb{C});\mathbb{C})$ (for $n = 2$, $\omega_{ij} = \omega_{ji}$, and $d\log(z_j - z_i)$ restricted to a small loop of z_j around z_i integrates to $2\pi i$). The identity of forms gives the identity of classes.

Exercise 34.16 (easy)**.** Compute $\chi(F_k(\mathbb{R}^n))$ from Theorem 34.3, and compare with Proposition 33.19 applied to the open manifold $\mathbb{R}^n$ via the Fadell–Neuwirth fibrations.

Source: original

Solution. Evaluate $\prod_{j=1}^{k-1}(1 + jt^{n-1})$ at $t = -1$: for n even, $\prod(1 - j) = 0$ ($k \geq 2$); for n odd, $\prod(1 + j) = k!$. From the fibrations: $\chi(\mathbb{R}^n \setminus \{j \text{ pts}\}) = \chi(\bigvee_j S^{n-1}) = 1 + j(-1)^{n-1}$, and $\chi(F_k) = \prod_{j=0}^{k-1}(1 + j(-1)^{n-1})$, the same.

Exercise 34.17 (medium)**.** In $H^4(F_3(\mathbb{R}^3);\mathbb{Z})$, express $\omega_{13}\omega_{23}$ in the basis of Theorem 34.3, and check that $\omega_{12}\omega_{13}\omega_{23} = 0$.

Source: original

Solution. The proof of Step 2 is valid for every n: $\omega_{13}\omega_{23} = \omega_{12}\omega_{23} - \omega_{12}\omega_{13}$. Multiplying by ω_{12} and using $\omega_{12}^2 = 0$: $\omega_{12}\omega_{13}\omega_{23} = \omega_{12}(\omega_{12}\omega_{23} - \omega_{12}\omega_{13}) = 0$. (Also $H^6(F_3(\mathbb{R}^3)) = 0$ for degree reasons, which is consistent: the triple product has degree 6.)

Exercise 34.18 (medium)**.** Use Theorem 34.9 to compute the ring $H^*(F_2(T^2);\mathbb{Z})$: write $H^*(T^2) = \Lambda(a, b)$ and give Δ explicitly.

SOURCE: original

Solution. With $\mu = ab$ the orientation class and the dual basis $1 \leftrightarrow \mu$, $a \leftrightarrow b$, $b \leftrightarrow -a$ (since $a \cdot b = \mu$, $b \cdot (-a) = \mu$), $\Delta = 1 \times \mu + \mu \times 1 - a \times b + b \times a$ (signs from $(-1)^{|b_i|}$). $H^*(F_2(T^2))$ is $\Lambda(a_1, b_1, a_2, b_2)/(\Delta)$ with $a_1 = a \times 1$ etc.: ranks $1, 4, 6-1=5, 4-2=2, 1-1=0$ in degrees $0, \dots, 4$, using that (Δ) has rank $1, 2, 1$ in degrees $2, 3, 4$. In particular $a_1 b_1 a_2 b_2 = \pm(\mu \times 1)\Delta = 0$ in $F_2(T^2)$: the open manifold has no top class.

Exercise 34.19 (hard)**.** For M as in Theorem 34.9, the swap τ of $M \times M$ preserves $F_2(M)$. Show $\tau^*\Delta = (-1)^m \Delta$. Deduce, for $M = S^m$, that $H^*(B_2(S^m); \mathbb{Q}) = H^*(\mathbb{RP}^m; \mathbb{Q})$, and check with Exercise 33.24.

SOURCE: original

Solution. τ preserves the diagonal pointwise and acts on its normal bundle TM by -1, which multiplies the Thom class by $(-1)^m$; hence $\tau^*\Delta = (-1)^m\Delta$. For $M = S^m$, $H^*(F_2(S^m)) = \langle 1, x \times 1 \rangle$ with $x \times 1 \equiv (-1)^{m+1}\, 1 \times x$ modulo $\Delta = x \times 1 + (-1)^m 1 \times x$; so $\tau^*(x \times 1) = 1 \times x = (-1)^{m+1} x \times 1$: the class in degree m is invariant iff m is odd. So $H^*(B_2(S^m); \mathbb{Q})$ is $\mathbb{Q}$ in degree 0, plus $\mathbb{Q}$ in degree m if m is odd: exactly $H^*(\mathbb{RP}^m; \mathbb{Q})$.

34.7 Connections and further reading

Arnold [60] and F. Cohen [40] computed $H^*(F_k(\mathbb{R}^n))$; Orlik–Solomon [102] generalised Arnold's algebra to complements of complex hyperplane arrangements. Totaro [109], Cohen–Taylor [75] and Kriz [92] are the sources for the spectral sequence and the model; Félix–Thomas [81] compare it with the Bendersky–Gitler spectral sequence and relate its higher differentials to Massey products of M (Chapter 37).

Chapter 35

Unordered configuration spaces: rational homology

35.1 The principle

For every manifold M and field $\mathbb{F}$ of characteristic 0 or $> k$, Theorem 32.7 gives

$$H^*(B_k(M);\mathbb{F}) \cong H^*(F_k(M);\mathbb{F})^{\mathfrak{S}_k}.$$

So, rationally, unordered configuration spaces are a problem in the *representation theory* of $\mathfrak{S}_k$ acting on $H^*(F_k(M);\mathbb{Q})$: one needs the multiplicity of the trivial representation. In characteristic $p \leq k$ this fails, and the difference is the subject of Chapter 36. This chapter collects what is known rationally, with examples computed by hand.

35.2 Examples

Example 35.1 ($\mathbb{R}^n$)**.** By Theorem 34.7, $H^*(B_k(\mathbb{R}^n);\mathbb{Q}) = \mathbb{Q}$ (n odd) or $\mathbb{Q}\oplus\mathbb{Q}[-(n-1)]$ (n even), for $k \geq 2$. For $n = 2$: Br_k has the rational cohomology of a circle; the class in degree 1 is the exponent sum (Corollary 33.16).

Example 35.2 ($B_3(\mathbb{R}^2)$ by hand)**.** $H^2(F_3(\mathbb{R}^2);\mathbb{Q})$ has basis $e_1 = \omega_{12}\omega_{13}$, $e_2 = \omega_{12}\omega_{23}$ (Theorem 34.3). With the convention $\sigma^*\omega_{ij} = \omega_{\sigma^{-1}(i)\sigma^{-1}(j)}$ and $\omega_{ij} = \omega_{ji}$ ($n = 2$): the transposition (12) fixes ω_{12} and exchanges ω_{13}, ω_{23}, so $e_1 \leftrightarrow e_2$. The 3-cycle $c = (123)$ sends $\omega_{12} \mapsto \omega_{13}$, $\omega_{13} \mapsto \omega_{23}$, $\omega_{23} \mapsto \omega_{12}$, so $c^*e_1 = \omega_{13}\omega_{23} = e_2 - e_1$ (by Step 2 of the proof of Theorem 34.3) and $c^*e_2 = \omega_{13}\omega_{12} = -e_1$. The traces are $2, 0, -1$ on the identity, transpositions and 3-cycles: this is the 2-dimensional irreducible (standard) representation, which has no invariants. Hence $H^2(B_3(\mathbb{R}^2);\mathbb{Q}) = 0$, as Theorem 34.7 predicts.

Theorem 35.3 (S^2)**.** *$H^*(B_k(S^2);\mathbb{Q})$ is $\mathbb{Q}$ in degrees 0 and 2 for $k = 1$; $\mathbb{Q}$ in degree 0 for $k = 2$; and $\mathbb{Q}$ in degrees 0 and 3 for $k \geq 3$.*

This is recorded with references in [105, §1.1]; for $k = 2$ it follows from $B_2(S^2) \simeq \mathbb{RP}^2$ (Exercise 33.24), and for $k = 3$ one can check it by hand from $F_3(S^2) \simeq \mathbb{RP}^3$ (Exercise 33.27) and the transfer. The free parts of the integral groups computed in Section 36.2 confirm the statement for $k \leq 11$.

Theorem 35.4 (Schiessl)**.** *For $k \geq 2$ the rational Betti numbers of $B_k(T^2)$ are*

$$b_0 = 1, \quad b_1 = 2, \quad b_i = 2i - 1 \ (2 \leq i < k), \quad b_k = \begin{cases} \frac{3k-4}{2} & k \text{ even} \\ \frac{3k-1}{2} & k \text{ odd} \end{cases},$$

$$b_{k+1} = \begin{cases} \frac{k-2}{2} & k \text{ even} \\ \frac{k+1}{2} & k \text{ odd} \end{cases},$$

and $b_i = 0$ for $i > k+1$.

See [106], where the computation uses the Félix–Thomas complex (Theorem 34.14); Drummond-Cole and Knudsen [76] computed the Betti numbers of all surfaces by a different method (Theorem 35.5). Three consistency checks: $\sum (-1)^i b_i = 0 = \chi(B_k(T^2))$ (Proposition 33.19); for $k = 2$ the values $1, 2, 1, 0$ agree with the hand computation of Exercise 35.8; and the values $b_2 = 1$ $(k = 2)$, $b_2 = 3$ $(k \geq 3)$, $b_3 = 4$ $(k = 3)$, $b_3 = 5$ $(k \geq 4)$, $b_4 = 4$ $(k = 4)$, $b_4 = 7$ $(k \geq 5)$ are those quoted by Church [72] from Napolitano's tables, illustrating that the rational stable range $k > i$ (Theorem 39.4) is sharp.

35.3 Lie algebra methods

Theorem 35.5 (Knudsen)**.** *For any d-manifold M (possibly with boundary), $\bigoplus_k H_*(B_k(M); \mathbb{Q})$ is isomorphic to the Lie algebra homology of an explicit graded Lie algebra built from the compactly supported cohomology of M; the homology of each $B_k(M)$ is a specified weight-graded piece of the Chevalley–Eilenberg complex.*

See [90]. The proof uses factorisation homology (Chapter 40). The theorem extends the results of Bödigheimer–Cohen–Taylor and Félix–Thomas (Theorem 34.14) and gives a combinatorial proof of rational homological stability (Chapter 39). Its moral: rationally, $B(M) = \coprod_k B_k(M)$ behaves like the "free e_d-algebra" on M, and e_d-algebras are rationally governed by shifted Lie algebras (Koszul duality, Chapter 30).

35.4 What changes in positive characteristic

Example 35.6. $H^*(F_2(\mathbb{R}^n); \mathbb{F}_2) = H^*(S^{n-1}; \mathbb{F}_2)$ has $\mathfrak{S}_2$-invariants in degrees 0 and $n-1$ only, but $H^*(B_2(\mathbb{R}^n); \mathbb{F}_2) = H^*(\mathbb{RP}^{n-1}; \mathbb{F}_2)$ is $\mathbb{F}_2$ in every degree $0, \dots, n-1$. The extra classes come from $H^*(\mathfrak{S}_2; \mathbb{F}_2) = \mathbb{F}_2[w]$: in the Cartan–Leray spectral sequence $E_2^{p,q} = H^p(\mathfrak{S}_2; H^q(S^{n-1}; \mathbb{F}_2))$, the bottom row is $H^*(B\mathfrak{S}_2; \mathbb{F}_2)$, which is not killed until degree n.

Remark 35.7. Over $\mathbb{Q}$, the Cartan–Leray spectral sequence of $F_k(M) \to B_k(M)$ collapses onto its column $p = 0$, which is the invariants (Corollary 32.8). Over $\mathbb{F}_p$ with $p \leq k$, the columns $p > 0$ contribute, and their interaction with the differentials is the source of torsion. This is why rational models (Chapter 37) can see neither the torsion of $H_*(B_k(M); \mathbb{Z})$ nor its mod-p homology.

35.5 Exercises

Exercise 35.8 (medium)**.** Compute $H^*(B_2(T^2); \mathbb{Q})$ from Theorem 34.9 and Exercise 34.18, by computing the invariants of the swap τ on $\Lambda(a_1, b_1, a_2, b_2)/(\Delta)$.

SOURCE: original

Solution. τ exchanges $a_1 \leftrightarrow a_2$, $b_1 \leftrightarrow b_2$, and $\tau^*\Delta = \Delta$ (Exercise 34.19, $m = 2$). Taking invariants is exact in characteristic 0, so $\dim(\text{quotient})^\tau = \dim \Lambda^\tau - \dim(\Delta)^\tau$ in each degree. Degree 1: invariants $a_1 + a_2$, $b_1 + b_2$; $b_1 = 2$. Degree 2: in Λ^2, τ exchanges $a_1b_1 \leftrightarrow a_2b_2$ and $a_1b_2 \leftrightarrow a_2b_1$, and negates a_1a_2 and b_1b_2; invariants have dimension 2; the ideal in degree 2 is spanned by the invariant Δ; so $b_2 = 1$. Degree 3: in Λ^3, τ exchanges $a_1b_1a_2 \leftrightarrow a_1a_2b_2$ and $a_1b_1b_2 \leftrightarrow b_1a_2b_2$ (check signs: $\tau(a_1b_1a_2) = a_2b_2a_1 = a_1a_2b_2$), so invariants have dimension 2; the ideal in degree 3 is spanned by $a_1\Delta = a_2\Delta$ and $b_1\Delta = b_2\Delta$, both invariant; so $b_3 = 0$. Degree 4: the quotient is 0. Result: $1, 2, 1, 0$, matching Theorem 35.4.

Exercise 35.9 (easy)**.** Show directly that $H^*(B_2(\mathbb{R}^3); \mathbb{Q}) = \mathbb{Q}$ and $H^*(B_2(\mathbb{R}^4); \mathbb{Q}) = \mathbb{Q} \oplus \mathbb{Q}[-3]$.

SOURCE: original

Solution. $B_2(\mathbb{R}^n) \simeq \mathbb{RP}^{n-1}$ (Example 32.12); $H^*(\mathbb{RP}^2; \mathbb{Q}) = \mathbb{Q}$ and $H^*(\mathbb{RP}^3; \mathbb{Q}) = H^*(S^3; \mathbb{Q})$ since $\mathbb{RP}^3$ is orientable.

Exercise 35.10 (medium)**.** Check Theorem 35.3 against Proposition 33.19 for all k.

SOURCE: original

Solution. $\chi(B_k(S^2)) = \binom{2}{k}$. For $k = 1$: $1 + 1 = 2 = \binom{2}{1}$. For $k = 2$: $1 = \binom{2}{2}$. For $k \geq 3$: $1 - 1 = 0 = \binom{2}{k}$.

Exercise 35.11 (medium)**.** Using Theorem 35.4, list the Betti numbers of $B_3(T^2)$ and $B_4(T^2)$, check that their Euler characteristics vanish, and compare b_3 for $k = 3$ and $k = 4$.

SOURCE: original

Solution. $k = 3$: $b_0, \dots, b_4 = 1, 2, 3, 4, 2$; $\chi = 1 - 2 + 3 - 4 + 2 = 0$. $k = 4$: $b_0, \dots, b_5 = 1, 2, 3, 5, 4, 1$; $\chi = 1 - 2 + 3 - 5 + 4 - 1 = 0$. So b_3 jumps from 4 to 5: in degree 3 the Betti numbers stabilise only from $k = 4$ on, i.e. for $k > i$ (Theorem 39.4).

Exercise 35.12 (hard)**.** Let $\mathfrak{S}_2$ act on $V = \mathbb{F}_2^2$ by swapping coordinates. Show that $V \to V/\langle(1,1)\rangle$ is surjective but the induced map on invariants is not. Explain why this breaks the argument "invariants of a quotient = quotient of invariants" used in Exercise 35.8 when $\mathbb{F} = \mathbb{F}_2$.

SOURCE: original

Solution. $V^{\mathfrak{S}_2} = \langle(1,1)\rangle$ maps to 0, while $V/\langle(1,1)\rangle \cong \mathbb{F}_2$ with trivial action has invariants $\mathbb{F}_2$. So invariants is not right exact: $H^1(\mathfrak{S}_2; \langle(1,1)\rangle) = H^1(\mathfrak{S}_2; \mathbb{F}_2) = \mathbb{F}_2 \neq 0$ obstructs it. In characteristic 0 (or prime to $|G|$), averaging $v \mapsto \frac{1}{|G|}\sum gv$ splits every surjection equivariantly, and invariants is exact.

35.6 Connections and further reading

Church [72] is the clearest introduction to the representation-theoretic point of view; Knudsen [90] and Drummond-Cole–Knudsen [76] give the most complete rational results for surfaces; Pagaria [103] computed the rational cohomology ring of $B_k(T^2)$.

Chapter 36

Integral homology and torsion

36.1 Where torsion can come from

Proposition 36.1. *Let M be a manifold and p a prime.*

(a) If $p > k$, the transfer $H_*(B_k(M);\mathbb{Z}_{(p)}) \to H_*(F_k(M);\mathbb{Z}_{(p)})$ is split injective. In particular, p-torsion in $H_*(B_k(M);\mathbb{Z})$ for $p > k$ can only come from p-torsion in $H_*(F_k(M);\mathbb{Z})$.

(b) If $H_*(F_k(M);\mathbb{Z})$ is torsion-free (e.g. $M = \mathbb{R}^n$, Theorem 34.3), every torsion element of $H_*(B_k(M);\mathbb{Z})$ has order dividing $k!$; so only primes $p \le k$ occur.

Proof. (a) $p_* \circ \mathrm{tr} = k!$, a unit in $\mathbb{Z}_{(p)}$; so $(k!)^{-1}p_*$ is a left inverse of tr. (b) is Theorem 32.7(b). □

So $H_*(\mathrm{Br}_k;\mathbb{Z})$ has only p-torsion with $p \le k$. We shall see that the truth is sharper: the first p-torsion appears at $k = 2p$.

36.2 The Fuks–Vainshtein cell complex

The one-point compactification $B_n(\mathbb{C})^+$ has a cell structure (Fox–Neuwirth [82], Fuks [83]): project a configuration to the real axis; the image is a finite set $c_1 < \cdots < c_s$, and the numbers n_i of points over c_i form a *composition* $[n_1,\dots,n_s]$ of n. The configurations with a given composition form an open cell of dimension $s+n$ (s real parts, n imaginary parts), and together with the point at infinity they give a CW structure on $B_n(\mathbb{C})^+$. Since $B_n(\mathbb{C})$ is an oriented $2n$-manifold, Poincaré duality for Borel–Moore homology gives $H^i(B_n(\mathbb{C})) \cong \widetilde{H}_{2n-i}(B_n(\mathbb{C})^+)$. Reindexing, a cell $[n_1,\dots,n_s]$ contributes to cohomological degree $r = n - s$.

Theorem 36.2 (Fuks, Vainshtein; signs as in [105])**.** *$H^*(B_n(\mathbb{C});\mathbb{Z}) = H^*(\mathrm{Br}_n;\mathbb{Z})$ is the cohomology of the cochain complex $A_n^\bullet$ with $A_n^r = \mathbb{Z}\{\text{compositions of } n \text{ of length } n-r\}$ and*

$$\delta[n_1,\dots,n_s] = \sum_{l=1}^{s-1}(-1)^{l-1}P(n_l,n_{l+1})\,[n_1,\dots,n_{l-1},n_l+n_{l+1},n_{l+2},\dots,n_s],$$

where $P(x,y) = 0$ if x,y are both odd and $P(x,y) = \binom{\lfloor x/2\rfloor+\lfloor y/2\rfloor}{\lfloor x/2\rfloor}$ otherwise.

The coefficient $P(x,y)$ is the incidence number of the cell in which two adjacent columns with x and y points merge: the y-coordinates of the two columns are shuffled, and each shuffle contributes its sign.

Lemma 36.3 (Signed shuffles)**.** $\sum_w (-1)^{\mathrm{inv}(w)} = P(x,y)$*, where w runs over the words with x letters a and y letters b and* $\mathrm{inv}(w)$ *is the number of pairs (b before a).*

Proof. Pair positions $(1,2),(3,4),\dots$. If w has a pair with two different letters, swap the letters of the first such pair: this changes inv by exactly ±1 and is an involution (the first such pair is unchanged), so these words cancel. The remaining words consist of blocks aa and bb (and one final single letter if $x+y$ is odd). If x and y are both odd there are none. Otherwise they correspond to arrangements of $\lfloor x/2\rfloor$ blocks aa and $\lfloor y/2\rfloor$ blocks bb; each crossing of a bb block before an aa block contributes 4 inversions, and the final single letter is preceded by complete blocks of the other letter, contributing an even number. So each surviving word has sign $+1$ and there are $\binom{\lfloor x/2\rfloor+\lfloor y/2\rfloor}{\lfloor x/2\rfloor}$ of them. □

Example 36.4 (Br_4 by hand)**.** $A_4^0 = \langle[1,1,1,1]\rangle$, $A_4^1 = \langle[2,1,1],[1,2,1],[1,1,2]\rangle$, $A_4^2 = \langle[2,2],[3,1],[1,3]\rangle$, $A_4^3 = \langle[4]\rangle$. Using $P(1,1) = P(1,3) = P(3,1) = 0$, $P(1,2) = P(2,1) = 1$ and $P(2,2) = 2$:

$$\delta[1,1,1,1] = 0, \quad \delta[2,1,1] = [3,1], \quad \delta[1,2,1] = [3,1] - [1,3], \quad \delta[1,1,2] = -[1,3],$$

$$\delta[2,2] = 2[4], \qquad \delta[3,1] = \delta[1,3] = 0.$$

So $H^0 = \mathbb{Z}$; $\ker\delta^1$ is spanned by $[2,1,1] - [1,2,1] + [1,1,2]$, so $H^1 = \mathbb{Z}$; $\ker\delta^2 = \langle[3,1],[1,3]\rangle = \operatorname{im}\delta^1$, so $H^2 = 0$; and $H^3 = \mathbb{Z}/2$, generated by $[4]$. In Smith normal form (Theorem 4.18), $\delta^2 = (2\ 0\ 0)$: the torsion coefficient 2 is $P(2,2)$. Thus $H^*(\mathrm{Br}_4;\mathbb{Z}) = (\mathbb{Z},\mathbb{Z},0,\mathbb{Z}/2)$, and by the universal coefficient theorem $H_*(\mathrm{Br}_4;\mathbb{Z}) = (\mathbb{Z},\mathbb{Z},\mathbb{Z}/2,0)$: the Schur multiplier of Br_4 is $\mathbb{Z}/2$.

Theorem 36.5 (Arnold; Fuks; Vainshtein)**.** *(a) $H^0(\mathrm{Br}_n;\mathbb{Z}) = \mathbb{Z}$, $H^1(\mathrm{Br}_n;\mathbb{Z}) = \mathbb{Z}$ ($n \geq 2$), all other groups are finite, and $H^i(\mathrm{Br}_n;\mathbb{Z}) = 0$ for $i \geq n$.*

(b) $H^i(\mathrm{Br}_{2m+1};\mathbb{Z}) = H^i(\mathrm{Br}_{2m};\mathbb{Z})$, and $H^i(\mathrm{Br}_n;\mathbb{Z})$ is independent of n for $n \geq 2i-2$.

(c) (Fuks) $\bigoplus_{n,d} H_d(B_n(\mathbb{R}^2);\mathbb{F}_2)$ is the polynomial algebra $\mathbb{F}_2[x_0,x_1,x_2,\dots]$, x_i of degree 2^i-1 and weight $n = 2^i$, the product being induced by juxtaposition of configurations.

(d) (Vainshtein) For $r \geq 2$, $H^r(\mathrm{Br}_n;\mathbb{Z})$ has no elements of order p^2 for any prime p: it is a direct sum of groups $\mathbb{Z}/p$, namely $\bigoplus_p \widetilde{\beta}_p H^{r-1}(\mathrm{Br}_n;\mathbb{F}_p)$.

Parts (a)–(b) are Arnold's [61], as recorded in [105, §1]; (c) is [83] in the form of [89, Thm. 3.1]; (d) is stated by Vainshtein and proved in [105, Thm. 2.11]. In (a), the vanishing above degree $n-1$ is visible in the complex: $A_n^r = 0$ for $r \geq n$. Part (b) is visible too: $A_n^\bullet$ splits as a direct sum of subcomplexes according to the number of odd entries, and inserting 1's identifies the summands with complexes for smaller n [105, §2.3].

Remark 36.6 (Bocksteins). Part (d) says that the Bockstein spectral sequence (Theorem 9.22) of Br_n collapses at E_2 for every prime: Bockstein cohomology $\ker\beta/\operatorname{im}\beta$ of $H^*(\mathrm{Br}_n;\mathbb{F}_p)$ has dimension equal to the rank of $H^*(\mathrm{Br}_n;\mathbb{Z})$ (which is 1 in degrees 0 and 1). This is the criterion [1, Cor. 3E.4] for the absence of elements of order p^2. In contrast, $H^2(B_3(S^2);\mathbb{Z}) = \mathbb{Z}/4$ (below): for configuration spaces of closed surfaces, higher Bocksteins do occur.

36.3 Machine computation

The program `code/fuks.py` builds $A_n^\bullet$, checks $\delta^2 = 0$, and computes its cohomology by Smith normal form (`code/snf.py`, Corollary 4.19). Its output for $n \le 11$:

$n\backslash i$	0	1	2	3	4	5	6	7	8	9
2,3	$\mathbb{Z}$	$\mathbb{Z}$								
4,5	$\mathbb{Z}$	$\mathbb{Z}$	0	$\mathbb{Z}/2$						
6,7	$\mathbb{Z}$	$\mathbb{Z}$	0	$\mathbb{Z}/2$	$\mathbb{Z}/2$	$\mathbb{Z}/3$				
8,9	$\mathbb{Z}$	$\mathbb{Z}$	0	$\mathbb{Z}/2$	$\mathbb{Z}/2$	$\mathbb{Z}/6$	$\mathbb{Z}/3$	$\mathbb{Z}/2$		
10,11	$\mathbb{Z}$	$\mathbb{Z}$	0	$\mathbb{Z}/2$	$\mathbb{Z}/2$	$\mathbb{Z}/6$	$\mathbb{Z}/6$	$\mathbb{Z}/2$	$\mathbb{Z}/2$	$\mathbb{Z}/5$

$H^i(\mathrm{Br}_n;\mathbb{Z})$ (blank: 0).

This agrees entry by entry with [105, Table 3]. One reads off: the first 3-torsion is at $n = 6$, the first 5-torsion at $n = 10$, i.e. at $n = 2p$ (compare the weight $2p^i$ of the Fuks–Vainshtein generators); every torsion summand has prime order (Theorem 36.5(d)); stability holds for $n \ge 2i - 2$; and $n = 2m + 1$ repeats $n = 2m$.

36.4 The sphere

Napolitano [101] extended the Fuks complex to closed surfaces. For $S^2 = \mathbb{C} \cup \{\infty\}$, separating configurations that contain ∞ from those that do not gives a complex $B_n^r = A_n^r \oplus A_{n-1}^{r-2}$ with differential $(a, b) \mapsto (\delta a,\ \delta b + (-1)^{n-r} D a)$, where

$$D[n_1,\dots,n_s] = \sum_i Q(n_i)(-1)^{n_1+\cdots+n_{i-1}}[n_1,\dots,n_i-1,\dots,n_s], \quad Q(m) = \begin{cases} 0 & m \text{ odd}, \\ 2 & m \text{ even} \end{cases}$$

(terms with $n_i - 1 = 0$ are dropped); D commutes with δ, and the complex is the mapping cone of D [105, §3]. Since $D \equiv 0 \bmod 2$, $H^r(B_n(S^2);\mathbb{F}_2) \cong H^r(B_n(\mathbb{C});\mathbb{F}_2) \oplus H^{r-2}(B_{n-1}(\mathbb{C});\mathbb{F}_2)$. `code/fuks.py` computes:

$n\backslash i$	0	2	3	4	5	6	7	8	9
2	$\mathbb{Z}$	$\mathbb{Z}/2$							
3	$\mathbb{Z}$	$\mathbb{Z}/4$	$\mathbb{Z}$						
4	$\mathbb{Z}$	$\mathbb{Z}/6$	$\mathbb{Z}\oplus\mathbb{Z}/2$						
5	$\mathbb{Z}$	$\mathbb{Z}/8$	$\mathbb{Z}\oplus\mathbb{Z}/2$	0	$\mathbb{Z}/2$				
6	$\mathbb{Z}$	$\mathbb{Z}/10$	$\mathbb{Z}\oplus\mathbb{Z}/2$	$\mathbb{Z}/2$	$\mathbb{Z}/6$				
7	$\mathbb{Z}$	$\mathbb{Z}/12$	$\mathbb{Z}\oplus\mathbb{Z}/2$	$\mathbb{Z}/2$	$\mathbb{Z}/6$	$\mathbb{Z}/2$	$\mathbb{Z}/3$		
8	$\mathbb{Z}$	$\mathbb{Z}/14$	$\mathbb{Z}\oplus\mathbb{Z}/2$	$\mathbb{Z}/2$	$\mathbb{Z}/2\oplus\mathbb{Z}/6$	$\mathbb{Z}/2$	$\mathbb{Z}/2$		
9	$\mathbb{Z}$	$\mathbb{Z}/16$	$\mathbb{Z}\oplus\mathbb{Z}/2$	$\mathbb{Z}/2$	$\mathbb{Z}/2\oplus\mathbb{Z}/6$	$\mathbb{Z}/2$	$(\mathbb{Z}/2)^2$	$\mathbb{Z}/3$	$\mathbb{Z}/2$

$H^i(B_n(S^2);\mathbb{Z})$; $H^1 = 0$ for all n; blank: 0.

(Also computed for $n = 10, 11$; all agree with [105, Table 4].) Note $H^2 = \mathbb{Z}/(2n-2)$, which by the universal coefficient theorem is equivalent to $H_1 = \mathbb{Z}/(2n-2)$ (Exercise 33.25); this group has elements of order p^2 whenever $p^2 \mid 2n-2$, so Theorem 36.5(d) fails for S^2 in degree 2. Schiessl proves that for $r \ge 4$ the groups $H^r(B_n(S^2);\mathbb{Z})$ are finite with no elements of order p^2 [105, Thm. 1.3].

36.5 The torus: what is known, and of what kind

For T^2 the rational Betti numbers are known (Theorem 35.4), and $B_k(T^2)$ is aspherical (Theorem 33.13), so its homology is that of the torus braid group. The integral picture is incomplete. It is essential to distinguish what kind of statement each result is.

Statement	Kind	Source / status
$H_1(B_k(T^2);\mathbb{Z}) = \mathbb{Z}^2 \oplus \mathbb{Z}/2$, $k \geq 2$	integral, one degree, all k	Exercise 33.26; proved
$H^*(B_k(T^2);\mathbb{Z})$ for $k \leq 6$; all torsion is 2-primary; e.g. $H^5(B_6(T^2)) = \mathbb{Z}^9 \oplus (\mathbb{Z}/2)^8$	complete integral groups, finitely many k	Napolitano [101], as quoted in [88]
$\dim H_*(B_k(T^2);\mathbb{F}_2)$ for all k	mod-2 dimensions; does not determine exponents	Bödigheimer–Cohen–Taylor [63]
$\dim H_i(B_k(T^2);\mathbb{F}_p) = b_i$ for odd p, $k \leq p$	absence of p-power torsion, all degrees, $k \leq p$	Chen–Zhang [71]
no p-torsion for $p < k < 2p$ (p odd)	absence, all degrees, a range of k	Idrissi–Roca i Lucio [88], preprint (2026)
$H_{2p-2}(B_{2p}(T^2);\mathbb{Z})$ has p-torsion iff $p \geq 5$	existence of one class, one degree, one k per p	same preprint
H_{2p}, H_{2p+1} of $B_{2p}(T^2)$ have no p-torsion	absence, two degrees	same preprint

Points to note.

- A mod-p dimension count determines the *number* of cyclic p-primary summands of $H_*(\,\cdot\,;\mathbb{Z})$ (given the Betti numbers), but not their *orders*: $\mathbb{Z}/p$ and $\mathbb{Z}/p^2$ contribute equally. Exponent statements need Bocksteins (Remark 36.6).
- "No p-torsion for $k \leq p$" is a statement about all degrees but a bounded range of k; Theorem B of [88] concerns a single degree at the threshold. Neither says anything about elementary decompositions in general.
- The torsion of the once-punctured torus is better understood: Bianchi–Stavrou [67] compute $H_*(B_k(\Sigma_{g,1});\mathbb{F}_p)$ for odd p. Filling the puncture can kill torsion: by [88], the $\mathbb{Z}/3$ in $H_4(B_6(\Sigma_{1,1}))$ dies in the closed torus, while for $p \geq 5$ the corresponding $\mathbb{Z}/p$ survives.
- The prediction that the first p-torsion occurs at $k = 2p$ (made in the arXiv version of [71]) is therefore false for $p = 3$ and, by [88], true for $p \geq 5$. Status as of October 2026; the 2026 results are in a preprint that had not appeared in a journal at that date.

36.6 Limits of the methods

- *Rational models* (Chapter 37) see no torsion at all.
- *Cell complexes* of Fuks–Napolitano type compute integral cohomology exactly, but their size grows exponentially (2^{n-1} cells for $B_n(\mathbb{C})$); Napolitano's torus computations stop at $k = 6$.

- *Mod-p methods* (Bianchi–Stavrou; the point-set models of Idrissi–Roca i Lucio with their `uconf` package) give dimensions, and Bocksteins require extra work.
- *Homological stability* (Chapter 39) fixes the degree and lets $k \to \infty$; the torsion discussed above sits on the "diagonal" (degree growing with k), beyond its reach.

36.7 Exercises

Exercise 36.7 (medium)**.** Compute $H^*(\mathrm{Br}_5;\mathbb{Z})$ by hand from Theorem 36.2 and confirm the recurrence of Theorem 36.5(b).

Source: original

Solution. Cells: length 5: $[1^5]$; length 4: the four compositions with one 2; length 3: six with two 2's or one 3 (namely $[2,2,1],[2,1,2],[1,2,2],[3,1,1],[1,3,1],[1,1,3]$); length 2: $[4,1],[1,4],[3,2],[2,3]$; length 1: $[5]$. Since $P(1,1)=0$, $\delta[1^5]=0$: $H^0=\mathbb{Z}$. On length 4 the only nonzero coefficients are $P(2,1)=P(1,2)=1$, mapping into $[3,1,1],[1,3,1],[1,1,3]$; the alternating sum $[2,1,1,1]-[1,2,1,1]+[1,1,2,1]-[1,1,1,2]$ is a cycle and spans the kernel, so $H^1=\mathbb{Z}$. On length 3: $\delta[2,2,1]=2[4,1]-[2,3]$, $\delta[2,1,2]=[3,2]-[2,3]$, $\delta[1,2,2]=[3,2]-2[1,4]$, $\delta[3,1,1]=0$, $\delta[1,3,1]=0$, $\delta[1,1,3]=0$ (using $P(3,1)=P(1,3)=P(1,1)=0$, $P(2,1)=P(1,2)=1$, $P(2,2)=2$). On length 2, with $P(4,1)=P(1,4)=1$ and $P(3,2)=P(2,3)=\binom{2}{1}=2$: $\delta[4,1]=\delta[1,4]=[5]$ and $\delta[3,2]=\delta[2,3]=2[5]$. So δ^3 is onto and $H^4=0$. $\ker\delta^3$ has basis $u_1=[4,1]-[1,4]$, $u_2=[3,2]-2[1,4]$, $u_3=[2,3]-2[1,4]$, and the image of δ^2 is spanned by $2[4,1]-[2,3]=2u_1-u_3$, $[3,2]-[2,3]=u_2-u_3$ and $[3,2]-2[1,4]=u_2$, i.e. by $u_2,u_3,2u_1$: so $H^3=\mathbb{Z}/2$. The three nonzero images under δ^2 are independent, so $\ker\delta^2=\langle[3,1,1],[1,3,1],[1,1,3]\rangle$, and this is the image of δ^1 ($\delta[2,1,1,1]=[3,1,1]$, $\delta[1,1,1,2]=[1,1,3]$, $\delta[1,2,1,1]=[3,1,1]-[1,3,1]$): $H^2=0$. Result $(\mathbb{Z},\mathbb{Z},0,\mathbb{Z}/2,0)$, equal to $H^*(\mathrm{Br}_4)$, as the program also reports.

Exercise 36.8 (medium)**.** Using Theorem 36.5(c), list $\dim H_d(\mathrm{Br}_6;\mathbb{F}_2)$ for all d, and check it against the integral table via the universal coefficient theorem.

Source: original

Solution. Monomials of weight 6 in x_0 (weight 1, degree 0), x_1 (weight 2, degree 1), x_2 (weight 4, degree 3): x_0^6 ($d=0$), $x_0^4x_1$ (1), $x_0^2x_1^2$ (2), x_1^3 (3), $x_0^2x_2$ (3), x_1x_2 (4). So the dimensions are $1,1,1,2,1$ in degrees 0–4. Integrally, $H^*(\mathrm{Br}_6)=(\mathbb{Z},\mathbb{Z},0,\mathbb{Z}/2,\mathbb{Z}/2,\mathbb{Z}/3)$, so $H_*=(\mathbb{Z},\mathbb{Z},\mathbb{Z}/2,\mathbb{Z}/2,\mathbb{Z}/3,0)$ (torsion shifts down by one). Mod 2: $H_d\otimes\mathbb{F}_2\oplus\mathrm{Tor}(H_{d-1},\mathbb{F}_2)$ gives $1,1,1,1+1,0+1,0$: i.e. $1,1,1,2,1,0$. They agree.

Exercise 36.9 (medium)**.** From the table, find the degrees in which $H^*(\mathrm{Br}_6;\mathbb{F}_3)$ is nonzero, and identify the Bockstein $\beta\colon H^4\to H^5$.

Source: original

Solution. By the universal coefficient theorem $H^r(\,\cdot\,;\mathbb{F}_3)=H^r\otimes\mathbb{F}_3\oplus\mathrm{Tor}(H^{r+1},\mathbb{F}_3)$: $\mathbb{F}_3$ in degrees $0,1$ (from $\mathbb{Z}$), 4 (from $\mathrm{Tor}(\mathbb{Z}/3,\mathbb{F}_3)$) and 5 (from $\mathbb{Z}/3\otimes\mathbb{F}_3$). The integral class of order 3 in H^5 is $\widetilde{\beta}$ of the degree-4 class, so $\beta=\rho\widetilde{\beta}\colon H^4\to H^5$ is an isomorphism $\mathbb{F}_3\to\mathbb{F}_3$. Bockstein cohomology is $\mathbb{F}_3$ in degrees $0,1$, which equals the rank: the Bockstein spectral sequence collapses at E_2.

Exercise 36.10 (easy)**.** Show without the Fuks complex that $H_*(\mathrm{Br}_k;\mathbb{Z})$ has no p-torsion for $p>k$, and that $H_*(\mathrm{Br}_k;\mathbb{Z})\otimes\mathbb{Z}[1/k!]$ is $\mathbb{Z}[1/k!]$ in degrees $0,1$ and zero otherwise (using Theorem 34.7).

SOURCE: original

Solution. $H_*(F_k(\mathbb{R}^2);\mathbb{Z})$ is free (Theorem 34.3), so Proposition 36.1(b) applies. After inverting $k!$, the transfer identifies $H_*(\mathrm{Br}_k)[1/k!]$ with the coinvariants of $H_*(P_k)[1/k!]$, a free $\mathbb{Z}[1/k!]$-module whose rank is the rational dimension: 1 in degrees 0 and 1 by Theorem 34.7. (The Fuks complex shows the sharper fact that p-torsion starts at $k = 2p$.)

Exercise 36.11 (medium)**.** Explain the row $n = 3$ of the sphere table: $H^*(B_3(S^2);\mathbb{Z}) = (\mathbb{Z}, 0, \mathbb{Z}/4, \mathbb{Z})$.

SOURCE: original

Solution. $H_1 = \mathbb{Z}/4$ by Exercise 33.25 ($2k-2=4$), so $H^1 = \mathrm{Hom}(H_1,\mathbb{Z}) = 0$ and $H^2 \supseteq \mathrm{Ext}(H_1,\mathbb{Z}) = \mathbb{Z}/4$. To see the rest, identify $F_3(S^2)$ with $PGL_2(\mathbb{C})$ via $g \mapsto (g(1), g(\zeta), g(\zeta^2))$, $\zeta = e^{2\pi i/3}$. Permuting the three points is right multiplication by the subgroup $S \cong \mathfrak{S}_3$ of rotations of the sphere permuting $1, \zeta, \zeta^2$, which lies in $PSU(2) \cong SO(3)$. The Iwasawa decomposition $PGL_2(\mathbb{C}) = AN \cdot PSU(2)$ (with AN contractible) is compatible with right multiplication by S, so $B_3(S^2) \cong AN \times \big(SO(3)/S\big) \simeq SO(3)/S$, a closed orientable 3-manifold. Hence $H^3 = \mathbb{Z}$, and by Poincaré duality $H^2 \cong H_1 = \mathbb{Z}/4$.

Exercise 36.12 (hard)**.** Show from the description of the sphere complex that $H^r(B_n(S^2);\mathbb{F}_2) \cong H^r(\mathrm{Br}_n;\mathbb{F}_2) \oplus H^{r-2}(\mathrm{Br}_{n-1};\mathbb{F}_2)$, and verify it for $n = 3$, $r = 2, 3$.

SOURCE: original; the statement is [105, Cor. 3.1]

Solution. Mod 2 the operator D vanishes ($Q(m) \in \{0, 2\}$), so $B_n^\bullet \otimes \mathbb{F}_2 = (A_n^\bullet \oplus A_{n-1}^\bullet[-2]) \otimes \mathbb{F}_2$ as complexes, and the cohomology splits. For $n = 3$: $H^*(\mathrm{Br}_3;\mathbb{F}_2) = \mathbb{F}_2$ in degrees $0, 1$ and $H^*(\mathrm{Br}_2;\mathbb{F}_2) = \mathbb{F}_2$ in degrees $0, 1$; so $H^2(B_3(S^2);\mathbb{F}_2) = 0 \oplus \mathbb{F}_2$, $H^3 = 0 \oplus \mathbb{F}_2$. From the integral groups $(\mathbb{Z}, 0, \mathbb{Z}/4, \mathbb{Z})$: $H^2(\,;\mathbb{F}_2) = \mathbb{Z}/4 \otimes \mathbb{F}_2 \oplus \mathrm{Tor}(H^3,\mathbb{F}_2) = \mathbb{F}_2$ and $H^3(\,;\mathbb{F}_2) = \mathbb{F}_2 \oplus \mathrm{Tor}(H^4,\mathbb{F}_2) = \mathbb{F}_2$. They agree. Note that the mod-2 answer cannot distinguish $\mathbb{Z}/4$ from $\mathbb{Z}/2$.

36.8 Connections and further reading

Fuks [83] and Vainshtein [110] for braid groups; Napolitano [101] for surfaces; Schiessl [105] for a complete integral answer for S^2; Bianchi–Stavrou [67], Chen–Zhang [71] and Idrissi–Roca i Lucio [88] for current work on odd torsion. Bendersky–Miller [66] give conditions under which the p-torsion of $H_i(B_k(M))$ is independent of k (Chapter 39).

Chapter 37

Rational and real models of configuration spaces

37.1 What a model should do

A (rational) model of a space X is a commutative differential graded algebra quasi-isomorphic to $A_{\mathrm{PL}}(X)$ (Chapter 24); over $\mathbb{R}$ one may use the de Rham algebra. A model of $F_k(M)$ built *functorially from a model of* M proves that the rational (or real) homotopy type of $F_k(M)$ depends only on that of M — a rational answer to the homotopy invariance question of Chapter 38 — and makes computations possible. The three main results are: Kriz's model for projective varieties, the Lambrechts–Stanley model (proved by Idrissi), and the graph-complex models of Campos–Willwacher.

37.2 Formality of $F_k(\mathbb{R}^n)$

Theorem 37.1. *$F_k(\mathbb{R}^n)$ is formal: $A_{\mathrm{PL}}(F_k(\mathbb{R}^n))$ is quasi-isomorphic to its cohomology, the Arnold algebra.*

For $n = 2$ this goes back to Arnold's observation that the forms $\frac{1}{2\pi i} d\log(z_j - z_i)$ satisfy the Arnold relation *on the nose* (Exercise 34.15): the subalgebra of closed forms they generate is a sub-cdga of the de Rham algebra (with zero differential) mapping isomorphically to cohomology. Hence the inclusion is a quasi-isomorphism from H^* to the forms — a formality quasi-isomorphism over $\mathbb{C}$, which descends to $\mathbb{Q}$ [34]. For general n the same works with suitable forms on the Fulton–MacPherson compactification (Kontsevich, Lambrechts–Voli; Chapter 40). The following weaker consequence is elementary.

Corollary 37.2. *All Massey products in $H^*(F_k(\mathbb{R}^n);\mathbb{Q})$ vanish (contain zero).*

Proof. Proposition 31.9 applied to Theorem 37.1. For $n = 2$ directly: in the sub-cdga generated by the forms ω_{ij}, a product of representatives that is zero in cohomology is zero on the nose (the subalgebra maps isomorphically to H^*), so the defining systems can be chosen to be 0. □

37.3 Kriz's model and two points

Example 37.3 (Kriz's model in the smallest case)**.** For $k = 2$ and M a smooth projective variety of complex dimension m, Kriz's model (Theorem 34.13) is $E =$

$H^*(M)^{\otimes 2} \otimes \Lambda(G)$ with $|G| = 2m-1$, $G\cdot(x\otimes 1 - 1\otimes x) = 0$, $dG = \Delta$. As a complex, $E = H^*(M\times M) \oplus H^*(M\times M)/(x\otimes 1 - 1\otimes x)\cdot G \cong H^*(M\times M) \oplus H^*(M)\,G$, and $d(xG) = (x\otimes 1)\Delta$. Its cohomology is $H^*(M\times M)/(\Delta) \oplus \ker\big(x \mapsto (x\otimes 1)\Delta\big)G$, and the kernel is zero: this is Theorem 34.9 over $\mathbb{Q}$. For $M = \mathbb{CP}^m$ one recovers Example 34.10(c).

Theorem 37.4 (Lambrechts–Stanley, two points)**.** *If M is a closed 2-connected manifold, the rational homotopy type of $F_2(M)$ depends only on the rational homotopy type of M, and there is an explicit Sullivan model of $F_2(M)$ built from a Poincaré duality model of M.*

See [95]; the survey [89, §5.2(3)] records that the conclusion also holds for simply connected even-dimensional closed manifolds, by later work. The model is the chain-level version of Theorem 34.9: one needs a model A of M satisfying Poincaré duality *on the nose*, so that the diagonal class $\Delta_A \in A\otimes A$ makes sense; in the notation of Construction 37.5, $G_A(2)$ is the mapping cone of the injective map $a \mapsto (a\otimes 1)\Delta_A$ and is therefore quasi-isomorphic to $A\otimes A/(\Delta_A)$, exactly as in the proof of Theorem 34.9.

37.4 The Lambrechts–Stanley model

Construction 37.5. Let A be a Poincaré duality cdga (a finite-dimensional cdga with an orientation $\varepsilon\colon A^m \to \mathbb{Q}$ making $A^i \otimes A^{m-i} \to \mathbb{Q}$ nondegenerate) and $\Delta_A \in (A\otimes A)^m$ its diagonal class. The *Lambrechts–Stanley cdga* is

$$G_A(k) = \Big(A^{\otimes k} \otimes H^*(F_k(\mathbb{R}^m))\Big)\Big/\big(p_i^*(a)\,\omega_{ij} = p_j^*(a)\,\omega_{ij}\big), \qquad d\omega_{ij} = p_{ij}^*(\Delta_A),$$

with the differential of $A^{\otimes k}$ on the first factor. It is the Totaro–Kriz E_m-page (Theorem 34.12) with $H^*(M)$ replaced by A.

Theorem 37.6 (Idrissi; conjectured by Lambrechts–Stanley)**.** *Let M be a smooth, simply connected, closed manifold and A a Poincaré duality model of M (over $\mathbb{R}$; one exists by Lambrechts–Stanley). Then for every k, $G_A(k)$ is a model of the real homotopy type of $F_k(M)$. In particular the real homotopy type of $F_k(M)$ depends only on the real homotopy type of M. If moreover M is framed and $\dim M \geq 4$, the models are compatible with the action of the Fulton–MacPherson (little disks) operad.*

See [86, Thm. 1, Cor. 2, Thm. 3]; in dimension ≤ 3 the only such M are spheres, treated directly there. Lambrechts and Stanley conjectured the theorem [94]; for $k = 2$ it was known for M 2-connected (Theorem 37.4) or even-dimensional. The result is over $\mathbb{R}$: as Idrissi points out, there is no general descent argument to $\mathbb{Q}$ for models with nonzero differential, and the rational statement is conjectural. The proof goes through Kontsevich's formality of the little disks operad and graph complexes (Chapter 40). Campos–Willwacher [70] construct graph-complex models valid more generally (including non-simply-connected cases, where a Poincaré duality model need not exist), at the cost of being larger.

Remark 37.7. These theorems are over $\mathbb{Q}$ or $\mathbb{R}$ and for simply connected M. They are compatible with Longoni–Salvatore (Theorem 38.4), whose lens spaces are not simply connected. Whether the *integral* homotopy type of $F_k(M)$ depends only on that of M for simply connected closed M is open (Chapter 42).

37.5 Massey products and the spectral sequence

The higher differentials of the Totaro–Cohen–Taylor spectral sequence, when M is not formal, are related to Massey products in $H^*(M)$.

Theorem 37.8 (Félix–Thomas)**.** *Let $\mathbb{F}$ be a field, M an oriented closed connected manifold.*

(a) The E_2-terms of the Cohen–Taylor–Totaro spectral sequence for $H^*(F_k(M);\mathbb{F})$ and of the Bendersky–Gitler spectral sequence (dualised and suspended) are isomorphic as $H^*(M)^{\otimes k}$-modules; one collapses at E_2 iff the other does.

(b) If $k \le 3$, both collapse at E_2.

(c) If $\operatorname{char}\mathbb{F} = 0$, M is simply connected, and there are indecomposable $a,b,c,d \in H^*(M;\mathbb{F})$ with all pairwise products zero, the triple Massey product $\langle b,c,d\rangle$ a nonzero indecomposable element, and a not in the span of $b,c,d,\langle b,c,d\rangle$, then the Cohen–Taylor–Totaro spectral sequence for $F_4(M)$ does not collapse at E_2.

See [81] (Theorems 1–3 there). Part (c) answers a question of Totaro, who asked for a manifold with a nonzero higher differential.

37.6 Exercises

Exercise 37.9 (medium)**.** Write down $G_A(2)$ for $M = S^m$ with $A = H^*(S^m) = \Lambda(x)/(x^2)$ (a Poincaré duality model, as S^m is formal) and compute its cohomology. Compare with $F_2(S^m) \simeq S^m$.

Source: original

Solution. $G_A(2)$ is spanned by $1, x_1, x_2, x_1x_2$ and $\omega, x_1\omega = x_2\omega, x_1x_2\omega = x_1^2\omega = 0$; so additively $\langle 1, x_1, x_2, x_1x_2\rangle \oplus \langle \omega, x_1\omega\rangle$, with $|\omega| = m-1$. The diagonal class is $\Delta = x_1 + (-1)^m x_2$ (up to sign), so $d\omega = \Delta$ and $d(x_1\omega) = x_1\Delta = (-1)^m x_1x_2$. The differential is injective on $\langle \omega, x_1\omega\rangle$, and the cohomology is $\langle 1, x_1, x_2, x_1x_2\rangle / \langle x_1 + (-1)^m x_2, x_1x_2\rangle$: one class in degree 0 and one in degree m. This is $H^*(S^m)$.

Exercise 37.10 (medium)**.** Prove Theorem 37.1 for $n = 2$ over $\mathbb{C}$ in detail: show that the $\mathbb{C}$-subalgebra $\mathcal{A}$ of the de Rham algebra of $F_k(\mathbb{C})$ generated by the forms $\eta_{ij} = \frac{1}{2\pi i} d\log(z_j - z_i)$ has zero differential, and that $\mathcal{A} \to H^*(F_k(\mathbb{C});\mathbb{C})$ is an isomorphism.

Source: original

Solution. Each η_{ij} is closed, so $d = 0$ on $\mathcal{A}$ (Leibniz). The map $\mathcal{A} \to H^*$ is surjective since the η_{ij} represent the ω_{ij}, which generate. The η_{ij} satisfy $\eta_{ij}^2 = 0$ (odd degree) and the Arnold relation on the nose (Exercise 34.15), so $\mathcal{A}$ is a quotient of the Arnold algebra A, and $A \to \mathcal{A} \to H^*$ is the isomorphism of Theorem 34.3. Hence $\mathcal{A} \to H^*$ is injective too. The inclusion $(\mathcal{A}, 0) \hookrightarrow \Omega^*(F_k(\mathbb{C});\mathbb{C})$ is then a quasi-isomorphism from $(H^*, 0)$.

Exercise 37.11 (hard)**.** Let $p\colon X \to Y$ be a map with a section $s\colon Y \to X$ ($ps = \mathrm{id}$), and let $\langle a,b,c\rangle$ be a triple Massey product in $H^*(Y)$ not containing 0. Show that $\langle p^*a, p^*b, p^*c\rangle$ does not contain 0. Explain how Longoni–Salvatore use this for the forgetful map $\widetilde{F}_n(L_{7,2}) \to \widetilde{F}_2(L_{7,2})$.

SOURCE: original

Solution. Massey products are natural: for any map f, $f^*\langle u,v,w\rangle \subseteq \langle f^*u, f^*v, f^*w\rangle$ (apply f^* to a defining system). Note that the right side can be strictly larger, since its indeterminacy involves all of the cohomology of the source; so pulling back a nontrivial Massey product along an arbitrary map can produce one containing 0. With a section: if $0 \in \langle p^*a, p^*b, p^*c\rangle$, then $0 = s^*0 \in s^*\langle p^*a, p^*b, p^*c\rangle \subseteq \langle s^*p^*a, s^*p^*b, s^*p^*c\rangle = \langle a,b,c\rangle$, a contradiction. This is the argument of [98, §5]: the forgetful map $\widetilde{F}_n(L_{7,2}) \to \widetilde{F}_2(L_{7,2})$ has a section, so the nontrivial Massey product on $\widetilde{F}_2$ pulls back to a nontrivial one on $\widetilde{F}_n$.

37.7 Connections and further reading

Kriz [92], Lambrechts–Stanley [94, 95], Idrissi [86] and Campos–Willwacher [70] are the main papers; Idrissi's lecture notes [87] are an introduction to real models of configuration spaces through operads. Félix–Thomas [81] relate non-formality of M to the spectral sequence.

Chapter 38

Homotopy invariance and its failure

38.1 The question

Question 38.1. *If M and N are homotopy equivalent closed manifolds, are $F_k(M)$ and $F_k(N)$ homotopy equivalent?*

For non-compact manifolds the answer is obviously no ($F_2(\mathbb{R})$ versus $F_2(\mathbb{R}^2)$), and configuration spaces are not functorial for non-injective maps (Caution 33.4), so there is no reason to expect a positive answer. On the other hand many invariants of $F_k(M)$ *are* homotopy invariants of closed M:

- the Euler characteristic (Proposition 33.19);
- the integral cohomology ring of $F_2(M)$, for M oriented with torsion-free cohomology (Theorem 34.9);
- the rational, resp. real, homotopy type of $F_k(M)$ for smooth projective varieties, resp. simply connected closed smooth manifolds (Theorems 34.13 and 37.6).

The question was described by Levitt as long-standing [98, Conj. 1.1].

38.2 Positive results

Theorem 38.2 (Levitt)**.** *For a closed manifold M, the based loop space $\Omega F_k(M)$ depends only on the homotopy type of M. If M is 2-connected, $F_2(M)$ itself depends only on the homotopy type of M.*

See [97], as quoted in [59, Introduction] and [98, §1].

For an unbased space Y let $\Sigma^j Y = (* \times S^{j-1}) \cup (Y \times D^j)$, glued along $Y \times S^{j-1}$ (up to homotopy, the join of Y and S^{j-1}).

Theorem 38.3 (Aouina–Klein)**.** *Let M and N be homotopy equivalent closed PL manifolds of dimension d, and assume M is r-connected ($r \geq 0$).*

(a) $\Sigma^\alpha F_k(M) \simeq \Sigma^\alpha F_k(N)$ *for* $\alpha = \alpha(k,d,r) = \max\big((k-2)d - r + 3,\, 0\big)$.

(b) If M and N are simple homotopy equivalent, the same holds with $\beta(k,d,r) = \max\big((k-2)d - r + 2,\, 0\big)$.

(c) Consequently the suspension spectrum $\Sigma^\infty F_k(M)_+$ is a homotopy invariant of a connected closed PL manifold.

(d) $\Sigma^3 F_2(M)$ is a homotopy invariant of connected M, and $\Sigma F_2(M)$ is a homotopy invariant of 1-connected M.

See [59, Theorems A, C, Corollaries B, D]. The proof combines Poincaré embedding theory with fibrewise homotopy theory: $F_2(M)$ is the complement of the diagonal, and its homotopy type is governed by the "Poincaré embedding" of the diagonal, which is determined up to suspension by the homotopy type of M. Aouina and Klein mention that Cohen and Taylor, in unpublished work, had obtained stable homotopy invariance for smooth manifolds by other methods, with weaker bounds but applying also to unordered configuration spaces.

38.3 The counterexample

Recall that the lens space $L_{m,q} = S^3/\mathbb{Z}_m$, with the generator acting by $(x_1, x_2) \mapsto (e^{2\pi i/m}x_1, e^{2\pi iq/m}x_2)$, and that $L_{7,1}$ and $L_{7,2}$ are homotopy equivalent but not homeomorphic (Warning 7.19, Exercise 38.8).

Theorem 38.4 (Longoni–Salvatore)**.** *$F_k(L_{7,1})$ and $F_k(L_{7,2})$ are not homotopy equivalent for any $k \geq 2$, and neither are $B_k(L_{7,1})$ and $B_k(L_{7,2})$.*

Outline of the proof, following [98]. Step 1: universal covers. Removing the diagonal (codimension 3) from $L \times L$ does not change π_1 (Exercise 38.7), so $\pi_1 F_2(L_{7,q}) = \mathbb{Z}_7 \times \mathbb{Z}_7$, and the universal cover is the *orbit configuration space* $\widetilde{F}_2(L_{7,q}) = \{(x, y) \in S^3 \times S^3 : x \neq gy \text{ for all } g \in \mathbb{Z}_7\}$. A homotopy equivalence of configuration spaces lifts to one of universal covers, so it suffices to distinguish the covers.

Step 2: the split case. For $q = 1$ the action is left multiplication by the unit complex numbers $\zeta^j \in \mathbb{C} \subset \mathbb{H}$ on $S^3 \subset \mathbb{H}$, and $(x, y) \mapsto (xy^{-1}, y)$ is a homeomorphism $\widetilde{F}_2(L_{7,1}) \cong (S^3 \setminus \mathbb{Z}_7) \times S^3 \simeq \left(\bigvee_6 S^2\right) \times S^3$ (Exercise 38.6). A product of wedges of spheres is formal, so all Massey products vanish (Proposition 31.9).

Step 3: the non-split case. For $q = 2$, projection to the first factor makes $\widetilde{F}_2(L_{7,2})$ a bundle over S^3 with fibre $S^3 \setminus \mathbb{Z}_7 \simeq \bigvee_6 S^2$ and a section; its cohomology is additively the same as for $q = 1$ ($H^2 \cong \mathbb{Z}^6$, $H^3 \cong \mathbb{Z}$, $H^4 = 0$, $H^5 \cong \mathbb{Z}^6$). Longoni and Salvatore compute a triple Massey product of classes in H^2 geometrically, via intersections of explicit submanifolds of $S^3 \times S^3$ Poincaré dual to the classes (the complements of the seven "diagonals" $\Delta_k = \{(x, \zeta^k x)\}$), and show that $\langle a_4, a_1, a_2 + a_6\rangle$ contains $a_2 \cup \iota \neq 0$ and not 0 [98, Thm. 4.1].

Step 4: more points. The forgetful map $\widetilde{F}_k(L_{7,q}) \to \widetilde{F}_2(L_{7,q})$ has a section, so the nontrivial Massey product pulls back to a nontrivial one (Exercise 37.11); for $q = 1$ a similar product decomposition shows all Massey products on H^2 vanish. The unordered spaces have the same universal covers. □

Remark 38.5 (Consistency). The theorem is consistent with everything in the previous section: $L_{7,q}$ are not simply connected (so Theorem 37.6 does not apply), not 2-connected (so Theorem 38.2 does not apply), and by Theorem 38.3(a) $\Sigma^3 F_2(L_{7,1}) \simeq \Sigma^3 F_2(L_{7,2})$: Massey products vanish in suspensions. $L_{7,1}$ and $L_{7,2}$ are not simple homotopy equivalent, so part (b) does not apply.

38.4 Status (October 2026)

According to Kallel's survey (version of June 2025) [89, §5.1]: whether configuration spaces of *simply connected* closed manifolds are homotopy invariant is not known; nor is it known whether there exist homotopy equivalent but non-homeomorphic closed manifolds M, N with $F_k(M) \simeq F_k(N)$ for all k. Stable homotopy invariance is known (Theorem 38.3). We found no later result settling either question. See Chapter 42.

38.5 Exercises

Exercise 38.6 (medium)**.** Prove Step 2: with S^3 the unit quaternions and $\mathbb{Z}_7 \subset S^1 \subset \mathbb{C} \subset \mathbb{H}$ acting by left multiplication, show that $(x, y) \mapsto (xy^{-1}, y)$ is a homeomorphism from $\{(x, y) : x \neq gy \ \forall g \in \mathbb{Z}_7\}$ onto $(S^3 \setminus \mathbb{Z}_7) \times S^3$, and that S^3 minus 7 points is homotopy equivalent to $\bigvee_6 S^2$. Check that left multiplication by $\zeta \in \mathbb{C}$ on $\mathbb{H} = \mathbb{C} \oplus \mathbb{C}j$ is $(x_1, x_2) \mapsto (\zeta x_1, \zeta x_2)$, i.e. the action defining $L_{7,1}$.

Source: [98, Prop. 2.1], details original

Solution. $x \neq \zeta^j y$ iff $xy^{-1} \neq \zeta^j$; the inverse map is $(z, y) \mapsto (zy, y)$. Both maps are continuous (quaternion multiplication and inversion are). S^3 minus a point is $\mathbb{R}^3$; $\mathbb{R}^3$ minus 6 points deformation retracts onto a wedge of 6 spheres. For $q = x_1 + x_2 j$, $\zeta q = \zeta x_1 + \zeta x_2 j$ (as ζ multiplies on the left), so the action is diagonal multiplication by ζ: this is $L_{7,1}$. For $L_{7,2}$ the action $(\zeta x_1, \zeta^2 x_2)$ is not left multiplication by a unit quaternion, and the trick fails.

Exercise 38.7 (medium)**.** Let X be a manifold and $Z \subset X$ a closed submanifold of codimension ≥ 3. Show that $\pi_1(X \setminus Z) \to \pi_1(X)$ is an isomorphism. Deduce $\pi_1 F_2(M) \cong \pi_1(M)^2$ for a manifold M of dimension ≥ 3.

Source: original; standard transversality

Solution. Surjectivity: a loop in X can be perturbed (transversality, $\dim S^1 + \dim Z < \dim X$) to miss Z. Injectivity: a null-homotopy $D^2 \to X$ of a loop in $X \setminus Z$ can be perturbed rel boundary to miss Z, since $2 + \dim Z < \dim X$. For $F_2(M)$, Z is the diagonal, of codimension $\dim M \geq 3$ in $M \times M$.

Exercise 38.8 (medium)**.** Use the classification of lens spaces — $L_{p,q} \simeq L_{p,q'}$ iff $qq' \equiv \pm n^2 \pmod p$ for some n, and $L_{p,q} \cong L_{p,q'}$ iff $q' \equiv \pm q^{\pm 1} \pmod p$ — to check that $L_{7,1} \simeq L_{7,2}$ but $L_{7,1} \ncong L_{7,2}$.

Source: original; the classification is quoted from [1, §2.1 and §3.E], see Warning 7.19

Solution. $qq' = 2$, and $3^2 = 9 \equiv 2 \pmod 7$: homotopy equivalent. The inverses of ± 1 mod 7 are ± 1, and $2 \not\equiv \pm 1$: not homeomorphic.

Exercise 38.9 (easy)**.** Compute $\alpha(k, d, r)$ and $\beta(k, d, r)$ of Theorem 38.3 for $k = 2$, and for $k = 3$, $M = L_{7,q}$. Explain why the result does not contradict Theorem 38.4.

Source: original

Solution. For $k = 2$: $\alpha = \max(3-r, 0)$, $\beta = \max(2-r, 0)$. So 3 suspensions suffice for connected M; for simply connected M (where homotopy equivalences are simple), 1; for 2-connected M, none — which is Levitt's theorem. For $k = 3$, $d = 3$, $r = 0$: $\alpha = 3 - 0 + 3 = 6$. Suspensions kill cup products and Massey products, the invariants used by Longoni–Salvatore, so there is no contradiction.

Exercise 38.10 (medium)**.** Show that $\widetilde{F}_2(L_{7,q}) \to S^3$ (first coordinate) is a fibre bundle with fibre $S^3 \setminus \mathbb{Z}_7$ and a section, and deduce the additive cohomology stated in Step 3 for both q.

Source: original; the statement is in [98, §4]

Solution. The fibre over x is $S^3 \setminus \{\zeta^j \cdot_q x\}$, seven points removed. Local triviality: the group $SU(2)$, or simply the translation argument of Theorem 33.9 applied to the seven points (which move together continuously with x), trivialises it locally. A section: $x \mapsto (x, y(x))$ with $y(x)$ a point close to x but off the orbit, e.g. $y(x) =$ the point at a fixed small distance along a nowhere-vanishing vector field (on S^3, $v(x) = ix$ in quaternions), which is not in the orbit of x for small distance. The base S^3 is simply connected and the fibre is $\simeq \bigvee_6 S^2$, so the Serre spectral sequence has $E_2 = H^*(S^3) \otimes H^*(\bigvee_6 S^2)$, concentrated in degrees $0, 2, 3, 5$; there is no room for differentials ($d_3 \colon E_3^{0,2} \to E_3^{3,0}$ vanishes because of the section, which makes $H^3(S^3) \to H^3(E)$ injective). So $H^0 = \mathbb{Z}$, $H^2 = \mathbb{Z}^6$, $H^3 = \mathbb{Z}$, $H^5 = \mathbb{Z}^6$, others zero.

38.6 Connections and further reading

Levitt [97], Aouina–Klein [59], Longoni–Salvatore [98]; for real and rational invariance, Chapter 37; for the general framework, Kallel [89, §5].

Chapter 39

Homological and representation stability

39.1 Stabilisation for open manifolds

Let M be a connected manifold of dimension $d \geq 2$ which is the interior of a compact manifold $\overline{M}$ with nonempty boundary. Choose a point $E \in \partial\overline{M}$ and a collar. "Pushing the configuration slightly into M and adding a new point near E" defines *stabilisation maps* $s\colon B_k(M) \to B_{k+1}(M)$, well defined up to homotopy. For $M = \mathbb{R}^2$ one can take $\{z_1, \dots, z_k\} \mapsto \{z_1, \dots, z_k, 1 + \max_i \operatorname{Re} z_i\}$, which on fundamental groups is the inclusion $\mathrm{Br}_k \hookrightarrow \mathrm{Br}_{k+1}$.

Theorem 39.1 (McDuff; Segal; Randal-Williams)**.** *For M as above, the map $s_*\colon H_i(B_k(M);\mathbb{Z}) \to H_i(B_{k+1}(M);\mathbb{Z})$ is split injective for all i, and an isomorphism for $2i \leq k$. If $d \geq 3$, s_* is a rational isomorphism for $i \leq k$.*

McDuff proved stability [99], Segal gave a stable range $k \geq 2i + 2$ [107, Appendix], and Randal-Williams [104, Thms. A, B] proved the range above, with labels in any path-connected space. Arnold had proved stability for braid groups (Theorem 36.5(b)).

Example 39.2. The table of Chapter 36 shows $H^i(\mathrm{Br}_k;\mathbb{Z})$ constant for $k \geq 2i-2$, e.g. H^5: $\mathbb{Z}/3$ for $k = 6, 7$, then $\mathbb{Z}/6$ for $k \geq 8$. In homology, $H_4(\mathrm{Br}_k)$ (the torsion of H^5) is $\mathbb{Z}/3$ for $k = 6, 7$ and $\mathbb{Z}/6$ for $k = 8, \dots, 11$: it becomes constant at $k = 8 = 2 \cdot 4$, in accordance with the range $2i \leq k$.

Idea of split injectivity. There is a "transfer" $H_*(B_{k+1}(M)) \to H_*(B_k(M))$ sending a configuration to the sum of all its k-element subsets (a map to a symmetric product, followed by a projection); composing with s_* gives an upper-triangular matrix with 1's on the diagonal in a suitable filtration. See [104, §7], where it is attributed to Nakaoka and Dold. □

Theorem 39.3 (McDuff's scanning theorem)**.** *Let M be a closed manifold and $\dot{T}M \to M$ the fibrewise one-point compactification of its tangent bundle. There is a map $B_k(M) \to \Gamma_k(\dot{T}M)$ to the space of sections of degree k which is a homology isomorphism in the range in which $s\colon B_k(M \setminus \mathrm{pt}) \to B_{k+1}(M \setminus \mathrm{pt})$ is one.*

See [99, Thm. 1.1], as stated in [66, Thm. 2.1]. The map is "scanning": a configuration is viewed through a small window at each point $x \in M$; the window sees either

nothing (the point at infinity of the fibre) or one point (a vector in T_xM). Scanning identifies stable homology with that of a section space, which is computable by obstruction theory and rational homotopy theory (Haefliger's models for section spaces).

39.2 Closed manifolds

For closed M there is no stabilisation map, and integral stability fails: $H_1(B_k(S^2)) = \mathbb{Z}/(2k-2)$ (Exercise 33.25).

Theorem 39.4 (Church; Randal-Williams)**.** *Let M be a connected orientable manifold of finite type (finite-dimensional rational cohomology), $\dim M \geq 2$. Then for each i and $k > i$, $H_i(B_k(M);\mathbb{Q}) \cong H_i(B_{k+1}(M);\mathbb{Q})$, induced by the correspondence $B_{k+1}(M) \Rightarrow B_k(M)$ sending a configuration to all its k-element subsets.*

See [72, Cor. 3]; Randal-Williams gave an independent proof [104, Thm. C], and Bendersky–Miller a third, by localisation [66]. The range $k > i$ is sharp (Theorem 35.4: b_3 of $B_k(T^2)$ is 4 for $k = 3$ and 5 for $k \geq 4$).

Theorem 39.5 (Randal-Williams)**.** *If $\dim M$ is odd and $\mathbb{F}$ is any field, or $\dim M$ is even and $\mathbb{F} \in \{\mathbb{F}_2, \mathbb{Q}\}$, then $\dim_{\mathbb{F}} H_i(B_k(M);\mathbb{F})$ is independent of k in the range in which $H_i(B_k(M \setminus \mathrm{pt});\mathbb{F})$ is stable. This is false in the remaining cases.*

See [104, Thm. C]. In the remaining cases ($\dim M$ even, $\mathbb{F} = \mathbb{F}_p$, p odd) the homology is eventually *periodic* in k rather than stable: by Nagpal, Cantero–Palmer and Kupers–Miller, $H_i(B_k(M);\mathbb{F}_p) \cong H_i(B_{k+p}(M);\mathbb{F}_p)$ for k large compared with i (as recorded in [105, §1.1]; sharpened in [93]). The sphere table of Section 36.4 illustrates this.

Theorem 39.6 (Bendersky–Miller, torsion)**.** *Let M be a smooth connected closed d-manifold, p a prime and $i \leq \min(k/2, j/2)$. The p-torsion subgroups of $H_i(B_k(M);\mathbb{Z})$ and $H_i(B_j(M);\mathbb{Z})$ are isomorphic if one of the following holds: (1) M is parallelisable and k/j is a unit in $\mathbb{Z}_{(p)}$; (2) d is odd and $p \geq d/2 + 3/2$; (3) d is even, $p \geq d/2 + 3/2$ and $(2k - \chi(M))/(2j - \chi(M))$ is a unit in $\mathbb{Z}_{(p)}$; (4) $d = 2$, M orientable and $(2k - \chi(M))/(2j - \chi(M))$ is a unit in $\mathbb{Z}_{(p)}$.*

See [66, Thm. 4.7]. For $M = S^2$ ($\chi = 2$): $(2k-2)/(2j-2)$, so the p-torsion of $H_1(B_k(S^2)) = \mathbb{Z}/(2k-2)$ agrees for k, j with $v_p(k-1) = v_p(j-1)$, as it should.

39.3 Representation stability and FI-modules

Ordered configuration spaces do not stabilise ($H^1(F_k(\mathbb{R}^2)) = \mathbb{Q}^{\binom{k}{2}}$), but their cohomology stabilises *as a representation*. Recall that irreducible $\mathbb{Q}\mathfrak{S}_k$-modules are indexed by partitions of k; for a partition $\lambda = (\lambda_1, \dots, \lambda_\ell)$ of m and $k \geq m + \lambda_1$, let $V(\lambda)_k$ be the irreducible module of the partition $(k-m, \lambda_1, \dots, \lambda_\ell)$. So $V(0)_k$ is trivial, $V(1)_k$ is the standard $(k-1)$-dimensional representation, and so on.

Definition 39.7 (Church–Farb)**.** A sequence of $\mathbb{Q}\mathfrak{S}_k$-modules V_k with maps $\phi_k \colon V_k \to V_{k+1}$ that are $\mathfrak{S}_k$-equivariant (with $\mathfrak{S}_k \subset \mathfrak{S}_{k+1}$) is *uniformly representation stable* for $k \geq N$ if, for $k \geq N$: ϕ_k is injective; the $\mathfrak{S}_{k+1}$-span of $\phi_k(V_k)$ is V_{k+1}; and for every λ the multiplicity of $V(\lambda)_k$ in V_k is independent of k.

Theorem 39.8 (Church)**.** *For a connected orientable manifold M of finite type and each i, the sequence $H^i(F_k(M);\mathbb{Q})$ (with the maps induced by forgetting the last point) is uniformly representation stable for $k \geq 2i$ if $\dim M \geq 3$, and for $k \geq 4i$ if $\dim M = 2$.*

See [72, Thm. 1]. Taking the multiplicity of the trivial representation and using the transfer recovers rational stability for $B_k(M)$ (with a worse range, improved to $k > i$ in [72] by a separate argument).

Definition 39.9. FI is the category of finite sets and injections. An *FI-module* over $\mathbb{Q}$ is a functor $V\colon \mathrm{FI} \to \mathrm{Vect}_{\mathbb{Q}}$; $V_k = V(\{1,\dots,k\})$ is a $\mathbb{Q}\mathfrak{S}_k$-module. V is *finitely generated* if there are finitely many elements whose images under all maps $V(f)$ span V.

An injection $f\colon \{1,\dots,k\} \to \{1,\dots,l\}$ induces $F_l(M) \to F_k(M)$, $(x_j)_j \mapsto (x_{f(i)})_i$, hence $H^i(F_k(M)) \to H^i(F_l(M))$; so $k \mapsto H^i(F_k(M);\mathbb{Q})$ is an FI-module.

Theorem 39.10 (Church–Ellenberg–Farb)**.** *A finitely generated FI-module V over $\mathbb{Q}$ is representation stable; its character is given, for k large, by a polynomial in the cycle-counting functions independent of k; in particular $\dim V_k$ is eventually a polynomial in k. For a connected oriented manifold M (of finite type, $\dim \geq 2$), $H^i(F_k(M);\mathbb{Q})$ is a finitely generated FI-module.*

See [74]. Example: $H^1(F_k(\mathbb{R}^2);\mathbb{Q})$ has dimension $\binom{k}{2}$, a polynomial in k of degree 2; it is the permutation module on 2-subsets, $V(0)_k \oplus V(1)_k \oplus V(2)_k$ for $k \geq 4$ (Exercise 39.11).

39.4 Secondary stability

In the range where $H_i(B_k(M))$ is not yet stable, there is further structure. Miller and Wilson [100] defined *secondary representation stability* for ordered configuration spaces of non-compact manifolds: the stabilising operation adds a pair of orbiting points, relating different homological degrees. Himes [85] proved *secondary homological stability* for unordered configuration spaces of connected manifolds, including closed ones, by constructing a chain-level stabilisation map. These are instances of the secondary stability phenomena discovered by Galatius, Kupers and Randal-Williams for mapping class groups.

39.5 Exercises

Exercise 39.11 (medium)**.** Let P_k be the permutation representation of $\mathfrak{S}_k$ on 2-element subsets of $\{1,\dots,k\}$, $k \geq 4$. Show that $P_k \cong V(0)_k \oplus V(1)_k \oplus V(2)_k$ by computing $\langle \chi_{P_k}, \chi_{P_k} \rangle = 3$ and exhibiting the trivial and standard summands.

Source: original

Solution. $\langle \chi_P, \chi_P \rangle$ is the number of $\mathfrak{S}_k$-orbits on pairs of 2-subsets (A,B), classified by $|A \cap B| \in \{0,1,2\}$: three orbits (for $k \geq 4$, all occur). So P_k is a sum of 3 distinct irreducibles. The sum of all subsets spans a trivial summand. The map $P_k \to \mathbb{Q}^k$, $\{i,j\} \mapsto e_i + e_j$, is equivariant and surjective for $k \geq 3$, and $\mathbb{Q}^k = V(0) \oplus V(1)$; so $V(1)$ occurs. The remaining summand has dimension $\binom{k}{2} - 1 - (k-1) = k(k-3)/2$, the dimension of $V(2)_k$ (partition $(k-2,2)$, by the hook length formula).

Exercise 39.12 (easy)**.** Show that $H_1(B_k(S^2);\mathbb{Z}) \cong H_1(B_{k+1}(S^2);\mathbb{Z})$ for no $k \geq 2$, but that $H_1(B_k(S^2);\mathbb{Q}) = 0$ for all k, in agreement with Theorem 39.4.

SOURCE: original

Solution. $\mathbb{Z}/(2k-2) \ncong \mathbb{Z}/2k$. Both are finite, so rationally 0.

Exercise 39.13 (medium)**.** Show that $\dim H^i(F_k(\mathbb{R}^n);\mathbb{Q})$ is a polynomial in k for each fixed i, of degree $2r$ when $i = r(n-1)$, using Theorem 34.3. Compare with Theorem 39.10.

SOURCE: original

Solution. The coefficient of $t^{r(n-1)}$ in $\prod_{j=1}^{k-1}(1+jt^{n-1})$ is the elementary symmetric function $e_r(1,2,\dots,k-1)$, a polynomial in k (by Newton's identities in terms of power sums $\sum_{j<k} j^s$, each a polynomial in k of degree $s+1$) of degree $2r$, with leading term $k^{2r}/(2^r r!)$. E.g. $e_1 = \binom{k}{2}$, $e_2 = \frac{(k-2)(k-1)k(3k-1)}{24}$.

Exercise 39.14 (medium)**.** Check Theorem 39.6(4) against the integral sphere table of Section 36.4 for $i = 1$ and $p = 2, 3$.

SOURCE: original

Solution. $H_1(B_k(S^2)) = \mathbb{Z}/(2k-2)$; its p-torsion is $\mathbb{Z}/p^{v_p(2k-2)}$. Condition (4): $(2k-2)/(2j-2)$ a p-local unit, i.e. $v_p(2k-2) = v_p(2j-2)$, exactly when the p-torsion agrees. E.g. $p=3$: $k = 2, 3$ ($2k-2 = 2, 4$, $v_3 = 0$) give no 3-torsion; $k = 4$ (6) gives $\mathbb{Z}/3$; $k = 10$ (18) gives $\mathbb{Z}/9$. The hypothesis $i \leq \min(k,j)/2$ requires $k, j \geq 2$.

39.6 Connections and further reading

McDuff [99] and Segal [107] for scanning; Church [72], Church–Farb [73] and Church–Ellenberg–Farb [74] for representation stability; Randal-Williams [104], Bendersky–Miller [66], Kupers–Miller [93], Knudsen [90] for stability via different methods; Miller–Wilson [100] and Himes [85] for secondary stability.

Chapter 40

Compactifications, little disks and factorisation homology

40.1 Why compactify

$F_k(M)$ is an open manifold. Integration of differential forms (to build models, Chapter 37), Poincaré–Lefschetz duality, and the operad structure of configuration spaces all require a compact version with controlled boundary. The naive closure M^k is useless: it forgets *how* points collide. The Fulton–MacPherson compactification remembers the directions and relative speeds of colliding points.

40.2 The Fulton–MacPherson–Axelrod–Singer compactification

Construction 40.1. Let $M \subset \mathbb{R}^N$ be a smooth closed manifold. Consider

$$F_k(M) \to M^k \times \prod_{i \neq j} S^{N-1} \times \prod_{(i,j,l)\ \text{distinct}} [0,\infty], \qquad x \mapsto \Big(x,\ \frac{x_j - x_i}{|x_j - x_i|},\ \frac{|x_i - x_j|}{|x_i - x_l|}\Big).$$

This is an embedding; the closure of the image is the *Fulton–MacPherson compactification* $\mathrm{FM}_M(k)$ (in Sinha's formulation [108]).

Theorem 40.2. *$\mathrm{FM}_M(k)$ is a compact manifold with corners whose interior is $F_k(M)$; the inclusion $F_k(M) \hookrightarrow \mathrm{FM}_M(k)$ is a homotopy equivalence (indeed $\mathfrak{S}_k$-equivariant). Its strata are indexed by collections of nested subsets of $\{1,\dots,k\}$ (equivalently, rooted trees with leaves labelled by $\{1,\dots,k\}$), the subsets recording which points have collided "at the same scale"; the codimension of a stratum is the number of nontrivial subsets.*

See [108], which gives global coordinates and identifies the strata, building on Fulton–MacPherson [84] (algebraic) and Axelrod–Singer (real). That the inclusion of the interior of a compact manifold with corners is a homotopy equivalence is a collar argument.

Construction 40.3 (FM_n)**.** For $M = \mathbb{R}^n$, divide out translations and positive dilations first: $\mathrm{FM}_n(k)$ is the closure of the image of $F_k(\mathbb{R}^n)/(\mathbb{R}^n \rtimes \mathbb{R}_{>0})$ in $\prod_{i\neq j} S^{n-1} \times \prod [0,\infty]$. It is compact, of dimension $nk - n - 1$ for $k \geq 2$ [86, §1.4], and $\mathrm{FM}_n(2) = S^{n-1}$. A codimension-one boundary stratum, where the points of a subset T ($2 \leq |T| < k$) collide, is $\mathrm{FM}_n(k - |T| + 1) \times \mathrm{FM}_n(|T|)$: the macroscopic configuration with T shrunk to one

point, and the infinitesimal configuration of T. These inclusions of boundary strata are the composition maps of an operad FM_n, weakly equivalent to the little n-disks operad D_n (Example 25.5).

40.3 Formality of the little disks operads

Theorem 40.4 (Kontsevich; Tamarkin; Lambrechts–Voli; Petersen; Fresse–Willwacher)**.** *The little n-disks operad is formal over $\mathbb{R}$: the operad of chains $C_*(\mathsf{D}_n;\mathbb{R})$ is connected by a zig-zag of quasi-isomorphisms of dg operads to its homology $\mathsf{e}_n = H_*(\mathsf{D}_n;\mathbb{R})$ (Theorem 25.7); the same holds in the setting of Hopf cooperads (commutative dg algebras). Over $\mathbb{Q}$ formality also holds.*

Kontsevich's proof [91], detailed by Lambrechts–Voli [96], integrates forms over the fibres of the forgetful maps $\mathrm{FM}_n(k+l) \to \mathrm{FM}_n(k)$ along graphs: to a graph Γ with k external and l internal vertices one assigns the form $\int_{\text{fibre}} \bigwedge_{\text{edges } (i,j)} \pi_{ij}^* \mathrm{vol}_{S^{n-1}}$ on $\mathrm{FM}_n(k)$. These forms span a cdga $\mathrm{Graphs}_n(k)$, and the zig-zag is

$$\Omega^*_{\mathrm{PA}}(\mathrm{FM}_n) \xleftarrow{\sim} \mathrm{Graphs}_n \xrightarrow{\sim} \mathsf{e}_n^\vee = H^*(\mathrm{FM}_n),$$

the second map killing graphs with internal vertices. The descent from $\mathbb{R}$ to $\mathbb{Q}$ uses a separate argument (Guillén Santos–Navarro–Pascual–Roig; see the remarks in [86]). For $n = 2$, formality of $F_k(\mathbb{R}^2)$ alone is Exercise 37.10; the content of Theorem 40.4 is compatibility with the operad structure.

40.4 Graph complexes

Definition 40.5. Kontsevich's graph complex GC_n is spanned by isomorphism classes of connected graphs with all vertices of valence ≥ 3, with an orientation datum depending on the parity of n (ordering of the edges for n even; ordering of the vertices and orientation of the edges for n odd), graded so that a graph with v vertices and e edges has degree $n(v-1)-(n-1)e$ (one common convention), and with differential splitting vertices.

Theorem 40.6 (Willwacher)**.** *$H^0(\mathrm{GC}_2)$ is isomorphic to the Grothendieck–Teichmüller Lie algebra $\mathfrak{grt}_1$. The homotopy derivations of the Gerstenhaber operad e_2 are parametrised by $\mathfrak{grt}_1$ up to one class (or two, depending on conventions); more generally, homotopy derivations of e_n are expressed through the cohomology of a graph complex.*

See [111]. Graph complexes govern the *automorphisms* of the formality zig-zag, hence the ambiguity in the real models of configuration spaces; the Campos–Willwacher models (Chapter 37) are built from decorated graph complexes.

40.5 Factorisation homology

Let A be an E_n-algebra (an algebra over D_n) in spaces or chain complexes, and M a framed n-manifold. Factorisation homology $\int_M A$ is the homotopy colimit of $A^{\otimes k}$ over embeddings of disjoint unions of k disks into M. It is characterised by Eilenberg–Steenrod-type axioms (excision for collar gluings, value A on $\mathbb{R}^n$) [62].

Example 40.7. For the free E_n-algebra on a space X, $\int_M \mathrm{Free}_{\mathsf{E}_n}(X) \simeq \coprod_k F_k(M) \times_{\mathfrak{S}_k} X^k$, the configuration space with labels in X (for framed M). With $X = \mathrm{pt}$ this is $\coprod_k B_k(M)$. Knudsen's theorem (Theorem 35.5) computes the rational homology of this via the rational Koszul duality between E_n-algebras and shifted Lie algebras (Chapter 30).

Theorem 40.8 (Nonabelian Poincaré duality; Salvatore, Segal, Lurie)**.** *For an $(n-1)$-connected based space X and a framed n-manifold M, $\int_M \Omega^n X \simeq \mathrm{Map}_c(M, X)$, the space of compactly supported maps.*

[62] give a concise proof. For $M = \mathbb{R}^n$ this is a tautology; for $M = S^1 \times \mathbb{R}^{n-1}$ it gives the free loop space $L\Omega^{n-1}X$. Combined with Example 40.7 and the approximation theorem $C(\mathbb{R}^n; X) \simeq \Omega^n\Sigma^n X$ for connected X (May, Segal; Theorem 25.6), it is the modern form of McDuff's scanning (Theorem 39.3).

40.6 Exercises

Exercise 40.9 (easy)**.** Show that $\mathrm{FM}_n(2) = S^{n-1}$ and that $\mathrm{FM}_M(2)$ is the complement of an open tubular neighbourhood of the diagonal in $M \times M$, up to homeomorphism; i.e. the blow-up of $M \times M$ along the diagonal, whose boundary is the unit sphere bundle $S(TM)$.

Source: original

Solution. $F_2(\mathbb{R}^n)/(\mathbb{R}^n \rtimes \mathbb{R}_{>0}) \cong S^{n-1}$ via $x \mapsto (x_2 - x_1)/|x_2 - x_1|$, already compact (no ratios for $k = 2$). For M: points of $\mathrm{FM}_M(2)$ are pairs (x_1, x_2) with a direction $u \in S^{N-1}$ which is $(x_2 - x_1)/|x_2 - x_1|$ if $x_1 \neq x_2$, and a unit tangent vector at x_1 if $x_1 = x_2$ (the limits of secant directions are tangent directions). So the boundary is $S(TM)$, and a collar of it corresponds to a tubular neighbourhood of the diagonal (whose normal bundle is TM) minus the diagonal.

Exercise 40.10 (medium)**.** Describe $\mathrm{FM}_2(3)$: its dimension, the homotopy type of its interior, and its codimension-one boundary strata. Check $\chi(\partial \mathrm{FM}_2(3)) = 0$.

Source: original

Solution. $\dim = 2 \cdot 3 - 2 - 1 = 3$. Interior: $F_3(\mathbb{C})/(\mathbb{C} \rtimes \mathbb{R}_{>0})$; normalising $z_1 = 0$, $|z_2| = 1$ gives $\{(0, u, z) : u \in S^1, z \neq 0, u\} \cong S^1 \times (\mathbb{C} \setminus \{0, 1\}) \simeq S^1 \times (S^1 \vee S^1)$. Codimension-one strata: one for each $T \subset \{1, 2, 3\}$ with $|T| = 2$, each $\mathrm{FM}_2(2) \times \mathrm{FM}_2(2) = S^1 \times S^1$. (No stratum for $|T| = 3$: that is the whole space after rescaling.) Two distinct 2-element subsets of $\{1, 2, 3\}$ are neither nested nor disjoint, so there are no strata of codimension 2 and the three tori are disjoint: $\partial \mathrm{FM}_2(3) = \coprod_3 T^2$ has $\chi = 0$, as the boundary of a compact odd-dimensional manifold must.

Exercise 40.11 (medium)**.** Using Exercise 40.10, show that $\chi(\mathrm{FM}_2(3)) = 0$ in two ways: from its interior, and from the formula $\chi(\partial W) = (1 - (-1)^{\dim W})\chi(W)$ for compact manifolds with boundary.

Source: original

Solution. From the interior, $\chi(S^1 \times (S^1 \vee S^1)) = 0 \cdot (-1) = 0$. From the formula, $\dim W = 3$, so $\chi(\partial W) = 2\chi(W)$; with $\chi(\partial W) = 0$ this gives $\chi(W) = 0$.

40.7 Connections and further reading

Fulton–MacPherson [84], Sinha [108]; Kontsevich [91], Lambrechts–Voli [96] and Fresse's books [43] for formality; Willwacher [111] for graph complexes; Ayala–Francis [62] and Knudsen [90] for factorisation homology; Idrissi's lectures [87] for an introduction aimed at configuration spaces of manifolds. Kallel [89, §9] surveys compactifications and their use in knot theory (configuration space integrals).

Chapter 41

A computational laboratory

41.1 Purpose and ground rules

Computation is part of the subject: Napolitano's and Schiessl's integral tables, the Betti numbers of $B_k(T^2)$, and the current work on odd torsion all rely on machine calculations. This chapter describes three short programs that accompany the book, what they compute, how their output was checked, and how to extend them. The ground rules are those of the rest of the book: a computation is evidence, not proof, unless the algorithm is proved correct and the program is checked; every table below states which it is.

Program	Computes	Checked against
`code/snf.py`	Smith normal form, integral homology of a chain complex	hand computations of Chapter 4
`code/fuks.py`	$H^*(\mathrm{Br}_n;\mathbb{Z})$, $H^*(B_n(S^2);\mathbb{Z})$	[105, Tables 3, 4], Theorem 36.5
`code/arnold.py`	graded dimensions and $\mathfrak{S}_k$-invariants of the Arnold algebra	Theorems 34.3 and 34.7
`code/checks.py`	the consistency checks of Exercises 41.2 to 41.4	—

Run them with `python3 code/fuks.py 11`, `python3 code/arnold.py 6` and `python3 code/checks.py` from the book directory (the last one takes a few seconds and prints `True` for each check).

41.2 Smith normal form

`snf.py` implements the algorithm in the proof of Theorem 4.18: repeatedly move an entry of least absolute value to the pivot, clear its row and column by integer row and column operations, and restore divisibility. By Corollary 4.19, $H_n \cong \mathbb{Z}^{\dim C_n - \operatorname{rank} d_n - \operatorname{rank} d_{n+1}} \oplus \bigoplus_i \mathbb{Z}/e_i$, where the $e_i > 1$ are the invariant factors of d_{n+1}. Pure-Python integer arithmetic is exact, and the matrices in this book are small, so coefficient explosion is not an issue; for large complexes one would use modular methods or a library such as PARI or Sage.

41.3 The Fuks complex

`fuks.py` enumerates compositions, builds the matrices of δ (Theorem 36.2) and of the sphere complex (Section 36.4), *checks* $\delta^2 = 0$ and $D\delta = \delta D$ for every generator, and computes cohomology by Smith normal form. Running time for $n \le 11$ is a few seconds. The output is reproduced in Chapter 36; it agrees with Schiessl's tables in every entry ($n \le 11$), and with Theorem 36.5: finiteness above degree 1, vanishing for $i \ge n$, the recurrence $n = 2m + 1 \leftrightarrow 2m$, and the absence of elements of order p^2.

Caution 41.1. The signs in δ matter for the integral answer: with all signs replaced by $+$, $\delta^2 \neq 0$ in general. Mod 2 the signs are irrelevant, which is why Fuks' mod-2 computation is the easiest. (`checks.py` finds 33 compositions with $n \le 7$ for which the unsigned differential squares to a nonzero map.) The program checks $\delta^2 = 0$, which is a necessary but not sufficient test of the sign conventions; the agreement with an independently published table is the real check.

41.4 The Arnold algebra

`arnold.py` builds $\Lambda[\omega_{ij}]/(\omega_{ij}^2, \text{Arnold relations})$ degree by degree as a quotient of the exterior (for n even) or square-zero commutative (for n odd) algebra on the ω_{ij}, computes the rank of the ideal in each degree by exact Gaussian elimination over $\mathbb{Q}$, and computes the dimension of the $\mathfrak{S}_k$-invariants of the quotient as the rank of the image of the averaging operator. Output ($k \le 6$, $n = 2, 3$; about two minutes for $k = 6$):

k	graded dimensions (both parities of n)
2	$1, 1$
3	$1, 3, 2$
4	$1, 6, 11, 6$
5	$1, 10, 35, 50, 24$
6	$1, 15, 85, 225, 274, 120$

These are the coefficients of $\prod_{j<k}(1 + jt)$, confirming the counting step of Theorem 34.3. The invariants are 1-dimensional in degrees 0 and $n - 1$ for $n = 2$, and only in degree 0 for $n = 3$, for every $k \le 6$: Theorem 34.7 in this range.

41.5 Projects

Exercise 41.2 (medium)**.** Modify `fuks.py` to compute $H^*(\mathrm{Br}_n; \mathbb{F}_p)$ directly (reduce the matrices mod p and compute ranks over $\mathbb{F}_p$), and check the result against the universal coefficient theorem applied to the integral output, for $p = 2, 3, 5$ and $n \le 11$.

SOURCE: original

Solution. Over $\mathbb{F}_p$, $\dim H^r = \dim A^r - \operatorname{rank}_p \delta^r - \operatorname{rank}_p \delta^{r-1}$. The universal coefficient theorem gives $\dim H^r(\,;\mathbb{F}_p) = \operatorname{rank} H^r + t_p(H^r) + t_p(H^{r+1})$, where t_p is the number of cyclic p-primary summands. Since by Theorem 36.5(d) every summand is $\mathbb{Z}/p$, $t_p(H^r)$ is the number of invariant factors of δ^{r-1} divisible by p, which equals $\operatorname{rank}_{\mathbb{Q}} \delta^{r-1} - \operatorname{rank}_p \delta^{r-1}$; the two formulas agree identically. `checks.py` confirms the agreement for $p = 2, 3, 5$ and $n \le 11$. For instance for $n = 6$, $p = 3$ one finds $\mathbb{F}_3$ in degrees $0, 1, 4, 5$ (Exercise 36.9).

Exercise 41.3 (medium)**.** Write a program counting the monomials of Theorem 36.5(c) by weight and degree, and compare with the $\mathbb{F}_2$-Betti numbers of Exercise 41.2 for $n \le 11$.

SOURCE: original

Solution. The generating function is $\prod_{i\ge 0}(1-t^{2^i-1}z^{2^i})^{-1}$ (t for degree, z for weight); expanding to z^{11} and comparing coefficient by coefficient with the mod-2 ranks is a finite check; `checks.py` performs it and it succeeds for $n \le 11$. (For $n = 6$ this is Exercise 36.8.)

Exercise 41.4 (hard)**.** Extend `arnold.py` to compute the full character of $\mathfrak{S}_k$ on each $H^{r(n-1)}(F_k(\mathbb{R}^n);\mathbb{Q})$ (the trace of a permutation is computed in the quotient using the pivots of the elimination). Verify that for n odd the total cohomology is the regular representation and for n even it is $2\ \mathrm{Ind}_{\mathfrak{S}_2}^{\mathfrak{S}_k}\mathbf{1}$, as stated in [89, §5.5.3], for $k \le 5$.

SOURCE: original; the statements are classical (Lehrer, Cohen–Taylor), quoted from the survey

Solution. The trace of σ on the quotient $Q = \Lambda/I$ is $\mathrm{tr}(\sigma|\Lambda) - \mathrm{tr}(\sigma|I)$; with an echelon basis of I one computes the matrix of σ on I by reducing the images of the basis vectors. For n odd, the total character is $k!$ at the identity and 0 elsewhere (the regular character); for n even, it is $2\cdot\#\{\text{fixed points of } \sigma \text{ on } \mathfrak{S}_k/\mathfrak{S}_2\}$. `checks.py` computes the characters and confirms both identities for $k \le 5$; e.g. for $n = 2$, $k = 4$ the character is $24, 4, 0, 0, 0$ on the classes $1^4, 1^2 2, 13, 2^2, 4$. The second is consistent with Exercise 34.16: at the identity, $2\cdot k!/2 = k!$ is the total dimension $\prod_{j<k}(1+j)$.

41.6 Further tools

For larger computations: SageMath (homology of simplicial complexes, group cohomology via GAP/HAP), Kenzo (effective homology), and the `uconf` SageMath library of Idrissi and Roca i Lucio for mod-p homology of unordered configuration spaces [88]. Discrete models of configuration spaces of graphs (Abrams, witkowski) and discrete Morse theory make configuration spaces of graphs and low-dimensional complexes accessible to computers.

Chapter 42

A research map

42.1 How to read this chapter

Each entry states a question, what is known, and the basis of the status claim. "Open" means: stated as open in the cited source, and we found no later work settling it in our (non-exhaustive) search in October 2026. Results from preprints are marked as such. The reader should re-check the status of any entry before working on it.

42.2 Homotopy theory of configuration spaces

Question 42.1 (Homotopy invariance for simply connected manifolds)**.** *If $M \simeq N$ are simply connected closed manifolds, is $F_k(M) \simeq F_k(N)$?*

Known. False without simple connectivity (Theorem 38.4); true after $\beta(k, d, r)$ suspensions (Theorem 38.3); true for $k = 2$ and M 2-connected (Theorem 38.2); true over $\mathbb{R}$ (Theorem 37.6). *Status.* Open according to [89, §5.1] (June 2025).

Question 42.2 (Rational descent)**.** *Is the Lambrechts–Stanley cdga $G_A(k)$ a* rational *model of $F_k(M)$ for simply connected closed M?*

Known. Real version [86]; Idrissi conjectures the rational one and explains that there is no general descent argument for models with nonzero differential. *Status.* Conjecture in [86].

Question 42.3 (Lens spaces)**.** *Do the homotopy types of configuration spaces distinguish lens spaces up to homeomorphism?*

Known. Asked in [98, §5]; studied for F_2 of 3-dimensional lens spaces by Evans-Lee and Saveliev (arXiv:1502.03408, using Cheeger–Simons characters and Massey products). *Status.* We have not verified the current state; partial results.

42.3 Integral homology and torsion

Question 42.4 (Odd torsion on the torus)**.** *For an odd prime p, what is the p-torsion of $H_*(B_k(T^2); \mathbb{Z})$? In particular, where does the first 3-torsion occur?*

Known (Section 36.5). No p-torsion for $k < 2p$; for $p \geq 5$, p-torsion in $H_{2p-2}(B_{2p}(T^2))$; no 3-torsion for $k \leq 6$, so the first 3-torsion (if any) is at weight

≥ 7. *Status.* The last three statements are from the preprint [88] (v2, August 2026). Exponents and full decompositions beyond Napolitano's range $k \leq 6$ are unknown.

Question 42.5 (Exponents)**.** *For closed surfaces Σ, is the p-primary torsion of $H_i(B_k(\Sigma);\mathbb{Z})$ of exponent p for i large? For which (Σ, p)?*

Known. Yes for $\mathbb{R}^2$ (Vainshtein, Theorem 36.5(d)); for S^2 in degrees ≥ 4 [105, Thm. 1.3], but not in degree 2; for $B_{2p}(\Sigma_{1,1})$ in the relevant degree [67]. *Status.* Not settled in general in the sources we read.

Question 42.6 (Graphs)**.** *Is the homology of ordered configuration spaces of graphs torsion-free? Is there odd-primary torsion in the homology of unordered configuration spaces of graphs?*

Known. Torsion-freeness is known for restricted classes (e.g. trees with loops, Chettih–Lütgehetmann); the general statement is described as a folklore conjecture, and no odd torsion is known for unordered graph configurations, in the introduction of [88] (2026).

42.4 Spectral sequences and Massey products

Question 42.7 (Higher differentials)**.** *When does the Cohen–Taylor–Totaro spectral sequence collapse? Are its higher differentials determined by (matric) Massey products of M, as conjectured by Bendersky and Gitler?*

Known. Collapse for $k \leq 3$, and non-collapse examples for $k = 4$ from triple (matric) Massey products (Theorem 37.8). *Status.* Conjecture as recorded in [81] (2004); we have not traced later progress.

42.5 Stability

Question 42.8 (Secondary and periodic phenomena)**.** *Describe the unstable range of $H_*(B_k(M);\mathbb{Z})$ for closed M: secondary stability, mod-p periodicity, and their interaction with torsion.*

Known. Rational stability for $k > i$ (Theorem 39.4); mod-p periodicity [93]; torsion stability under the conditions of Theorem 39.6; secondary stability [85, 100]. The torus results above show that the torsion on the "diagonal" $i \approx k$ is not controlled by stability.

42.6 Where to start

For a first project, the computational ones are the most accessible: reproduce a published table (Chapter 41), extend it, and compare with theory. For a reading project: Kallel's survey [89], then Church [72], Idrissi [86], and the 2026 preprint [88] with its bibliography, which is an up-to-date entry point to the torsion questions.

A guide to the literature

This chapter says what the standard references are good for; full data are in the bibliography.

Foundations (Parts I–II). Hatcher [1] is the default reference and the source of many exercises of Parts I–III (with exact numbers in the exercise headers). May [4] is terse and structural, with the best short treatment of cofibrations and fibrations; Miller's notes [21] follow the MIT courses whose problem sets are used in this book. For homological algebra: Weibel, Cartan–Eilenberg and Mac Lane's *Homology*; the algebra needed here is in Chapters 8 and 9.

Spectral sequences and operations (Part III). Hatcher's spectral sequence notes [16] for running the Serre spectral sequence and for Serre classes; McCleary [8] for the general theory, the Bockstein spectral sequence (Ch. 10) and the Cartan–Leray and Lyndon–Hochschild–Serre spectral sequences (Ch. 8bis); Steenrod–Epstein [14] and Mosher–Tangora [15] for operations; Milnor–Stasheff [10] and Hatcher [9] for characteristic classes; Steenrod [11] and Whitehead [12] for obstruction theory.

Abstract homotopy theory (Part IV). Mac Lane [19], Riehl [20], Dwyer–Spaliński [26], Hovey [24], Hirschhorn [25], Goerss–Jardine [27], Lurie [31], Riehl–Verity [32].

Rational homotopy and operads (Parts IV–V). Félix–Halperin–Thomas [17] is the standard reference; Félix–Oprea–Tanré [36] for geometric examples; Loday–Vallette [42] and Fresse [43] for operads.

Stable homotopy (Part V). Adams [48], Hovey–Shipley–Smith [49], Schwede [51], Ravenel [47].

Configuration spaces (Part VI). Kallel's survey [89] is a broad map of the field; Idrissi's Peccot lectures [87] for real homotopy of configuration spaces of manifolds; Fadell–Husseini [77] for the classical fibration methods; Cohen–Lada–May [40] for the homology of configuration spaces of Euclidean spaces. Further references are given at the end of each chapter of Part VI.

Results stated without proof. Theorems quoted but not proved in this book are marked "recalled" or come with an explicit "proof sketch" and a precise reference at the point of use. Among the deepest: Adams' theorem on Hopf invariant one [13]; the homotopy classification of lens spaces (Warning 7.19); Milnor–Moore [41]; the Serre–Cartan theorem on the Steenrod algebra and Serre's computation of $H^*(K(\mathbb{Z}/2, n))$;

Brown's theorem on $H^*(BSO(n);\mathbb{Z})$ [44]; Sullivan's realisation theorem [34]; and, in Part VI, the main theorems of the research literature surveyed there.

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